DEVELOPMENTS IN INTEGRATED ENVIRONMENTAL ASSESSMENT

VOLUME 1A

INTEGRATED AND PARTICIPATORY WATER RESOURCES MANAGEMENT: THEORY

by

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Introduction

Take care of the earth and the water:
they were not given to us by our fathers,
but loaned to us by our children.

A nomadic shepherd's saying from Kenya

The issue of water

It will be water that shapes the new century, just as petroleum shaped the one that has just passed. Over the last century the population of the planet has tripled, while water consumption has increased by six or seven times. Consumption of water has increased at double the rate of the population, and as a consequence 30% of humanity does not have sufficient water and each year 7 million people die from diseases caused by polluted water. The forecasts are that in 2025 the world population will be about 8 billion and that the fraction with water scarcity will rise to 50% (Rosegrant et al., 2002). The deficit will be particularly severe in Asia and in sub-Saharan Africa, that is in those very countries that are today among the poorest in the world, but will occur also in regions that today are neither arid or even semi-arid.

In developing countries, where agriculture is an important component of the economy, irrigation uses from 75 to 90% of the fresh water derived from rivers or pumped from aquifers, but also in developed countries, where agriculture employs less than 5% of the inhabitants, agricultural water consumption is still very high, between 50 and 65% of the total. This means that the competition for water between agriculture and the other sectors is very high and destined to increase with population growth (Bonell and Askew, 2000).

It is predicted that the expansion of water demand will produce, each year, in the most critical months, the drying up of many rivers before they reach the mouth. The phenomenon is not new, however. Already, in the summer of 1972, the Huang Ho (Yellow River) ran completely dry for a few weeks close to its mouth (Brown et al., 1998). This phenomenon was repeated occasionally in the following years, but from 1985 it has occurred regularly every year. In 1997 the mouth of the river was dry for 226 days and for many months the river flow did not even reach Shandong, the last province that the river crosses on its journey to the sea. This is not the only case. The Colorado River rarely reaches the Gulf of California, because its waters are totally withdrawn to satisfy Arizona's thirst, and above all California's. The water volume that the Nile dumps into the Mediterranean is negligible by now, just like the volume that the Ganges brings to Gulf of Bengal (Brown, 2001).

For poor countries, to be able to have enough safe water is an essential condition for getting out of poverty. In rich countries the recovery of the water quality in rivers and lakes, which has been sacrificed in the past to economic progress, is an essential condition for

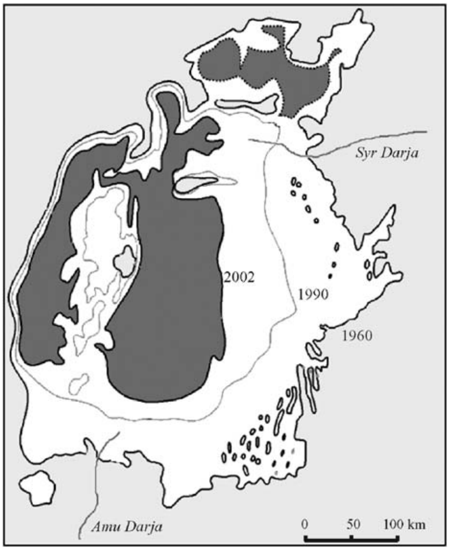

Figure 1: The progressive drying out of Lake Aral.

improving the quality of life. But to improve the quality and quantity of water available for human and environmental aims is a difficult task, since these objectives are often in conflict with each other. To increase availability, consumption can be rationalized (the volume of water dumped each time that a westerner uses the toilet is the volume that the average inhabitant of the third world uses in a day to drink, wash him/herself, wash, and cook) or the exploitation of the resource can be made more efficient.

In both cases the growing demand will increase the competition among water users at local, national and international levels. Ismail Serageldin, at the time vice-president of the World Bank and chairman of the Global Water Partnership, has declared more than once, and in no uncertain terms, that *the wars of the XXI century will be fought over water* (Homer-Dixon, 1996). He has been criticized a great deal for his thesis, but just as many have supported it (Starr, 1991; Bulloch and Darwish, 1993; Ohlsson, 1995). In 2003, at the Third World Water Forum, Klaus Toepfer, executive director of UNEP (United Nations Environment Programme), in presenting the Atlas of International Freshwater Agreements (Wolf, 2002b) indirectly confirmed this thesis. *The work* he revealed *shows the need to monitor, to adopt scientific rigor and diplomatic energy to assure that cooperation between the states be maintained and extended. Although 3000 treaties and agreements have been signed in the last century in 100 transnational river basins, another 158 are still without one*. These river basins collect 60% of the world's fresh water and host 40% of the population, and their number increases with political instability. The dissolution of the USSR, for example, made the Dnieper, the Don and the Volga international, and the Lake Aral river basin, in Central Asia, was left divided between five ex-Soviet republics. Tensions rose very quickly between these young nations over the sharing out of the waters of the Amu Darya and the Syr Darya, the two rivers that feed the lake and over the interventions aimed at improving the environmental and human disaster produced by its drying out (see Figure 1), which was the result of 40 years of massive derivations from these rivers to cultivate cotton in the deserts of Central Asia.

Until 1950 only one war generated by a dispute over water had been recorded, but in the following fifty years one quarter of such disputes have been hostile. This is a clear sign of growing tensions. In most cases the hostilities do not go beyond the verbal level, but unresolved tensions over water have nevertheless exacerbated relations and fuelled other reasons for hostility. In 37 cases military action has been taken, mostly limited to the destruction of dams. Almost all of these conflicts have developed in the same way: the construction of a big dam or a big project has created a prolonged period of regional instability and hostility, followed by a long and difficult process of dispute resolution (Postel and Wolf, 2001).

Competition is clearly not limited to nations, but arises also within them, between regions and economic sectors. Throughout the world, agriculture, cities, industry and environment compete for water and this competition affects in turn the relationships between political entities (cities and provinces) and their neighbours. For example, in Pakistan there is currently a bitter conflict between the regions of the Punjab and the Sind for the water of the lower Indus; and in Thailand between the north and the south of the country for the Chao Phraya, which feeds Bangkok. To satisfy the thirsty cities, water is often taken from agriculture. Sometimes, the farmers, who can no longer irrigate, react violently, as happened in 2000 in Shandong, when thousands entered into a bloody conflict with the police to block repairs to a large dam on the Huang Ho, whose leakage they had been using for some time to irrigate their fields. But even when it does not come to conflicts, the consequences are not the best because the farmers abandon the fields and go to increase the numbers of the unemployed masses in the overflowing and thirsty cities. This has happened in Pakistan, where the crisis of irrigated agriculture has produced an enormous emigration to the big cities, which in turn has led to repeated explosions of ethnic violence.

What can be done?

Unfortunately water cannot be produced in significant volumes within acceptable costs. The quantity of fresh water available is essentially invariable in time, which means that today, to satisfy 6 billion people, we have the same amount of annual flow (nearly 34 000 billion cubic metres per year) that was available 4000 years ago, when, in China and in Mesopotamia, the first great irrigation empires were formed and the population of the planet did not exceed 100 million inhabitants.

If we cannot increase the overall flow, we could try to increase the fraction of it that we use; however, this also is a blind alley, as today we capture little more than half (about 54%), and the residual is very difficult to acquire.

The possibility remains to reallocate the resource, both in space and time. To achieve this we need canals and reservoirs. In the last half century, the creation of these structures has proceeded at a frenetic pace. While in 1950 there were only 5000 ‘large sized’[1] reservoirs in the world, in 1994 there were more than 38 000 and together they intercepted 16% of the total flow of the rivers on the planet, with very significant economic and environmental effects (Silvester et al., 2000). It is not possible to construct a significantly greater number of dams, also because the marginal yield of the investment decreases rapidly, given that the best sites have already been used.[2]

[1] Defined as reservoirs formed by a dam higher than 16 metres.

[2] This situation is well reflected by the fall in financing for the construction of dams. For example, while in the period 1970–1985 financing from the World Bank for new dams represented 3% of the overall financing provided by the bank, the percentage drops to 2% in the ten years from 1986 to 1995, to decrease further to 0.9% in 2001 [www.worldbank.org].

Only one possibility remains: manage the water that we already have better. In other words, the conflicts will have to be resolved by moving the resource between bordering regions or between economic sectors. But, to avoid producing new conflicts, it must be done only with the agreement of the interested parties, in a collaborative way. Some have already begun to pioneer this way. The municipalities of some big cities that are particularly thirsty, such as for example Los Angeles and Beijing, subsidize the reduction of leaks in the irrigation system in peripheral agricultural areas, in exchange for the water that is recovered. In this way the farmers continue to farm and the cities obtain additional water at reasonable cost. Cooperation replaces competition.

The solution to the crisis will not come either from the 'State', or the 'market' alone, but, as the Second World Water Forum (GWP, 2000) concluded, from a change in the management paradigm with which water resources have been managed until now:

> *There is a need for profound change in the way water is managed, if we are to achieve any sense of sustainable water use in the near future. The empowerment of people at the local level to manage their water resources – the 'democratization' of water management – is essential.*

This is therefore the path to follow for a better future: to adopt participatory decisions. We must transform the drawing up of the Plan or the Project from a technical act to a *political* process, in the noble sense of the word. A process in which all the *Stakeholders* must be involved: the institutions, organizations and individuals that are interested in the decisions to be taken; because, directly or indirectly, they will feel their effects, or because they have power to influence or thwart the decision. But this is not yet sufficient. Besides the plurality of the *U*sers, two other *U*s must be considered (as an effective slogan created by UNESCO says): the *Uncertainty*, that is intrinsic to the dynamics of water resources, and the complexity of its *Uses*. In other words, it is necessary to consider and integrate the physical aspects (hydrological, climatological, ecological), as much as the non-physical ones (technical, sociological, economic, administrative, legal), considering them from all the points of view from which the different users judge them. And since water does not respect administrative and political boundaries, management must be carried out at the level of the river basin, which is the natural hydrological and geological unit, crossing boundaries if it is transnational, and it must be extended to all the basins that are involved when inter-basin water transfer is being considered. For the same reason, the quality and the quantity of the resource must be considered jointly and simultaneously, because they are not two different problems, but simply different aspects of the same problem.

In summary: the point of view must be holistic and decisions integrated and participatory, so that they enjoy a wide consensus from the Stakeholders, or, as they say in jargon, *consensus is built* around them. This is the management paradigm that is proposed today and that is synthesized in the acronym IWRM: Integrated Water Resources Management (GWP, 2003). To put this into practice it is necessary to activate a decision-making process that

> *...promotes the coordinated development and management of water, land and related resources, in order to maximize the resultant economic and social welfare in an equitable manner, without compromising the sustainability of vital-ecosystems.* (GWP, 2003)

Such a process must begin from the periphery and move towards the centre, from the particular towards the general, briefly *bottom–up*, and not *top–down* as is traditional, to construct a viewpoint that is holistic and shared, which embodies all the viewpoints of the individuals as partial, but equally considered, viewpoints.

The adoption of the IWRM paradigm encounters a serious difficulty, which has been summarized by UNESCO in the following way:

Water management policy is generally based on outdated knowledge and technology. In many cases, procedures are followed where Stakeholders are unaware of what technical alternatives are available and scientists do not realize what is required. This 'Paradigm Lock' has come about because the two main groups have become isolated: scientists by the lack of proven utility of their findings, and Stakeholders by legal and professional precedents and disaggregated institutions. (Bonell and Askew, 2000)

To overcome this impasse UNESCO and the WMO launched the HELP programme[3] which has the aim of creating a global network of river basins in which the new paradigm is tested, demonstrating its utility in a concrete way.

Good intentions and examples are still not enough to get IWRM applied. Two other conditions must be satisfied: legislation must provide a normative framework that supports it and scientists must collect data, formulate procedures and make all of this available in information systems that allow the paradigm to be applied.

The appropriate legislation is absolutely necessary, even if alone insufficient, to manage our waters in a participatory way (Wolf, 2002a). The European Union has now equipped itself and in 2000 it enacted the Water Framework Directive (WFD) [Directive/2000/60/EC], which is totally centred around the new paradigm. For example, Article 13, dedicated to the 'River basin management plans', establishes:

1. *Member States shall ensure that a river basin management plan is produced for each river basin district lying entirely within their territory.*
2. *In the case of an international river basin district falling entirely within the Community, Member States shall ensure coordination with the aim of producing a single international river basin management plan ...*
3. *In the case of an international river basin district extending beyond the boundaries of the Community, Member States shall endeavour to produce a single river basin management plan ...*

Article 14 requires that the said plans be created through participation:

Member States shall encourage the active involvement of all interested parties in the implementation of this Directive, in particular in the production, review and updating of the river basin management plans.

As a consequence of participation:

- the capabilities of the Stakeholders are mobilized;
- advantages can be taken by their knowledge and experience of the system, making it easier to respond to the challenge of complexity;
- they are made actors, instead of passive subjects;
- they are made aware of the 'whys' and therefore become conscious individuals.

In order to make participatory decisions we need procedures and software systems that allow us to:

- evaluate the effect of the decisions ex-ante;
- facilitate dialogue and exchange viewpoints among the Stakeholders;

[3] http://icm.landcareresearch.co.nz/Library/project_documents/HELP~strategy~document.pdf.

- identify their alliances and conflicts;
- negotiate, i.e. to look for a compromise between the opposing needs, that might improve the conditions for everyone (a so-called *win-win alternative*).

The identification and reciprocal recognition of different viewpoints, the construction of a shared model with which to evaluate the effects of decisions for each viewpoint, and the dialogue created by these activities are often much more important than the decision itself, because they activate a social learning process.

What this book is about

The aim of this book is to introduce the reader to the water resource planning and management according to the IWRM paradigm, the adoption of which requires the use of dedicated software systems called *Multiple Objective Decision Support Systems*, commonly denoted with the acronym MODSS. Since the IWRM paradigm cannot actually be applied without the help of an MODSS the aim of this book is to describe the nature and structure of such a system in order to teach the reader how to design them.

The creation of an MODSS is an activity at the crossroads of the following three worlds:

1. The world of the physical, biological, economic, and social sciences which study and describe the processes that occur in a water system, both physical (hydrological, climatological and ecological) and non-physical (technical, sociological, economic, administrative and legal).
2. The world of methodologies, mathematical and non-mathematical, that allow us to describe those processes in a quantitative way and to define the decision-making and management procedures to govern them: System Analysis, Optimal Control Theory, Operations Research, Decision Theory and Alternative Dispute Resolution (ADR).
3. The world of Information Communication and Technology (ICT), which allows us to create the software system in which an MODSS takes material shape.

We will assume that the reader knows enough about the first point to follow the subject matter. Actually, we do not expect very much. Since our aim is to introduce the project methodology, we will simply consider the most elementary case, which is extremely frequent, in which water is used for irrigation, domestic and industrial supply. It is enough to know that water runs downstream and that both the scarcity of water (drought), and the excess of water (floods), cause damage. We will concentrate on the second and third points in the above list. On the second, in order both to illustrate how the models are built and how they are embedded in a decision-making procedure, and to define the latter, without entering into the more behavioural and psychological aspects, such as how the Analyst, i.e. the 'manager' of the decision-making process, should interact with the Stakeholders and facilitate the negotiations among them (only a few glimpses will be provided in Appendix A10). We will concentrate on the third point in order to show that the software system design has to be done not after but simultaneously with the identification of the procedure and the models, given that the nature of the available MODSS conditions the nature of the applicable procedure.

Since the designing of MODSS is carried out at the interface of these three worlds, we will not go into great detail on any one of them, but will concentrate on their relationships, showing where these worlds begin and which their more typical technical aspects are. The reader may find these *technicalities* developed further in the specialist treatises suggested through the text.

To be able to reach this goal, and contain the work within reasonable limits, reluctantly, some issues have had to be totally excluded.

What this book is not about

The reader will not find anything about water demand projections or population growth forecasts, two issues that certainly cannot be ignored when drawing up a Plan or Project. In fact, the Plan or the Project is written today but its effects come about in a future from which we are usually separated by many years (for example in the case of a dam, on average not less than ten years will pass between its commissioning and its coming into service). There will be no mention of water quality, certainly not because it is not important (this is the subject of the WFD!). We will simply consider the most basic intervention actions (delivery from a dam, construction of a reservoir or a diversion, definition of the MEFs[4]), completely ignoring more complex interventions, such as the rationalization of consumption, the upgrading of a river, or the establishment of an early warning system for floods or drought. As for the water systems that are considered, we will only deal with surface water (catchments, rivers, lakes, and canals) completely ignoring groundwater and coastal waters or distribution systems with pressure networks. We will not even speak about the design of the information system or data collection networks. The reason for all these exclusions is because it is not possible in an introductory book, to take for granted that the reader has the knowledge necessary to describe these systems and processes, and to evaluate the effects that the proposed interventions will have on them. At the same time, providing this information would render this work disproportionate and the thread of the presentation would be lost.

In any case these exclusions do not constitute a limit, as the reader will see that what is learned can be extended to include all that has been excluded, provided that the necessary notions for describing those systems, processes, interventions, and their effects, are given. This is possible because the decision-making procedure is totally independent from the water system and the interventions that are being considered, and can therefore be explained only using the simplest systems and interventions as examples. To convince the reader of the truth of this statement we prepared Appendix A1, in which we demonstrate how the baseline scenario (BLS) for the Seine-Normandy river basin could have been created accordingly to the WFD requirements using the PIP procedure. The reader will thus see that the notions provided in the following chapters can be useful even when water quality issues and spatially distributed (*distributed-parameters*) systems are considered.

The book goes into the methodological aspects of water resource planning and management in the light of the IWRM paradigm. Explanations are furnished with references to real Projects,[5] which are introduced through dedicated boxes, in which the problem and the solution are framed. We are conscious of the fact the theoretical exposition alone will not allow

[4]Minimum Environmental Flow.

[5]For didactic reasons, when necessary, we freely modified the system, the problems, the clients or the events with respect to reality, eliminating details that would have distracted the reader or adding elements that make the example more interesting. Therefore the Projects should be considered realistic, but not real, and what is presented, should not be attributed in any sense either to the real system or to the actors involved.

the reader to understand the complex articulations of the decision-making procedure and the practical problems that the Analyst must face. For this reason we have prepared a second volume: *Integrated and Participatory Water Resources Management: Practice* (Soncini-Sessa et al., 2007): in the following it will be referred to simply as PRACTICE. We advise, even if it is not strictly required, that it be read in parallel with this book, of which it could be considered an appendix for frequent consultation.

Who this book is for

When writing a book it is necessary to keep in mind the reader who it is being addressed. Our reader is a professional or a university student who has the basics for understanding problems of water resource planning and management. It is not strictly necessary, but it would be ideal, if the reader had some basic knowledge of hydrology. Further, we will ask for the basic elements of statistics and mathematics, which are usually provided in first year university courses, and lastly, basic knowledge of System Analysis and Mathematical Programming. Since all these notions are not always offered in scientific faculties, the reader who does not have some of them can fill this gap by reading Appendices A2–A4, A6–A8 on the CD that accompanies this book.

Reading pathways

The book is structured like a *matryoshka* and has a *top–down* order: the issues are at first presented at a high level and then developed in detail.

For this reason we suggest that readers begin with Chapter 1, which illustrates the decision-making process with which a Project is developed: from the formulation to the implementation of the alternative to be carried out. Even expert readers should not skip Section 1.3, in which the decision-making procedure (PIP) upon which the whole work is based is described. It is, in fact, not only the procedure that we propose for the development of a Project according to the IWRM paradigm, but also the key to reading the whole book, as it is explained in that section. Once the reader knows it, (s)he can proceed in a non-sequential way, if that is what (s)he wants to do.

Following the *matryoshka* structure, Chapter 2 provides an overall view of all the problems that the adoption of the PIP procedure poses: in other words, it is a conceptual map of the entire work. In particular Section 2.8 explains the subdivision of the book into parts and the organization of Appendices. Each Part opens with an introductory chapter that provides a high level description of it.

Going deeper into detail, Part B is essentially dedicated to modelling aspects. As one can easily understand, since not less than eighty percent of the literature is dedicated to it, this part could easily have dominated the others. To avoid making this mistake, we gave it a double, precise aim: to give the reader the cultural instruments to move in the vast world of modelling literature to search for what (s)he might need in the particular Project that is of interest to him/her, and to help him/her thoroughly to understand the links that run between modelling and the rest of the decision-making procedure. The consequence could be the dissatisfaction of those readers that are searching for a specific model of a given component, because they might not find it here. Appendices A5–A8 aim at mitigating this possible disappointment a little.

Part C is totally dedicated to the design of alternatives in the case where there is just one Project objective, while Part D is concerned with that case in which, as is usually the case in practice, there are multiple objectives. This division was created because not only it is didactically more efficient to introduce many concepts for the Single-Objective case and then extend them to the general case but mainly because Multi-Objective Design Problems are always led to Single-Objective Design Problem. In Part D the evaluation and the comparison of the alternatives find their place (Chapters 20 and 21).

Finally, Part E is dedicated to information technology. Just as for Part B, the aim of this Part is to provide a high level view of the particular requirements that the structure of an MODSS must respect to be effective, rather than to enter into great detail about the MODSS that exist today.

Parts B, C and E can be read autonomously from each other. Given that all of the terms are defined only when they first appear in the text a non-sequential reading could cause difficulties for the reader. To avoid this we prepared an analytical index in which the first reference is to the definition of the terms.

Didactic use

This book is born from the experience of more than twenty years of teaching undergraduate and post-graduate courses on *Natural Resource Management* at the Politecnico di Milano for the degrees in Environmental Engineering and Information Technology Engineering. It can be adopted as a reference text for both undergraduate and post-graduate courses. In undergraduate courses, Chapters 1–10, 16 and 17 can be used and the remaining chapters used in post-graduate courses. Thereby, at the undergraduate level the issues relative to structuring the problem and modelling the system are treated and some mention is made of techniques for the design, evaluation and comparison of the alternatives. These latter techniques are presented at the post-graduate level together with aspects of ICT. For courses in Information Technology, Part C can be used as a good example of an application of Optimal Control techniques in conditions of uncertainty and risk and Part E as an example of designing the architecture of an MODSS.

The attached CD

The CD is subdivided into the following sections:

- *Appendices* (A1–A10): these can be consulted by the reader to fill any methodological gaps or for a closer examination of particular issues.

- *Exercises*: these are divided into
  - *Theoretical Questions* (E1), which are subdivided for
    - *undergraduate* courses: mainly on Phases 0–3 of the PIP procedure and on the formulation of the Design Problem, i.e. on the material of Chapters 1–10, 17 and relevant Appendices;
    - *graduate* courses: mainly on Phases 4–7, i.e. on the material of Chapters 11–22, and on the more technical aspects of modelling presented in Appendices A1–A8.
  - *Simple applications of the algorithms* (E2), only for the graduate courses, mainly on the material of Chapter 12.

- *Projects* (P1–P10): the most 'artistic' part of a Project, the one that most heavily depends on the Analyst's experience, is the formulation of the Design Problem. To provide students with a more complete preparation and give some food for thought to professionals, a series of Projects is presented. The Projects are not completely developed in all their detail, because a complete description of all their phases would require an entire book (such as the book PRACTICE that deals with just one Project), but, after the presentation of the goal and the system that each of them is concerned with, the focus is the formalization of the Design Problem.

WEB site

To facilitate the updating of this work, a WEB site is available (www.elet.polimi.it/people/soncini) where the reader may find didactic material, updates and the errata–corrigenda prepared on the basis of the comments and suggestions provided by the readers.

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I would be grateful to anyone who would like to send me comments and suggestions to improve this work and I hope that no MODSS created with this book will ever be used to legitimize decisions that have been taken by someone a priori.

Milano, 30 June 2006

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YS	Yona Shamir	The Israel Center for Negotiation & Mediation, Haifa, Israel

Translator's note

In order to avoid boring forms like (s)he and his/her, which are used to acknowledge the fact that both men and women can be found performing all the roles described in this book, we chose to allocate (in a subjective, but seemingly fair way) each of our characters with a gender right from the beginning, and have been consistent throughout.

Valerie Cogan

Chapter 1

Making decisions: a difficult problem

AC, AN and RSS

The evolution of a natural system that is subject to anthropic pressure is well described by the DPSIR framework, proposed by the European Environmental Agency (EEA) and reproduced in Figure 1.1 (EEA, 1999, see also OECD, 1994 and UNCSD, 1996). The *Drivers* generate *Pressures* that change the *State* of the system. This variation produces *Impacts* on society, which reacts by devising and implementing *Responses*, which can be directed at the Drivers, as well as the Pressures, the State or the Impacts themselves.

The following example is useful to clarify the framework[1]: consider an enchanting lake, surrounded by fields, forests, a fishing village and a few small hotels.[2] The Drivers are the agricultural, industrial and domestic practices. They produce a flow (Pressure) of nitrogenous substances that reaches the lake, through agricultural land run off, or through direct or indirect discharge from the sewage system. It follows that there is an increase in the trophic level of the lake, which induces algal blooms, anoxic conditions and mass fish death, and so, a variation in the State of the lake. In this way two Impacts are produced: a reduction in fishing activity and a loss of the lake's appeal to tourists. In order to respond to the fishermen's and hotel-keepers' discontent, the Environmental Agency (EA) must design an intervention (Response). It can choose among different forms: issue a regulation regarding the use of nitrogenous fertilizers in agriculture (arrow 1 in Figure 1.1), create a stage for the removal of phosphorous in the treatment plant that purifies the sewage prior to discharge (2), collect the algae when necessary or inject oxygen at a certain depth to prevent lake waters from becoming anoxic (3), or simply introduce a monetary compensation (4) for the damage. In general terms, the EA is not limited to choosing only one of these *interventions*, each of which can be realized in different forms and degrees: it can also select a combination of them, in an integrated and coordinated package, that we will call *alternative*.[3]

[1] An example of real world application of the DPSIR framework within the WFD context is outlined in Appendix A1.

[2] In what follows we will refer to this example as the 'enchanting lake' example.

[3] A specification for readers who are familiar with negotiation theory: in the literature (see for example Raiffa et al., 2002) there is a distinction between the alternatives that a Party can pursue alone (i.e. without reaching a negotiated agreement), and those that are subject to negotiations, because they contain actions that can be carried out only after an agreement has been reached. The term 'alternative' is reserved for the first, while the term 'option' is used for the second. We will not make this distinction (except for Appendix A10 on the CD) and we will use the term 'alternative' to designate both of them.

In some contexts the term 'programme of measures' is used instead of 'alternative'.

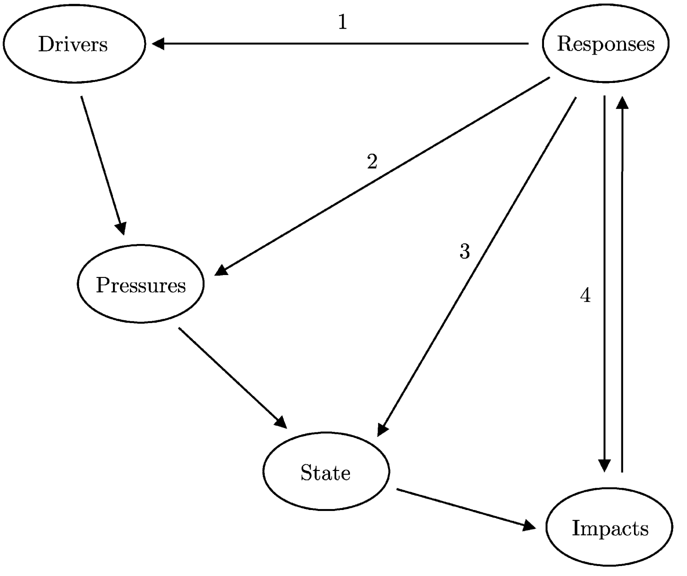

Figure 1.1: The DPSIR framework.

In practice, the same system will require different Responses as time goes by and the context changes. Therefore, very rarely the alternative is established once and for all, more often interventions occur in cycles, as in Figure 1.2. Each cycle is a sequence of events: a *planning* phase, in which an alternative is chosen, its *implementation*, and the *management* of the modified system for a period of time afterwards, during which the system behaviour, i.e. the Impacts produced, is monitored (*monitoring*). When such impacts require a new response, a new cycle begins.

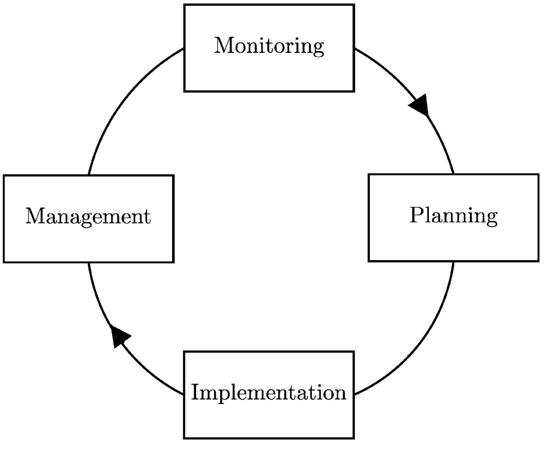

Figure 1.2: The intervention cycle.

The choice of an alternative constitutes a *Planning Problem*, the solution of which requires a *procedure* to be established. This chapter is dedicated to the individuation of the latter, but before proceeding, an effort is made to clarify what kind of interventions we are talking about and what relationship they have with actions and decisions.

1.1 Interventions, actions and decisions

The interventions we are concerned within this book are generally those that are defined by a *Project* (i.e. the choice of an alternative that permits the achievement of given goals), but they could also be those that make up a *Plan* (substantially a project, but with a broader range, that usually deals with a wider area and considers larger scale decisions, but at a lower level of detail), a *Policy*[4] (less detailed than a Plan and at a level that is more strategic and less operative) or a *Program* (a set of Projects, organized in time, through which a Plan or Policy is implemented). In practice, the boundaries among these terms are not well defined and are therefore subjective. For this reason, sometimes Projects may seem like Plans, or Policies may seem like Projects. Independently of the name that is adopted, the essential nature of the thing does not change. In the text that follows the term *Project* will (almost) always be used. As is common practice, it will sometimes be employed also to denote the decision-making process.

We are now ready to give a more precise idea of the types of interventions that we are thinking about when we speak of Projects: these are the different approaches that can be used to reach the Goal that has been set. In practice, interventions can vary greatly from case to case and be quite case-specific. We give the following general guidelines for identifying possible interventions in any given case: do not exclude any intervention without first evaluating it, even if at first glance it is not very 'orthodox'; and avoid, as much as possible, interventions that for sure will preclude others in the future. As it would be impossible to make a list of all the possible interventions, we prefer to exemplify with two projects.

1.1.1 A first example: the Egyptian Water Plan[5]

The Egyptian system and the reasoning behind the Plan are described in the box on page 6. In brief, its aim is to improve the country's quality of life by targeting water availability, bearing in mind the need for environmental sustainability. The Plan covers a period of fifty years into the future.

Given the significant extension of the time horizon, it is necessary to consider all the activities that may affect the availability or quality of water resources in the long run. For example: the increase of urbanized areas, which follows from population growth and takes fertile land away from agriculture, and industrial development, which increases the demand for water and depletes the available resources through pollution. The 'quantity' of the resource is in fact strictly connected to its 'quality': for water to be usable, appropriate standards must be passed, which depend on its use. Not only must the quantity and quality of water in bodies of water be considered, but also the effects of water on soils (salinization, desertification), on crops (increase or loss of productivity, bioaccumulation of toxins), on human health (diffusion of water-borne pathogenic agents) and on the natural state and biodiversity of the environment. Finally, given that the largest portion of the demand is due to irrigation,

[4] Not to be confused with the meaning of the term that will be defined in Section 2.1.1.2.

[5] Following Nardini and Fahmy (2005), with a few modifications for didactic requirements.

The Egyptian Water Plan

System description For Egypt, water is the most important natural resource, and at the same time the one at the greatest risk. Egypt's groundwater reserves are in fact very limited. The water required for domestic use, industrial activities and irrigation comes from the Nile. This is regulated by the High Aswan Dam that creates Lake Nasser, which collects the flow from a vast catchment. It is expected, however, that in following years the flow will decrease, not only because of the climate change that is probably occurring, but mostly because many towns in the catchment area foresee an increase in the amount of water drawn from the Nile to meet the needs of a growing population. Problems come not only from the limited quantity of the resource (the entire volume supplied by Lake Nasser is consumed before it reaches the sea), but also from its quality, which is depleted by many sources of pollution. It is legitimate therefore to affirm that the quantitative and qualitative scarcity of water resources is threatening to become the principal factor limiting the future development of the country.

Project proposal The water resource problem in Egypt has been a subject of study for many years (Elarabawy et al., 1998; NAWQAM, 1999) and a great number of proposals have been put forward. They are, nevertheless, primarily sectorial and the positive effects that they produce in a given sector are often accompanied by negative effects in others. For example, using recycled (polluted or brackish) water for irrigation makes it possible to reallocate precious water volumes from agriculture to domestic uses. However, if this practice is abused the fertility of the land is reduced in the medium to long term. The development or intensification of agricultural practices in one zone can bring about a reduction in the availability of water in another. The introduction of more efficient irrigation techniques can slow the recharge rate of aquifers and therefore reduce their usability downstream or in the future. In order to consider this complicated tangle of factors, the Egyptian Government decided to prepare a National Water Plan, which defines the terms and timescale for the interventions to be carried out and the measures designed to guarantee the water resources that the country will need in the medium to long term. The Egyptian Government's objective is to improve quality of life for the country over the next 50 years. The Government also wants the plan to be economically efficient, environmentally sustainable, financially and politically feasible, and socially equitable so that it guarantees national security.

among the interventions to be considered one should include all those which affect farmers' choices. These can be laws about land use, provisions for water saving, economic incentives and disincentives, and fixing prices of production commodities and foodstuffs.

Below is a summary of the options for intervention that can be considered within Egypt itself, excluding political agreements that could be established with the States of the Nile basin to improve the quantity and quality of water that reaches Lake Nasser. This summary should not be considered either absolute or exhaustive. In a different geographic situation or with other aims it would be appropriate to eliminate some of the headings and add others.

- **Interventions to locally increase water availability**
  - constructing desalinization plants for brackish and marine water;
  - constructing structures for exploiting rainfall and flash floods;
  - constructing waste water recycling plants;
  - constructing water treatment plants to provide drinking water;
  - constructing pumping stations for superficial or deep aquifers;
  - installing pumps for lifting water from interceptor canals into the irrigation network;
  - improving irrigation network efficiency, e.g. by coating the earth canals;
  - improving drinking water distribution efficiency, e.g. by reducing seepage and leakage;
  - extending the irrigation network into lands reclaimed from the desert;
  - changing irrigation method: a shift from the flooding method, which is extremely water consumptive, to the sprinkler or drip methods.[6]

- **Interventions to safeguard the quality of the environment**
  - constructing domestic and industrial wastewater treatment plants;
  - improving the drainage network to counteract the salinization of soils;
  - enacting laws that establish quality standards for effluents or receiving bodies.

- **Regulation policies**[7]
  - defining regulation policies for Lake Nasser, the only surface reservoir that serves Egypt;
  - defining regulation policies for the aquifers (define the extractable volumes from each of them according to the month of the year and the condition of Lake Nasser);
  - defining distribution policies between land districts.

[6] One may ask why not simply decide to change over to the drip method at one fell swoop, which would drastically reduce consumption and might finally allow Egypt's thirst to be satisfied. There are several reasons. First of all, the high costs of the equipment, and the fact that not all soils are suited to or able to sustain the crops that are compatible with drip irrigation. In addition, the surplus water required by the flooding method is not all lost. A great deal filters into the water table, from where it evaporates less than from the canals, and from where it is then pumped to be reused downstream. Lastly, there is cultural inertia. The agricultural community is structured around the extremely dense network of irrigation and interceptor canals, the very structure of the flooding method, which delimits the plots of land. The people's way of life is linked to this structure. The modification of this state of affairs requires a cultural education programme which can be a slow process. This should be considered in the context of an integrated intervention.

[7] A regulation policy is a procedure that defines the rate of flow to release from a reservoir or to divert at a diversion dam, or to pump from an aquifer (see page 40) as a function of the data acquired from the information system.

- **Interventions aimed at guiding behaviour**
  - sizing areas used for particular crops; imposing crop rotation in the first years, in lands reclaimed from the desert; imposing taxes/subsidies on some crops;
  - setting limits to urban expansion;
  - improving irrigation efficiency through education programs and economic incentives;
  - setting taxes/subsidies for the use of chemical products;
  - setting quotas, taxes/subsidies for importing/exporting certain foodstuffs;
  - setting prices for products controlled by the government;
  - defining tariff schemes[8] for water service users, both domestic and industrial;
  - defining incentives for the settlement of farmers in reclaimed lands.

Note that all these interventions have (and must have) something in common: they can all be carried out and managed by the organization that commissions the Plan: the Egyptian Government.

1.1.2 A second example: controlling hydraulic risk[9]

The management of hydrogeological instability and especially the risk of flooding are of vital importance in many countries. Traditionally, the interventions that are considered are:

- **Structural interventions for hydraulic protection and regulation**
  Among the most common are the construction of flood detention areas,[10] detention basins,[11] dry dams,[12] dikes, riverbank defences, flood diversion canals and rectification of the river channel.

All these interventions have reduced the naturalness and beauty of the water courses and the landscape, they have impoverished biodiversity and, albeit surprising, they often have not reduced the risk of flooding. This is due to two reasons: the first is that in the 'safety' zones land use has intensified, increasing the potential damage. The second is that the interventions have increased the flood peaks in downstream transects, thus worsening the situation, because they have reduced the time of concentration and the detention capacity of the river channel. Experience has demonstrated that these interventions are too often neither sufficient, nor effective, nor efficient, nor sustainable. As a result, in the last decade in many countries a new approach is emerging, which aims at a generalized renaturalization and is based on the following actions:

- **Interventions to reduce the potential damage**
  Avoiding and eliminating the presence of assets in flood risk zones, establishing binding building regulations and/or promoting the delocalization of settlements already present through regulations and economic incentives.

[8] The rules that establish how much each user must pay for the water used as a function of the volume of water withdrawn, of the withdrawal period, and of the category to which the user belongs.

[9] This section is based upon Nardini (2005).

[10] Portions of land that are dedicated to the temporary storing of the volumes of water that overflow from a water course, thus lowering the peak level of the flood and thus alleviating the flooding risk for areas downstream.

[11] These are flood control structures similar to detention areas, but artificially controlled: they typically include banks, which separate the river channel from the flood zone, a spillway that regulates the intake of water from the river channel, and a drainage system.

[12] A non-regulated dam that, in normal conditions, holds back no water and allows the river to flow freely. During periods of intense flow, which would otherwise cause flooding downstream, the dam temporary holds back the excess water, releasing it downstream at a controlled rate.

- **Interventions to increase storage capacity**

  Modifying the general situation of the territory: reafforesting; constructing retention basins for storm water in urban zones; reducing the impermeability of urban ground, creating draining surfaces in parking lots, in squares, and on the roofs of buildings (*green roofs*); and establishing incentives/disincentives to drive the actions of the private sector in this direction.

  Reestablishing space for rivers to flood, to modify morphology and to meander: moving or eliminating banking, rectifications, and riverbank protection where possible, particularly in the minor network; substituting the interventions to protect areas of a low intrinsic value (e.g. agricultural and treed areas) with mechanisms for damage compensation.

- **Interventions for recovering geomorphologic equilibrium**

  In the basin: afforestation to stabilize the slopes, monitoring and controlling fires and grazing to reduce erosion, identifying feeder basins for refurnishing the river beds with sediments, eliminating dams and dredging reservoirs (releasing the sediments into the river downstream).

  In agricultural zones: giving incentives for suitable agricultural practices and crops, and regulating land management.

  In the river channel: enhancing the riverbank vegetation, forbidding the extraction of gravel, and limiting the artificial protection of riverbanks from erosion as much as possible, given that the possibility to sediment and erode is key to maintaining the equilibrium of the river channel.

- **Interventions for living with risk**

  Interventions to raise responsibility: informing, sensitizing and educating the public; inducing people to participate in decision-making and assume responsibility.

  Planning the management of emergencies: setting up efficient warning systems for flooding events and appropriate emergency plans; equipping the zones to face these events.[13]

Note that also in this case all these interventions can be decided upon by the administration responsible: a river basin authority, a regional authority, or a local body.

1.1.3 Actions and decisions

Every option for intervention should be then broken down into one or more *actions*, which are characterized by the fact that each one of them can be completely and precisely identified through the specification of the values assumed by a set of attributes (parameters and/or functions). In the 'enchanting lake' example that opens the chapter, the normative intervention can be, for instance, specified by the maximum load [kg/ha/year] of nitrogen allowed in field fertilization, while the construction of the water treatment plant can be defined by its location and by the percentage of nitrogen removed. The *decisions* that the Project must take are concerned with the options for intervention to consider, the type of actions by which

[13] For example, construct buildings on piles; provide openings below flood level with watertight doors; construct retaining walls around buildings; make basements and ground levels floodable without incurring damage; avoid locating residences on the ground floor; provide the sewage system with one way valves; locate the electricity plants, telephone systems and heating above flood level.

to realize them and the values to assign to their attributes. The attributes must be defined in such a way that it is always possible to leave things as they are, i.e. to choose non-action.

As we have already said, an *alternative* is an integrated and coordinated package of actions. The purpose of the Project is to identify the alternative (or alternatives) which permits the achievement of the overall Project Goal among the alternatives being considered. The set of these must always include the *Alternative Zero* (denoted with $A0$), which is composed by non-actions and therefore is often described as *business-as-usual*.

1.1.4 Classifying actions

The actions, and as a consequence the decisions that are concerned with them, can be classified in various ways.

1.1.4.1 Structural and non-structural actions

The first distinction is between *structural actions* and *non-structural actions*: the first are concerned with physical modifications of the system, as, for example, the location and dimensions of the structures for the collection, transportation, distribution and use of the resource. The second either modify the system only functionally or they alter the effects that the system produces. Examples of structural actions are: the construction of a dam or a canal; the installation of an irrigation system; the construction of a waste water treatment plant; and the renaturalization of a river that has been rectified in the past. Examples of non-structural actions are: a regulation that introduces quality standards for effluents; setting tariffs for water services; an incentive programme for farmers to encourage 'virtuous' behaviour (e.g. adopting crops that need less water or planting woody buffer strips); and the regulation policy of a reservoir.

The assignment of an action to one class or the other is not always univocal: for example, the US Army Corps of Engineers classifies the action 'raising a building on piers' as non-structural. The reason is that such an action does not modify the functioning of the system (the river flooding), while it does influence the effects that it produces. Also note that a non-structural action can indirectly produce structural actions. For example, an incentive programme for farmers can encourage them to plant buffer strips along water courses, so modifying the flood regime. Therefore the border between the two classes of actions can be, at times, very elusive, but the classification is useful just the same.

1.1.4.2 Planning and management actions

A second distinction is made between *planning* actions and *management* actions. The discriminating factor is the time step with which the actions are decided. An action is a *planning* action when it is decided once and for all (a typical example is the construction of a dam). An action is a *management* action when it is decided upon frequently or periodically.

There are two outstanding characteristics of management actions. First: when one decides on the next action, up to date information about the system is available so that the evaluations that were carried out to take the previous decision can be updated. Even the decision-making method can be reviewed on the basis of this update. Second: the decision is *recursive* when the system is dynamical.[14] This means that every action will be decided by considering the decisions that will have to be taken in the future on the basis of the

[14] For the precise meaning of this term see Appendix A3.

states that today's decision will have produced. In some ways it is more difficult to choose management actions than planning actions, because it is necessary to evaluate not only the current decision, but also all the decisions that will need to be taken in the future, while taking into account the effects that the first ones will have induced. However, a planning problem can be conceptually more difficult than a management problem, because it may incorporate management problems (think of the Egyptian Water Plan).

We can summarize the twofold classification that has been introduced above with a 'sample' of actions, which is a far from exhaustive list, accompanied by the principal attributes that define them:

- **Planning actions**
  - Structural actions: construction of
    - *Reservoirs*[15]: location, size of the dam, characteristics of the outlets;
    - *Curtains* for pathogenic control or fixing nutrients in natural lakes and reservoirs: location and maximum reachable depth;
    - *Aerators* in natural lakes in anoxic conditions or those requiring destratification: location, depth, power;
    - *Diversions*: location, regulability, maximum derivable flow;
    - *Canals*: location, layout, minimum and maximum flow, presence or absence of coating;
    - *Irrigation systems*: location/extension, irrigation/drainage technique;
    - *Pumping stations*: location, capacity, head;
    - *Aqueducts*: layout, average and maximum flows, losses;
    - *Hydropower plants*: location, intake and outlet points, maximum and minimum flows of the turbines.
  - Non-structural actions: definition of
    - *Management criteria for reservoirs and diversions*:
      - *minimum environmental flow* (MEF), i.e. the minimum flow that must be released to the river downstream from a reservoir whenever the inflow exceeds the MEF[16];
      - storage constraints (regulation range);
      - constraints on the dam operation;
    - *Land use regulations*: zoning, limits to expansion, urban regulation;
    - *Regulation and/or water distribution policies*[17];
    - *Economic instruments*: tariff schemes for water services, insurance plans against the risk of flooding;
    - *Information and education campaigns* to increase awareness: program, people involved, budget, means employed.

- **Management actions**
  - Structural actions
    - *Maintaining storage structures*: volume and location of sediments to be removed in reservoirs and diversion dams;

[15] In the text that follows we will use the term reservoir to refer to both artificial reservoirs and regulated lakes; in other words, we will use it as a synonym for regulated storage facilities.

[16] When the inflow is less than the MEF it is never compulsory to release more than the inflow. This does not exclude that a reservoir could be used for low-flow augmentation.

[17] See page 40.

    - *Planting woods*: number, type of plants, and frequency and zones of afforestation.
  - Non-structural actions
    - *Releasing water from reservoirs*: flow rate;
    - *Aerating natural lakes and reservoirs*: intensity;
    - *Operating curtains*: depth;
    - *Reviewing economic tools* in the light of contingent conditions: degree of variation;
    - *Broadcasting alarms and pre-alarms for floods*: area involved.

Note finally that another attribute, which must be considered for all these actions, is the time at which its effects will come about: for example, the time a dam becomes operational or the time a regulation comes into action. This attribute is of particular interest not only for Programs, but also for Projects, when short-term (transient) effects are considered (see Section 3.2.2.1).

1.1.4.3 Other classification criteria

Another classification criterion is based on the aim of the actions. This is how we have classified the actions in the two examples that open this paragraph. A further possibility is to classify the actions according to the Decision Maker (DM) that can take them, or according to the decision-making level at which they are established. For example, a public administrator, a single farmer and an individual citizen all operate at different levels: the first can decide on all the actions listed in Section 1.1.1, the second only on crop rotation, irrigation techniques and agricultural practices, while the third decides how to save water when brushing his teeth or taking a shower.

1.2 Difficulties and keys to their solutions

The projects that we are concerned with show two fundamental characteristics: they involve many individuals (even sometimes many DMs), and they require a decision from an Agency or a public body (the choice of the alternative to be implemented or simply the authorization to implement an alternative proposed by others). For this reason we talk about decisions in the public realm. Such decisions are usually taken by following the procedure described in the diagram in Figure 1.3. In the diagram the term *Decision Maker* (DM) refers both to the commissioner of the Project, who oversees the first phases (e.g. the Egyptian Ministry of Public Works that wants to produce the Water Plan), and the final DM, who must approve it (in the example the Government or Parliament). The *Analyst* is (s)he who actually conducts the necessary studies and draws up the Project on paper. It can be a technical office of the same administration, but, more often, it is a consultancy that has been entrusted with the job after putting in a tender. The *Stakeholders*[18] are either all those (people, institutions, organizations) that experience the effects of the Impacts for which a Response[19] is being sought, or those that could be influenced by the options for interventions considered for implementing the Response. The horizontal arrows indicate the moments in which the *actors* (DMs, Analyst, and Stakeholders) interact and their direction leads from those that pose the questions to those that must respond.

[18]Some prefer to call them 'rightholders', which is a broader category, according to those who support the idea, than Stakeholders. In actual fact, however, often there are Stakeholders whose rights are not recognized.

[19]According to the terminology of DPSIR scheme in Figure 1.1.

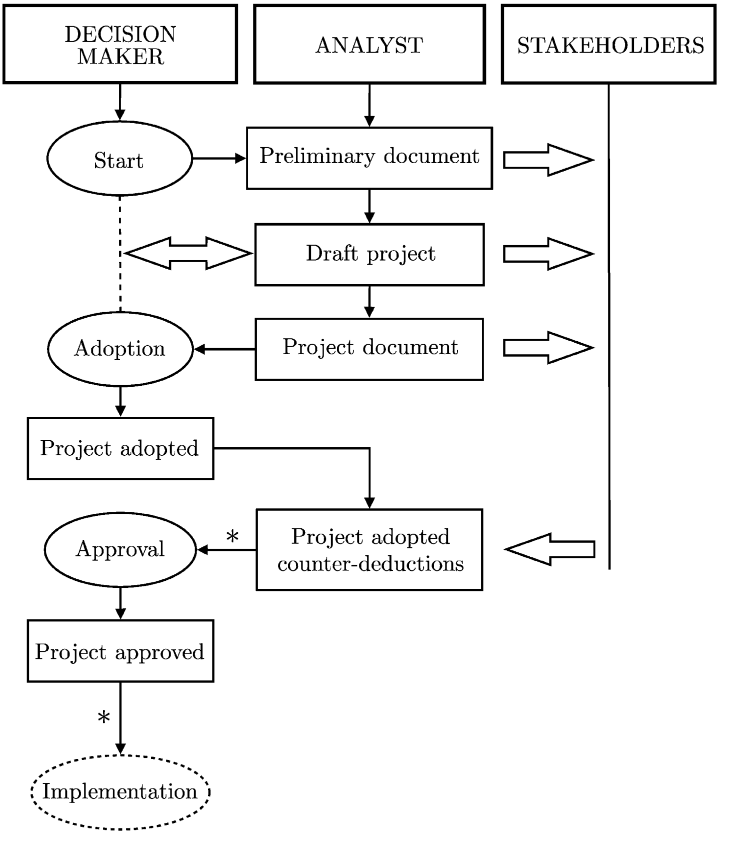

Figure 1.3: The standard procedure for a planning process.

Often the two steps that are indicated with an asterisk are protracted for an excessive amount of time, sometimes even for years, and, since they are not transparent, it is not always very clear what is happening. Informal negotiations, lobbying and political battles are likely to develop. There are two principal difficulties that provoke these delays:

- the *conflicting interests*[20] are often of great importance, especially economic importance and when not adequately managed, can lead to lobbying, opposition and boycotts;

- the *inadequacy of the approach* that is top–down, i.e. a Decision–Announcement–Defence approach. The administration decides, communicates the decision and defends it from the Stakeholders' reactions; the latter, having had very little influence on the decision, are almost always against it. Furthermore, the approach emphasizes the political component of the decision, which while it is certainly essential for orienting the evolution of the system in the long-term, by its nature does not provide the

[20]"Perceived divergence of interests, or belief that the Parties' current aspirations cannot be achieved simultaneously" (Rubin et al., 1994).

transparency, the credibility and the explanation of the choices that the Stakeholders demand. In addition, it risks being technically inefficient and unsustainable because the information-modelling and decision-support tools that are available today are too often not really integrated into the approach, but only juxtaposed, at times, acting purely as frills.

In order to achieve good planning and good management it is absolutely necessary to overcome these difficulties by conducting a *participatory, integrated and rational decision-making process*. Let us examine these three adjectives one by one.

1.2.1 Participation

1.2.1.1 Awareness

If we do not manage to awaken or instill profound human values, such as a sense of belonging to the territory, a love for the environment, and a responsibility for its evolution, we will not be able to get very far. Therefore, education is fundamental and must be structured at many levels, from children to administrators and politicians, including technicians and the general public. In this book we will not linger over the ways to increase awareness of the problems, especially environmental ones, but that does not mean that this issue can be neglected.

1.2.1.2 The participatory process

The top–down approach of the standard planning procedure needs to be reversed, proposing and launching a participatory process (Renn et al., 1993; Renn, 1995; Budge, 1996; Delli Priscoli, 2004), that is 'bottom–up' and based on the management of participation: a process of participation that begins from the proposal of the Project, the recognition of the need for the process itself, and continues with the choice of the alternative to be implemented, right through to the monitoring of the effects after its implementation. This process should not be limited to providing the Stakeholders with information (*Informative Participation*), nor to just asking them for information (*Consultation*), but should also involve the Stakeholders in the design and evaluation of the alternatives (*Co-designing*) and ideally even in the final choice (*Co-deciding*)[21] (Mostert, 2003; Hare et al., 2003).

In this way, a process of *social learning* is created, in which the Stakeholders become aware of the problem, of the alternatives, and of the viewpoints of others; they take responsibility and together they develop the alternative to be carried out (Renn, 1995). The key ingredients are information, transparency, repeatability, and the active involvement of the Stakeholders. In order for the process to be effective, it must be supported by an open and flexible decision-making procedure to accommodate the unforeseen events and elements that are introduced by the Stakeholders. At the same time this process must be structured, so that it does not degenerate into a 'condominium meeting' (see Renn et al., 1995, for an

[21] With surprising speed legislation has already adapted to this need: the Århus Convention, signed in 1998 and in force since 2001, recognizes citizens' rights to "have access to information and be enabled to participate in the decision-making process with regard to the environment", based on the principle that only participation can make sustainable development possible. In escort to that convention, the Directive 2003/35/EC (European Commission, 2003b) was issued by the European Parliament which establishes that "the public concerned shall be given early and effective opportunities to participate in the environmental decision making procedures" right from the initial phases, so that they have an effective possibility to influence the choices. The Water Framework Directive (WFD) [Directive/2000/60/EC (European Commission, 2000)] anticipated this position.

evaluation of different approaches to participation). As it will be shown in Section 1.3, the synthesis between these antithetic characters can be achieved with an accurate, explicit and shared definition of the phases of the process and professional guidance of the moments in which the interaction and negotiation between Stakeholders take place.

A participatory process takes time and it is this aspect in particular that often discourages its implementation. The final goal should be made clear: if it is the 'actual implementation of Responses', then a participatory process is almost always quicker and more efficient than a non-participatory one (the duration is measured between the beginning of the Project and the end of the implementation phase). If, instead, the goal is just to 'draft the Project', then the non-participatory process is faster, because it avoids many tiring phases of negotiations (in this case the duration is measured from the beginning to the approval of the Project). However, drawing up a Project which almost surely will not be implemented because it is not shared, is a real waste of public money.

Here are some *guidelines* to stick to when developing a participatory process:

1. Share the idea that everyone is working together to solve a problem and improve the quality of life.

2. Participation, not just communication: accept that the participation really influences the final decisions.

3. Always stick to what was agreed upon in the previous meetings.

4. Create responsibility through agreement.

5. Try to transform the difficulties into opportunities,[22] for example, looking for alternatives so that no one is, if possible, worse-off than before[23] (*win–win alternatives*).

6. Look for equity by identifying who bears the costs and who reaps the benefits; look for interventions that make those that impose costs on others responsible for their actions.[24]

7. Guarantee flexibility, but maintain rigor, to avoid 'houses of cards'. One should not proceed by taking decisions on a weak basis that can crumble afterwards. A well conducted process gains the confidence of Stakeholders, who will then be under pressure to participate for fear of being excluded from decision-making.

8. Look for agreement at every step, but at the same time accept that differences and uncertainty are integral and inevitable parts of the process. Do not force the Parties to rush at an agreement, but accept that, in order to reach one, time and interaction are required.

[22] Often it is only a matter of imagination: if for example a group of farmers are opposed to a project for a river park because they fear that they will have to change their activities and be dispossessed of their land, the opposition could be overcome by also including among the actions of the Project, to be developed with those same farmers, a reconversion action of cropping patterns: e.g., substituting the production of maize with biological crops or high value herbs that can be cultivated and certified, thanks to the very fact that the park exists. For this reason, also interventions for training and technical assistance have to be considered in the project, and potentially subsidies. Furthermore, instead of expropriating the lands, one can think about a form of contract to use land for reciprocal advantage.

[23] Excluding, clearly, those that started in an illegal condition.

[24] For example, by imposing a tax on dumping polluted waste.

9. Provide guarantees instead of demonstrations. It is useless to insist that there will not be undesirable effects; it is much more convincing to sign a commitment to a corrective action in the eventuality that the effects that the Stakeholders fear do occur.

10. Recognize that the public has a role to play in monitoring and carrying out the project.

11. Accept the existence of different points of view.

12. Distinguish facts from value judgements, which are the product of subjective preferences.

13. Keep the decision-making process transparent and repeatable, guaranteeing access to information.

14. Respect the role of the political DM, but make sure that she explicitly clarifies the reasons for her choices.

And here are a few *sine qua non* conditions that the Analyst must respect, if the participatory process is to be successful:

1. State the criteria and the rules to follow and respect them. The 'statute of participation' (Connor, 1997) is a useful tool: it is a document that states principles, intentions and rules of the decision-making process.

2. Define the decision space clearly, and clarify the 'power of the participation machine' and the relations with the Administration.

3. Have the DM participate in the meetings with the Stakeholders, ideally in person, or at least by sending a delegate.

4. Act in a way that gains the trust of all the Parties and never betrays that trust.

5. Be very careful to uncover any misunderstandings that are produced by the terminology and have the patience to take the time and energy to resolve them.

6. Evaluate often the state of the participatory process.

1.2.1.3 Evaluation for negotiations

It is almost always impossible to identify an alternative that produces the best possible effects for all the Stakeholders. Each alternative is a particular compromise between the interests[25] at stake and thus it is essential that its effects be evaluated from the viewpoint of each Stakeholder, so that each one can express his/her opinion about it and negotiate the *best compromise*.

1.2.2 Integration

The decision-making process must be founded in the *principle of integration*, which manifests itself at many levels:

[25]By the term *interest* we mean the needs, desires, worries and fears, concerns, and more generally, whatever reason that encourages a Party to negotiate.

- among the parts that compose the system;
- between rationality and emotionality;
- among Stakeholders and political DMs;
- among the Stakeholders themselves, particularly between those who benefit and those who incur damage;
- among the evaluation approaches: Cost Benefit Analysis (CBA), Cost Effectiveness Analysis (CEA), Multi-Attribute Value Theory (MAVT), Environmental Impact Assessment (EIA), Strategic Environmental Assessment (SEA);
- between environmental policies and sectorial policies (applying the *precautionary principle*[26]);
- between technical approaches to solution and decision-making techniques (*integrated technical approach*);
- between planning and implementation (plan/project; SEA/EIA) and between strategic and tactical scales (applying the *subsidiarity principle*[27]).

In our view, the keys to integration are the correct identification of impacts and the correct identification of the indicators that quantify them, and so two conditions must be respected:

(a) the Stakeholders whose interests will be, or might be, affected must be clearly identified: not only those who could be disadvantaged by the decisions being considered, but also those who may benefit;

(b) the values that the Stakeholders attribute to the impacts must be made explicit (Keeney, 1992).

In this way, the evaluation can focus on the reasons why an alternative is preferable to another, rather than fall into a sterile conflict of positions, in which some defend an alternative to the utmost while others attack it.

1.2.3 Rationalization

We have therefore understood that the decision-making process must be rationalized through the adoption of a precise *decision-making procedure* that allows the *best compromise alternative* to be identified. This is done by way of a participatory negotiation process that is structured and transparent, and whose core is an integrated evaluation that allows each Stakeholder to evaluate the effects that will result from each alternative, and explicate the political compromise between conflicting interests upon which every choice that is made is founded.

[26] Environmental phenomena are very complex and in many cases we do not know how to predict their effects, especially in the long-term. It is therefore advisable not to take actions whose effects we are unable to evaluate or, if it is really necessary to take such actions, to do so only with appropriate security measures.

[27] To each administrative level (e.g., Ministry of the Environment), leave only the decisions that cannot be taken at a lower level (e.g., Province) so that procedures are quicker and more responsibility is given to the lower levels.

The first, fundamental condition to identify an alternative that enjoys a wide agreement is that the very decision-making process get the consent of the Stakeholders. To this aim the sought-for procedure must:

- break the process down into phases and establish the sequence in which they are executed; and
- specify the aim of each phase and the technical means (algorithms and procedures) by which it will be achieved.

In addition, it must give a concrete form to principles 11–14 stated above. More specifically:

- accepting different points of view translates into the fact that each phase (except for the last) can close with a plurality of outputs, all equally adequate;
- maintaining the distinction between facts and value judgements does not mean that subjective preferences are suppressed, because they too guide the choices, but that they are kept distinct from that which is 'objective';
- the transparency and repeatability are realized by making the information available to all the Stakeholders and the DMs, documenting and distributing the results of all the phases;
- and lastly, respecting the role of the final DM means that the final choice (and therefore the last phase of the procedure) is reserved for her and she is guaranteed the right not to choose one of the alternatives that have emerged from the decision-making process. However, the procedure must ask her to justify the choice with the same instruments (indicators and criteria) that were defined by the Stakeholders during the course of the process. If the process was well conducted, the DM's different choices should be justified only by the different relative importance she gave to the evaluation criteria.

The definition of the decision-making procedure is the subject of the next section.

1.3 Planning: the PIP procedure

Sometimes there is more than one DM, as would be the case if the 'enchanting lake', in the example that opens this chapter, and its inlet were to define the border between two countries. In that case, neither of the DMs (the Environmental Agencies of the two countries) could assume efficient decisions autonomously. Even when only one DM is concerned, we have seen (Section 1.2.1.2) that it is advisable to choose the best compromise alternative by taking into account the Stakeholders' viewpoints. For this reason, the decision-making procedure should not be limited to considering information collected from the Stakeholders (*Consultation*), but they should, instead, be treated as if they were DMs that must negotiate a compromise alternative (*Co-deciding*). The goal of the decision-making process is to reach an agreement that is acceptable to them all, to which they remain committed, and which is actually implemented. Only the last, decisive step of the procedure (the formal choice of the alternative that is to be implemented) is in most cases the reserved responsibility of the DM (or DMs) that has (have) the institutional power and responsibility to make the choice.

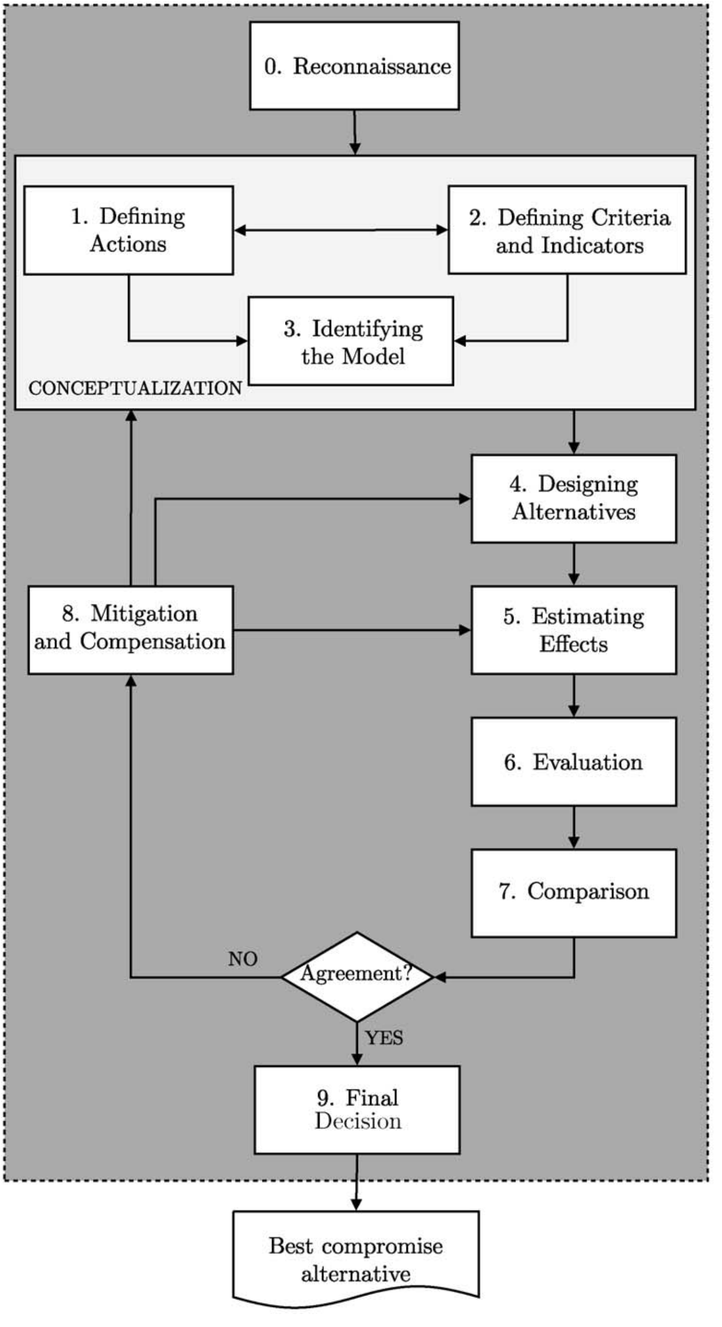

Figure 1.4: The phases of the PIP procedure.

In Figure 1.4 the flow diagram of the procedure that we are proposing is shown. We call it PIP procedure, for Participatory and Integrated Planning procedure.[28] The diagram is also a conceptual map of the modules of the software systems used to implement the decision-making procedure, called *Multiple Objective Decision Support Systems* (MODSS). This structure is also reflected in the succession of the parts and of the chapters in this book. The description of the component phases is only a preview of what will be presented in detail in the following chapters. Many new concepts appear in it and some may, inevitably, seem obscure. If this is so, the reader should not be discouraged, because such doubts are completely normal and will be dispelled further on. What we wish him/her to capture is just a view of the whole procedure. Keep in mind that all the terms in italics will be picked up and carefully defined in the following chapters. This paragraph can be seen both as the beginning point and the conclusion of the book. When the reader has completed the book, our advice is to read it again. If at that point the contents seem evident and meaningful it will mean that our efforts have been successful.

1.3.1 The phases

Phase 0 – Reconnaissance

Statisticians consider two types of errors: type I error (rejecting a true hypothesis) and type II errors (accepting a false hypothesis). The aim of *Reconnaissance* phase is to avoid "type III errors": solving the wrong problem (Raiffa et al., 2002). The work is concentrated on defining the *Project Goal*, the (spatial and temporal) boundaries of the system being considered,[29] the normative and planning context in which the procedure operates, the data available, and the information that needs to be collected.

One must start off from the identification of the Stakeholders involved and their needs, expectations, fears and perceptions, in a word, their *interests*.[30] In fact, the definition of the Project Goal closely depends on the interests being considered and on the hopes and expectations that one wants to fulfil.

Then the PIP procedure has to be explained to and accepted by, or, if necessary, negotiated among all the actors (Stakeholders and DMs).

At this point it becomes possible to define the *Goal* that the Project must pursue. It is derived from the DM's strategic goals, from the Stakeholders' interests and from the regulatory and planning context. In the case of the Egyptian Water Plan the specific Goal of the Plan ('to improve the quality of life for Egyptians by targeting water availability') was derived from the government's strategic goal 'to improve the quality of life for Egyptians'. It is useful to translate the Goal for each Stakeholder into a *vision* that visualizes, with words, or better still, with a picture, the condition that the Project aims for. For example, the vision for the civil users of the Egyptian Water Plan could be: "no longer water only in the evening from 17:00 to 20:00 one day in three, but a continuous reliable supply, which is not too expensive ..."; while a project for upgrading a river system could be expressed by the pictures in Figure 1.5. The choice of a good vision is important

[28] Even though it was devised autonomously, it can be interpreted as a variation of the PROACT scheme proposed by Hammond et al. (1999), which has been suitably modified to take into account that, in the case we here examine, the decision-making process is targeted at consensus building and actions include management, i.e. recursive decisions.

[29] These two points are often referred to as *scoping*.

[30] "It is crucial for the legitimacy of a planning process to start dialogue as early as possible in the phase of problem definition" (European Commission, 2003b).

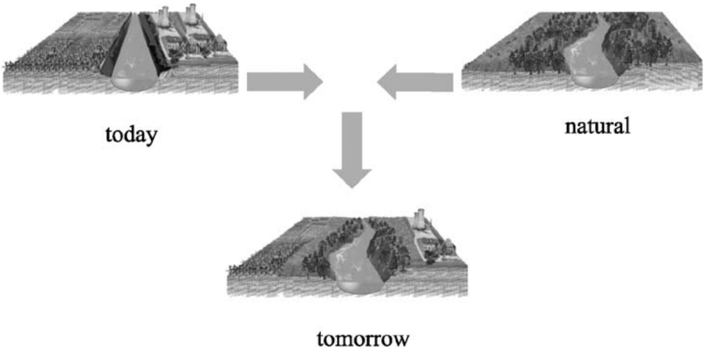

Figure 1.5: The *vision* of a river upgrade project (from CIRF, 2004).

when the Stakeholders are not very motivated to take part in the decision-making process or resistant, or unable to express their own goals.

An extensive knowledge of the system is the fundamental support to all these activities and it is acquired by: analysing the regulatory and planning context of the Project; collecting the information and data available; identifying the missing information; and finally, filling the information gaps by conducting hydrological, economic and social surveys. The actors should share all the available information, agree upon its validity (this is crucial!) and the potential need for further investigations. In other words, when necessary, even the validity and availability of the information must be negotiated.

An example of Phase 0 is presented in Chapter 2 of PRACTICE.

Phase 1 – Defining Actions

In this phase, the options for intervention that are supposed to achieve the Project Goal must be identified, bearing in mind the interests of the different Stakeholders. This is not a simple operation because the opinions can be very discordant. For example, for some the obvious solution for the 'high water' problem in Venice would be the construction of the MOSE[31] at the mouths of the lagoon, for others it is the construction of gate-ships,[32] which are less complicated to construct and which would be more adaptable to the bradyseism of the lagoon bottom; others say that the only sensible option would be the reduction in the green house gas emissions that are responsible for the eustatic sea[33] in front of the lagoon. From this initial, decidedly disorganized, collection of ideas, which are in part silly, in part gifted with incredible wisdom, good ones always emerge. It may

[31] A system of submerged hollow steel gates, hinged at the bottom of the lagoon and installed at each of the lagoon's three openings to the Adriatic sea. When 'high water' is foreseen, they can be raised by pumping compressed air into them and creating a sea barrier.

[32] Two ships, whose length is about half the width of the mouth of the lagoon, which are hinged to the two offshore piers that mark the mouth's boundaries. The free end of each ship is equipped with a propeller that allows them to position themselves across the mouth opening. When the ships are in this position they flood the compartments that make up their hulls so that they sink, creating an insurmountable barrier for the sea. When the high water event is over, the mouth of the lagoon is reopened by reversing the operation. The major advantage with respect to the MOSE is that the construction and the maintenance are done on dry land and the mouth of the lagoon would need to be modified only slightly.

[33] The phenomenon of sea level rise in the long-term in response to geological and climatic changes, such as, for example, the melting of the polar ice caps, produced by the green house effect.

seem strange to begin with brain storming; however, it is essential to promote creative decision-making that considers more than just a set of interventions given a priori, and is able to open new perspectives and discover unexpected alternatives. If an intervention is really useful it will emerge in the following steps, and if all the Stakeholders' suggestions are considered and processed, they are more prone to collaborate since they feel they are being "taken seriously" (WFD, Annex VII, point A9, see European Commission, 2000). Moreover, how can there be a participatory process if one does not listen carefully to the ideas and proposals of the actors?

This first, creative phase must necessarily be followed by a phase of 'sedimentation', in which we separate what can be decided and what cannot, and thus, the effective decision space becomes apparent. It is in fact useless passionately to debate aspects that cannot be decided in the Project. But note carefully that this is not the time to discriminate the useful interventions from the useless ones, or the sensible ones from the less sensible. This will be a task for the next phases. Here, we only aim at obtaining a list of the interventions 'that can be decided upon' in the context of the Project, e.g. in the case of the Egyptian Water Plan one obtains an extract from the long list of actions in Section 1.1.1.

Each intervention is finally broken down into one or more (*meta-*)*actions*, i.e. into elementary interventions that can be fully and easily defined by specifying the values of their attributes (see Section 1.1.3), that is by specifying who is doing what, how and when. In this way a meta-action is transformed in an *instantiated* action. Technically, this transformation is carried out by assigning values to the parameters and/or the functions that describe the attributes of the meta-action. The specification of these values is a matter for a future phase (Phase 4: *Designing Alternatives*), but their feasibility sets have to be defined in this phase, thus fixing the meta-actions to be considered. The instantiated actions are the 'building blocks' from which the alternatives will be constructed later. In the following, as we have done so far, we will use the term 'action' to denote both the meta-actions and the instantiated actions. The actual meaning will be clear from the context.

A complete example of this phase is described in Chapters 3 and 5 of PRACTICE.

Phase 2 – Defining Criteria and Indicators

To evaluate and compare the effects of the alternatives on the system it is necessary to identify, together with the Stakeholders, a set of *evaluation criteria* that reflect the characteristics of the problem and the values that are at the base of the judgements that the Stakeholders express. The criteria do not have to pertain only to the Project Goal, but to all the positive and negative effects that the Stakeholders hope for or fear: in other words they must express their interests.[34] In particular, the criteria for sustainable development will be proposed by the Agencies and by the Environmental Associations which must always be included in the Stakeholder group.

Not every evaluation criterion is necessarily expressed in an operative way, i.e. it may not spontaneously define a procedure that allows us to ascertain how much a given alternative satisfies that criterion. This is why an *index* must be defined, that is a *procedure* which associates the criterion with a value expressing its satisfaction (see Chapter 3). This is done through the identification of relationships between the evaluation criterion and the

[34] Objectives are the same as interest: unfortunately the negotiation theorists and the decision-making theorists have not agreed on a common term. The first talk about 'interests' and the second about 'criteria and objectives'; but they mean the same thing. We will adopt the jargon of the second group.

variables (e.g. lake level and water release, river and channel flows) that describe the system condition. In practice, one proceeds by first splitting the evaluation criterion into lower level criteria and, in turn, splitting those into even lower level criteria, until it is possible to associate each one of the criteria at the lowest level (*leaf criteria*) with an *indicator*, i.e. a function of the trajectories of the variables describing the system condition. In this way a hierarchy of criteria is obtained for each evaluation criterion. The definition of a criterion and of its hierarchy ought to encompass either thresholds (e.g. minimum environmental flow or the level above which a flood occurs) or a Stakeholder's wish (*leitbild*, see Egger et al., 2003), which is often related to the performance level (s)he demands (e.g. a preferred flow).

It is necessary to dedicate a great deal of time and attention to interactions with the Stakeholders and to studying their points of view, because it is essential that each Stakeholder sees that his/her interests are expressed in at least one of the indicators. If this does not happen, negotiations in Phase 7 will inevitably fail.

In the next phase we will see that very often the system is affected by *random inputs* (either *stochastic* or *uncertain*). It follows that also the values assumed by the indicators are not generally deterministic. When this occurs, it is necessary to take account of the *risk aversion* that the DMs and Stakeholders may have. This can be expressed through the classical approach of utility functions, proposed by Keeney and Raiffa (1976), but more often it is translated through *criteria*[35] from which the most frequently adopted are the *Laplace criterion* (expected value) and the *Wald criterion* (worst case) (French, 1988). We will deal with these in Chapter 9.

Like all the phases of the PIP procedure this one must also be participatory: the evaluation criteria should be forthcoming from the Stakeholders and the definition of the indicators must have their contribution and approval. This last step is, however, often very technical and so, as for the technical steps in the following phases, the Stakeholders can be supported by *Experts*.

Phase 3 – Identifying the Model

In order to quantify the effects that the different alternatives would produce on the different indicators if they were to be implemented, it is necessary to provide a *model* that describes the cause–effect relationships within the system. Such a model can take the form of an Expert, who, on the basis of his experience, is able to estimate the effects that each alternative will produce (see for instance the MÖLL Project (Muhar and Schwarz, 2000; Jungwirth et al., 2000)). Alternatively, it can take the form of a mathematical model, which is the type of model considered for the most part in this book. The choice of the level of detail in which the model must describe the phenomena is strictly connected to the indicators defined in Phase 2 and to the actions that are considered. In the case of the 'enchanting lake' described in the introduction, the regulations could be planned by describing the system by a set of algebraic equations, while an executive project for artificial aeration would require the system to be described by differential equations.

The input variables of the model must include the parameters that quantify the attributes of the actions (e.g. the maximum nitrogen supply allowed in agricultural practices and

[35]Take care not to confuse this meaning of the term criterion (Stakeholders' attitude towards risk, see Section 9.1), with the one previously introduced (judgement category). We will encounter semantic ambiguities of this type on other occasions: they occur when the subject of discourse is on the frontier of different disciplines, each of which has independently developed its own jargon.

the nitrogen removal efficiency of the treatment plant) as well as all the variables that allow the future conditions of the system to be described (e.g. the precipitation in the catchment area and the users' water demand). The choice of values to attribute to the former constitutes the subject of the Project, while the values assigned to the latter describe the context within which the alternatives are evaluated and are therefore called the *scenario*.[36] Both alternatives and scenarios have to be quantitatively specified before the model can be run. Note that there can be more than one scenario: in the 'enchanting lake' example we might be interested in evaluating what would occur in a 'high' or in a 'low' rainfall scenario. Moreover, the scenario is not compulsorily deterministic: very often it can be random.

The scenario(s) may be chosen by Experts or it (they) can be obtained by running models, if they are available, that describe the processes that produce the driving forces. In the 'enchanting lake' example the future scenario of rainfall can be generated by a climate change model, while the future scenario of agricultural practices can be suggested by an Expert. When all the processes are stationary, the *historical scenario*, i.e. the trajectories registered in the past, is often adopted for the reasons that will become apparent in Phase 5. In any case, the time horizon of the scenario should be sufficiently long to capture all the types of significant events the system may face. It is a common practice to adopt different scenarios for the design of the alternatives (*design scenario*) and for the estimate of the effects (*evaluation scenario*, sometimes also called *baseline scenario* (BLS), see Appendix A1), which are generally fixed in the corresponding phases.

To facilitate a *social learning* process and help the Stakeholders share a quantitative understanding of the system it is important that they go through the same thinking process and be exposed to the same information and arguments as the Analyst. For this reason, the implicit assumptions of the models should be made explicit and the modelling activity should be supported by an MODSS that must be flexible enough to identify models through a participatory process. Only in that way can the Stakeholders share a common interpretation of the system behaviour (model), which is necessary for them to be able to trust the effects that are estimated with the model. Agreeing on the same model does not prevent them from having different perceptions (indicators) of these effects.

We will deal with these issues in Chapters 4–6.

The set of Phases 1–3 constitutes the *conceptualization* of the Project.

Phase 4 – Designing Alternatives

Unfortunately, very often, it is common practice to consider only the alternatives that are prompted by the Analyst's experience and the suggestions from the Stakeholders. These alternatives make a good starting point, but we believe that it would be a mistake to limit the choice to them alone. More correctly, remembering that an alternative is an integrated package of actions, all the alternatives that can be obtained by combining the actions identified in Phase 1 in all possible ways should be considered. Often the number of alternatives that follows is so high that it would be impossible to examine them all in the following phases and so it is necessary to select only the 'most interesting' ones.

[36]Dictionaries give the term scenario the following definition: "a possible set of future events". Brought into our context the term lends itself to three different possible meanings: (1) synonymous to alternative (e.g. *business-as-usual scenario* means *Alternative Zero*); (2) the set of effects that an alternative produces; (3) the time series of input variables that are not controlled by the DM. We will strictly adhere to this last meaning.

However, these must be chosen still following the Stakeholders' criteria, which were identified in Phase 2, rather than the Analyst's preferences.

In more complicated projects, such as those that this text is concerned with, where a higher level of mathematical formalization is required, to identify the 'most interesting' alternatives it is necessary to define a mathematical problem (a Mathematical Programming Problem or an Optimal Control Problem), called *Design Problem*, which selects the *efficient alternatives* with respect to *objectives*, accordingly called *design objectives*. These are defined on a *design horizon*, given the *design scenario* and taking into account only a subset of the evaluation indicators: the *design indicators*. This simplification is introduced when the considering of all the evaluation indicators would make the Design Problem unsolvable in acceptable computing times; it does not excessively polarize the result if the design indicators are carefully chosen, since in Phases 5 and 6 the alternatives will be evaluated with respect to the complete set of indicators.

The characteristics of the system appear in the Design Problem as *constraints*, while other elements that define the design scenario (e.g. user demands, produce prices) contribute, along with the structural and normative actions, to determine the value of parameters that appear in the constraints and in the objectives of the Problem. Solving the Problem through an appropriate *algorithm*, provides the set of alternatives that will be examined in successive phases. To these the *Alternative Zero* ($A0$) is always added, that is the alternative that assumes that nothing is done and everything remains the way it is (business as usual).

Chapters 7–18 are dedicated to the definition of the Design Problem and to the study of the algorithms that solve it.

Phase 5 – Estimating Effects

Once the alternatives have been identified, the *effects* that each produces must be estimated: in other words, it is necessary to compute the values that the indicators assume as a result of each of the alternatives being implemented. When the system is not dynamical, the evaluation is immediate. When the system is dynamical this estimation requires that each alternative be simulated over a time horizon (*evaluation horizon*) long enough to make extreme events (e.g. droughts or floods) likely to occur, in order to avoid the risk of estimating the effects in 'average conditions' only. In both cases it is necessary to feed the model with an appropriate input: the actions of the alternative considered and one or more *evaluation scenarios*. The alternatives will be compared to single out the 'best' one on the basis of the effects estimated in correspondence with one of these scenarios, the most probable for example; the effects estimated with the others will be useful for evaluating what would happen if the scenario did not occur and in order to adopt a precautionary viewpoint. The choice of the scenario(s) to adopt can be critical, and all DMs and Stakeholders must agree, otherwise, the following phases would fail.

The adoption of an *historical scenario* (i.e. of a situation that was historically recorded) has an advantage in that it allows the comparison between what happened and what would have happened if the alternative had been implemented at the beginning of the historical horizon being considered. This information has a heightened significance for the Stakeholders and DMs because it provides a more immediate perception of the effects of a given alternative, when they have, as they often do, a direct memory of those events. If the historical horizon is too short, artificially generated scenarios can be used, provided that they are as probable as the historical one. By doing so, the estimate of the effects is statistically more reliable, but the psychological significance is lost. Both of these ways to proceed are meaningful only when one can reasonably assume that the processes that generate the scenario (e.g. the meteorological system and land use) remain unchanged into the future. If not, the scenarios have to be generated with models that describe the expected changes.

At the end of this phase the values that have been obtained for the indicators are organized in a matrix, called *Matrix of the Effects*, whose columns correspond to the alternatives and whose rows to the indicators.

In Chapter 19 we will describe these issues in detail.

Phase 6 – Evaluation

An indicator measures, in physical units, the effect produced by an alternative on a particular leaf criterion (see page 23). Nevertheless, the 'value' that the Stakeholders attribute to an alternative, in other words the satisfaction that they get from it, is not always directly proportional to the value assumed by the *evaluation indicator*. In the 'enchanting lake' example, the 'value' that the fishermen attribute to the catch grows very rapidly for low catch yield, but very slowly at high catch yield, i.e. it 'saturates' when the fishermen feel satisfied. To account for this effect, it is necessary to translate each indicator (sometimes a group of indicators) into the 'value' assigned by the Stakeholders. This can be done by means of a *partial value function* that has to be identified through interviews with the Stakeholders. Once all the indicators are transformed into 'values', a Stakeholder (or a DM) can express the overall satisfaction that (s)he assigns to an alternative through a dimensionless *index*, whose value can be computed from the attained 'values'. Therefore, it is possible to sort the alternatives by decreasing values of the index, thus identifying the alternative that the Stakeholder (or the DM) prefers (the first alternative in the ranking).

If there is only one Stakeholder (or DM), the *optimal alternative* is found and the decision-making process is concluded. We will study this in detail in Chapter 20.

When, instead, as is almost always the case, there is more than one Stakeholder (or DM), by working in the aforesaid manner a different ranking is obtained for each one of them. The choice of an alternative then requires the expression of a judgement about the relative importance of the involved Parties (Stakeholders or DMs), i.e. requires that the Parties negotiate between themselves or that a DM (or a Super-DM) expresses her preferences among the Parties (Stakeholders or DMs). Since, however, preferences and negotiations concern subjective aspects, dealing with them is postponed to the successive phases to maintain the distinction between facts and value judgements, as required by guideline 12 in Section 1.2.1.2.

Phase 7 – Comparison

The aim of this phase is the identification of an alternative that is judged to be an *acceptable compromise* by all the Parties and so does not encounter opposition from anyone. Clearly a *win–win* alternative, i.e. an alternative that improves all the Parties' indices with respect to the Alternative Zero, would be the ideal solution for the decision-making process. Unfortunately, such an alternative does not always exist. In a case of irresolvable conflict between the interests of different Parties, the phase concludes with the identification of the alternatives that obtain wide *agreement* from them and listing the supporting and opposing Parties for each of them. We call these alternatives *reasonable alterna-*

tives[37] (or *compromise alternatives*[38]). Thus, with this term we refer to the alternatives that are supported by at least one Party, are admissible (because they satisfy physical, technical and legal constraints), are economically feasible and are Pareto-efficient, i.e. they are such that it is impossible to improve the satisfaction of one Party without worsening that of another (see Section 18.2).

To achieve this result, first of all a series of activities is promoted, which help each of the Parties to know and understand the others' points of view, and, if such exist, the negative effects that the alternative (s)he prefers produce for the others. Once this information has been shared, the heart of the phase is the search for a compromise through *negotiations* among the Parties.

The negotiation process can take place with different procedures that we will study in Chapter 21. Sometimes it is necessary to suspend negotiations and move back to Phase 4 to design other alternatives, in view of what has been understood of the needs, aspirations and requests of the Parties; the effects of the new alternatives should be then estimated (Phase 5), evaluated (6) and brought to negotiations (7). In this way an iteration between Phases 4–5–6–7 is established; an example of this can be found in Sections 5.4 and 10.5 of PRACTICE.

Phase 8 – Mitigation and Compensation

If an alternative enjoys the agreement of the majority of the Parties, but not all them, it is important to explore whether or not it is possible to enlarge the agreement and satisfy some of the unsatisfied Parties through *measures* (meta-actions) of mitigation or compensation. To do this it is necessary to identify new (meta-)actions to include in the alternative, which act specifically on the criteria of unsatisfied Parties. Once these (meta-)actions have been identified, they must be instantiated into actions (Phase 4) and their effects estimated (5); then they must be evaluated (6) and compared (7) with the reasonable alternatives previously identified, in order to see if they actually broad the agreement. In this way one obtains a new set of reasonable alternatives, which could be examined in their turn to find new mitigation measures. Mitigation will be analysed in Chapter 22.

Here a recursion is established between Phases 4–5–6–7–8 (Figure 1.4), which sometimes also includes Phases 1–2–3, during which the whole set of alternatives is 'sifted' in order to single out the reasonable alternatives. Sifting ends when a reasonable alternative is identified, which is accepted by all Parties; or when it is no longer possible to identify mitigation measures or new measures that make it possible to enlarge the agreement; or simply when the time available for the decision-making process has run out. By construction, each alternative obtained in this way has the support of at least one of the Parties. All of them are presented in the summary document of the study (see for example Chapter 15 of PRACTICE), which sums up the entire development of the Project and its results. This document is the material needed to begin the next and last phase.

[37] This term is taken from art. 5 of the Directive 2001/42/EC (European Commission, 2001) about Strategic Environmental Assessment (SEA).

[38] They are given this name because they emerge from a process of negotiation in which an attempt is made to find a compromise among different points of view. However, it is not necessarily possible to achieve this, so the term seems equivocal to us and we prefer the first.

Phase 9 – Final Decision

This phase is put into practice only when there are one or more DMs, at a higher level than the Parties who sifted the alternatives, who are responsible for the final decision about which alternative will be implemented. It is therefore up to these DMs to choose the *best compromise* alternative from the reasonable alternatives, where 'best compromise' means the alternative that best reconciles the different interests, or simply the one upon which they manage to agree. In many cases, this phase is simply a comparison (if there is only one DM) or a negotiation process (if there is more than one) of the reasonable alternatives, which is often carried out with less formalized methods than those used in Phase 7, taking account of the customs and local culture. Sometimes, however, the DM(s) feels the need to explore new alternatives or introduce new criteria. In that case the phase is transformed into a new cycle of Phases 1–8.

The last three phases are the core of the decision-making process and, together, they are framed in greater detail in Chapter 16, before each phase is addressed separately in the chapters that follow.

1.3.2 Remarks

Often the importance of phases that have an engineering or modelling character (in particular *Identifying the Model* (Phase 3) or *Designing Alternatives* (Phase 4)) is emphasized, at the expense of more socio-political phases, like *Defining Criteria and Indicators*, *Evaluation* and *Comparison*. This is a mistake, since a correct decision can be taken only when the expectations, desires, images, knowledge, problems and fears of the Stakeholders are as well described and understood as the physical, technical and economic aspects of the system. Therefore, not only are Phases 3 and 4 of equal importance to the others, and must be considered as such, but the participation of Stakeholders should be full and continuous in all the phases, because only in this way will negotiations in Phase 7 be successful. We will never tire of repeating that if the Stakeholders do not believe in the index values that are shown them in that phase, they will never be willing to negotiate. Actually, they will probably not decline to participate in negotiations, but these latter will develop laboriously, with Stakeholders that listen passively or react aggressively and the result, even if formally it can be achieved, will not really change anything in the existing conflict.

Not all the phases are always necessary. If, for example, in Phase 2 only one criterion is identified, the decision-making process concludes with Phase 4 (or at best with Phase 5). If, instead, there is only one DM and she does not intend to activate a participatory decision-making procedure, the process concludes with Phase 6. If there is no DM above the Parties that participated in negotiations, it makes no sense to go through Phase 9.

It is important to underline that the real development of the decision-making process is not serial as Figure 1.4 might lead one to think. Besides the recursion between Phases 4–5–6–7–8, which is explicitly highlighted in the figure, many others can appear. For example, the criteria cannot actually be correctly identified if one does not known the actions being considered, since these latter produce the effects that the Stakeholders endure. On the other hand, it is not possible to identify the actions without knowing the interests at stake, and therefore the criteria. The presence of recursions is essentially due to the fact that in carrying out the decision-making process new information is produced, because it is a process of *social learning* (Renn, 1995). In view of the new information that is acquired, it is then necessary to re-examine the conclusions of the phases that were considered to be already finished and, when necessary, to modify them. In one sense the aim of the decision-making process is to increase the actors' understanding about the Project, so that they can formulate more and more precise requests and justified opinions.

The PIP procedure has to be supported: except for the phase of *Recognition*, all the phases must be handled by an appropriate set (*toolbox*) of ICT (Information and Communication Technologies) tools, which its users must perceive as part of a unique and coherent system, i.e. a Multi-Objective Decision Support System (MODSS). In the literature this term is sometimes used only in relation to Phases 6 and 7, but we consider this to be inappropriately restricted.

The PIP procedure has to be managed: in all the phases the Stakeholders must be assisted by the Analyst, for the more technical aspects, and helped by a Facilitator as far as organizational and relational aspects are concerned (see Appendix A10).

Finally, it is important to underline that there is a difference between a phase and the method used to implement it in a given context. The phase defines a methodology, i.e. a set of methods and the rules to choose between them. Thus there is not a one-to-one, but a one-to-many relationship between phases and methods. The aim of the chapters in Parts B, C and D of this book is to illustrate these very relationships.

1.3.2.1 The FOTE paradigm

Negotiations and relations among the Parties are easy or difficult in relation to the degree to which they share and exchange information about the problem, the system and their own personal interests; that is the degree to which they adopt a paradigm of *Full Open Truthful Exchange* (FOTE) (Raiffa et al., 2002). This is why the PIP procedure does not begin with negotiating the alternatives, but instead with an information exchange (Phase 0). It goes on with a participatory definition of the actions (Phase 1), the enunciation of the interests (Phase 2), the identification of a shared model of the system (Phase 3). Not always do the Parties involved agree to adopt the FOTE paradigm; however, the Analyst must always suggest its adoption because, otherwise, the Parties might not make the most of their potential synergies. With FOTE paradigm certain basic concepts such as efficiency and equity, along with the reservation values that each side has, gain clarity and crispness of definition, but even when the paradigm is not completely satisfied the analysis made on its basis is still useful, particularly when the Parties agree to tell the truth but not necessarily the whole truth, for example when they are reluctant to disclose their reservation values (BATNA, see Chapter 21). It takes two to tango: it may happen that idealistically one Party is willing to negotiate in a FOTE style, but they are not able to trust the other Party. In that case the FOTE approach is impracticable, but even then FOTE remains a benchmark by which to judge the line of action.

1.3.2.2 Data

Do not underestimate the essential role that the data, and the methodologies for their acquisition play in the decision-making process. The data are used, in a qualitative way, in the phases of *Recognition* and *Defining Actions*, and in a quantitative form in Phases 2, 3, 4, 5 and 8. Their quantity, availability and accuracy are essential to the success of the decision-making process, but even more important is that all the Parties believe that the data are valid and meaningful. The credibility of the model, which is the basis for the credibility of the evaluation of the alternatives, is founded upon this belief, without which the negotiation process is a fruitless exercise. In simple terms: the entire decision-making process depends on the social acceptability of the data.

1.3.2.3 Uncertainty

Ignorance is being unaware that our knowledge is imperfect. An imperfect knowledge implies uncertainty and uncertainty generates apprehension. For this reason, DMs often have the tendency to remove the problem of uncertainty: they want scenarios to be deterministic and models to provide exact estimates, so that their evaluations will be perfect. However, hiding uncertainty is none other than a form of ignorance. Thus, in many phases of the PIP procedure the problem of treating *uncertainty* arises. Uncertainty is produced by corrupt, insufficient or scarce information, and by the errors that are committed unknowingly. We will see in the following that all these causes can be represented as the effects of *disturbances*, which can assume different forms: disturbances are *stochastic* when we know or we can estimate their probability distributions; they are *uncertain* when we know only the set of values that we guess they might assume. The form of a disturbance depends upon the source that generates it. For example data collection generates the most common uncertainty: *measurement errors*, which are always described as stochastic and afflict all the phases in which the data are used. In Phase 1 uncertainty appears also in the description of the actions, since the way in which they will actually be implemented is not always certain (implementation uncertainty); this uncertainty is not only due to implementation aspects, but also to institutional inertia. In Phases 3 and 6 one must account for the disturbances that can make the design and evaluation scenarios uncertain: it is when confronted with this type of uncertainty about the future that DMs and Stakeholders reveal their aversion to risk, which we discussed in the description of Phase 2. In Phase 3 it is necessary to take into account the so-called *process errors*, i.e. the eventuality that the model does not perfectly describe reality.

The effects produced by all these disturbances combine to generate the uncertainty that afflicts the indicator values that make up the Matrix of effects. One must keep this fact in mind in Phases 6–9. The methods with which to do this will be described further on. In particular, Chapter 23 is devoted to a recapitulation of the different types of uncertainty that affect the decision-making procedure and of the techniques that have been proposed to handle such uncertainty.

1.3.3 The Project scheme

The very concept of a Project requires the definition of a mental scheme of the system and its evolution in time, subject to the joint action of the events that would naturally occur, and the implementation of an alternative. The scheme that is commonly adopted is shown in Figure 1.6, where the system is described on the vertical axis, following the DPSIR framework that was introduced at the beginning of the chapter, and time is represented on the horizontal axis.

The decision-making process is considered to be instantaneous, because its duration is negligible with respect to the duration of the following phases, which generally covers several decades. It is placed in the origin of the time axis. Decision-making is followed by the *implementation* period (see also Figure 1.2), during which the chosen alternative is implemented. This period can last from a few months to many years, but the most frequent case is the second, and it ends with the beginning of the *management* period, which is assumed to

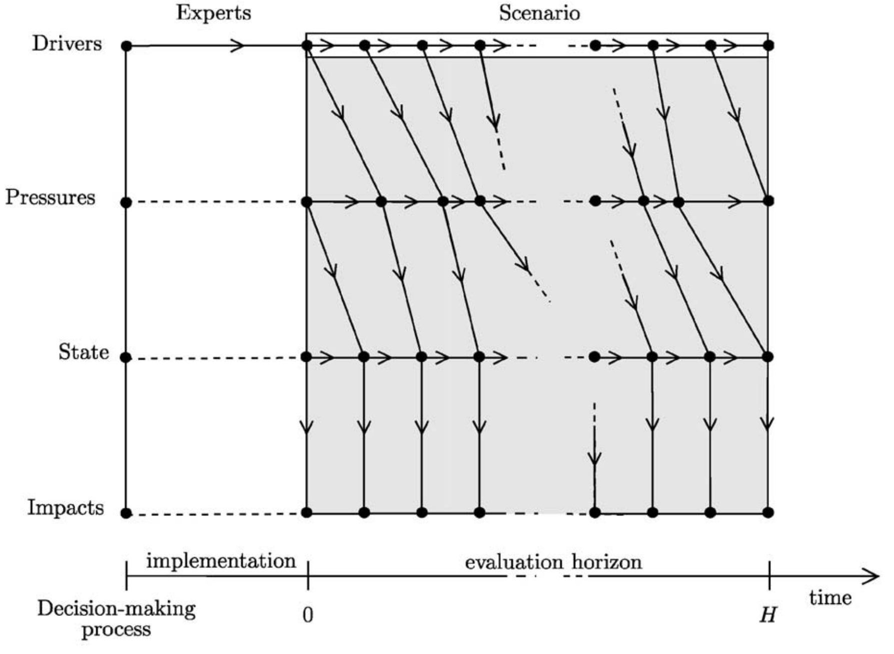

Figure 1.6: The Project scheme for a dynamical system.

extend over a time horizon called *evaluation horizon*. Decision-making requires the estimate of the effects that each alternative would induce over the whole evaluation horizon. During this period the system is subject to the action of the Drivers (in the 'enchanting lake' example, with which we opened the chapter, these are agricultural, industrial and civil practices that produce nitrogenous substances that reach the lake, as well as the rain that generates the lake inflow). As the figure shows, not only does the evaluation horizon extend over time in a significant way, but its starting point is separated from the time instant at which the decision-making process is developed by the duration of the implementation, which, as we have said, is often many years. It follows that it is hardly ever possible to adopt the Drivers' pattern that was recorded in the past, and therefore it is necessary to ask the Experts to predict its future evolution, i.e. to provide a *scenario*. Developing this scenario is not easy, and this is why often the Experts provide several *alternative scenarios*. To generate each of these the Experts can use the behaviour of other variables, termed *external or exogenous*, and they can make use of models (in the 'enchanting lake' example they could use a climate change model to provide the rainfall scenario, in which case the exogenous variable would be the behaviour of the greenhouse gas emissions). By doing so, however, the system being studied has clearly been enlarged (in the example, the climate system that determines the rainfall pattern has been added to the lake) and the external variables are none other than the Drivers of the enlarged system. It would thus seem opportune to redefine the system but, when actions that influence the added system (in the example, the Kyoto protocol which acts on the climate system) are not considered, it is advisable to keep the original description of the system (i.e. consider the lake alone).

The system being studied is almost always a dynamical system and, as such, its condition (state) at the beginning of the evaluation horizon is the result of what will happen to the system during the implementation period. Nevertheless, this is taken into account only in the development of the executive Project, when also the so-called *construction phase* (i.e. the implementation period) is considered. Instead, when the alternatives are screened to find the best compromise alternative, that phase is ignored, since it would be too onerous to take it into account. The consequence of this omission is that the Experts must also provide an estimate of the state of the system at the beginning of the evaluation horizon (the so-called *initial state*).

For the whole evaluation horizon it is necessary to explain how the variation of the Pressures is influenced by the variation of the Drivers, how the former influence the State which, in turn, produces the Impacts. It is the task of the *system model* to describe these transformations. The *actions* of which the alternative is composed play a very important role in such transformations, and so it is essential that the model permit an accurate description of them. When the system is proper and dynamical[39] a variation in the value of the pressure does not result instantly in a variation of the state and, in turn, the state varies progressively in time not only owing to the action of the pressure, but also as a function of its current value (this is the actual meaning of the term 'dynamical'). The same thing occurs very often between Drivers and Pressures, while almost always the Impacts are in a non-dynamical relation with the state. In other words the scheme of the cause–effect relationships among these variables is the one described by the arrows that appear in the grey rectangle in Figure 1.6, which individuates the space–time domain described by the model of the system. Sometimes, however, to simplify the description, one assumes that the system is not dynamical, so that the cause–effect relations become those described in Figure 1.7.

Very often the system is not just dynamical, but also extended over space (i.e., in jargon, it is a *distributed-parameters* system), as is, for example, a river (see also Appendix A1). In that case, the state of the system is at every moment characterized by the spatial patterns of the quantities that characterize it (flow, level and concentrations, in the example of the river) and, since these patterns evolve over time, the evolution of the state is represented by a surface as in Figure 1.8 (see Rinaldi et al., 1979).

The description of these systems is particularly onerous (it requires partial differential equations rather than total ones) and this is why often one looks for a way to simplify their description. The most common solution is to consider the system only in *steady-state conditions*, i.e., when its state has reached an equilibrium. This occurs when the spatial trend of the quantities that describe it does not change over time and so the evolution of the state is described by a surface whose temporal sections are all equal (see Figure 1.9a). Clearly, this condition occurs only when the Drivers do not vary in time, so that each of them is defined by a single value. In steady-state conditions the system is often described by the viewpoint of an observer who moves through space: for example, in the case of a river, the concentration pattern is described from the point of view of an observer who travels downstream on a boat that is carried by the current (see Figure 1.9b). By doing this the description of the system is brought back to the sphere of models that are defined by full differential (or difference) equations, i.e. of *lumped-parameters* models, in which the independent variable is space, or better still the flow time, i.e. the time that has passed from the moment when the observer left the first section. When one decides to consider the system in steady-state conditions, the

[39]The terms are defined in Appendix A3.

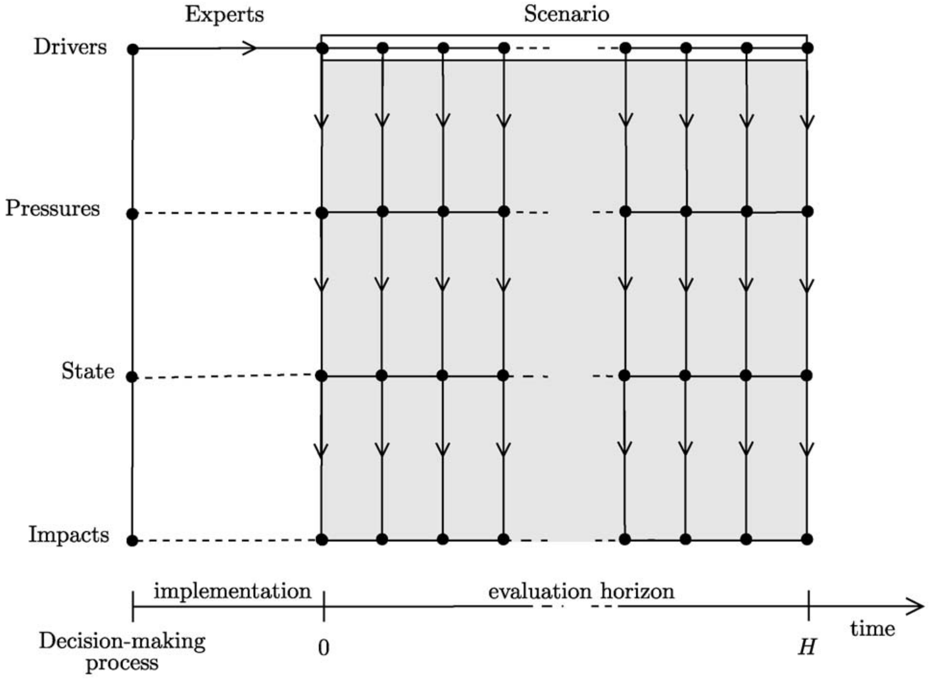

Figure 1.7: The Project scheme for a non-dynamical system.

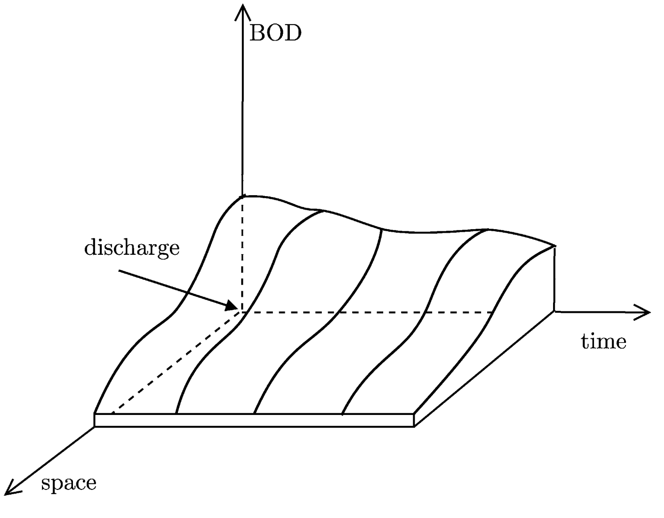

Figure 1.8: The evolution over time, downstream of an effluent point, of a component of the state (concentrations of BOD) of a river, whose initial flow varies because of the regulation (Driver) of a reservoir.

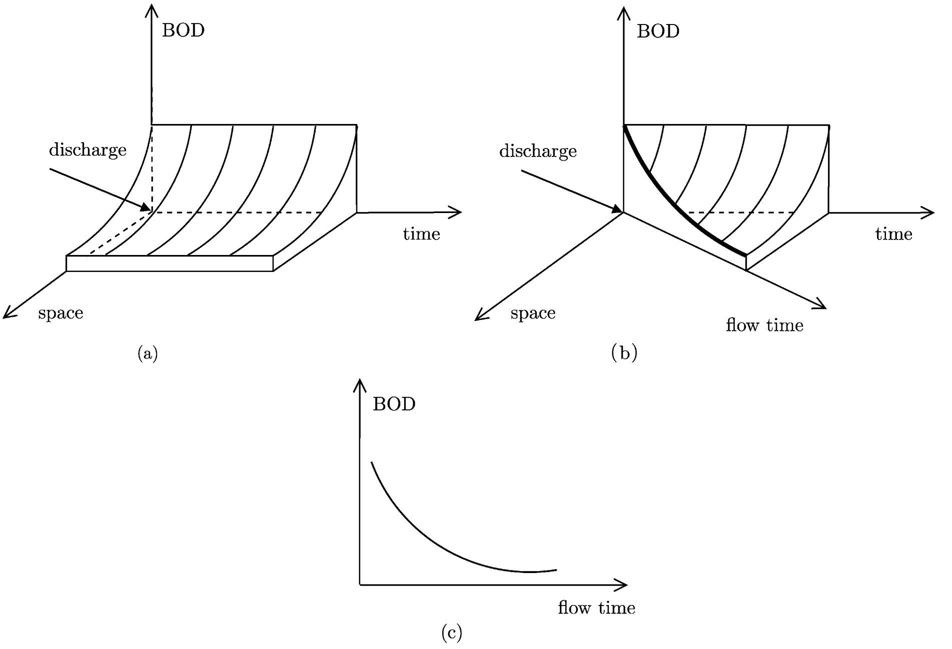

Figure 1.9: The temporal evolution of a state component (concentration of BOD) in the river of Figure 1.8, when the reservoir release is constant over time and the system is in steady-state conditions (a); the same from the viewpoint of an observer who travels down the river in a boat and defines the position of the boat by the time passed from the beginning of the journey (flow time) (b, the bold line); the same as it is represented by the observer (c).

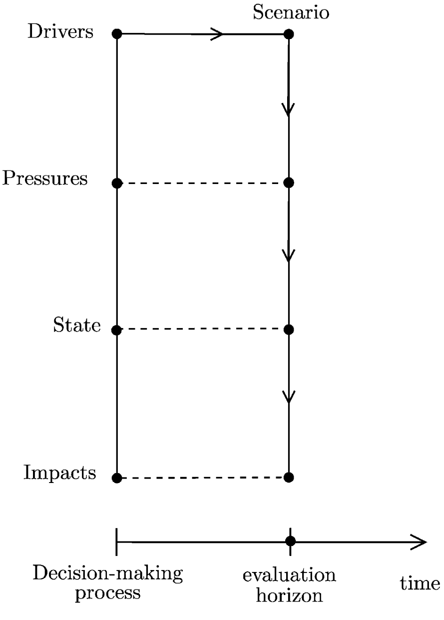

Figure 1.10: The Project scheme when only steady-state conditions are considered.

evaluation horizon contracts to a point and the diagram in Figure 1.6 is transformed into the one in Figure 1.10. An example of this case is described in Appendix A1.

1.3.4 Evaluating Alternative Zero

It is not always evident a priori whether the Alternative Zero ('no actions'), would be sufficient in and of itself to attain the Project Goal. In this case, before beginning the decision-making procedure, the Analyst is often tempted to ascertain if this is true by using a shortcut: he fixes the evaluation scenario and on that basis simulates the effects of the Alternative Zero. By doing so he can see whether it is necessary or not to intervene. The following example may help the reader to better conceive this problem.

The WFD requires that by 2015 all European rivers will have achieved "good status" and that by 2009 each member state will have defined a "plan of measures" (i.e. the alternative, according to the terminology of this book) that are necessary for attaining that Goal. Before designing the plan, the Seine-Normandy River Basin Authority in France clearly wants to ascertain whether the actions that have already been programmed along the Seine from now until 2015 would be sufficient by themselves to bring the river to 'good status'. This is why they decide to estimate the presumed conditions of the sources of pollution in 2015 (i.e. the baseline scenario), with which they simulate the condition of the river in that year. Thereby, they obtain an estimate of the future 'status' of the water system (see Appendix A1).

In our opinion this way of proceeding may create problems in the event that the Alternative Zero proves to be unable to meet the goal. The reader should remember, in fact, that the scenario is none other than the behaviour of the Drivers over the entire evaluation horizon. These Drivers are those input variables of the system model upon which the DM cannot act, i.e. the variables that are not influenced by any of the actions considered. The definition of the scenario, therefore, cannot be done without defining, at least implicitly, the actions that are being considered (Phase 1) and the criteria (Phase 2) by which their effects are evaluated. From these last two definitions follow the definitions of the input and output variables of the model with which the system is represented (Phase 3). From them and from the definition of the actions, the Drivers with which one wants to predict the system behaviour emerge, i.e. the scenario emerges (Phase 5). If these phases are not carried out explicitly, the Analyst runs the risk of discovering a posteriori that the scenario was identified incorrectly. This may emerge once the Alternative Zero has proved to be insufficient for meeting the Project Goal, and all the choices that were previously implied are made explicit, in implementing the PIP procedure. If this should happen, the Analyst would be seriously embarrassed.

It is true, however, that, when it is not evident a priori, ascertaining whether the Alternative Zero alone would be sufficient to attain the Project Goal is clearly the first thing to do. But the correct way to do this is to carry out a first iteration of the PIP procedure, during which in Phase 1 only the actions that have already been deliberated are considered, thus being an integral part of the Alternative Zero; Phases 2 and 3 are carried out as stated in the PIP procedure; Phase 4 is omitted; in Phase 5 the scenario is defined and the effects that the Alternative Zero produces are estimated. These effects are evaluated in Phase 6; Phase 7 is limited to verifying whether the Project Goal has been achieved and, if the response is negative, one returns to Phase 1 to identify actions that are suitable to attain it. This example is developed in greater detail in Appendix A1.

1.4 Management

Once the best compromise alternative has been selected (Phase 9 of the PIP procedure), it has to be implemented (see Figure 1.2): this is achieved by implementing the structural and normative actions it includes, and applying the regulation policy, if this has been designed, at the scheduled time instants. Since this policy may leave some degree of freedom to the Regulator (see Section 10.1.1 on *set-valued policies*), a decision-making problem must be formulated at the *management level* as well, though the degree of freedom that the policy allows is much less than the degree that the DM had when the problem was formulated at the *planning level*, i.e. at the level that we have been considering until now. The best compromise alternative is therefore at the same time the conclusion of the decision-making process at the planning level and the starting point for the decision-making process, which is renewed periodically, often daily, at the management level, on the basis of the new information which is obtained as the time goes on.

1.5 Monitoring

Once the best compromise alternative has been implemented and management has begun, the effects that are produced should be monitored (see Figure 1.2) continuously over time, in order to make sure that the real effects are actually those that were foreseen. Should they not be, it is necessary to open a new cycle of intervention. We will not deal with monitoring in this book, but the interested reader may consult UNECE (1990) and Wiersma (2004).

Chapter 2

From the decision-making procedure to MODSS

AC and RSS

In the previous chapter we saw how the decision-making process should unfold. We would now like to identify the critical points of the process, to give the reader a general vision of the issues that will be dealt with in this book. We will introduce many concepts that will be taken up and brought into focus in the following chapters. We invite the reader not to dwell too much on the details and to concentrate on the line of thinking. Moreover, some aspects have been simplified a little to provide the reader with an overview of the entire content of the book.

2.1 Planning and management

In Section 1.1.4 we saw that planning actions, such as the construction of a reservoir or the installation of a network of remote rain gauges, are implemented once only, while management actions, which are mostly concerned with distributing resources in time and space (such as releasing water from a reservoir or distributing flows among canals), are implemented periodically.

Planning decisions are made within a Project, in which different alternatives are compared to identify the one that most satisfies the *Decision Makers* (DMs) and the *Stakeholders*. Management decisions, on the other hand, must be taken periodically, often every day, and therefore, by their nature, it seems that they must be left to the intuition of the *Regulator*, who decides what to do on the basis of the available information. However, this is not a rational way to proceed.

To understand the reason, we consider a water system in which the outflow from a catchment feeds an irrigation district, where maize is grown. Let us suppose that the outflow pattern is the one typical of Mediterranean coastal plains, and therefore does not satisfy the farmers, because flows are abundant during spring rains and scarce in the dry summer, just when the maize water requirement is high. To satisfy the farmers a reservoir could be constructed (Figure 2.1), in which a part of the spring flows could be stored, in order to release it in summer. The morphology of the land suggests only one location where a reservoir could be constructed and the geological conditions univocally define its capacity.

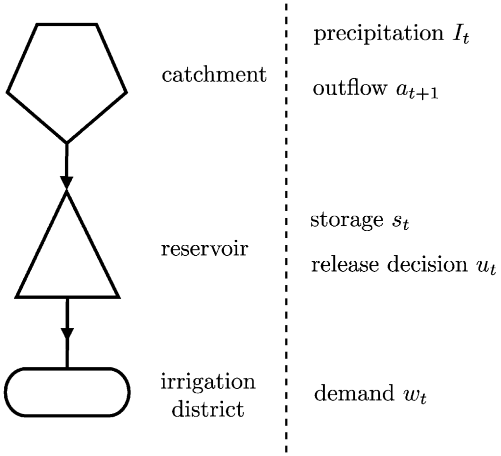

Figure 2.1: A simple water system and its most important variables.

Thus, the only decision to be taken is whether to build the reservoir or not. In order to discriminate between these two alternatives, we must compare the effects that each of them would produce. In absence of the reservoir, the average annual supply deficit with respect to the water demand of the maize can easily be computed because the release equals the outflow and therefore its time pattern is known. However, what would be the deficit in the opposite case? The reader can easily understand that it is impossible to answer the question without knowing how the reservoir will be regulated. If, for example, the water is never detained and it is allowed to run through the reservoir's bottom outlet, the deficit would be the same as if there were no reservoir. If, instead, the reservoir were used to collect the outflow produced by the spring rains and distribute it during the summer, the deficit might diminish, but to say by how much, we must establish the way in which water is stored and released every day. Therefore, we must know which *release decisions* (note that these are management decisions) will be taken, or even better, how they will be taken. In this case, the planning process requires that not only is a planning decision taken (to build the reservoir or not), but also that the management decisions are specified. We will see how this can be done shortly. For now it is sufficient to point out that a planning decision cannot be evaluated without dealing with the management, whenever this second is implied by the first (as it is in our example) or the first modifies the context in which the second operates (as would be the case in our example if the reservoir already existed and we wanted to evaluate whether or not to install a remote rain gauge in its catchment).

In these cases it is necessary to *plan the management*. How can we do this? How can we structure and then solve this problem, which we will call a *Management Problem*?

2.1.1 Planning the management

To define the instrument with which to formulate correctly the Management Problem it is above all necessary to specify the decision that has to be assumed. We will use the example above once again to exemplify. Once the reservoir has been constructed, every morning its Regulator has to decide the volume of water u_t (*release*) to deliver to the irrigation district over the following twenty-four hours.

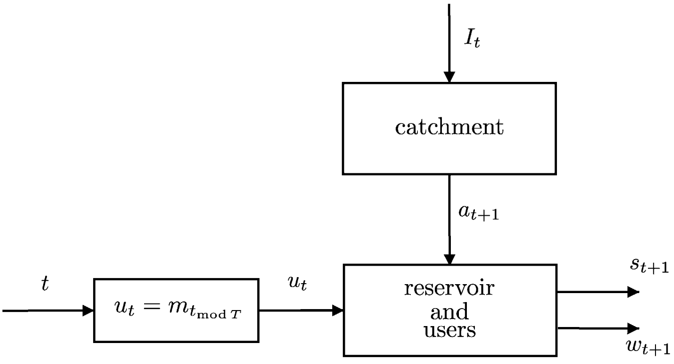

Figure 2.2: Open-loop control scheme.

2.1.1.1 The release plan

The first instrument conceived to avoid having such management decisions entrusted completely to the intuition of the Regulator was the *release plan*. It is defined as a sequence of releases

$$m_0, m_1, \dots, m_{364} \tag{2.1}$$

one for every day of the year, to be executed 'in normal conditions'. In other words, one assumes that[1]

$$u_t = m_{t_{\text{mod }T}} \tag{2.2}$$

'whenever possible'. With this assumption the reservoir is controlled according to the scheme in Figure 2.2, which is called an *open-loop* control scheme. The reason for this name will become clear shortly.

Clearly, a good dose of indeterminacy is still present. What do the phrases 'in normal conditions' and 'whenever possible' mean? In practice, during the actual management process, the reservoir Regulator feels free to deviate from the release plan whenever necessary or when he thinks it is appropriate. Nevertheless, establishing when and how much to deviate is not a simple decision. In fact, even if it is evident a priori that the larger the deviation from 'normal conditions', the larger the deviation from the release plan must be, it is not clear how the Regulator might quantify this deviation.

To assist the Regulator, a reference *trajectory*[2] ($s_{\cdot}^*$) of the reservoir storage was associated to the release plan. It was called *rule curve*. This trajectory can be defined as the trajectory followed by the reservoir's storage when the release pattern follows the release plan and the hydrological year is 'normal'. The rule curve can easily be calculated with a

[1] With the notation $t_{\text{mod }T}$ we denote the remainder of the division of t by T, T being the duration of the period of the periodic process that we are considering. In our example, as is almost always the case in water related problems, T is equal to 365 days, i.e. a year. Given any day t (for example the 413th), $t_{\text{mod }T}$ is the number (48) of that day with respect to the beginning of the year to which it belongs (the second).

[2] From here on we will adopt the following notation: s_t denotes the value assumed at time t by the variable s, whose value changes in time. The trajectory of the variable s is the path of the values that it assumes over time and it will be denoted with $s_{\cdot}$ (see the box on page 41). When we want to show that the trajectory in the interval $[t_1, t_2]$ is to be considered, we write $s_{[t_1,t_2]}$, while if time t_2 is excluded we write $s_{[t_1,t_2)}$.

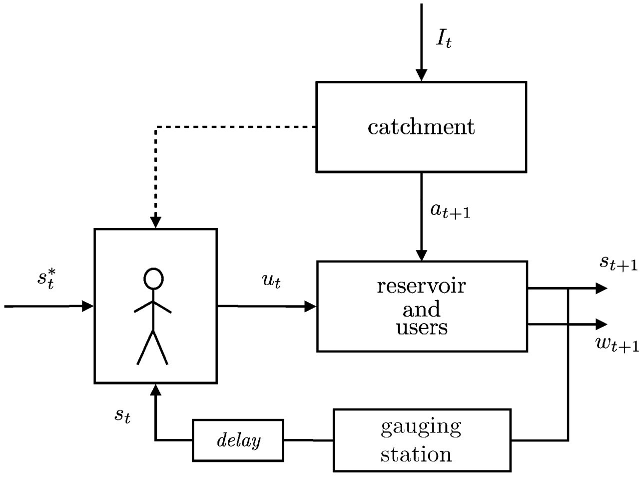

Figure 2.3: The role of the Regulator.

simple simulation, using a model of the reservoir and of the catchment. Once the rule curve is plotted, the release decision u_t can be taken with the following rule:

> *today's release should be such as to bring tomorrow's storage as close as possible to what is specified by the rule curve.*

In this way, as Maas et al. (1962) observed a little ironically, the Regulator "spills water when the storage s_t in the reservoir exceeds the quantity s_t^* specified by the rule curve and hopes for rain when it falls below". The corresponding control scheme[3] is shown in Figure 2.3.

Often, however, because of the stochastic nature of the outflow from the catchment, the releases produced by following the rule curve can deviate so much from the release plan as to be unacceptable. When this occurs the Regulator is left alone again with his decision-making problem and may rely on his own intuition, good sense and experience. Acquiring this experience is a long and costly process and, as a consequence, quite often the systems that are managed with a release plan are exploited well below their potential.

2.1.1.2 The regulation policy

At the beginning of the 1970s it was understood (Maas et al., 1962) that the only rational solution to the Management Problem is that a decision be taken at every time instant on the basis of the information available at the time. For example, if a storage s_t is being recorded, the decision should have the form

$$u_t = m_t(s_t) \tag{2.3}$$

where $m_t(\cdot)$ is a monotonic increasing function of s_t, called *control law*. In this way, the decision is not determined a priori for time t, regardless of the conditions that might occur

[3]When making a decision the Regulator takes into account the storage s_t, which is present at that time in the reservoir. The delay that appears on the line of the gauging station in the figure shows that the storage s_{t+1} that, as a consequence, will be produced can be observed only on the following day.

An important notational convention *If the variable y is a function of the variable x we will write*

$$y = f(x)$$

to denote the value that y assumes in correspondence with the value x. Therefore $f(x)$ denotes a value. We will write

$$y = f(\cdot)$$

instead to denote the relationship between y and x. Therefore, $f(\cdot)$ is a function. For example $m(s_t)$ denotes the decision assumed in correspondence with the storage s_t, while $m(\cdot)$ denotes the control rule. If the function $f(\cdot)$ changes through time we will write $f(\cdot, t)$ in the case of continuous time, and $f_t(\cdot)$ in the case of discrete time. For example, the total precipitation [mm] *that fell in day t in the catchment in Figure* 2.1 *is denoted with I_t, while if we were interested in the intensity of the precipitation I* [mm/h] *at each time t of the day we would denote it with $I(t)$.*

(as is the case in the release plan), but depends on the condition that does occur at that moment (the storage s_t). This condition depends in turn on the decision that was taken at the previous time step, and thus there is a recursive loop. Therefore, the control scheme (2.3) is said to be *closed-loop* (comparing this scheme, shown in Figure 2.4, with the one in Figure 2.2 reveals the origin of the name). The loop is closed by the function $m_{\cdot}(\cdot)$, which is almost always a periodic function with period T (usually a year). This function is called *regulation policy* (or control policy) and is indicated with the letter p. From here on we will simply call it *policy*. A policy p is therefore defined[4] as a periodic time sequence of *control*

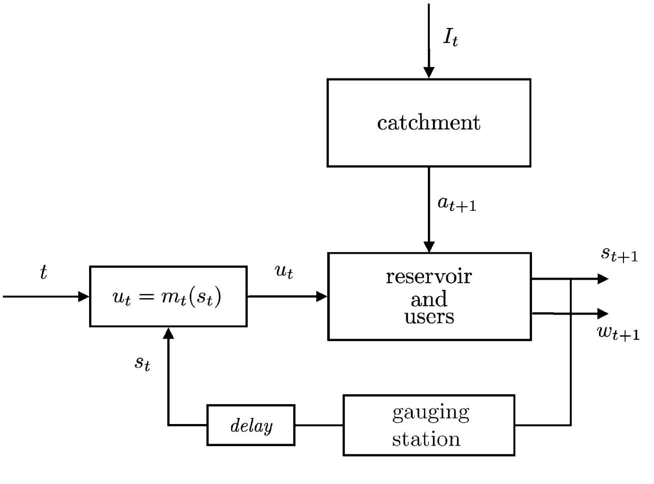

Figure 2.4: Closed-loop control scheme.

[4]The symbol $\triangleq$ stands for 'equal by definition to'.

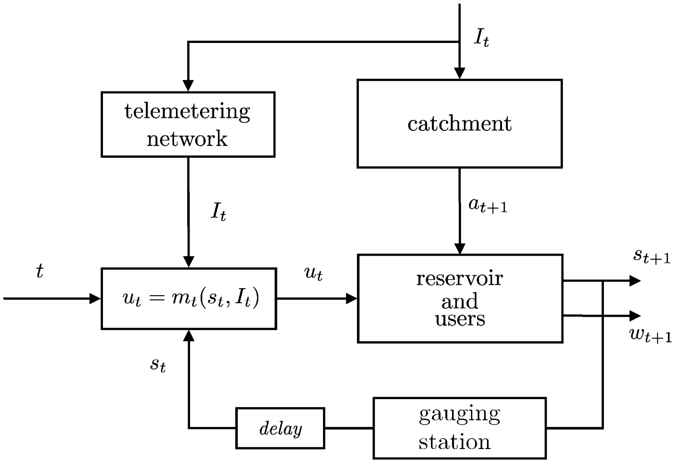

Figure 2.5: Closed-loop control scheme with compensation.

laws

$$p \triangleq \{m_0(\cdot), \ldots, m_{T-1}(\cdot), m_0(\cdot), \ldots\} \tag{2.4}$$

each of which specifies, for a given time t, the release decision u_t as a function of the current storage s_t of the reservoir.

Examining Figure 2.4 makes it evident that the management of the system could be more effective if the outflow a_{t+1} into the reservoir, which acts as a *disturbance*, were known beforehand. This is clearly impossible, but the effects of the outflow can be anticipated by using the hydro-meteorological information I_t that is collected in the catchment. In this way the release decision u_t is given by

$$u_t = m_t(s_t, I_t) \tag{2.5}$$

and the control scheme takes the form shown in Figure 2.5. We say that it includes a *compensation line*, because the policy attempts to 'compensate', i.e. to reduce, the effect of the disturbance a_{t+1}.[5]

If the water demand w_t can deviate significantly from its nominal value, as for example occurs frequently for irrigation districts, one ought to take these variations into account explicitly. The demand w_t is then included among the policy arguments, so that the decision u_t takes the form

$$u_t = m_t(s_t, I_t, w_t) \tag{2.6}$$

The corresponding scheme[6] is shown in Figure 2.7.

[5] Often one assumes that the policy has the form (2.5), but one implements it through a cascade connection of two blocks as shown in Figure 2.6. The first block is a outflow forecaster that, on the basis of the information I_t, generates a forecast $\hat{a}_{t+1}$. The second is a policy that has the couple $(s_t, \hat{a}_{t+1})$ as its argument, so that the decision u_t is given by $u_t = m_t(s_t, \hat{a}_{t+1})$.

[6] This is characterized by a double loop. The first closes on the reservoir, the second on the irrigation district. Reservoir and district are both in fact influenced (in parallel) by the decision u_t and so the control loop should be closed on each of them.

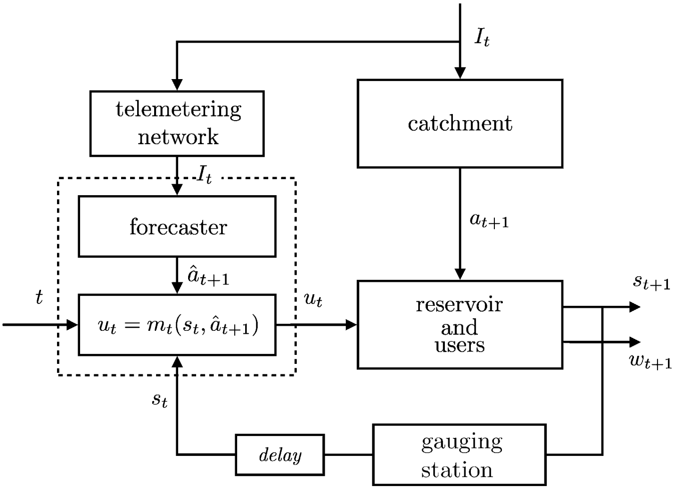

Figure 2.6: Closed-loop control scheme with compensation, achieved by a outflow forecaster.

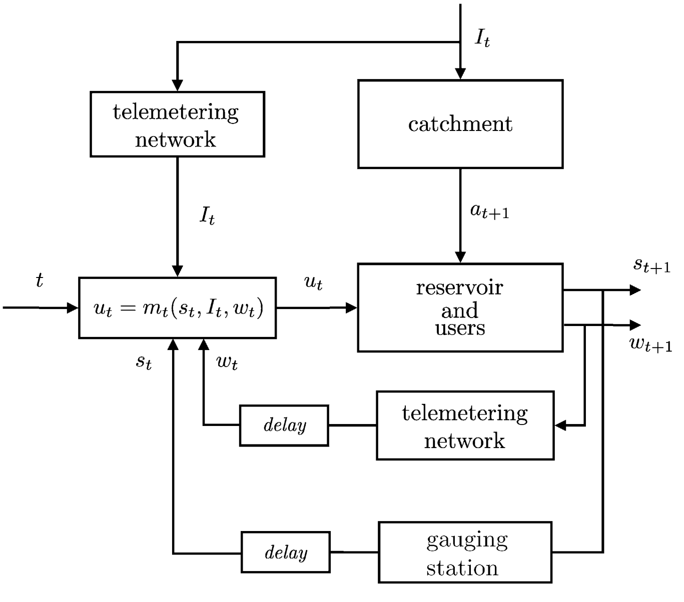

Figure 2.7: On-line use of agricultural demand.

2.1.1.3 The set-valued policies

Even if the policy-based approach seems very rational, it has been unsuccessful, since the applications in which it has been actually adopted are very few. The reasons for this failure are many and complex. For now we will illustrate just one which is linked to the very definition of policy.

The definition that we have given to the term is the one that is commonly adopted in Control Theory, where the task of the policy is to 'automatically' control a given system. When the policy is used in the management of a natural resource, such as a water reservoir, it is not expected that it manage the system 'automatically', but it should help the Regulator, by suggesting valid decisions and leaving the last word about the one to choose to him. In this perspective, with the information $\Im_t$ available at time t (that in the last example corresponds to the triple (s_t, I_t, w_t)), it seems more rational that the policy proposes not just one control, $m_t(\Im_t)$, but the set $M_t(\Im_t)$ of all the equivalent controls (Aufiero et al., 2001), i.e. the set of all the controls that provide, in the long-term and from the Regulator's point of view, 'equivalent' system performances. In this way, at time t the Regulator can choose the control, from the set $M_t(\Im_t)$, which he believes best corresponds to the current situation. We should therefore substitute the concept of a point-valued policy with the concept of a *set-valued policy*, defined as a succession of set-valued control laws $M_t(\Im_t)$

$$P \triangleq \left\{M_0(\cdot), \ldots, M_{T-1}(\cdot), M_0(\cdot), \ldots\right\} \tag{2.7}$$

2.1.2 Designing the management

Compare the two schemes in Figures 2.3 and 2.4. They are structurally similar, but the human Decision Maker that appears in the first is substituted by the policy in the second. Once a given policy $\bar{p}$ (or $\bar{P}$) has been specified, i.e. the sequence (2.4) (or (2.7)) has been completely defined, the average annual irrigation deficit that it creates can be calculated by simulating the behaviour of the reservoir controlled by $\bar{p}$ (or by $\bar{P}$). Then the DM can compare various policies in order to identify the one that satisfies her the most. The policy is therefore the means by which the difficulties posed by management decisions are overcome, and which at the same time helps to eliminate the uncertainty generated by the subjective and non-transparent presence of a human decision maker.

2.1.3 Generalizations

Until this point we have referred to a very simple water system, but what has been said can easily be generalized. The system could be, in fact, composed of many reservoirs and many users, interconnected by a network of canals. In this case the decision would be represented by a vector $\mathbf{u}_t$, whose components are the volumes to be released from each reservoir in day t (or in the time step being considered) and the flows to be diverted into each canal, and, potentially, also the binary decisions, such as, for example, whether or not to announce a flood alarm in a given zone.

2.1.3.1 The distribution policy

A particular, but common, problem is concerned with a water system consisting of a network of canals and diversions, without reservoirs, that distributes the outflow from one or more catchments among various users. A completely equivalent case is that in which such a network is fed by releases from one or more reservoirs that are not under the control of the Regulator. In both cases the Regulator's problem is to distribute the resource among the users. This is a Management Problem since the flows that have to be diverted into the different canals must be established every day. The solution is, as we have seen, a policy, called *distribution policy*, that defines the flow to be diverted into each canal as a function of the flows that have entered the network. When the models that describe the users are not dynamical (see Appendix A3), the problem is a non-dynamical problem, because today's decision does not depend on the decision taken yesterday. The policy can then be designed by determining the control laws for different days separately. However, in the case that we considered before, in which the resource can be allocated over time by storing it temporarily in a reservoir, all the control laws have to be designed together, since the decisions are interlinked. This difference in structure means that designing a distribution policy is computationally much simpler than designing a regulation policy. We will analyse this aspect in Chapter 15.

2.2 Decision making under full rationality conditions

In the previous section we learned how it is possible to plan management, i.e. how the Management Problem can and must be traced back to a Planning Problem. In doing this we have taken for granted that the latter problem can be formulated and solved easily, but since in reality this is not the case, in this section we will give an idea of what can be done. Once again we will use an example.

The map in Figure 2.8 shows the Cabora Bassa reservoir in Mozambique, which was created by damming a stretch of the River Zambesi with an arch dam. Thus a reservoir with a capacity of 56 billion cubic metres (km^3) was created that now regulates the river's flows, which are on average 77 km^3 per year. This volume serves[7] an irrigation district

Figure 2.8: The River Zambesi and the Cabora Bassa reservoir.

[7] The reasons for which the dam was constructed actually included hydropower production. The plant at the foot of the dam has an installed power of 3870 MW and it is the biggest in Africa, but, because of didactic reasons,

located downstream on the plain, in which cereals and cassava are produced. Today the area extends for only 20 000 ha, but it will probably extend to 1 700 000 ha in the future. The Cabora Bassa reservoir was designed in the 1960s, but its construction was slowed by the civil war for Mozambique's independence and its operation started only in 1977. Let us suppose now that it is 1960 and that we have to design the dam's capacity and its regulation policy on the basis of the following criterion: "satisfy the district's irrigation demand as much as possible".

To develop our understanding about how to proceed, let us suppose for a moment, even if the hypothesis is clearly absurd, that the inflow to the reservoir is deterministic.

2.2.1 In a deterministic world

Even if we assumed we are in 1960, given that the inflows are deterministic, their trajectory $\{a_1, \ldots, a_h\}$ is exactly known over any time horizon[8] H following the start-up of reservoir operations; for example over the amortization horizon, let us say 1975–2074. We are then able to simulate the behaviour of the reservoir that corresponds to every given value of its *capacity* c (see page 138) and to each regulation policy

$$p \triangleq \left\{m_0(\cdot), \ldots, m_{T-1}(\cdot), m_0(\cdot), \ldots\right\}$$

Thus, we obtain the trajectory $q_{[0,h]}$ of the supply that will be made available for irrigation, on a daily basis, over the whole horizon H. Given the capacity c, an engineering company can provide us with the estimated cost of construction $C(c)$ for the corresponding reservoir. The farmers (or more precisely the Experts they have nominated) can estimate the benefit[9] $B^{\mathrm{Moz}}(q_{[0,h]})$ produced by selling all the harvests yielded over the horizon H given the supply pattern $q_{[0,h]}$.

With H being the hypothesized write-off period, the net benefit i from the management of Cabora Bassa is given by

$$i(c, p) = B^{\mathrm{Moz}}(q_{[0,h]}) - C(c)$$

It follows that it is rational to choose the pair capacity–policy (c^*, p^*) that maximizes the value of i. This implies that the ambiguous expression 'as well as possible' that appears in the management criterion ("satisfy the district irrigation demand as well as possible"), has to be understood to mean 'so that the net benefit produced during the write-off period is maximized'.

Note that the values that the capacity c can assume will be limited to a set $\mathcal{U}^c$, which depends on the geological characteristics of the gorge where we want to construct the dam. Analogously, the volume that we will be able to actually release in day t will be limited both by the storage s_t that is available in the reservoir that day, and by the characteristics of the intake.[10] In other words, the release decision u_t will be restricted to a set of values $\mathcal{U}(s_t)$.

we will not consider this aspect. More generally in this chapter the River Zambesi and the States that it passes through are used only as a scenario and actors in a didactic example, that as such, allows us to manipulate reality, ignoring aspects that could not be left out in a real project.

[8] The horizon H is composed of the succession of days numbered from 0 to h.

[9] Since the Cabora Bassa dam is a public good, the Project should be structured with a Cost Benefit Analysis, and so the benefit is the willingness of the farmers to pay for the irrigation supply $q_{[0,h]}$. If, instead, the dam and the irrigation district belonged to a private owner, a Cost Recovery approach would have to be adopted and then the benefit would be the revenue from the sale of harvest.

[10] For the sake of simplicity we assume that there is only one way to build it.

Moreover, on day $t+1$, the storage s_{t+1} will be univocally defined by the storage s_t available the previous day, the release decision u_t taken and the outflow a_{t+1} that will reach the reservoir in the 24 hours between time t and time $t+1$. In other words

$$s_{t+1} = f(s_t, u_t, a_{t+1})$$

and as a first approximation (see Section 5.1 for more details) we can assume that

$$s_{t+1} = s_t + a_{t+1} - u_t$$

Lastly, given the release u_t, we can compute the supply q_t to the district, once the characteristics of the diversion dam that diverts water from the river have been defined; that is

$$q_t = h(u_t)$$

It follows that the optimal pair (c^*, p^*) is the solution to the following problem

$$\max_{c,\,p} i(c, p) \tag{2.8a}$$

$$s_{t+1} = f(s_t, u_t, a_{t+1}) \qquad t = 0, \dots, h-1 \tag{2.8b}$$

$$q_t = h(u_t) \qquad t = 0, \dots, h \tag{2.8c}$$

$$u_t = m_{t_{\text{mod}\,T}}(s_t) \qquad t = 0, \dots, h \tag{2.8d}$$

$$u_t \in \mathcal{U}(s_t) \qquad t = 0, \dots, h \tag{2.8e}$$

$$p \triangleq \{m_0(\cdot), \dots, m_{T-1}(\cdot), m_0(\cdot), \dots\} \tag{2.8f}$$

$$c \in \mathcal{U}^c \tag{2.8g}$$

$$\{a_1, \dots, a_h\} \qquad \text{given scenario} \tag{2.8h}$$

which in the literature is known as *Optimal Control Problem*.[11]

2.2.1.1 The PIP phases

It is very useful to reinterpret what we have done up until this point, in the light of the PIP procedure which we introduced in Section 1.3 (see Figure 1.4). Above all we have defined the Project Goal: to maximize the net benefit produced in the write-off period. The reader who is familiar with economics will have understood that we chose to formalize the project by adopting the approach known as *Cost Benefit Analysis* (CBA), but clearly this choice was not the only possibility.

Subsequently, the verbal expression that we used to define the Goal was transformed into a quantitative indicator, i.e. a functional of the actions considered (that in our case are specified by c and $u_{[0,h]}$) and of the trajectory, over the whole horizon H, of the variables ($q_{[0,h]}$) that these actions influence. The indicator is identified by consulting both the Stakeholders (the farmers) and the Experts (the engineering company). Both groups provide information about the effects of the decisions. The farmers provide an estimate of the benefit $B^{\text{Moz}}(q_{[0,h]})$ that they would expect with a given trajectory of supply ($q_{[0,h]}$), while the Experts provide an estimate of the construction costs $C(c)$ for a reservoir with capacity c.

[11] The name derives from the fact that such a problem was originally formulated in a mechanical-electronic context with the aim of determining the best 'control' for a device such as an engine. Among its first and most famous applications was the design of the control systems of the space ships, used in the Gemini and Apollo project, that in 1969 sent a man to the Moon.

The next phase is the identification of the model for the water system considered in the Project, which quantitatively describes the links between all the variables involved. This model is made up of a catchment model (the time series (2.8h)), a reservoir model (2.8b), (2.8e) and a model of the distribution network (2.8c), as well as the geological limits to the capacity of the reservoir (2.8g).

Finally, the two phases, *Designing Alternatives* and *Evaluation* are carried out at the same time by formulating and solving problem (2.8), which provides the optimal alternative. How this problem can be solved in practice is a mere technicality, and we will deal with it in Chapters 12–14. For now the reader should not worry about it.

2.2.1.2 The MSS

In order to carry out the phases of the PIP procedure, data must be stored and manipulated and models must be specified and run, both descriptive models (such as the reservoir model (2.8b) and the distribution network model (2.8c)), and decision-making models (such as problem (2.8)).

These activities can be carried out only with the support of an information system. Such a system may be constituted by a set of programs, or, in the simplest cases, by a simple spread sheet. The system should be structured in a way that makes the operations as simple as possible, both to minimize the risk of error, and above all, to allow the Analyst and the DM to concentrate on the analysis of the alternatives, by making the information and modelling aspects as transparent as possible. Information systems are designed for this purpose and called Modelling Support Systems (MSS).

2.2.2 In an uncertain world

The hypothesis that the outflow is deterministic, i.e. it is known beforehand, is, however, completely unrealistic and no serious project could be based upon it. Therefore it is necessary to let this hypothesis lapse, even if the consequences are not exactly trivial, for two kinds of reasons.

The first is that it is not sufficient just to say that the future inflows, i.e. the ones that will materialize after the reservoir operational start-up, are not known beforehand. If they were completely unknown, no problem could be formulated and therefore no decision could be taken. If we deem that we are able to decide rationally it is because we presume that we can forecast and evaluate, to some extent, what will occur in the future. But this is possible only if some of the properties that were observed in the past are preserved into the future. Indeed, only in such a case is it possible to make a prediction. Even if it is not sensible to assume that the future inflow sequence is known exactly, it is however necessary to assume that at least the statistical properties of the process that generates the inflows will remain unchanged over time (*the steady-state paradigm*). In the simplest case this means that we must assume that the mean inflow does not change. So, it will be necessary to identify a model with such properties (a *stochastic model*) to replace the deterministic series (2.8h) in problem (2.8). The identification of such a model, however, requires the steady-state hypothesis to be satisfied. We will discuss this assumption in Section 2.5.

The second reason is that the uncertainty[12] of the inflows generates a corresponding uncertainty in the value of the *net benefit*, i.e. in the value of the objective (2.8a) of the

[12] In this chapter we will use, as is custom in common language, the adjectives uncertain, stochastic, and random, as if they were synonymous. Later on we will see that technically speaking they are not.

Optimal Control Problem. In other words, the net benefit also becomes a stochastic variable and, as such, it can no longer be the problem objective: maximizing a stochastic variable has no meaning.

Even if it is impossible to maximize a stochastic variable, it is however possible to maximize any one of its statistics, which, by definition, are deterministic variables. For example, one could think that the farmers are interested in maximizing the *expected value of the net benefit*. In so doing, the problem (2.8) takes the following form

$$\max_{c,\, p} E_{\{a_1,\dots,a_h\}}\big[i(c, p)\big] \tag{2.9a}$$

$$s_{t+1} = f(s_t, u_t, a_{t+1}) \qquad t = 0, \dots, h-1 \tag{2.9b}$$

$$q_t = h(u_t) \qquad t = 0, \dots, h \tag{2.9c}$$

$$u_t = m_{t_{\mathrm{mod}\, T}}(s_t) \qquad t = 0, \dots, h \tag{2.9d}$$

$$u_t \in \mathcal{U}(s_t) \qquad t = 0, \dots, h \tag{2.9e}$$

$$p \triangleq \big\{m_0(\cdot), \dots, m_{T-1}(\cdot), m_0(\cdot), \dots\big\} \tag{2.9f}$$

$$c \in \mathcal{U}^c \tag{2.9g}$$

$$a_{t+1} \text{ is provided by the stochastic model} \quad \forall t \tag{2.9h}$$

where $E[\cdot]$ is the expected value operator with respect to the future inflow pattern.

Nevertheless, it is not certain that the farmers are actually interested in maximizing the expected net benefit. The reason is that the expected net benefit over the horizon H is an estimate of the average net benefit that one would obtain if the management over that horizon were repeated an infinite number of times, each time with a different inflow sequence generated by the stochastic model. These sequences are in fact by construction equally probable. The farmers however, will not experience the horizon H an infinite number of times: they will experience it only once. Thus the expected net benefit may have a very weak relationship to the benefit that will be actually produced. There might be a very dry inflow pattern that produces a drastic reduction of the harvest. It is therefore plausible that the farmers would be more interested in the minimum benefit value than in the expected benefit value, i.e. they might be interested in the benefit produced in correspondence with the worst possible inflow sequence. By maximizing that minimum value they would know that, once the dam was constructed, the benefit that they would receive would never be any less. For this reason the minimum value is often called the *certain performance*.

Since these are very important ideas, we will try to make them clearer with a simple example. Let us suppose that we have two alternative investment proposals and that the choice will be offered to us just once. Both of the alternatives require an investment of €100, as a result of which the first alternative (A) provides a sure return of €120, and the second (B) of €121. There is no doubt that anyone would choose alternative B. Let us now suppose that the return from the second alternative, instead of being deterministic, were stochastic: there is a 90% probability that we will lose our investment, and a 10% possibility that we will get a return of €1210. Now it is not so easy to say which the DM would choose. If the adopted criterion was the maximization of the expected return she would have to choose alternative B, for which the expected return[13] is €121, just as in the

[13] By definition the expected return is obtained by calculating the sum of possible returns, each being weighted with the probability of its realization

$$0 \times 90\% + 1210 \times 10\% = 121$$

previous case. We doubt, nevertheless, that all the readers would choose B. Many would surely be frightened by the very high probability (90%) of losing the whole investment and much prefer the more secure alternative A. Choosing alternative A means that they are not thinking about the expected benefit, but about the worst case benefit. Not everyone reasons in this way however, some, even if only a few, are attracted by the small probability of achieving a very high return and for this reason opt for B. The person who makes this kind of choice is almost always a person that is attracted by games of chance, because (s)he is excited by the risk.

In a case where the investment can be repeated and the DM has a large amount of capital, the decision could be different. The number of people who would opt for alternative B would be higher, because statistics show that, as we get closer to an infinite number of repetitions, the frequency tends towards the average. Not everyone, however, would choose B; there would certainly be those that would still opt for alternative A, observing that alternative B provides a sure return that is superior to alternative A's only if one has infinite capital. In fact, nothing excludes the possibility that a long series of unfavorable cases, sufficient to consume any finite capital, could occur. If the investment can be repeated, the decision becomes even more complicated because it then depends not only on the DM's risk aversion, but also on his capital.[14]

From the above it emerges that, when there is uncertainty, it is not sufficient just to substitute the inflow sequence with a stochastic model, it is also necessary to identify the DM's risk aversion. We have seen that the statistic through which to define the objective changes according to the intensity of such aversion. In the previous example we considered either the expected value or the minimum value, but these are not the only two possibilities. Some believe that the DM's risk aversion can be expressed through a function (called the *utility function*), that can be identified by interviewing the DM. We will come back to this topic in Chapter 9.

2.2.2.1 The PIP phases

The phases are the same as those described for the deterministic case (see Section 2.2.1.1), the only difference being that in the phase where the indicators are identified (Phase 2) one must estimate the Stakeholders' and DM's risk aversion, and in the phase where the model is identified (Phase 3) one must identify the stochastic inflow model.

2.2.2.2 The disturbances

Uncertain inflows are one of the most common manifestations of an aspect that we should never neglect: the DM is not omniscient. A moment's reflection is enough to realize that this lack of total knowledge will have an influence on the decision, not only through the lack of knowledge about future inflows, but in many other ways. We are not sure that in the future the sale prices for agricultural products will be those that the farmers used to estimate benefit $B^{\text{Moz}}(q_{[0,h]})$. The same thing can be said about the construction costs for the reservoir; and we do not know if the cloud cover above the irrigation district will be that which the farmers expect, while the intensity of solar radiation available to the crops for their growth is a result of it. These and many other factors contribute to making the net benefit obtained on the horizon H uncertain. For this reason they are called *disturbances*.

In planning the reservoir and its management we must therefore take account of all the disturbances that we can realistically consider.

[14] A psychologist would observe that the level of risk aversion strictly depends upon the available funds.

We can describe the disturbances that we have considered until now, such as the inflow and the prices, by using a stochastic model (in the simplest case a probability distribution function), and this description appears to be meaningful. Not all the disturbances are of this type, however. There are others for which the stochastic description is in fact impossible, or is without meaning. In order to identify them, we resume with our example.

Mozambique is not a producer of petroleum and the 'petroleum bill' weighs heavily on its accounts. Other states in the same situation, Brazil for example, have solved the problem by favoring the substitution of petroleum products, used in motors and vehicles, with methanol (an alcohol), which can be obtained from the fermentation of some agricultural products. In Brazil sugar cane, of which the country is a big producer, is used. In Mozambique Indian millet could be used. This shift would give the country two main advantages: the possibility to avoid heavy outlays for the purchase of petroleum, and the creation of a new market for millet, which would boost its agricultural sector. If Mozambique were to decide to adopt methanol, many farmers in the agricultural district would be interested in producing millet instead of cereals and cassava, and that would have an influence on the benefit estimate. The adoption of methanol is therefore a disturbance that acts on the system, but it is a disturbance that is different from the others. First of all, while it is relatively simple to associate a probability to every inflow value, it is very difficult, if not impossible, to estimate the probability that Mozambique decides to change from petroleum to methanol. In the second place, if a given inflow value does not occur today, it could occur in one of the following days, but if the country changes to methanol the decision would certainly not be modified in a short time frame. In other words, both the inflow and the type of fuel are stochastic variables (even if the former assumes value in a continuous set, while the latter is a binary variable) and therefore their values are conceptually equivalent to the result of a lottery. However, the fundamental difference is that the lottery corresponding to the inflow is played many times, while the other is only played once. Technically, one says that on the first variable many 'experiments' are performed, while only one is performed on the second. Therefore, it does not make sense to treat the fuel type in the same way as the other disturbances, considering the expected effects or the worst effects, because only one of the two alternatives will actually take place. In a certain sense this variable describes a *scenario*, a background against which all the other events that we are considering are set. It is a different type of disturbance: an *uncertain disturbance*.

Where there is a deterministic disturbance that can assume many values (*scenarios*) one should determine the optimal alternative that corresponds to each one of them. In the case of Mozambique one would identify the best reservoir–policy pair when petroleum was adopted and another when methanol was adopted.

2.3 Decision making under partial rationality conditions

A look at Figure 2.9 shows that the water system of the Cabora Bassa reservoir, which we have considered until now, is only a part of the greater system of the River Zambesi. The inflow that reaches the Cabora Bassa is not natural, but regulated by two other large reservoirs, Kafue and Kariba. The first is in Zambia, the second on the border that divides Zambia and Zimbabwe. For didactic reasons, let us assume that it is completely in Zimbabwe. Also for didactic reasons, we will assume that each one of these dams is managed only to produce hydropower energy in a plant placed at the foot of the dam, following a regulation policy that is fixed by the State to which it belongs.

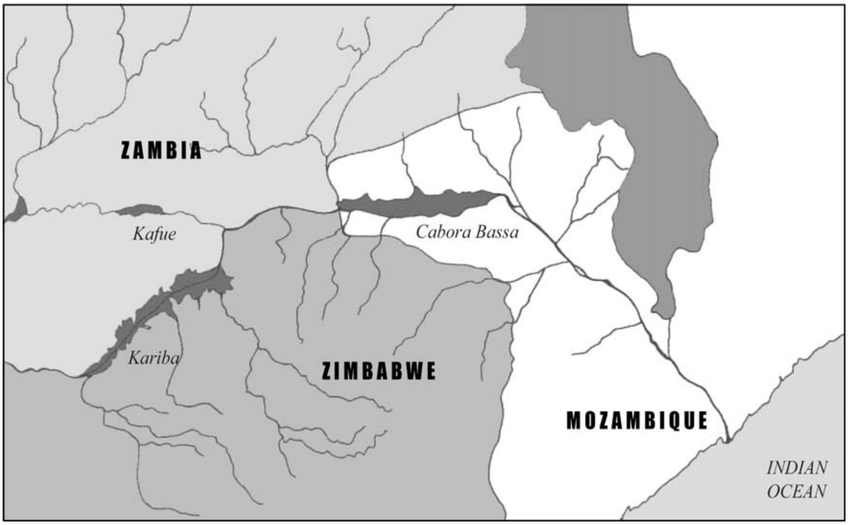

Figure 2.9: The Zambesi water system and its three reservoirs.

The presence of these reservoirs raises a doubt: the previously adopted approach for the Cabora Bassa Project might not be the best, since the effects on the Cabora Bassa reservoir of the inflows, which are produced by the management of the upstream reservoirs, could be significant. As long as Kafue's and Kariba's regulation policies remain those that influenced the inflow series used in the Project, the approach is correct and the results obtained are valid. If, instead, Zambia and/or Zimbabwe were to modify their reservoirs' regulation policies, this approach would no longer be suitable for the Cabora Bassa Project. If the regulation policies were changed, tensions would grow between those two States and Mozambique, which might rightly be afraid of the negative consequences that such a decision could have on its agriculture.

The system in Figure 2.9 is very different from the one that we have been considering, not only from a hydrological point of view but also from the decision-making point of view. In the first system there was only one DM, the Mozambique Government, that acted in favour of a homogeneous group of Stakeholders (the farmers), and therefore there was just one well-defined evaluation criterion: maximization of the expected net benefit.[15] We were, in fact, in the condition that Decision-Making Theory terms *full rationality*. Now the system is more complex, not so much because is composed of several reservoirs, but because there are several DMs: three States that have different interests. This means that we are dealing with *partial rationality*, because, even though we are still in a rational environment (one presumes that the States act to achieve ends), the objective that is being pursued is no longer well defined.

To understand what happens in conditions of partial rationality, let us assume once again that the inflow is deterministic (by now we know how this hypothesis can be removed) and

[15] By formulating the criterion in this form we implicitly assume that the farmers are *neutral to risk* (the exact meaning of the term will be defined in Section 9.1.3), but this hypothesis does not influence the ideas that we are about to consider. Note that the level of risk aversion cannot be freely established by the Analyst, since it is something that is part of the particular DM's or Stakeholder's nature, and can therefore only be acknowledged.

modify the problem that we posed. We will no longer consider the design of the Cabora Bassa reservoir and its policy, but, instead, we will suppose that the reservoir has already been constructed and that now Mozambique asks us to study the possibility of proposing to the other two states that a new Agency be created: the Zambesi Water Authority (ZWA). This proposed ZWA would be responsible for regulating all three reservoirs.

The ZWA would have to be responsible for the interests of the three states that constituted it, which can be summarized as follows: Zambia and Zimbabwe are interested in the benefits[16] produced by the hydropower generated in their plants, while Mozambique is interested in agricultural benefits. Any policy p^{ZWA} that the ZWA might adopt for the regulation of the entire system would consist in the succession of control laws $m_t^{\text{ZWA}}(\cdot)$, each of which specifies, for a given time t, the vector $(u_t^{\text{Kaf}}, u_t^{\text{Kar}}, u_t^{\text{Cab}})$ of the release decisions from each reservoir, as a function of the vector $(s_t^{\text{Kaf}}, s_t^{\text{Kar}}, s_t^{\text{Cab}})$ of their storages.

In order to assess the effects that each policy p^{ZWA} could induce, a time horizon H has to be fixed, over which the trajectories $\{a_1^{\text{Kaf}}, \ldots, a_h^{\text{Kaf}}\}$ and $\{a_1^{\text{Kar}}, \ldots, a_h^{\text{Kar}}\}$ have to be fixed for the inflows to the Kafue and Kariba reservoirs. With this information, for each policy it is possible to calculate, via simulation, the series of volumes run through the turbines in each of the two hydropower plants and the volumes supplied to the irrigation district. Once these series are known, the managers of the two powerplants and the farmers' Experts are able to assess the benefits B^{Zam}, B^{Zim}, and B^{Moz} that follow. We could then associate to each policy p^{ZWA} the vector of the three benefits it produces and visualize it in the space of these same benefits (see Figure 2.10).

Unlike the previous case, it is not possible to define an optimal policy. There is no policy that is considered optimal by all three States, since the maximization of hydropower generation benefits requires timing releases in such a way that agricultural needs cannot be satisfied.

In the actual situation, without the ZWA, each State fixes the regulation policy for its own reservoir autonomously. The topology of the system implies that Zambia's choice is not affected by choices made by the other states, since the releases from the Kariba and Cabora Bassa reservoirs do not influence the inflows to the Kafue reservoir. We can say the same for Zimbabwe. Zambia therefore adopts the policy $p_{\text{opt}}^{\text{Kaf}}$ that produces the maximum value possible $(B_{\text{opt}}^{\text{Zam}})$ for its benefit. Such a policy can be identified by solving a problem similar to problem (2.8). Analogously, Zimbabwe adopts the policy $p_{\text{opt}}^{\text{Kar}}$ that determines the maximum benefit $(B_{\text{opt}}^{\text{Zim}})$ possible for it. Mozambique's choice is, however, affected by the decisions taken by the two upstream States, because the inflow that its reservoir receives depends on them. It has no choice but to adopt the policy $p_{\text{con}}^{\text{Cab}}$ that produces the maximum value that can be obtained for its irrigation benefit with those inflows, i.e. the maximum value conditional on those inflows. We will denote this value with $B_{\text{con}}^{\text{Moz}}$. It follows that the Zambesi system is actually managed with the policy $p_{\text{act}}^{\text{ZWA}}$, which corresponds to the adoption of the three policies $p_{\text{opt}}^{\text{Kaf}}$, $p_{\text{opt}}^{\text{Kar}}$ and $p_{\text{con}}^{\text{Cab}}$ for the three reservoirs. This produces point A with coordinates $(B_{\text{opt}}^{\text{Zam}}, B_{\text{opt}}^{\text{Zim}}, B_{\text{con}}^{\text{Moz}})$ in Figure 2.10.

Given the way in which it is obtained, point A corresponds to the best possible condition for Zambia and Zimbabwe. These two States are well aware of this. Mozambique, however, does not approve of A, since it feels that its neighbours have abused their upstream position to decide to their own advantage. From here it is a short step to a claim for damages and it is for this reason that Mozambique has asked us to study the possibility of proposing the ZWA.

[16] Just as for Mozambique, the benefits for Zambia and Zimbabwe are defined as the citizens' willingness to pay for the provision of the energy produced.

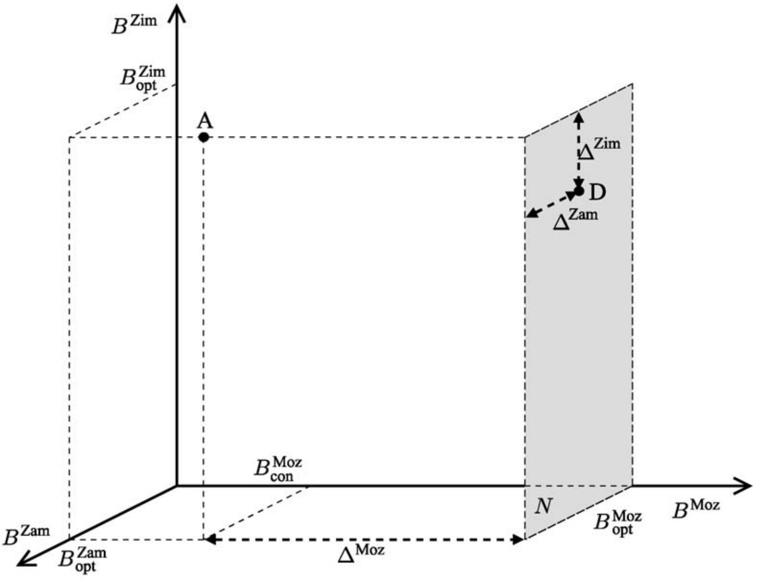

Figure 2.10: The benefit space for the three States of the Zambesi.

Actually, Zambia and Zimbabwe's policies, which produce point A, are not necessarily damaging for Mozambique. To verify whether or not they are, it is necessary to optimize the management of the Cabora Bassa reservoir. This is done by solving the problem that is obtained if we modify problem (2.8) to take into account the fact that the Cabora Bassa reservoir is already constructed (and so its capacity c is given and the cost of construction $C(c)$ is zero), and we adopt the trajectory $\{a_1^{\text{Cab,nat}}, \ldots, a_h^{\text{Cab,nat}}\}$ of natural inflow to the Cabora Bassa[17] in equation (2.8h). By 'natural inflow' we mean the inflow trajectory that would occur if the two upstream reservoirs did not exist. This trajectory can easily be computed, given the pair of trajectories $\{a_1^{\text{Kaf}}, \ldots, a_h^{\text{Kaf}}\}$ and $\{a_1^{\text{Kar}}, \ldots, a_h^{\text{Kar}}\}$, by simulating the system with the condition that at each time instant the volumes released from the Kafue and Kariba reservoirs are equal to the inflow volumes. The solution to the problem, reformulated in this way, provides the benefit $B^{\text{Moz}}_{\text{nat}}$ that Mozambique would obtain if the Zambesi were in natural conditions upstream from the Cabora Bassa. Only if that value were greater than $B^{\text{Moz}}_{\text{con}}$, could Mozambique consider itself to have incurred damage from its neighbours. If, instead, $B^{\text{Moz}}_{\text{nat}}$ were less than $B^{\text{Moz}}_{\text{con}}$, this information in itself would diminish the tensions between the three States, since it would show Mozambique that the two upstream reservoirs contribute (though involuntarily) to increasing the benefit to its agriculture. This is a first example of the positive role played by an MODSS in the resolution of conflicts.

Nevertheless, even if this were to be the case, Mozambique would always have reason to believe that its own benefit could be improved if the Kafue and Kariba reservoirs were regulated *also* according to its interest. Clearly, the maximum benefit value $B^{\text{Moz}}_{\text{opt}}$ would be obtained by regulating the Kafue and Kariba reservoirs with *exclusive* consideration for Mozambique's requirements. A policy $p^{\text{ZWA}}_{\text{Moz}}$ that produces this benefit value can be identi-

[17] To simplify, we will assume that the natural inflow is produced only by the releases from the two upstream reservoirs.

fied by solving the following problem

$$\max_{p^{\text{ZWA}}} B^{\text{Moz}}(q_{[0,h]}) \tag{2.10a}$$

$$\mathbf{s}_{t+1} = f\left(\mathbf{s}_t, \mathbf{u}_t, a_{t+1}^{\text{Kaf}}, a_{t+1}^{\text{Kar}}\right) \qquad t = 0, \ldots, h-1 \tag{2.10b}$$

$$q_t = h\left(u_t^{\text{Cab}}\right) \qquad t = 0, \ldots, h \tag{2.10c}$$

$$\mathbf{u}_t = m_{t_{\text{mod}\,T}}^{\text{ZWA}}(\mathbf{s}_t) \qquad t = 0, \ldots, h \tag{2.10d}$$

$$\mathbf{u}_t \in \mathcal{U}(\mathbf{s}_t) \qquad t = 0, \ldots, h \tag{2.10e}$$

$$p^{\text{ZWA}} \triangleq \left\{m_0^{\text{ZWA}}(\cdot), \ldots, m_{T-1}^{\text{ZWA}}(\cdot), m_0^{\text{ZWA}}(\cdot), \ldots\right\} \tag{2.10f}$$

$$\left\{a_1^{\text{Kaf}}, \ldots, a_h^{\text{Kaf}}\right\} \text{ and } \left\{a_1^{\text{Kar}}, \ldots, a_h^{\text{Kar}}\right\} \qquad \text{given scenarios} \tag{2.10g}$$

in which $\mathbf{s}_t$ and $\mathbf{u}_t$ are the vectors of the storages and the decisions at time t in the three reservoirs, and (2.10b) is the model of the entire system.

By solving problem (2.10) one obtains not only the policy $p_{\text{Moz}}^{\text{ZWA}}$ that Mozambique dreams of, but also the maximum value $B_{\text{opt}}^{\text{Moz}}$ that its benefit can reach. If it were only slightly higher than $B_{\text{con}}^{\text{Moz}}$, one would conclude that Mozambique would obtain very little from the creation of the ZWA and our task would be completed by illustrating this fact to our Client. In the opposite case, if the value $B_{\text{opt}}^{\text{Moz}}$ were significantly higher than $B_{\text{con}}^{\text{Moz}}$, we would have still a lot to do. For didactic purposes, let us assume that this is the case.

Thinking about what might be the next step, the first idea that comes to mind is to identify Zambia and Zimbabwe's benefits in correspondence with the policy $p_{\text{Moz}}^{\text{ZWA}}$. In other words we could identify the point that this policy produces in the benefit space, which is shown as D in Figure 2.10. Clearly, this point lies on the plane N that is normal to axis B^{Moz} at the point $B_{\text{opt}}^{\text{Moz}}$. Once D has been plotted, it is possible to determine the benefit losses Δ^{Zam} and Δ^{Zim} that Zambia and Zimbabwe would incur if the ZWA were to adopt the policy $p_{\text{Moz}}^{\text{ZWA}}$. These estimates are very important, because they give an idea of the magnitude of the compensation that Mozambique would have to offer Zambia and Zimbabwe, to convince them to create the ZWA and to adopt the policy $p_{\text{Moz}}^{\text{ZWA}}$. Mozambique could evaluate whether it was worth proposing the creation of the ZWA, by comparing that amount of compensation with the increase $\Delta^{\text{Moz}} = B_{\text{opt}}^{\text{Moz}} - B_{\text{con}}^{\text{Moz}}$ that it would obtain.

If you think for a moment, you will realize that the idea is correct, i.e. the evaluation of variations Δ of the benefit values must be carried out, but that the policy according to which they have to be computed is not policy $p_{\text{Moz}}^{\text{ZWA}}$. This in fact was determined by solving problem (2.10), whose only objective is Mozambique's benefit. That problem ensures that the value $B_{\text{opt}}^{\text{Moz}}$ obtained by solving it is Mozambique's maximum obtainable benefit, but does not guarantee that the benefit of one of the other States, for example Zambia, could not be increased with respect to the one produced by policy $p_{\text{Moz}}^{\text{ZWA}}$, while maintaining Mozambique's benefit at the optimal value $B_{\text{opt}}^{\text{Moz}}$. In other words, it is not possible to exclude the existence of policies that produce points, in plane N, whose ordinates with respect to B^{Zam} are better than D's (i.e. points such as E in Figure 2.11). If such points were to exist it is clear that the evaluation of variations Δ in the benefit values would have to be carried out for each one of the policies that produce the maximum value of B^{Zam}, which can be identified by solving the following problem

$$
\begin{aligned}
&\max_{p^{\mathrm{ZWA}}} B^{\mathrm{Zam}}\left(u^{\mathrm{Kaf}}_{[0,h]}\right) && \\
&\mathbf{s}_{t+1} = f\left(\mathbf{s}_t, \mathbf{u}_t, a^{\mathrm{Kaf}}_{t+1}, a^{\mathrm{Kar}}_{t+1}\right) && t = 0, \dots, h-1 \\
&q_t = h\left(u^{\mathrm{Cab}}_t\right) && t = 0, \dots, h \\
&\mathbf{u}_t = m^{\mathrm{ZWA}}_{t_{\mathrm{mod}\,T}}(\mathbf{s}_t) && t = 0, \dots, h \\
&\mathbf{u}_t \in \mathcal{U}(\mathbf{s}_t) && t = 0, \dots, h \\
&p^{\mathrm{ZWA}} \triangleq \left\{m^{\mathrm{ZWA}}_0(\cdot), \dots, m^{\mathrm{ZWA}}_{T-1}(\cdot), m^{\mathrm{ZWA}}_0(\cdot), \dots\right\} && \\
&\left\{a^{\mathrm{Kaf}}_1, \dots, a^{\mathrm{Kaf}}_h\right\} \text{ and } \left\{a^{\mathrm{Kar}}_1, \dots, a^{\mathrm{Kar}}_h\right\} && \text{given scenarios} \\
&B^{\mathrm{Moz}}\left(q_{[0,h]}\right) = B^{\mathrm{Moz}}_{\mathrm{opt}} &&
\end{aligned}
$$

A priori, we cannot exclude the possibility that E might coincide with D, i.e. it is not given that it will be possible to improve Zambia's benefit without worsening Mozambique's.

The above reasoning can be repeated, with appropriate changes, in relation to Zimbabwe and therefore, by solving a problem analogous to the one above, point F can be determined.

The points E and F are the 'extremes' of a line[18] which lies on the plane N and is characterized by points that all have the following property: for each one there exists a policy p^{ZWA} which can produce it, and, at the same time, there is no policy that improves the benefits for both Zambia and Zimbabwe without reducing the benefit for Mozambique (i.e. maintaining $B^{\mathrm{Moz}}(q_{[0,h]}) = B^{\mathrm{Moz}}_{\mathrm{opt}}$). Such a line always exists, because at worst it is reduced to just one point, when E and F coincide. It is called the *Pareto Frontier*, after the

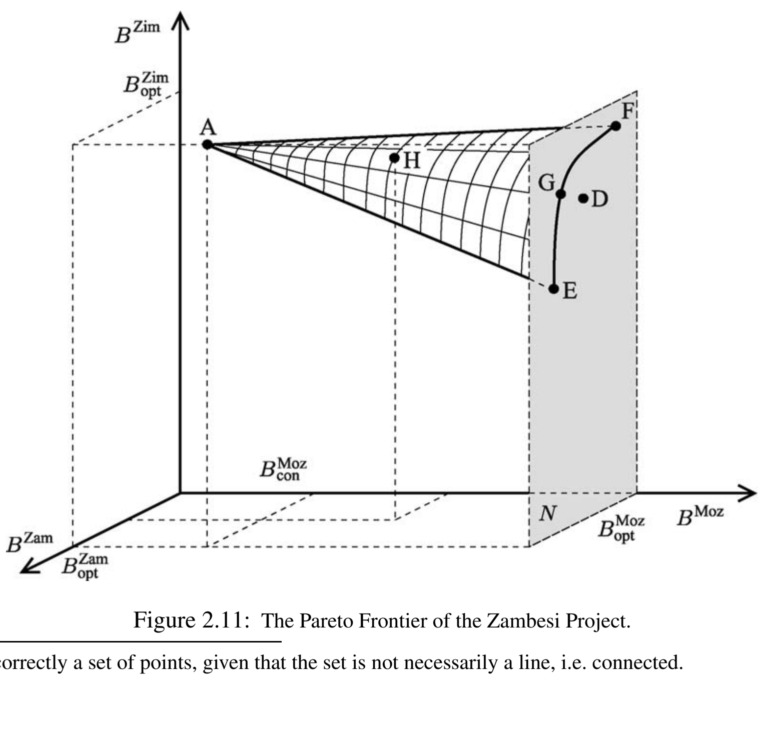

Figure 2.11: The Pareto Frontier of the Zambesi Project.

[18]More correctly a set of points, given that the set is not necessarily a line, i.e. connected.

economist that first discovered it, and it will be studied thoroughly in Section 18.2. Here we will simply give the idea that, in a certain sense, it is the solution to the following problem

$$\max_{p^{\text{ZWA}}}\left[B^{\text{Zam}}\left(u^{\text{Kaf}}_{[0,h]}\right), B^{\text{Zim}}\left(u^{\text{Kar}}_{[0,h]}\right)\right] \tag{2.11a}$$

$$\mathbf{s}_{t+1} = f\left(\mathbf{s}_t, \mathbf{u}_t, a^{\text{Kaf}}_{t+1}, a^{\text{Kar}}_{t+1}\right) \qquad t = 0, \dots, h-1 \tag{2.11b}$$

$$q_t = h\left(u^{\text{Cab}}_t\right) \qquad t = 0, \dots, h \tag{2.11c}$$

$$\mathbf{u}_t = m^{\text{ZWA}}_{t_{\text{mod}\,T}}(\mathbf{s}_t) \qquad t = 0, \dots, h \tag{2.11d}$$

$$\mathbf{u}_t \in \mathcal{U}(\mathbf{s}_t) \qquad t = 0, \dots, h \tag{2.11e}$$

$$p^{\text{ZWA}} \triangleq \left\{m^{\text{ZWA}}_0(\cdot), \dots, m^{\text{ZWA}}_{T-1}(\cdot), m^{\text{ZWA}}_0(\cdot), \dots\right\} \tag{2.11f}$$

$$\left\{a^{\text{Kaf}}_1, \dots, a^{\text{Kaf}}_h\right\} \text{ and } \left\{a^{\text{Kar}}_1, \dots, a^{\text{Kar}}_h\right\} \qquad \text{given scenarios} \tag{2.11g}$$

$$B^{\text{Moz}}(q_{[0,h]}) = B^{\text{Moz}}_{\text{opt}} \tag{2.11h}$$

in which optimization is carried out with respect to the vector of benefits $[B^{\text{Zam}}, B^{\text{Zim}}]$, in a way that will be precisely defined in the above-mentioned paragraph.

Just as points E and F are the two extremes of the Pareto Frontier E–F with respect to Zambia and Zimbabwe when Mozambique's benefit must be maximized (i.e. $B^{\text{Moz}}(q_{[0,h]}) = B^{\text{Moz}}_{\text{opt}}$), point A and the E–F Frontier are the extremes of a three-dimensional Pareto Frontier A–E–F (see Figure 2.11), when one analyses the problem of the ZWA in the entire benefit space, i.e. without the constraint that Mozambique's benefit must lie on plane N.

The A–E–F Frontier is therefore the solution to the following problem

$$\max_{p^{\text{ZWA}}}\left[B^{\text{Zam}}\left(u^{\text{Kaf}}_{[0,h]}\right), B^{\text{Zim}}\left(u^{\text{Kar}}_{[0,h]}\right), B^{\text{Moz}}(q_{[0,h]})\right] \tag{2.12a}$$

$$\mathbf{s}_{t+1} = f\left(\mathbf{s}_t, \mathbf{u}_t, a^{\text{Kaf}}_{t+1}, a^{\text{Kar}}_{t+1}\right) \qquad t = 0, \dots, h-1 \tag{2.12b}$$

$$q_t = h\left(u^{\text{Cab}}_t\right) \qquad t = 0, \dots, h \tag{2.12c}$$

$$\mathbf{u}_t = m^{\text{ZWA}}_{t_{\text{mod}\,T}}(\mathbf{s}_t) \qquad t = 0, \dots, h \tag{2.12d}$$

$$\mathbf{u}_t \in \mathcal{U}(\mathbf{s}_t) \qquad t = 0, \dots, h \tag{2.12e}$$

$$p^{\text{ZWA}} \triangleq \left\{m^{\text{ZWA}}_0(\cdot), \dots, m^{\text{ZWA}}_{T-1}(\cdot), m^{\text{ZWA}}_0(\cdot), \dots\right\} \tag{2.12f}$$

$$\left\{a^{\text{Kaf}}_1, \dots, a^{\text{Kaf}}_h\right\} \text{ and } \left\{a^{\text{Kar}}_1, \dots, a^{\text{Kar}}_h\right\} \qquad \text{given scenarios} \tag{2.12g}$$

which is a Multi-Objective *Optimal Control Problem*. It becomes a stochastic problem, of the same type as problem (2.9), when the untenable hypothesis of deterministic inflow is removed.

2.3.1 Negotiations

Determining the Pareto Frontier A–E–F requires that all the phases mentioned in Sections 2.2.1 and 2.2.2 be completed. Once this Frontier has been determined, we have everything needed to begin negotiations among the three States for the constitution of the ZWA, more precisely to carry out the *Comparison* phase of the PIP procedure (Figure 1.4), of which the *negotiation* is the most important activity. Negotiations can be carried out in different ways, but they can all be traced back to the following:

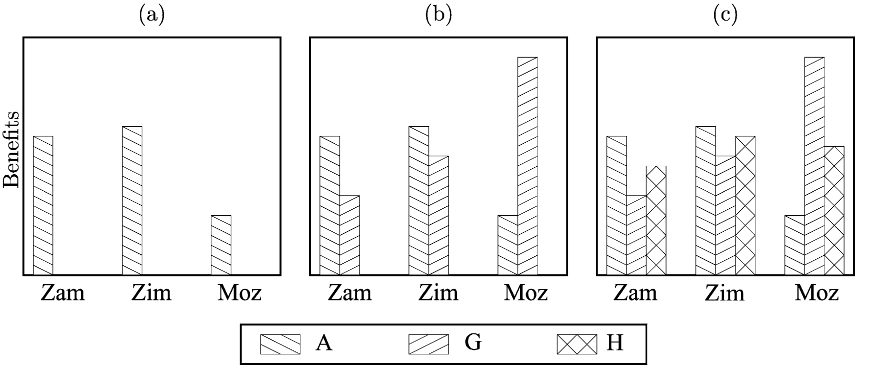

Figure 2.12: The benefit bars that the Facilitator shows during negotiations. The letters A, G and H refer to the points in Figure 2.11; the three cases are commented in the text.

1. The Facilitator shows the histogram in Figure 2.12a to the negotiators who represent the three States. The three bars correspond to the benefits of the three States in the actual situation, i.e. to point A in Figure 2.11. He explains how this situation came to be, and the reason why Mozambique judges it disadvantageous, and proposes the constitution of the ZWA (i.e., the hope of creating a new situation, as for example the one described in Figure 2.12b). Moreover, he tells the negotiators that, among all the possible solutions, this is the one that Mozambique would prefer.[19] He finishes by showing that an increase in Mozambique's benefit would result in an inevitable decrease in Zambia's and Zimbabwe's benefits, and that Mozambique is well aware of this. It is for this very reason that the proposed negotiations are being launched. The floor is then left to the negotiators.

2. Most probably, Zambia and Zimbabwe would react by affirming that the reductions in their benefit that have been suggested are unacceptable. Mozambique would respond by offering compensation, which could take the form of an annual monetary reimbursement, or the provision of free energy produced by its thermoelectric plants, or tax exemptions for the goods that are unloaded at its ports and destined for Zambia and Zimbabwe.

3. Zambia and Zimbabwe might accept, or raise their requests, or, more probably, declare that there cannot be compensation for the loss of energy. If this happens, the Facilitator would have to ask each one of the States to define the amount of loss for which it believes there could be compensation. Once he has the answer, he shows the histogram in Figure 2.12c, which is obtained from Figure 2.12 by adding the bars corresponding to a point on the A–E–F Frontier that induces the said loss (as for example point H in Figure 2.11).

4. Clearly, this point leads to a reduction in Mozambique's benefit, which must then review the compensation that it is willing to offer. The third State at this point could…

[19]The benefits that appear in Figure 2.12b beside the corresponding benefits to point A are those that correspond to a point on the E–F Frontier that would represent a compromise between E and F, as for example point G in Figure 2.11.

The reader may now have an idea on how negotiations will continue: one moves along the A–E–F Frontier step-by-step, showing the benefits which correspond to the points that are as close as possible to what the negotiators propose. Negotiations are conducted by the negotiators; the Facilitator provides them with information about how the situations that they propose could be realized and what the consequences would be for the other States, hypothesizing that the available resources would always be used in an *efficient* way. Note the attribute 'efficient', which is crucial and will be precisely defined in Section 18.2. In the course of negotiations an idea of what is possible, and what is not, slowly forms in the minds of the negotiators, and in this way the compromise solution takes form.

2.3.2 The MODSS

This way of carrying out negotiations requires not only that the Analyst be able to generate the points of the Pareto Frontier that are requested during the development of negotiations, but also that he be able to provide the Facilitator with an estimate of the effects that these points would produce for each of the Stakeholders involved (the three States). The visualization of these effects might not be limited to the histograms shown in Figure 2.12 alone: it can be extended to include all the details that the negotiators may request. This is the task of a good MODSS that consists in a MSS (see page 48) coupled with a negotiation support system for managing the *Comparison* phase.

By adopting an MODSS we can overcome a difficulty that water resource management has been struggling with over the last few decades. In the 1970s when the idea of policy was first introduced, its synthesis seemed, from the Analyst's point of view, a well structured problem, whose only difficulties were algorithmic ones, limited to the speed and memory of the computers that were available at the time. In fact, the rapid increase in computer performance during the seventies led to the development of ever more complicated algorithms (ReVelle et al., 1969; Heidari et al., 1971; Su and Deininger, 1974; Tauxe et al., 1979; Sniedovich, 1979; for a summary Yeh, 1985) for solving problems that were nothing more than simplifications of a single original problem: a periodic, multi-objective, stochastic, Optimal Control Problem. Nevertheless, in the 1980s it was slowly understood that, despite the fact that many cases had been studied, only a few of them were concluded with an effective realization and application of the proposed solution (Rogers and Fiering, 1986). The cases in which the designed policy was actually implemented had been few. It seemed that the policy-based approach had failed, just as the release-plan approach had failed in its turn, and conferences dealing with the theme *Closing the gap between theory and practice* proliferated (see for example Loucks and Shamir, 1989).

The reasons for these shortcomings seemed to be many and complicated, but two of them are particularly meaningful. The first is the intrinsic falseness of the hypothesis upon which the new approach was founded: the hypothesis that a Project could always be formulated assuming full rationality. To be convinced of the intrinsic weakness of this hypothesis it is enough to think about the case of the River Zambesi. The second reason is the very nature of the models: they are almost always a rough approximation of reality. For example, the structure of the distribution network is often drastically simplified and the inflow formation process is assumed to be purely random. From the Analyst's point of view these simplifications are justified by the fact that they are often unavoidable, if the Optimal Control Problem that results is to be solved with the available computational resources. However, these simplifications may be perceived by Stakeholders and DMs as false and deceiving. This perception, together with the fact that neither the Stakeholders nor the DMs dared to express them verbally, since they did not feel able to counter the argument with which the Analyst would defend his position, led to the fact that very often they formally accepted the suggested solution but did not implement it in practice.

The second reason for failure is therefore the lack of communication between Analyst, Stakeholders and DMs, which can be overcome by involving them directly in the study, as is proposed today. We see that already in 1985 several forerunners believed that "a direct involvement of the DM herself in the modelling process is the only way to make credible models: these can in fact be built only by people who are familiar with both the problem and the institutional setting in which the problem is to be addressed" (Loucks et al., 1985). But this involvement can be achieved only if an MODSS is available.

2.3.3 Many objectives, one Decision Maker

The partial rationality that characterizes the analysis of the Zambesi system is not confined to the case with more than one DM, but also occurs when a single DM is confronted with a plurality of evaluation criteria, as would occur, for example, if the three reservoirs along the River Zambesi were all in one State. Their Regulator would not have an univocal view of the problem, and he could not define with certainty what *optimal* would mean, because he would have to look for a compromise between hydropower production and irrigation supply. Even in this case, it is essential to formulate the problem in such a way that the conflict between Stakeholders is not hidden and the subjective choice that a single DM will take is made transparent.

2.4 Managing

Until this point we have been focusing on problems in which the decisions were, or were traced back to, planning decisions. There are cases, however, in which such decisions have already been taken in the past or simply do not have to be taken, and the only decisions to take are management ones. Think for example of the case in which the ZWA is already established and a set-valued policy (see page 44) assigned: every day the Regulator of the system must choose, from all the decisions that the policy suggests, the one he believes to be the best for the current situation. To make this choice, he must on the one hand take into account the interests of the Stakeholders, and on the other have a system to evaluate the inflow trajectory in the near future; in fact this trajectory is an essential part of the scenario that the Regulator must consider in making his choice. The first point reveals that an MODSS may be necessary also for management, while the second shows that an essential part of an MODSS for management consists in *predictive models*.

2.5 The steady-state paradigm

Until just a few years ago the hypothesis that hydrological processes were steady-state processes was accepted without question in the field of water resource planning. It was clear that, on a geological time scale, the hypothesis was not confirmed, i.e. it was evident that hydrology changed, in response to climate changes, with the passing of eras. However, on a historical time scale (the one of interest in a Project), it was taken for granted that changes in hydrology were caused only by anthropic actions. With this term we mean the more traditional effects that humans impose on the water system: changes in land use, and damming or deviating water courses. As a consequence, the most important hydrological variables (such as for example precipitation) were sampled over a period of a few decades and the resulting statistics were used, as we have seen in Section 2.2.2, in designing water systems and their policies.

From the middle of the 1980s, however, scientists became aware of the existence of inter-annual oscillations in precipitation, of the effects produced by El Niño[20] and by La Niña on extreme precipitation and drought events, of the dependency relationship between the monsoon season in Australia and Asia and El Niño, and of the fact that all these phenomena are connected to variations in the cloud cover over Siberia. These relationships, and others that may be discovered in the future, can generate seasonal distortions in the statistics. But that is not all: it seems that the intensity of the fluctuations in these phenomena may vary on a ten-year time scale and be significantly influenced by the *greenhouse effect*.

It follows that the steady-state paradigm is highly questionable but, nevertheless, it is still used in current planning practice because an alternative paradigm, legitimized by concrete results, and thus legally applicable, is lacking. This difficulty cannot be avoided however, and so several international institutions, such as UNESCO for example, have launched pilot initiatives to identify new project methodologies.

We will return to these issues in Section 4.11.

2.6 The decision-making levels

We have already underlined several times the fact that there are both planning and management decisions. More generally, there are three levels at which decisions are taken in a Water Agency:

- the planning level;
- the management level;
- the operational control level.

Moving from one level to another changes not only the subject of the decision (Figure 2.13), but also the time horizon that it is concerned with, and so the horizon over which the system is observed and objectives are defined.

At the *planning level*, once the Goal of the Project is fixed, the appropriate options for intervention are chosen (such as constructions, directives, norms and regulation policies) in order to achieve it. The *time horizon* over which the performance of the system is evaluated is therefore a long-term one.

At the *management level*, once the planned structural actions have been implemented and the norms and directives issued, decisions are taken (regarding, for example, release and distribution) to achieve the efficient use of the resource, both in the short and medium terms, according to the policy stated at the planning level.

At the *operational control level* one establishes which actions to undertake in enacting the decisions taken at the management level. For example, one must establish how to operate the control gates in the course of the day, so that at the end of that period the volume determined at the management level has actually been released.

[20] www.elnino.noaa.gov.

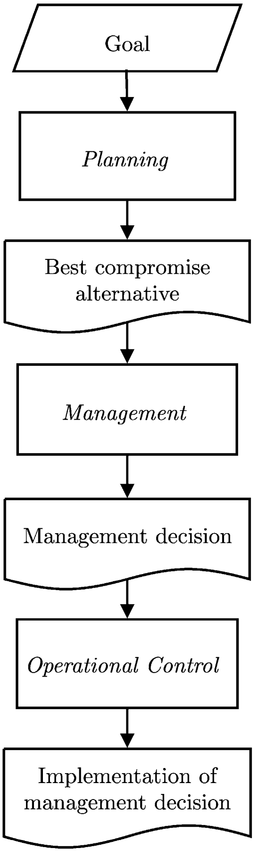

Figure 2.13: The decision-making levels in a Water Agency.

A water system with reservoirs presents decision-making problems at all three levels and these must be dealt with and solved in the correct order. The first to be considered is therefore the planning level and the greater part of this book has been dedicated to it. At this level, in a system that contains reservoirs, one cannot disregard the choice of regulation policy. However, one should not think that the management level is embodied within, and terminates with, the planning level. In fact, the *release* and *distribution* decisions that are taken on a daily basis must conform to the directives that were defined at the planning level, but they must also adapt to the contingent and unforeseeable situations that may occur, such as the closure of a canal for extraordinary maintenance. These types of events cannot be and should not be inserted in the planning process, but they cannot be ignored at the management level, when they do occur.

From the above, we can see that the decision-making level considered, and therefore the decisions that must be taken, have an influence on the model used to describe the system and on the simplifying hypotheses assumed to identify it.

Finally, it is useful to remember that all three levels must have input from a monitoring and data collection system, which includes the validation and pre-treatment processes of the data. In particular, data availability is crucial at the management level to take daily decisions through the regulation policy.

2.7 Functions and architecture of a MODSS

The architecture of a *Multiple Objective Decision Support System* (MODSS[21]) must be based upon three guidelines.

1. The architecture must reflect the structure of its intended user: an Agency that plans and manages a water system, in which the regulation policy and the actions are designed at the planning level, the daily decisions for release and distribution are taken at the management level, and the operational level decisions can be almost always taken automatically.

2. It must be a decision support system for the first two levels and not an automatic control system. It must help to evaluate the consequences of decisions, both at the planning level and at the management level, but it neither takes or enacts those decisions.

3. It must allow the user to retrace the decision-making process that was followed to reach a given decision, so that 'the how' and 'the why' of having taken it are transparent for all the Stakeholders, not just for the Analyst, who took it.

2.7.1 The levels of the MODSS

From the three guidelines that have just been stated, it follows that at the planning level specific criteria are required to evaluate the performance of the system and to choose the appropriate (structural, normative and management) actions to improve it. When there is more than one evaluation criterion, the Analyst cannot identify the optimal decision autonomously. In fact, as we have seen, an optimal alternative does not exist because the decision sought is inevitably a *compromise solution* between the Stakeholders, which must emerge from their negotiation.

The aim of the management level is, instead, to use the resources in an efficient way in the short to medium term, according to the directives expressed at the planning level, taking into account the contingent situation that is presented (as when, for example, the measures of rain and inflow provided by a monitoring system signal the imminence of a flood). At this level the decisions are much more structured. The planning level has in fact already defined a compromise between the Stakeholders, which has been translated into constructions, norms and policies, to which decisions taken at the management level must adhere. We have also seen (Section 2.1.1) that it is advisable that policies are set-valued, which means that they suggest to the Regulator not just a single control, but the set of all the controls that are equivalent in the medium to long run, with regard to the priorities established in the compromise solution. In doing so, however, the possibility of conflict among Stakeholders persists, because in the transient period these controls could be more advantageous to some than to others. Therefore, also at the management level, the Regulator must search for a compromise between the different interests.

[21] Remember that MODSS is the acronym we have adopted for the DSSs (Decision Support Systems) that are used to take decisions in a way that is coherent with the IWRM paradigm.

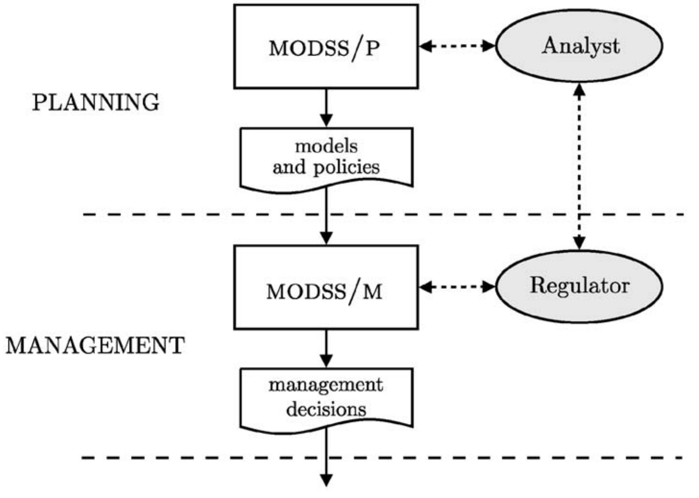

Figure 2.14: The two decision-making levels of TWOLE (Soncini-Sessa et al., 1990).

Finally, the operational level must establish how to enact the management decision; e.g. how to operate the sluice gates of a reservoir in order to obtain the release that was established by the Regulator at the management level. At this level, the decision-making problem is completely structured, in the sense that the criterion with which one chooses the operational actions is perfectly and univocally defined. Thus the Control Problem that is formulated and solved at this level is a deterministic problem, with only one objective and, as such, it has only one solution or all its solutions are equivalent. Decisions at the operational level can therefore be allocated to a control system (regulator) that automatically defines the best solution, without human intervention, as is now common practice in most hydropower plants. The first two levels, however, require the intervention of a human DM and therefore they must be assisted by an MODSS that helps the DM in making decisions, taking account of the plurality of viewpoints, but that does not substitute her in making the decisions. An Agency will thus need two MODSS: one to use at the planning level (MODSS/P), whose user is the Analyst, and one to use at the management level (MODSS/M), whose user is the Regulator.

It is not, however, advisable, from the technical point of view, that the two MODSS be independent monads because the policies produced and the models identified with the first (MODSS/P) have to be utilized in the second (MODSS/M). This is why one ought to think of an MODSS on two levels (Soncini-Sessa et al., 1990). The acronym TWOLE (TWO LEvels) given to the system we have developed comes from this idea (Figure 2.14). A system of this type has another advantage in that the models and data are used daily, so that in the future, if it is necessary to return to the planning level to take new decisions, they will be up to date and available.

2.7.2 Functions

Let us now examine how the two-level structure reflects on the functions that a MODSS must include.

In Section 1.3 the phases of the decision-making process at the planning level were described. They follow one another according to the diagram of the PIP reproduced in the upper frame in Figure 2.15. This diagram can easily be completed, as in the figure, to include the management level and the monitoring system.

All the phases that appear in the diagram must be supported by the MODSS. This system is therefore composed of modules that are specialized in carrying out each of the phases and that share the same information environment. At the planning level the MODSS helps to separate the water system into the set of components of which it is composed. Each component is then described by the data series and/or by the parameters that characterize it, by the indicators that are used to measure its performance, by the actions that can be applied to it, and by the norms to which it is subordinated or to which one wants it to be. The MODSS must then provide tools to identify the model of each component, on the basis of this information, and tools to interconnect these models, according to the topology of the system provided by the user, in order to create a global model of the water system.

Once this model has been created, the MODSS helps to design the alternatives, to quantitatively define the structural and normative actions and the regulation (or distribution) policies. The Analyst can thus determine the set of the *Pareto-efficient* alternatives, i.e. the set of alternatives that map into the *Pareto Frontier*, which constitutes the solution of the Multi-Objective Optimal Control Problem that translates the Design Problem. Once the Pareto-efficient alternatives have been determined, their effects can be evaluated by simulating the regulated system (i.e. the system subject to the regulation and/or distribution policy) in which the planned interventions were implemented. It is possible to visualize the trajectories of the most important hydrological quantities (such as levels and releases) that have been determined in this way and calculate the values of all the indicators. The Analyst and the Stakeholders can evaluate the performance of the alternatives and identify the zones of the Pareto Frontier that should be explored in more detail and/or the sectors that need further study, if necessary by increasing the accuracy of the models of some of the components. In this way, all the necessary information is produced for the next phase (*Comparison*), which is executed by the Stakeholders and/or the DM(s), with the help of the Analyst who applies the modules dedicated to that phase (for example the negotiation support system). Eventually, the *best compromise alternative* is identified.

Once this alternative has been identified, the Analyst transfers the set of models with which it has been designed and the regulation policy that characterizes it from the MODSS/P to the MODSS/M. The Regulator can thus use the MODSS/M on a daily basis to get suggestions for the daily release decision and to simulate the transient effects of the alternative decisions, given the inflow forecast(s) produced by the predictive model(s).

2.7.3 Architecture

In order to achieve the above, the architecture of the MODSS should permit the Control Units of the two levels to access a set of shared resources: the database, the knowledge base and the computational tools that perform the functions described in the previous paragraph.

Figure 2.16 shows how the interaction with the user occurs: the Graphic User Interface (GUI) allows the user to interact with the Control Unit (CU), which interacts with the Knowledge Base (KB), which in turn can draw information from external data sources (EXS). The Control Unit manages the use of the computational tools, which the user chooses from a "toolbox" that changes according to the phase of the decision-making process.

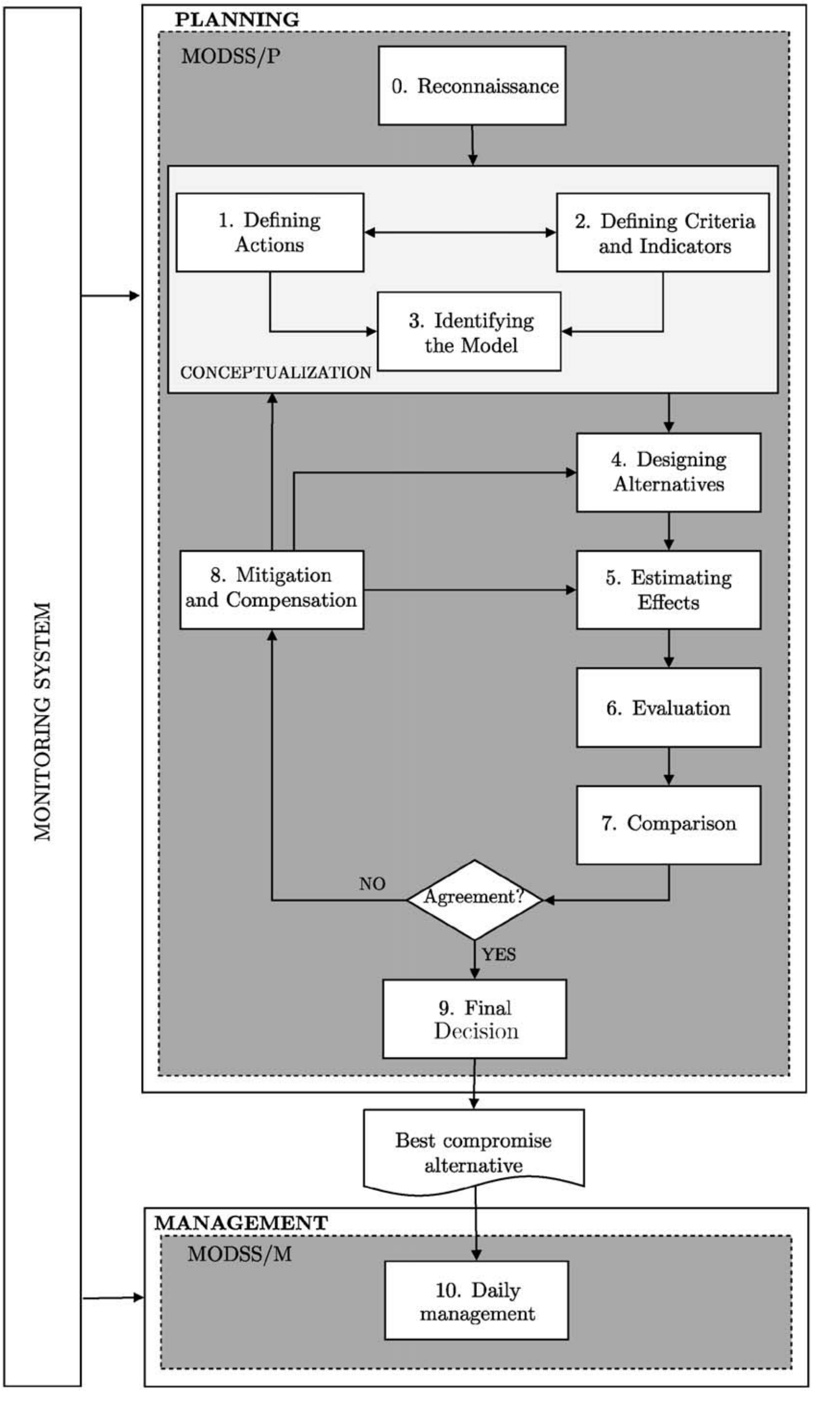

Figure 2.15: The phases of the two-level decision-making process.

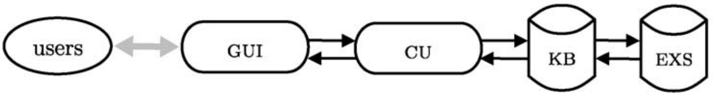

Figure 2.16: The architecture of TWOLE.

In a traditional DSS the GUI provides access both to models and to the computational tools, which both draw from an undifferentiated database. This structure is, however, too poor for an MODSS, whose database (Knowledge Base) must satisfy the following fundamental needs:

1. The water system model identified by the Analyst must not only be valid from a scientific point of view, but must also win the trust of the Stakeholders. In other words, it is essential that the Stakeholders be firmly convinced that at least one of the values of the indicators, which are calculated with the model, expresses the system performance, in correspondence with a given alternative, in a realistic way and *from their point of view*. Only when this trust exists are the Stakeholders really prepared to negotiate with each other, on the basis of the performance estimates produced by the model.

In order to satisfy this first need, it is essential that the Analyst work interactively with each Stakeholder, trying and retrying, refining and modifying both the models of the components the Stakeholder is concerned with, and the definition of the indicators that describe his/her point of view. The interaction with a Stakeholder is concluded only when (s)he is completely satisfied and believes that the model accurately describes the components (s)he is interested in, and that the indicators effectively express the way that (s)he evaluates the system's performance. For this it is essential that models and indicators be easily exchangeable, and thus that they communicate with the other models, and with the data from which they are fed, through well-defined, immutable interfaces. Technically, we say that the system's architecture must be *object oriented*.

2. At any point of the process the Analyst must be able to reply to questions such as: "if, instead of x, we assign the value y to the quantity W, by how much would the value of that indicator change?", or "if we were to use model B, instead of model A, how would the value of that indicator change?". It is essential for the success of negotiations that the Analyst be able to respond to this type of question quite quickly. These questions are almost always posed by Stakeholders who fear that a certain result that has been shown to them might depend significantly on the assumption of the value x (or on the adoption of the model A), about which they have some reservations or of which they do not feel certain. The question must be answered, because the rest of the negotiations depends on it.

To satisfy this second need it is necessary not only that the first be satisfied, but also that a record be kept of the computational tools that were used, of the models that they were applied to, and of the order in which they were employed. In brief, the history of the *experiments* that were conducted from the moment that the value x (or the model A) was assumed, until the moment that the dubious result was obtained, must be recorded. In this way it will be possible to repeat these experiments after substituting value y for value x (or model B for model A), to ascertain what the effect of the substitution would be on the result being debated.

To respond to these two needs the database, more correctly called a Knowledge Base, given the role it plays in our vision, must be structured in four distinct parts (see Chapter 24):

- The *Domain Base*: this part of the knowledge base contains the *domains*. A domain is a structure that organizes the data describing the characteristics, the measures and the time series relative to or relevant for an entity, such as a canal or a measurement station. The domain is the first level of abstraction from reality, and does not yet require any hypothesis about the mathematical relationships that exist among the variables, but requires only that we state which data are available and how they are represented.

- The *Model Base*: this part contains the mathematical models that describe the components of the system and the indicators that have been defined for them. The models are identified and operate by accessing data through the Domain Base. Each component can be described with alternative models and indicators, so that it is easy to experiment with alternative modelling strategies.

- The *Tool Base*: this part contains the tools for operating upon the domains and models.

- The *Experiment Base*: this fourth part memorizes the formulations of the modelling and decision-making problems, and the tools used for their solution. A modelling problem is for example the calibration or the validation of a model (see Chapter 4). A decision-making problem consists of the union of a model and a set of objectives and constraints, such as for example problem (2.12). Through an appropriate tool (for example an estimator, an optimizer or a simulator) the solution of the posed problem is determined. This base helps to keep track of how each problem was solved and how the problems were concatenated to form the decision-making process, making it possible to repeat the process and modify the problems when requested.

2.8 Organization of the book

The information that is needed for designing an MODSS cannot be presented in a linear sequence, because it forms a network. This is why we decided to proceed with successive developments and dedicated the first two chapters to a global vision. In this way, we hope that the reader, when dipping into the more specialized parts, will not take on the myopic and local vision that often, too often, characterizes the specialist view. We encourage him/her to maintain the broad outlook of a generalist, who stays aware of the links between the single parts and the whole.

The book is structured in parts, which correspond to phases, or groups of phases, of the PIP procedure (Figure 1.4) and so also to the modules of the MODSS (Figure 2.15).

Part B provides the material used in Phases 2 and 3: *Defining Criteria and Indicators* and *Identifying the Model*. One chapter (Chapter 3) has been dedicated to the first phase, while three others deal with the second: the first (Chapter 4) provides a panorama of modelling problems and modelling techniques; the second (Chapter 5) illustrates the models that are most commonly used to describe the components that constitute a water system (catchments, reservoirs, canals, diversions and users); and the third (Chapter 6) shows how these elementary models can be aggregated to create a model of the whole water system.

Part C deals with Phase 4, *Designing Alternatives*, and is limited to the case of full rationality, i.e. to the case in which there is only one DM that evaluates the alternatives with a single well-defined criterion. This case is propaedeutic to the more complex case of partial rationality, in which there are several evaluation criteria and several DMs and/or Stakeholders. Chapter 7 introduces the design issues and provides a reading guide to Part C. The two following chapters are dedicated to the description of the algorithms that allow the Design Problem to be solved in the case we call *Pure Planning*, that is when the project deals with structural and normative actions only and is not concerned with management actions. More precisely, in Chapter 8 we will consider the decision-making process in deterministic conditions, while in Chapter 9 we will consider how to deal with uncertainty. The following chapters are dedicated to the design of policies. Their technical description is presented in Chapters 10 and 11, for point-valued and set-valued policies respectively. In the last chapters of Part C we will present the descriptions of the algorithms for the design of *off-line policies*, both point-valued and set-valued (Chapters 12 and 13), for the design of *on-line policies* (Chapter 14) and *distribution policies* (Chapter 15).

Part D extends the design of alternatives to the case of partial rationality. The particularities that characterize the decision-making process in this case are presented in Chapter 16, which introduces the chapters that follow. In Chapter 18 algorithms for solving Multi-Objective Design Problems are studied, while in Chapter 19 we will look at those used for estimating the effects that an alternative will induce. Chapter 20 is dedicated to the evaluation of these effects, which leads directly to the identification of the best compromise alternative, when there is only one DM, or produces the necessary elements for the *Comparison* phase, when there are several DMs. This latter phase is examined in Chapter 21. Part D concludes with an examination of the *Mitigation* phase (Chapter 22) and an ample recapitulation of how to cope with uncertainty (Chapter 23).

In contrast to the first four parts of the book, which are concerned with procedural or algorithmic aspects, Part E studies the information systems that support them.

The accompanying CD contains ten appendices that complete and develop a number of the issues dealt with in the text, principally mathematical ones, and also briefly provide the prerequisite notions that in this text are assumed to be known to the reader. More precisely, Appendix A1 has already been described in Chapter 1. Appendix A2 deals with Statistics and Stochastic Process Theory. Appendices A3 and A4 recall some elements of Systems Theory and of State/Parameters Estimation; they are therefore useful aids for reading Chapters 4, 5 and 6. The different types of models that are presented in these last three chapters are exemplified or developed in Appendices A5–A8, in which examples of a mechanistic model, empirical linear models PARMAX, Data-Based Mechanistic (DBM) models and Artificial Neural Networks (ANN) are presented in this order. Appendix A9 provides an overview of the Mathematical Programming methods that can be used to solve the Design Problems described in Chapters 8 and 9. Finally, Appendix A10 introduces sociological and psychological aspects of negotiations and is thus the natural companion to Chapter 21.

Chapter 3

Actions, criteria and indicators

AC, FP and RSS

3.1 From Reconnaissance to actions

A Project begins with analysing the purposes assigned to it, delimiting the system it involves, identifying the Stakeholders and formalizing its Goal (the phase of *Reconnaissance*); it continues with defining the options for intervention with which the Goal might be pursued, and resolving such interventions into actions (the phase of *Defining Actions*). The Analyst must carry out these activities in collaboration with the Stakeholders.

To make this discussion more concrete, we will base it on the case of Lake Maggiore described in the box[1] on page 74. It is a regulated lake whose shoreline inhabitants have watched the frequency and intensity of floods increase with the passage of time. Their protests inspired the launching of a project with the aim of identifying interventions that would mitigate the lake's flooding.

The lake's release is used to feed a group of hydropower plants and three irrigation districts. The first require a high flow in winter, the second in summer, which are the two seasons in which water is naturally scarcest. To satisfy the needs of these users, the lake was transformed into a reservoir, through the construction of a regulation dam at the lake's mouth. This dam makes it possible to store water during the seasons in which flows are abundant (and therefore flooding is more likely) and release it in the dry seasons. In order to limit opposition, interventions aimed at reducing flooding must be prepared carefully so that they damage the interests of the irrigators and the hydropower companies as little as possible. The Project Goal might therefore be articulated more precisely as: "to identify a way to mitigate flooding, without provoking opposition". The non-specificity of the second part of the phrase is due to the fact that the irrigators, the hydropower plant operators and the shoreline lake dwellers mentioned above are not the only Stakeholders interested in the lake level: there are many others, such as fishermen and navigation companies.

However, the Goal is not yet well defined. It is not just the lake dwellers who are complaining about the regulation of the lake; there are others who are dissatisfied. For example, tourist operators would like a 'postcard lake' that is always at the same level, the optimal one for their guests; environmentalists complain about the destruction of nests caused by

[1] For more details see Chapter 2 of PRACTICE.

The Verbano Project

System description Lake Maggiore, locally called Verbano, has a surface area of 211 km^2, 170 of which are in Italy and the rest in Switzerland (see Figure 3.1). The lake is fed by a 6598 km^2 catchment, which is characterized by a pluviometric regime typical of sublittoral alpine zones, which produces scarce inflows in winter and summer, and high inflows in autumn and late spring. However, the pluviometric regime is not the only determining factor for lake inflows. Beginning in 1911, a series of alpine hydropower reservoirs has been created over the whole catchment, the most recent of which began operating in 1973. These reservoirs drain 16% of the catchment. Between late spring and autumn they retain a volume of water which is on average equal to double the operational capacity of the lake; this volume is then returned to the lake in the months from October to April, when electricity has a higher value.

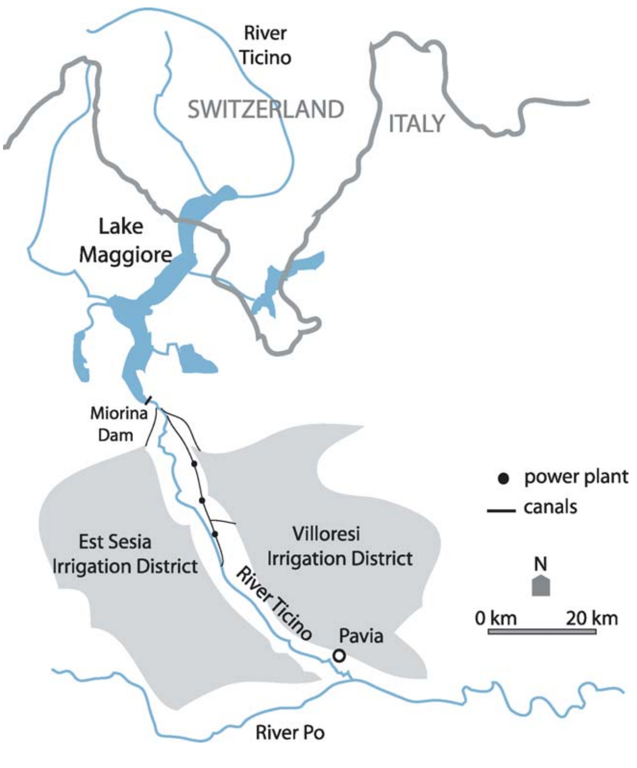

Figure 3.1: The Lake Maggiore water system.

Continued

Lake Maggiore's main effluent is the River Ticino, which flows into the River Po, just downstream from the city of Pavia. On the banks of the Ticino there are important urban centres, including Pavia itself and Vigevano, and zones of great environmental value. The Ticino feeds a dense network of irrigation ditches and canals that has contributed over the centuries to the development of a rich agricultural system, based on the production of rice, maize and wheat. The waters of the Ticino are not only used for irrigation, they are also used to produce hydropower, through run-of-river power plants.

Since 1943, when the Miorina dam was built at the lake's outlet, on the River Ticino, the outflow has been regulated with the aim of satisfying the water demand of the downstream users, even in the periods when inflows are scarce. An international agreement signed by Italy and Switzerland defines the *regulation licence*, in which it is stated that the Regulator can arbitrarily operate the dam only when the lake water level is within a specific range (*regulation range*). When the upper bound of this range is reached, the Regulator is obliged to completely open the dam gates, thus allowing the maximum possible release,[a] with the aim of reducing the risk of flooding on the lake's shores.

Project proposal It is common opinion among the lake's shoreline communities, particularly in the city of Locarno, that the regulation of the lake has increased the frequency and intensity of floods. The disastrous flood events in 1993 and 2000 sharpened the population's sensitivity to the problem and the requests for resolute intervention are ever more insistent. Let us suppose that in response to these requests the Swiss Government decides to finance a project aimed at evaluating the effects of the interventions proposed by the shoreline inhabitants to reduce flooding, and at getting the technical indications needed for deciding whether to open discussions with the Italian Government. At first glance, one of the most interesting intervention proposals is to increase the release capacity at the outlet, so that in flood conditions the lake can be emptied more quickly. This could be achieved by dredging the outlet, given that the release capacity depends on the elevation of its threshold and on the form of its cross section (the lower the threshold and the wider the section, the greater the flow that can be released).

[a]Note that, when the dam was built, the lake outlet was excavated to increase the release capacity.

wide variations of the water level, both of the lake and of its effluent River Ticino; and there are many others. Any proposal to modify the lake regime causes these dissatisfied people to come forward, asking that their interests also be taken into consideration. Therefore, the Goal must be extended again and expressed as follows: "to identify a way to produce a regulated lake regime that increases the overall satisfaction of the Stakeholders involved". Note that this broad, generic formulation could be applied, more or less unchanged, to any system; it acquires an operative meaning only when the interested parties express their judgement about the alternatives and an agreement is achieved through direct negotiations among them.

Once the Goal has been defined (at least tentatively) the spatial and temporal boundaries of the water system have to be defined. The definition of the system boundaries is a tricky operation and requires the full involvement of the Stakeholders and a careful examination of the Project Goal. As for the spatial boundaries, one must take into account hydrological, geological and geomorphological information, along with socio-economic aspects in order to identify a portion of territory, outside which one can assume that the effects one wants to study vanish. In the Verbano Project, on the basis of this information, one concludes that this portion is constituted by the Lake Maggiore catchment, by the territory irrigated with the River Ticino water, as well as the entire course of the River Po, from the confluence with the Ticino down to its delta. This latter is included because the flood waves of the Ticino may synergically interact with those of Po. However, its inclusion creates practical difficulties, since it implies considering the whole Po valley down to the Adriatic sea. Anyway, this is just a tentative definition that can and must be re-examined in the phase *Defining Criteria and Indicators*.[2]

As far as the temporal boundaries are concerned, they can be specified by taking into account the life-time of the options for intervention considered and the extent to which it is possible for the Experts to define future scenarios. The life-time of the dredging of the outlet is of the order of one century, while it is difficult for the Experts to specify scenarios significantly longer than a couple of decades. Therefore in the Verbano Project the temporal boundaries were fixed to 25 years.

Once the Goal and the system boundaries have been defined (at least tentatively) the options for intervention must be identified, and resolved into elementary actions, by which the Goal can be pursued. The progressive refinements with which we succeeded in defining the Goal now become useful. First of all, one must identify the interventions to reduce flooding: for example, a change in the regulation policy, or a modification in the dam and/or the outlet of the lake, with the aim of allowing a greater release at any given level, so that the lake rises more slowly when there is a huge inflow. Other than these types of intervention, whose effects act on the whole lake, one can also consider local interventions, such as the creation of a system of mobile floodgates to stop the flood waters from reaching particularly valuable areas, like the centre of Locarno. Other interventions, such as flood warnings, aim at reducing damage without changing the maximum water level, so that the inhabitants can secure their most precious objects or activate personal protection systems, such as watertight bulkheads on the doors of their houses. Once the census of the interventions that might mitigate the effects of the lake's flooding has been completed, one proceeds to identify the interventions that are interesting from the perspective of the other dissatisfied Stakeholders. The irrigators, for example, propose raising the maximum storage level, i.e. the level at which the Regulator is obliged to completely open the dam gates, because this would result in a greater water reserve. The environmentalists ask for an increase in the minimum environmental flow (MEF) that is currently guaranteed to the Ticino effluent. Finally, all the Stakeholders agree that if any of these interventions is realized, the regulation policy for the lake should be redefined.

In order to individuate the options for intervention, it is always advisable to begin with the proposals from the Stakeholders; by doing so the Analyst will not only get the benefit of their experience, but will also facilitate interactions with them. Each Stakeholder has in fact his/her own idea about how the performance of the system could be improved, at least from

[2] A more detailed description of the recursive process through which the boundaries of the Lake Maggiore water system were actually defined is available in Sections 2.1 and 4.2.2.1 of PRACTICE.

his/her point of view, and wants it to be taken into consideration. In general, these proposals are not mutually exclusive: some come about as counterproposals to others (for example the irrigators' proposal is clearly a countermeasure to the request that the lake be emptied more quickly, which they consider to be damaging to their interests) and so they must be all evaluated together. In this way, it is possible to identify the range of options for intervention and then the actions to be evaluated. Moving from options to actions is a delicate operation and, since it depends closely on the system and the interventions considered, it cannot be described in an abstract way; the reader will find an example of this process in Chapters 3 and 5 of PRACTICE. However, it is not sufficient to have defined the actions: it is also necessary to single out the ones that can be excluded. In principle, none of the actions should be excluded a priori, and actions that are useless, too expensive or infeasible for technical reasons should be discarded only during the decision-making process itself. Nevertheless, evaluating an action has a cost, even if nothing more than the time required to estimate its effects, and so the useless or impossible actions should be identified as soon as possible. Clearly, the Analyst must always remember, and make the Stakeholders aware, that the results obtained are conditional on the exclusion of these actions, and he should be prepared to reconsider them later if necessary. To understand the last point a little better we will go back to the Verbano example. To reduce the conflict between the shoreline dwellers and the irrigators, one can modify the irrigation technique. If a method that could reduce current irrigation demand were found, it would be easier to reduce the lake levels without damaging irrigators' interests. The installation of drip irrigation systems for all the crops that could use it should be included among the actions to be considered in the Project. However, not only this action is very costly, but it is also very difficult to estimate how the irrigation demand would vary once the action had been implemented. Therefore, it ought to be excluded and considered only if no agreement can be reached among the Stakeholders with the other actions.

Note that the interventions we have discussed until now are not proper actions, but *meta-actions*. A meta-action is transformed (*instantiated*) into an action by establishing exactly and in quantitative terms how it has to be implemented. Thus, an action is defined by a vector whose elements specify the values of the attributes that define the meta-action. More precisely, it is not always sufficient to specify the values of some parameters to define an action, but sometimes is necessary to specify a function; we will suppose, however, that even in these cases it is always possible to specify a function by means of a finite vector. For example, in the case where the meta-action being considered is the dredging of the lake outlet, we can assume that a given excavation action is completely defined by the storage–discharge relation that it produces, the section in which the intervention is carried out, the ways in which the material is removed and transported, etc. All these elements, except for the storage–discharge relation, are spontaneously defined by scalars or vectors. However, even the storage–discharge relation can be reduced to a vectorial representation, by specifying the parameters that define it in a class of given functions or by describing it with a look-up table.

As we highlighted on several occasions in the preceding chapters, planning actions should be distinct from management actions. The latter can also be described with a vector: for example the daily release decision for Verbano can be specified by the volume that one wants to release in the following 24 hours, or equally, by the positions that are assigned to the dam gates. Unlike planning decisions, management decisions do not have a time-invariant value; they can assume a different value at each time instant. In order to distinguish one from the other we will denote *planning decisions* with $\mathbf{u}^p$ and *management*

The Verbano Project: results after Phase 1

System boundaries The system consists of Verbano (Lake Maggiore), its catchment, the Ticino effluent, the areas irrigated with water from the Ticino, and the areas at risk of flooding along its course and along the course of the River Po from the confluence with the Ticino down to its delta.

Stakeholders The Stakeholders are all the people and the biota, whose interests and whose existence are influenced by Verbano's level and releases, and whose dynamics would be modified by the options of intervention considered. These Stakeholders can be divided into the following groups: the shoreline population, environmentalists concerned with the lake, tourist operators, navigation companies that operate on the lake, fishermen, the owners of hydropower plants downstream from the lake, irrigation consortia, the inhabitants along the banks of the Ticino and the Po (downstream from the confluence with the Ticino), environmentalists concerned with the river, and river tourist operators.

Goal The Goal is to single out a way to produce a regulated lake regime that increases the overall satisfaction of the Stakeholders involved.

Options for intervention The options include dredging the outlet; regulating lake releases; modifying the maximum and minimum storage values (regulation range) established in the regulation licence; and changing the value of the minimum environmental flow (MEF).

decisions with $\mathbf{u}_t$. The first are made *una tantum* (once and for all), while the second are made every day. However, as we explained in Section 2.1, management decisions can be traced back to a planning decision by choosing a regulation policy p, which, as we will see in Section 10.1, can be represented by a matrix, and thus, by a vector.[3]

It is not very often that only a single action is considered; more often a mix of planning and management actions is considered. Therefore, in general, an alternative A is defined by a pair $(\mathbf{u}^p, p)$ (when a point-valued policy is considered) or $(\mathbf{u}^p, P)$ (when a set-valued policy is considered). For the sake of simplicity, in the following chapters of the book we will often refer only to the point-valued case, but what we will say holds also for the set-valued case.

3.2 Criteria and indicators

An alternative corresponds to a pair $(\mathbf{u}^p, p)$. The Project must thus individuate the 'best' pair; but to be able to speak about 'best' we must define an evaluation criterion, and so the Stakeholders come into play once again.

An *evaluation criterion* is an attribute, or a factor, with which the Decision Maker (DM) or a Stakeholder judges the performance of an alternative from the viewpoint of one of

[3] A matrix can be represented by a list of numbers, and thus by a vector, by listing its column elements, column after column; this is how a matrix is represented in a computer.

his/her interest. It is not always expressed in an operational way, i.e. it does not automatically define a procedure for determining how much it is satisfied by a given alternative. For this reason, it is necessary to define an *index*, i.e. a *procedure* that associates the criterion with a value[4] expressing the degree to which the criterion is satisfied.

The index must make it possible to compare alternatives, i.e. to solve pairwise comparisons: given two alternatives $A1$ and $A2$, it must be possible to single out the best one with respect to criterion C, by comparing the values that its associated index I_c assumes in correspondence with $A1$ and $A2$. By reiterating pairwise comparisons, the index thus makes it possible to rank the alternatives with respect to the criterion it expresses. In conclusion, an index is a function of the alternative that describes the preferred direction of change embedded in the evaluation criterion, and can be defined either on an *ordinal* scale (*qualitative index*) or a *cardinal* scale[5] (*quantitative index*).

3.2.1 Index and indicators

Sometimes the index value for an alternative can be *qualitatively* estimated by interviewing the Stakeholders or, given that often it is not possible to interview all the Stakeholders in person, by referring to an Expert who represents them (Figure 3.2a). Assume, for example, that the aesthetic impact of a new dam must be evaluated: a series of pictures of the site, both in its actual state and after the construction of the dam, are shown to the Stakeholders. By comparing the images the Stakeholders can express their opinions on an appropriate scale. This procedure has two defects however:

- the evaluation might appear too subjective, particularly to those that feel disfavoured by its result; therefore it could easily be a controversial point;

- when there are many alternatives, the task of the Expert is arduous and tiredness and boredom may affect his judgement as the work proceeds.

For these reasons, with the Expert's help, the Analyst should identify a procedure to estimate automatically the value of the index (Figure 3.2b). The procedure must 'incorporate', 'reproduce', and 'contain' the Expert's experience in an operational and repeatable way, and be identifiable through interviews.

Very often the alternative acts on a system whose quantities vary naturally through time: think for example of the levels in a reservoir and the flows in the rivers and in the canals. The behaviour of the system in a given time horizon (the one over which we want to evaluate the effect of the alternative, called *evaluation horizon*) is described by the trajectories[6] of these quantities which we will call system *outputs* (see Section 4.1.2.3). An alternative modifies the system and, as a consequence, also the trajectories of its outputs.

We can now subdivide the procedure for estimating the index into two steps:

- The simulation of the behaviour of the system under the alternative, to obtain the output trajectories that it produces. Note that this requires a mathematical model of the system.

[4] In practice, however, the term *index* is used to denote both the function, i.e. the procedure, and the value that it produces.

[5] With both scales pairwise comparisons are possible.

[6] By the term *trajectory* we mean the time pattern of a variable over a given horizon, i.e. the series of values that it assumes at all the time instants of the horizon.

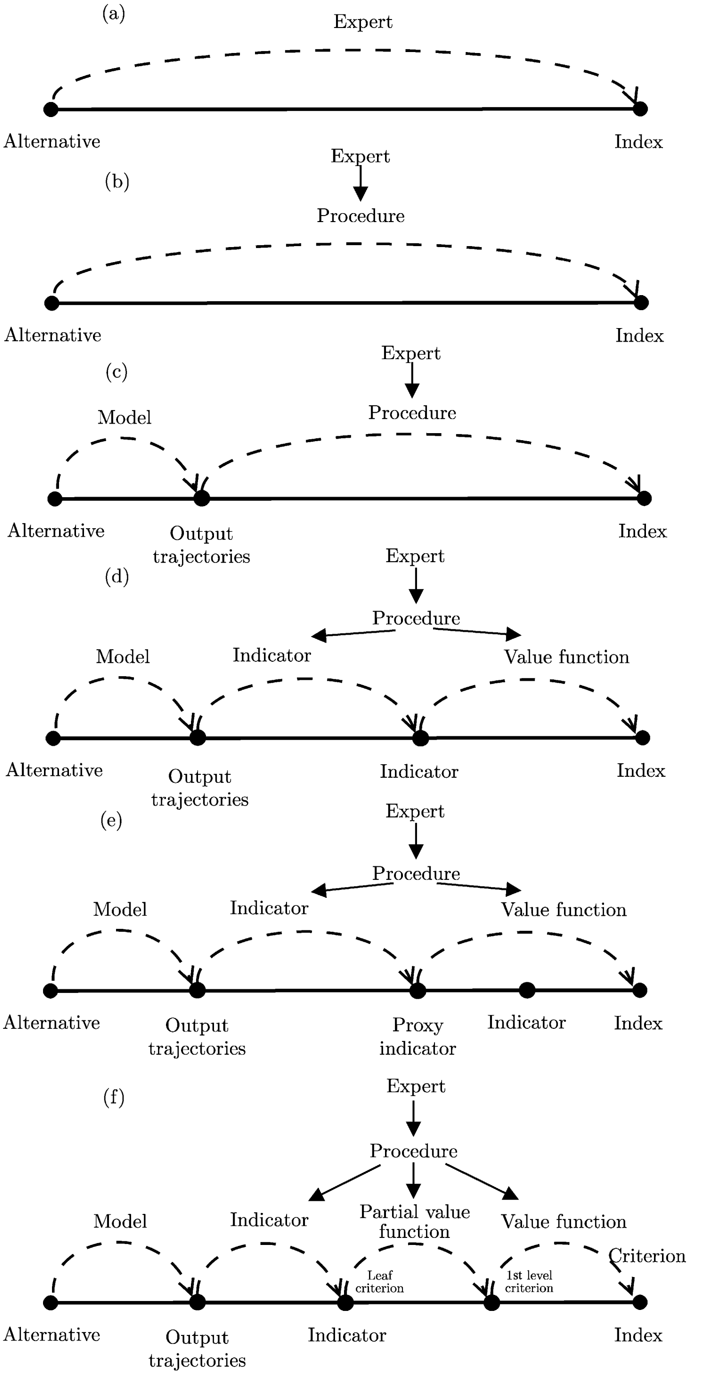

Figure 3.2: Different ways to associate an index to an alternative (Soncini-Sessa et al., 2003).

- The calculation of the index value through a procedure, usually expressed by a functional, which defines the index as a function of the output trajectories.

This approach is illustrated in Figure 3.2c.

The functional should be defined by the Expert, but he is not always able to do this. To simplify this task, the passage from the output trajectories to the index is broken down into two steps, by introducing an *indicator*, i.e. an 'intermediate' quantity between the trajectories and the index (Figure 3.2d), which can be measured directly or estimated with a procedure,[7] making it easier to evaluate the satisfaction of the corresponding criterion. The indicator must be chosen in such a way that, on the one hand, it is easy to single out the functionals through which it is calculated, once the output trajectories are obtained, and on the other hand it is possible for an Expert to pass from the indicators to the index (Beinat, 1997). Therefore, a good practice is to choose the indicators in such a way as to make the distance between indicators and index as short as possible, so that the space for the Expert's subjective opinion and thus the triggers for contention are minimized. Unfortunately, the complexity of reality, along with limits to knowledge, data availability and computing time, make it impossible, in most cases, to completely nullify this distance.

The passage from indicator to index is technically covered by the *value function*, which we will discuss in Chapter 20. Here we simply anticipate that it is a mathematical relationship that 'reproduces' the Expert's judgement and that it is identified through interviews and experiments.

3.2.1.1 Proxy indicator

Sometimes, however, it is not possible to associate an indicator to a criterion. When this happens, a *proxy indicator* should be used, i.e. a variable in a logical relationship with the criterion and related to the effects of the alternatives through a functional, objective and potentially quantifiable link, even if, in reality, this relationship is not quantified, because it exists only in the mind of the Expert (Keeney and Raiffa, 1976). For an example of this, see Section 4.3.2 of PRACTICE. The proxy indicator thus assumes the role of an *indirect ordinal estimator of the criterion* and the degree of satisfaction of such a criterion can be evaluated through the value assumed by the proxy indicator. This is possible because *in the mind of the Expert* there is a relationship between the proxy indicator and the criterion. Note that, since the proxy indicator is an ordinal quantity, the values that it assumes can be compared only with each other, and it will not be admitted to perform algorithmic operations upon them. We can interpret all of this in light of Figure 3.2 which shows that the proxy indicator is none other than the indicator we are looking for. It is, in fact, the input of the value function and the old indicator is completely unused: its role was only to help us to identify the proxy indicator.

3.2.1.2 Hierarchy of criteria

If an index must be expressed in quantitative terms, its definition may be difficult when one tries to formulate it in just one step. The task becomes easier if the criterion is broken down into a set of *sub-criteria*. If it is possible to associate an indicator to each of these criteria, and thus work back from them to the index, one can adopt the set of criteria in place of the original evaluation criterion. Otherwise, one must proceed with a new breakdown. With this

[7] Like index, the term *indicator* also is used in practice to denote both the procedure and the estimated value that results from applying it.

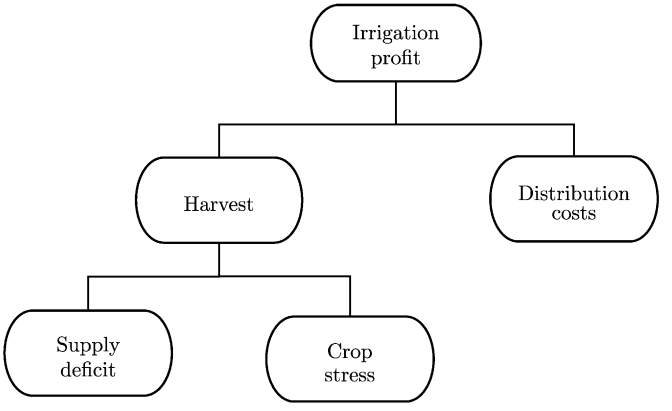

Figure 3.3: The hierarchy of criteria for the *Irrigation profit*.

procedure a partially ranked set of criteria is individuated, which is called a *hierarchy of criteria*. In this hierarchy the criteria are classified according to different hierarchical levels, beginning with the criterion that was originally proposed (called *root criterion*) and working down to those criteria (called *leaf criteria*) to which the indicators[8] are associated.

For example, if one wants to evaluate the impact a new dam will have on irrigation, the evaluation criterion to consider is the *Irrigation profit*. The index that measures how much this criterion is satisfied must clearly depend on the trajectory of the release to the irrigation district. Expressing this dependence in quantitative terms, however, is not easy: the trajectory of the flow in fact influences both agricultural production (and therefore the proceeds) and also the water distribution costs. Therefore it is advisable to articulate the criterion *Irrigation profit* in two criteria which specify it more thoroughly, such as *Harvest* and *Distribution costs*.

The effect of an alternative with respect to the criterion *Distribution costs* is easily measured, once the release trajectory is known; for an example of an indicator of this type see Section 4.5 of PRACTICE. With regard to the *Harvest*, the most natural indicator would be the biomass of the harvest, which, however, is difficult to calculate. One may thus wish to break the criterion *Harvest* down even further, into two criteria, the *Supply deficit* and the *Crop stress*, to which is easy to associate two indicators that take into account the overall supply volume and its distribution over the course of the year respectively. In conclusion, the hierarchy of criteria for *Irrigation profit* is the one represented by the tree in Figure 3.3.

Analogously to what we have said above, the satisfaction associated with each leaf criterion is measured through an index, i.e. a function of the corresponding indicator (through a *partial value function*, see Section 20.2). The satisfaction of a criterion is expressed through an index obtained by combining the indices of the criteria on the level below (Figure 3.2f). This aggregation step can also be carried out using an automatic procedure. However, the definition of this procedure is often not a strictly technical operation, as was the definition

[8] In the following we consider a case where each (leaf) criterion is associated to only one indicator. However, in some particular cases, in order to measure the degree to which a criterion is satisfied, it may be necessary to use more than one indicator. In such cases, the so-called *mutual preferential independence condition* (see Section 20.3) does not hold among the indicators. However, since it is a necessary condition for adopting the MAVT method in moving from indicators to indices (see Chapter 20), in this text we will assume that it is always verified.

of the indicators, because it may involve extremely subjective choices.[9] We will return to this issue in Section 3.4.

Until now we have assumed that all the Stakeholders are in agreement about considering a single criterion with which to evaluate the alternatives. Almost always, however, there are multiple interests in play and so it is necessary to consider a plurality of evaluation criteria from the beginning, and to identify a hierarchy for each of them. Each hierarchy is identified by interviewing only the Stakeholders who are interested in the corresponding criterion, but when these hierarchies have all been identified, they can been seen as the branches of a single hierarchy (*Project Hierarchy*), which structures the whole set of the Project's evaluation criteria. The root criterion of this overall hierarchy is the Project Goal, which we spoke about in the first section.

3.2.2 Properties of the criteria

Keeney and Raiffa (1976) suggest that the criteria should respect the following properties:

- *Completeness*: at a given hierarchical level, a set of criteria is complete if it is adequate for evaluating the extent to which the criteria at the next higher level of the hierarchy have been satisfied.

  To determine a complete set, one can recursively apply the *importance test* suggested by Ellis (1970): a criterion is *important* if the Experts believe that its presence can influence the choice of the best compromise alternative. When the Experts are no longer able to suggest important criteria, the set can be considered complete. In particular, completeness requires that for each action at least one criterion exist by means of which its effects can be evaluated, and that for each Stakeholder there be at least one criterion that expresses his/her viewpoint.

- *Operability*: the criteria must be chosen so that the evaluation of an alternative is easy and comprehensible to the Experts and the Stakeholders. It is thus necessary that these people collaborate with the Analyst to define the criteria.

- *Decomposability*: the set of criteria must be decomposable into a number of subsets, such that the evaluation of the criteria of each subset can be carried out independently of the others. If this condition does not hold, decision-making problems in which the number q of criteria is even modestly high (e.g. $q = 5$), become extremely difficult to handle.

- *Non-redundancy*: the same effect should not be considered twice, i.e. aspects which are actually the results of the same effect should not be considered separately.

- *Minimum size*: to avoid increasing the complexity of the problem unnecessarily, the number of criteria must be kept as low as possible, clearly with respect for the properties of completeness and operability.

[9] This does not mean that, in reality, the indicators' definition, as well as any operation we have described so far, is actually free from subjective elements.

3.2.2.1 Transient or steady-state?

We have seen that the definition of a criterion helps to evaluate the effects of an alternative with respect to a specific interest, but we have not specified which effects we are speaking of. Two different types can be considered:

- the effects in the construction phase, which means in the period in which the alternative is implemented, and in the *transient* (or short-term) period, during which the system adjusts to the new condition;

- the *steady-state* (or long-term) effects, i.e. those that occur in the period (which is often considered to be never-ending) following the initial transient.

Returning to the example of the Verbano Project, consider the action of dredging the mouth of the lake. In the construction phase there will be an increase in suspended solids, in the noise produced by the dredgers and in the disturbances to traffic caused by the trucks transporting the removed material. Once the excavation has been completed, the regime of lake levels will have been modified and the lake ecosystems will require a certain period to adapt. For example, the waves might lap at the roots of the reed beds more often, causing their regression, which in turn will modify the possibilities for migratory birds to nest. After several decades, the reed beds will probably have reached a new equilibrium and the bird life will have adapted to it: some species may have modified their rhythm of life, others may have abandoned the area. The steady-state effects appear in this new situation.

Interest in transient effects or in steady-state effects, or in both of them depends on the Project Goal. When interventions are planned whose effects extend into the distant future, interest is normally focussed on the steady-state effects, since they are the ones which last the longest. Clearly, this is reasonable only if the transient effects are not so devastating that the steady-state effects result to be completely uninteresting. Furthermore, the evaluation of the transient period requires a more detailed analysis than the evaluation of the steady-state conditions: for example, to identify the steady-state effects it is sufficient to say that the excavation of the lake outlet will result in a given storage–discharge relation. To evaluate the transient effects, however, one must specify the machinery that will be used and the season in which the dredging will occur, where the removed material will be dumped, how many trucks will be used to remove it, which precautions will be taken to reduce the diffusion of suspended solids, and so on. Without these specifications the estimate of the effects in the construction phase cannot be made. Therefore, it is evident that considerations regarding the time and cost of the Project impose a limit to the number of alternatives for which the transient effects can be evaluated. In order to identify these alternatives, one should first compare the whole set of alternatives by considering only steady-state effects. The alternatives that emerge from this first selection must then be analysed in detail with respect to the transient period. This will increase their number because rarely will there be only one option available for construction. The alternatives found must then be subjected to a new decision-making process, in which both the transient effects and the steady-state ones are taken into account.

3.2.3 Factors influencing indicator choice

The reader will probably have formed the idea that the choice of the evaluation criteria and the definition of the indicators are guided essentially by the Project Goal. Ideally, this is how it should be, but actually the choice of both criteria and indicators is influenced by the conditions imposed upon the use of the indicators in the following phases of the procedure.

In Section 1.3 we explained that a model of the water system is required both for designing and evaluating the alternatives. In particular, as we explained in Section 3.2, we are interested in getting the trajectories of the variables that are required to compute the indicators. The choice of the latter thus poses conditions on the structure of the model that can be adopted. But the relation should also be considered in the opposite direction: indicators, whose computation requires knowledge of variables that we cannot simulate, should not be adopted.

Other than these general considerations, there are other more specific elements that must be taken into account. In Chapter 2 we saw that a regulation policy is often a component of the alternatives of the problem. The design of a policy entails that the regulation objectives should be among the indicators, which means that when choosing the indicators the regulation objectives must be taken into account. This is reasonably obvious. Less obvious is the fact that the algorithms used to design regulation policies impose conditions on the form of the indicators that can be used (see Section 12.1.2). These conditions should also be taken into consideration when the indicators are defined; sometimes this is easy, other times it is difficult, if not impossible, without altering the corresponding criterion.

In particular, one of these conditions is that the indicator must be a combination of the instantaneous values of a quantity (called *step indicator*) in the whole evaluation horizon. At every time instant, this value depends only on variables relative to that same time: in mathematical terms, the indicator is a functional (e.g. the summation) of the step indicators, each of which is a function of the system outputs at a given time instant.

This condition is expressed by the so-called *hypothesis of separability* and an indicator that respects it is a *separable* indicator. In order that this hypothesis hold true, the value of the step indicator at time t must not depend on the value assumed by that same indicator at preceding time instants; in other words, the step indicator cannot be the state of a dynamical system.

When the hypothesis of separability is proven to be true, the identification of the indicator is simple. When it is not, it is always possible to trace the problem back to a case in which it does hold, provided that the state of the system is suitably enlarged (see Section 10.2.2.2). For example, if the step indicator at time t depends on its value at time $t-1$, it is sufficient to include a *state transition function* in the model of the system (see Section 4.1.2.3) that expresses the dynamics of the indicator.

Difficulties arise only when the hypothesis of separability does not hold and we do not want to satisfy it in the way we have just described. An example of what happens in this case is described in Section 4.3.4 of PRACTICE. The attempt to avoid using a dynamical description is not due to laziness, but to two serious considerations. The first is the desire to reduce the cost of model identification, which is never negligible: often, a simple well-chosen non-dynamical relation is much more precise and reliable than a dynamical model that has been badly calibrated to save money or time. The second reason is that including a dynamical system has severe consequences for the design of the regulation policy. Such a policy, in fact, should have as arguments not only the state of the physical system, but also the states of all the systems that appear in the description of the indicators. To understand why this happens, consider for example the *Irrigation* criterion. If the crop is a dynamical system (as in reality it is), the release decision cannot be made without considering its state. Think of the extreme case in which the crop is dead: clearly it is useless to irrigate it and the flow rate which would have been allocated to it can, more usefully, be used somewhere else. Thus, knowledge of the state is essential for the correct description of the management problem, but this becomes more complicated and costly to solve. It is up to the Analyst to guess when an increase in complexity is justified by the advantages that can be obtained from it and when, instead, it is better to simplify the description.

Lastly, when the indicators to be associated to the leaf criteria are chosen, it is always advisable to verify whether for the particular aspect considered there are already existing indicators that are frequently used. Many international organizations[10] have defined reference frameworks to classify and provide list of environmental, economic and social indicators (the EEA's DPSIR framework, for example, which we discussed at the beginning of Chapter 1) that are accepted in those contexts. When possible it is preferable to use these indicators rather than redefining new ones; this has the advantage that it makes the results of the analysis easier to compare.

3.2.4 Validation of the indicators

As we have seen, every indicator i_h must allow us to compare the alternatives with each other, with respect to the satisfaction of the criterion C_h to which it is associated. We have also seen that an indicator is a procedure that allows us to associate each alternative A with the value $i_h(A)$ that it assumes in correspondence with that indicator.

Once indicator i_h is defined, it is necessary to *validate* it, i.e. to ascertain that it effectively does allow alternative comparison. The validation is carried out by showing the Expert (or the Stakeholder), who defined the indicator, the values $i_h(A1)$ and $i_h(A2)$ that it assumes in correspondence with the two alternatives $A1$ and $A2$. On the basis of these values (s)he is asked which (s)he prefers. Then, (s)he is shown the effects produced by the two alternatives (for example, the trajectories of several quantities that are significant for criterion C_h) and (s)he is asked if the preference that (s)he expressed earlier is confirmed. This experiment is repeated with other pairs of alternatives; if each time the second judgement confirms the first, the indicator is considered to be validated; if not, the choice of the indicator, or of the leaf criterion itself, should be reconsidered. When possible, the validation is repeated after a period of time, to make sure that the judgements are stable and consistent. If they are not, the choice of the indicator should be reviewed, but only after ascertaining that the instability is not due to a lack of motivation on the part of the Expert.

An example will help clarify these concepts. Consider a Project aimed at reducing the flooding due to high tides (*acqua alta*) in the city of Venice. One of the criteria considered is *Tourist satisfaction*. Since it is not possible to associate an indicator directly to this criterion, it is resolved into the following two sub-criteria: *Duration* and *Intensity* of the floods. The first criterion is associated with the indicator *Average annual number of days in which the level of the water exceeds the flood threshold*, denoted with ($i_{\text{dd\_flooding}}$); the second is associated with the indicator *Average height of the peak* in the same period. In order to validate the first indicator, two alternatives are shown to the Expert, $A1$ and $A2$, which respectively produce the following values: $i_{\text{dd\_flooding}}(A1) = 7$ and $i_{\text{dd\_flooding}}(A2) = 4$. Naturally, he says that he prefers alternative $A2$ and confirms his judgement when he is shown the trajectories (Figure 3.4a) of the water level that the two alternatives produce: the same trajectories from which the aforesaid values were derived. Then he is posed a new comparison, between alternatives $A3$ and $A4$, whose corresponding values are $i_{\text{dd\_flooding}}(A3) = 9$

[10] OECD (1994), UNCSD (1996).

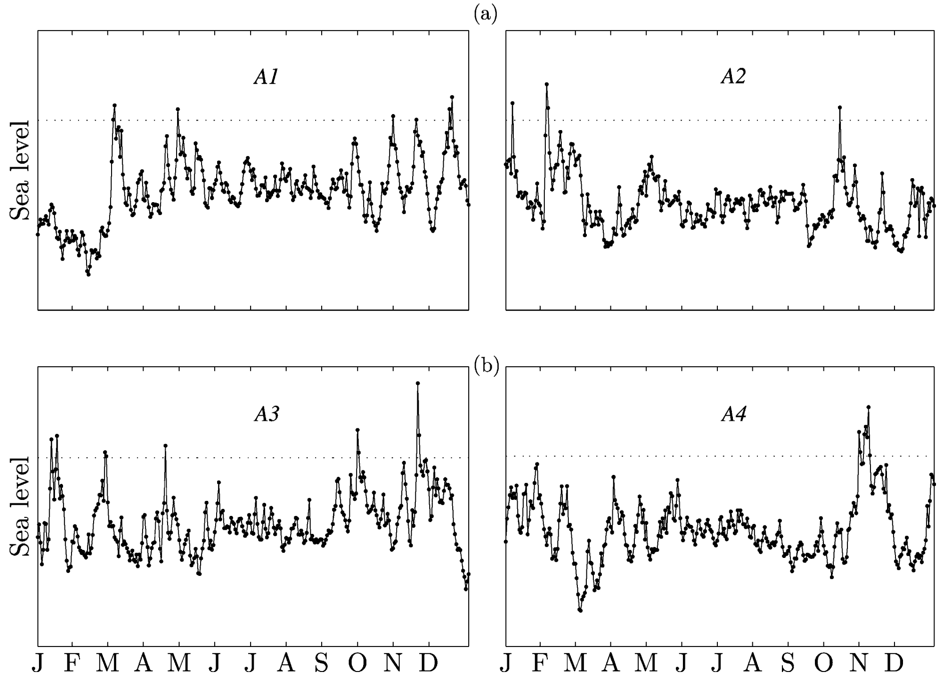

Figure 3.4: Comparison of the sea level trajectory in the city of Venice produced by four different alternatives. The dashed line represents the flooding threshold.

and $i_{\text{dd\_flooding}}(A4) = 7$. The Expert declares that he prefers $A4$ over $A3$, but when he is shown the corresponding trajectories (Figure 3.4b), he changes his mind and says that he prefers $A3$. He justifies this decision by stating that a prolonged period of high water, such as the one produced by alternative $A4$, causes tourists to flee the city, while the floods produced by alternative $A3$, even if they are more numerous overall, cause less irritation, because they are brief and sporadic and can even be considered attractive and part of the local folklore. From this it does not necessarily follow that the indicator $i_{\text{dd\_flooding}}$ is incorrect, but only that it is insufficient for evaluating the satisfaction of the criterion *Duration*. Therefore, it will have to be coupled with another indicator that measures the maximum (or average) duration of the floods. This means that the hierarchy must be reformulated, by subdividing the criterion *Duration* into a set of lower-level criteria.

Note that to compare the alternatives the Expert must have some idea of the existing relationship between the value of the indicator and the satisfaction of the corresponding criterion. This relationship will be formalized only in Phase 6 of the PIP, through the definition of the partial value function (see Chapter 20). However, to validate the indicators it is not necessary to have an analytical expression of that function[11]: it is sufficient that the Expert has a qualitative idea of its trend.

The reader may have noticed some incongruence in the reasoning developed until this point. In fact we have said that, in order to design the alternatives, it is necessary to identify

[11] As we will see in Chapter 20, an analytic expression can be determined only after the phases of *Designing Alternatives* and *Estimating Effects*.

a set of indicators; and that before using those indicators, it is necessary to validate them by using a set of alternatives. Fortunately, this vicious circle is easily overcome: the indicators can be validated without comparing trajectories generated by real alternatives, which at the beginning of the Project are not available, but instead by comparing trajectories that were recorded in the past. For example, the trajectories of the levels in Figure 3.4 could be those recorded in four different years. Then, in the phase of *Designing Alternatives*, when the first alternatives are available, they can be employed to repeat the validation of the indicators, in order to make sure that these latter are really valid before proceeding any further with the design. This is why we will return to the subject of validating the indicators in Section 19.4, after the phase of *Designing Alternatives*.

3.3 An example: the Egyptian Water Plan

We resume the example of Egypt's Water Plan (see Section 1.1.1) to illustrate how the hierarchies of criteria are constructed and the indicators are defined.

In the phase of *Reconnaissance* (see the box on page 89) the Stakeholders were identified. They can be subdivided into seven homogenous groups: the farmers, the urban population and the environment[12] from both Upper and Lower Egypt, and the fishermen on the coastal lakes. Each of these groups was asked to express a criterion with which they would judge the alternatives, and even if they all nominated the same criterion, the *Quality of life*, each group gave this concept a different meaning, thus resulting in a variety of criteria.

To identify the index for each criterion, the Stakeholders who proposed it were questioned. Since examining all the responses would require much time, we present only the results of the interview with the farmers from Lower Egypt as an example. It emerges that the quality of life varies among the different districts of the delta and this determined the first branching in the hierarchy (Figure 3.5). Let us consider, for example, the responses provided by the farmers from the Bagona district, in which the crop is rice. For them, the quality of life depends on the health conditions (*Health*); the economic conditions (*Income*); and the conditions of the drinking water supply (*Drinking water supply*). This is the second level of the hierarchy. They believe that health is influenced by the *State of nutrition*, the *Incidence of schistosomiasis*[13] and *Infant diarrhea*. With the help of Experts they suggest that the *State of nutrition* be measured with the following indicator[14]:

$i_{\text{LEfarm\_h\_1}}$ ($\downarrow$) the daily average percentage [%] of caloric deficit with respect to a reference standard.

and they propose the following formula to compute it[15]

$$i_{\text{LEfarm\_h\_1}} = \frac{1}{365 \cdot N} \sum_{t \in H} \left[\frac{(\bar{k} - k_t)^+}{\bar{k}} \right]$$

[12] Here and in the following we will use the term 'environment' as if it were a Stakeholder; with this use we are actually referring to all those (public institutions, associations, private citizens, etc.) who for various reasons are concerned with environmental conservation.

[13] A parasitic disease caused by trematodes which live in the water of the rice paddies, penetrate the skin of the farmers that work there, and enter their circulatory system.

[14] The symbol ($\downarrow$) indicates that the satisfaction of Stakeholders increases as the value of the indicator decreases.

[15] The symbol $(\cdot)^+$ denotes an operator that returns the value of the argument, when it is positive, or zero in the opposite case.

The Egyptian Water Plan: results after Phase 1

System boundaries The inflows to Lake Nasser come from a catchment that covers nine countries. As a consequence, the water availability in Egypt is inevitably affected by these countries' policies for exploiting the waters of the Nile. Thus the system that should be considered is the entire Nile River catchment basin. However, the Egyptian Government does not want the study to have this breadth and, despite the fact that the Analyst underlines that this would be the only correct way to structure the Plan, the Government requests that the study be limited to the national territory.

Stakeholders The phase of *Reconnaissance* highlights that water availability should be evaluated with respect to the quality of life and the quality of the environment, and that the first is interpreted differently by farmers, fishermen and the urban population. Moreover, the interests of farmers, urban populations and environment are not homogeneous over the whole area, since the conditions in the narrow valley of Upper Egypt and in the wide delta of Lower Egypt are very different. Therefore, seven groups of Stakeholders must be defined: *Upper Egypt* (UE) *Farmers*, *Lower Egypt* (LE) *Farmers*, UE *Urban Population*, LE *Urban Population*, UE *Environment*, LE *Environment* and *Fishermen*.

The Goal From the meetings held by the Analyst with the DM (the Egyptian Government) and the Stakeholders it emerges that the Goal of the Plan must be to improve the quality of life in Egypt through the identification of interventions that are sustainable from an environmental and economic perspective.

Options for intervention All the options for intervention listed in Section 1.1.1 are considered to be suitable. Coherently with the limits established for the study, they do not include political agreements that Egypt might try to establish with the other countries of the Nile basin.

where k_t is the average caloric supply that an inhabitant of the district receives on day t, $\bar{k}$ the reference standard, H the *evaluation horizon*, and N the number of years that it contains. Furthermore, they propose the following indicators to quantify the *Incidence of schistosomiasis* and *Infant diarrhea*:

$i_{\text{LEfarm\_h\_2}}$ ($\downarrow$) percentage [%] of farmers in the district that are infected with *schistosomiasis* each year;

$i_{\text{LEfarm\_h\_3}}$ ($\downarrow$) percentage [%] of children in the district between 0 and 5 years who suffer from diarrhea for more than thirty days of the year.

For brevity's sake we will not report their analytic expression here. Since all these criteria were associated with indicators they are *leaf criteria*.

The criterion *Income* is strictly linked to the *Average annual rice harvest*; this criterion can immediately be associated with an indicator. Note, however, that the value of this indicator can be estimated only if a model of the rice crop in the district is available. If the identification of this model is too expensive, one can use a proxy indicator of the harvest: the *Average annual supply stress*, an indicator that we will define precisely in Section 5.6.2.

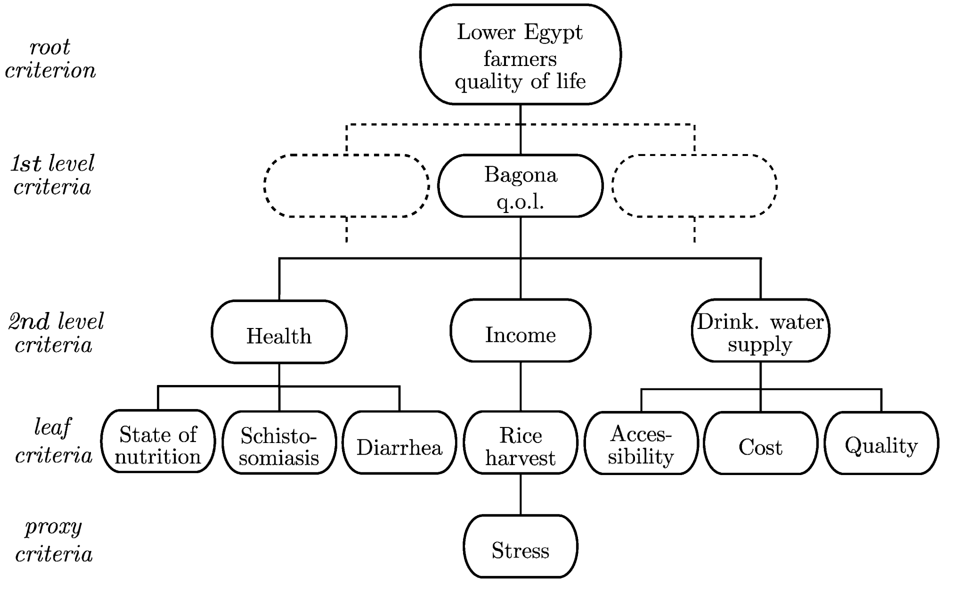

Figure 3.5: The hierarchy of criteria produced by the farmers from Lower Egypt.

Finally, the *Drinking water supply* is resolved into water *Accessibility*, *Cost* and *Quality*. The explanation of how these criteria can be quantified would take us too far off track and so we will stop here, assuming that the reader has already understood how to proceed.

3.4 Project and sector indices

The effect of a given alternative with respect to an evaluation criterion can thus be measured by the corresponding indicator. When the values of all the indicators for all the alternatives are known, one must

- quantify the satisfaction for the effect that is measured by the indicator, i.e. compute the value of the index associated to the criterion;
- aggregate the values of indices in order to evaluate the 'global satisfaction' associated to an alternative.

The latter is measured using a synthetic index, called *Project Index*. It provides a simple criterion for the comparison of the alternatives: the 'best' alternative is the one that maximizes that index.

The aggregation operation that allows us to work up from the set of indicators to get to the Project Index, which is called *evaluation*, is, however, anything but trivial. It will be discussed in detail in Chapter 20. Here we simply note that the value of the Project Index must depend on the values of the indicators associated to all, and exclusively, the criteria that are of interest for the person who carries out the evaluation. The aggregation of these values requires that (s)he express relative value judgements about the corresponding criteria. To do this, one can use the hierarchy of criteria once again: the relative weights of the criteria

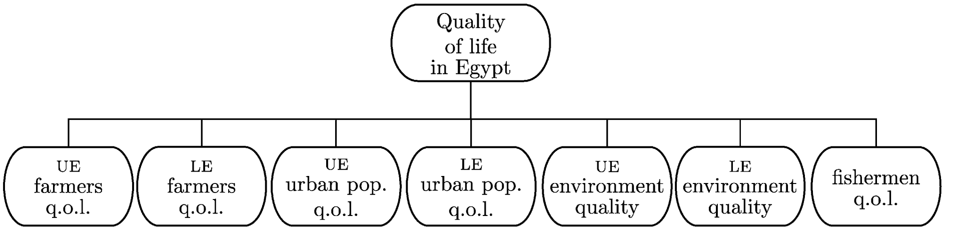

Figure 3.6: A first tentative Project Hierarchy for the Egyptian Water Plan.

can be fixed by comparing the criteria that descend from the same node and by repeating this operation for all the nodes. This procedure is simpler and faster than comparing all the criteria directly.

To clarify these concepts, once again we return to the example of the Egyptian Water Plan. The Government is the only DM and its Goal is "to improve the quality of life in Egypt". The set of criteria, into which this objective is resolved, was identified by the Analyst through interviews with the Stakeholders, as we explained in Section 3.3. Thus seven hierarchies were obtained, one for each of the seven homogeneous groups of Stakeholders (see box on page 89), each of which is like the one shown in Figure 3.5 and resolves one of the following seven criteria: *Quality of life* (*q.o.l.*) *of farmers* in UE and in LE; *Quality of life of urban population* in UE and in LE; *Quality of the environment* in UE and in LE; and *Quality of life of fishermen*. These seven hierarchies can be considered as branches of the Project Hierarchy shown in Figure 3.6, where the seven leaf criteria are the seven root criteria of the seven hierarchies: for example, the hierarchy in Figure 3.5 is attached to the second block on the bottom left of Figure 3.6 (LE *farmers q.o.l.*). It may be that the Government finds it difficult to express the relative importance of the seven root criteria in the definition of the quality of life in Egypt and that it would find it easier to express the relative importance of the UE and LE farmers' quality of life in defining the quality of life for the farmers in the whole country. By proceeding in the same way for the urban population and the environment, at the end, it would need to compare only four criteria: the *Quality of life of the farmers*, *of the population*, *of the fishermen*, and the *Environmental quality*. This way of doing things is expressed by the hierarchy in Figure 3.7. However, it might not be the only interesting way to perform the evaluation: it might happen that, as the analysis goes on, the Government realizes that it would be easier to compare first the three criteria concerning Upper Egypt, then those of Lower Egypt, and finally to conclude by comparing the quality of life in Upper, in Lower Egypt and of the fishermen. In other words, it could happen that the Government might want to adopt the hierarchy in Figure 3.8.

During the *Evaluation* phase, this type of transformation is useful to determine, from all the possible hierarchies, the one that helps the DM the most in its task of criteria comparison. Therefore, it is important that these transformations be managed with ease. This is not difficult, since the hierarchies in Figures 3.7 and 3.8 are none other than two different partitions of the same set of criteria (note that they share the same leaf criteria), as shown in Figure 3.9.

We want the reader to understand that the evaluation is not just a technical operation, on the contrary it inevitably includes subjective judgements. Beginning with the same set of criteria, different people may arrive at very different definitions of the Project Index and,

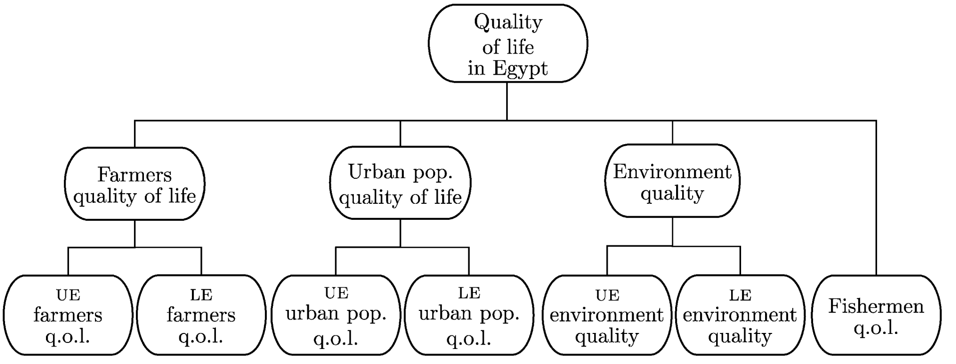

Figure 3.7: A hierarchy of criteria expressed by the Egyptian Government.

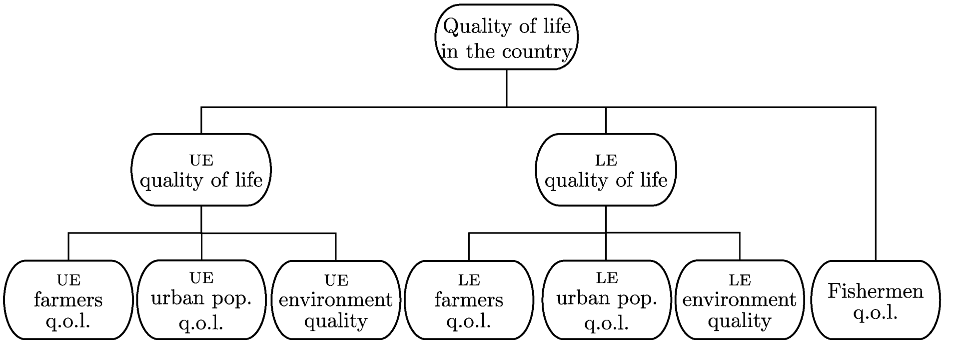

Figure 3.8: A hierarchy of criteria alternative to the one in Figure 3.7.

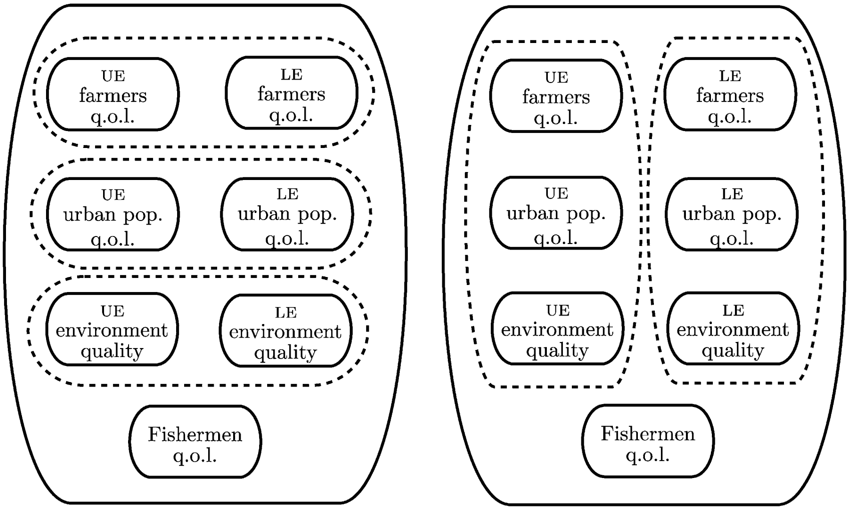

Figure 3.9: Two different partitions of the same set of criteria.

as a consequence, different rankings of the same alternatives: the result of the evaluation depends on the contingent choices that are made by each person.

When the numbers of Stakeholders involved and of evaluation criteria are high, it is necessary to limit the potential for subjective choices as much as possible. An agreement on the aggregation procedure must be found, at least for the low level criteria: very often, in fact, aggregation at this level is more technical than subjective. In the Egyptian Water Plan, for example, we implicitly assumed that the hierarchies defined by the seven Stakeholder groups to individuate their own criteria would be used also during the evaluation. We have already mentioned that the seven hierarchies individuated by the Stakeholders (an example of which is shown in Figure 3.5) are hooked onto the leaves of the hierarchies in Figures 3.7 and 3.8. Thus, this makes it possible to define seven indices, each one being a function only of the indicators associated to the leaves of the corresponding hierarchy. The DM (the Egyptian Government) plays a part only after these operations have been completed by the Stakeholders, and decides how to aggregate these indices, without questioning the way in which they were defined.

We can formalize this procedure by defining a *sector* as a sub-tree of the hierarchy on which there is an agreement about the aggregation method. A *sector index* is associated to the root criterion of that sub-tree, and it is an aggregation of the indices of the criteria below. The value of that index is, by definition, unquestionable, because the procedure for calculating it is shared, whether it is the fruit of an Expert's technical evaluation or of a negotiation process among Stakeholders (see Section 21.2). The concept of sector is very useful because it allows us to divide the evaluation into two phases. In the first, an index is defined for each sector by considering only the criteria inherent to it, and involving only the Stakeholders interested in those criteria. In the second phase, the evaluation is based only on the *sector criteria*, i.e. the criteria at the root of the sub-tree that constitutes the sector, and the sector indices are aggregated to define the Project Index. In practice, the term 'sector' is used to denote both the sub-tree, whose aggregation method a group of Stakeholders (even composed by only one Stakeholder) agrees on, and the group of Stakeholders itself.

Finally, let us consider who should conduct these operations. In the example of the Egyptian Water Plan the sector indices are constructed by groups of Stakeholders, each of which is interested in one of the seven sectors individuated by the leaves of the hierarchy in Figure 3.7 (or in Figure 3.8). In order to obtain the Project Index, the aggregation of the sector indices is carried out by the DM, i.e. the Egyptian Government. This way of proceeding is typical of participatory decision-making processes with only one DM: the Stakeholders are involved in the more technical part of the evaluation and cooperate to define the sector indices; the aggregation of the sector indices, in which political considerations prevail, is carried out by the DM. If there is more than one DM or the Stakeholders are given a co-deciding role, this approach cannot be adopted. Therefore, after the phase of *Evaluation* a phase of *Comparison* must follow, which is conducted through negotiation. We will discuss this theme in Chapter 21. The introduction of the sectors proves to be an advantage also in this case, because it allows the set of variables to be considered in negotiations to be reduced, by moving from the set of indices associated to the leaf criteria to the set of sector indices.

From what has been said, we deduce that the definition of the sectors should emerge during the phase of *Evaluation*. In practice, however, it is often possible to determine the sectors, at least tentatively, in the *Reconnaissance* phase, by individuating homogenous interest groups, within which the definition of a sector index is presumed to generate no conflicts among the Stakeholders. If during *Evaluation* this hypothesis is disproved, there are two possible cases: the sector that was hypothesized to be a single sector must be exploded into several sectors, because an agreement cannot be reached within it; or, sectors that were hypothesized as being distinct can be fused into a single sector, because the agreement is wider than expected.

Chapter 4

Systems, models and indicators

AC and RSS

In Chapter 3 we looked at the role that indicators play in the PIP decision-making procedure. We also saw how, at least in the cases that this book is concerned with, an indicator is specified by a *functional* of the *trajectories* of the system output. Towards the end we mentioned the fact that these trajectories can be generated in correspondence with the different alternatives when one has a model of the system under examination. The aim of this chapter is to illustrate the structure of such a model and the way to identify it. We will also definitively clarify the relationships between model and indicators. The presentation will be kept at a general, in a certain sense procedural, level, because we want to illustrate how to proceed in structuring a model: it is a complex process, full of recursions, in which different disciplines are intertwined together: those relative to the water system that is being studied, and those relative to the definition and manipulation of models. We will describe modelling as a cognitive activity, concentrating on aspects of encapsulation and reproduction of knowledge, and avoiding the technical aspects as much as possible: their presentation alone would require an entire volume. Given, though, that these aspects cannot be completely ignored, nor can it be taken for granted that they are known to the reader, we will view them in an elementary way in the following two chapters and in Appendices A3–A8 on the CD which accompanies this book. This chapter and the two following ones are dedicated to the phase *Identifying the Model*, the third phase of the PIP procedure (Figure 1.4).

4.1 From the water system to its model

The systems in which one operates are always systems whose quantities (as for example the levels and the flows) vary in time. Their behaviour in response to a given alternative is therefore described by the trajectories of these quantities in the *evaluation horizon*. Thus, to evaluate an alternative it is necessary to know the trajectories that it produces, but since none of the alternatives, except for the Alternative Zero ($A0$), have ever been implemented, these trajectories cannot have been registered in the past. Clearly it is not even possible to obtain them from experiments performed on the real system. This is prevented not only for reasons of cost and security, but also due to the fact that, to be significant, the evaluation of the effects must extend over a horizon of several decades, and one could certainly not wait that long to take a decision. There is no other solution but to obtain the trajectories from experiments conducted on a copy of the system, whose behaviour is as close as possible to the real system, and on which it would be simpler and less expensive to conduct experiments. Such a copy is called a *model*[1] and the calculation, made with it, of the trajectories that the real system would follow in correspondence with an alternative is called *simulation*.

We think that it is worthwhile to clarify these fundamental ideas with an example. We take up once again the Verbano Project, which we introduced in Section 3.1 and to which we dedicated the boxes on pages 74 and 78. From 1943 the Lake Maggiore (Verbano) has been regulated by a dam (Miorina dam) built at its outlet. The general opinion of the inhabitants of the city of *Locarno*, which sits on the bank of the lake, is that the regulation has exacerbated the flooding of the city. Let us suppose that before engaging us in the Verbano Project, the Swiss Government had asked us to ascertain whether this opinion was substantiated. To do so we have to compare two alternatives. The first ($A0$) considers what actually happened, while the second ($A1$) considers the situation in which the lake is not regulated. To evaluate the degree of flooding we must define an indicator i, for example the number of days in which, in the time horizon 1943–2000 (that we will denote with H), the flood threshold $h_{\text{fl}}^{\text{Loc}}$ in the city was exceeded. It is expressed by the following function

$$i = \sum_{t \in H} g_t\left(h_t^{\text{Loc}}\right) \tag{4.1}$$

where g_t assumes either the value 1, if on day t the level h_t^{Loc} of the lake at Locarno exceeds the flood threshold $h_{\text{fl}}^{\text{Loc}}$, or the value 0, if this is not the case, i.e.

$$g_t\left(h_t^{\text{Loc}}\right) = \begin{cases} 1 & \text{if } h_t^{\text{Loc}} > h_{\text{fl}}^{\text{Loc}} \\ 0 & \text{otherwise} \end{cases} \tag{4.2}$$

To estimate the value $i(A0)$ that the indicator assumes in correspondence with the first alternative, it is sufficient to assume that in (4.1) the levels h_t^{Loc} are those that correspond to the trajectory $\{h_t^{\text{reg}}\}_{t \in H}$ which was historically recorded during the horizon H at the Locarno hydrometric station. To estimate the value $i(A1)$ that the indicator assumes in correspondence with the second alternative, we must have the trajectory $\{h_t^{\text{nat}}\}_{t \in H}$ of the levels of the lake at Locarno in its natural regime. This information is unavailable, because the natural lake ceased to exist in 1943. We can, however, reconstruct it by identifying a model of the non-regulated lake and using it to simulate the trajectory that the level of the natural lake would have followed, given the inflow sequence that was recorded in the horizon H, i.e. the same inflow sequence that characterizes the alternative $A0$. By doing this, the difference between the two alternatives is only the presence of the dam and the regulation of the lake, exactly as it must be in order to respond to the question that was posed to us. What we need then is a model of the lake in its natural condition.

The example allowed us to clarify the reason why the models are necessary; we must now explain how they can be *identified*. Let us suppose that we have ascertained that the

[1] In this text we consider only *mathematical models*, that is to say models in which the system is described by mathematical objects, for example graphs and equations. The *physical models*, in which reality is 'physically' reproduced on a reduced scale, are of no use in planning and management, since they are not able to describe vast systems such as those that have to be dealt with. Physical models are useful, for the most part, in hydraulics and fluid dynamics applications, to study particular local phenomena, as for example the flow profile generated by the opening of the gates of a dam.

construction of the Miorina dam at the lake outlet has worsened the flooding and that, following this result, the Swiss Government has engaged us to execute the Verbano Project. To simplify matters, let us assume that:

(1) the only action considered is the modification of the regulation policy;

(2) instead of the set of hydropower reservoirs, mentioned on page 74, there is only a single hydropower plant in the lake catchment, whose turbines discharge directly into the lake;

(3) the effects of the regulation on the River Po can be neglected and therefore the system considered includes the Lake Maggiore catchment and the territory irrigated with the Ticino water.

4.1.1 From the system to its components

The reason why we construct the model and the way in which we plan to use it are clear, but to proceed to its identification we must above all define:

(1) the level of detail in which to describe it;

(2) its components.

We will examine these elements in order.

The lake is strongly interconnected with other systems: the catchment, the distribution network, the users. The union of all these systems constitutes the system that we are interested in, and of which each of the listed systems, including the lake itself, can be seen as a subsystem, i.e. a *component*. Each component can be broken down into other subsystems (for example the irrigation users are organized into districts) and these, in their turn, into even smaller ones (the districts are composed of properties and the properties of crops, the crops of fields, and the fields of plants). We must decide upon the level to which to limit our analysis, i.e. which are the components that we assume to be elementary, whose union and interaction will constitute the system we are modelling. Almost always one settles at a high level: the catchment, the lake or regulated reservoir, the canals, the diversion dams, the powerplants and the irrigation districts. For example, the scheme in Figure 4.1 shows the components of the Verbano water system. This approach is very advantageous, since it permits us to represent a complex system as an aggregate of simple components. Each of these components serves a precise function and the decomposition of the system must be based on the identification of these functions. A model of a component is a formal simplified representation of the real component and so it can capture only some of its aspects, which must be those that are most relevant from the point of view of the Project for which the model is being constructed. In our example, the essential function of the catchment is to generate the inflow to the lake, the function of the lake is to store the inflow, and the function of the diversions to distribute the water among the users; while we are not interested in tourist activities in the catchment or in the turbidity of the lake. For this same reason, sometimes *logical components* are considered, i.e. components that do not have a direct physical referent, but which describe in an aggregated way some of the functions that are carried out in the real system. For example, to describe the Verbano water system it is preferable to substitute the scheme in Figure 4.1 with the one in Figure 4.2, where the two diversion dams in the first scheme are replaced by three logical nodes. As was explained

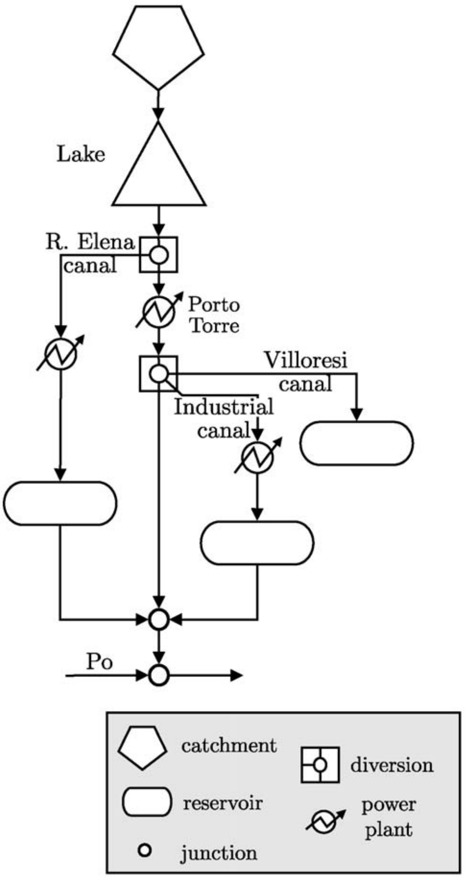

Figure 4.1: The components of the Verbano water system.

in greater detail in Section 6.7 of PRACTICE, these nodes describe the logical phases of the lake Regulator's decision-making process: above all, complying with the MEF in the River Ticino (node 1 in Figure 4.2), then deciding about delivering a flow larger than the MEF to the Ticino in order to improve the environmental objectives (node 2), and eventually, allocating the residual flow (node 3) between the irrigation (Regina Elena and Villoresi) canals and the hydropower (Industrial) canal. When all the components in the scheme have physical matches, they must satisfy the continuity equation and the arcs between them represent physical fluxes (flow rates). In the opposite case, the arcs represent information fluxes (signals) and the logical components may, if necessary, violate the continuity equation. For instance, in Figure 4.2 node 5 does not actually split the flow, but channels the same signal (value of the flow downstream from node 4) to nodes 6 and 7.

Once the components and the *topology* of their interconnection have been determined, the strategy for identifying the system model (*aggregated model*) is to first identify the models of each component (*component model*), and then aggregate them according to the topology that was previously identified.

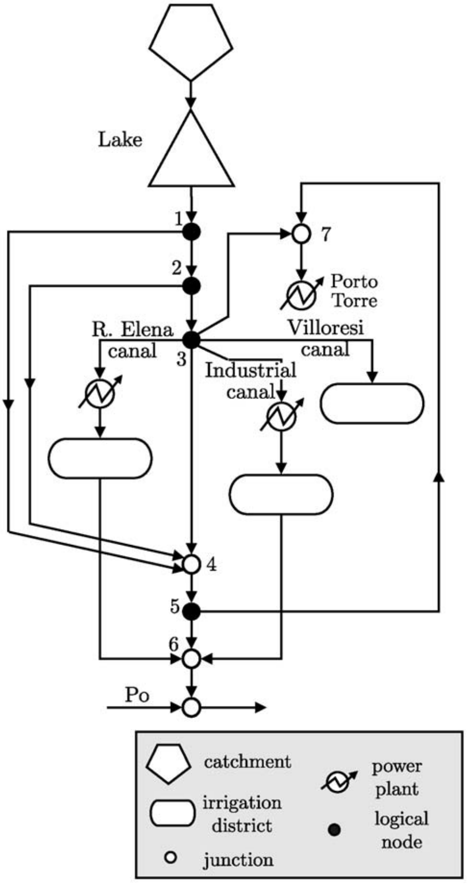

Figure 4.2: Schematization of the Verbano water system adopting logical components.

4.1.2 From the component to its model

4.1.2.1 Domain

Decomposing the system into interconnected components and the determining of the topology of their interconnections is the first level of formalizing reality: it is not yet a model, but just a partition of knowledge (forethought, theories, information, data relative to the system) into sets of information called *domains*, each of which contains the information that concerns a component. The domain is an embryo of a model. It groups together the attributes of reality in a way that anticipates future modelling choices and at the same time serves the purpose of logically separating the real world from the world of models, since it contains the information from the real world that feeds the models. It is the first level of abstraction from reality that does not yet require hypotheses about the mathematical relationships among the variables, but simply a definition of the data which will be used and how they will be represented. Many different models can be associated with the same domain.

Let us consider for example the Lake component in the diagram in Figure 4.2. A lake is a complex system, in which a number of physical and chemo-physical processes take place. The lake domain is the set of all the quantities and information that pertain to it, including the sources of the latter:

- inflow
- release
- level
- chemical characteristics of the water
- biota
- algae
- ...
- bathymetry
- topography
- stage–discharge relation at the lake outlet
- ...
- the Hydrobiological Research Center of Pallanza
- the Po River Authority
- ...

The model of the lake is, instead, a simplified representation of reality that should reproduce the features that are relevant from the point of view of the Project for which it is being created. In our example the aim is flood reduction and therefore the most relevant feature of the lake is the dynamics of its level. What we need to describe these dynamics must therefore be selected from the information contained in the domain: the quantities that contribute either directly or indirectly to forming the levels and the characteristics of the physical system that influence them. From the above domain we choose the following headings:

- inflow (a)
- release (r)
- level (h)
- bathymetry
- stage–discharge relation at the lake outlet.

The first three are quantities and so they are natural candidates for *variables* of the model. Thus, it is convenient to associate each one with a symbol that denotes it: we will use the symbol that appears after them in the list above. Let us look at them a little more closely: are these quantities well defined? which inflow are we talking about? at what point in the lake is the level measured? at what time? If we did not respond to these questions, the variables would not be well defined, and if we tried to employ them to define a model, it would in fact be unusable. We must therefore define them with care, and this means that we must make some choices for which we often cannot anticipate the consequences. We have no other possibility but to proceed on a trial and error basis. We choose in the way that seems best, and then return to the decision if it was mistaken. The process of creating a model is a *recursive process*, in which the choices are made and re-made until the model seems logical and coherent.

Returning to our case, we can, for example, establish that the inflow a denotes the total inflow to the lake, produced by the complex of all its tributaries and the hydropower reservoir that releases into the lake. We can establish that the level h is the one measured at the Locarno hydrometric station, read every day at 8 o'clock in the morning. As a consequence, we ought to define the inflow a and the release r as the volumes that enter and exit from the lake in the time interval between 8 o'clock one day (time t) and 8 o'clock the following day (time $t+1$), a time interval denoted then with $[t, t+1)$. Nevertheless, a moment's reflection suggests that we should subdivide the inflow a into the flow ε from the catchment and the flow w released from the hydropower reservoir directly into the lake. The first, in fact, is known only at the end of the day, at time $t + 1$, while the second can be known right away from time t, because it is at that time instant that the Regulator of the hydropower reservoir establishes the production for the day and thus the volume that will be released.

Note that all these operations still concern the setting up of the domain; this indeed, is not usually perfectly formed on the first attempt, but is constructed slowly and with difficulty, through successive iterations. The definitions that we have assumed, and the simplifications that we have introduced (e.g., the decision to adopt the total inflow, instead of considering the flows of the single tributaries separately) must be communicated to the Stakeholders, and understood and shared by them, in such a way that in the following phases their meaning is always clear. For this to be possible, it is also important that the terminology be established in the setting up of the domains. Thus, to conduct this phase of the process it is useful to have the support of an information system, and in fact the domains are the first 'object' to be defined in a MODDS (see Section 24.3.2).

4.1.2.2 Model

When the definitions of all the selected variables from the domain of the lake seem to be well posed, we can proceed with the construction of the model. However, a priori we cannot be completely sure that a definition is well posed: we can only assume that it is, and be ready to change our mind if, in the following phases of creating the model and using it, the need emerges. Like every conceptual structure produced by science, a model (and a domain is an embryo of a model) can only be "falsified" (Popper, 1959).

The first step in the definition of the model is to complete the definition of the symbols that denote the variables: note in fact that they do not yet specify the time that the variable refers to. One can use different notation systems; we believe that the one defined in box on page 102 is very useful and therefore we will adopt it in this book.

In the second step the cause–effect relationships that link the variables must be identified. Once again, the process develops by trial and error, working with the Experts of the component that we want to describe and the affected Stakeholders. When the system is not well known (for didactic purposes we will pretend, for a while, that the lake is not) the best way to proceed is to construct, by trials, a *causal network*. Let us see what this means in practice.

The level h_{t+1} on day $t + 1$ depends on the net volume of water that enters the lake in the interval $[t, t + 1)$, i.e. on the difference between the inflow a_{t+1} and the release r_{t+1} in the same interval, which in its turn is produced by the release decision u_t taken by the

Notation *In the symbol of a variable the subscript denotes the time at which it assumes a deterministic value. For example, since the level of the lake is deterministically known at time t, it is denoted by h_t. On the contrary, the inflow in the interval $[t, t+1)$ is not deterministically known at time t, since at that time it has not yet occurred; it will be known only at time $t+1$. So the inflow is denoted with the symbol a_{t+1}. Before time $t+1$, for example at time t, the inflow a_{t+1} is not a deterministic but an uncertain variable. Note that with this notation the subscript is not only used to define the time instant to which the variable refers, but it also makes it possible to understand its nature (deterministic or uncertain). The latter is not, in fact, an invariable property, but depends on the temporal position (τ) of the observer with respect to the variable: the variable x_t is uncertain if observed from any time instant τ prior to t, and deterministic for any τ equal to or greater than t* (Piccardi and Soncini-Sessa, 1991).

Regulator at time t. In turn, the inflow a_{t+1} is the sum of the outflow ε_{t+1} and the release w_t from the hydropower reservoir. We can represent these cause–effect relationships with the causal network in Figure 4.3, in which the variables are interconnected by oriented arcs, which originate at the cause and point towards the effect.[2]

Let us now observe the network. Can it be considered a good model of reality? To ascertain this we will perform a mental experiment. Let us suppose that we are at a given time t and that the variables acknowledged as causes (a_{t+1} and u_t) have been measured, and let us ask ourselves if these data are sufficient to quantify the effects they induce (r_{t+1} and h_{t+1}). A moment's reflection reveals that the answer is negative. If the Regulator of the lake were to decide to release a very large volume u_t, the volume r_{t+1} actually released would differ according to whether the lake were full or empty (if for example the lake were empty r_{t+1} would be zero). Therefore, r_{t+1} is influenced not only by the decision u_t, but also by the storage s_t of the lake at the time t of the decision. Analogously, we can observe that h_{t+1} is not completely defined once a_{t+1} and r_{t+1} are known, because it also depends on the storage s_t. We must therefore extend the causal network as in Figure 4.4 and subject it to a new mental experiment: now there do not seem to be errors[3] and the network can be

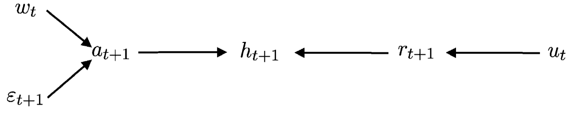

Figure 4.3: The causal network of the lake at the first attempt.

[2]This type of representation guarantees transparency in the construction of the model and aids in creating it with and/or explaining it to the Stakeholders, since it reflects an intuitive cognitive procedure which is inductive and which everyone understands.

When the Stakeholders are not used to the formal language of graphics, the network can be constructed manually with the them by writing the names (not the symbols!) of the variables on pieces of paper and asking them to arrange the pieces of paper so that the causes precede the effects (Hodgson, 1992).

[3]Actually, the reader could rightly observe that it is not in fact satisfactory, since we have neglected the volume of water that evaporates from one day to the next. We will consider this more complex network in Section 5.1. But even this latter could be considered unsatisfactory, because the level at a point on the lake shores, such as Locarno,

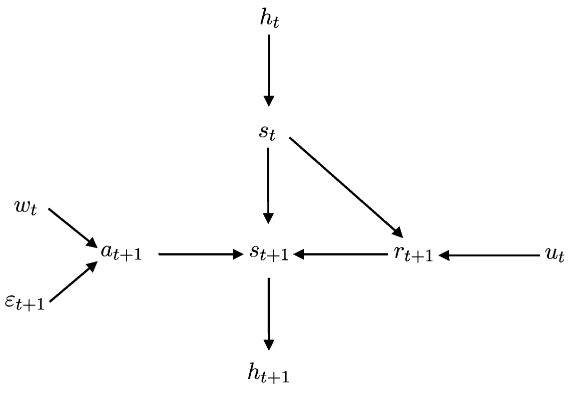

Figure 4.4: The causal network of the lake at the second attempt.

considered to be formally satisfactory. Furthermore, it achieves our aim: to predict the level h_{t+1}. Before using the network a final control is necessary: to verify whether it is acyclic (Jensen, 2001), since for a variable to be the cause of itself is clearly impossible and should never happen.

Let us look at the network more closely: s_{t+1} appears to be the direct cause of h_{t+1} but, if we apply the network at the next time instant, h_{t+1} turns out to be the cause of s_{t+1}. Therefore s and h appear to be in one-to-one correspondence, and in fact, they can be derived from each other if both the bathymetry of the lake and the altimetry of its banks are known. So we can redesign the network as in Figure 4.5. This network is functionally equivalent to the previous one, but it has the advantage that it demonstrates (if one looks at the framed portion) that the core of the dynamic of the lake is the storage. Note that from the storage s_t at time t we can calculate the storage at time $t + 1$ and from this derive the level h_{t+1} that we are looking for. When, as in this case, a variable depends on its value at a previous time, it is called *state* variable, since it describes the state, the present condition of the system, which is determined by the past history and influences the future. In fact, to know the future (the storage s_{t+1} and the level h_{t+1}), it is necessary to know the storage s_t, i.e. the state that has been inherited from the past. The level h_t is the variable that we want to observe and so it is called system *output*. Note that it depends on the storage s_t.

The variables w_t, ε_{t+1} and u_t do not depend upon the other system variables, but they have to be externally provided, and so they are called *exogenous variables*, or *inputs*. The exogenous variables can be of two different types:

1. Some are under our direct control and so they are called *control variables*, or, in short, *controls*. In our case the decision u_t is a control, given that, do not forget, we want to redesign the regulation policy and thus the way to take the decisions u_t. Through the control we want to modify the dynamics of the system in such a way

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would not be equal to the level at the lake's outlet (see Section 6.6.3 in PRACTICE). From these observations we may deduce that a causal network can never be satisfactory in an absolute way, but only relative to the desired accuracy. However, the accuracy of a causal network cannot be established a priori, but only a posteriori, through experiments on the model that is constructed on the basis of it. The decision on when to stop in the development of a network can thus be based only on the Analyst's intuition, and the correctness of his intuition is verified afterwards on the basis of the accuracy of the estimates that the model obtained from it provides.

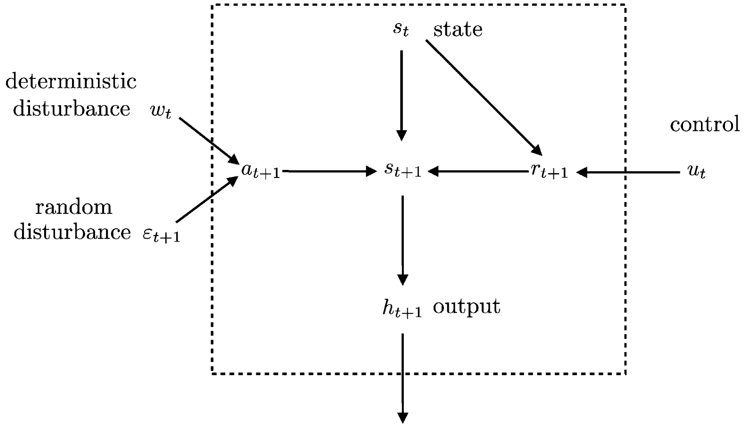

Figure 4.5: The causal network of the lake in the definitive version.

that the output trajectory would be more satisfactory from the Stakeholders' point of view. When the control is established by a regulation policy the system is said to be *controlled*.

2. Other inputs, such as the natural outflow ε_{t+1} and the release w_t from the hydropower reservoir, cannot be established by us and so they are called *disturbances*. We experience their values and above all their effects. In their turn disturbances can be:

   - *deterministic*, because their value is known at time t when the Decision Maker (DM) must decide the control, e.g. the hydropower release w_t;

   - *random*,[4] because they occur during the time interval $[t, t+1)$ and so they are not known at time t. An example is the natural outflow ε_{t+1}.

There are finally other variables, such as the total inflow a_{t+1} and the release r_{t+1}, that do not fall into any of these categories. These are called *internal variables*, because they serve only to retain, during the calculations, a partial result to be used afterwards. For example, the release r_{t+1} is calculated only as a step towards determining the state s_{t+1}.

Note that a single variable can be classified in different ways according to the aim of the model, with the exception of the state variables, which remain as such regardless of the aim. For example, if the aim had been to study the effects of the regulation of the

[4]In this chapter the attribute 'random' is understood as a synonym of non-deterministic. Stochastic and uncertain, which in common language are also its synonyms, will be used, instead, to denote two different types of randomness, as will be explained in Section 5.7.

In literature one does not usually encounter deterministic disturbances, and so the term disturbance is commonly given the meaning 'random disturbance'. The reason for this is that in artificial systems, which most of the literature deals with, all the inputs that are known at time t are by nature controllable and controlled by a single DM. On the contrary, in environmental systems these inputs can be in the hands of different DMs and so, from any one DM's viewpoint, the controls managed by the others are not controls. They are not, however, random disturbances, because each DM can always ask the others to communicate the values of their controls, once they have been decided. To describe this particular type of input variables we have created the class of deterministic disturbances.

lake on the downstream water users, the output variable would have been the release r_{t+1}, and the level h_t would not even have appeared in the causal network. If, instead, the aim had been to evaluate the effects of the regulation of the hydropower reservoir, the control variable would have been w_t, and u_t would have assumed the role of a deterministic disturbance.

Generally, the control u_t cannot assume any value, but only those that are feasible at that time, given the condition, i.e. the state, of the system. For example, in the case of the lake, it is not possible to decide any water release if the storage is empty.

It is usual to represent all this information with two equations and one constraint:

1. The first equation expresses the future state as a function of the present state and the inputs

$$s_{t+1} = f(s_t, u_t, w_t, \varepsilon_{t+1}) \tag{4.3}$$

   and the function $f(\cdot)$ which defines it is called, for obvious reasons, *state transition function*. The constraint

$$u_t \in \mathcal{U}(s_t)$$

   that defines the feasible controls as a function of the storage is associated to it.

2. The second equation expresses the output as a function of the state

$$h_t = h(s_t) \tag{4.4}$$

   and the function $h(\cdot)$ which defines it is called *output transformation function*.

It is important to note that the forms of the functions $f(\cdot)$ and $h(\cdot)$ are not yet known: their identification, either explicit or implicit, is the aim of the next step in the modelling process. All that we know for now is that the network in Figure 4.5 assures us that these functions exist and that their arguments are those indicated. Note lastly that these two equations do not encompass all the information contained in the network in Figure 4.5: there is more information, since the network also reveals the existence of internal variables, such as the release r_{t+1}, and the cause–effect relations that link these with the other variables.

4.1.2.3 General structure of a model

What we have understood from this simple example has a general validity. Any model, regardless of its mathematical formulation, is described by variables that can be classified as state ($\mathbf{x}_t$), output ($\mathbf{y}_t$) and input variables. These last, in their turn, are subdivided into control variables (actual controls $\mathbf{u}_t$, and planning decisions $\mathbf{u}^p$ that are controls whose values do not change in time) and disturbances, which are subdivided into deterministic $\mathbf{w}_t$

and random $\boldsymbol{\varepsilon}_{t+1}$. Since, generally, the variables that belong to each of these categories can be more than one, it is useful to interpret these symbols as vectors and this is why they are in bold type. Their dynamics are described by the two equations that we have already encountered (Kalman et al., 1969):

1. The *state equation*

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \tag{4.5a}$$

   defined by the *state transition function*, to which the following conditions are associated

$$\mathbf{u}^p \in \mathcal{U}^p \tag{4.5b}$$

$$\mathbf{u}_t \in \mathcal{U}_t\left(\mathbf{x}_t, \mathbf{u}^p\right) \tag{4.5c}$$

   which define the planning decisions and controls feasible at time t.

2. The *output transformation*, defined by the *output transformation function*,[5] which assumes the following form

$$\mathbf{y}_t = h_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t\right) \tag{4.5d}$$

   when the random disturbance $\boldsymbol{\varepsilon}_{t+1}$ is not among its arguments, while, in the opposite case, it becomes

$$\mathbf{y}_{t+1} = h_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \tag{4.5e}$$

   Note that in this last case the subscript of $\mathbf{y}$ is $t+1$: when the random disturbance directly influences the output, this is deterministically known only at time $t+1$ and so, with the notation we have adopted, the output must be denoted with $\mathbf{y}_{t+1}$.

The forms of the two functions $f(\cdot)$ and $h(\cdot)$ are more general than those we encountered in the lake example, not only because of the two alternative expressions for the output transformation, but also because they can be time-varying, in which case the model is said to be *time-varying*, and because the output transformation can depend not only on the state, but also on the input. When this transformation does not depend on the input, the model is said to be *proper*,[6] while it is said to be *improper* in the opposite case.

Sometimes the output can coincide with the state

$$\mathbf{y}_t = \mathbf{x}_t$$

given that nothing prohibits equation (4.5d) from being an identity. Or, some of the output components may coincide, up to a factor of scale, with some of the state components: this occurs for example when the component is a canal (see equation (5.19) in Chapter 5).

Equation (4.5) defines a *dynamical system*[7] that communicates with the outside world only through its inputs and outputs: everything which occurs inside the system is not, by definition, interesting if it does not influence the output. A system is therefore an 'object', in

[5] As is common practice we have denoted that function with $h_t(\cdot)$, but it should not be confused with the symbol h_t, which represents the level.

[6] The reason for the term is explained in Appendix A3 on the CD.

[7] In modelling the terms system and model are often used as synonyms: in the end the real system is not directly knowable.

the programming language sense, as we will see in Section 24.2. Not always is it necessary to introduce state variables to describe the dynamics of the system variables. For example (Section 5.4), a diversion is described only by input variables (the incoming volume ε_{t+1} and the diversion decision u_t) and output variables (the volume y_{t+1} diverted). The dynamics of the second is completely specified by an output transformation function

$$y_{t+1} = h_t(u_t, \varepsilon_{t+1})$$

Systems of this type are called *non-dynamical*.

Note the fact that the term 'dynamical system' has exactly the meaning given above and it does not mean 'a system whose variables change their value through time', given that this property also belongs to non-dynamical systems. Neither does the term 'time variable system' have that meaning, given that, as we have already seen, the term denotes a system whose state and output transformations depend explicitly on time. We can now clarify the precise meaning of the term *simulation*, which we have used until now, trusting that its meaning was intuitive. Simulation is defined as determining the trajectories followed by the state and the output of the system starting from a given initial state and in response to a given sequence of inputs. More precisely, given a (*time*) *horizon* H, which begins with an initial time instant conventionally called 0, the state $\mathbf{x}_0$ at that time (*initial state*) and the *input trajectories* ($\{\mathbf{w}_t\}_{t\in H}$, $\{\boldsymbol{\varepsilon}_{t+1}\}_{t\in H}$ and $\{\mathbf{u}_t\}_{t\in H}$) for the entire horizon, the aim of the simulation is to determine the trajectories $\{\mathbf{x}_t\}_{t\in H}$ and $\{\mathbf{y}_t\}_{t\in H}$ of state and output. When equation (4.5) is quantitatively defined, the state trajectory can be calculated by recursively solving equation (4.5a) from time 0 up to the end of horizon H. At each time step (*simulation step*) the variables that appear in its right-hand member are known, so that it is easy to obtain the value $\mathbf{x}_{t+1}$ of the future state. This value is then transformed into the corresponding output, through the output transformation (4.5d) (or (4.5e)), in order to obtain the output trajectory. In the next step, the same value $\mathbf{x}_{t+1}$ feeds the right-hand member of equation (4.5a).

4.1.2.4 Conclusion

We have seen how, starting from a *domain*, one proceed to identify a *mental model* of the component that is expressed through a *causal network*. From the structure of the network we then deduced the general forms of the state transition and output transformation functions. In this way we arrived at the frontier of quantitative modelling. The next step will be to define the two functions, either implicitly or explicitly.

A model can be classified according to two alternative frameworks. The first, more traditional, considers the mathematical characteristics of the two functions that define a model and the properties that follow. The second considers the quantity of a priori information[8] about the internal processes of the component to be modelled, which is used in identifying the two functions. In the following section we will follow this second framework, because it is better suited to the aim of this book. Nevertheless, since it is not possible to ignore the knowledge of the mathematical structures with which models are built, they are presented, very succinctly, in Appendix A3 on the CD.

In the following presentation, we will continue to use the case of the lake as an example, but since the aim of the chapter is to introduce methodological ideas, we will simplify, quite drastically, the description of the processes concerned, to reduce the risk that the reader

[8] Deduced from scientific disciplines, such as Hydrology in the case of a catchment, or from the experience of Experts.

might have difficulty following discussion. We will do the same with the other examples that it will be necessary to introduce. The models developed therefore contain imprecisions, which will be readdressed in Chapter 5, when we will deal with the effective development of the models.

Now, it remains only to clarify a linguistic point. As is usual practice, in the following sections, the term *system* will be used to refer to the component for which one wants to develop a model. The term has, in fact, a broad meaning and is used to denote the subject to which attention is paid time after time. Depending on the context, it might refer to the whole system that is the object of the Project, as well as one of its components. Correspondingly, also the meaning of 'component' shifts. When the system denotes a component, the lake in our example, the term component is used with reference to one of its parts, as for example the lake's outlet. Further, as was already underlined in a preceding note, the term 'system' is very often used as a synonym of model, because one tends to confuse the real system with its representation.

4.2 Bayesian Belief Networks

We now have a causal network that describes the system for which we want to create a quantitative model. To obtain this model we have to transform the qualitative cause–effect relationships that appear in the network into quantitative relationships. In some cases there are theories, or empirical knowledge is available, which suggest the form of the relationships we are looking for, while in other cases we do not have any information that helps us. The way to formalize the relationships is different in the two cases. In this section we will deal with the second, while we will deal with the first in the following section. The lake falls into the first category, given that one can draw on support from many scientific disciplines, such as Hydraulics and Hydrology, to construct the model. Nevertheless, for didactic purposes, we will ignore them, so that we can continue to use the same example in this section.

If no theory can help us, the only information that we can hope to obtain is observations of a statistical nature such as: "thirty per cent of the times that the inflow and the storage are high, and the release scarce, there is a flood the following day". These observations can be either subjective evaluations, provided by Experts or Stakeholders, or quantitative evaluations, derived from counting conditional frequencies in time series of recorded data. These time series, at least in a subjective form, are always available, and it is thanks to them that the causal network is constructed.

To be able to use them in the construction of a model it is first necessary to discretize[9] all the variables that make up the causal network, assuming that each one of them can only assume values in a finite and discrete set, e.g. the set of values that appear in the observations. Table 4.1 shows an example of these sets[10] for the network in Figure 4.5. Once the discrete sets have been fixed, we can quantify the cause–effect relationships, by associating a table $\phi(z|w_1, \ldots, w_r)$ to each variable z, which is influenced by others (and therefore to each of the variables in the network, except for the inputs). The table expresses the conditional probability of z, given the values of the variables $w_1, \ldots, w_r$ that influence it.[11] For example, the release r_{t+1} in the network in Figure 4.5 is described by the table (c)

[9] We will deal with the discretization of the variables in Section 10.5.

[10] Clearly we are dealing with didactic discretization. In a real model the number of values considered for each variable varies from several scores to several hundreds.

[11] Such probability may assume values different from zero and one only when z is affected by disturbances that do not explicitly appear among the entry variables $w_1, \ldots, w_r$.

Table 4.1. The discretization of the variables that appear in the network in Figure 4.6

Variable	Discretization	
w_t	*S*carce	*A*bundant
ε_{t+1}	*S*carce	*A*bundant
a_{t+1}	*S*carce	*A*bundant
s_t, s_{t+1}	*S*carce	*A*bundant
h_t	*L*ow	*H*igh
u_t	*S*carce	*A*bundant
r_{t+1}	*S*carce	*A*bundant

in Figure 4.6, which has the form $\phi(r_{t+1}|s_t, u_t)$. Each element in this table expresses the probability that the release assumes the value r_{t+1}, which is associated to the row to which the element belongs, when the storage in the lake assumes the value s_t and the decision assumes the value u_t which appear in the corresponding column. For example, 0.8 is the probability that the release is abundant (A) when the storage is abundant (A) and the release decision scarce (S). In such conditions, 80% of the time, the release is abundant because the Regulator opens the spillways, but for 20% of the time this is not necessary because evaporation (a disturbance that is not described by the model, see footnote 3 on page 102) is high and therefore the release is scarce. Tables of this type are called *conditional probability tables* and are denoted with the acronym CPT.

When a CPT has been associated to each variable in the network, except the inputs, as in Figure 4.6, we have obtained a quantitative model of the system that is called a *Bayesian Belief Network* (BBN) (Pearl, 1988; Jensen, 1996, 2001). The attribute 'Bayesian' derives

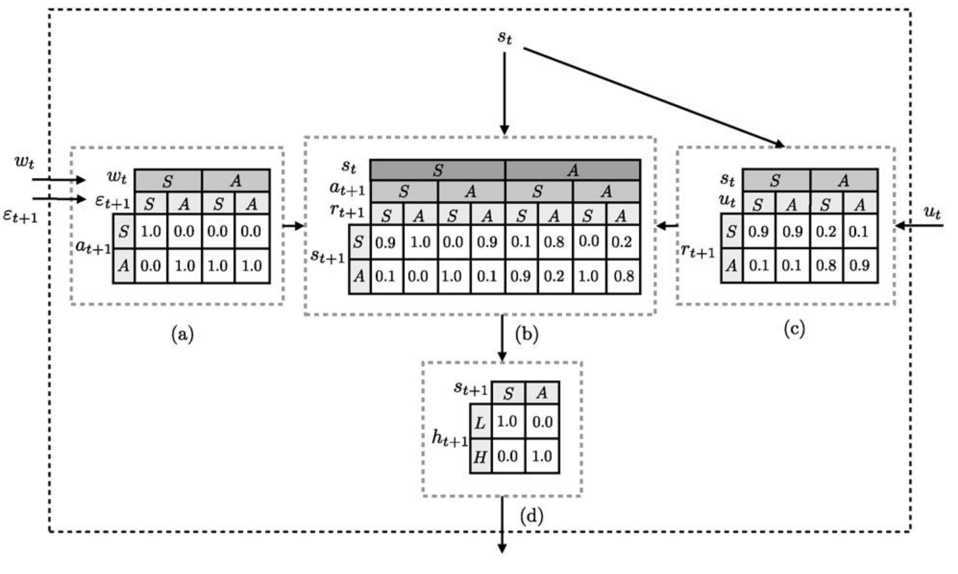

Figure 4.6: The BBN that describes the lake.

s_t	S								A							
w_t	S				A				S				A			
ε_{t+1}	S		A		S		A		S		A		S		A	
u_t	S	A	S	A	S	A	S	A	S	A	S	A	S	A	S	A
s_{t+1} S	0.9	0.9	0.1	0.1	0.1	0.1	0.1	0.1	0.7	0.7	0.2	0.2	0.2	0.2	0.2	0.2
s_{t+1} A	0.1	0.1	0.9	0.9	0.9	0.9	0.9	0.9	0.3	0.3	0.8	0.8	0.8	0.8	0.8	0.8

Figure 4.7: The CPT that expresses the state transition function of the lake.

from the fact that the use of the network requires the Bayes theorem, while the term 'belief', which will be generally omitted, reminds us that the observations with which the CPTs are filled are for the most part subjective.

The CPT that expresses the conditional probability $\phi(h_t|s_t)$ of the level h_t with respect to the storage s_t is the output transformation $h_t(\cdot)$ that we are looking for. This statement may surprise the reader, since the conditional probability $\phi(h_t|s_t)$ might not seem to describe the one-to-one level–storage relationship that one would expect after the remark made in the previous section. Note, however, that a CPT can express also a deterministic relationship. For this it is sufficient that in each of its columns only one non-zero element appears and that this element assumes a value equal to one.[12] That is what actually happens in the CPT that expresses the output transformation in our example (see table (d) in Figure 4.6). However, the output transformation function does not have to be a deterministic function; indeed, equation (4.5e) affirms that the output can also depend on random disturbances, as occurs for example when the level of the lake is affected by an error of measurement, so that the relationship between h and s becomes non-deterministic.

Now let us consider the three CPTs that express the variables a_{t+1}, s_{t+1} and r_{t+1} as a function of the variables that condition them. They can be properly concatenated[13] in such a way as to obtain the conditional probability $\phi_t(s_{t+1}|s_t, u_t, w_t, \varepsilon_{t+1})$, which is quantitatively expressed by the CPT in Figure 4.7. This is actually the state transition function $f(\cdot)$ that we were looking for.

In conclusion, a BBN implicitly defines the state transition and the output transformation functions of the model. These can be obtained in an explicit form by concatenating the CPT in such a way as to obtain two tables that have the arguments of the desired functions as input (entry) variables.

BBNs are very useful for representing systems, such as social systems, for which we do not have quantitative theories. We will see an example in Section 5.6.2. They are, however, not very well suited to representing systems, such as the lake, in which there are known deterministic relationships and/or the number of values that the each variable can assume can be high. An example of the first case is the laborious description of a simple arithmetic operation such as the sum of w_t and ε_{t+1}, which is expressed with the CPT (a) in Figure 4.6. To understand the difficulties that emerge in the second case, think of the model of the lake adopted in the Verbano Project. The number of discrete values considered for each of the variables s_t, u_t, w_t and ε_{t+1} is 120, 150, 25 and 25, respectively (Section 7.8 in PRACTICE);

[12]The sum of the elements in a column of a CPT is always equal one, because it is assumed that for each combination of the conditioning (entry) variables, the conditioned one cannot assume a value which is not included among the values considered, and that one and only one value will certainly occur.

[13]Concatenation is performed by means of probability calculus (Mood et al., 1974).

it follows that if that model were represented in the form of a BBN the CPT that defines the state transition function would have 1 350 000 000 elements ($120 \times 120 \times 150 \times 25 \times 25$).

4.3 Mechanistic models

When a priori knowledge about the system to be modelled is available, i.e. when there are theories explaining its internal *mechanisms* in a quantitative way, it is rational to use them in formulating a quantitative expression of the cause–effect relationships that appear in the causal network.

For example, in the case of the lake, Physics suggests the relationship that links tomorrow's storage s_{t+1} with today's storage s_t, with the outflow ε_{t+1} from the catchment, with the release w_t from the hydropower reservoir, and with the release r_{t+1} from the lake: it is the mass conservation equation

$$s_{t+1} = s_t + w_t + \varepsilon_{t+1} - r_{t+1} \tag{4.6}$$

Hydraulics teaches us that, when the dam is completely open, the release r_{t+1} is linked to the storage s_t by a functional relationship (Figure 4.8)

$$r_{t+1} = N(s_t) \tag{4.7}$$

called *storage–discharge relation in free regime*, which generally assumes the following form

$$N(s_t) = \alpha s_t^{\beta} \tag{4.8}$$

where α and β are two positive parameters. To regulate the release, the Regulator operates the dam gates so that the release is equal to the decision u_t whenever it is physically possible and the legal constraints permit it. More precisely, the Regulator cannot exceed the maximum release (expressed by the storage–discharge relation) and he must completely open the dam when the storage of the lake gets to the value $\bar{s}$, established for precautionary reasons, in the Regulation Licence.[14] In conclusion, the relationship that expresses the release as a

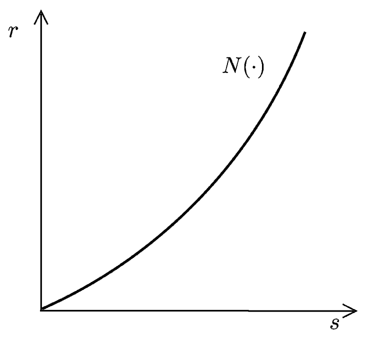

Figure 4.8: The storage–discharge relation of the lake.

[14] That is the concession act for regulation that states the range within which it can be applied.

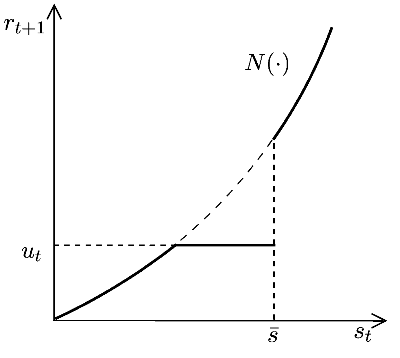

Figure 4.9: The relationship that expresses r_{t+1} as a function of u_t and s_t.

function of the storage s_t and of the decision u_t is the following

$$r_{t+1} = \begin{cases} N(s_t) & \text{if } s_t > \bar{s} \\ \begin{cases} N(s_t) & \text{if } u_t > N(s_t) \\ u_t & \text{otherwise} \end{cases} & \text{otherwise} \end{cases} \tag{4.9}$$

which is represented in Figure 4.9.

Finally, the relationship that links the level to the storage is derivable from knowledge about the bathymetry of the lake and the altimetry of its banks

$$h_t = h(s_t) \tag{4.10}$$

For example, if we knew that the lake had very steep banks, we could assume that it were a cylinder, from which follows that

$$h_t = s_t/A \tag{4.11}$$

where A is the area of the lake's surface. Note how in all these cases a priori information is incorporated in the model.

The output transformation (4.5d) of the model is given by (4.10), while the state equation (4.5a) is represented by (4.6), when the release r_{t+1} that there appears is expressed by (4.9). Thus we have obtained the model we were seeking. Given the way it is derived, this type of model is called *mechanistic* or *conceptual*, since it is based on the conceptualization of the processes, or of the 'mechanisms' that operate in the system.

4.3.1 Stochastic models

The reader might notice that even if a BBN and a mechanistic model are identified with two different procedures, they should provide two equivalent representations if they describe the same system. However, this would not seem to be the case for our lake. In fact, we have seen that the BBN in Figure 4.6 indirectly defines a stochastic state transition function, while the mechanistic model defined by equations (4.6), (4.9) and (4.10) is deterministic, once the disturbances w_t and ε_{t+1} are known.

This observation is correct and the difference between the two models is due to the fact that we have defined a *deterministic mechanistic model*. This in fact is the most common practice, but not the only one possible, nor the most correct. Let us re-examine the previous relationships. There is no doubt that the mass conservation law, established by Physics, has a general validity, but equation (4.6) that we have derived from it implicitly assumes that the storage and inflow measures are perfect. If they were affected by errors, as is most probably the case, equation (4.10) would not be exactly verified and would have to be replaced by the following

$$h_t = h(s_t) + v_{t+1}^m \tag{4.12}$$

where v_{t+1}^m is a random disturbance that expresses an error in the measurement of the level and is thus called *output noise*.[15] Analogously the reader who is well versed in Hydraulics may have noticed that by adopting the storage–discharge relation we have committed an additional error. Indeed, Hydraulics affirms that such a relation exists only in steady-state conditions, i.e. when the storage does not vary in time, while our model certainly could not operate exclusively in that condition, because it would be completely useless. In unsteady state conditions, i.e. when the storage varies in time, the value given by the storage–discharge relation is not exact and the actual release differs from that value by a quantity that, as a first approximation, we can assume is a random disturbance. So, equation (4.6) has to be replaced by the following

$$s_{t+1} = s_t + w_t + \varepsilon_{t+1} - r_{t+1} + v_{t+1}^p \tag{4.13}$$

where the disturbance v_{t+1}^p in it is called *process noise*, because it represents the error that affects the description of the process.[16]

The *mechanistic model* described by equations (4.13), (4.9) and (4.12) is a *stochastic model*, because random disturbances appear in its equations. Its state transition function is therefore qualitatively analogous to the one provided by a BBN.

From the above we can make two important considerations. First of all, the BBN includes the effect of the output and process errors in an implicit way, while in a mechanistic model they must be introduced explicitly. Secondly, it appears that, as we have already noticed for the structure of a causal network, the structure of a model is never satisfactory in an absolute way. The deterministic mechanistic model defined by equations (4.6), (4.9) and (4.10) is not 'mistaken', just conceptually less satisfactory than the stochastic model defined by equations (4.13), (4.9) and (4.12); which, in its turn, is less satisfactory than another that describes the storage–discharge relationship in a way which fits more closely the Laws of Hydraulics. Note that we have used the expression "conceptually less satisfactory" and not less reliable or less precise. The reason is that precision and reliability do not depend solely on the structure of the model, but also on how it is calibrated (see Section 4.9.2) and so also on the value of the parameters that specify the models of the output and process noises (see Section 4.6). If these parameters were badly calibrated, the deterministic model might be more precise and reliable than the stochastic one.

4.4 Empirical models

An idea is born out of all this. Would it be possible to identify the functions of the model directly, as pure mathematical expressions, without going through the analysis of the chain of causal relationships that generate them?

[15] In Information Theory jargon, the term 'noise' is used as a synonym for 'error'. This use derives from the field of telephonic communications: on a telephone line the disturbances take the form of noise.

[16] In modelling the term 'process' is used to denote the 'mechanism' through which one variable influences another. By extension it can assume a meaning close to 'system'.

In many cases the construction of a BBN, or of a mechanistic model, proves to be an operation that is too costly for the goal we are trying to achieve. Maybe the clearest case is the catchment: the processes that take place within it and that transform precipitation into outflow are very complex and so the BBNs or the mechanistic models that describe them will be equally complex (see Appendix A5 on the CD). When the alternative being evaluated does not directly affect the catchment, as in the lake example, one may be more interested in a model that produces precise and reliable outflow estimates (output) in response to precipitation measurements (input), rather than an accurate and realistic description of the processes that take place in the catchment. In these cases it is appropriate to simply identify a model that describes the relationship between the input and the output of the system (*input–output relationship*), without worrying about explaining the internal mechanism. In other words we could simply describe the dynamics of the output through a relationship such as the following

$$\begin{aligned} \mathbf{y}_{t+1} = \mathsf{y}_t\big(&\mathbf{y}_t, \ldots, \mathbf{y}_{t-(p-1)}, \mathbf{u}_t, \ldots, \mathbf{u}_{t-(r'-1)}, \mathbf{w}_t, \\ &\ldots, \mathbf{w}_{t-(r''-1)}, \boldsymbol{\varepsilon}_{t+1}, \ldots, \boldsymbol{\varepsilon}_{t-(q-1)}\big) \end{aligned} \tag{4.14}$$

which explains the output $\mathbf{y}_{t+1}$ at time $t+1$ on the basis of the values that input and output assumed in a suitable number of preceding time steps. Equation (4.14) is called *input–output form*[17] (or *external form*), while in contrast the pair of equations (4.5a)–(4.5d) is called *state-space form* (or *internal form*). However, a function $\mathsf{y}_t(\cdot)$ able to describe the very catchment being considered does not always necessarily exist for finite values of the parameters p, r', r'' and q.

It is certain that the input–output form exists whenever the state-space form can be manipulated in such a way as to obtain it. For example, it is easy to see that in the case of the lake there is the following representation

$$h_{t+1} = \mathsf{y}(h_t, u_t, w_t, \varepsilon_{t+1}) \tag{4.15}$$

both because it can be obtained by manipulating equations (4.6), (4.9) and (4.10), and, more simply, because the causal network in Figure 4.4[18] shows it. Systems Theory helps us by providing the necessary and sufficient conditions so that a model of a given class has an input–output form. Nevertheless, both the first consideration and the conditions provided by the Theory are of no practical use, since they require that we know the mechanistic model, or at least the class that it belongs to. In other words, they require that we know exactly what we do not know and are trying to avoid the identification of.

Thus, in practice, one assumes 'empirically' that the input–output form (4.14) exists, fixes a priori the values of the parameters (p, r', r'', q), which define its *order*, and attempts to individuate it with the techniques that we will present in a while. If the operation does not succeed there are two possibilities: either the input–output form does not exist, or the order is mistaken. One tries again with a new quadruple (p, r', r'', q) in which the sum of the four parameters is greater than it was in the preceding attempt, i.e. one tries with a *higher order*, as is said in jargon. One iterates this procedure until the sought relationship is identified, or until such a high order is reached that the model, even if it were to exist in correspondence with superior orders, would be useless in practice, for reasons that will be presented in Section 12.2.1.

[17] Instead of 'form' the term 'representation' is sometime used.

[18] It shows, in fact, that it is possible to calculate h_{t+1} given h_t.

This way of proceeding is founded on the idea that the input–output form is completely sufficient for our aim. We do not try to understand how the system works (scientific aim), but only to predict the behaviour of its output in response to a given input, with the aim of identifying the actions that can make its behaviour more satisfactory from the Stakeholders' point of view (engineering aim). To achieve this aim it is sufficient that the model correctly reproduces the behaviour of the output, but it is not necessary, nor does one expect, that it reproduces the system's internal processes. That is the reason why empirical models are also called *black-box models*.

Since one does not have any information about the structure of the system, the input–output form has to be fixed a priori in a class of functions, such that a particular function can be identified by assigning values to a finite (and small) number of parameters. When all the variables are scalar, one often assumes that equation (4.14) is a linear relationship

$$\begin{aligned} y_{t+1} = {} & \alpha_t^1 y_t + \cdots + \alpha_t^p y_{t-(p-1)} + \\ & + \beta_t^{1'} u_t + \cdots + \beta_t^{r'} u_{t-(r'-1)} + \\ & + \beta_t^{1''} w_t + \cdots + \beta_t^{r''} w_{t-(r''-1)} + \\ & + \varepsilon_{t+1} + \gamma_t^1 \varepsilon_t + \cdots + \gamma_t^q \varepsilon_{t-(q-1)} \end{aligned} \tag{4.16}$$

because, under that hypothesis, as we will see in Section 4.9, there are powerful algorithms to estimate the values of the parameters $\alpha_t^1, \ldots, \alpha_t^p, \beta_t^{1'}, \ldots, \beta_t^{r'}, \beta_t^{1''}, \ldots, \beta_t^{r''}, \gamma_t^1, \ldots, \gamma_t^q$. This class of models is denoted with the acronym PARMAX and is described in Appendix A6. For example, in the case of the lake we can assume

$$h_{t+1} = \alpha h_t - \beta u_t + \gamma (w_t + \varepsilon_{t+1}) \tag{4.17}$$

The linear relationship is not necessarily the most suitable, however. In the example, it would not give good results, since the strong non-linearity of equation (4.9) means that the relationship between h_{t+1} and u_t would be in its turn strongly non-linear. Since, in general, we cannot know when this happens, we ought always to assume, a priori, relationships that can also be non-linear. Until a few years ago, the adoption of non-linear relationships would have required computing times too long for the estimation of their parameters and so they were not used, and attention was concentrated only on the PARMAX models. Today, very flexible relationships are known (i.e. relationships that can represent a broad class of forms), for which powerful algorithms for the estimation of the parameters are available. They are known as *Neural Networks*. Some notions about their structure are offered in Appendix A8. So, it is better to assume a neural network to represent the function $\mathsf{y}(\cdot)$ in (4.15).

The two empirical models of the lake that we have considered are deterministic, as they produce a deterministic output value when the input values are deterministically known. Nevertheless, given that we cannot have any guarantee that the true model of the lake belongs to the class that we are considering, it is advisable to assume also a random *noise*[19] e_{t+1} among the inputs, as we have already done in equation (4.13). With this hypothesis equation (4.17) assumes the following form

$$h_{t+1} = \alpha h_t - \beta u_t + \gamma (w_t + \varepsilon_{t+1}) + e_{t+1} \tag{4.18}$$

[19] Since the internal structure of the model is no longer defined, it is now impossible to distinguish between process noise and measurement noise.

or more generally, the following

$$h_{t+1} = \alpha h_t - \beta u_t + \gamma(w_t + \varepsilon_{t+1}) + e_{t+1} + \delta_t^1 e_t + \cdots + \delta_t^q e_{t-(q-1)} \tag{4.19}$$

if we think that the noise dynamics is more complex. The model now contains two random disturbances.

This idea can be applied in general and so equation (4.14) should be rewritten as

$$\begin{aligned}\mathbf{y}_{t+1} = \mathrm{y}_t\big(&\mathbf{y}_t, \ldots, \mathbf{y}_{t-(p-1)}, \mathbf{u}_t, \ldots, \mathbf{u}_{t-(r'-1)}, \mathbf{w}_t, \\ &\ldots, \mathbf{w}_{t-(r''-1)}, \boldsymbol{\varepsilon}_{t+1}, \ldots, \boldsymbol{\varepsilon}_{t-(q'-1)}, \mathbf{e}_t, \ldots, \mathbf{e}_{t-(q''-1)}\big) + \mathbf{e}_{t+1}\end{aligned} \tag{4.20}$$

The input–output form can be identified only if time series of the input and output variables are both available and long enough to allow an estimation of the parameters with the techniques that we will mention in Section 4.9. It is because the parameters are identified from historical data that the empirical models cannot, and should not, ever be used when the actions being considered could modify the internal structure of the system, and therefore the input–output relationship. For example, in the case of the lake, this happens when a modification of the lake's outlet is included among the considered actions.

Finally, we ought to underline the fact that the representation capacity of an empirical model strongly depends on the class of functions that have been adopted to identify it.

4.5 Data-Based Mechanistic models

One of the drawbacks of mechanistic models is that they are too complicated and they often describe particulars that are discovered a posteriori to be irrelevant for the relationship between input and output. The drawback of empirical models is that their class must be fixed a priori and the result strongly depends on this blind choice. Recently, a new model identification approach was proposed (Young and Beven, 1994; Young, 1998) to overcome these two obstacles, and it is proving to be very promising. The approach's founding idea is to use a mechanistic model, whose form, however, is not to be guessed on the basis of a priori knowledge, but directly inferred from the data. By identifying the form of the model in this way, only the relevant elements of the input–output relationship are identified. This approach is called Data-Based Mechanistic (DBM) approach.

As one can guess from the fact that DBM modelling is the most recent approach that has been proposed, the mathematics that supports it is much more complex than that of the other approaches, and so we cannot describe it here. Even in Appendix A7 on the CD, which is dedicated to it, we simply give a feeling of the essential ideas. We can, however, show an example of the results that it allows us to obtain.

In Figure 4.10 the series of the flow (y_t) from February 1985 to January 1987 is shown for a section of the Canning River in Western Australia along with the corresponding series of average rainfall (w_t) in the previous 24 hours in the catchment upstream of that section. A quick look shows that the rainfall–flow relationship is strongly non-linear, since the extreme rainfall event at the end of 1985 does not produce any flow. A PARMAX model, as for example the following

$$y_{t+1} = \alpha y_t + \beta w_t + \varepsilon_{t+1} \tag{4.21}$$

cannot therefore give good results, as Figure 4.11 shows. Even linear models of a higher order do not provide appreciable improvements.

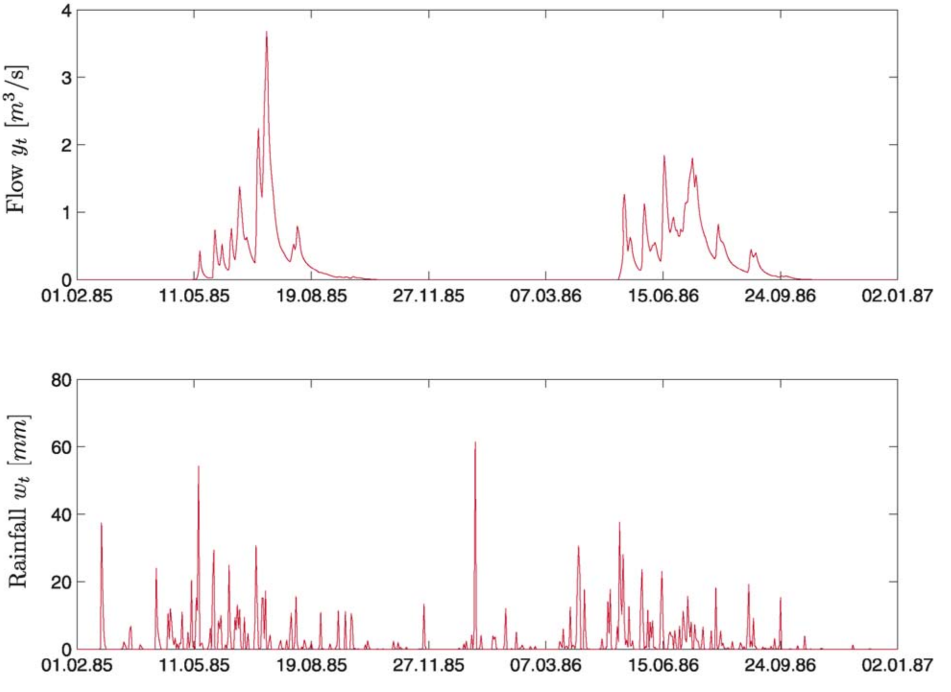

Figure 4.10: The historical flow and rainfall series for the Canning river.

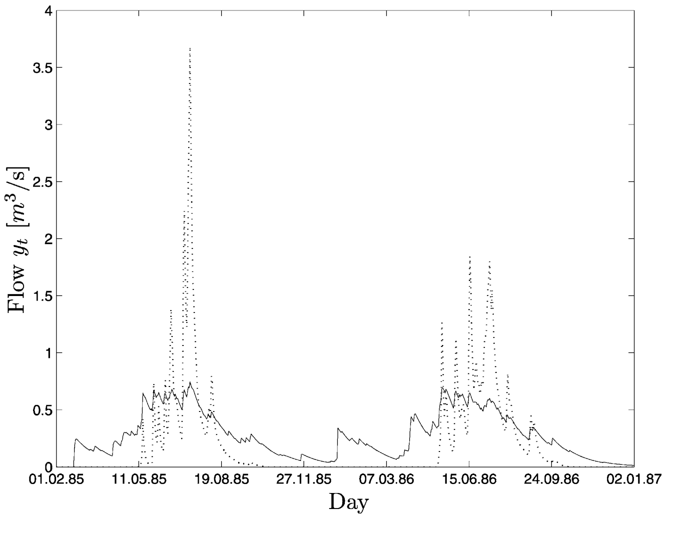

Figure 4.11: The recorded flow (dashed line) and the one calculated with the PARMAX model (4.21) (continuous line).

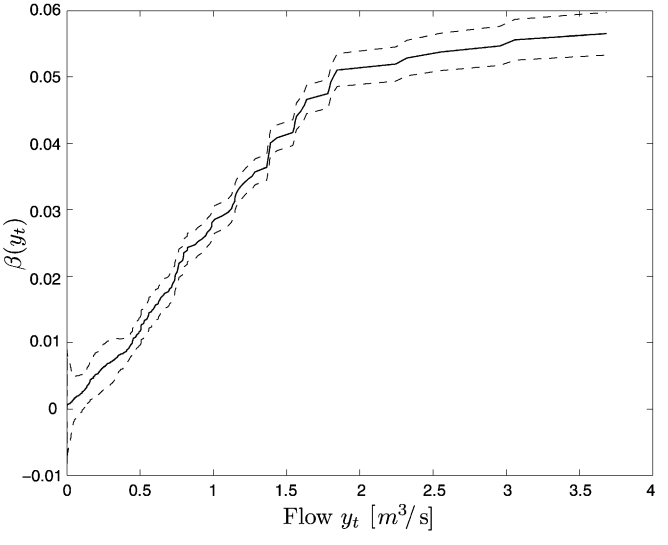

Figure 4.12: The parameter $\beta(y_t)$ as a function of the flow y_t. The dashed lines delimit the confidence interval.

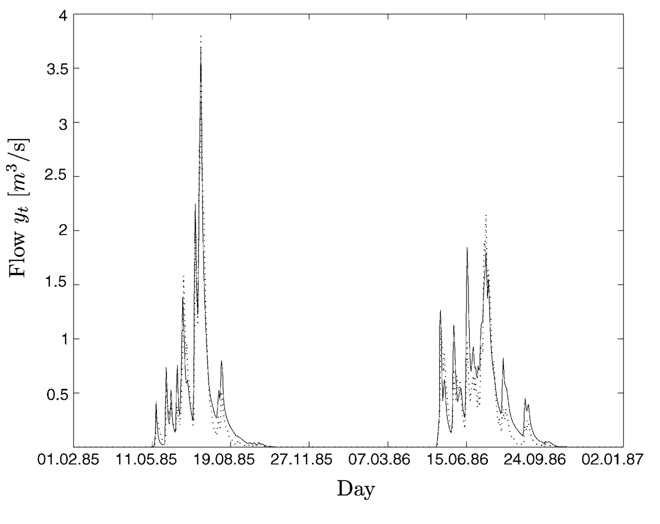

Figure 4.13: The flow produced by the model (4.22) (continuous line) and the recorded flow (dashed line).

The DBM approach shows that, instead, the model must have the following form

$$y_{t+1} = \alpha y_t + \beta(y_t)w_t + \varepsilon_{t+1} \tag{4.22}$$

where $\beta(y_t)$ is a parameter whose value depends on the condition of the system (*State-Dependent Parameter*, SDP), which can be related to the output values (as in this case) or the input values. The shape of the SDP is identified from the two data series through ad hoc algorithms. In the Canning case (Figure 4.12) its form shows that the contribution of rainfall to flow decreases as the flow decreases, and this fact can easily be interpreted: when the ground is very dry, as it is in summer (remember that the Australian summer is from December to March), the rain is absorbed by the ground and its contribution to the flow is slight, but as the rains gradually soak the ground its contribution gradually increases. We do not have a direct index for the humidity of the ground, but the inflow is an excellent indirect index, as is proved by the result shown in Figure 4.13.

The DBM approach is particularly useful also for another reason. As we will see (Sections 12.2.1 and 19.2) the *Designing Alternatives* phase requires the use of 'simple' models, i.e. models whose state is of a small dimension. These models are often called *screening models*. Instead, the *Estimating Effects* phase does not pose this limit and more complex models (*evaluation models*) can be adopted. When this is the case a good approach would be to build the first as a *parsimonious* version of the second, and the DBM modelling approach provides a way of doing this.[20]

4.6 Models of the disturbances

The final purpose for which we construct a model is to simulate the behaviour of the system in correspondence with the alternatives that we want to evaluate and with appropriate input trajectories. Relative to the latter, it is clear how one can obtain the control trajectories: they are produced by the same alternatives that we want to evaluate, since, if there is a control, the alternative certainly includes within its actions the design of the policy that defines it. If this were not so, the control would not be classified as such, but as a disturbance upon which it is not possible to act. Think again about the Verbano Project: the release from the hydropower reservoir was classified as a disturbance for this very reason, while the release from the lake is a control, since the aim of the Project is to identify a 'good' regulation of the lake. On the contrary, it is not so easy to understand how to obtain the disturbance trajectories: indeed, since they are disturbances it is not in our power to fix them.

The easiest solution is to adopt the trajectory that was historically recorded. Moreover, it is also the best possible solution, since it was not we who chose it. However, if its horizon were to be too short, and therefore insufficient to evaluate what might happen in the system in the long-term, it might be judged unsuitable. It would then be necessary to lengthen the recorded series, but we certainly cannot wait until this occurs naturally through the passage of time. The only remaining solution is to identify a *model of the disturbance* with which to generate the trajectories that we need.

If in turn the model of the disturbance has one or more input disturbances, the problem shifts to the modelling of them. At times this shift is completely useless. For example, if in the Verbano Project we were to describe the outflow ε_{t+1} from the catchment with a model that explained it as an effect of precipitation, we would find that we would have

[20] Alternatively, complexity reduction techniques can be adopted (Hooijmeijer et al., 1998).

to model the precipitation without deriving any advantage from it. On the other hand, the shift is advantageous when it allows us to use available information, and this happens, most often, when the disturbance is deterministic. For example, in the Verbano Project the release w_t from the hydropower reservoir could be described as the output of a model of such a reservoir, whose input is the outflow from the catchment that feeds it. If we were to know the regulation policy of that reservoir, we could include it in the model, obtaining a good description of w_t. Also in this case, however, the problem of modelling the disturbance is shifted, because to describe a disturbance we have introduced a new one (the flows to the hydropower reservoir), which we must describe with a new model in its turn.

It is thus clear that sooner or later the disturbance will have to be explained without introducing new inputs, and therefore only on the basis of the values that it has assumed in the past and, possibly, of the values of the state ($\mathbf{x}_t$) and/or control variables ($\mathbf{u}_t$) of the model that describes the rest of the system. The model of the disturbance could then in general be empirical of the form

$$\varepsilon_{t+1} = \mathsf{y}_t\big(\varepsilon_t, \ldots, \varepsilon_{t-(p-1)}, \mathbf{u}_t, \ldots, \mathbf{u}_{t-(r'-1)}, \mathbf{x}_t, \ldots, \mathbf{x}_{t-(r''-1)}, e_t, \ldots, e_{t-(q'-1)}\big) + e_{t+1} \tag{4.23}$$

where e_{t+1} is the noise. However the formula is ambiguous, inasmuch as ε_{t+1} is the output and not the input of the system that we want to describe: the system that generates the disturbance. It is therefore advisable to shift the focus onto that system and rewrite equation (4.23) as

$$y_{t+1} = \mathsf{y}_t\big(y_t, \ldots, y_{t-(p-1)}, \mathbf{u}_t, \ldots, \mathbf{u}_{t-(r'-1)}, \mathbf{x}_t, \ldots, \mathbf{x}_{t-(r''-1)}, \varepsilon_t, \ldots, \varepsilon_{t-(q-1)}\big) + \varepsilon_{t+1} \tag{4.24}$$

where y_{t+1} denotes the disturbance that we want to model and the noise is now denoted by ε_{t+1}. We have used the symbol of the random disturbance, but it could also be a deterministic disturbance.

If we study equation (4.24) carefully, we note that the problem is not yet solved, because the model still has a disturbance in its input: the noise. It seems that we are trapped in a vicious circle.

There is one possible way to get out of it[21]: in equation (4.24) the disturbance ε_{t+1} does not have to be described by a dynamical model, for the simple reason that it is not explainable, since it is a *purely random* disturbance, also known as *white noise*.[22] A purely random disturbance is a very simple object, but its definition is subtle and very important. To make it very clear, we have to take a few steps back and re-examine the notion of a model, which until now we have taken for granted.

The original information about reality is a set of time series of data. Science analyses this data with the hope of succeeding in explaining them through the recognition of their internal structure. If one does identify a structure, one can in fact abandon the long and cumbersome data series and remember only the structure, which is much simpler and compact, because its description requires less data with respect to the original series. Such a structure is a model. As Science has progressed, more sophisticated and compact structures have been discovered, able to explain enormous series with only a few parameters, using surprisingly compact formulas. The classic example is Newton's laws that explain a long series

[21] Actually, there is also another that we will analyse in Section 6.4.1.

[22] In what follows the terms 'purely random' and 'white' will be considered as synonymous.

of planetary positions recorded by Tycho Brahe in a long life of daily (or better, nightly) uninterrupted observations.

If a series of n data admits a model that allows us to reconstruct it on the basis of a number of parameters which is smaller than n, one says that it is *algorithmically compressible* (Barrow, 1991). Let us suppose that there is a data series that is not algorithmically compressible, which means that the only way to represent it is to supply the series itself. A series of this type is called a *purely random*[23] (or *white*) series.

A purely random disturbance is therefore an algorithmically incompressible series of values ε. This implies that the value ε_{t+1} is, for each t, independent from the values that ε assumed in all the preceding instants, which in its turn requires as a necessary condition that the *autocorrelogram of ε is identically equal to zero*.[24] It follows that ε is completely described by its probability distribution[25] $\phi_t(\cdot)$ at every instant; a condition that we denote with

$$\varepsilon_{t+1} \sim \phi_t(\cdot) \quad \forall t \tag{4.25}$$

Note that equation (4.25) describes the statistical properties of ε, but it does not allow us to reconstruct an observed trajectory. Despite this fact, we will say that it is the model of ε, because it represents all of the understanding that can be had of it.

We can therefore conclude that a stochastic disturbance y_{t+1} is completely and correctly defined when it is described by a model of the form (4.24)–(4.25), in which ε is a white noise. This statement, however, is rather ambiguous, because y is no longer a disturbance, being at this point explained by an empirical model that has the disturbance ε as input. Only this last is a disturbance. More correctly, we conclude that to explain a disturbance, one must introduce another disturbance, and then, if necessary, yet another one, until one runs into a purely random disturbance. Then and only then will we have finished describing the system, since by definition, only a purely random disturbance cannot be described further. The stochastic disturbance is therefore completely described by equation (4.25).

We should now explain what would be the description of the purely random disturbance in the case that the disturbance y_{t+1} is not stochastic, but we will postpone this topic to Section 5.7.

As for all models, in order for equation (4.25) to describe particular disturbances effectively it is necessary to estimate (*calibrate*) the values of the parameters that define its distribution. The procedures to do this are described in Appendix A2. Since now the disturbance has its own model too, it can and must be interpreted as the output of a component that generates it and it will be considered as such in the following chapters.

To conclude, we may finally respond to the need which prompted this section: to provide the trajectories of the disturbances with which the behaviour of the system can be simulated in the long-term. They can be obtained by extracting (generating) at every time t a value from the probability distribution (4.25). This last operation is called *generation of synthetic series* of disturbances and we will examine it in greater detail in Section 6.4.1.

[23] It is precisely because our sensations and collected data are not purely random that the mind is able to enter into contact with the world. The brain has the capacity to compress complex sequences in a simpler form and from this capacity thought and memory are born (Barrow, 1992).

[24] The autocorrelogram is the function that expresses the correlation coefficient ρ_τ between $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\varepsilon}_{t-\tau}$ as a function of τ, see Appendix A2. The necessary condition turns out to be also sufficient in case of Gaussian disturbances.

[25] Expressed by a Probability Density Function (PDF), when the variable is continuous, or a Discrete Density Function (DDF), when the variable is discrete (see Appendix A2).

4.7 Markov chains

Markov chains cannot yet be clearly described at this point because the reader is lacking several notions that will be introduced in Chapter 10. Their formal description is therefore postponed to Section 11.1. However, we believe that it is important to provide some ideas about the nature of Markov chains, in order to present a complete framework of the classes of available models, though the reader will be able to completely appreciate what follows only after having read the two chapters cited above.

The utility of this new class of models emerges only when we examine how the model can be used to evaluate the effects that the alternative produces. We already know that this will require the simulation of the system. When a mechanistic model (or a BBN) is fed by a stochastic white noise[26] the state $\mathbf{x}_t$ of the system is at every instant (except for the initial one) a stochastic variable that has to be described by its probability distribution $\boldsymbol{\pi}_t$. This can be determined in a purely numerical way, by performing a great number of simulations of the model in correspondence with different trajectories of the random disturbance, which can be generated though the model of the disturbance. This way of proceeding is called *Monte Carlo method*, in honour of the Principality's casino in which the outputs of the roulette wheel are supposed to be perfectly random. We learned in the preceding section that the generation of a trajectory of the disturbance is achieved by randomly extracting, at every time instant, a value from the probability distribution (4.25) that describes it. The numerical methods to extract it require, in the final analysis, the generation of an equiprobable number in a given set of numbers, exactly as a (honest) roulette generates an equiprobable number between 0 and 36. As one can imagine, this method is, however, computationally expensive in terms of time.[27]

To avoid this difficulty one can have recourse to the following idea: when the state $\mathbf{x}_t$ is discrete, as we have seen in the BBN (Section 4.2), instead of using a description of the system based on that state, we can use a description that assumes that the state is the probability distribution $\boldsymbol{\pi}_t$ of $\mathbf{x}_t$. The dynamics of the latter is in fact described by a simple, compact, linear model,[28] named *Markov chain*,[29]

$$\boldsymbol{\pi}_{t+1}^T = \boldsymbol{\pi}_t^T \mathbf{B}_t \tag{4.26}$$

where the superscript T denotes the transposition of the vector to which it is applied. The matrix $\mathbf{B}_t$ has a simple meaning: its element b^{ij} is the probability that the state of the system passes from its ith value at time t to its jth value at time $t+1$, when the disturbance that acts on the system is the white noise considered and the system is controlled by a given policy (we are evaluating an alternative, so the policy, if it does exist, is certainly defined).

To clarify, consider the following simple example: let us suppose that

- the system is described by the Markov chain that corresponds to the BBN in Figure 4.6,

[26] In Section 6.4.1 we will show that almost always, in practice, one may only consider white (stochastic) disturbances. The case in which the disturbance is not stochastic will be examined in Chapter 16.

[27] The distribution of the variable of the (pseudo-)random series produced in this way tends towards the distribution with which it was generated only when the number of time instants considered tends towards infinity. If the series is finite there is always a non-zero probability that the two distributions differ significantly.

[28] For the definition of linearity see Appendix A3.

[29] From the name of the mathematician A.A. Markov (1856–1922) who first proposed it (Markov, 1887).

- the two disturbances w_t and ε_{t+1} that appear in it are both stochastic and white, and their values are equiprobable,
- the adopted policy establishes that the supply decision is 'scarce' (S) when the storage is 'scarce' (S), and 'abundant' (A) when the storage is 'abundant' (A).

Then equation (4.26) is the following

$$\begin{vmatrix} \pi_{t+1}^S & \pi_{t+1}^A \end{vmatrix} = \begin{vmatrix} \pi_t^S & \pi_t^A \end{vmatrix} \begin{vmatrix} 0.29 & 0.71 \\ 0.32 & 0.68 \end{vmatrix}$$

where π_t^S and π_t^A represent the probabilities that, at time t, the storage will be scarce and abundant respectively. Compare the compactness and simplicity of this structure with the intricacy of the BBN in Figure 4.6 or even of the mechanistic model defined by equations (4.13), (4.9) and (4.12). It is easy to see that, as time goes on, the probability $\boldsymbol{\pi}_t$ tends to the following condition of equilibrium

$$\boldsymbol{\pi} = \begin{vmatrix} 0.31 \\ 0.69 \end{vmatrix}$$

which shows that, with the adopted policy, 31% of the time the storage will be 'scarce' and 69% 'abundant'; a result with a great informative value that would have been only very laboriously obtained operating directly on the BBN in Figure 4.6.

As the example reveals, the matrix $\mathbf{B}_t$ can easily be derived, with the formulae that we will show in Section 11.1, once the following are known:

- the state transition equation of the model (mechanistic or BBN),
- the models of the disturbances,
- the adopted regulation policy.

Note that this is the same information that is required for simulating with the Monte Carlo method.

By adopting a Markov chain, the mechanistic system with a stochastic state $\mathbf{x}_t$ is described by a model with a very simple structure, which has the appearance of an empirical model whose state $\boldsymbol{\pi}_t$ is deterministic. The simulation of this system is very easy because, as one can see, it is an autonomous system, i.e. a system without inputs. The analysis of the characteristics of the matrix $\mathbf{B}_t$ allows us also to recognize several important properties of the system controlled by the regulation policy adopted (for example whether it tends towards an equilibrium probability and what that would be, as we have seen in the example).

Through the Markov chain the simulation is notably simplified, but at the price of an increase in the dimension of the state of the system to be simulated, inasmuch as the components of the vector $\boldsymbol{\pi}_t$ are as many as the values that the state $\mathbf{x}_t$ can assume. The number of elements that describe the matrix $\mathbf{B}_t$ is therefore very high, in contrast to what happens for the CPT of the BBN. The elements that compose it do not have to be directly provided by the Analyst, but can be calculated, given a mechanistic model (or a BBN) and a model of the disturbance. Hardly ever is a model directly described in the form of a Markov chain but, rather, it is first formulated in one of the three forms that we presented previously (BBN, mechanistic models, and empirical models) and then a Markov chain is derived.

4.8 The time step

We have, until now, taken the value Δ of the *modelling time step*[30] for granted, assuming uncritically that it was a day. In reality, most of the variables (levels, disturbances, flows, ...) are continuous through time. Only the management decision, i.e. the control, is taken in discrete instants of time. The control cannot be changed with continuity through time, because the subsystems that it affects cannot adapt instantly to its changes in value. For example, when the release from Lake Maggiore (Verbano) is increased, the positions of the sluice gates of the diversions that feed the irrigation canals must be modified only when the wave, with which the flow variation propagates, reaches them, and the travelling speed of the wave is only a few kilometres per hour. That is the reason why a decision is modified only when one recognizes the need for it, as a consequence of significant changes of the state of the system. When the state changes quickly the decisions also become more frequent. In the Verbano case the supply decisions are not taken any more frequently than once a week during the long summer droughts, but, if a flood occurs and the level of the lake rises rapidly, they are taken even twice a day.

The duration of the time interval between one decision and the next is called the *decision step* and from what has been said it would seem to vary with the state of the system. In reality, if we think carefully, we realize that it cannot be so. There seem to be only two possibilities:

1. The decision step is uniform and the decisions follow on with a certain regularity. It often happens however, as for Verbano in summer, that from one decision time to the next the state of the system does not vary, or varies only a little, and so the decision remains unchanged. But even so, a decision is taken, even if so rapidly that one does not notice: a quick look, "everything OK, leave things as they are". Only the time instants in which the decision changes are evident and appear to be irregularly spaced. That is why even the Regulator himself will often say that the decision step depends on the situation. But this is not so, the decision 'not to change a decision' is in itself a decision.

2. The decision step is irregular and its duration is established contextually to every decision: "We have decided for today, we'll look at the decision again in a week, before then would be useless".

If we think carefully about the second possibility we realize that it is not realistic and that things could not really happen in that way. If during that week the level of the lake rose rapidly, the Regulator would quickly reevaluate the decision. This means that at regular intervals, someone, or some automatic system, verifies the state of the system and 'decides' that the decision does not need to be modified. Thus we are still dealing with the first case, which is really the only one that exists: the management decisions are taken periodically and the decision-making step is constant (or periodic).

The modelling step must be equal to the decision-making step: to take a decision it is necessary to estimate the state of the system at the time the next decision will be taken (Section 12.1).

[30] The time step is the time interval $[t, t+1)$.

To fix the duration Δ of the time step we have to reconcile two opposing needs. The first is that Δ be short enough to adapt the decision to a variation in the conditions of the system in a timely way. For this Δ must be sufficiently short, so that between one decision and the next the state of the system cannot ever change too much. We will see how it is possible to deduce from the model of the system a condition for Δ that takes this need into account (Sections 5.1.4 and 6.2.1). The second is the need to keep the decision unchanged for a period of time that is long enough to allow all the physical and economic processes that it influences to adapt to its last value. We have already mentioned that implementing a variation of the control takes time; we would like now to add that one must also pay attention to the social and economic costs that result. For example, in an irrigation system a value of Δ that is too short would create problems in the distribution system, both for physical/economic reasons (the users would have to modify the position of the sluice gates on the canals very often), and for social reasons (the distribution almost always occurs in rounds that last several days and, to be equitable, everyone should receive the same volume during a turn). In choosing Δ it is therefore fundamental to identify a good compromise between these opposing needs.

When one intervenes on a system that is in operation, a good compromise is always suggested by the Regulator's customary conduct. If in fact there is already a value for Δ, it means that not only will it adapt well to the users' needs, but it is also suitable for describing the system. If it were not, the Regulator would have difficulty operating since the state of the system would change too quickly. Thus the value of Δ that has been adopted in the past is almost always able to describe both the physical system and the decision-making system correctly.

When, instead, one is designing a system completely from scratch, the decision step must be defined by taking into account the limits posed by the dynamics of the system (Sections 5.1.4 and 6.2.1), the frequency with which the essential hydrological variables are measured, and the Stakeholders' requirements for stability. These last two must be described by indicators that express the cost induced by excessively rapid variations in the control, so that that cost can be taken into account not only in the choice of Δ, but also in the policy design.

Note, lastly, that the requirement for a constant decision step can be attenuated by accepting it to be time-varying and periodic. This is necessary whenever the value of Δ one would like to adopt is not a submultiple of the period T of the system. This case is anything but infrequent, since in practice T almost always represents a year and so any values of Δ that are greater than a day (for example a week or ten days) are not submultiples of the period. To get around this difficulty, one can define a week as an interval of seven days for the first 52 weeks and of one or two days at the end of the year. When one considers a ten day period, one should define it as a period of 10 days, when it begins the first or the eleventh day of the month, and of a length equal to the complement to the month (10, 11, 8 or 9 days) when it begins the twenty-first day. When the decision step is a month, clearly its duration will vary. The reader should lastly note that the problem of the step's periodicity is actually even more complicated, since the year itself does not have a fixed length due to leap years. In the applications it is therefore absolutely necessary, in order to avoid defining

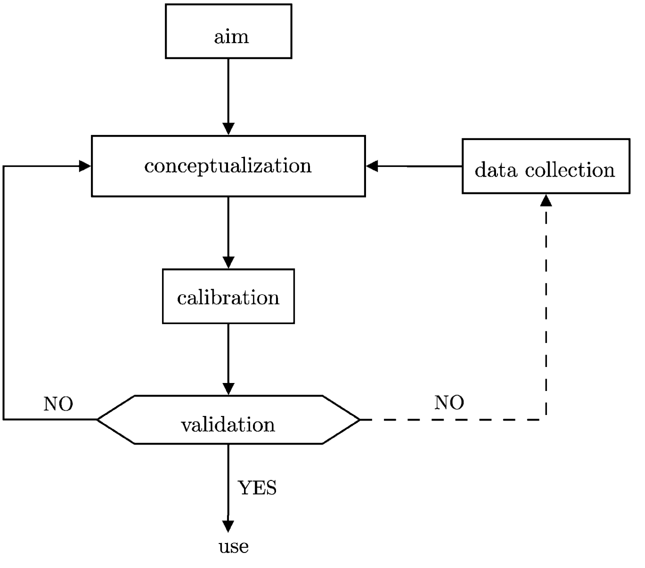

Figure 4.14: The modelling process.

T as a period of 28 years,[31] to devise an artifice that allows us to 'ignore', in a certain sense, the 29th of February in leap years. We will illustrate one in Section 8.1.3.

Sometimes in the same component there are periodic phenomena that have different periods. When this occurs, one must assume the lowest common multiple of the two periods, but this might be too long to be actually useful. This happens, for example, for irrigation districts in which the intensity of solar radiation has an annual periodicity, but agricultural activities have a weekly periodicity. Or, for hydropower plants, given that energy demand is the composition of two periodic functions: one that has an annual period, modulated by fluctuations in the temperature, and one that has a weekly period due to the weekly working cycle of the industries. In both cases, since a week is not a submultiple of a year, the artifice suggested at the end of the last paragraph provides the solution.

4.9 The modelling process

The process with which a model is created is described in Figure 4.14. The Pole Star for the whole process is the aim for which the model is being built. Until now we have assumed that it consists in the estimation of the effects that a given alternative will produce, once it is implemented. But the aim might also be to forecast future events on the basis of the available data at a given time, for example forecasting future flows in order to prevent floods.

[31] This value is valid only for the period between 1901 and 2099. In fact the leap year was introduced to keep the civil year and the cycle of the seasons synchronized. These occur with a period equal to the tropical year (the time that elapses between two spring equinoxes), i.e. every 365.2422 days. A civil year of 365 days would then produce a progressive, slow delay in the seasons. By inserting one leap year of 366 days every four civil years, the discrepancy becomes smaller, but in the opposite direction. To compensate for this fact every four centuries three leap years must be removed and it was decided to do this in coincidence with centennial years. Therefore a centennial year is a leap year only when its first two digits are divisible by 4, e.g. 1600 and 2000 are leap years, but not 1700, 1800 and 1900. In this way the discrepancy between the civil year and the tropical year is always less than one day.

4.9.1 Conceptualization of the system

The first step in the modelling process is the *conceptualization* of the system. This is the step that we have been describing in this chapter up until this point. It begins with the identification and the precise definition of the relevant variables, continues with the recognition of the cause–effect relationships that exist between them and the identification of the causal network that describes them, and finishes with the quantitative representation of these relationships. As we have seen, the types of models differ in the mathematical form of their representations, which is chosen on the basis of the available information. If we only have statistical observations (both subjective and objective) the best model is the Bayesian Belief Network (BBN, Section 4.2). If we have theories or information from which to deduce the relationships it is advisable to use a mechanistic model (Section 4.3). If we are interested in reproducing only the input–output (external) behaviour of the system and we have a sufficient amount of data, empirical models and data-based mechanistic models are preferred over the others (Sections 4.4 and 4.5). For more details see Castelletti and Soncini-Sessa (2006).

With the previous section we have completed the description of the types of models. Once the right type of model has been identified and all the cause–effect relationships have been described, the object that we have created is not yet a model (except, sometimes, when one is using a BBN), but only a structure of a model, a *meta-model*. In fact, it is potentially able to describe not only the specific real system, on the basis of which it has been developed, but all the systems that belong to the class of systems of which the one we are considering is an instance.[32] The reason for this is that the meta-model contains variables that are introduced ad hoc, called *parameters*, whose values are not yet specified. To clarify these ideas, think about the two meta-models that we have developed for the lake. In the mechanistic model (4.13), (4.9) and (4.12) the parameters are the coefficients α and β that define the storage–discharge relation (4.8) and the threshold $\bar{s}$ that appears in (4.9). In the empirical model they are the coefficients α, β and γ that appear in (4.18).

Once the meta-model has been obtained, one should analyse its properties, such as the existence of steady states, its stability and its parametric sensitivity (see Appendix A3 on the CD). These analyses are very important, because they are the only rational basis for validating the meta-model. If the comparison between the results of the analysis and the data and/or the foreknowledge of the behaviour of systems that are typical of the class we are modelling does not satisfy us, there are two possibilities: the structure of the model must be modified, or new data are necessary. In the second case one has to look for data that have already been collected, but not yet used, or, if there are no such data, one must design data collection campaigns taking into account the characteristics of the real system already identified in the meta-model (Rinaldi et al., 1979).

To transform, or to *instance* the meta-model into a model of the real system that we are interested in (the Verbano in the example), we must assign a precise numeric value to each of its parameters. The phase in which the value of the parameters is determined is called *calibration* or *parameter estimation*. The variety of names mirrors the variety of scientific areas in which it is used. In the following text we will use the term calibration when we pay attention to the model, and parameter estimation when attention is being paid to the parameters.

[32] An element that is realized in a set of elements that could be potentially realized is called an instance.

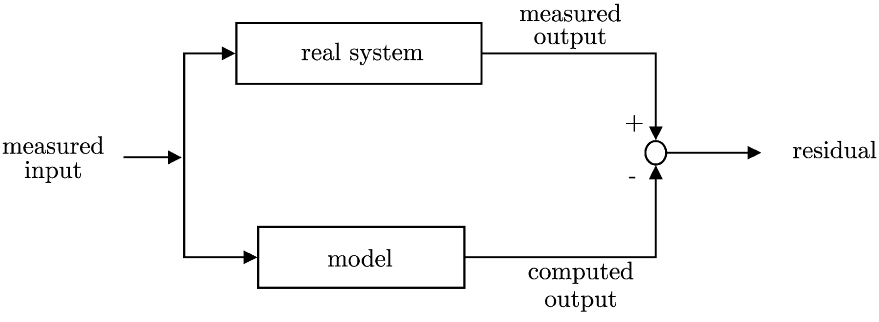

Figure 4.15: The calibration scheme.

4.9.2 Calibration

In the calibration phase one adopts different procedures according to whether the system to be modelled has yet to be created or already exists. In the first case the parameter values are inferred from the project, assuming that it will be carried out exactly to plan and by adopting an *a priori model* of the system. To explain the meaning of this last term consider the case in which one wants to evaluate the probability of winning a game of dice. To estimate the outcome probability of a number we can assume that the die is perfectly balanced (this is its a priori model) and so the outcome probability of a given value is $1/6$. If however, we have to play with a die provided by our adversary who is not a very trustworthy individual, it is more prudent not to trust an a priori model of the die, but, given that the die exists, to identify an *a posteriori model*, on the basis of the data collected in the past or in ad hoc experiments. We could, for example, record the results of the throws of the die made in a game that was already played or record the results of test throws that we make ourselves. From the series of results obtained we will estimate, with an appropriate algorithm (an *estimator*), the outcome probability of each number. These probabilities are the parameters that define the a posteriori model of the die with which we will play, i.e. of a particular instance of dice.

When the calibration concerns the model of a system that already exists, several parameters could be directly measured, such as the threshold $\bar{s}$ in (4.9), but the greater part of them cannot, because they do not have a precise physical correspondent. Their values must be deduced from observations of input–output pairs recorded from the real system; more precisely, from the comparison of the output values that are measured in the real system with the output values computed by the model when it is driven by the inputs that are contextually measured in the real system (Figure 4.15). For example, to estimate the parameters α and β that define the storage–discharge relation (4.8), one can use synchronous observations of the storage s_t and release r_t and fix the values of the two parameters in such a way that the sum of the squares of the residuals (i.e. the differences between the storages measured and those calculated) is as small as possible. This way of proceeding is based on the idea that the sum of the squares of the residuals is, in some sense, a measure of the "distance between the model and reality". It is not the only measure possible: there are others from which one chooses according to the hypotheses that are made about the nature of the measure and/or of the process noises that appear in the equations of the model (as for example in the pair (4.13)–(4.9) or in (4.18)). The determination of the parameter values that minimize the "distance between the model and reality" requires the use of an algorithm, called *estimator*. In the previous case, the estimator is the *least square estimator*, which is presented in Appendix A4. If the system being considered is not dynamical, such as the storage–discharge relation in the example, the input–output observations used in the estimation can be non-consecutive in time, while if the system is dynamical, they have to be consecutive. For example, consecutive observations are required to estimate the parameters α and β of the mechanistic model of the lake ((4.13), (4.9) and (4.12)) on the basis of input $(u_t, w_t, \varepsilon_{t+1})$ and output (h_t) data series.

Conceptually, the *estimation of the parameters* of a given system can be traced back to the *estimation of the state* of an appropriate dynamical system. This second problem is posed when one wants to know the state of a dynamical system, for which one has a model and a sequence of synchronous observations of input and output variables. Since this problem will become very important in a successive phase (page 293) and because the estimation of the parameters is also an interesting problem, the two notions are elaborated in Appendix A4.

4.9.3 Validation

Conceptualization and calibration, seen as a single phase, are called *identification*. Once the identification has been completed, the model must be *validated*, i.e. tested to see if it is able to reproduce input–output observations that were not used for its calibration. The result of the *validation* may be judged to be so insufficient that it may be necessary to return to the conceptualization, in order to review the structure of the model. Sometimes failure can be due to insufficient data for calibration; in that case it is necessary to collect new data (Figure 4.14). The input–output observations used in calibration and in validation should cover, in a sufficiently uniform way, the set of values through which one expects the input and output will vary in the conditions that the model is called upon to reproduce. When this occurs one says the model is used as an *interpolator* and this is the most correct way to use it. Sometimes, however, it is not possible to have data that correspond to the conditions that one expects the system will operate in, when the alternatives that one wants to evaluate will be implemented. As there are no other possibilities, the model must still be used, but since it operates as an *extrapolator* the estimates provided by it must be used with a lot of caution.

When the model has passed the validation phase it can be utilized.

4.10 The indicators

4.10.1 Form

In Section 4.1 we saw that the aim of the model is to provide the trajectories of the variables that are required for computing the indicators that we talked about in Section 3.2. In those sections we explained what indicators are and what role they play. We will now take up those ideas to formalize the definition of indicator.

Each *quantitative indicator*[33] i is a function of the trajectories[34] $\mathbf{x}_0^h$, $\mathbf{u}_0^{h-1}$, $\mathbf{w}_0^{h-1}$ and $\boldsymbol{\varepsilon}_1^h$ of the state, of the control and of the disturbances that act on the system in a time horizon $H = \{0, \dots, h\}$, termed *design horizon* or *evaluation horizon* according to the phase in

[33] Qualitative indicators are not, in general, expressed by functionals, even though they are functions of the alternative, and they are usually estimated by Experts (Section 3.2). In the rest of this part of the book and in Part C we will consider only quantitative indicators.

[34] The symbol $\mathbf{z}_0^h$ denotes the trajectory of the variable $\mathbf{z}_t$ in the horizon $\{0, \dots, h\}$, i.e. the series $\{\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_h\}$.

which the indicator is used (Section 1.3). It depends also on the planning decisions $\mathbf{u}^p$ and so we can write

$$i = i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right) \tag{4.27}$$

Note that among the arguments of equation (4.27) the output trajectory $\mathbf{y}_0^h$ does not appear. The reason is that the output is linked by the output transformation (4.5d) to the other arguments of (4.27), and therefore it is not necessary to consider it. This does not mean that, when writing the expression for an indicator, it is not often simpler to write it as a function of the output; it means only that, when reasoning about the dependence of the value of the indicator on the variables that describe the system, we can avoid, and this is simpler, considering the output.

The horizon H can be either finite or infinite (see Section 8.1.2.1). To have an idea of what might be in practice the form of the functionals $i(\cdot)$ the reader can look at Section 4.5 of PRACTICE. In doing so he will note that almost always the functional $i(\cdot)$ is *separable* in time, i.e. it can be expressed as a combination, through an appropriate operator Ψ (very often the sum) of elementary functions. More precisely of h functions $g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$, called *step indicators* (or more familiarly *step costs*), and of the function $g_h(\mathbf{x}_h, \mathbf{u}^p)$ of the final state $\mathbf{x}_h$, called *penalty*, which is sometimes missing. The step indicator associated with time t expresses the 'cost' produced in the transition from $\mathbf{x}_t$ to $\mathbf{x}_{t+1}$, and its arguments are all variables related to the time interval $[t, t+1)$. Note, they are relative only to that time interval, and not others. The penalty on the final state expresses the 'cost' paid for the fact of being at the end of the horizon in the state x_h. The word 'cost' is used here as a synonym of 'measure of performance' and one does not necessarily have to attribute to it the meaning expense, or damage. In fact, it could also be a 'negative cost', i.e. a 'benefit'. When an indicator i is separable it can then be expressed as

$$i = \Psi\left\{g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right),\ t \in [0, h-1];\ g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right\} \tag{4.28}$$

The step indicator g_t is a function $g_t(\cdot)$ that takes values in $\mathbb{R}$. As equation (4.28) shows, its value, besides depending on the state, can depend on the planning decisions, the control and the deterministic disturbance at time t, even on the random disturbance $\boldsymbol{\varepsilon}_{t+1}$, whose value is known only at time $t+1$. When the value of g_t effectively depends on $\boldsymbol{\varepsilon}_{t+1}$, it cannot be known at time t and so it is a random variable.

Moreover, observe that $\mathbf{x}_{t+1}$ depends on the same variables that the step cost depends on (remember the state transition function), so g_t can, in a perfectly equivalent way, be thought as a function of the tuple $(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \mathbf{x}_{t+1})$, i.e. as a function of the initial and final states, besides the decisions adopted and the deterministic disturbance.

In general, the function $g_t(\cdot)$ is time-varying and bounded, i.e.

$$g \leqslant g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \leqslant G \quad \forall\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \tag{4.29}$$

with $g, G \in \mathbb{R}$. When it is time-varying, we limit our attention to the case in which it is periodic, of period T

$$g_t(\cdot) = g_{t+kT}(\cdot), \quad t = 0, 1, \ldots;\ k = 1, 2, \ldots \tag{4.30}$$

To specify a (periodic) indicator i in an operative way, it is necessary to define, besides the T functions $g_t(\cdot), t = 0, 1, \ldots, T-1$ and the penalty $g_h(\cdot)$, also the operator Ψ and the time horizon H. For the operator Ψ, there are two forms that are most frequently used, the sum and the maximum, and we will limit our attention to these. In the first case the indicator i turns out to be the sum of the step costs along the whole time horizon and it is therefore said to be *integral*. In the second case the indicator is the maximum of the costs and is said to be *point wise*. We will postpone the description of the possible time horizons to Sections 8.1.2 and 10.2.2.1.

It is possible that the step indicators' natural arguments are not the variables of the tuple $(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$, but, even when this happens, they can always be reduced to them. Think for example about the indicator that expresses the average area [km^2/yr] that is flooded each year around the banks of a lake. The step indicator represents the area that is flooded daily, which is naturally expressed as a function of the level of the lake. This latter, as we have seen on page 103, is not the state of the lake, but it can be traced back directly to it, through the output transformation (4.4).

There is a fundamental reason for the fact that very often the functional that expresses an indicator is separable. The indicator is a measure of the performance of the system with respect to a given criterion and this performance cannot be other than an appropriate combination of day by day performances. One should not, however, deduce from this that the operator Ψ, with which the combination is made, must necessarily be the sum, because it could also be another operator, as for example the maximum, if we were interested in the maximum flooded area. It immediately follows that the step indicator $g_t(\cdot)$ is nothing but one of the outputs of the water system. So g_t is a component of the output of a system component and the function $g_t(\cdot)$ is therefore one of the components of its output transformation. In the example the component is the lake and the output is the flooded area on day t.

It remains to be explained however, why, if the preceding observation is true, the functional $i(\cdot)$ is not always separable, at least in the form in which it would be spontaneously defined, and what to do in that case. We will postpone the explanation to Section 10.2.2.2, but we anticipate that the indicators can always be expressed in a separable form, provided that the state of the system is suitably chosen. Nevertheless, we are not always interested in using the model that contains that state, for reasons that will become apparent much further on.

We can conclude by affirming that the step indicators can, and must, always be seen as the output transformations of a model, both dynamical or non-dynamical,[35] that describe the interests of a sector (group of Stakeholders). From this it follows that it is advisable always to include sectors among the components of the system.

4.10.2 Random indicators

As we have noted above, each time that the system is affected by a random disturbance $\boldsymbol{\varepsilon}_{t+1}$ and the step indicator $g_t(\cdot)$ depends explicitly on it, the value of the indicator is not deterministically known at the time in which the decision is taken. This happens also when the step indicator depends indirectly on the disturbance through the state or the control u_t (that in its turn depends on the state through the regulation policy), given that the presence of a disturbance makes the future values of the state random. Whenever the values of the step indicators are random, the value of the indicator i is also random.

However, a random indicator is not suitable, except in very particular cases, for ranking the alternatives. It is not possible to rank two random variables with respect to their value, i.e. to establish which of the two is greater, except in the rare case in which all the values

[35] In the second case the model is reduced to the step indicator only, while in the first the step indicator will be accompanied by the state transition function of the sector.

that one of them can assume were to be greater than (or less than) all the values of the other. So every time the system is affected by random disturbances, the indicator, if it is going to be useful, must be defined in such a way that its value is certainly deterministic. For this it is sufficient to adopt as indicator, not the random indicator i that spontaneously we would want to consider, but a statistic i of it with respect to all the possible realizations of the trajectory $\{\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \dots, \boldsymbol{\varepsilon}_h\}$ of the disturbances. According to the cases we could, for example, assume

$$\mathrm{i} = \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} \left[i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{4.31}$$

or

$$\mathrm{i} = \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{4.32}$$

or also

$$\mathrm{i} = \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{VAR} \left[i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{4.33}$$

The choice of one statistic over another depends on the interests of the Stakeholder that defines the indicator, since it makes a big difference whether one considers what is expected on average (equation (4.31)), or the worst possible case (equation (4.32)), or an index (equation (4.33)) of the dispersion of values that i can assume. Clearly, given a random indicator i, nothing prevents more than one indicator i from being defined, by using different statistics. For example, a Stakeholder might think that all three the indicators defined above were interesting, and might want to consider them at the same time.

In jargon the statistic is called *criterion*, implying a *filtering criterion* of the disturbance. The criteria most commonly considered are the *Laplace criterion*, which corresponds to adopting the expected value operator

$$\underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} [\cdot]$$

and the *Wald criterion*, which considers the worst case. This last is expressed by the operator

$$\max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} [\cdot]$$

when the indicator i represents an actual ‘cost’, and by the operator

$$\min_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} [\cdot]$$

when it represents a benefit. When the disturbance is uncertain and so has a *set-membership* description, (Section 5.7.1.3) only the *Wald criterion* can be adopted, since only this does not require that we know the probability distribution of the disturbance.

Pay attention to the term ‘indicator’, which is used to mean both the random indicator i, expressed by equation (4.27) or by equation (4.28), and also the indicator i that one obtains from it by filtering the disturbances with an appropriate criterion (equations (4.31)–(4.33)). Unfortunately, common practice has not yet coined distinct terms to distinguish one (i) from the other (i), so that there is a potential ambiguity. We will mostly adopt current use, employing the term ‘indicator’ in both the accepted meanings of the word. Whenever it could be ambiguous, we will accompany the term with the attribute *random*, to clearly denote the first meaning (i), or *evaluation*, to denote the second (i). Finally, for the reasons that we will present in Section 18.6, the indicators used in the *Designing Alternatives* phase may be different, both in number and in form, from the indicators used in the *Evaluation* phase. In order to distinguish them, we will term the first *design indicators*, before they are filtered by a criterion, and (design) *objectives*, afterwards; we will denote these latter with the symbol J whenever it is necessary to emphasize their difference from the evaluation indicators. When there are not random disturbances the terms 'design indicators' and 'objective' are synonyms. The application of a criterion does not solve all the problems and sometimes it can create new ones. We will take up this issue again in Section 9.1.

4.10.3 Identification of the indicators

In the PIP procedure the second phase is explicitly dedicated to the definition of the criteria and the indicators. In Chapter 3 we described the actions that are carried out in this phase, as well as the methods and the criteria to be used in the choice of the indicators. We have shown in the previous section how the step indicators are none other than the output transformation of the models that describe the sectors.

If, as we have supposed in the previous sections and as we will continue to suppose throughout this book, the system being studied is periodic, from the above it follows that to identify an indicator of the form (4.28) it is not necessary to identify h step indicators, but only T, since from the periodicity of the system it follows that

$$g_{t+kT}(\cdot) = g_t(\cdot), \quad t = 0, 1, \dots, T-1; \; k = 1, 2, \dots \tag{4.34}$$

The *identification of the indicators*, meaning the complete definition of the functionals that specify them, is therefore an action that always begins in Phase 2 (*Defining Criteria and Indicators*), but that is often concluded in Phase 3 (*Identifying the Model*). In Phase 2 the variables that one intends to adopt, the indicators, and the form of the functionals with which they will be calculated are chosen. These functionals are generally defined up to the value of the parameters they contain, which are measured (or obtained with interviews, or estimated from other quantities that have been measured) at a later time. To clarify this point, consider two simple examples taken from PRACTICE. A proxy indicator of the average annual loss of harvest in an irrigation district is the *average annual crop stress*, defined by equations (4.7), (4.9) and (4.10) in Section 4.3.4 of PRACTICE (which will be taken up again later on, in this volume, see equations (5.34a)–(5.34b)). The water demand w_t that appears in those equations is none other than a time-varying parameter. A second example: the riparian inhabitants of Lake Maggiore (Verbano) consider that the flooded area S_{fl} is a good indicator of the effects produced by floods. The flooded area at a given time t is a function of the lake level h_t, i.e. $S_{\text{fl}}(h_t)$, and this function can be assumed to belong to a fixed class. For example, we may assume that it belongs to the class of third-order polynomials in h_t, and therefore we have to fix (estimate) four parameters in order to specify a polynomial univocally. The water demand can be determined through interviews with the farmers, while the parameters of the polynomial can be estimated from a set of pairs (h, S_{fl}) recorded in the field, or deduced from a DEM.[36] In this last case the estimation of the parameters is carried out, generally, in Phase 3 (*Identifying the Model*), since it is convenient and appropriate that they be estimated together with the models of the components, of which the step indicator constitutes a part of the output transformation. In fact, the methodologies used are the same as those that are employed in the estimation of the parameters of the models.

[36] Digital Elevation Model: a digital model of the terrain.

4.11 Stationary or non-stationary?

The wise choice of structure and the adoption of a suitable parameter estimator should guarantee the identification of a model that has good predictive capacities. Nevertheless, to insure that this actually happens, and that the model retains its validity through time, it is necessary that the system, including in the extension of this word also the process that generates the disturbances, retain its characteristics unvaried through time. Only if this condition is verified can the data collected in the past contribute to explain the future.

In other words, the system must be stationary, i.e. its parameters must be time-invariant. Only when the system is such is the future statistically indistinguishable from the past. This is a necessary condition for the model of the system, which is based upon data from the past, to be able to provide useful estimates of the conditions that will occur in the future.

Consider the example of the dice introduced on page 128. If the internal structure of the die were to change through time, the estimate of the outcome probability for the different faces could not provide any estimate of the probability of winning a future game.

A moment's reflection is enough to realize that it is not strictly necessary that the system be stationary: it is sufficient that it be periodic,[37] i.e. that its parameters are functions of time, periodic with period T. Note that, when the system is periodic, it can appear to an observer as stationary if the observer *samples* it with step T. A periodic system can always be thought of as a set of T stationary systems, each one of which operates at time $t_{\mathrm{mod}\,T}$, $t_{\mathrm{mod}\,T}$ being the remainder[38] of the division of t by T. Therefore, all the properties of stationary systems hold also for periodic systems and the calibration of a periodic system can be traced back to the calibration of T stationary systems.

In reality, when the observed system is a dynamical system, as it almost always is in the management of water resources, the condition that the system be stationary is not, strictly speaking, sufficient to allow the system to be correctly calibrated. We need a stricter condition: all the statistics (such as the expected value) of the variables involved should be computable from the recorded time series. For example, the expected value of each variable should be equal to the temporal average of the values that it assumed over a sufficiently long period of time (in theory infinite). For a system to have this property it is necessary, but not sufficient, that it be stationary. The necessary and sufficient condition is that it be *ergodic* or *cycloergodic*. We cannot provide a formal definition of *ergodicity* or of *cycloergodicity* here, because we would exceed the mathematical limits that have been posed on this work, but we will give a sense of their meaning in Section 19.1.3. Fortunately, natural processes are (almost) always cycloergodic, so their models can in practice be calibrated.

In Section 2.5 we saw that it seems plausible that the hydrological system on the planetary scale (i.e., the system that generates the input disturbances for any water system) is not actually stationary. If this is true it follows that we cannot calibrate any model. What then should we do?

If the characteristics of the disturbance changed with a multi-annual period T (for example if El Niño were a periodic phenomenon with a period of 4 years) it would be sufficient

[37] We could have said *cyclostationary*. Cyclostationary and periodic are synonyms: both denote the fact that the parameters that define the system vary with periodicity in time. The attribute 'periodic' is used in Systems Theory, 'cyclostationary' in Statistics. The attribute 'cyclostationary' is used mostly in reference to the model of random disturbance, when this is stochastic, because some model parameters are statistics of the probability distribution of the disturbance.

[38] The values that the remainder can assume vary from 0 to $T - 1$, i.e. they are as many as the values that a discrete and periodic function of time, with period T, can assume.

to assume T as the period of the system, and the difficulty disappears. However, it does not seem that the phenomena that we are talking about are periodic, and even if they were, the period might be so long (several centuries) that it would no longer be useful for us to take it into account. Only two possibilities remain.

The first is to identify a model that explains the evolution of the disturbance. This is the path followed by all the projects that want to take into account all the phenomena listed in Section 2.5. For example, to predict the consequences of the greenhouse effect, these projects try to identify a dynamical model that provides, on a daily basis, the precipitation at every point on the planet (or in one of its two hemispheres) as a function of its hydro-meteorological state and the energy exchanges with the rest of the universe. A model of this kind is clearly stationary, but its complexity can produce phenomena that seem aperiodic or chaotic. Its state is, however, very large and therefore it is almost impossible to use in a Design Problem, as we will see in Section 12.2.1.

Hence only the second possibility remains, which, however, can only be adopted when the Project actions do not include the sizing of constructions. It consists in

(1) identifying the best possible model of the disturbance with the hypothesis that it is stationary (or cyclostationary in a short period, such as a year);

(2) designing the policy (or the norms) on the basis of that model;

(3) utilizing the policy (or the norms) thus obtained until a change is ascertained in the disturbance generation process; one then returns to point (1) to identify a new model of the disturbance.

We will talk about this design scheme in Chapter 14.

4.12 Realization and state estimation

In Section 2.1.1.2 we saw that the rational solution to the Management Problem is to define a regulation policy, namely a succession of functions (called *control laws*) whose argument is the state of the controlled system. In Chapter 12 it will be proved that a regulation policy can be designed when all the models of the components of the system are in state-space form, that is they are expressed by equations of the form (4.5a)–(4.5d). We can then conclude that, to be able to design and use a policy, two conditions must be satisfied:

(1) the state must be measurable (*perfect state information*),

(2) all the models of the components must be expressed in state-space form,

but unfortunately these two conditions are not always met.

Let us consider the first. Inputs and outputs are measurable by definition, but not the state. For example, in the case of a lake, we can measure the level (output), but not the storage (state). Therefore, when the model is expressed in an input–output form (4.20), the state is not even defined. Note, however, that, in the case of a lake, even though the storage cannot be measured directly, it can be estimated through equation (4.10), given a measurement of the level, i.e. a measurement of the output. This observation suggests a way to overcome the difficulty. If it were possible to estimate the value of the state through a procedure (*state estimator*), given the measures of the inputs and of the outputs, we could use this estimation in place of the actual state value in the control laws. This scheme performs well in practice (Bertsekas, 1976; Luenberger, 1979).

Let us now examine the second condition. We have already shown (page 110) that it is possible to express a BBN in state-space form and we have seen that a mechanistic model is naturally written in that form. It remains to explain what one must do when the model that describes the component is empirical (equation (4.20)). The empirical model has to be substituted by an equivalent model in the state-space form. It must be equivalent in the sense that, when the two models are fed by the same input trajectories, there always exists an appropriate initial state for the second model, which produces an output trajectory equal to that of the first. A model that satisfies this condition is called *realization* and the problem of identifying it is called the *realization problem*. This problem rarely has a single solution, because in general there can be multiple alternative ways to define the state of the realization. In order to realize this, imagine that we have found a realization M^1 of a given model and that its state has n^1 components. Define now a new model M^2 that is totally equal to M^1, except for the fact that it has an extra state that does not influence the output. Clearly M^2 is equivalent to M^1 and so it is equivalent to the given model. From the point of view of the description of the component, we do not have any reason to prefer one realization over the other, since, by definition, they are equivalent. However, from the point of view of the computing time required for solving the Design Problem, model M^1 is preferable to M^2, because the computing time increases exponentially with the number of the model's state variables. Therefore, among all the possible realizations we are interested in identifying the one that has the minimum number of state variables, i.e. the *minimum realization*.

In conclusion, to be able to design and implement management policies we must be able to solve the following problems:

1. Given the recorded input and output trajectories of a state-space model, estimate the value of its state, when this is not directly measurable (*problem of state estimation*).
2. Given an input–output model, find an equivalent state-space model with the minimum possible number of state variables (*problem of minimal realization*).

These two problems, of which here we have provided only a general idea, are identified in a mathematically more rigorous way in Appendices A4 and A6 on the CD.

4.13 Conclusion

In this chapter we have provided the fundamental concepts for the construction of a model, describing their different types and the steps by which a model is identified. In the following chapters we will retrace this series of ideas, firstly analysing the models of the *components* (Chapter 5), then assembling them (Chapter 6) to form the model of the whole water system (*aggregated model*). We will examine only a few components: the catchment, the reservoir, the canal, the diversion, and the users. These are the essential components of any water system. For some components we will describe the models most commonly used, for example for the reservoir and the diversions the mechanistic model; for the users a BBN. For others we will propose some alternative models. The reader will be able to understand what we have seen in this chapter better and in greater detail.

The procedures with which the calibration is carried out are by nature very technical and the subject of specialized studies. Therefore, they will not developed further, but are resumed in Appendix A6, where attention is concentrated mostly on empirical models, since, due to their nature, they are the ones for which parameter estimation is unavoidable.

Chapter 5

Modelling the components

RSS and EW

In the last chapter we saw how the *global model* of the system can be obtained by aggregating the models of the individual components and how these components must be first conceptualized and then calibrated. Conceptualization is carried focusing on the specific component of the system that is to be described, but this operation does not result in the model of that component, instead, a meta-model is obtained, of which the model that we want is a specific instance. It is the successive calibration that instantiates the meta-model into the model of the specific component. Thus, it is not necessary to begin ex-novo with the conceptualization of the components in every application, but with the use of a meta-model base, which must always be included in an MODSS (Chapter 24), one may simply establish which class the component that is to be modelled belongs to and instantiate its model by calibrating the meta-model. Therefore, the conceptualization phase needs to be carried out just once for each class of components, and so this chapter is dedicated to the development of the meta-models for the components which are essential to the description of a water system: the reservoir, or regulated lake; the river basin; the canals; the diversions; the water users; and the disturbances. We will consider only the most common water users: hydropower plants and irrigation districts. As we conceptualize these components we will also take the opportunity to present some examples of the application of the three types of models that we have seen in the preceding chapter: BBN, mechanistic models and empirical models.

Once the models of the components have been identified, they must be aggregated to form the global model of the system. We will deal with this in the next chapter. It is important to point out right away that the distinction between components and the global system can be viewed at different levels: a component could be composed in turn of subcomponents, or the global system could be considered as a component of a greater system. We will encounter examples of this issue of scale in this chapter when we look at the catchment and the irrigation district.

The literature on the models of components is vast and the number of books and articles dedicated to this topic is twice or three times the number of works that have been written on management. We do not want to, nor can we, provide an exhaustive view of water system modelling, but much more modestly, we would like to provide some key ideas that the reader must have in order to understand the following chapters of this work.

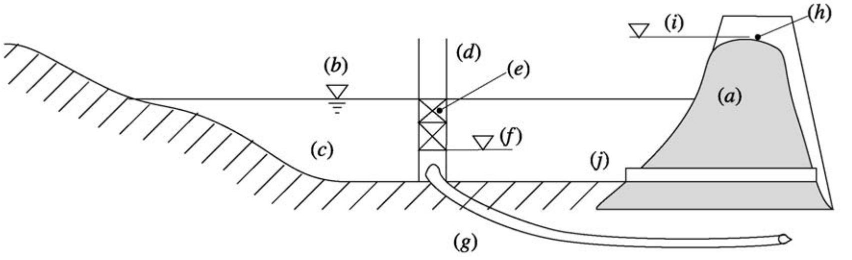

Figure 5.1: The cross section of a dam showing: the barrier (*a*), the level of the water surface (*b*), the storage impounded upstream (*c*), the tower intake structure (*d*) with intakes (*e*), the minimum intake level (*f*), the feeder pipe or penstock (*g*), the surface spillway (*h*), the crest of which defines the elevation (*i*) of the maximum storage and the bottom outlet (*j*).

5.1 Reservoirs

When we use the term *reservoir* we mean a storage and regulation structure that can be either an artificial or natural lake which is regulated by an artificial barrier.

A reservoir is created by constructing a *dam* across a narrow section of a river valley. In this way, the water level behind the dam is raised in order to create a storage that permits water volumes to be carried over in time, by acting opportunely on the discharge from the dam, i.e., in jargon, by "regulating" the reservoir. The barrier is called a *dam* when the increase in water level is such that the water invades an area that is greater than that occupied previously by the river channel, and *diversion* when the storage that is formed is completely contained in the river channel. We will now deal with modelling a dam, while we consider diversions in Section 5.4.

From a management perspective, a reservoir is characterized by (see Figure 5.1):

- the *active* or *live storage*, which is the volume of water between the minimum level (*f* in Figure 5.1), at which water can be drawn, and the maximum storage level.[1] Over this level, to prevent the dam from overflowing (a very dangerous event for the integrity of the barrier, particularly if the dam is made of earth as in the figure), the water flows freely through one or more *surface spillways* (*h*) (also called *dischargers*);

- the global *stage–discharge relationship* of the spillways, which is the relationship that describes how the volume that flows through the spillways varies as a function of the water elevation (see Figure 5.2);

- the *stage–discharge relationship of the intake structure*, with which the release is regulated. In artificial lakes this is sometimes a tower (*d*), which is separated from the body of the dam (*a*), and has intakes (*e*) that direct the water into the *feeder pipe* (which is called *penstock* when it feeds a power plant) (*g*), through which the flow is regulated by sluice gates. In regulated lakes the holding structure consists in gates or other mobile structures fixed in the dam or which actually form the dam itself (for an example of the latter case see Figure 2.10 in PRACTICE).

[1] Note that despite the name, the maximum storage is not the maximum water volume that can be present in the reservoir at a given time, because when the surface spillway is active, the storage is greater. The volume of maximum storage is also called reservoir *capacity*.

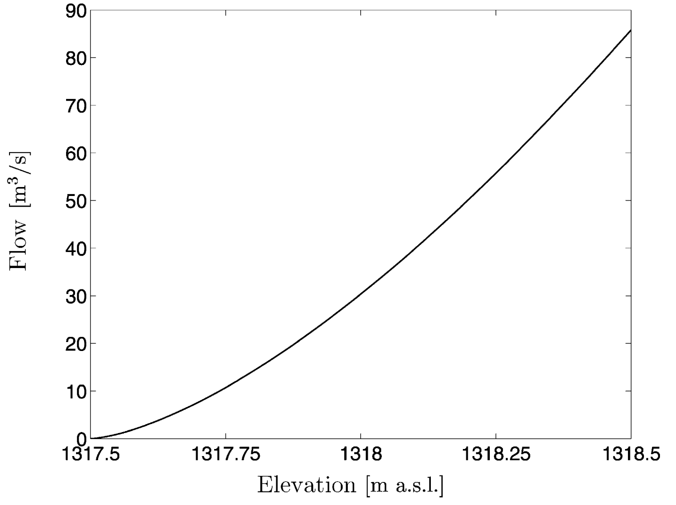

Figure 5.2: Stage–discharge relationship for the Campotosto dam 'morning glory' spillway.

Finally, an artificial dam has a *bottom outlet* (*j*) that allows the reservoir to be completely emptied for exceptional interventions, but does not have any function in normal management.

5.1.1 The causal network

The water volume present in the reservoir at time t, which we will call *storage* and denote with the symbol s_t, varies from one point in time to the next in response to the inflow, the amount of evaporation, and the volume delivered through regulation and spillage[2] in the time interval $[t, t+1)$. The *inflow* a_{t+1} to the reservoir is the cumulative water volume added by its tributaries, the distributed runoff along its banks, and the direct precipitation on the water surface. The *evaporated volume* E_{t+1} is proportional to the *area* S_t *of the water surface* and the *specific evaporation* volume e_{t+1}. The first depends on the storage, and the second on a complex multitude of factors (water temperature, air temperature, relative humidity, atmospheric pressure, wind velocity, relative insolation,[3] etc.) which are the inputs to the model that describes e_{t+1}, just as rainfall is an input to the model that describes the inflow a_{t+1}. The released volume, which we call *release* and denote with r_{t+1}, is the entire volume released through the intake structure[4] and the spillways. It is because of these spillways that the release r_{t+1} cannot be considered (Krzysztofowicz and Jagannathan, 1981, and Loaiciga and Mariño, 1986) as the control variable for the reservoir. Instead, the control variable is the *release decision* u_t (which in the following we will simply call *control*) that coincides with r_{t+1} when the spillways are not operating. The volume of spillage depends

[2]We do not consider infiltration, given that in artificial dams the phenomenon does not exist and in regulated lakes it is almost always negligible.

[3]The ratio of the real sunshine duration to the maximum sunshine duration for the day and location considered.

[4]In the following text we assume that there is only one intake structure, but a reservoir can have multiple intake structures, for example, one to feed a hydropower plant and another for an irrigation district. The model presented in the following text can easily be generalized to these cases by considering the stage–discharge relation and the release decisions relative to each structure.

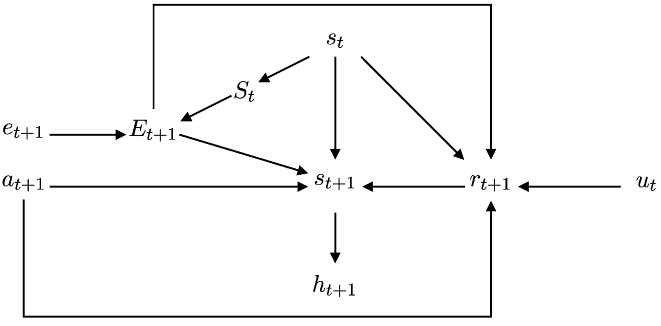

Figure 5.3: The causal network of a reservoir.

on inflows and the evaporation in the relevant time interval, or, more precisely, on the trend of these volumes in the interval. Clearly, this dependence cannot be expressed by describing the reservoir with a discrete time step but at the same time, it cannot be ignored, since its effects are always considerable. We will describe how to overcome this difficulty in the following paragraphs.

This information allows us to design the causal network in Figure 5.3, in which h_{t+1} is the water level that corresponds to the storage s_{t+1}. We already discussed its role and relationship with the storage in Section 4.1.2. Also note that the network in Figure 5.3 is an extension of the one in Figure 4.5 in which, to simplify the presentation, we did not consider evaporation E_{t+1}, or the dependency between the release r_{t+1} and the inflow a_{t+1}.

5.1.2 The mechanistic model

In order to obtain a mechanistic model we must express the cause–effect relationships that appear in the network in Figure 5.3 in a quantitative way, i.e. express each one of the five quantities s_{t+1}, E_{t+1}, h_t, S_t and r_{t+1} as a function of the variables that cause it. The preceding analysis provides us with a description of two of these relationships

$$s_{t+1} = s_t + a_{t+1} - E_{t+1} - r_{t+1} \tag{5.1a}$$

$$E_{t+1} = e_{t+1} S(s_t) \tag{5.1b}$$

We know only the arguments of the other three

$$h_t = h(s_t) \tag{5.1c}$$

$$S_t = S(s_t) \tag{5.1d}$$

$$r_{t+1} = R_t(s_t, u_t, a_{t+1}, e_{t+1}) \tag{5.1e}$$

and therefore a more detailed analysis is required to identify them. The attentive reader might expect, quite correctly, that in equation (5.1e) E_{t+1} would appear in place of e_{t+1}. Nevertheless, one must note that s_t and e_{t+1} univocally define E_{t+1}, and so the pair (s_t, e_{t+1}) is equivalent to the pair (s_t, E_{t+1}). For reasons that will emerge in box on page 141, it is best to use the first pair.

An important convention *Since the storage s_t is a volume* $[\mathrm{m}^3]$, *from equation* (5.1a) *it follows that the other three variables that appear in it* (a_{t+1}, E_{t+1} *and* r_{t+1}) *must have the dimension of a volume per time step* $[\mathrm{m}^3/\Delta]$; *e.g.* $[\mathrm{m}^3/\mathrm{d}]$ *when* Δ *is a day. Nevertheless, expressing an inflow or a release in this unit of measurement is not very meaningful, because we usually think in terms of average daily flow* $[\mathrm{m}^3/\mathrm{s}]$, *i.e. the quantity obtained by dividing the value of the three variables by the number of seconds in a time step* (86 400 *when* Δ *is a day*). *To settle this quarrel between physics and intuition, in this book we assume that the dimension of the symbols that denote flows, such as a_{t+1} and u_t, is* $[\mathrm{m}^3/\mathrm{d}]$. *However, when their numeric value is reported, it is expressed in* $[\mathrm{m}^3/\mathrm{s}]$, *as if the flows were expressed in terms of average daily flow. To simplify the presentation, we will use '(daily) volume' and '(daily) flow rate' as if they were synonymous, often omitting the attribute 'daily', while the expression 'instantaneous flow rate' will be used with its standard meaning.*

5.1.2.1 The water balance equation

Equation (5.1a) implicitly assumes that the size of the reservoir makes the flow transfer times from each tributary to the intake structure negligible with respect to the modelling time step that has been adopted.

Taking into account equations (5.1b) and (5.1e), from equation (5.1a) one can derive the following equation

$$s_{t+1} = s_t + a_{t+1} - e_{t+1}S(s_t) - R_t(s_t, u_t, a_{t+1}, e_{t+1}) \tag{5.2}$$

which describes the dynamics of the storage as a function of the control u_t and the inputs a_{t+1} and e_{t+1}.

When the reservoir is cylindrical, which means the area S does not vary with the storage, for reasons that will be illustrated later on, equation (5.2) can be reduced to the following, simpler expression

$$s_{t+1} = s_t + n_{t+1} - R_t(s_t, u_t, n_{t+1}) \tag{5.3}$$

where $n_{t+1} = a_{t+1} - e_{t+1}S$ is called *effective inflow* and expresses the net inflow, net of evaporation, in the time interval $[t, t+1)$. This expression is particularly convenient whenever the inflow is not directly measured, and must be estimated using the water balance equation itself, given the storage at two successive time instants, and the release r_{t+1} that occurs in the time interval between those instants

$$n_{t+1} = s_{t+1} - s_t + r_{t+1}$$

This last equation allows us to estimate the effective inflow even when the hypothesis of a cylindrical reservoir is not satisfied. Nevertheless, in this case the estimated inflows cannot be used for the designing of alternatives, because when the reservoir is not cylindrical, the evaporation varies with the storage, which in turn depends on the adopted policy. It follows that the value of the effective inflow varies with the regulation policy and therefore cannot be considered as a priori given data for the project.

5.1.2.2 The level and area of the lake surface

Equation (5.1c) expresses the relationship that exists between the storage and the level and it implicitly assumes that the water surface is always horizontal, because only in that case can a relationship exist between the level measured at a given point and the storage. This hypothesis is valid if the section of the reservoir is big enough to disregard the hydraulic gradient caused by the flow that runs between the tributaries and the intake structure, a condition which, with the exception of big floods, is almost always satisfied. When the hypothesis is valid, not only is the form of equation (5.1c) acceptable, but we are also certain that the relationship is one-to-one, since for each storage there is one and only one corresponding level. Therefore, there exists the inverse relationship $s = h^{-1}(h)$, which is of great practical use since it can be employed to determine the storage value from the water level, which is the only measure that can be actually taken.

Equation (5.1c) is easily derived when one can assume that the area S of the water surface does not vary with the level (and so also with the storage), as is often the case in regulated lakes (where the range of level variation is generally small, within a metre or so), so that any increase in the area as a result of an increase in the water level can be disregarded. This hypothesis is equivalent to assuming that the storage is a cylinder and so we can write

$$s = S \cdot h + s_{\inf} \tag{5.4}$$

where $s_{\inf}$ is a constant, which is usually introduced to assign the value zero to the storage corresponding to a selected level. The value of $s_{\inf}$ can be arbitrarily chosen since it is the variation of the storage that is meaningful, rather than its absolute value. Equation (5.4) can also provide negative storage values, which, contrary to an objection that is sometimes raised, do have a precise physical meaning: they express the missing volume required to bring the water surface up to the level that corresponds to the zero storage (for example up to the level of the hydrometric zero when $s_{\inf} = 0$). Inverting equation (5.4) gives an explicit expression for equation (5.1c)

$$h = (s - s_{\inf})/S \tag{5.5}$$

If, instead, the range of level variation is significant and the banks of the reservoir are not vertical, equation (5.4) is no longer valid and one must proceed in a different way to obtain an explicit expression for equation (5.1c). There are two possible approaches, which are chosen depending on whether we have the bathymetry of the reservoir (preferably digitalized in a DEM (*Digital Elevation Model*)) or a historical data series of storage–level pairs. The first approach is the most commonly used and can be adopted just as well for a reservoir that is being designed as for a reservoir that is already in operation. Given the bathymetry, one can determine the function $s = h^{-1}(h)$ point by point, numerically computing how the volume s, contained between the bottom and the surface of the water body, varies as a function of h. As in the last case, the inversion of this function provides the relationship we want.

A series of pairs (s_t, h_t), of which the first is always an estimate and the second a measure, is generally available when the reservoir is artificial and already in operation. From this series one can immediately identify the desired relationship by estimating the parameters that express it in a given class, such as a polynomial or a generalized parabola. For example, in the Vomano Project (see the description on page 147) this method was used for the Campotosto reservoir to find the relationship between the elevation h of the water surface (a.s.l.) and the storage

$$s_t = 160.51h_t^2 - 409\,102.33h_t + 260\,593\,865.62$$

which explains the historical series very well.

Also the storage–surface area relationship (5.1d) is generally derived from a DEM, with which the surface area S can be calculated for various levels h, thus obtaining the relationship $S = S(h)$ which, combined with equation (5.1c), provides the relationship we want.

5.1.2.3 The release function

Conceptually, the relationship (5.1e), which is called a *release function*, is the most complicated relationship to identify, because it describes processes that are continuous in the time interval $[t, t+1)$, whose effects, however, we want to describe with a time-discrete model (Piccardi and Soncini-Sessa, 1991). To identify this relationship we use[5] $\tilde{a}_{[\xi,\xi+\Delta)}(\cdot)$ and $\tilde{e}_{[\xi,\xi+\Delta)}(\cdot)$ to denote the trajectories of the instantaneous inflows and the specific instantaneous evaporation during the continuous time interval[6] $[\xi, \xi+\Delta)$ which corresponds to the discrete time interval $[t, t+1)$. The release r_{t+1} is necessarily bounded by the following inequality

$$\mathsf{v}\big(s_t, \tilde{a}_{[\xi,\xi+\Delta)}(\cdot), \tilde{e}_{[\xi,\xi+\Delta)}(\cdot)\big) \leqslant r_{t+1} \leqslant \mathsf{V}\big(s_t, \tilde{a}_{[\xi,\xi+\Delta)}(\cdot), \tilde{e}_{[\xi,\xi+\Delta)}(\cdot)\big) \tag{5.6}$$

where the functionals $\mathsf{v}(\cdot)$ and $\mathsf{V}(\cdot)$ denote the minimum and the maximum volumes, respectively, that can be feasibly released in the interval $[t, t+1)$, when the storage at time t is s_t and $\tilde{a}_{[\xi,\xi+\Delta)}(\cdot)$ and $\tilde{e}_{[\xi,\xi+\Delta)}(\cdot)$ are the trajectories of the instantaneous inflows and the specific instantaneous evaporation in the interval $[\xi, \xi+\Delta)$. The functional of minimum release $\mathsf{v}(\cdot)$ (maximum release $\mathsf{V}(\cdot)$) is defined by assuming that the regulation gates of the intake structure are completely and permanently closed (open) and accounting for the possible contribution from the spillways in correspondence with very high storage or inflow values.

However, the constraint (5.6) cannot be used in practice, given that the trajectories $\tilde{a}_{[\xi,\xi+\Delta)}(\cdot)$ and $\tilde{e}_{[\xi,\xi+\Delta)}(\cdot)$ are never known in real cases, where the only information available is the value of their integrals over the same time interval $[\xi, \xi+\Delta)$, i.e. the volumes a_{t+1} and e_{t+1}. We must therefore assume that the trajectories are qualitatively similar in every time interval; the simplest hypothesis is to assume that the instantaneous inflow and evaporation are constant over the whole interval

$$\tilde{a}(\zeta) = \frac{a_{t+1}}{\Delta} \qquad \tilde{e}(\zeta) = \frac{e_{t+1}}{\Delta} \qquad \forall\zeta \in [\xi, \xi+\Delta)$$

With this hypothesis the functionals $\mathsf{v}(\cdot)$ and $\mathsf{V}(\cdot)$ become two functions $v(\cdot)$ and $V(\cdot)$ that provide the minimum and maximum volumes that can be feasibly released in the interval $[t, t+1)$, given the storage s_t and the volumes of inflow a_{t+1} and specific evaporation e_{t+1}. Equation (5.6) therefore assumes the form

$$v(s_t, a_{t+1}, e_{t+1}) \leqslant r_{t+1} \leqslant V(s_t, a_{t+1}, e_{t+1}) \tag{5.7}$$

Given the *minimum and maximum instantaneous storage–discharge relationships* $\tilde{N}^{\min}(\cdot)$ and $\tilde{N}^{\max}(\cdot)$, which correspond to the complete closure and the complete opening of the

[5] Here and in the following text the symbols with the tilde denote instantaneous values.

[6] Remember that Δ denotes the *modelling time step* (see Section 4.8).

regulation gates, respectively, the functions $v(\cdot)$ and $V(\cdot)$ can be calculated by means of the differential equation

$$\frac{d\tilde{s}}{d\zeta} = \tilde{a}(\zeta) - \tilde{e}(\zeta)S(\tilde{s}) - \tilde{r} \tag{5.8}$$

which describes the continuous dynamics of the storage. For example, the function of minimum release $v(\cdot)$ is determined by numerically integrating the following differential equation, in correspondence with each triple (s_t, a_{t+1}, e_{t+1})

$$\frac{d\tilde{s}}{d\zeta} = \frac{a_{t+1} - e_{t+1}S(\tilde{s})}{\Delta} - \tilde{N}^{\min}(\tilde{s}) \tag{5.9a}$$

in the continuous time interval $[\xi, \xi + \Delta)$, starting from the initial condition $\tilde{s}(\xi) = s_t$. In this way, one obtains the trajectory $\tilde{s}_{[\xi,\xi+\Delta)}(.)$ followed by the storage in that time interval, from which the minimum release is derived

$$v(s_t, a_{t+1}, e_{t+1}) = \int_{\xi}^{\xi+\Delta} \tilde{N}^{\min}\big(\tilde{s}_{[\xi,\xi+\Delta)}(\zeta)\big)\, d\zeta \tag{5.9b}$$

The maximum release $V(s_t, a_{t+1}, e_{t+1})$ is obtained in a similar way, substituting $\tilde{N}^{\min}(\tilde{s})$ with $\tilde{N}^{\max}(\tilde{s})$ in the two previous relationships. Note that when the reservoir is cylindrical, the term $a_{t+1} - e_{t+1}S$ in equation (5.9a) is the volume of the effective inflow n_{t+1}. In this case the functions $v(\cdot)$ and $V(\cdot)$ are functions of only two arguments: s_t and n_{t+1}.

The functions $v(\cdot)$ and $V(\cdot)$ are time-varying when either the modelling step Δ, or the minimum and maximum instantaneous storage–discharge relations $\tilde{N}^{\min}(\cdot)$ and $\tilde{N}^{\max}(\cdot)$, are time-varying. The first condition occurs when the modelling step is not a submultiple of the period (think for example of one month with respect to one year). The second occurs when specific normative conditions, which we will illustrate shortly, impose time-varying minimum or maximum releases. Since in such cases both the time step and the two relations $\tilde{N}_t^{\min}(\cdot)$ and $\tilde{N}_t^{\max}(\cdot)$ are always periodic functions of period T, also the functions $v_t(\cdot)$ and $V_t(\cdot)$ are periodic, i.e.

$$v_t(\cdot) = v_{t+kT}(\cdot) \quad V_t(\cdot) = V_{t+kT}(\cdot) \quad t = 0, 1, \ldots;\ k = 1, 2, \ldots$$

Finally, to define the release function, let us assume that the Regulator has *normal behaviour*, in the sense that he operates the regulation gates in such a way that the release r_{t+1} coincides with the release decision u_t if possible, and, if not, takes the closest feasible value. With this hypothesis, from equation (5.7) the following expression for the *release function* (5.1e) can be derived

$$\begin{aligned} r_{t+1} &= R_t(s_t, u_t, a_{t+1}, e_{t+1}) = \\ &= \begin{cases} v_t(s_t, a_{t+1}, e_{t+1}) & \text{if } u_t < v_t(s_t, a_{t+1}, e_{t+1}) \\ V_t(s_t, a_{t+1}, e_{t+1}) & \text{if } u_t > V_t(s_t, a_{t+1}, e_{t+1}) \\ u_t & \text{otherwise} \end{cases} \end{aligned} \tag{5.10}$$

Two sections of this function are shown in Figure 5.4, along with the functions $v_t(\cdot)$ and $V_t(\cdot)$. From the periodicity of $v_t(\cdot)$ and $V_t(\cdot)$ the periodicity of the release function follows.

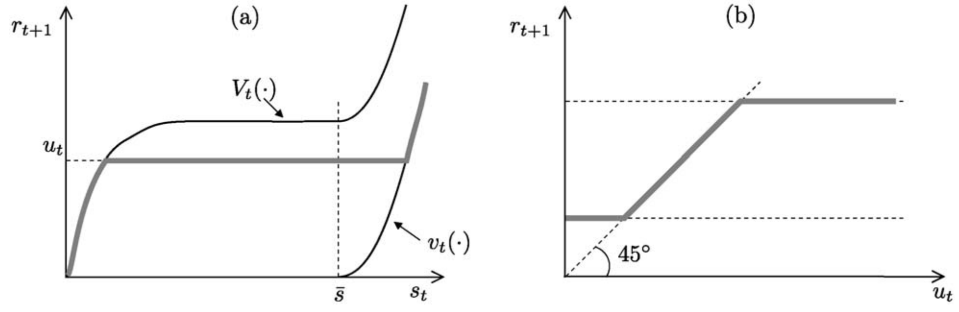

Figure 5.4: The minimum and maximum release functions $v_t(\cdot)$ and $V_t(\cdot)$ and two sections (heavy line) of the release function $R_t(\cdot)$: (a) with respect to the storage s_t, given a_{t+1}, e_{t+1} and u_t; (b) with respect to the release decision u_t, given s_t, a_{t+1} and e_{t+1}.

Very often the management of a reservoir must comply with normative constraints. These can include, for example, the obligation to release not less than a given value q_t^{MEF}, the *minimum environmental flow* (MEF), when the inflow a_{t+1} is higher than that value, and to release not less than the inflow value itself, if it is less than the MEF. Normative constrains may also prohibit the intentional retention of a storage over $\bar{s}_t$, called *maximum operative storage*. Constraints of this type have to be incorporated in the definition of the functions $v_t(\cdot)$ and $V_t(\cdot)$, and therefore, by way of these, also in $R_t(\cdot)$. This can easily be done by substituting the storage–discharge relation $\tilde{N}^{\min}(\tilde{s})$ in equation (5.9a) with a new function $\tilde{N}_t^{\text{min,nor}}(\tilde{s}, a_{t+1})$ that takes such conditions into account. For example, constraints such as MEFs and maximum operative storage are internalized in $v_t(\cdot)$ by setting

$$\tilde{N}_t^{\text{min,nor}}(\tilde{s}, a_{t+1}) = \begin{cases} \min\{q_t^{\text{MEF}}/\Delta, a_{t+1}/\Delta\} & \text{if } \tilde{N}^{\min}(\tilde{s}) < q_t^{\text{MEF}}/\Delta \\ \left.\begin{matrix} \tilde{N}^{\max}(\tilde{s}) & \text{if } \tilde{s} > \bar{s}_t \\ \tilde{N}^{\min}(\tilde{s}) & \text{otherwise} \end{matrix}\right\} & \text{otherwise} \end{cases}$$

One proceeds in a similar way when normative constraints impose conditions on the maximum release.

In a regulated lake there is often a specific type of normative constraint. When the lake level is contained within an interval of values, called *regulation range* and defined by a pair of levels $(h_t^{\inf}, h_t^{\sup})$, called *limits of active storage*, the Regulator can freely establish the release decision. Once the lower limit $h_t^{\inf}$ is reached, he is obliged to keep the release below the inflow, so that the level does not drop any further. When the level reaches the upper limit $h_t^{\sup}$, the Regulator must completely open the regulation gates, to prevent any further increase in the lake level as much as possible. The lower limit $h_t^{\inf}$ is fixed taking account of the hygienic and environmental conditions of the lake, tourism interests, and requirements for navigation. The upper limit $h_t^{\sup}$ aims, instead, at containing the floods in the lake. When the level of the lake is higher than $h_t^{\sup}$ we say that the lake is in *free flow regime*, since regulation is no longer active. When there is a regulation range, the storage–discharge relations $\tilde{N}_t^{\text{min,nor}}(\cdot)$ and $\tilde{N}_t^{\text{max,nor}}(\cdot)$ are usually defined as a function of the level $\tilde{h}$ (Figure 5.5) rather than the storage $\tilde{s}$. However, through equation (5.1c) they can easily be transformed them into a function of $\tilde{s}$, as required by equation (5.9a). The reader will find additional useful information about regulated lakes in Section 6.6 of PRACTICE.

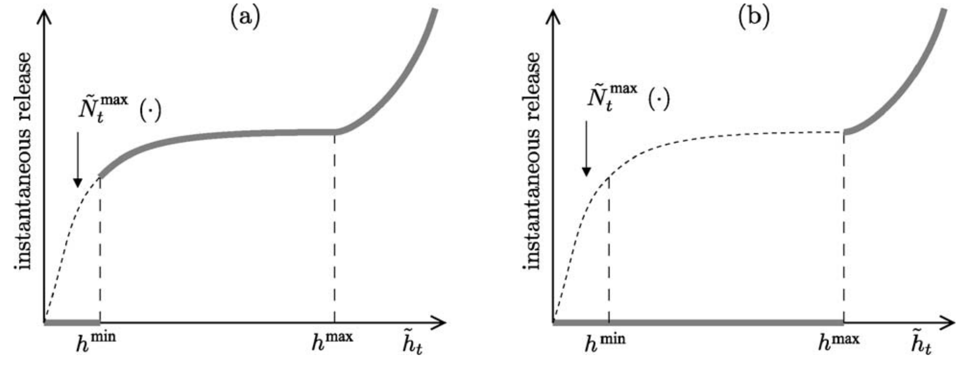

Figure 5.5: The functions (a) $\tilde{N}_t^{\max,nor}(\cdot)$ and (b) $\tilde{N}_t^{\min,nor}(\cdot)$ for a regulated lake that has a regulation range.

5.1.2.4 An example: the Campotosto reservoir

The Campotosto reservoir was constructed for hydropower purposes (see the description on page 147) and is equipped with a 'morning glory' spillway and an intake tower that conveys water through a penstock to the Provvidenza power plant, whose maximum turbine flow rate is 61.8 m^3/s. There are no normative constraints on regulation. In Figure 5.7 the minimum and maximum instantaneous storage–discharge relationships are shown. The minimum release is zero until the storage reaches the maximum operative storage[7] (217 Mm^3), at which point the water level reaches the spillway crest elevation, whose storage–discharge relation, shown in detail in Figure 5.2, determines the release for greater storage values. Since the zero storage is defined at the level of the lowest intake of the intake structure, it follows that negative storage values cannot be generated intentionally. They could only occur as a consequence of high evaporation rates that coincide with very low storage levels or, if the bottom outlet of the reservoir is opened, a condition that is not described in the model, however. The maximum release is thus zero for negative storage values. For positive storage values, the maximum release increases sharply, until it has saturated the capacity of the penstock, after which it remains constant until the storage reaches its maximum operative value (217 Mm^3). When this value is reached, the function $\tilde{N}_t^{\max}(\cdot)$ rises suddenly because the excess water is routed through the spillway. With the above procedure, one obtains the functions of minimum and maximum release from the storage–discharge relationships. Some sections of these are shown in Figure 5.8, with respect to the storage and in correspondence with different inflow values. Note that, when the inflow is zero, the function $v(\cdot)$ shows a sharp variation when the reservoir reaches its maximum operative storage (217 Mm^3). However, as the inflow increases, the same variation occurs at lower storage values because the spillway begins to discharge before the end of the time step.

5.1.3 The set of feasible controls

Because of the way that the release function is defined, it is not necessary to impose any constraint on the control u_t, given that both the physical and normative constraints are incorporated in $R_t(\cdot)$, which transforms any decisions which are physically impossible or

[7] When there are no legal restrictions this is the same as the maximum storage.

The Vomano Project

System description The Vomano river basin (Figure 5.6) covers almost all of the Teramo Province (Abruzzo) and hosts a system of reservoirs and reversible hydropower plants. The main reservoir is the Campotosto lake (active storage 217 million m^3), from which water is first conveyed, through a system of penstocks, to the Provvidenza power plant, which discharges into the Provvidenza reservoir (active storage 1.6 million m^3). Then flows are directed to the Piaganini reservoir (active storage 0.9 million m^3) passing through the S. Giacomo power plant, and finally outflows are discharged into the River Vomano passing through the Montorio power plant. The overall head is about 1200 metres. Slightly downstream from the Montorio power plant outlet, the River Vomano is dammed by a diversion, which supplies a 7000 ha irrigation district, managed by a Land Reclamation Consortium (Consorzio di Bonifica Nord, CBN in the following). Each of the reservoirs is fed, in addition to its own catchment, by one or more interceptor canals which divert the streams flowing down from Monti della Laga and the northern slope of Gran Sasso.

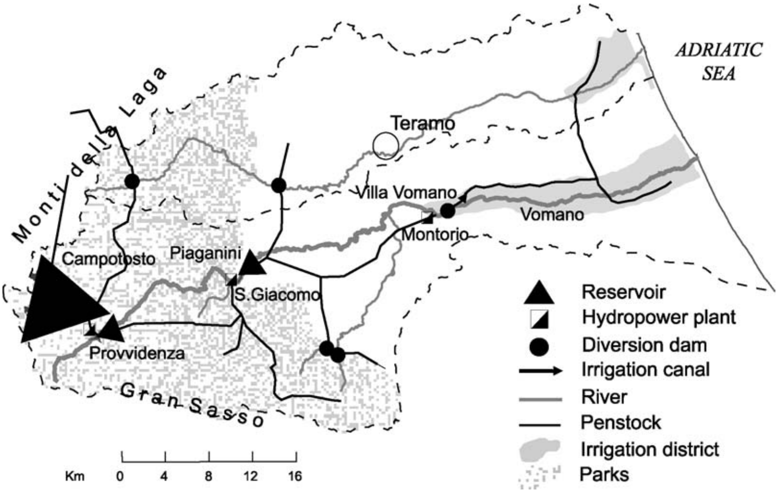

Figure 5.6: The Vomano water system.

Stakeholders Being equipped with reversible, high-power plants, Provvidenza and S. Giacomo power stations play a key role in the Italian electrical system. The hydropower company (ENEL) holds all the rights for the use of the water in the catchment and regulates the flow rates according to the company's business rationale. However, ENEL is not the only Stakeholder, since water from the River Vomano is used to supply the aqueducts that serve the urban centres in the Teramo province, for irrigation by the CBN, and is a fundamental resource for safeguarding the river ecosystems within the boundaries of two parks: the *Gran Sasso and Monti della Laga National Park* and the *River Vomano Territory Park*. The exploitation of water resources has a potential for conflict among the water users, but the existing water rights do not allow the conflict to become real.

Continued

Aim and interventions The CBN would like to extend the irrigated area and so it submitted a feasibility study to the Region of Abruzzo (the Water Authority). On the other hand, environmentalists are pushing for the definition of MEF values along the stretches of river that are intercepted by the hydropower plants and want this to be done in the very near future. The Region, for the moment, cannot withdraw the existing water rights, but can promote a *gentleman's agreement* among the parties, in view of a future water rights review. For the development of such an agreement, the Region wants to evaluate the effect on hydropower production of the two proposals and, in order to limit the consequent increase in water demand, it decides to couple these interventions with the introduction of financial incentives that should encourage the farmers to adopt more water-saving irrigation techniques.

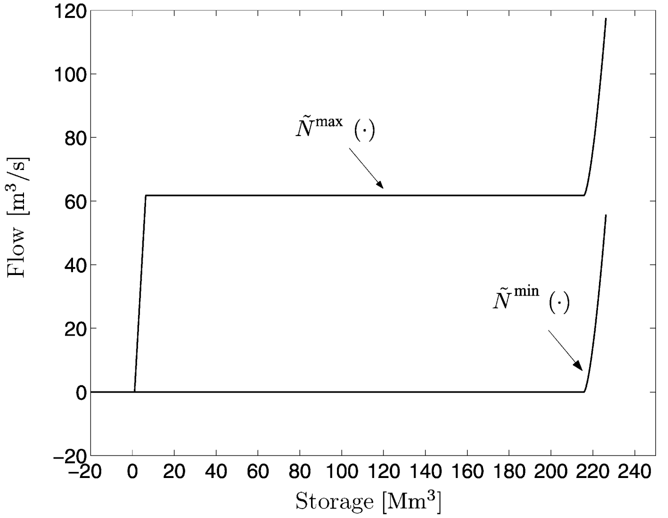

Figure 5.7: Minimum $\tilde{N}^{\min}(\cdot)$ and maximum $\tilde{N}^{\max}(\cdot)$ instantaneous storage–discharge relationships for the Campotosto reservoir.

forbidden into potential and permitted releases. From a planning viewpoint, it is not very productive to consider decisions that are not feasible, even if only for computational efficiency. Therefore, it is convenient to define, in correspondence with each storage s_t, the set of feasible controls. This set is the interval of values included between the minimum and maximum releases, defined by the functions $v(\cdot)$ and $V(\cdot)$. However, since a_{t+1} and e_{t+1} which appear in these functions are not known at time t, when the decision is made, this set cannot be known at the moment at which it would be needed. One must settle for a broader set $\mathcal{U}_t(s_t)$, which contains the former one and which can be defined, for example, by evaluating the two functions $v(\cdot)$ and $V(\cdot)$ in correspondence with extreme cases, i.e. assuming

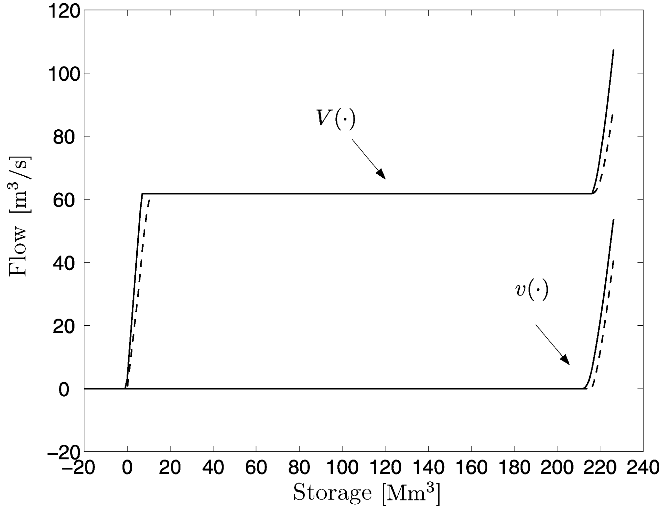

Figure 5.8: Sections of the minimum $v(\cdot)$ and maximum $V(\cdot)$ release functions from the Campotosto reservoir for zero inflow (dashed line) and an inflow of 50 m^3/s (continuous line).

$$\mathcal{U}_t(s_t) = \big\{u_{t+1}\colon v_t\big(s_t, \min a_{t+1} - \max e_{t+1} S(s_t)\big) \leqslant u_t \leqslant \\ \leqslant V_t\big(s_t, \max a_{t+1} - \min e_{t+1} S(s_t)\big)\big\}$$

where the maximums and minimums are considered to be in the sets of values that a_{t+1} and e_{t+1} can assume in the interval $[t, t+1)$, which are provided by the models that define the dynamics of the two variables. Sometimes $\mathcal{U}_t(s_t)$ is further constrained by management considerations. For example, when the reservoir is used exclusively for hydropower purposes, only the flow rate values that are interesting for turbine efficiency are considered. Despite the fact that $\mathcal{U}_t(s_t)$ does not contain all and/or only feasible controls, we will call it the *set of feasible controls*.

5.1.4 The time step and the time constant

When the reservoir is not controlled, the instantaneous release $\tilde{r}$ which appears in its time-continuous model (5.8) is defined by a storage–discharge relation $\tilde{N}(\tilde{s})$. If it is controlled, it is defined by a control law $m(\tilde{s})$. The two cases are formally equivalent, so we can examine just one of them, the first. Even if the specific evaporation is zero, equation (5.8) cannot be a linear equation because the storage–discharge relation $\tilde{N}(\tilde{s})$ is not linear, both for physical reasons (think about the causes that generate the non-linearity of the curve in Figure 5.7), and, above all, because linearity would require that the reservoir were without upper and lower bounds. Nevertheless, given a storage $\check{s}$, we can assume, with the exception of a few very specific values, that, in its neighbourhood, the storage–discharge relation $\tilde{N}(\tilde{s})$ can be approximated by the following linear equation

$$\tilde{N}(\tilde{s}) \simeq \alpha\tilde{s} + \beta \tag{5.11}$$

From this equation we can derive an important condition, which can be considered approximately valid also for the real reservoir.

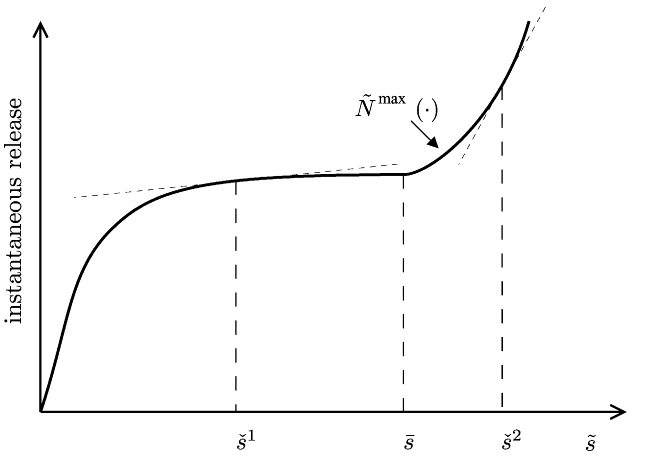

Figure 5.9: Linearizing the storage–discharge relation in correspondence with various storage values $\check{s}$. Note the sharp increase in the slope when $\check{s}$ is greater than the maximum operative storage $\bar{s}$.

Under the hypothesis that equation (5.11) is valid and evaporation is zero, equation (5.8) becomes a linear equation

$$\frac{d\tilde{s}}{d\zeta} = \tilde{a} - (\alpha\tilde{s} + \beta) \tag{5.12}$$

that can be integrated with the Lagrange formula (see Appendix A3 on the CD). Let us integrate it over the continuous time interval $[\xi, \xi + \Delta)$, which corresponds to the discrete time interval $[t, t + 1)$, starting from the initial condition $s(t)$. One obtains the following expression for $s(t + 1)$

$$s(t+1) = s(t)e^{-\alpha\Delta} + \int_{\xi}^{\xi+\Delta} \big(\tilde{a}(\zeta) - \beta\big)e^{-\alpha(\xi+\Delta-\zeta)}\, d\zeta$$

which shows that the initial storage plays a relevant role in determining $s(t + 1)$ if $\Delta \ll \alpha^{-1}$, and that it is irrelevant if $\Delta \gg \alpha^{-1}$. It follows that, in order that the time discrete representation of the reservoir can specify s_{t+1} as a function of s_t, the modelling step Δ must be smaller than α^{-1} by at least one order of magnitude. Note that α^{-1} is the time constant of the linearized model of the reservoir (see Appendix A3). We have thus obtained a very useful condition on the basis of which we can choose the modelling time step as a function of the time constant. We derived it in an intuitive way, but it is formally proved by Shannon's famous *Sampling Theorem* (Shannon, 1949).

Figure 5.9 shows that, with the exception of low storage values, the value of the parameter α increases with an increase in the storage $\check{s}$ around which the storage–discharge relation scale is linearized. It follows that the modelling step should decrease with an increase in the storage, but a step that varies with the state is not easy to deal with. For now, we will not take this observation into account, but we will reconsider it later in Section 14.5.

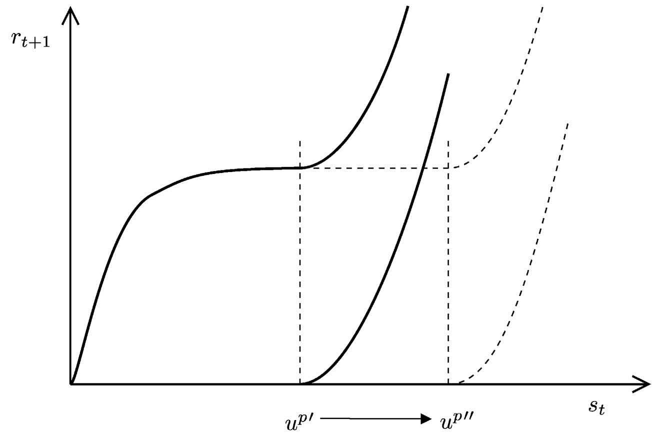

Figure 5.10: The functions of minimum and maximum release change as the capacity u^p varies from $u^{p\prime}$ to $u^{p\prime\prime}$.

5.1.5 The model of the reservoir

In conclusion, the model of an existing reservoir is described by the following equations

$$s_{t+1} = s_t + a_{t+1} - e_{t+1}S(s_t) - R_t(s_t, u_t, a_{t+1}, e_{t+1}) \tag{5.13a}$$

$$u_t \in \mathcal{U}_t(s_t) \tag{5.13b}$$

$$r_{t+1} = R_t(s_t, u_t, a_{t+1}, e_{t+1}) \tag{5.13c}$$

$$h_t = h(s_t) \tag{5.13d}$$

$$S_t = S(s_t) \tag{5.13e}$$

$$E_{t+1} = e_{t+1}S(s_t) \tag{5.13f}$$

Equation (5.13a) is the state transition function, equation (5.13b) defines the feasible controls for a given event, i.e. a pair (t, s_t), and lastly, equations (5.13c)–(5.13f) define the output transformation.

On the other hand, when the reservoir is being designed, its model must be parametric in the project variables. For example, let us assume that the storage–discharge relation of the spillway has already been defined and that only the reservoir capacity (u^p) has to be dimensioned. As u^p varies, the functions of minimum and maximum release change as shown in Figure 5.10 and, as a consequence, the release function $R_t(\cdot)$ and the set of feasible controls $\mathcal{U}_t(\cdot)$ also become functions of u^p. If the capacity is zero, the reservoir does not exist and therefore its storage must always be zero. It follows that model (5.13) is transformed into the following

$$s_{t+1} = \begin{cases} s_t + a_{t+1} - e_{t+1}S(s_t) - R_t(s_t, u_t, a_{t+1}, e_{t+1}, u^p) & \text{if } u^p > 0 \\ 0 & \text{if } u^p = 0 \end{cases}$$

$$h_t = h(s_t)$$

$$S_t = S(s_t)$$

$$E_{t+1} = e_{t+1}S(s_t)$$

$$
\begin{aligned}
r_{t+1} &= R_t\big(s_t, u_t, a_{t+1}, e_{t+1}, u^p\big) \\
u_t &\in \mathcal{U}_t\big(s_t, u^p\big) \\
u^p &\in \mathcal{U}^p
\end{aligned}
$$

The last constraint establishes the feasible values for the capacity, which depend upon both hydrographical and geological conditions, and technical considerations (e.g. very low values could be technically unfeasible). In any case the set $\mathcal{U}^p$ must contain the element zero, because among the alternatives the Alternative Zero (not to construct the reservoir) must always be taken into account.

In the same way, one can define a model of the reservoir to design the spillway (in this case the design parameters are those that define its storage–discharge relation) or any other element.

5.2 Catchments

A drainage basin, also called catchment basin or simply *catchment*, is the portion of land that collects the precipitation which flows through a section, which is called the (catchment) *outlet*, of the watercourse that drains it. This section is always located in a position that has some relevance for the Project, for example a point in which the watercourse flows into a reservoir. A characteristic feature of catchment models is that they do not contain controls, given that, by definition, the catchment is the component that produces the flows that one wants to regulate. However, planning decisions can be included among their inputs.

What was said in the introduction of this chapter about the amount of literature on hydrological modelling is even more relevant for catchment modelling literature, to which enormous research efforts have been dedicated for over one and a half century. It is not our intention to present a review of existing models, but we do want to provide the fundamental ideas that the reader will need to understand the following chapters in this book and at the same time take the opportunity to show how the analysis and identification of the model of a complex component can be carried out.

First of all, let us see which phenomena the model must reproduce. For each rainfall event there is a corresponding *hydrograph* of flow rates at the catchment outlet (Figure 5.11), which shows how the flow rate, after reaching its peak, decreases slowly until it returns to the pre-existing *base flow* before the rainfall event. The elements that characterize the hydrograph are:

- the highest value reached by the flow (*peak*);
- the time at which the peak occurs (*time to peak*) at the end of the *rising* limb;
- the delay between the beginning of the rain event and the peak (*lag time*);
- the filtering (*detention capacity*) of the rapid variations in the intensity of the rain, which is produced by the difference in the length of the paths followed by the single particles to reach the catchment outlet;
- the long tail (*recession curve*) of the hydrograph that is sustained by the groundwater contribution (*base flow*).

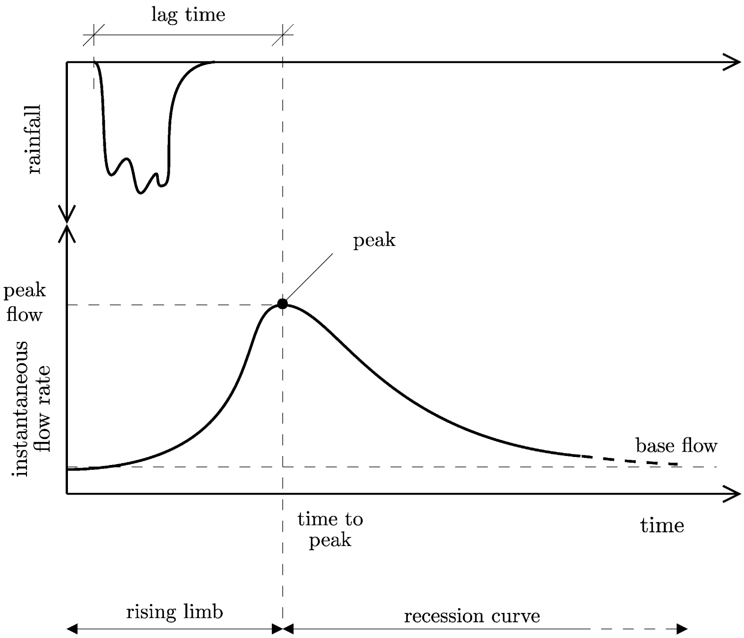

Figure 5.11: A hydrograph in response to a rainfall event, described by its hyetograph.

Even more complex phenomena appear when the soil is covered by snow, which detains the rainfall until the snow pack is saturated with water, or when the precipitation itself is solid (*snow*) and accumulates on the ground until an increase in temperature causes it to melt.

5.2.1 The block diagram

When the component to be modelled is complex, such as a catchment, the causal network that describes it can rarely be plotted all at once. Instead, it is constructed in successive steps, firstly by identifying its sub components, and then by creating models of them. Thus, one uses the same procedure that was adopted in the previous chapter (Section 4.1.1), to model the entire system. The instrument used to conduct this analysis is called the block diagram, which, like causal networks, is used to describe the cause–effect relationships among the variables. Unlike the causal network, however, it implicitly assumes that the relationship between the variables is due to a complex process, which is represented by a block and given a depictive name. Therefore, the aim of the block is to state explicitly that there may be other variables hidden inside it, which at present are unknown. Each block is then analysed using a similar process, decomposing it into sub-blocks, which in turn are decomposed until each of the resulting blocks can easily be described by a causal network: this occurs when all the variables have been identified. When the causal network that describes each block has been identified, one could, by reversing through the procedure, obtain the network for the entire component, but this is hardly ever done, as it is preferable to build models for each of the blocks and then aggregate them, just as we proposed for the global system.

Let us analyse the catchment with this procedure. The inputs to the model of the catchment are meteorological variables, such as the volume P_{t+1} of *precipitation* in the interval

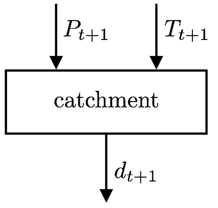

Figure 5.12: The initial block diagram of the catchment.

$[t, t+1)$, as well as the relative insolation, the air temperature and humidity, the atmospheric pressure, the wind velocity, etc.; in other words, all the variables that describe and modulate the energy exchange between the earth and the atmosphere.[8] To simplify the description, here we will only consider the temperature T_{t+1}. Our aim is to use these two inputs (precipitation and temperature) to describe the outflow volume d_{t+1} in the interval $[t, t+1)$. We have defined thus the input and output variables (Figure 5.12) of the catchment model.

Let us analyse the block in Figure 5.12. Precipitation does not always reach the ground[9]: it can be in the form of snow and accumulate, or, if it is rain, be held in the snow pack when it exists and is not saturated with water. For this reason, it is advisable to highlight a process (a block that we will label *snow pack*) which transforms the precipitation into the flow q^s_{t+1} that reaches the ground (*flow to the ground*). This flow is produced by the rain that reaches the ground and by the water released by the snow pack, which melts due to energy exchange with the atmosphere. The flow to the ground does not run downstream immediately: in part it returns to the atmosphere through evaporation, in part it is absorbed by the vegetation, and in part it is detained in the ground, while the rest percolates towards the water table, or during heavy precipitation, streams over the ground. This set of processes, that we will label *ground*, originates an outflow q^g_{t+1} which enters into the *drainage network*, whose output is d_{t+1}. This series of processes is described in the block diagram in Figure 5.13.

Next, we would analyse each of these blocks to identify and describe the processes contained within them, but we will stop at this point since we just wanted to illustrate the method. Let us now concentrate on another aspect of modelling a catchment. Precipitation does not only vary in time, but also in space: in some zones it can be intercepted by the snow pack, but not in others, since generally there will not be snow cover over the whole catchment. Also, the snow pack does not melt at the same time everywhere so, since temperature varies with altitude, melting will begin at lower altitudes, and then spread to the higher altitudes as the season proceeds. In short, until now we have considered the catchment as if it were homogeneous, both in horizontal and vertical dimensions, but rarely does this occur. In order to take account of this we must decompose the system down into blocks, not only functional ones as we have done up to now, but also spatial ones. For example, the dynamics of the snow melting can be represented by subdividing the territory into a number m of zones (*bands of altitude*) which, a priori, can be considered sufficiently homogeneous with respect to the phenomena that we have cited. The snow pack block in Figure 5.13 is thus exploded as shown in Figure 5.14. This approach can also be used for the other blocks, resulting in the type of diagram shown in Figure 5.15.

[8] Just as for reservoir models, the variables that represent flows are measured in volumes, but they are often thought of as average flow rates in the time step. Similarly, the intensive variables, such as temperature, are represented as the average value in the time step.

[9] Here we will overlook the fraction that is detained by vegetation, which, however, in certain conditions can be significant.

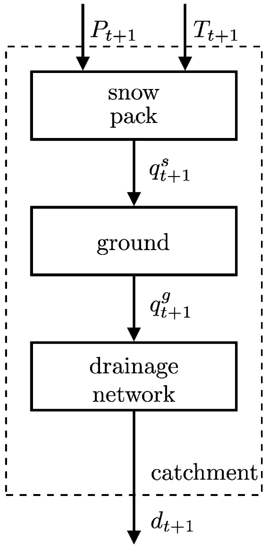

Figure 5.13: The block diagram of the catchment after the first iteration.

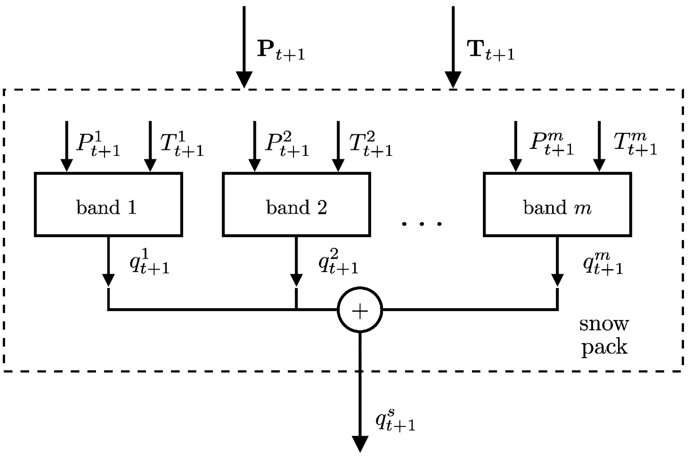

Figure 5.14: The block diagram of the snow pack when the catchment is described with m bands of altitude.

5.2.2 Mechanistic models

Mechanistic models are obtained by proceeding with the previous analysis until the model for each block is identified. However, describing such an analysis would be just an exercise in hydrological modelling and would not contribute to achieving the aim of this book. For this reason, we will terminate the discussion at this point. To satisfy the curious reader, we have prepared Appendix A5, in which we explain how a mechanistic model was built for the Lake Como catchment.

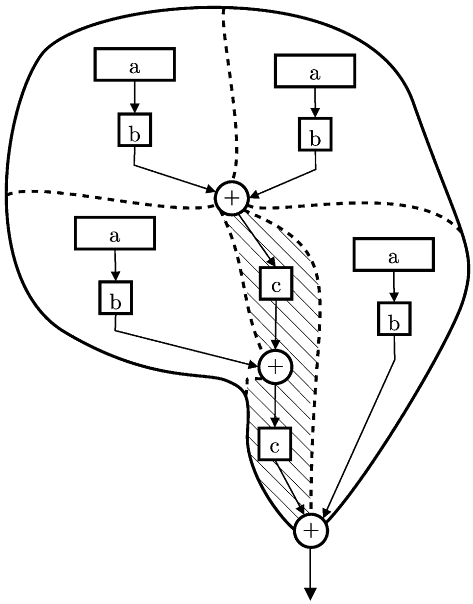

Figure 5.15: Example of a block diagram of the catchment when its hydrological characteristics cannot be considered uniform. Note the blocks that describe the ground (a), and the drainage network distinguished into the network of sub-basins (b) and the network of the major watercourse (c).

5.2.3 Empirical and Data-Based Mechanistic models

We saw in Section 4.4 that the aim of an empirical model is to explain the interrelationship between inputs and outputs without trying to describe the processes from which it originates. Data-based mechanistic models (Section 4.5) have a similar aim, even if pursued with different means. The way in which these models are identified is thus independent of the particular component that is being modelled and can be explained in general terms, applicable to all components. The reader will find these models described in Appendix A7.

Nevertheless, the reader that has some understanding of hydrological modelling may object, because we are neglecting models which are widely used, such as Sherman's hydrograph, for example. This objection deserves some attention and so we will examine some classic rainfall–runoff models to show that they are none other than empirical models, and that the interpretations with which they are generally introduced are purely verbal exercises.

We will consider three models, three benchmarks of hydrological modelling progress: Mulvaney's rational method, Sherman's unit hydrograph method, and the Nash model.

5.2.3.1 The rational method

The first attempt to describe the relationship between rainfall and runoff in quantitative terms was made by the Irish hydrologist Mulvaney (1850), whose observations can be synthesized as follows:

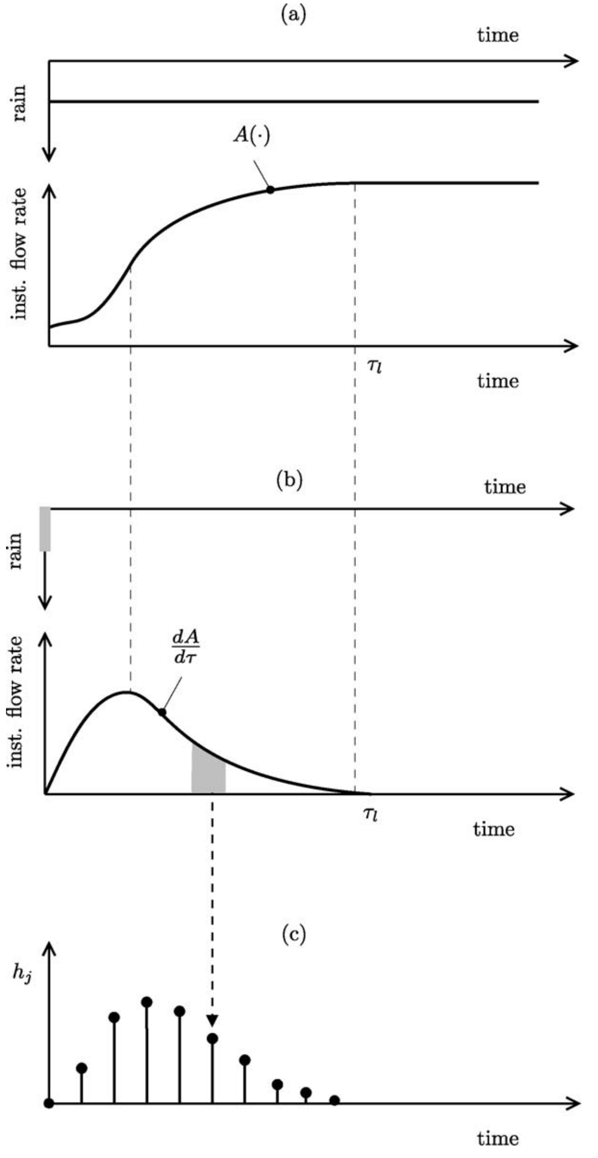

Figure 5.16: The catchment response to a step rainfall event (a), to a unit impulse rainfall event (b) and the discretization of the latter (c).

1. The response to a rainfall event described by a step function (a rainfall event that begins at the same time (zero) throughout the catchment and continues to fall uniformly over space and time for an indefinite period) is a hydrograph with the form of function $A(\cdot)$ in Figure 5.16a. For a given time τ, this function provides the area $A(\tau)$ of the portion of catchment that contributes at that time to the flow in the outlet section, i.e. of the portion of the catchment whose points are characterized by *travel*

times[10] less than or equal to τ. In this hydrograph the *lag time* (τ_l) is equal to the longest travel time within the catchment (*time of concentration*).

2. The response to a unit impulse of rainfall (a violent storm of infinitesimal duration which occurs at the same time and with equal intensity throughout the catchment) is a hydrograph that has the same form as the first derivative of $A(\tau)$ (Figure 5.16b). Its duration is therefore equal to the lag time τ_l in the previous case, i.e. to the time of concentration. The reason for this is clear: being the rainfall impulsive the last drop (the one with the longest travel time) passes through the outlet section at time τ_l.

Since the function $A(\tau)$ can be quantitatively deduced from a geographical map, even if only by a rough approximation, the Mulvaney method is considered to be the first quantitative model of a catchment.

5.2.3.2 The Sherman model: unit and instantaneous unit hydrograph

A little less than a century later, an American hydrologist, Sherman (1932), observed that:

(1) in a given catchment, the duration of the surface runoff is equal for precipitation events of equal duration, regardless of their total volume;

(2) at time t, from the beginning of the event, the instantaneous flow rates produced by any two rainfall events, which are distributed equally in time, have the same ratio as the total volumes produced by those events;

(3) the runoff trajectory is independent from the previous history of the catchment.

From these three observations, which he used as axioms, Sherman deduced that the outflow d_{t+1} in the interval $[t, t+1)$ is linked to rainfall by the following relationship

$$d_{t+1} = \sum_{j=0}^{\infty} P_{t-j+1} h_j \tag{5.14}$$

where h_j, $j = 0, 1, \ldots$, represents the outflow in a time step, which is recorded after $0, 1, \ldots$ steps from the time the unit impulse rainfall occurs (Figure 5.16c). As a consequence, the trajectory $\{h_t\}_{t=0}^{\infty}$ is termed *Instantaneous Unit Hydrograph* (IUH). If the outflow at time 0, at which the rainfall impulse occurs, is zero, equation (5.14), commonly used in hydrological texts, takes the following form, which is handiest for operative purposes

$$d_{t+1} = \sum_{j=0}^{t+1} P_{t-j+1} h_j \tag{5.15}$$

5.2.3.3 The Nash model

Remembering what was said in Section 5.2.1 with regard to decomposing a catchment into sub-catchments, Nash (1957) proposed that a catchment be described as a cascade of n elementary sub-catchments, the ith of which is described by a reservoir that collects the fraction γ_i of rainfall and is characterized by a linear storage–discharge relation with slope κ_i.

[10] The travel time associated with a point in the catchment is the time a particle of water takes to reach the catchment outlet.

The dynamics of its storage s_t^i is thus described by the following equation

$$s_{t+1}^i = \kappa_{i-1} s_t^{i-1} + (1-\kappa_i) s_t^i + \gamma_i P_{t+1}$$

for $i = 2, \ldots, n$. The equation that describes the dynamics of the first reservoir is similar, but it lacks the first term, which expresses the contribution from the upstream reservoir. By calling $\mathbf{x}_t$ the state of the n reservoirs, i.e. the vector $|s_t^1, \ldots, s_t^n|$, the state transition equation is the following

$$\mathbf{x}_{t+1} = \mathbf{F}\mathbf{x}_t + \mathbf{g} P_{t+1} \tag{5.16a}$$

and the output transformation is

$$d_{t+1} = \mathbf{h}^T \mathbf{x}_t \tag{5.16b}$$

where

$$\mathbf{F} = \begin{vmatrix} 1-\kappa_1 & 0 & 0 & \cdots & 0 \\ \kappa_1 & 1-\kappa_2 & 0 & \cdots & 0 \\ 0 & \kappa_2 & 1-\kappa_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1-\kappa_n \end{vmatrix} \qquad \mathbf{g} = \begin{vmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \vdots \\ \gamma_n \end{vmatrix}$$

$$\mathbf{h}^T = |0 \quad 0 \quad 0 \quad \cdots \quad \kappa_n| \tag{5.16c}$$

with the constraint that the sum of the parameters γ_i must be unitary.[11] The response to a rainfall impulse is thus given by (see Appendix A3)

$$d_{t+1} = \begin{cases} 0 & \text{if } t = 0 \\ \mathbf{h}^T \mathbf{F}^{t-1} \mathbf{g} & \text{otherwise} \end{cases}$$

5.2.3.4 Conclusion

The differences among the three models are only outward, as they all are linear models. Mulvaney's model is linear because the impulse response is the derivative of the step response. In Sherman's model, equation (5.14) is none other than the convolution integral to which the Lagrange formula is reduced when the initial state is zero (see Appendix A3). Further, Sherman's three observations are properties which only linear systems can possess. Finally, the Nash model is explicitly hypothesized as a linear model. Because all three are linear, these models can be traced back to the input–output relationship

$$d_{t+1} = \alpha^1 d_t + \cdots + \alpha^p d_{t-(p-1)} + \beta^0 P_{t+1} + \\ + \beta^1 P_t + \cdots + \beta^r P_{t-(r-1)} \tag{5.17}$$

where p and r assume different values depending on the model and β^0 is zero in the Nash model. As the reader can easily verify, equation (5.17) is actually a PARMAX model, which was introduced in Section 4.4 and are described in detail in Appendix A6. Therefore, the objection from which we began this section has no foundation. The following statement

[11] From the form of equation (5.16b) one infers that the Nash model, unlike the preceding ones, is a proper model and so the flow rate in the time interval $[t, t+1)$ should be denoted with the symbol d_t. However, we preferred not to modify the notation to allow a more immediate comparison of the three models.

can thus be inferred: what matters is only the mathematical structure of the model, and the same structure can be derived from hypotheses which are verbally different, but which are equivalent in practice because they give rise to the same structure. All the differences which are based on such verbal interpretations are thus merely semantic exercises. The three classic models and the PARMAX models differ only in the calibration algorithm which is traditionally used with one or the other.

Therefore, we can conclude that nowadays it no longer makes sense to use these classic models, since we have an efficient and rigorous estimation procedure to calibrate a model from the class to which they belong: the PARMAX models are preferable, when calibrated with the procedure described in Appendix A6. This is not, however, the only solution, nor the best one, since the numerous non-linearities that are present in the processes being considered[12] make the class of empirical linear models, i.e. the PARMAX models, not the most appropriate a priori choice. Neural networks (see Appendix A8) are more suitable, and above all the data-based mechanistic models (Section 4.5) described in Appendix A7.

5.3 Canals

The various components in a water system are interconnected by a network of artificial canals, channels and/or by natural watercourses. We must therefore describe the elementary component of the network, which we will call *canal*. We want it to be representative of an artificial canal as well as a stretch of river. Examining the canal will allow us to see how to deal with a component which, even though it has a very simple, almost trivial, causal network, needs a physical–mathematical description so complex that it requires a distributed-parameter model, i.e. a model described by partial differential equations.

As usual, we will first describe the phenomenon that we want to model in words. Suppose that a variation in flow rate (*wave*), of a significant size, occurs in the initial section of the canal and that it then depletes rapidly. We can observe the effects of it on the canal in two different ways.

The first way is to register the hydrographs in different sections of the canal: in this way we will notice that the propagation of the wave is described by a series of bell curves (Figure 5.17) and that as the distance of the section considered from the initial section increases, the time it takes to reach the peak (*translation time*) grows and the shape of the bell curve widens (*detention capacity*). Since the area subtended by each curve is the water volume transported by the wave (which, without lateral contributions or leakage, remains constant), the widening of the curve indicates a diminishing flow rate at the peak.

The second way to observe this phenomenon is to trace the flow rate profile along the course of the canal at different times (Figure 5.18). Again, one obtains a series of bell curves, which subtend the same area, because it represents the water volume of the wave entering the canal. The curves are asymmetrical and steeper on the front slope (right-hand side in the figure): the velocity of the propagation of the peak is faster than the velocity of the stream and the difference between the two velocities increases with the depth of the canal.

From a management perspective, we are more interested in the first way, since we want to know what happens in particular sections (*control sections*), placed where the flooding risk is high or where there is a diversion or a confluence. We can, therefore, describe the canal as a succession of branches, each of which has an *upstream section* and a *downstream section* and concentrate our attention on one of these elementary branches.

[12] The reader is invited to discover these non-linearities by reading Appendix A5.

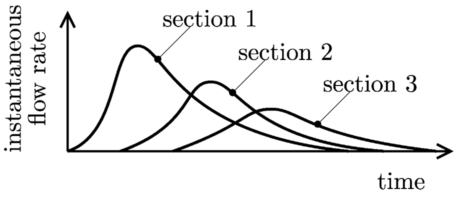

Figure 5.17: Hydrographs which correspond to three sections (1, 2 and 3), positioned at an increasing distance along a canal, in response to an impulsive flow rate in the initial section.

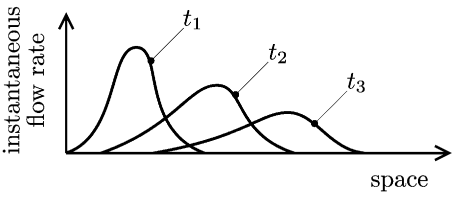

Figure 5.18: The flow rate profile along the canal at three different times.

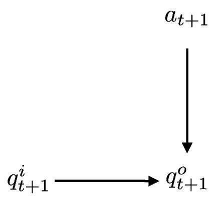

Figure 5.19: The causal network of a canal.

5.3.1 The causal network

At first glance the variables that describe the canal seem to be simple: the inflow q^i_{t+1}, which enters the upstream section, the outflow q^o_{t+1}, which flows through the downstream section, and any inflow a_{t+1} from tributaries distributed along its course. With the third variable, we can also describe any water loss due to seepage, by attributing a negative sign to its value. The only difficulty is the choice of subscripts: are we sure that the outflowing volume is influenced by the volume which enters the upstream section at the same time? This seems strange. Maybe this model is not as simple as it first seemed, but, if we decided to postpone the choice of subscript values for a moment, the causal network describing a canal would be the one shown in Figure 5.19.

5.3.2 Mechanistic models

We could dispel our doubts by assuming that the transfer of the flow rate occurs through rigid modules (*plug flow*), i.e. we assume that the canal does not have any detention capacity. The relationship between input and output volumes is then purely a time lag

$$q^o_{t+1} = q^i_{t-\tau+1} + a_{t+1} \tag{5.18}$$

where τ is the *translation time* of the canal (τ being a whole positive number). The equation thus describes the translation as if it occurred along a railway line: at each point in time a train leaves from the upstream section carrying a quantity q_t^i of water, and arrives at the downstream section after τ time steps.[13] This visualization shows us that the state $\mathbf{x}_t$ of the canal at time t can be represented by a vector that describes the loads of each of the trains which have already left, but have not yet reached their destination, i.e.

$$\mathbf{x}_t = \left| q_t^i \quad q_{t-1}^i \quad \cdots \quad q_{t-\tau+1}^i \right|^T$$

so that the input–output relationship (5.18) corresponds to the state-space description

$$\begin{aligned} \mathbf{x}_{t+1} &= \begin{vmatrix} 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \\ \vdots & & & & \\ 0 & 0 & \cdots & 1 & 0 \end{vmatrix} \mathbf{x}_t + \begin{vmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{vmatrix} q_{t+1}^i \\ q_{t+1}^o &= \left| 0 \quad 0 \quad \cdots \quad 0 \quad 1 \right| \mathbf{x}_t + a_{t+1} \end{aligned} \tag{5.19}$$

When τ is equal to zero, equation (5.18) becomes a pure algebraic relationship, thus the state disappears and equation (5.19) loses meaning. As we will see in Part C of this book, the computation times for solving the Design Problem increase exponentially with the number of state variables. It follows that, when possible, the modelling time step Δ has to be chosen so that τ can be assumed to be zero.

An estimate of τ is the time lag of the peak of cross-correlogram (see Appendix A2) between two synchronous series of upstream and downstream flow rate measures. The peak, however, is often not very marked, so that it is difficult to identify a precise value of τ. This happens since neither of the two flow rate series is generated by a white noise process, but by an autocorrelated process. The autocorrelation can be eliminated by *whitening* the two series, as one says in jargon, which means substituting each of them with the residual of a PARMA model (i.e. of a PARMAX model without the exogenous part) identified by means of that series. Therefore the value of τ is determined by analysing the cross-correlogram of the two whitened series.

When there is leakage, i.e. when the distributed contribution is negative and its value is basically invariant with the flow rate,[14] equation (5.18) is a good description of what happens but, if the leakage is proportional to the flow rate, the canal must be described by the following equation

$$q_{t+1}^o = (1-\alpha) q_{t-\tau+1}^i \tag{5.20}$$

where α is the fraction of lost flow rate. Unlike equation (5.18) the latter does not run the risk of providing negative values for q_{t+1}^o when the flow rate $q_{t-\tau+1}^i$ is very low.

Equation (5.18) was written on the basis of common sense alone, but we must ask ourselves under which hypothesis it would be legitimate to assume the type of flow to be *plug flow*. To determine these conditions, it is necessary to analyse what Hydraulics can tell us about the relationship that links the exiting flow rate to the flow rate that enters a canal.

[13] Observe that, just as with the Nash model, the *plug-flow* model can be considered as an empirical model.

[14] As happens, for example, when the leakage occurs through percolation from the bottom of the canal and the bottom is always completely submerged, as in a canal with a rectangular cross-section.

5.3.2.1 The propagation of waves: the de Saint Venant equations

A canal is said to be *one-dimensional* when variation among the values of a quantity associated to the flow (such as the concentration of a pollutant or the depth or the velocity of the stream), measured at different points of a single cross section, are negligible with respect to variations among the values of the same quantity measured along the canal. In a one-dimensional canal, the transport of any quantity is described by the following equation (*one-dimensional balance equation*)

$$\frac{\partial(Ap)}{\partial t} + \frac{\partial(Apv)}{\partial l} - \frac{\partial}{\partial l}\left(DA\frac{\partial p}{\partial l}\right) = AS_p \tag{5.21}$$

where l is the distance from the initial section, p the quantity considered, A the area of the cross-section, v the velocity of the stream, D the dispersion coefficient and S_p the distributed sources of the quantity, and all the variables are functions of time t and the space l.

Let us now assume that p is the density ρ of the water (which is constant in space and time) and define the flow rate Q as vA. Now, S_p is the flow rate (in mass [kg/s]) of the net lateral distributed inflow and equation (5.21) becomes

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial l} = AS$$

called *continuity equation*, where the term for source $AS = AS_p/\rho$ is the volumetric flow rate per metre [m^2/s] of net lateral distributed inflow (contributions minus losses). To simplify the discussion let us assume now that S is zero, so that the continuity equation becomes

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial l} = 0 \tag{5.22}$$

This equation has two variables and so, in order for it to be integrated, it must be coupled with a second equation. The first possibility is to assume that there is a one-to-one relationship between the depth h, which is the water level with respect to the bottom, and the flow rate Q in each existing section l, i.e. that there exists the *stage–discharge relationship*

$$Q = Q^*(h, l) \tag{5.23}$$

which we encountered for the reservoir model. The stage–discharge relationship is a convex function with respect to h; it describes the real behaviour of the stream flow very well when the flow rate varies slowly in time. From equation (5.23) and from knowledge of the geometry $A(h, l)$ of the cross-sections, we can obtain the relationship

$$Q(l, t) = \tilde{Q}\big(A(l, t), l\big) \tag{5.24}$$

which is also convex with respect to A. If the continuity equation (5.22) is multiplied by the velocity

$$w = \frac{\partial \tilde{Q}}{\partial A}$$

which is well defined thanks to equation (5.24) and which, as we will show shortly, represents the transfer velocity of the peak flow, one obtains the following equation

$$\frac{\partial Q}{\partial t} + w\frac{\partial Q}{\partial l} = 0 \tag{5.25}$$

which is called *kinematic model* or kinematic approximation (of a pair of equations which we will soon encounter). This model states that the flow rate is constant along the characteristic line $dl/dt = w$. In other words, the wave is propagating downstream at velocity w without losing its form: in fact, an observer moving with the wave would see no variation of the flow rate

$$\frac{\partial Q}{\partial t} + w\frac{\partial Q}{\partial l} = \frac{\partial Q}{\partial t} + \frac{dl}{dt}\frac{\partial Q}{\partial l} = \frac{dQ}{dt} = 0$$

Thus, from a functional point of view, the canal behaves as a pure delay: an impulsive variation of the flow rate in an upstream section generates an impulsive variation of the flow rate in a downstream section, situated at distance L from the first, with time lag $\tau = L/w$, just as equation (5.18) states. Further, equation (5.25) correctly describes the dependence of the velocity of the peak flow on the depth h: in fact, since $\tilde{Q}(A, l)$ is a convex function with respect to A and A is increasing with h, also w increases with h. This equation does not describe, however, the detention capacity effects.

If for this deficiency the kinematic approximation is deemed unacceptable, the logical solution is to couple equation (5.22) with the *momentum balance equation* which can be obtained from equation (5.21) if we neglect the dispersion D and pose $p = \rho v$. In this way one obtains the following equation

$$\frac{\partial(Av)}{\partial t} + \frac{\partial(Av^2)}{\partial l} = \frac{AS_p}{\rho} \tag{5.26}$$

The term S_p on the right side accounts for the force of gravity in the direction of the stream and for the forces produced by hydrodynamic friction (known as *load losses*), which are the distributed sources and losses of momentum, respectively. If one assumes that the load losses are independent of Q and h, one can show that the pair of equations (5.22) and (5.26) degenerates into equation (5.25), which is actually the kinematic approximation. In fact, if the load losses are independent of Q and h, even intuitively, one can understand that there must be a one-to-one relationship between these two variables in each section, i.e. there is a stage–discharge relationship. In order for the model to describe the detention capacity, we cannot therefore assume that the load losses are constant; instead, they must be expressed through empirical formulae as a function of Q and h. By introducing one of these formulae, and neglecting the variations in kinetic energy over space and time with respect to the variations in potential energy due to the water level and the specific energy losses, one obtains an approximation known as *parabolic model*. We will not describe it here, but refer the interested reader to a classic hydraulics text (for example Chow, 1959 or Chow et al., 1998), because we will not be using this model in the following. We would just like to note that, by manipulating it, one arrives at the following equation

$$\frac{\partial Q}{\partial t} + w\frac{\partial Q}{\partial l} = D^*\frac{\partial^2 Q}{\partial l^2} \tag{5.27}$$

where the velocity w and the diffusion D^* are convenient positive functions of Q and h. This equation shows that when moving at velocity w, the total derivative of Q has the sign of $\partial^2 Q/\partial l^2$. This means that a bell shaped impulse which moves downstream with a velocity w eases off (at the peak it is clearly $\partial^2 Q/\partial l^2 < 0$) and, as a consequence, widens. Thus the parabolic model describes the detention capacity.

The two equations (5.22) and (5.26) are known as *de Saint Venant equations*, because they were obtained for the first time by Barré de Saint Venant in 1871, and they are still often used in simulation studies, despite three significant disadvantages. The first is that these equations can be integrated only if a good approximation of the river channel geometry, described by the function $A(h, l)$, is known, but this information is very costly to obtain and it is also ephemeral, since erosion and the movement of gravel produced by floods quickly change the form of the river channel. The second disadvantage is that the integration of the pair of equations requires a significant computation effort (Greco and Panattoni, 1977), since in order to control the numerical errors produced by the integration scheme very small spatial and temporal integration steps (hundreds of metres and minutes) have to be adopted. This means that the simulation of long river channels (one or more hundreds of kilometres) and, above all, of long periods of time (scores of years) requires a prohibitive computation time. The third is that the load losses are generally unknown and must thus be expressed with empirical formulae (such as the Chezy and Darcy–Weisbach formulae) which include parameters whose values are estimated on the basis of the geometry of the sections and the roughness of the bottom, for short homogenous canals, or on the basis of several hydrographs, for long irregular canals.

For all these reasons it is understandable why, when the detention capacity cannot be ignored, it would be preferable to make use of empirical models.

5.3.2.2 Detention areas

On page 8 we saw that among the structural interventions for hydraulic protection and regulation are the creation of detention areas, detention basins and dry dams. The effect produced by these constructions is to create a storage, in which part of the outflow is temporarily impounded, when the flow rate of the canal is particularly high. With the first two structures this is achieved by increasing the cross-section of the canal above a predetermined flow rate, while dry dams achieve the same goal by narrowing the cross-section. It follows that all three structures can easily be described by the de Saint Venant equations, given that their effect is expressed in a natural way through the function $\tilde{Q}(A, l)$. They can, however, be represented also as an aggregation of two elementary models, a canal and a reservoir. For example, the model of a canal with a detention area is constituted by a canal (Figure 5.20a) with an inflow q^i_{t+1} and an outflow q^o_{t+1}, inside which the flow rate a_{t+1} moves into the reservoir when the level h^c_{t+1} in the canal exceeds the spillway threshold $\bar{h}$ (Figure 5.20b). The level h^c_{t+1} is determined by the flow rate q^i_{t+1} and, therefore, the flow rate a_{t+1} is a function of the flow rate q^i_{t+1}, of the threshold $\bar{h}$, and also of the storage s_t impounded in the reservoir, since as the last grows the outflow from the canal to the reservoir decreases. When the inflow begins to decrease, the flow a_{t+1} changes sign (Figure 5.20c) and the reservoir empties progressively, feeding the canal. In conclusion, when the translation time is negligible the model is the following

$$\begin{aligned} q^o_{t+1} &= q^i_{t+1} - a\big(q^i_{t+1}, \bar{q}, s_t\big) \\ s_{t+1} &= s_t + a\big(q^i_{t+1}, \bar{q}, s_t\big) \end{aligned}$$

where $\bar{q}$ is the flow rate at which the level of the canal reaches the spillway crest elevation $\bar{h}$. If one can assume

- that the stage–discharge relation of the canal is linear for flow rate values q^i_{t+1} greater than $\bar{q}$, i.e. that $(q^i_{t+1} - \bar{q}) = \alpha(h^c_{t+1} - \bar{h})$;

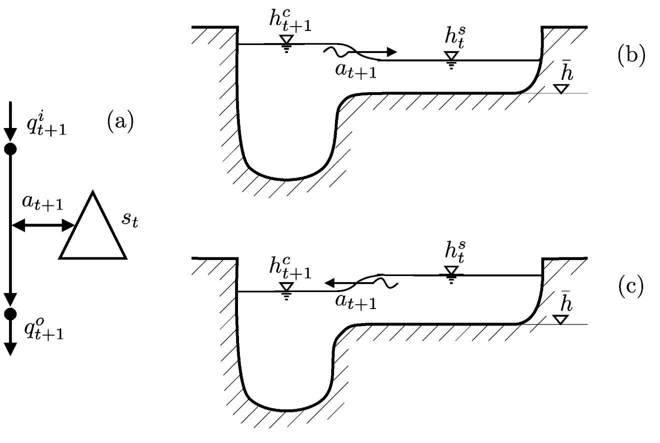

Figure 5.20: A canal with a detention area: (a) diagram; (b) section during the rising of the flood; (c) section during the recession of the flood.

- that the reservoir is cylindrical, i.e. the level h^s_t of the reservoir with respect to the spillway crest and the storage s_t are linked by the relationship $s_t = \beta h^s_t$;
- that the stage–discharge relation between canal and reservoir is also linear in the difference of the level between the two, i.e. $a_{t+1} = \gamma[(h^c_{t+1} - \bar{h}) - h^s_t]$;

then the function $a(\cdot)$ is defined by the following expression

$$a\big(q^i_{t+1}, \bar{q}, s_t\big) = \begin{cases} 0 & \text{if } s_t = 0 \text{ and } q^i_{t+1} \leqslant \bar{q} \\ \gamma\big(\frac{q^i_{t+1} - \bar{q}}{\alpha} - \frac{s_t}{\beta}\big) & \text{otherwise} \end{cases}$$

5.3.2.3 Canals under construction

When a canal must be designed, its model must contain a parameter u^p (or a vector of parameters) with which the alternatives that one wishes to consider can be expressed. The most common case is when the canal must be sized, which means that one must define the maximum flow rate u^p that it can convey. If the description provided by equation (5.18) is sufficient, under the hypothesis that there are no distributed inflows (a very frequent case in artificial canals), the model is the following

$$q^o_{t+1} = \min\big\{u^p, q^i_{t-\tau+1}\big\}$$

Note that when $u^p = 0$, the equation describes the Alternative Zero: not to construct the canal. Each time that a model is constructed, one must remember to verify that there is a value for u^p which expresses such an alternative.

5.3.3 Empirical and Data-Based Mechanistic models

The observation that was made regarding the catchment models is still valid: the methodology for the identification of models of these classes does not depend on the component being considered. Nonetheless, many empirical models are often introduced with verbal interpretations, which create the illusion that they are mechanistic models.

We could, for example, describe a canal as a series of n cascade reservoirs: the input to the first is the flow rate entering the canal, the release from the last is the flow rate leaving the canal. This is the idea upon which the well known model proposed by Nash (1957) is based. This model differs from the model (5.16), which he proposed for a catchment, because it is a continuous-time model and because the parameters γ_i are all zero, with the exception of the first one, which is always equal to 1 (the flow, in fact, enters, only and entirely, the first reservoir). By modifying equation (5.16) as above,[15] one obtains the following formula,[16] which expresses the flow rate in the downstream section in response to a flow rate impulse in the upstream one

$$q^o_{t+1} = \frac{t^{n-1}}{(n-1)!}\kappa^n e^{-\kappa t} \tag{5.28}$$

where n is the number of cascade reservoirs. From equation (5.28) one can easily derive the time to peak τ^* and the value of peak flow q^*, which are given respectively by

$$\tau^* = (n-1)\kappa^{-1}$$
$$q^* = \kappa\frac{(n-1)^{n-1}}{(n-1)!}e^{-(n-1)}$$

These formulae are often used to estimate the values of κ and n, given the values observed for τ^* and q^*.

In any case, the remark already made on the catchment model by Nash is still valid: Nash's model of a canal is none other than a linear model, i.e. a model of the most common class of empirical models.

If the canal is very long, the number n of reservoirs to be considered often becomes too high to be acceptable. To reduce this number Dooge (1959) proposed a description of the canal as a succession of reservoirs alternated with rapids, the ith of which is described as a pure time lag τ^i. Representing the model in the frequency space[17] shows immediately that this model, apparently much more complicated and realistic than Nash's, is none other than Nash's model, in which the output (or the input) is lagged by $\tilde{\tau}$, with

$$\tilde{\tau} = \sum_{i=1}^{n} \tau^i$$

Therefore, we could conclude with observations which are completely analogous to those made in the previous section, but which it is useless to repeat.

There are other empirical models for canals. The interested reader will find an example of a neural network model in Section 6.9.1 of PRACTICE.

5.3.4 The step indicator

As with all the other components, a step indicator can be associated also with canals

$$g_{t+1} = g_t\left(q^i_{t+1}\right)$$

[15] To simplify we also will assume that the parameters κ_i, $i = 1, \ldots, n$, all have the same value κ.

[16] See equation (A3.10) in Appendix A3.

[17] The reader does not have to know the meaning of this operation.

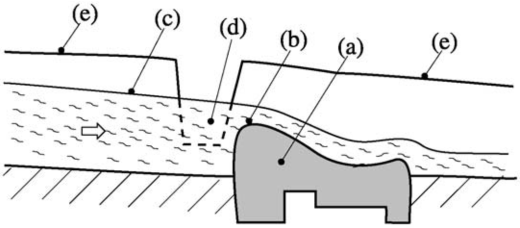

Figure 5.21: Longitudinal section of a non-regulated diversion showing the dam (a) with the spillway crest (b), the back-flow profile (c) and the inlet (d) of the diversion canal cut into the bank (e) of the water course.

which is a function of the inflow q^i_{t+1}. The step indicator g_{t+1} can, for example, express the cost of a flood on the banks of the canal, or an environmental cost which occurs when the flow rate is so low that the river biota is at risk.

5.4 Diversion dams

The branch points in a water network are hardly ever natural points, but usually artificial ones. They are created through a *diversion dam*, which is a small dam, placed across the axis of the water course. Thanks to the *back flow* (increase in water level) that is produced upstream, it is possible to entirely or partly channel the flow in a *diversion canal*, which is almost always artificial, and whose *inlet* is alongside the diversion dam (see Figure 5.21). Since the rise in water level is slight, the storage that is created upstream is insignificant for management purposes and is never considered from this point of view: the aim is just to create the *head* (water height) required by the canal. To this end the diversion dam can be *non-regulated*, when there is no possibility to modify the upstream levels as in Figure 5.21, or *regulated*, when it has mobile parts (usually *sluice gates*) with which it is possible to modify the level and thus modify the flow channelled into the canal. The diversion canal almost always has a lateral spillway that limits the diverted flow to the maximum flow that the canal can convey, in order to avoid flooding along its course. Along the stretch of river downstream of the diversion dam there is often a *minimum environmental flow* (MEF) imposed, i.e. the minimum flow value below which the residual flow should never drop, as long as the flow that reaches the diversion is not inferior.

5.4.1 The causal network

A regulated diversion dam is described by the causal network in Figure 5.22, in which q^i_{t+1} is the inflow, u_t the *diversion decision*, q^d_{t+1} the diverted flow rate and q^r_{t+1} the residual downstream flow and all the flow rates are relative to the time interval $[t, t+1)$.

When the diversion dam is non-regulated the causal network is simplified since the diversion decision disappears.

5.4.2 Mechanistic models

Given their simplicity, diversions are always described by a mechanistic model. Since the storage that is formed upstream is negligible with respect to the volumes that pass through,

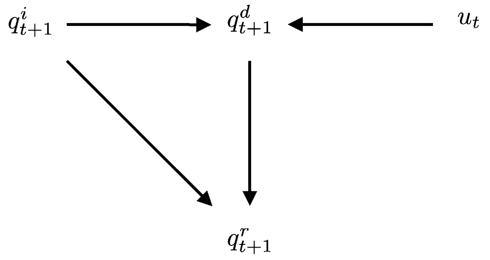

Figure 5.22: The causal network of a regulated diversion.

in the time step Δ, a regulated diversion is described by the following algebraic relations

$$q^d_{t+1} = \min\left\{u_t, \left(q^i_{t+1} - q^{\text{MEF}}_t\right)^+, q^{\max}\right\} \tag{5.29a}$$

$$q^r_{t+1} = q^i_{t+1} - q^d_{t+1} \tag{5.29b}$$

$$u_t \in \mathcal{U}_t \tag{5.29c}$$

where the operator $(\cdot)^+$ returns its argument, if this is higher than zero, while zero in the opposite case; $q^{\min}$ and $q^{\max}$ are the minimum[18] and maximum flow rates that can be diverted, and q^{MEF}_t the minimum environmental flow. Analogously, a non-regulated diversion is described by the following pair of relations

$$q^d_{t+1} = \min\left\{\left(q^i_{t+1} - q^{\text{MEF}}_t\right)^+, q^{\max}\right\} \tag{5.30a}$$

$$q^r_{t+1} = q^i_{t+1} - q^d_{t+1} \tag{5.30b}$$

The presence of a regulated diversion in a water system does not always imply that a *distribution policy* must be designed for it. The control u_t could already be set by a norm (for example, the flow to be diverted could be a percentage of the upstream flow), or could be established at each moment by others and would then be configured as a disturbance, either deterministic or random according to whether or not the decision is communicated at the moment it is taken.

5.5 Confluence points

The point of confluence between two canals or two river reaches is an extremely simple, but essential, component of a water system. Its model is constituted by the algebraic balance equation between the inflow rates $q^{i,j}_{t+1}$, $j = 1, \dots, n$ and the outflow rate q^o_{t+1}

$$q^o_{t+1} = \sum_{j=1}^{n} q^{i,j}_{t+1}$$

[18]Often a canal has a minimum operative flow rate $q^{\min}$, which is required for its correct hydraulic functioning or because with lower flow rates the inlets of the canals that it serves cannot be fed.

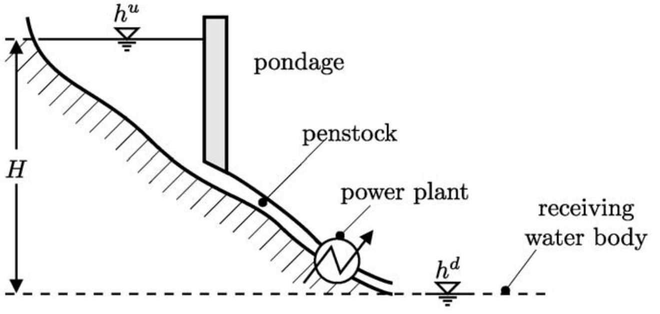

Figure 5.23: Section of a run-of-river power plant.

5.6 Stakeholders

In Section 4.10 we saw that a step indicator is a component of the model's output, which describes a Stakeholder (or a sector). Unlike the components that have been considered until this point, there are various types of Stakeholders and so there is no one model that describes them all. We must therefore settle for a couple of examples. We have chosen the two types of Stakeholder that are most frequently encountered: the hydropower plant and the irrigation district. The interested reader will find examples of other Stakeholders in Chapter 4 of PRACTICE: flood areas, the lake environment, the river environment, fishing, navigation, tourism, and mosquitoes (This last one seems a bit strange, does it not? Have a look at the documentary on the DVD of PRACTICE and you will learn more about it.)

5.6.1 Hydropower plants

There are different types of power plants. The simplest one cannot store significant quantities of water at or above its site and therefore uses only the water flowing in a stream to generate energy (*run-of-river power plant*). It is described by the section in Figure 5.23 and by the diagram in Figure 5.24. We can see that the power plant is situated at the foot of a *penstock* (with a maximum flow rate $q^{\max}$), which is fed by a short-term storage called *pondage*. The latter is serviced by a diversion canal, into which runs a flow q^d_{t+1}, that is diverted from a water course with a flow rate q^i_{t+1}, through a diversion dam following the *diversion decision* u_t. The *hydraulic head* H is the difference between the water level in the

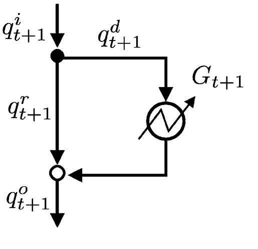

Figure 5.24: The diagram of a run-of-river power plant.

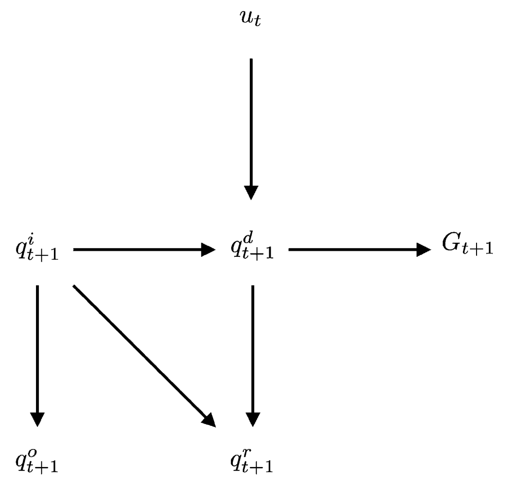

Figure 5.25: The causal network of a run-of-river power plant.

pondage and the water level in the *receiving water body*[19] (not the elevation of the turbine, given that this can, within certain limits, work in aspiration). The flow rate in the penstock is regulated so that it is always equal to the diverted flow rate, and therefore the water level of the pondage is constant. It follows that when the water level of the receiving water body is also time-invariant, the hydraulic head H is constant. The flow q^d_{t+1} that passes through the penstock moves a turbine whose drive shaft is integral with the drive shaft of an alternator. In this way, the group transforms the kinetic energy from the hydraulic stream that runs through the penstock into electric energy G_{t+1} which is delivered to the network. There is often a constraint q^{MEF}_t imposed on the residual flow rate q^r_{t+1} in the stretch of river downstream from the power plant, to ensure minimum environmental flows, as we saw when we dealt with the diversion. Sometimes a minimum value $q^{\min}$ is defined for the flow rate q^d_{t+1}, below which the turbine cannot be activated. In the end, the flow q^o_{t+1} returned to the water course downstream of the power plant is equal to the flow entering the diversion dam.

The causal network that describes the power plant is shown in Figure 5.25 and, given the above, can be translated immediately into the following mechanistic model

$$q^d_{t+1} = \begin{cases} 0 & \text{if } (q^i_{t+1} - q^{\text{MEF}}_t) < q^{\min} \\ \min\{u_t, (q^i_{t+1} - q^{\text{MEF}}_t)^+, q^{\max}\} & \text{otherwise} \end{cases} \tag{5.31a}$$

$$q^r_{t+1} = q^i_{t+1} - q^d_{t+1} \tag{5.31b}$$

$$G_{t+1} = \psi \eta^g g \gamma q^d_{t+1} H \tag{5.31c}$$

$$q^o_{t+1} = q^i_{t+1} \tag{5.31d}$$

$$u_t \in \mathcal{U}_t \tag{5.31e}$$

where

G_{t+1} is the energy produced [kWh] in the time interval $[t, t+1)$ whose duration is the modelling time step Δ [s];

[19]This is valid when the load losses in the penstock are negligible. If they are not, they must be subtracted.

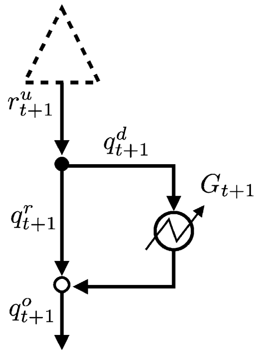

Figure 5.26: The diagram of a storage power plant.

ψ is a coefficient of dimensional conversion, whose value is $\Delta/(3.6 \cdot 10^6)$;
η^g is the turbine efficiency [–];
g is gravitational acceleration, equal to 9.81 $\mathrm{m/s^2}$;
γ is water density, equal to 1000 $\mathrm{kg/m^3}$.

A second type of power plant is characterized by the presence of a reservoir (*storage power plant*), from which the flow q^d_{t+1} is diverted through an intake structure. The power plant can still be described by the diagram in Figure 5.24 in which the reservoir takes the place of the diversion, but such a diagram would require the description of a reservoir with a model that is a little different from the one we adopted in equation (5.13), because in this equation the release r_{t+1} from the reservoir is defined as the sum of the flow diverted by the intake structure and the flow that passes through the spillway. Therefore, in order to be able to use the model (5.13) we prefer to substitute the diagram in Figure 5.24 with the diagram in Figure 5.26, in which a logical non-regulated diversion separates the spilled flow q^r from the diverted flow q^d.

The model of the power plant is thus the following

$$q^d_{t+1} = \begin{cases} 0 & \text{if } (r^u_{t+1} - q^{\mathrm{MEF}}_t) < q^{\min} \\ \min\{(r^u_{t+1} - q^{\mathrm{MEF}}_t)^+, q^{\max}\} & \text{otherwise} \end{cases} \tag{5.32a}$$

$$q^r_{t+1} = r^u_{t+1} - q^d_{t+1} \tag{5.32b}$$

$$G_{t+1} = \psi \eta^g g \gamma q^d_{t+1} H \tag{5.32c}$$

$$q^o_{t+1} = r^u_{t+1} \tag{5.32d}$$

where r^u_{t+1} is the release from the upstream reservoir. Note that the release decision no longer appears, because it is included in the reservoir model.

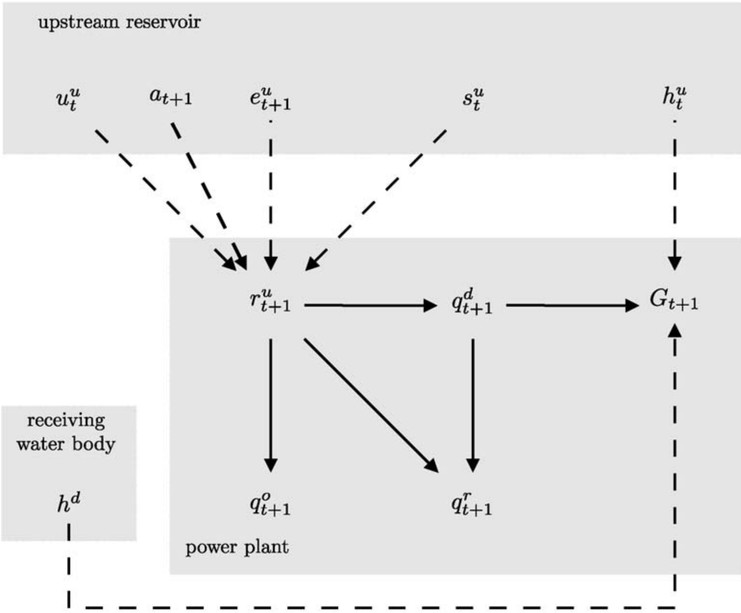

Figure 5.27: The causal network of a storage power plant.

In some power plants the penstock begins at the intake structure and the role of the pondage is performed by the reservoir. In this case the water level upstream of the penstock is no longer constant and the hydraulic head H becomes a function of the level h_t^u in the reservoir and, through it, also of the storage s_t^u; precisely

$$H = h_t^u\left(s_t^u\right) - h^d$$

where h^d is the level in the downstream receiving water body, which we assume to be constant. This dependence can never be ignored.

The two types of power plants that we have described are *generation plants* in which the power flows from the water, which passes through the penstock, to the electrical network. Others, called *reversible* power plants, allow the reverse process, i.e. the network provides power to the alternator, which, acting as a motor, turns the turbine to pump water from the receiving water body back to the reservoir. In some cases the pumping facility (motor and pump) can be physically distinct from the generation facility, but continues to share the same penstock. There are also purely *pumping* power plants. Reversible and pumping plants are used to accumulate the power supplied by thermal plants that cannot be shut down at night, when energy demand decreases, in the form of the potential energy held in the stored water.

We will now describe the model of a purely *pumping* power plant (Figure 5.28) and postpone describing the model of a reversible plant until Section 6.4.2, given that it is not a component, but an aggregate.

The flow rate q_{t+1}^{pot} that could potentially be pumped is not decided by the Regulator; instead, it is determined by the amount of water which can be lifted given the energy ε_{t+1}^p

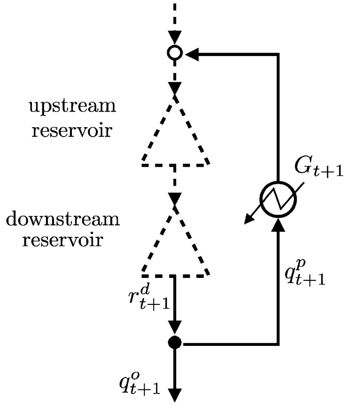

Figure 5.28: The diagram of a pumping power plant.

that the electrical network provides during the night. Because this amount is only known after the decision has been taken, ε^p_{t+1} is a disturbance and this explains the subscript $t+1$ of q^{pot}_{t+1}, while its superscript points out that it is only potential. Its value could be, in fact, greater than the capacity $q^{\max}$ of the penstock, or the flow rate r^d_{t+1} that can be pumped from the receiving water body, which is almost always a reservoir, or even greater than the maximum value that can be stored in the upstream reservoir, which is equal to the difference between its active storage $\bar{s}^u$ and the existing storage s^u_t. Therefore, the causal network of the plant is the one shown in Figure 5.29 and so, the model is the following

$$q^{\text{pot}}_{t+1} = \varepsilon^p_{t+1} / \left[\psi \eta^p g \gamma \left[h^u_t\left(s^u_t\right) - h^d_t\left(s^d_t\right)\right]\right] \tag{5.33a}$$

$$r^d_{t+1} = R^d_t\left(s^d_t, q^{\text{pot}}_{t+1}, r^u_{t+1}, e^d_{t+1}\right) \tag{5.33b}$$

$$q^p_{t+1} = \min\left\{q^{\text{pot}}_{t+1}, r^d_{t+1}, \left(\bar{s}^u - s^u_t\right)^+, q^{\max}\right\} \tag{5.33c}$$

$$G_{t+1} = \psi \eta^p g \gamma q^p_{t+1}\left[h^d_t\left(s^d_t\right) - h^u_t\left(s^u_t\right)\right] \tag{5.33d}$$

$$q^o_{t+1} = r^d_{t+1} - q^p_{t+1} \tag{5.33e}$$

The superscript u denotes the upstream reservoir, and d the downstream one; q^p_{t+1} is the flow rate that is actually pumped, q^o_{t+1} the flow rate that runs downstream from the downstream reservoir, and η^p is the efficiency of the pump. The hydraulic head is the difference between the water elevations h^u_t and h^d_t of the upstream and downstream reservoirs respectively. Note that the hydraulic head is negative and therefore also the energy G_{t+1} is negative: in fact, it must be provided.

The step indicator used for hydropower plants is often the energy produced (consumed, in the case of a pumping plant) and so the equation that expresses it is either (5.31c), (5.32c) or (5.33d) according to the particular case. Sometimes, however, it is preferable to consider economic quantities rather than physical ones: the return, the willingness to pay or the social cost; these last two indicators are for the most part adopted when the economic approach

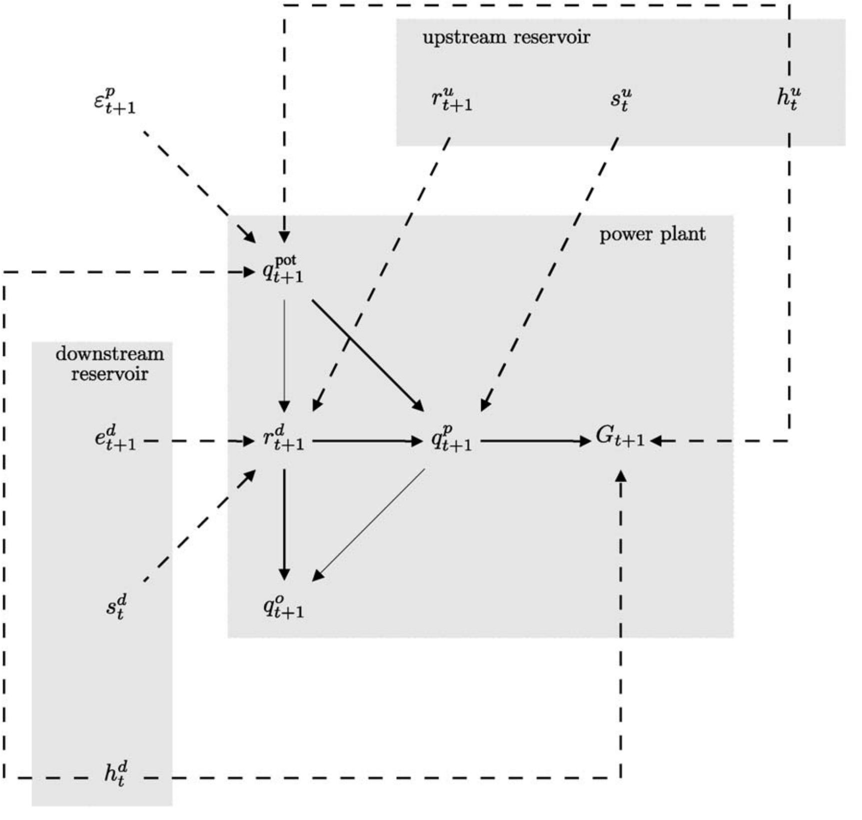

Figure 5.29: The causal network of a pumping power plant.

used in the Project is a Cost Benefit Analysis. The value of the first indicator is obtained by multiplying G_{t+1} by the price (which can also be a function of G_{t+1} itself). The values of the second and third are obtained by transforming G_{t+1} with the suitable functions.

5.6.2 Irrigation districts

The most natural indicator of annual production is the harvested biomass (*harvest*), or the lost harvest with respect to the potential harvest, given that from both it is easy to obtain other interesting indicators, such as the economic return. Nevertheless, since it is difficult to evaluate both of these indicators, one often relies on their proxy[20]: the *average annual potential damage* (*stemming*) *from the stress*, defined by the following expression

$$i_{\text{Irr}} = \frac{1}{N} \sum_{a=1}^{N} f(F_a) \tag{5.34a}$$

[20] For the definition of the term see page 81.

where $f(\cdot)$ is the function[21] expressing the potential damage from the stress and $\digamma_a$ denotes the *maximum stress* that occurred during the year[22] a. The latter is, in turn, given by

$$\digamma_a = \max_{t \in a} \frac{1}{\delta} \sum_{\tau=t-\delta}^{t} (w_\tau - q_\tau)^+ \tag{5.34b}$$

where the number of days δ considered in the sum depends on the *field capacity* of the soil,[23] and the *deficit* $(w_\tau - q_\tau)^+$ is the difference between the *water demand* w_τ and the water (*supply*) q_τ provided to the crop on day τ, when the demand is greater than the supply, while it is zero in the opposite case. The indicator i_{Irr} is clearly non-separable[24] and so, unlike what one would expect, a step indicator cannot be defined: this is a signal that equation (5.34) does not correctly describe a crop. We will tackle this important question when we get to Section 10.2.2.2. For now we will simply observe that, to be able to evaluate equation (5.34), the model of the irrigation district must provide the water demands w_t for all the crops for every time t.

5.6.2.1 The demand scenario

The simplest way to obtain these demands is to ask an Expert. For each crop he can provide the *demand scenario* $\{w_t\}_0^{T-1}$ that is expected in the irrigation district, according to his experience and taking into account the crop characteristics, the irrigation system, and the current agricultural practices in the area. Assuming then, that the demand is periodic, we can obtain its trajectory along the whole time horizon of the project: this is the simplest and most commonly used model.

Not always, however, is this solution correct. A moment's reflection is enough to realize that the description of a cultivation cannot be so simple: if, for several days it is not irrigated, its water demand becomes greater than the demand of a regularly watered crop. This means that the irrigation district is a dynamical system, whose state must include the state of the crops. Furthermore, if at the beginning of the year the farmers choose to plant dry crops, the supply amount would not have any influence on the harvest, just as the supply becomes useless if the farmers (even if they have planted wet crops) decide to stop irrigation because they think their crops will not be profitable enough. Therefore, the harvest depends on human expectations and decisions, as well as on the dynamics of the crop. We can thus understand that the trajectory $\{w_t\}_0^{T-1}$ provided by the Expert is the demand that the system presents when everything proceeds in a normal way, including the supply of water, which cannot deviate too much from its usual values. Thus the demand trajectory is a good model for small variations in supply, but it is not when the variations are significant, as can occur in drought conditions, or when a change in the status quo is planned. Therefore, if one thinks that one of these two eventualities may occur, it is necessary to use a more complex model, which we can identify by proceeding with successive block diagrams as was shown for the catchment (Section 5.2.1).

[21] The reader who is interested in its form will find it in Figure 4.22 of PRACTICE.

[22] So as not to make the notation too cumbersome, here and in the following we will use the symbol a to denote both the year, such as the number of the year, and also the set of days of that year.

[23] The *soil moisture* value at which the balance between capillary forces and the force of gravity results in no water percolation.

[24] For the definition of this term see page 85.

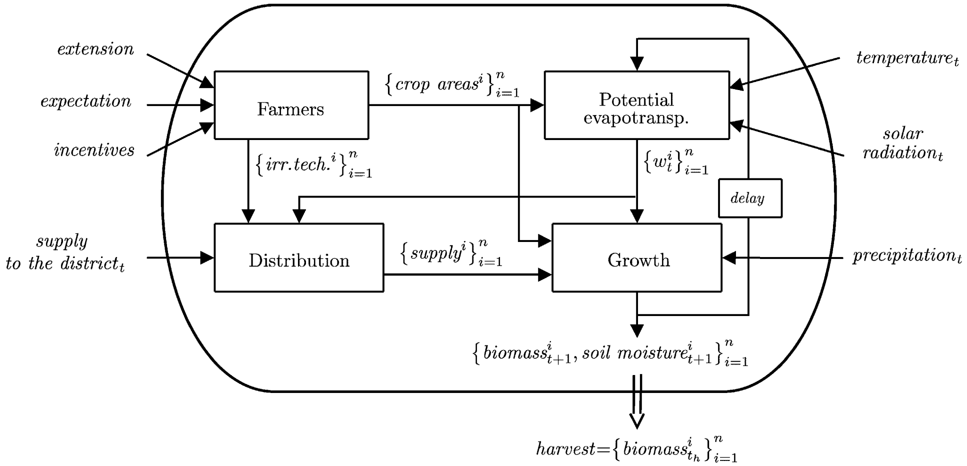

Figure 5.30: The block diagram of a district; the index i denotes the different crops.

5.6.2.2 The block diagram

To give the discussion a better focus, consider a real case study, which was dealt with in the Vomano Project (see the box on page 147): the Consorzio di Bonifica Nord (CBN) would like to assess the opportunities for extending its irrigation district from 7000 ha to 14 000 ha, and to create a new network of feeder canals. The CBN believes that in this way agriculture would get a big boost, because the farmers would be able to adopt wet crops, which are much more profitable that their present products. To feed the extended district, the CBN would have to increase its off-take, and come to an agreement with ENEL, which manages the hydropower plants upstream. This agreement would not be easy to achieve, since there is a strong conflict between irrigation and hydroelectric priorities and ENEL holds the rights for the water that the CBN would like to use. Therefore, in order to evaluate the Project we must identify the models of the reservoir network and of the hydropower plants, and also a model of the district, which we will focus upon now.

One begins by identifying its output (the harvest, i.e. the biomass at harvest time t_h) and its inputs: the meteorological variables (temperature, precipitation, solar radiation, etc.) and the variables that influence the behaviour of the farmers. This last group of variables includes three variables. The first is the *extension* of the district, given that if it were increased, the choice between wet crops and dry crops would be given to a greater number of farmers. The second is the *expectation* that farmers have for the supply, a qualitative variable which expresses how probable the farmers think it is, given ENEL's interests, that the demand of wet crops would be effectively satisfied: with a low expectation, few farmers would choose to plant wet crops. The third is the amount of *incentives* that will be offered to adopt more water-saving irrigation systems, if the irrigation infrastructure in the fields already exists, or to create it from scratch, if it does not.

Once inputs and outputs have been defined we try to identify the processes that link them. Through interviews with Experts, the existing literature, and the analysis of the problem, one may obtain the diagram in Figure 5.30. In response to the last three above-mentioned inputs, the first block (Farmers) provides the farmers' decision: the area to be cultivated with each individual wet crop and the mix of irrigation techniques (micro-irrigation and sprinklers) with which each crop will be served. At each time t, for each crop, the second block (Potential evapotranspiration) provides the demand w_t^i, taking into account *temperature*, *solar radiation*, the crop *biomass* and *soil moisture*. The third block (Distribution) describes how the supply to the district is distributed among the different crops, while taking the leakage in the distribution network into account. The fourth (Growth) provides, for each crop, the growth in the *biomass* and the variations in the *soil moisture* in response to the supply that it receives, the demand w_t^i and the *precipitation*. Lastly, the output is the harvest, i.e. the vector of the *biomass* values at the time t_h of the harvest. The model is thus dynamical and its state $\mathbf{x}_t$ is the vector of the *biomass* and *soil moisture* values of each crop. Furthermore, the model is time-varying because, other things being equal, the evapotranspiration and the growth depend on the phenological phase of each crop.

By studying the diagram in Figure 5.30 we understand better the previous observations about the demand scenario: the diagram shows very clearly how, at every time step, the demand of the ith crop depends on the behaviour of the farmers and on the supply that crop has received in the past, the effects of which are displayed through the actual value of its biomass and soil moisture. Assuming a demand scenario a priori is thus equivalent to assuming that the state of the crops will follow a predetermined trajectory, which can actually be followed only if the supply is sufficient. Therefore, if the latter satisfies the demand, the state follows the trajectory and the scenario description is acceptable; but, if at any given time the supply were insufficient and, as a consequence, the state $\mathbf{x}_t$ diverged from the trajectory, the scenario would lose its meaning. From that moment the demand could be estimated only by a dynamical model, which becomes necessary also when one must take account of variations in the structure of the system, such as variations in the extension of the district.

To identify the model we should now open the four blocks one by one, to identify the processes within them. It would be a useless and tedious operation, as the reader who has read Appendix A5 may understand, since it would be necessary to describe these processes, but such a description exceeds the purpose of this chapter. Thus, we will simply observe that for three of the blocks, Potential evapotranspiration, Distribution, and Growth, the analysis can be conducted in a similar way to the one in Appendix A5, but this is not so for the Farmers block, given that there are no physical laws that describe their behaviour. Thus, it is useful to examine this block in greater detail.

5.6.2.3 A BBN of the farmers' behaviour

When the farmers' choice is made between dry crops, among which it is not necessary to make a distinction, and a single wet crop, such as cauliflower, the causal network that describes this choice is the one shown in Figure 5.31.

The cultivated crop area is described by the variable S, the area planted with cauliflowers, which naturally must respect the constraint $S <$ *extension*, and the *irrigation technique* is described by the variable *%micro* which specifies the fraction of S that is served with micro-irrigation; by exclusion the remaining part of the arable surface is irrigated by sprinklers, given that we have assumed that one can choose between only two techniques. The type of model that is best suited for representing this network is a BBN (Section 4.2), because the only information that we can acquire are the probabilities that the farmers will opt for the different choices (see Castelletti and Soncini-Sessa, 2007a, 2007b).

To demonstrate how a BBN is defined in practice, we consider the slightly more complicated case, where there are two wet crops: cauliflower (*cau*) and the pair tomato–maize

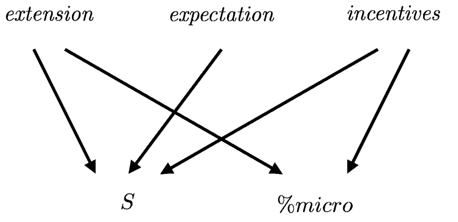

Figure 5.31: The causal network for the Farmers block when their choice is between dry crops and a single wet crop.

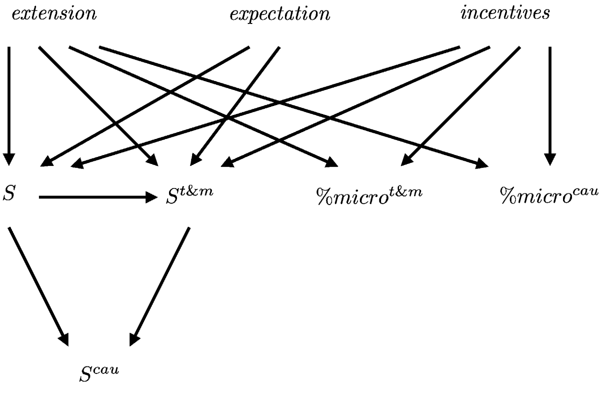

Figure 5.32: The causal network for the Farmers block when they must choose between two irrigated crops: cauliflower (*cau*) or the rotation between tomato and maize ($t\&m$).

($t\&m$), which can be grown in rotation, so that the soil is not overexploited. Figure 5.32 shows the causal network which describes this case, where S is the total area destined for irrigated crops, $S^{t\&m}$ and S^{cau} are the cultivated areas of tomato–maize and cauliflower and $\%micro^{t\&m}$ and $\%micro^{cau}$ the fractions of these that are served by micro-irrigation.

Among such variables the following constraints hold[25]: $S \leqslant extension$, $S^{t\&m} \leqslant S$ and $S^{cau} = S - S^{t\&m}$.

In a BBN the variables must be discrete and so, if they are continuous as in our case, they are discretized in classes. Let us hypothesize then that the variable *expectation* can assume[26] three values (low, medium and high), *extension* two (7000 ha and 14 000 ha), *incentives* three (none, 300 €/ha and 600 €/ha), that the areas S, $S^{t\&m}$ and S^{cau} assume values in five classes (0, 3500, 7000, 10 500 and 14 000 ha) and finally that the percentages of micro-irrigation $\%micro^{t\&m}$ and $\%micro^{cau}$ assume values in two classes (0–50% and 51–100%). With these hypotheses the BBN is described by the 5 conditional probability tables (CPT) shown in Figure 5.33 and is completely defined once values have been assigned to all their elements.

[25] The network in Figure 5.32 is not the only possible network: one could firstly specify $S^{t\&m}$ and S^{cau} and then derive S by summing.

[26] In order to keep the example simple, the number of classes must be very small, but this makes the values that they define extremely unrealistic; the reader should consider them as mere labels which serve to explain how the model is made and used.

(a)

	extension	7000			14000		
	expectation	low	medium	high	low	medium	high
S	0	0.9	0.7	0.5	0.5	0.3	0.1
	3500	0.1	0.3	0.5	0.4	0.4	0.3
	7000	0	0	0	0.1	0.2	0.4
	10500	0	0	0	0	0.1	0.1
	14000	0	0	0	0	0	0.1

incentives

(b)

	extension	7000			14000		
	expectation	low	medium	high	low	medium	high
$S^{t\&m}$	0	0.8	0.7	0.5	0.4	0.3	0
	3500	0.2	0.3	0.4	0.3	0.3	0.4
	7000	0	0	0.1	0.3	0.2	0.3
	10500	0	0	0	0	0.2	0.2
	14000	0	0	0	0	0	0.1

incentives

(c)

	extension	7000			14000		
	incentives	none	300	600	none	300	600
$\%micro^{t\&m}$	0-50	0.9	0.7	0.5	0.9	0.7	0.5
	51-100	0.1	0.3	0.5	0.1	0.3	0.5

(d)

	extension	7000			14000		
	incentives	none	300	600	none	300	600
$\%micro^{cau}$	0-50	0.9	0.5	0.2	0.9	0.5	0.2
	51-100	0.1	0.5	0.8	0.1	0.5	0.8

(e)

	S	3500					7000					10500					14000				
	$S^{t\&m}$	0	3500	7000	10500	14000	0	3500	7000	10500	14000	0	3500	7000	10500	14000	0	3500	7000	10500	14000
S^{cau}	0	0	1	1	1	1	0	0	1	1	1	0	0	0	1	1	0	0	0	0	1
	3500	1	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0
	7000	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	0
	10500	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0
	14000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Figure 5.33: The conditional probability tables (CPT) of the BBN that describes the farmers' choice between two irrigated crops. The dark grey cells correspond to violations of a constraint.

Remember that each element represents the probability that the dependent variable will assume the value that corresponds to the row, conditioned by the fact that the variables that influence it assume the values which are specified by the column; for example from the first table one deduces that 0.3 is the probability that the total irrigated crop area S falls in the first class when the *expectation* is 'medium' and the *extension* is '14 000 ha'.

Estimating the parameters We must now explain how such conditional probabilities can be estimated, i.e. how the model is calibrated, a phase that in jargon is referred to as 'populating the BBN'. When a table represents a deterministic relationship its elements are defined on the basis of that relationship: this is the case for table (e) in Figure 5.33, which expresses the relationship $S^{cau} = S - S^{t\&m}$, as one can see by observing the disposition of the values 1 and 0. Note how such a simple relationship is expressed by a relatively awkward table.

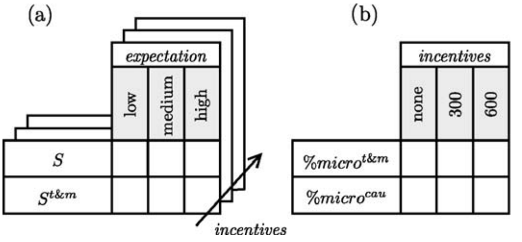

Figure 5.34: The tables for the interview with the farmers (simple approach).

The disadvantage of BBN is just this: the representation of algebraic relations is extremely onerous. The existence of constraints allows us to establish a priori that several probabilities (those corresponding to the dark grey cells in Figure 5.33) are either zero or irrelevant. They are zero when the constraint acts on the conditioned variable, such as in table (a), where the constraint $S \leqslant extension$ prevents the irrigated crop area S from assuming the values 10 500 and 14 000 when *extension* is equal to 7000. Instead, they are irrelevant when the constraint concerns only the conditioning variables, as for example in table (e), where the constraint $S^{t\&m} \leqslant S$ makes it impossible that pairs $(S^{t\&m}, S)$ with $S^{t\&m} > S$ could ever occur. Therefore, arbitrary values can be assigned to the elements in those columns, as long as the sum of those values is equal to one.[27] As far as the remaining values are concerned, either they are estimated by Experts according to their experience, or they are derived from interviews with farmers. The first possibility is self explanatory, while the second deserves a further remark.

When we interview a farmer, we know that his/her plot of land is either located in the existing district, or in the proposed enlargement, and so the responses to the questions that we ask are conditioned by this fact, i.e. by the value corresponding to the variable *extension*.[28] We can interview them in two different ways, a simple one and a more complicated one. When we use the simpler approach, we show the farmers the two tables in Figure 5.34 and ask them to fill them in.

The first requires the farmer to specify, for every possible value of *expectation*, the number of hectares that (s)he intends to dedicate to wet crops, and, in particular, to tomato–maize. The second table requires the farmer to specify, for every possible value of *incentives*, the percentage of surface area for each crop that (s)he intends to irrigate by means of the micro-irrigation. In the more complex approach the tables are formulated so that the farmer's reply could be a probability distribution, not just a value. For example, in Figure 5.35 the table for requesting the percentage of surface area to be devoted to wet crops is shown: corresponding to every *expectation* value a column of cells appears, to each of which a class of crop area is associated. The interviewee fills in each cell with a subjective estimation of the probability that his/her choice falls into that class. From the statistical elaboration of the data collected by means of one or the other approach, it is possible to obtain the conditional probability estimations with which the tables in Figure 5.33 are completed. Once the BBN has been identified, it provides the conditional probability distributions for its two

[27] This condition is conceptually irrelevant, but practically useful for the formal verification of the accuracy of the table in the computer codes.

[28] This holds only if in the enlarged district, in the case of a supply deficit, priority of supply is assured to the farmers of the existing district; if it were not, the responses from these farmers might differ according to whether their land is in the existing district or in the enlargement.

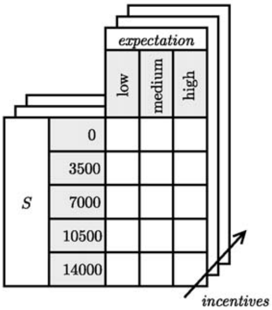

Figure 5.35: One of the tables for the interview with the farmers (complex approach).

output vectors (*irrigation technique* and *crop areas*, see Figure 5.30), which are conditioned on the values taken on by the three inputs. These outputs act as stochastic disturbances on the dynamical models that describe the blocks Potential evapotranspiration and Distribution. These models, just as the Growth model, can be created either with a BBN, or with other kinds of model, i.e. it is not necessary that all the models of a block diagram be of the same type, given that, as we demonstrated in Chapter 4, from any type of model it is possible to derive the state-space representation (4.5) from which one obtains the model of the entire system.

Validation Given the form of a BBN, its validation might seem difficult, but actually it is no different to that of any other model: with a pair of observed input and output trajectories one can verify whether the model, when it is fed with the observed input, produces an output trajectory 'not too far' from the one observed. This test can be conducted with a BBN just as with any other model. In the case of the irrigation district in the Vomano Project, for example, it was possible to derive the average flows supplied to the farmers in the years 1993–2002 (see the continuous curve in Figure 5.36) from the data provided by the CBN. By feeding the model represented by the diagram in Figure 5.30 with the historical series of meteorological variables in the same years and the values for *incentives*, *extension* and *expectations*, which correspond to the historical situation,[29] the trajectory of the average demand including all the crops was obtained (dashed curve in Figure 5.36). The comparison between the two curves shows that the model is satisfactory. Note in particular that the summer peak in the demand (4.5 m^3/s) is not covered by the supply, because the CBN's concession is only 4.1 m^3/s. Thus, the model suggests that there is a structural deficit in the summertime and this was confirmed by the CBN: further proof of the model's validity. Pay attention however: validation is always limited to those 'modes' of model behaviour that are excited in the validation experiment. For example, in the case in question we do not know anything about the effective capacity of the model to explain what might happen when the *incentives* are '600 €/ha', because this case never occurred in the historical period.

[29]These values were 'none', '7000 ha' and 'high' respectively. The historical *expectation* was 'high' and the comparison between the calculated demand and the supply to the farmers is significant, because the CBN claims to have always satisfied the demand within the limits of its possibilities: in fact, in critical moments, it purchased from ENEL the water volumes that it foresaw would have not been otherwise available.

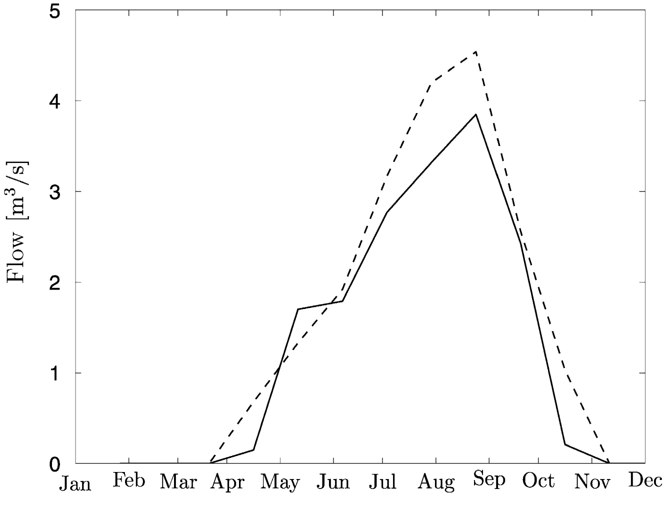

Figure 5.36: The calculated demand (dashed line) of the irrigation district and the historical supply (continuous line).

5.7 Disturbance

As we have stressed on many occasions, various disturbances act upon the different components of the system. The outflow from a catchment is, for example, influenced by precipitation; in a reservoir, the storage depends on the inflow and evaporation; the harvest of an irrigation district depends on solar radiation, temperature and precipitation; in a canal there are water losses due to seepage. Up to now, a disturbance has been defined as such with respect to the model of the component that is being considered, but the possibility that it can be explained by a suitable model as a function of the values taken on by other variables and by its past values is not to be excluded. For example, inflow is a disturbance in a reservoir model, but it is also the output of a catchment model. Note that in this way we are shifting our attention away from the component onto the system. If the disturbance of a component is explained by a model, it is no longer a disturbance for the system, but an internal variable of the global model, since it is the output of one component (i.e. the model of the disturbance itself) and the input of another. We will examine this aspect in detail in Section 6.1.3. From a global point of view, the variable that is a candidate disturbance is (if it exists) the input disturbance to the new model, in the example the rain on the catchment. In Section 4.6 we intuited that this chain would break when all of the disturbances in the global model were either deterministic or *purely random*.

From the above it follows that, while the state of a model of a component is certainly part of the state of the global model, the disturbance for a component is also a disturbance for the global model only if it is not explained by a model. More precisely, the disturbance should not be or cannot be explained: it should not be when it is a deterministic variable; and it cannot be when it is purely random. Ascertaining whether the disturbance is a deterministic variable is quickly done: just check if its value is deterministically known at every time t, as happens, for example, with decisions made by others that are communicated as soon as they are taken. Instead, to verify whether a disturbance is purely random a statistical *whiteness test* must be applied to one of its time series: if the test confirms the hypothesis, we can be sure that it is a disturbance for the global system and we build a model for it, as explained in the next paragraph; if the hypothesis is rejected, it means, instead, that there is an empirical model that explains that variable. The easiest solution is to identify it with the procedure described in Appendix A6, but there is also a second possibility that we will examine in Section 6.4.1, after we have seen how to assemble the global model.

Let us now suppose that we have ascertained that the candidate global disturbance variables are either deterministic or purely random and examine the models that describe them.

5.7.1 Empirical models

5.7.1.1 Deterministic disturbances

If the disturbance $\mathbf{w}_t$ has been classified as deterministic, it means that its value is deterministically known at time t. Thus, it can be described by a deterministic model, i.e. a model without stochastic inputs, but in that case it would no longer be a disturbance for the global model. Therefore, the last possibility is that it be described by a trajectory $\{\mathbf{w}_t\}_0^{h-1}$ over the time horizon that we are interested in.

5.7.1.2 Purely random stochastic models

The purely random stochastic disturbance $\boldsymbol{\varepsilon}_{t+1}$ is completely described when its probability distribution[30] $\phi_t(\cdot)$ is known (note that, because $\boldsymbol{\varepsilon}_{t+1}$ is a vector, such a distribution is the *joint distribution* of its components). Such probability is not conditioned by the value assumed by any other variable, exception made for planning decisions. Therefore

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{u}^p\right) \tag{5.35}$$

If the distribution is time-varying, we will assume that it is periodic of period T, i.e.

$$\phi_t\left(\cdot|\mathbf{u}^p\right) = \phi_{t+kT}\left(\cdot|\mathbf{u}^p\right) \quad t = 0, 1, \ldots; \ k = 1, 2, \ldots$$

The estimation of the distribution is made with the procedures described in Appendix A2.

5.7.1.3 Purely random uncertain disturbances

A disturbance is uncertain when one does not have sufficient knowledge to associate a probability (or a density of probability) to each of its possible occurrences, and one knows only that the values that it may assume are the elements of a set $\varXi_t$, which can depend on planning decisions

$$\boldsymbol{\varepsilon}_{t+1} \in \varXi_t\left(\mathbf{u}^p\right) \tag{5.36}$$

For this reason, we say that the disturbance has a *set-membership* description. As in the previous case, if the set $\varXi$ is time-varying we will assume that it is periodic of period T, which means

$$\varXi_t\left(\mathbf{u}^p\right) = \varXi_{t+kT}\left(\mathbf{u}^p\right) \quad t = 0, 1, \ldots; \ k = 1, 2, \ldots$$

[30] Expressed by a Probability Density Function (PDF), when the variable is continuous, or a Discrete Density Function (DDF), when the variable is discrete (see Appendix A2).

Chapter 6

Aggregated models

RSS and EW

In the last chapter we dealt with meta-models of the components; in this one we will show how they can be interconnected to create aggregated models, the first and most important of which is the *global model* of the system. However, the global model is not the only aggregated model since, as we will see, also distribution networks turn out to be very useful aggregated models.

6.1 Identification procedure

6.1.1 Decomposing the system

To explain the identification procedure for aggregated models we will use as example the Piave Project described in the box on page 187. The first step in creating the global model is to single out the system components. By elaborating the information in the box we can construct the block diagram in Figure 6.1.

The mountainous part of the system, which we call Upper Piave (UP), is made up of six catchments ($C1, \dots, C6$), three reservoirs ($R1$, $R2$, $R3$), two regulated diversions ($D1$, $D2$) and three hydropower plants ($H1$, $H2$, $H3$), as well as artificial canals, river stretches, and the confluence points which connect these components. The river stretches $S1$ and $S2$ are significant from an environmental perspective. In the real system the reservoir $R2$ has two intake structures: one feeds the penstock to the hydropower plant $H2$ and the other discharges into the Piave. The meta-model of the reservoir that we use has only one outlet, which must therefore be divided into two parts by the logical diversion $D2$, as is explained on page 172. The catchments of the tributaries Boite and Maè are described with two components (see Figure 6.2): the first ($C4$) represents that part of their catchments whose outflow feeds the power plant $H2$, while the second ($C3$) describes the remaining part, whose outflow reaches the Piave directly.

The downstream part of the system, which we will call Middle Piave (MP), has no reservoirs and includes four catchments ($C7, \dots, C10$), six irrigation districts ($I1, \dots, I6$), five hydropower plants ($H4, \dots, H8$) and six regulated diversions ($D3, \dots, D8$), as well

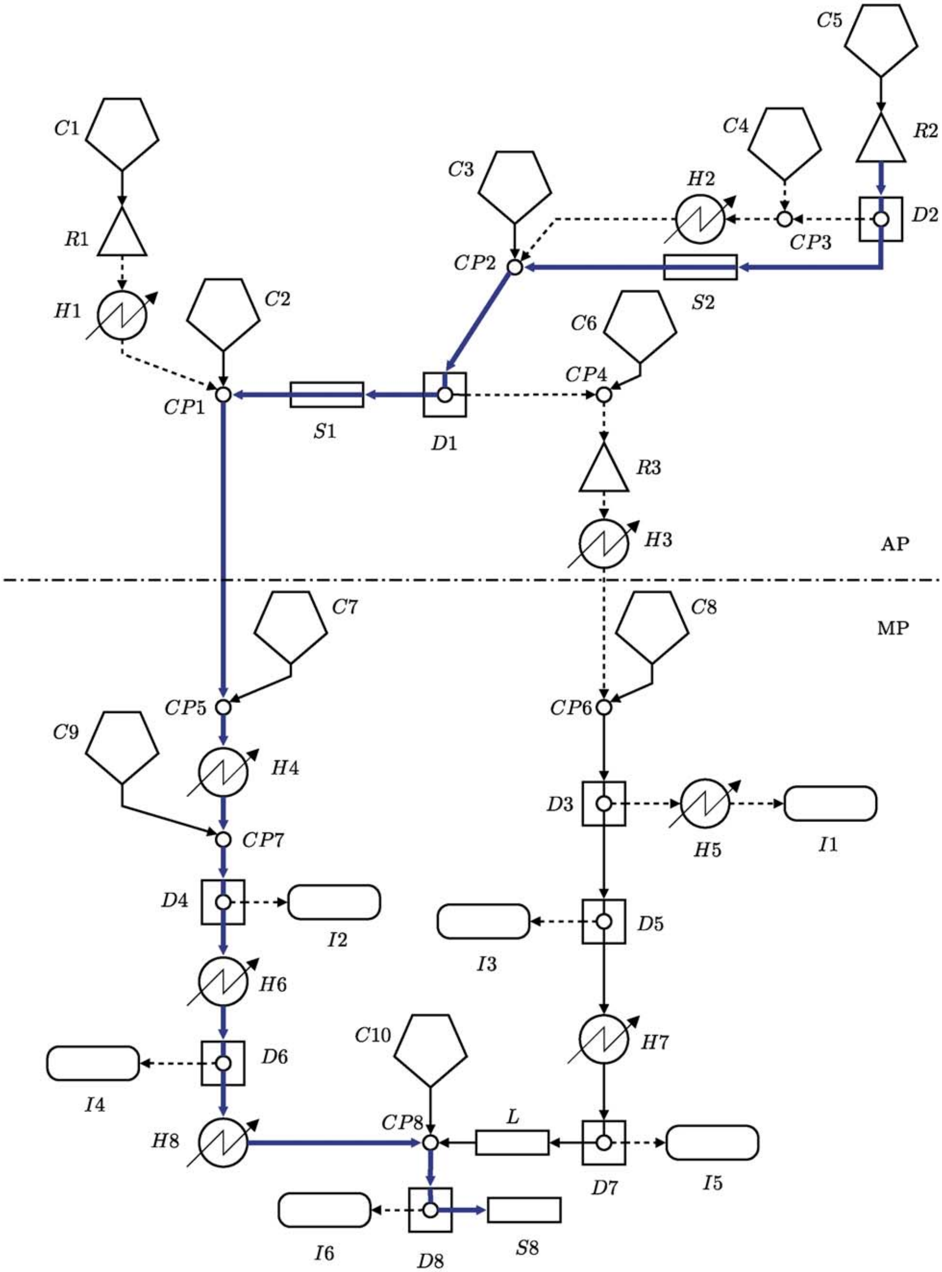

Figure 6.1: The block diagram for the Piave system. The dashed lines represent artificial canals, the bold ones represent the Piave river channel.

The Piave Project

System description In the Piave catchment there are many reservoirs and several regulated lakes (Figure 6.2), most of which, due to their relatively small capacity, can only operate on a daily water storage and release cycle. However, the three largest, the artificial reservoir Mis ($R1$) and the regulated lakes Pieve di Cadore ($R2$) and S. Croce ($R3$), which together constitute 80% of the total storable volume (215 Mm^3), are large enough to be operated on a seasonal cycle. For didactic purposes we will suppose that the shores of Lake S. Croce are subject to occasional flooding.

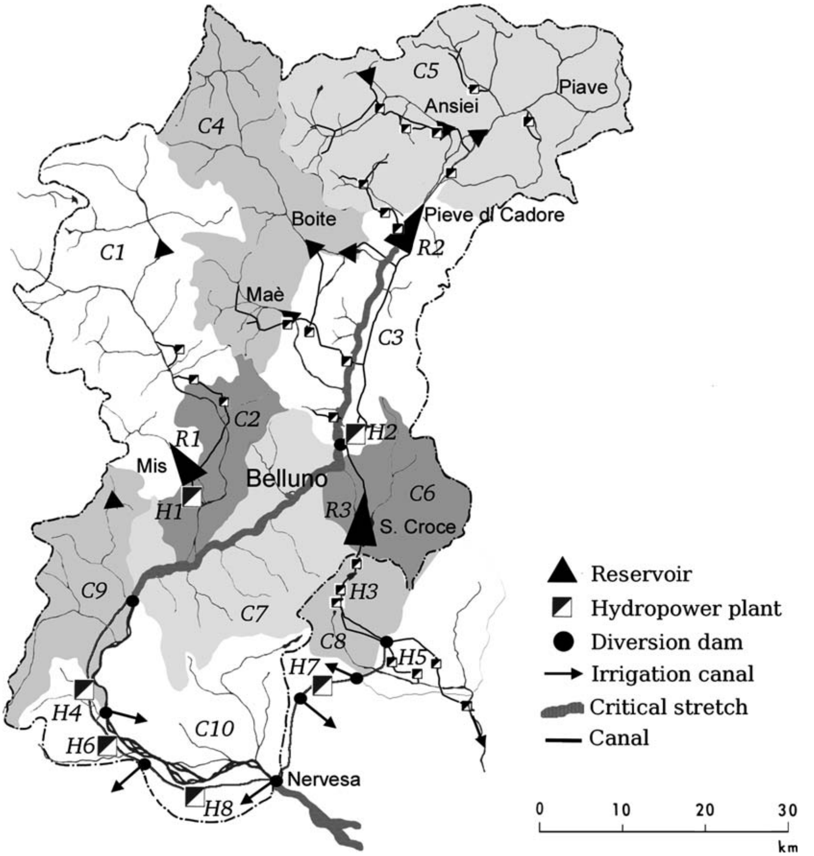

Figure 6.2: The Piave catchment.

The hydrological regime of almost all of the tributaries is strictly dependent on rainfall, except for the Cordevole ($C1$, $C2$), the Ansiei and the Piave upstream from Lake Pieve di Cadore ($C5$). The effect of glacial and snow ablation is not very relevant, as it is limited to the springtime and only to a few secondary catchments. Floods occur mostly in spring and autumn.

Continued

The low flows in summer are sometimes interrupted by sudden moderate flows produced by storms; the winter lows extend over the whole season and can become critical in the months of February and March.

Stakeholders In the Piave catchment there are agricultural, industrial and tourist activities. Beginning in the 1920s a complex network of reservoirs and canals was built progressively over time; it now feeds 40 hydropower plants (average production 3200 GWh/year, overall installed power 905 MW) and serves the pre-existing water users (mostly irrigation districts) through diversion dams. The total demand from these pre-existing water users is about 2 billion m^3/year, which is about 10 times more than the storage capacity of the reservoirs. In Figure 6.2 the positions of the most important diversion dams and the hydropower plants (or groups of plants) are shown. Given the position of the reservoirs and the diversions, the Piave catchment can be disaggregated into the ten sub-basins shown in the figure.

In the last few decades increased environmental awareness has brought the problem of the survival of riverine biocoenosis to the fore, particularly with regard to some critical stretches that the system's regulation often leaves almost dry. These stretches are highlighted in grey in the figure.

Aim and interventions The Piave Water Authority wants to define the minimum environmental flows (MEF) for the most critical stretches of the River, and so consultants were commissioned on this question. For each river stretch they provided several values corresponding to different levels of protection. Since the final choice is the Water Authority's responsibility, it would like to know the effects of each combination of MEF values on the hydropower production and the harvest in the irrigation districts. To conduct this evaluation one must bear in mind that, once the MEF values are imposed, the Regulator of the system would modify the regulation policy to account for them. The policy will therefore have to be redesigned in correspondence with each combination.

For didactic purposes we presume that the Authority would also like to evaluate the synergy between this normative intervention and some infrastructure proposals that have been submitted for approval: the creation of a new spillway to reduce flooding on the shores of Lake S. Croce ($R3$) and a vast tourism settlement in the catchment ($C5$) of Lake Pieve di Cadore ($R2$).

as confluence points and stretches of river and canals, one of which (L) we will assume is affected by a significant amount of leakage. The stretch of river ($S8$) that is critical from an environmental perspective is downstream from the Nervesa diversion ($D8$). Between $CP1$ and $CP8$ we merged the stretch of the Piave and the pipeline that runs parallel to it at a higher altitude; the distributed inflows to the Piave in this branch were represented by two catchments ($C7$ and $C8$), closed at the diversions that feed the penstock in this branch (Figure 6.2).

As is explained in the box, the Project Goal is to define some planning interventions and this requires redesigning the regulation policy for the entire system.

6.1.2 Choosing the time step

Once the components have been singled out, a model must be identified for each of them. Therefore, the modelling time step Δ must be identified; its choice is a compromise among various contrasting needs:

(a) the step must be the same for all of the components, because only in that way can the models be aggregated;

(b) the modelling step must be equal to the decision-making step and so, when fixing its value, it is necessary to keep management needs in mind (Section 4.8);

(c) a sufficiently long step allows translation times in the canals to be neglected (Section 5.3.2), and so significantly reduces the size of the state of the global model;

(d) the step must be such to allow all the components to be described correctly; and so the condition posed by the Sampling Theorem, which was stated in the description of the reservoir model (Section 5.1.4) and will be extended to the global model in Section 6.2.1, must be respected.

Note that condition (d) can be verified only once the global model has been built and thus only after all the models of the components have been identified. Choosing the time step is therefore a recursive operation, which is carried out by trial and error: its value is fixed, taking into account the first three conditions, and all the models of the components are identified with it. Then condition (d) is checked. If it is satisfied, the model can be utilized in the successive phases; if not, the time step is too long, therefore a smaller value is assumed, and the process starts again from the beginning. With a shorter step it may be that the translation times of some of the canals are no longer negligible and so they must be represented through dynamical models. The consequent increase in the size of the state increases the computing time required to solve the Design Problem; when this is unacceptable (as might be the case if computations required more than one week) one must adopt a heuristic solution: two different models are identified. The first (*screening model*), with a long time step and a small number of state variables, is used for the Design, even if it does not respect the condition of the Sampling Theorem and thus may provide imprecise estimates of the effects of an alternative. The second (*evaluation model*), with a shorter step and a greater number of state variables, is used to evaluate, with greater precision, the effects of the alternatives identified with the screening model, and so is used in the phase of *Evaluation* (see Section 19.2). Often the evaluation model adopts more detailed models not only for the canals, but also for many other components.

In the Piave case study, having contemplated the different requirements, we tentatively assumed a time step of seven days; more precisely, to respect the annual periodicity we assumed that a year is composed of 52 steps: in non-leap-years the first 51 are seven days long and the 52nd is eight days long, while in leap years the 9th step is also eight days long.

6.1.3 Identifying the models of the components

6.1.3.1 Modelling the actions

The reader may be surprised by the title of this subsection, because, according to the PIP procedure (Section 1.3), (s)he knows that the actions have to be defined in a phase (*Defining Actions*) that precedes the phase mentioned in the title of this section. However, the identification of the models cannot be structured if the actions have not been defined and modelled, and therefore we must specify them here, at least in general terms.[1]

The planning actions being considered by the Water Authority can be defined by the following five-dimensional vector

$$\mathbf{u}^p = \left| u^{\mathrm{MEF},S1} \quad u^{\mathrm{MEF},S2} \quad u^{\mathrm{MEF},S8} \quad u^{\mathrm{URB}} \quad u^{\mathrm{SP}} \right|^T \tag{6.1}$$

The first three components represent the values of the MEF in the three stretches of river that we are concerned with. For simplicity, we assumed that the biologists proposed time-invariant MEF values; if, instead, one (or more) of the MEF values changed with seasonal variation, in the place of a single value, the entire series of its values would be considered in the vector $\mathbf{u}^p$ and, as a consequence, the dimensions of $\mathbf{u}^p$ would increase. The fourth component u^{URB} is the surface area of the new urban settlement in the Ansiei catchment ($C5$), the last u^{SP} is a logical variable, whose value denotes the presence or absence of the morning glory spillway at Lake S. Croce ($R3$). The planning decisions must belong to a set $\mathcal{U}^p$

$$\mathbf{u}^p \in \mathcal{U}^p$$

deduced by researching the options for intervention. For example, the MEF values are those suggested by the consultant biologists, the urbanized surface area u^{URB} can only assume the values contained in the plan submitted and the value zero (Alternative Zero), and the decision u^{SP} for the spillway can only be 'true' or 'false'.

The management action requires that the following eleven-dimensional vector is defined every day

$$\mathbf{u}_t = \left| u_t^{R1} \quad u_t^{R2} \quad u_t^{R3} \quad u_t^{D1} \quad u_t^{D2} \quad \cdots \quad u_t^{D8} \right|^T \tag{6.2}$$

It includes the release decisions from the three reservoirs and the diversion decisions from the eight diversion dams.

6.1.3.2 Models

For the next step it is necessary to identify the models of the single components one by one. Clearly, we cannot describe this operation in detail, so we will simply highlight the structural aspects of this step that influence the aggregation process which follows. Therefore we will examine the models of the components one component type at a time. Their variables are denoted with the same symbols that appear in the corresponding meta-model in Chapter 5, with the addition of a superscript that indicates the component that it refers to.

Reservoirs The outputs of each reservoir model are the variables through which it interacts with the downstream components as well as any possible step indicators. The model is then completed by specifying the set of the feasible controls. The Piave's reservoir models are therefore the following

$$R1 \qquad s_{t+1}^{R1} = s_t^{R1} + a_{t+1}^{R1} - R_t^{R1}\left(s_t^{R1}, u_t^{R1}, a_{t+1}^{R1}\right) \tag{6.3a}$$

$$u_t^{R1} \in \mathcal{U}_t^{R1}\left(s_t^{R1}\right) \tag{6.3b}$$

$$r_{t+1}^{R1} = R_t^{R1}\left(s_t^{R1}, u_t^{R1}, a_{t+1}^{R1}\right) \tag{6.3c}$$

$$h_t^{R1} = h^{R1}\left(s_t^{R1}\right) \tag{6.3d}$$

[1] The interested reader will find an exhaustive description in Chapter 5 of PRACTICE.

$$R2 \qquad s_{t+1}^{R2} = s_t^{R2} + a_{t+1}^{R2} - R_t^{R2}\big(s_t^{R2}, u^{\mathrm{MEF},S2}, u_t^{R2}, a_{t+1}^{R2}\big) \tag{6.3e}$$

$$u_t^{R2} \in \mathcal{U}_t^{R2}\big(u^{\mathrm{MEF},S2}, s_t^{R2}\big) \tag{6.3f}$$

$$r_{t+1}^{R2} = R_t^{R2}\big(s_t^{R2}, u^{\mathrm{MEF},S2}, u_t^{R2}, a_{t+1}^{R2}\big) \tag{6.3g}$$

$$R3 \qquad s_{t+1}^{R3} = s_t^{R3} + a_{t+1}^{R3} - R_t^{R3}\big(s_t^{R3}, u^{\mathrm{SP}}, u_t^{R3}, a_{t+1}^{R3}\big) \tag{6.3h}$$

$$u_t^{R3} \in \mathcal{U}_t^{R3}\big(u^{\mathrm{SP}}, s_t^{R3}\big) \tag{6.3i}$$

$$r_{t+1}^{R3} = R_t^{R3}\big(s_t^{R3}, u^{\mathrm{SP}}, u_t^{R3}, a_{t+1}^{R3}\big) \tag{6.3j}$$

$$h_t^{R3} = h^{R3}\big(s_t^{R3}\big) \tag{6.3k}$$

$$g_t^{R3} = g_t^{R3}\big(h_t^{R3}\big) \tag{6.3l}$$

The model of the Mis reservoir ($R1$) has two outputs, the second of which is the water level, which we must know as $R1$ acts as a pondage for hydropower plant $H1$; it is best that the level is expressed with respect to sea level (i.e. it is the water elevation) so that the hydraulic head can easily be derived (see equation (6.8b) below). Lake S. Croce ($R3$) is in a similar relationship with plant $H3$ and, in addition, a step indicator (g_t^{R3}) is associated[2] with it, which quantifies flood damage. The value $u^{\mathrm{MEF},S2}$ of the MEF that is imposed upon the stretch $S2$ is included among the arguments of Lake Pieve's ($R2$) release function, because this reservoir must guarantee the fulfilment of that constraint. Thus the value $u^{\mathrm{MEF},S2}$ modulates the minimum instantaneous storage–discharge relationships $\tilde{N}^{\min,R2}(\cdot)$ that is used in the calculation of the minimum release $v^{R2}(\cdot)$, which appears in the definition of $R_t^{R2}(\cdot)$ (see Section 5.1.2.3). In a similar way, Lake S. Croce's ($R3$) release function is influenced by the decision u^{SP} to equip it with a new spillway.

Catchments In the Piave system there are 10 catchments. An analysis of their characteristics and the need to contain the time required to solve the Design Problem within acceptable limits suggest modelling the inflows from the smaller catchments as purely random disturbances and describing only the two largest catchments ($C1$ and $C5$) with dynamical models. The first ($C1$) can be modelled with an empirical model, which is always simpler and more economical to build, and often more precise than a mechanistic model. One must necessarily use a mechanistic model for the second ($C5$), because only this type of model can describe the effect that would be caused by the urbanization of a part of the catchment surface.[3] Thus, by denoting the states of the two catchments with $\mathbf{c}_t^{C1}$ and $\mathbf{c}_t^{C5}$, assuming the state-space representation (see Appendix A4 on the CD) for the empirical model of $C1$ and assuming a mechanistic model of the form described in Appendix A5 for $C5$, the two models take the following forms

$$C1 \qquad \mathbf{c}_{t+1}^{C1} = f_t^{C1}\big(\mathbf{c}_t^{C1}, \varepsilon_{t+1}^{C1}\big) \tag{6.4a}$$

$$d_{t+1}^{C1} = h_t^{C1}\big(\mathbf{c}_{t+1}^{C1}, \varepsilon_{t+1}^{C1}\big) \tag{6.4b}$$

$$C5 \qquad \mathbf{c}_{t+1}^{C5} = f_t^{C5}\big(\mathbf{c}_t^{C5}, u^{\mathrm{URB}}, T_{t+1}^{C5}, P_{t+1}^{C5}\big) \tag{6.4c}$$

$$d_{t+1}^{C5} = h^{C5}\big(\mathbf{c}_t^{C5}\big) + \varepsilon_{t+1}^{C5} \tag{6.4d}$$

[2] For information about how to define this step indicator see Section 4.3.2 of PRACTICE.

[3] For example, the effect of increasing the impermeable surface area due to urbanization can be described by reducing the values of parameters s^M and z^M within the mechanistic model described in Appendix A5.

The air temperature T^{C5}_{t+1}, precipitation P^{C5}_{t+1} and the two disturbances that appear in them must be described in turn, as we see later on in this chapter.

Diversion dams The models of the diversions are the following

$$Di \qquad q^{d,Di}_{t+1} = \min\{u^{Di}_t, q^{i,Di}_{t+1}, q^{\max,Di}\} \quad \text{for } i = 3, \dots, 7 \tag{6.5a}$$

$$q^{r,Di}_{t+1} = q^{i,Di}_{t+1} - q^{d,Di}_{t+1} \tag{6.5b}$$

$$u^{Di}_t \in \mathcal{U}^{Di}_t \tag{6.5c}$$

$$Dj \qquad q^{d,Dj}_{t+1} = \min\{u^{Dj}_t, (q^{i,Dj}_{t+1} - u^{\text{MEF},Sj})^+, q^{\max,Dj}\} \quad \text{for } j = 1, 2, 8 \tag{6.5d}$$

$$q^{r,Dj}_{t+1} = q^{i,Dj}_{t+1} - q^{d,Dj}_{t+1} \tag{6.5e}$$

$$u^{Dj}_t \in \mathcal{U}^{Dj}_t \tag{6.5f}$$

Note that the diverted flow rates $q^{d,Di}_{t+1}$ $(i = 1, \dots, 8)$ are those that run in the diversion canals that originate at the diversion, and are represented by dashed lines in Figure 6.1.

Canals Since the duration Δ of the time step (7÷8 days) is much greater than the time (about 40 hours) taken by the waves to travel across the whole system, each canal is described by an identity, which does not have to be expressed with an explicit equation that includes its variables, but can be formalized as an identity between the outputs and inputs of the components that it connects. The canals representing the stretches of river ($S1$, $S2$, $S8$) with environmental problems are an exception, because the step indicator g^{Si}_{t+1} that describes the environmental damage is defined[4] as a function of the flow rate that passes through them. Canal (L) is another exception because of leakage a^L_{t+1} that is not negligible. Therefore, these canals are described by the following models

$$Sj \qquad q^{o,Sj}_{t+1} = q^{i,Sj}_{t+1} \tag{6.6a}$$

$$g^{Sj}_{t+1} = g^{Sj}_t\left(q^{i,Sj}_{t+1}\right) \quad \text{for } j = 1, 2, 8 \tag{6.6b}$$

$$L \qquad q^{o,L}_{t+1} = q^{i,L}_{t+1} - a^L_{t+1} \tag{6.6c}$$

where a^L_{t+1} is a time-varying and periodic parameter.

Confluences The eight confluence points are described by simple water balance equations

$$CPj \qquad q^{o,CPk}_{t+1} = \sum_{k=1}^{n^{CPj}} q^{i,CPj,k}_{t+1} \quad \text{for } j = 1, \dots, 8 \tag{6.7}$$

where n^{CPj} is the number of branches entering into the jth confluence point.

Stakeholders There are two types of Stakeholders in the water system: hydropower plants and irrigation districts. The first group is described by the following models

$$Hj \qquad q^{d,Hj}_{t+1} = \begin{cases} 0 & \text{if } (q^{i,Hj}_{t+1} - q^{\text{MEF},Hj}_t) < q^{\min,Hj} \\ \min\{(q^{i,Hj}_{t+1} - q^{\text{MEF},Hj}_t)^+, q^{\max,Hj}\} & \text{otherwise} \end{cases}$$
$$\text{for } j = 1, \dots, 8 \tag{6.8a}$$

[4]For information about how to define this indicator see Section 4.3.3 of PRACTICE.

$$G_{t+1}^{Hj} = \begin{cases} \psi \eta^{Hj} g \gamma q_{t+1}^{d,Hj} (h_t^{u,Hj} - h^{d,Hj}) & \text{for } j = 1, 3 \\ \psi \eta^{Hj} g \gamma q_{t+1}^{d,Hj} H^{Hj} & \text{for } j = 2, 4, \ldots, 8 \end{cases} \tag{6.8b}$$

$$q_{t+1}^{o,Hj} = q_{t+1}^{i,Hj} \quad \text{for } j = 1, \ldots, 8 \tag{6.8c}$$

where the parameters ψ, η^{Hj}, g and γ have the same meaning as in equation (5.31). The hydraulic head is invariant in the power plants that are fed by pondage, while it depends on the differences between the water level in the upstream ($h_t^{u,Hj}$) and downstream ($h^{d,Hj}$) water bodies in the cases of the plants $H1$ and $H3$, whose penstocks begin at the intake of the reservoirs above them (Mis ($R1$) and Pieve ($R3$) respectively). Unlike equation (5.31), equation (6.8) does not consider the diversion decision, since the rule adopted in the hydropower plants along the Piave is to turbine as much as possible.

The irrigation districts may require dynamical models, as explained in Section 5.6. However, to simplify the model we suppose the districts may be described by a step indicator[5] only, within which the water demand w_t^{Ij}, which we assume is periodic of period T, is included. With this hypothesis the irrigation districts can be described by the following equations which express the water stress

$$Ij \qquad g_{t+1}^{Ij} = g_t^{Ij}\left(q_{t+1}^{i,Ij}, w_t^{Ij}\right) \quad \text{for } j = 1, \ldots, 6 \tag{6.9}$$

6.1.3.3 Interaction graph and disturbances

Having reached this point, all that remains to be described are the disturbances, which must always be considered last. Not all the disturbances that appear in the single models are in fact disturbances for the global model, only those whose processes were not considered in the models of the components. In other words, the disturbances that we are dealing with in this section are the disturbances of the global model and not those of the individual components. To identify them, a directed graph, termed an *interaction graph*, is constructed, whose nodes represent the components of the original block diagram (Figure 6.1). Each node has as many incoming arcs as the model of the component it represents has inputs, and as many outgoing arcs as that model has outputs. The inputs and outputs are interconnected using the topological relationships defined by the canals; thereby one obtains a graph such as the one shown in Figure 6.3 which represents the top left part of the block diagram in Figure 6.1.

By analysing this graph, the input variables that have not yet been described by a model can be recognized. In this way we identify the disturbances ε_{t+1}^{C1} and ε_{t+1}^{C5}, the air temperature T_{t+1}^{C5} and precipitation P_{t+1}^{C5} that appear in the models (6.4a)–(6.4b) and (6.4c)–(6.4d) of the catchments, as well as the inflow from the remaining catchments (for example $q_{t+1}^{i,CP1,2}$), which we decided to represent as purely random disturbances and which we will therefore denote with the symbol of a disturbance (for example $q_{t+1}^{i,CP1,2}$ becomes ε_{t+1}^{C2}).

In order to proceed, we must check that these 12 disturbances are effectively purely random, and that they are not cross-correlated. The whiteness test applied to their historical time series[6] reveals that the first hypothesis is acceptable for all of them except for temperature T_{t+1}^{C5}. The cross-correlograms of the same series show, instead, that ε_{t+1}^{C2} is significantly

[5] More details on the difficulty of defining this step indicator are provided in Section 5.6.2 and in Section 10.2.2.2.

[6] For T_{t+1}^{C5}, P_{t+1}^{C5} and ε_{t+1}^{Cj}, $j = 2, \ldots, 4, 6, \ldots, 10$ these are the series of the historically registered values, while for ε_{t+1}^{C1} and ε_{t+1}^{C5} they are the series of the residuals of the models (6.4a)–(6.4b) and (6.4c)–(6.4d) when these are applied to the time series of the recorded inflow values.

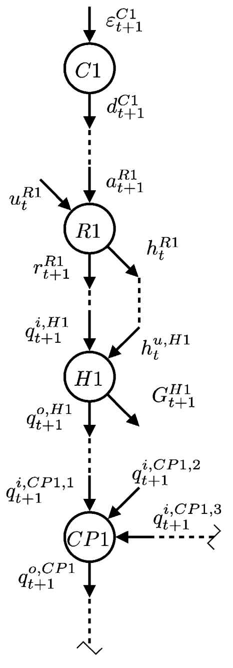

Figure 6.3: The part of the interaction graph that corresponds to the top left part of the block diagram in Figure 6.1.

correlated to ε_{t+1}^{C1}, as one might have expected, given that both are part of the Cordevole catchment (Figure 6.2). Moreover, the cross-correlogram shows that ε_{t+1}^{C4} and ε_{t+1}^{C3} are correlated for the same reason. We must therefore explain the variables T_{t+1}^{C5}, ε_{t+1}^{C2} and ε_{t+1}^{C4} with three empirical models, while the other disturbances can be described with their probability distributions, which can be identified from their historical series

$$E^{Cj} \qquad \varepsilon_{t+1}^{Cj} \sim \phi_t^{Cj}(\cdot) \quad \text{for } j = 1, 3, 5, \ldots, 10 \tag{6.10a}$$

$$P^{C5} \qquad P_{t+1}^{C5} \sim \phi_t^{P^{C5}}(\cdot) \tag{6.10b}$$

Through the historical series of the variables T_{t+1}^{C5}, ε_{t+1}^{C2} and ε_{t+1}^{C4} the following PARMAX models are identified

$$T^{C5} \qquad T_{t+1}^{C5} = \alpha^T T_t^{C5} + \eta_{t+1}^T \tag{6.11a}$$

$$E^{C2} \qquad \varepsilon_{t+1}^{C2} = \alpha^{C2} \varepsilon_{t+1}^{C1} + \eta_{t+1}^{C2} \tag{6.11b}$$

$$E^{C4} \qquad \varepsilon_{t+1}^{C4} = \alpha^{C4} \varepsilon_{t+1}^{C3} + \eta_{t+1}^{C4} \tag{6.11c}$$

The first model reveals that temperature is an autocorrelated Gaussian process of the first order: thus, it is a dynamical process. The other two are, on the other hand, non-dynamical models which express a spatial cross-correlation among the variables that appear in them.

Equations (6.10) and (6.11) are the models of the new components that are added to the interaction graph (Figure 6.3) in the proper places, thus obtaining the new graph shown in

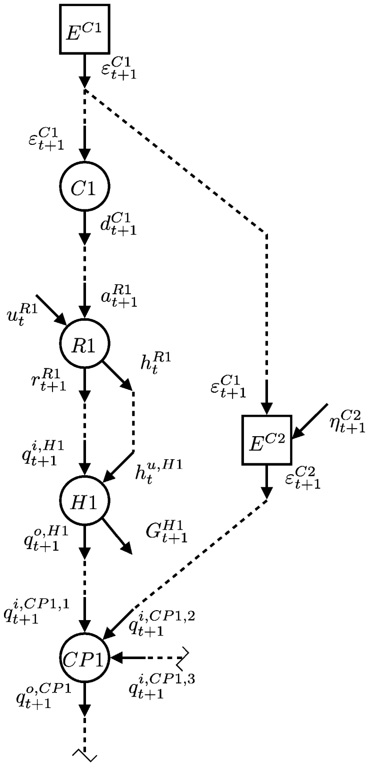

Figure 6.4: The part of the interaction graph shown in Figure 6.3 after all the new components that describe the disturbances have been added.

Figure 6.4. To help the reader we denoted these components with symbols which recall the name of the output variable.

By studying the interaction graph, updated with nodes that correspond to the new components (Figure 6.4), the disturbances that have not yet been explained are identified: they are the three disturbances that appear in the last three relationships. They must be described in turn. We know that the first disturbance is white (or it can be considered to be so) by construction; if it were not, the AR model would be of a higher order; the whiteness test confirms that the other two are also white. These disturbances can therefore be described by their probability distributions

$$ET^{C5} \qquad \eta_{t+1}^{T} \sim \phi_t^{\eta^T}(\cdot) \tag{6.12a}$$

$$EE^{C2} \qquad \eta_{t+1}^{C2} \sim \phi_t^{\eta^{C2}}(\cdot) \tag{6.12b}$$

$$EE^{C4} \qquad \eta_{t+1}^{C4} \sim \phi_t^{\eta^{C4}}(\cdot) \tag{6.12c}$$

The interaction graph is then updated and we look for any inputs that have not yet been explained. Since there are no others, the phase of describing the disturbances is terminated and with it the identification of the models of the components.

Finally, a *simplified interaction graph* is generated from the preceding one by substituting each group of parallel arcs with a single arc (see Figure 6.5). This new graph will be used in the following.

6.1.4 Aggregating the components

The last step in the identification of the global model is to interconnect the models of the single components, according to the topology defined by the interaction graph. Thereby we obtain a set of equations that can be subdivided into four subsets, which correspond respectively to the state transition functions, the output transformations of the global model, the equations that define the internal variables, and the description of the disturbances (deterministic and random). Finally, from the first two subsets the state and output vectors of the model are deduced. More precisely, using the simplified interaction graph, it is easy to single out the disturbances: they correspond to *root nodes*, i.e. nodes without any incoming arcs. Thereby the vectors of the deterministic disturbances ($\mathbf{w}_t$) and of the random disturbances ($\boldsymbol{\varepsilon}_{t+1}$) can be determined; in the case of the Piave the first does not exist, while the second is defined in the following way

$$\boldsymbol{\varepsilon}_{t+1} = \left| \varepsilon^{C1}_{t+1} \quad \eta^{C2}_{t+1} \quad \varepsilon^{C3}_{t+1} \quad \eta^{C4}_{t+1} \quad \varepsilon^{C5}_{t+1} \quad \eta^{T}_{t+1} \quad P^{C5}_{t+1} \quad \varepsilon^{C6}_{t+1} \quad \cdots \quad \varepsilon^{C10}_{t+1} \right|^T \tag{6.13}$$

Each of the components of the vector $\boldsymbol{\varepsilon}_{t+1}$ is completely defined by a probability distribution (if the disturbances were uncertain, they would be defined by set-membership instead), which is none other than the model of the corresponding component (see also equations (6.10) and (6.12))

$$\begin{aligned} \varepsilon^{C1}_{t+1} &\sim \phi^{C1}_t(\cdot) \\ \eta^{C2}_{t+1} &\sim \phi^{\eta^{C2}}_t(\cdot) \\ \varepsilon^{C3}_{t+1} &\sim \phi^{C3}_t(\cdot) \\ &\dots \end{aligned}$$

Thus, by reviewing all the nodes in the interaction graph, except for the root nodes, a list of equations (*LE*) is put together, by copying out the equations of the relative model for each node. As each equation is transcribed, each of the symbols of its input variables is substituted by the symbol that appears at the input side of the incoming arc to which the original symbol is associated in the graph in Figure 6.4. In other words, in the equations of each component the input variables are substituted with the output variables of the components that are upstream of it.

The first part of the list *LE*, which corresponds to the part of the interaction graph shown in Figure 6.4, is presented below.

$$\begin{array}{lrl} C1 & \mathbf{c}^{C1}_{t+1} &= f^{C1}_t\left(\mathbf{c}^{C1}_t, \varepsilon^{C1}_{t+1}\right) \\ & d^{C1}_{t+1} &= h^{C1}_t\left(\mathbf{c}^{C1}_{t+1}, \varepsilon^{C1}_{t+1}\right) \\ R1 & s^{R1}_{t+1} &= s^{R1}_t + d^{C1}_{t+1} - R^{R1}_t\left(s^{R1}_t, u^{R1}_t, d^{C1}_{t+1}\right) \\ & r^{R1}_{t+1} &= R^{R1}_t\left(s^{R1}_t, u^{R1}_t, d^{C1}_{t+1}\right) \\ & h^{R1}_t &= h^{R1}\left(s^{R1}_t\right) \end{array}$$

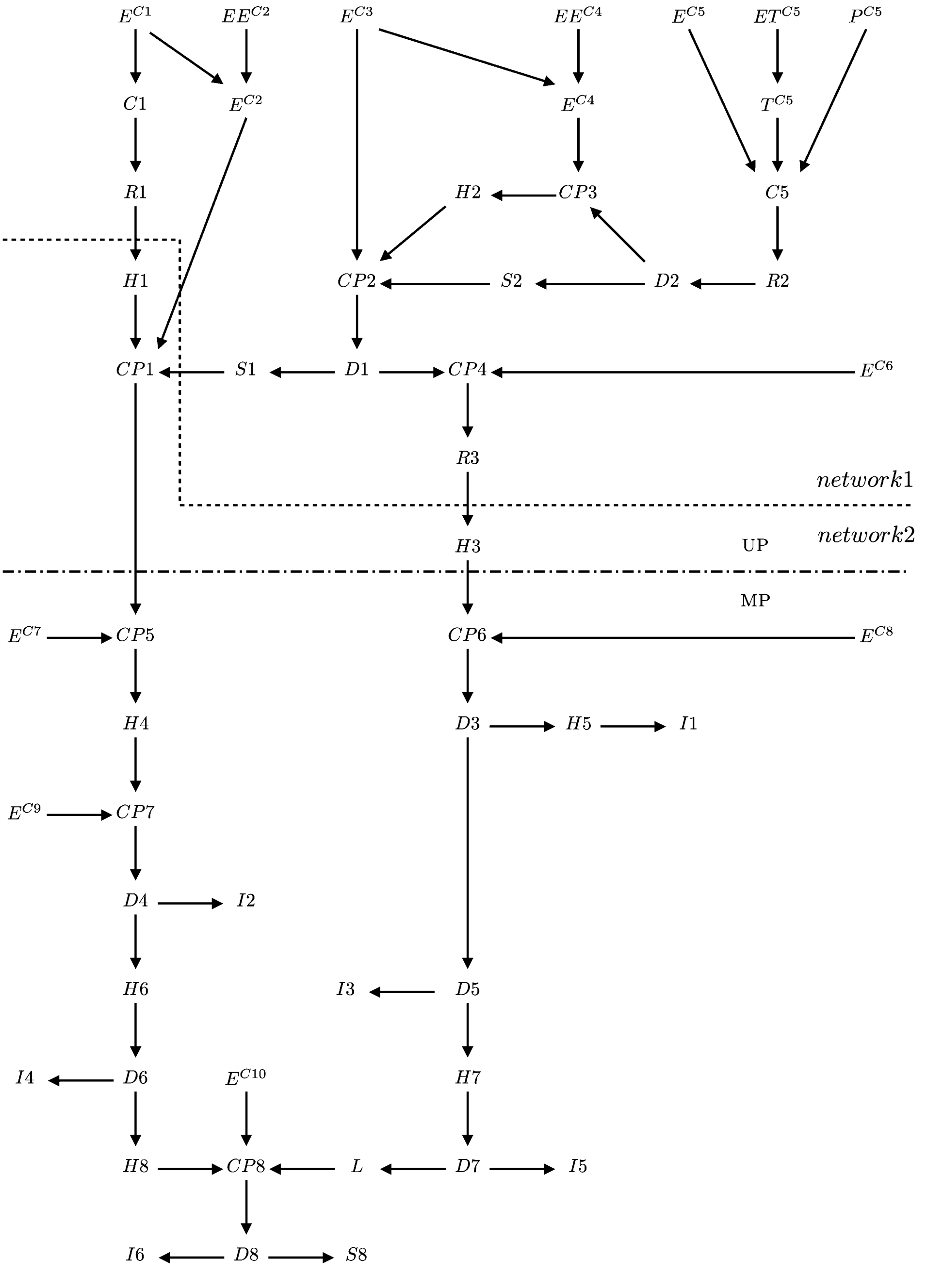

Figure 6.5: The (simplified) interaction graph of the Piave system. The meaning of dotted and dot-dashed lines is explained in the text.

$$
\begin{aligned}
H1 \qquad q_{t+1}^{d,H1} &= \begin{cases} 0 & \text{if } (r_{t+1}^{R1} - q_t^{\text{MEF},H1}) < q^{\min,H1} \\ \min\{(r_{t+1}^{R1} - q_t^{\text{MEF},H1})^+, q^{\max,H1}\} & \text{otherwise} \end{cases} \\
G_{t+1}^{H1} &= \psi \eta^{H1} g \gamma q_{t+1}^{d,H1} \big(h_t^{R1} - h^{d,H1}\big) \\
q_{t+1}^{o,H1} &= r_{t+1}^{R1}, \\
E^{C2} \qquad \varepsilon_{t+1}^{C2} &= \alpha^{C2} \varepsilon_{t+1}^{C1} + \eta_{t+1}^{C2} \\
CP1 \qquad q_{t+1}^{o,CP1} &= q_{t+1}^{o,I1} + \varepsilon_{t+1}^{C2} + q_{t+1}^{o,S1}, \\
& \dots
\end{aligned} \tag{LE}
$$

Once the list has been completed, one proceeds to identify all the recursive equations, i.e. the equations that, on the right-hand and left-hand side, contain the same variable in two different time instants, thus obtaining the following list

$$
\begin{aligned}
\mathbf{c}_{t+1}^{C1} &= f_t^{C1}\big(\mathbf{c}_t^{C1}, \varepsilon_t^{C1}\big) \\
s_{t+1}^{R1} &= s_t^{R1} + d_{t+1}^{C1} - R_t^{R1}\big(s_t^{R1}, u_t^{R1}, d_{t+1}^{C1}\big) \\
T_{t+1}^{C5} &= \alpha^T T_t^{C5} + \eta_{t+1}^T \\
\mathbf{c}_{t+1}^{C5} &= f_t^{C5}\big(\mathbf{c}_t^{C5}, u^{\text{URB}}, T_{t+1}^{C5}, P_{t+1}^{C5}\big) \\
s_{t+1}^{R2} &= s_t^{R2} + d_{t+1}^{C5} - R_t^{R2}\big(s_t^{R2}, u^{\text{MEF},S2}, u_t^{R2}, d_{t+1}^{C5}\big) \\
s_{t+1}^{R3} &= s_t^{R3} + q_{t+1}^{o,CP4} - R_t^{R3}\big(s_t^{R3}, u^{\text{SF}}, u_t^{R3}, q_{t+1}^{o,CP4}\big)
\end{aligned} \tag{6.14}
$$

These are all and exclusively the state transition functions of the component models and their set defines the state transition function of the global model. The state is thus given by the vector of the variables (or of the vectors) that appear on their left side

$$
\mathbf{x}_t = \left| \mathbf{c}_t^{C1} \quad s_t^{R1} \quad T_t^{C5} \quad \mathbf{c}_t^{C5} \quad s_t^{R2} \quad s_t^{R3} \right|^T \tag{6.15}
$$

Since the global model is to be used to evaluate the alternatives, the output vector is made up of the step indicators: in the case of the Piave these are the energy produced by the various power plants, the damage generated by flooding on Lake Santa Croce's shores, the environmental damage along the three critical stretches of the Piave, and the water stresses in the irrigation districts

$$
\mathbf{g}_{t+1} = \left| G_{t+1}^{H1} \quad \cdots \quad G_{t+1}^{H8} \quad g_t^{R3} \quad g_{t+1}^{S1} \quad g_{t+1}^{S2} \quad g_{t+1}^{S8} \quad g_{t+1}^{I1} \quad \cdots \quad g_{t+1}^{I6} \right|^T \tag{6.16}
$$

From the list *LE* we can therefore extract the output transformations that define these step indicators

$$
\begin{aligned}
& G_{t+1}^{H1} = \psi \eta^{H1} g \gamma q_{t+1}^{d,H1} \big(h_t^{R1} - h^{d,H1}\big) \\
& \dots \\
& g_t^{R3}\big(h_t^{R3}\big) \\
& g_{t+1}^{S1}\big(q_{t+1}^{r,D1}\big) \\
& \dots
\end{aligned} \tag{6.17}
$$

By deleting the equations that appear in the lists (6.14) and (6.17) from the list LE, one obtains the list of the relationships that define the internal variables of the global model

$$
\begin{array}{lll}
C1 & d_{t+1}^{C1} = h_t^{C1}\big(\mathbf{c}_{t+1}^{C1}, \varepsilon_{t+1}^{C1}\big) & \\
R1 & r_{t+1}^{R1} = R_t^{R1}\big(s_t^{R1}, u_t^{R1}, d_{t+1}^{C1}\big) & \\
 & h_t^{R1} = h^{R1}\big(s_t^{R1}\big) & \\
H1 & q_{t+1}^{d,H1} = \begin{cases} 0 & \text{if } (r_{t+1}^{R1} - q_t^{\text{MEF},H1}) < q^{\min,H1} \\ \min\{(r_{t+1}^{R1} - q_t^{\text{MEF},H1})^+, q^{\max,H1}\} & \text{otherwise} \end{cases} & \\
 & q_{t+1}^{o,H1} = r_{t+1}^{R1} & \\
E^{C2} & \varepsilon_{t+1}^{C2} = \alpha^{C2}\varepsilon_{t+1}^{C1} + \eta_{t+1}^{C2} & \\
CP1 & q_{t+1}^{o,CP1} = q_{t+1}^{o,H1} + \varepsilon_{t+1}^{C2} + q_{t+1}^{o,S1} & \\
 & \dots &
\end{array}
\tag{6.18}
$$

We will use the symbol $\mathbf{z}_{t+1}$ to denote the vector of these variables.

Remember finally that the controlled inputs of the model are the vectors $\mathbf{u}^p$ and $\mathbf{u}_t$ of the planning and regulation decisions defined by equations (6.1) and (6.2). They must respect the following constraints

$$\mathbf{u}^p \in \mathcal{U}^p \qquad \mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{u}^p, \mathbf{x}_t\big)$$

in which the set $\mathcal{U}^p$ defines the considered alternatives and the set $\mathcal{U}_t(\mathbf{u}^p, \mathbf{x}_t)$ is derived from the constraints on the controls which appear in the models, see for example equations (6.3b) and (6.5c). More precisely it is the product set of the feasible control sets for each component

$$\mathcal{U}_t\big(\mathbf{u}^p, \mathbf{x}_t\big) = \mathcal{U}_t^{R1}\big(s_t^{R1}\big) \times \cdots \times \mathcal{U}_t^{R3}\big(u^{\text{SF}}, s_t^{R3}\big) \times \mathcal{U}_t^{D1} \times \cdots \times \mathcal{U}_t^{D8}$$

Observe that the set of feasible controls for the diversions, and more generally, for all of the non-dynamical systems, does not depend on the state.

6.1.5 Remarks

6.1.5.1 Classification of the variables

Very often the classification of the input variables changes if the aim of the model is changed. For example, if the reservoir $R2$, the diversion $D2$ and the hydropower plant $H2$ were not the property of the Project promoter and as a consequence their regulation policies did not have to be designed, the variables u_t^{R2} and u_t^{D2} would not be controls, but deterministic disturbances of the global model. Therefore, in the phase of *Designing Alternatives* in which policies are calculated, these variables would have to be described by a time series or with models, defined by the components' regulation and distribution policies. More precisely, if these policies are point-valued (see Section 10.1.1) the models are defined by

$$u_t^{R2} = m_t^{R2}\big(s_t^{R2}\big) \qquad u_t^{D2} = m_t^{D2}\big(s_t^{R2}\big)$$

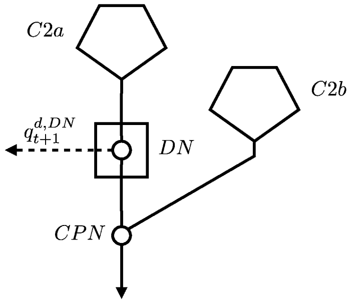

Figure 6.6: The components of the catchment $C2$ if an interceptor canal is being designed.

where $m_t^{R2}(s_t^{R2})$ and $m_t^{D2}(s_t^{R2})$ are the point-valued control laws of the two policies at time t; instead, if the policies are set-valued the models are the following

$$u_t^{R2} \in M_t^{R2}\big(s_t^{R2}\big) \qquad u_t^{D2} \in M_t^{D2}\big(s_t^{R2}\big)$$

6.1.5.2 Introducing new actions

The introduction of new actions can result in a structural modification of the model. For example, consider the case in which, in a subsequent phase of the study, the Water Authority asks for an evaluation of the effects of constructing an interceptor canal that, through a non-regulated diversion, collects part of the flow from a stream in catchment $C2$ and transfers it into the reservoir $R1$. In this case, a new planning variable u^{DC} must be introduced, which specifies the capacity of the interceptor canal, and the catchment $C2$ must be subdivided into two parts as shown in Figure 6.6.

The catchments $C2a$ and $C2b$ can be described with empirical models only if we have a historical series of flow rates registered at the point where the new diversion dam (DN) should be constructed; if we have no such series, it is necessary to identify a mechanistic model of $C2$ that allows us to estimate the flow in the section DN. By substituting the model $C2$ in the old global model with the models of the two catchments and the diversion DN, one obtains the global model with which to evaluate the construction of the new interceptor canal.

6.1.5.3 The distribution policy given

Sometimes the distribution policy for some of the diversions cannot be re-designed, because it is given by norms or traditions. The controls that correspond to such diversions are therefore defined on the basis of these rules and so they become internal variables of the model. If, for example, the irrigation district $I2$ had supply priority with respect to the power plant $H6$ and all the downstream users (see Figure 6.1), one would have to pose $u_t^{D4} = w_t^{I2}$. The control of a reservoir could also have the same fate, as we saw in the first remark mentioned above.

6.2 The global model

The procedure described in the previous paragraph is completely general and can be applied to any system. As we have seen, it allows us to identify the input ($\mathbf{u}^p$, $\mathbf{u}_t$, $\mathbf{w}_t$, $\boldsymbol{\varepsilon}_{t+1}$), state ($\mathbf{x}_t$)

and output ($\mathbf{y}_t$) vectors of the model of the whole system (*global model*); the constraints on the decisions $\mathbf{u}^p$ and $\mathbf{u}_t$; the probabilities or the set-membership of the random disturbance; and the functions that define:

- the state transition (the list (6.14) for the Piave system)

$$\mathbf{x}_{t+1} = \breve{f}_t\big(\mathbf{x}_t, \mathbf{z}_{t+1}, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.19a}$$

- the output transformation (the list (6.17) for the Piave system)

$$\mathbf{g}_{t+1} = \breve{g}_t\big(\mathbf{x}_t, \mathbf{z}_{t+1}, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.19b}$$

- the internal variables[7] (the list (6.18) for the Piave system)

$$\mathbf{z}_{t+1} = \breve{z}_t\big(\mathbf{x}_t, \mathbf{z}_{t+1}, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.19c}$$

  subject to the constraints

$$\mathbf{u}^p \in \mathcal{U}^p \tag{6.19d}$$

$$\mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{u}^p, \mathbf{x}_t\big) = \prod_{i \in \mathcal{N}} \mathcal{U}_t^i\big(\mathbf{u}^p, \mathbf{x}_t\big) \tag{6.19e}$$

  where $\mathcal{N}$ is the set of controllable components, and subject to the description of the random disturbances which can be stochastic

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\big(\cdot|\mathbf{u}^p\big) \tag{6.19f}$$

  or uncertain

$$\boldsymbol{\varepsilon}_{t+1} \in \Xi_t\big(\mathbf{u}^p\big) \tag{6.19g}$$

  according to the particular case.

This set of equations constitutes the global model of the system. Its form is not, however, the most compact, because internal variables $\mathbf{z}_{t+1}$ are only instrumental to the calculation and are not explicitly required to evaluate the alternatives. Therefore, it is convenient to substitute internal variables $\mathbf{z}_{t+1}$ within equations (6.19a)–(6.19b) with equation (6.19c), thus obtaining the *standard form of the global model*:

- when the random disturbances are stochastic it assumes the following form

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.20a}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{6.20b}$$

$$\mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{u}^p, \mathbf{x}_t\big) \tag{6.20c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\big(\cdot|\mathbf{u}^p\big) \tag{6.20d}$$

$$\mathbf{g}_{t+1} = g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.20e}$$

[7] The form of equation (6.19c) that defines the vector $\mathbf{z}_{t+1}$ in an implicit way (note that it appears on both sides) might be surprising at first glance, given that nothing similar occurs in the scalar equations in the list (6.18) that it represents. Nevertheless, even if none of the scalar equations are implicit, the vector equation is, because the components of the vector $\mathbf{z}_{t+1}$ appear on both sides of the scalar equations.

- when the disturbances are uncertain, its form is completely analogous to the previous one, except for the fact that equation (6.20d) is substituted with the following one

$$\boldsymbol{\varepsilon}_{t+1} \in \Xi_t\big(\mathbf{u}^p\big) \tag{6.20f}$$

Note that equation (6.20) is none other than the general expression of the state-space form of the system, i.e. the representation that we introduced in Section 4.1.2.3, to which equation (6.20d) (or (6.20f)) is added in order to describe the random disturbance that appears in it. Finally, note that *we assumed that the disturbances are either all stochastic or all uncertain*, which means that it is not acceptable that some be of one type and the rest of the other. The reason for this is that mixed cases create mathematical difficulties that exceed the limits of the present work.

The model (6.20) describes the non-regulated system. When the system is regulated, the control u_t is defined by the policy p and it is therefore an internal variable. In that case, when the disturbances are stochastic, the model takes the following form[8]

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.21a}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{6.21b}$$

$$\mathbf{u}_t = m_t(\mathbf{x}_t) \tag{6.21c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\big(\cdot|\mathbf{u}^p\big) \tag{6.21d}$$

$$\mathbf{g}_{t+1} = g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{6.21e}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \dots, h-1\big\} \tag{6.21f}$$

with h being the time horizon of the problem (i.e. the design or the evaluation horizon, depending on the use of the model). When the disturbances are uncertain the model is the same, except that equation (6.21d) is substituted by equation (6.20f).

Finally, to completely define the model, one must specify the sets $\mathcal{S}_{\mathbf{x}_t}$, $\mathcal{S}_{\mathbf{u}_t}$, $\mathcal{S}_{\mathbf{w}_t}$ and $\mathcal{S}_{\boldsymbol{\varepsilon}_t}$ in which state, control and disturbances, both deterministic and random, assume values. Such sets are, very often, bounded subsets of $\mathbb{R}^{n_\mathbf{x}}$, $\mathbb{R}^{n_\mathbf{u}}$, $\mathbb{R}^{n_\mathbf{w}}$ and $\mathbb{R}^{n_{\boldsymbol{\varepsilon}}}$, with $\mathbb{R}$ being the set of real numbers and $n_\mathbf{x}$, $n_\mathbf{u}$, $n_\mathbf{w}$ and $n_{\boldsymbol{\varepsilon}}$ the dimensions of the four vectors. Usually such sets are continuous sets.

6.2.1 Verifying the time step

We have already discussed the modelling time step several times (Sections 4.8, 5.1.4 and 6.1.2). Now we have to add that once the global model has been identified, one must verify the correctness of its value, i.e. that it does not generate any information losses in the sampling process. Nothing is lost when *the duration Δ is smaller, by about one order of magnitude, than the smallest time constant of the linearized model* in the neighbourhood of a state value that is held to be significant, such as the average system operating condition. This condition derives directly from the Sampling Theorem, an informal derivation of which is provided in Section 5.1.4.

[8]This form is valid when then the policy is point-valued. If, instead, the policy is set-valued, the expression is more complicated, as we will see in Section 19.2.2.

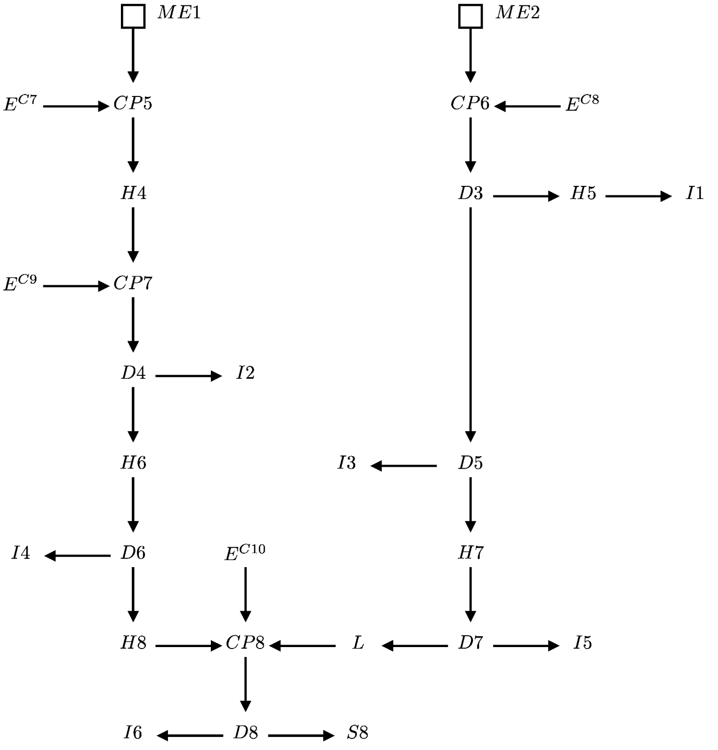

Figure 6.7: The simplified interaction graph of the system of the Middle Piave (MP): note how it does not contain reservoirs and how its flows do not influence any reservoirs.

6.3 The distribution network

Let us take up the Piave example once again. The state vector (6.15) allows us to single out the components described by dynamical models, in the following referred to as *dynamical components*. Observing their location in the graph in Figure 6.5 we notice that they are all situated above the dot-dashed line, in the Upper Piave (UP). The Middle Piave (MP) does not contain any reservoirs and therefore it is described by a non-dynamical model. A system that has this property is called *water network*. It might be that, as in this example, the outgoing flows from a water network do not influence any reservoir. We will call a network of this type *distribution network*.

The distribution network MP has the flows $q_{t+1}^{o,CP1}$ and $q_{t+1}^{o,H3}$ as inputs, which pass through the arcs $CP1$–$CP5$ and $H3$–$CP6$ by means of which it is connected to the rest of the system, i.e. to UP. It is easy to imagine that these flows come into the network through the nodes $ME1$ and $ME2$, which we will call entrance terminals of the network (Figure 6.7).

We can visualize MP as a component that produces the output vector

$$\mathbf{y}_{t+1}^{\text{MP}} = \left| G_{t+1}^{H4} \quad \cdots \quad G_{t+1}^{H8} \quad g_{t+1}^{S8} \quad g_{t+1}^{I1} \quad \cdots \quad g_{t+1}^{I6} \right|^T$$

in response to four input vectors:

(1) the vector of the flow rates entering MP through the terminals $ME1$ and $ME2$

$$\mathbf{e}_{t+1}^{\mathrm{MP}} = \left| q_{t+1}^{o,CP1}, q_{t+1}^{o,H3} \right|^T$$

(2) the vector of the controls

$$\mathbf{u}_t^{\mathrm{MP}} = \left| u_t^{D3} \quad \cdots \quad u_t^{D8} \right|^T$$

(3) the vector of the planning decisions

$$\mathbf{u}^{p\mathrm{MP}} = \left| u^{\mathrm{MEF},S8} \right|$$

(4) the vector of the disturbances

$$\boldsymbol{\varepsilon}_{t+1}^{\mathrm{MP}} = \left| \varepsilon_{t+1}^{C7} \quad \cdots \quad \varepsilon_{t+1}^{C10} \right|^T$$

Given the values of the components of the four input vectors, it is possible to calculate the value of the components of the output vector using the appropriate relationships selected from lists (6.17) and (6.18).

All the above can easily be extended to the general case: given a system S, the subpart that does not include dynamical components is called a *water network*. If the outgoing flows do not influence dynamical components the water network is called a *distribution network* and we denote it with D. Clearly, a distribution network does not always exist in a water system. Like all non-dynamical systems, the network D is generally characterized by an input vectors (composed by five vectors: $\mathbf{u}^{p\mathrm{D}}$, $\mathbf{u}_t^{\mathrm{D}}$, $\mathbf{w}_t^{\mathrm{D}}$, $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}$ and $\mathbf{e}_{t+1}^{\mathrm{D}}$); a vector $\mathbf{z}_{t+1}^{\mathrm{D}}$ of internal variables; and a vector $\mathbf{y}_{t+1}^{\mathrm{D}}$ of output variables. Each one of these vectors[9] is constituted by those components of the homonymous vectors of S which concern the components of the network. To these vectors one must add one more input: the vector of the network's incoming flows $\mathbf{e}_{t+1}^{\mathrm{D}}$. The vectors $\mathbf{z}_{t+1}^{\mathrm{D}}$ and $\mathbf{y}_{t+1}^{\mathrm{D}}$ can be calculated as a function of the five input vectors using the following relationships[10]

$$\mathbf{z}_{t+1}^{\mathrm{D}} = \breve{z}_t^{\mathrm{D}}\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}^{p\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \mathbf{w}_t^{\mathrm{D}}, \mathbf{e}_{t+1}^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}\right) \tag{6.22a}$$

$$\mathbf{y}_{t+1}^{\mathrm{D}} = \breve{h}_t^{\mathrm{D}}\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}^{p\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \mathbf{w}_t^{\mathrm{D}}, \mathbf{e}_{t+1}^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}\right) \tag{6.22b}$$

taking into account the description of the disturbance[11]

$$\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}} \sim \phi_t^{\mathrm{D}}\left(\cdot|\mathbf{u}^{p\mathrm{D}}\right) \tag{6.22c}$$

and the constraints

$$\mathbf{u}^{p\mathrm{D}} \in \mathcal{U}^{p\mathrm{D}} \tag{6.22d}$$

$$\mathbf{u}_t^{\mathrm{D}} \in \mathcal{U}_t^{\mathrm{D}}\left(\mathbf{u}^{p\mathrm{D}}\right) = \prod_{i \in \mathcal{N}^{\mathrm{D}}} \mathcal{U}_t^i\left(\mathbf{u}^{p\mathrm{D}}\right) \tag{6.22e}$$

where $\mathcal{N}^{\mathrm{D}}$ is the set of D's controllable components and $\mathcal{U}^{p\mathrm{D}}$ the set of feasible planning decisions concerning D.

[9] Clearly, sometimes some of these vectors may be missing, as is $\mathbf{w}_t^{\mathrm{D}}$ in the network MP.

[10] These may be obtained in a simple way from equations (6.19c) and (6.19b) respectively.

[11] If an uncertain description is adopted, equation (6.22c) would be substituted by $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}} \in \Xi_t(\mathbf{u}^{p\mathrm{D}})$.

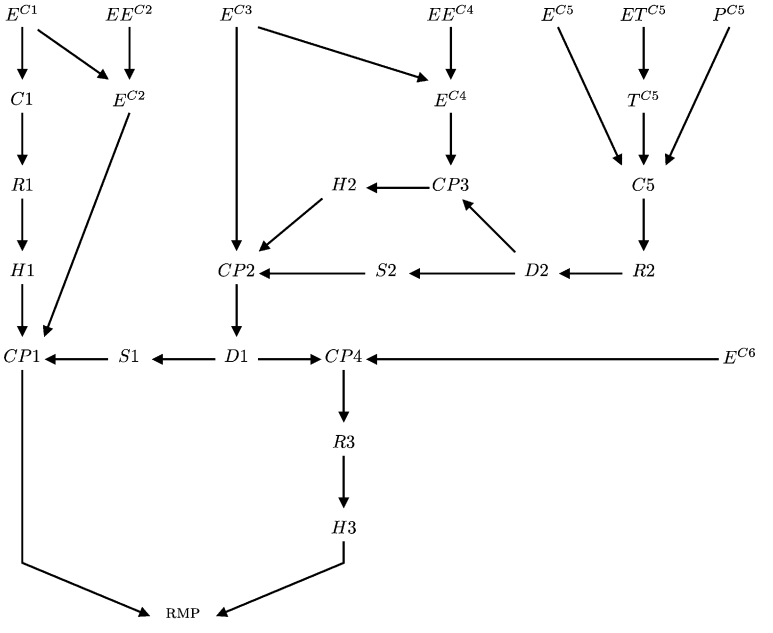

Figure 6.8: The simplified interaction graph of the Piave system when MP is described by an equivalent model.

By substituting equation (6.22a) in model (6.22b) one obtains the following expression

$$\mathbf{y}_{t+1}^{\mathrm{D}} = h_t^{\mathrm{D}}\big(\mathbf{u}^{p\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \mathbf{w}_t^{\mathrm{D}}, \mathbf{e}_{t+1}^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}\big) \tag{6.23}$$

which we will call an *equivalent model* of the network. This expression allows us to describe a network as if it were an *equivalent user* and avoid calculating the internal variables. Rarely can the equivalent model be obtained analytically; more often it is numerically obtained by calculating the values of $\mathbf{y}_{t+1}^{\mathrm{D}}$ in correspondence with each n-tuple of input values and recording them in a look-up table.

The equivalent model is particularly useful in the phase of *Designing Alternatives*, when the solution algorithms require that the values of $\mathbf{y}_{t+1}^{\mathrm{D}}$ be computed a huge number of times for the same input values. It is easy to understand that by using this model the description of the system S is simplified: this can be appreciated by comparing the interaction graphs of the Piave system with (Figure 6.8) and without (Figure 6.5) the equivalent model RMP of the distribution network MP.

When the network includes diversions for which a distribution policy is not defined, the vector $\mathbf{u}_t^{\mathrm{D}}$ of their controls is included among the inputs of the equivalent model. If, instead, the policy is defined, these controls are internal variables and, as a consequence, the number of the input vectors of the equivalent model decreases and with it the size of the look-up table that expresses it.

As we will show in detail in Chapter 15, when designing a regulation policy for reservoirs and diversions, such as in the Piave Project, the computing time needed to solve the Design Problem decreases quite significantly if one develops the policy design in three steps:

(1) design the distribution policy for the diversions in network D,

(2) obtain the equivalent model of the network,

(3) use it to design the regulation policy for the remaining part of the system.

By now, one can see why this is advantageous: if, in the case of the Piave, we designed the policy in a single step, we would have to determine 11 control laws (i.e. one for each control variable of the whole system). Instead, if we subdivide the policy design into three steps, in the first step a 5-control-law policy has to be determined for MP, and in the third a 6-control-law policy for UP. Since the computing time increases exponentially with the number of controls, the difference between the two ways of proceeding is enormous. To appreciate the difference, let us assume that each control can assume 10 values. The single step design would require an exhaustive search procedure with 10^{11} evaluations for each time instant, while the three-step design would require only $10^6 + 10^5$. Thereby the computation times would be reduced, in the case of the Piave, from almost 73 years to only seven hours, when the modelling step is one week; if the step is one day the times would be reduced from almost 500 years to 2 days.

In a water system it is almost always possible to identify more than one distribution network. For example, the system below the dashed line in Figure 6.5 (*network*2) is also a distribution network, since it neither contains, nor do its flows influence, any dynamical components. It is, however, connected to the rest of the system by 4 incoming arcs, so that its vector $\mathbf{e}^{\mathrm{D}}_{t+1}$ has 4 dimensions (each cut arc produces a flow), while the vector $\mathbf{e}^{\mathrm{D}}_{t+1}$ of MP has only 2 dimensions. As one can see, the time required to compute the table of the equivalent model grows exponentially with the dimension of the input vector: it follows that MP is much more interesting than *network*2, and so, in general, among the many alternative distribution networks, the one which corresponds to the smallest number of intersected incoming arcs is chosen.

Once a network D has been identified,[12] the system R which is complementary to it is univocally defined. By definition it contains all the dynamical components of S and it is described by the following model, which is obtained by eliminating from model (6.19) the relationships that describe D

$$\mathbf{x}_{t+1} = \breve{f}_t\left(\mathbf{x}_t, \mathbf{z}^{\mathrm{R}}_{t+1}, \mathbf{u}^{p\mathrm{R}}, \mathbf{u}^{\mathrm{R}}_t, \mathbf{w}^{\mathrm{R}}_t, \boldsymbol{\varepsilon}^{\mathrm{R}}_{t+1}\right) \tag{6.24a}$$

$$\mathbf{z}^{\mathrm{R}}_{t+1} = \breve{z}^{\mathrm{R}}_t\left(\mathbf{x}_t, \mathbf{z}^{\mathrm{R}}_{t+1}, \mathbf{u}^{p\mathrm{R}}, \mathbf{u}^{\mathrm{R}}_t, \mathbf{w}^{\mathrm{R}}_t, \boldsymbol{\varepsilon}^{\mathrm{R}}_{t+1}\right) \tag{6.24b}$$

$$\mathbf{y}^{\mathrm{R}}_{t+1} = \breve{h}^{\mathrm{R}}_t\left(\mathbf{x}_t, \mathbf{z}^{\mathrm{R}}_{t+1}, \mathbf{u}^{p\mathrm{R}}, \mathbf{u}^{\mathrm{R}}_t, \mathbf{w}^{\mathrm{R}}_t, \boldsymbol{\varepsilon}^{\mathrm{R}}_{t+1}\right) \tag{6.24c}$$

$$\boldsymbol{\varepsilon}^{\mathrm{R}}_{t+1} \sim \phi^{\mathrm{R}}_t\left(\cdot|\mathbf{u}^{p\mathrm{R}}\right) \tag{6.24d}$$

$$\mathbf{u}^{p\mathrm{R}} \in \mathcal{U}^{p\mathrm{R}} \tag{6.24e}$$

$$\mathbf{u}^{\mathrm{R}}_t \in \mathcal{U}^{\mathrm{R}}_t\left(\mathbf{u}^{p\mathrm{R}}, \mathbf{x}_t\right) = \prod_{i \in \mathcal{N}^{\mathrm{R}}} \mathcal{U}^i_t\left(\mathbf{u}^{p\mathrm{R}}, \mathbf{x}_t\right) \tag{6.24f}$$

where $\mathcal{N}^{\mathrm{R}}$ is the set of the controllable components in R and $\mathbf{u}^{p\mathrm{R}}$ the vector of feasible planning decisions concerning R. Depending on the case, the description of the disturbance is stochastic as in equation (6.24d) or uncertain. The definition of the variables in model (6.24) is evident, though it is worthwhile to note that the components of the vector $\mathbf{e}^{\mathrm{D}}_{t+1}$ of network D's incoming flows are none other than some of $\mathbf{z}^{\mathrm{R}}_{t+1}$'s components.

[12]For simplicity we assume that the system S contains a single network, but nothing prevents it from containing more.

6.4 More about disturbances

6.4.1 Generating synthetic series

In Section 4.6 we noted that the global model is built in order to evaluate the behaviour of the system for the alternatives that one would like to examine. This requires that input trajectories be available for a sufficiently long time horizon $[0, h)$. We saw that there are no difficulties in obtaining the trajectories of the control values, because once the policy (6.21f) has been defined, the control is an internal variable. Not even the deterministic disturbances create difficulties because, being such, their trajectories are known. Only for the random disturbances is it not clear what one must do, given that, since they are random, one cannot define their trajectories.

When the disturbances are purely random, i.e. when at every time instant their value does not depend on the value taken on by any other variable, we have seen that the difficulty can easily be overcome: just generate a series of values with their models (which do not have inputs!). This series is called a *synthetic series*. More precisely, when the disturbance is purely stochastic,[13] and therefore its model has the form (6.21d), one adopts the following procedure:

Procedure for synthetic generation (I): *Given the vector* $\mathbf{u}^p$ *of the planning decisions, for each time* $t \in [0, h)$ *extract a value* ε_{t+1} *from the distribution* $\phi_t(\cdot|\mathbf{u}^p)$. *Thereby the synthetic series* $\{\varepsilon_t\}_1^h$ *is obtained.*

This series, by construction, is equiprobable to every other series that the model can generate and therefore it is suitable for evaluating the alternatives. After that, the behaviour of the water system is simulated and the trajectories of all the variables that are necessary to evaluate the indicators are obtained; we will deal with this topic in Chapter 19.

The condition that the disturbance is purely random is therefore, as we saw in Section 4.6, a sufficient condition to evaluate the performance of an alternative by generating a synthetic series. We must, however, ask ourselves if this is necessary.

The answer is negative, and to convince the reader of this it is enough to suppose that the disturbance is not described by equation (6.21d), but by the following condition

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t\right) \tag{6.25}$$

With this condition, Procedure I can be started, because at time 0 the alternative that we are simulating is known, so $\mathbf{u}^p$, and $\mathbf{x}_0$ are known too; further, through the control law (6.21c), we can compute $\mathbf{u}_0$ and $\mathbf{w}_0$ is a deterministic, and therefore known, disturbance. Thus the probability distribution $\phi_0(\cdot|\mathbf{x}_0, \mathbf{u}^p, \mathbf{u}_0, \mathbf{w}_0)$ is known, from which $\boldsymbol{\varepsilon}_1$ is extracted. At the next step, however, the procedure stops, because we do not know $\mathbf{x}_1$. We can avoid this difficulty if we give up the idea of obtaining the synthetic series before the simulation, and, instead, we carry out the generation and the simulation simultaneously. We can thus define the following:

Procedure for synthetic generation (II): *Given the vector* $\mathbf{u}^p$ *of the planning decisions, for every time* $t \in [0, h)$

[13] Here and in the rest of the section we consider a stochastic disturbance, but what is said can easily be adapted to the case of an uncertain disturbance.

(1) *given the state* $\mathbf{x}_t$*, through the policy, determine* $\mathbf{u}_t$;

(2) *extract a value* $\boldsymbol{\varepsilon}_{t+1}$ *from the distribution* $\phi_t(\cdot|\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t)$;

(3) *once the tuple* $(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$ *is known, compute the state* $\mathbf{x}_{t+1}$ *with the state transition equation* (6.21a); *go back to step* (1).

Thereby the trajectories of all the variables that are necessary to evaluate the indicators are obtained. The difficulty is thus overcome, but this new procedure is clearly computationally more onerous that the preceding one.

We have thus proved how the condition that the disturbance be purely random is not strictly necessary, because the probability distribution in equation (6.25) is conditional. Now we must determine the necessary and sufficient condition, and for that purpose we must verify if there are any other conditioning variables to include in equation (6.25). Let us begin with the output. One can see that including the output is either useless or impossible: it is useless when the output $(\mathbf{y}_t)$ depends only on the three vectors $(\mathbf{x}_t, \mathbf{u}_t, \mathbf{w}_t)$, and impossible when the output $(\mathbf{y}_{t+1})$ depends also on $\boldsymbol{\varepsilon}_{t+1}$, because this creates a vicious circle. The same is true for any internal variable. There is then the possibility that the probability distribution is conditioned by one or more of the following variables: $\mathbf{x}_{t-1}$, $\mathbf{u}_{t-1}$, $\mathbf{w}_{t-1}$, $\boldsymbol{\varepsilon}_t$, or also by these same variables with even higher time lags. But, if this were so, at point (2) of Procedure II we would have to remember their values, and therefore they would have to be included in the state.

In conclusion, *a necessary and sufficient condition for the alternative evaluation in the presence of a random disturbance is that the value of the disturbance in the time interval* $[t, t+1)$ *does not explicitly depend on the values that the disturbance assumes at previous times, but may depend on the state and control values at time* t. We will see in Section 12.1.2 that the algorithms for the solution of the Design Problem are applicable only when this condition is satisfied.

From the above it follows that a variable does not necessarily have to be purely random to be a disturbance for the global model. Note, in fact, that when the distribution $\phi_t(\cdot)$ is conditioned by $\mathbf{x}_t$ (or $\mathbf{u}_t$), the disturbance $\boldsymbol{\varepsilon}_{t+1}$ is never purely random, since $\mathbf{x}_t$ (or $\mathbf{u}_t$) is not. Therefore, it is not strictly necessary that the whiteness test be carried out successfully, but if it is, everything is simplified.

In the models (6.20) and (6.21) we must therefore insert equation (6.25) in place of equations (6.20d) and (6.21d). However, in order to lighten the notation slightly, as it is already cumbersome, we did not do this, since the cases in which the probability distribution $\phi_t(\cdot|\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t)$ of the disturbance, or the set $\Xi_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t)$ of its possible values, effectively depend on one or more of the four variables $\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t$ are rare. In the following chapters we will always assume that the disturbance is white, noting, however, in the more crucial points that this condition is not actually necessary.

Now we will examine two cases in which equation (6.25) is, instead, very useful.

6.4.2 A reversible hydropower plant

The example that we will elaborate now allows us to demonstrate how disturbances of the form (6.25) can become necessary when an improper time step is adopted.

A reversible hydropower plant is described by the diagram in Figure 6.9.

The reservoir $R1$ feeds the penstock whose discharge is the only inflow to the reservoir $R2$. Suppose that the plant is already in operation, that the reservoirs are used exclu-

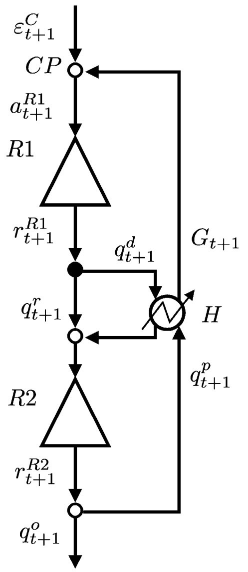

Figure 6.9: The diagram of a reversible power plant.

sively for hydropower purposes, and that the Project is concerned with the regulation policy for reservoir $R1$. In daylight hours the Regulator decides the flows u_t to release from $R1$ to generate energy, while during the night hours, using the energy supplied by the grid, a flow q^p_{t+1} is pumped from $R2$ to $R1$. The inflow a^{R1}_{t+1} to the reservoir $R1$ is thus the sum of the pumped flows q^p_{t+1} and of the natural inflow ε^C_{t+1} from its catchment. At night the release decision u_t is always zero, while during the day pumping is zero. The flow q^{pot}_{t+1} that could potentially be pumped is not decided by the Regulator, but it is created in real time by using the energy ε^p_{t+1} that the grid makes available. For this reason, even if q^{pot}_{t+1} plays the role of a decision from the perspective of $R2$, its subscript is $t+1$. The flow q^p_{t+1} that is effectively pumped does not always coincide with q^{pot}_{t+1}, because the available energy might permit the pumping of a flow q^{pot}_{t+1} that is either greater than the capacity $q^{\max}$ of the penstock, or greater than the actual storage s^{R2}_t in $R2$, or greater than the volume that can be stored in reservoir $R1$ (which is equal to the difference between its active storage $\bar{s}^{R1}$ and the actual storage s^{R1}_t).

From the above we deduce that the modelling step must be at least 12 hours and that the causal network that describes the power plant is the one inside the block H in Figure 6.10.

Note how this network corresponds to the union of the causal networks that describe the power generation plant and the pumping power plant (Figures 5.27 and 5.29). The figure also shows the causal networks of the upstream and downstream reservoirs, since only in this way does one appreciate the structure of the system, which is constituted by components, which, however, are so strongly integrated that the power plant can be thought of as a single component. The grey areas individuate the components, while the dashed lines interconnect the input and output variables. The whole figure is the causal network of the reversible

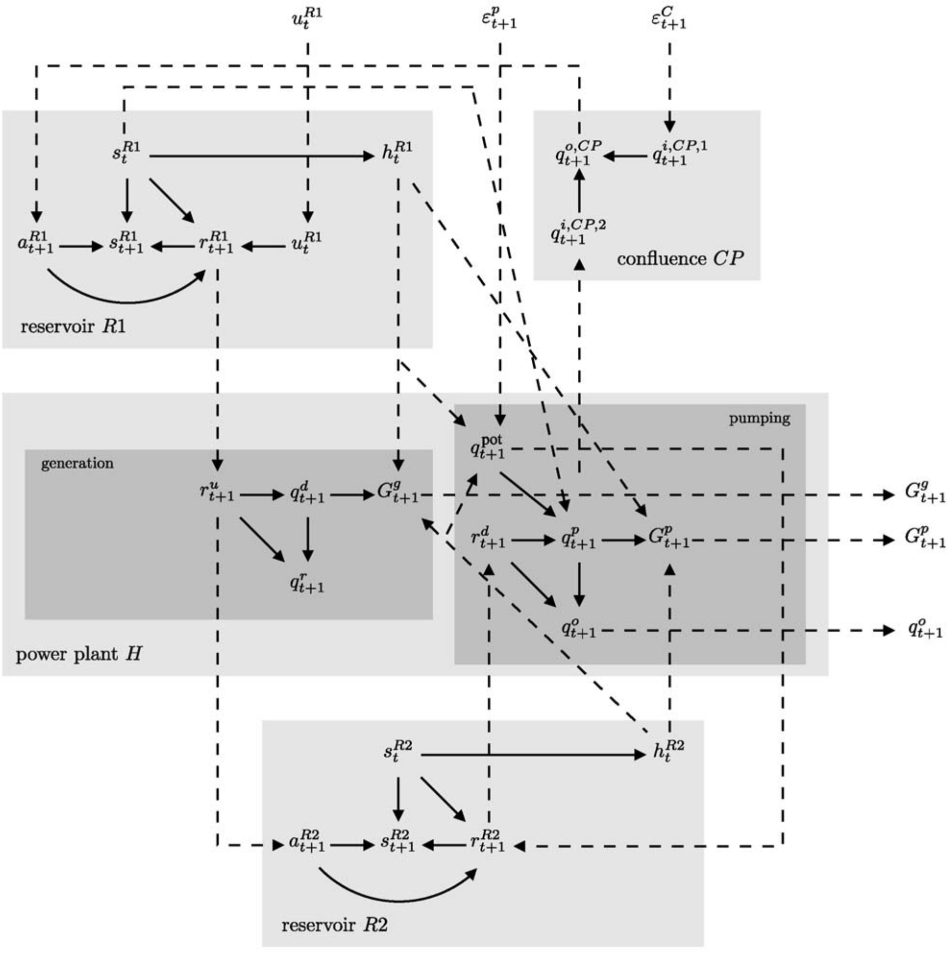

Figure 6.10: The causal network of a reversible power plant.

power plant. Examining this network allows us to identify the disturbances that act on the system (ε_{t+1}^{C} and ε_{t+1}^{p}) and to interconnect the equations that describe the components, thus obtaining the following list of equations

$$
\begin{aligned}
CP \qquad & q_{t+1}^{o,CP} = \varepsilon_{t+1}^{C} + q_{t+1}^{p} \\
R1 \qquad & s_{t+1}^{R1} = s_t^{R1} + q_{t+1}^{o,CP} - R_t^{R1}\big(s_t^{R1}, u_t^{R1}, q_{t+1}^{o,CP}\big) \\
& u_t^{R1} \in \mathcal{U}_t^{R1}\big(s_t^{R1}\big) \\
& r_{t+1}^{R1} = R_t^{R1}\big(s_t^{R1}, u_t^{R1}, q_{t+1}^{o,CP}\big) \\
& h_t^{S1} = h^{R1}\big(s_t^{R1}\big)
\end{aligned}
$$

$$
\begin{aligned}
H \qquad q_{t+1}^d &= \begin{cases} 0 & \text{if } (r_{t+1}^{R1} - q_t^{\text{MEF}}) < q^{\min} \\ \min\{(r_{t+1}^{R1} - q_t^{\text{MEF}})^+, q^{\max}\} & \text{otherwise} \end{cases} \\
G_{t+1}^g &= \psi \eta^g g \gamma q_{t+1}^d \left[h_t^{R1}\left(s_t^{R1}\right) - h_t^{R2}\left(s_t^{R2}\right)\right] \\
q_{t+1}^r &= r_{t+1}^{R1} - q_{t+1}^d \\
q_{t+1}^{\text{pot}} &= \varepsilon_{t+1}^p / \left[\psi \eta^p g \gamma \left[h_t^{R2}\left(s_t^{R2}\right) - h_t^{R1}\left(s_t^{R1}\right)\right]\right] \\
q_{t+1}^p &= \min\left\{q_{t+1}^{\text{pot}}, r_{t+1}^{R2}, (\bar{s}^{R1} - s_t^{R1})^+, q^{\max}\right\}, \\
G_{t+1}^p &= \psi \eta^p g \gamma q_{t+1}^p \left[h_t^{R2}\left(s_t^{R2}\right) - h_t^{R1}\left(s_t^{R1}\right)\right] \\
q_{t+1}^o &= r_{t+1}^{R2} - q_{t+1}^p \\
R2 \qquad s_{t+1}^{R2} &= s_t^{R2} + r_{t+1}^{R1} - R_t^{R2}\left(s_t^{R2}, q_{t+1}^{\text{pot}}, r_{t+1}^{R1}\right) \\
r_{t+1}^{R2} &= R_t^{R2}\left(s_t^{R2}, q_{t+1}^{\text{pot}} r_{t+1}^{R1}\right) \\
h_t^{R2} &= h^{R2}\left(s_t^{S2}\right)
\end{aligned}
\qquad (6.26)
$$

Generation

Pumping

The disturbances are ε_{t+1}^C and ε_{t+1}^p, the control is u_t^{R1}, the state is the vector $|s_{t+1}^{R1}, s_{t+1}^{R2}|^T$ and the outputs are the energy generated (G_{t+1}^g) and absorbed (G_{t+1}^p) as well as the flow q_{t+1}^o that exits the system. The state transition function and the output transformation of the model of the reversible power plant are derived from equation (6.26) with the procedure described in Section 6.1.4.

Very frequently, historical series with a 12-hour time step are not available: they usually have a daily time step, and the data about the pumped flows q_{t+1}^p are missing. One must necessarily describe the system with a daily step. If, as often happens, the reservoir $R2$ has an active storage that is much smaller that $R1$'s, and so its time constant is much shorter, its dynamics can no longer be described with this new time step, because this would violate the condition posed by the Sampling Theorem. In this case, $R2$'s storage varies significantly over the course of the day and this variation is unobservable if a daily step is adopted. In fact, very often its storage appears nearly constant if it is observed every day at the same hour, i.e. it appears to be in equilibrium. On the other hand, it is exactly by assuming that it is in equilibrium that we can ignore it.

Thus, the diagram that describes the power plant with a daily step is the one shown in Figure 6.11, which appears to be indistinguishable from the one of a pure generation power plant that is fed by a reservoir (Figure 5.26).

One can see, however, that from an operational perspective the two systems cannot be equal, since the first reuses the same water several times thanks to nighttime pumping. The difference must therefore be in the inflows. The causal network (Figure 6.12) that describes the reservoir $R1$ in Figure 6.11 is, in fact, different from the one of a normal reservoir (Figure 5.3).

The difference (aside from the lack of evaporation that was not considered intentionally) is the presence of the two grey dashed bold arcs in Figure 6.12, which make the disturbance a_{t+1}^{R1} dependent on s_t^{R1} and u_t^{R1}. This dependence is the consequence of the fact that pumping during the night hours is influenced by the release in the preceding 12 daylight hours,

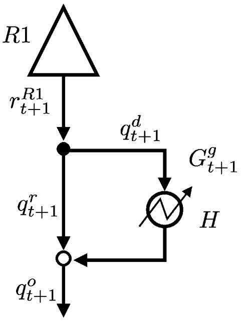

Figure 6.11: The diagram of a reversible power plant with a daily step.

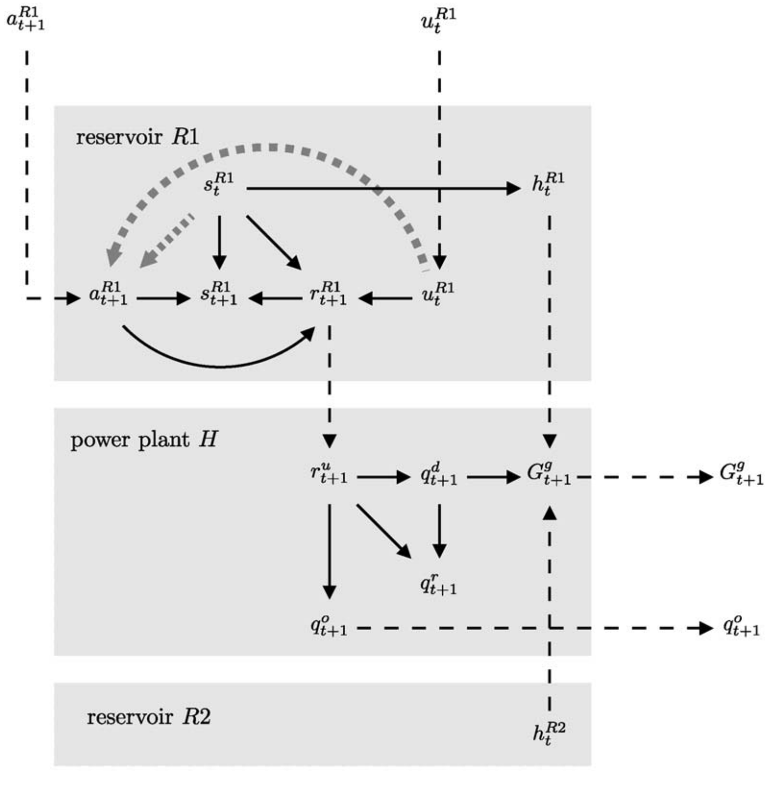

Figure 6.12: The causal network for a reversible power plant with a daily step.

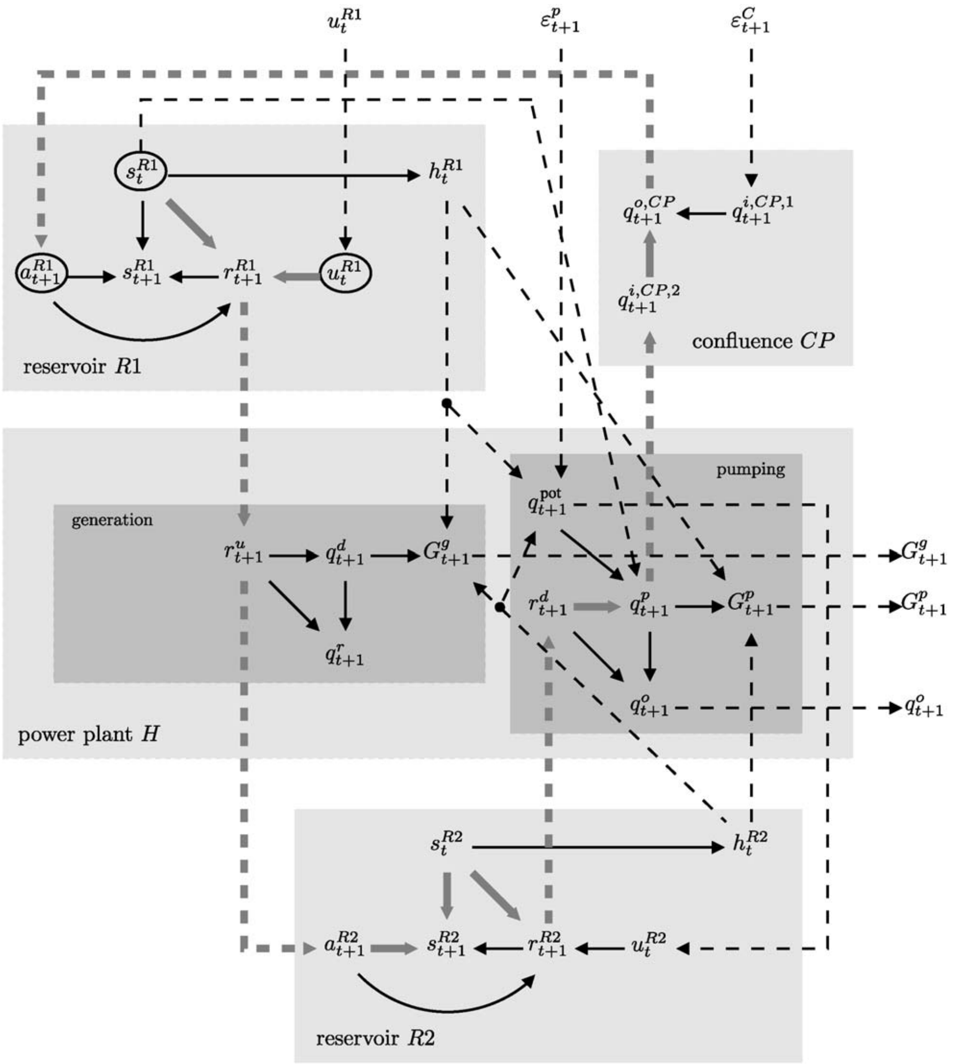

Figure 6.13: The causal network for a reversible power plant: the meaning of the arcs in grey dashed bold is explained in the text.

which in turn depends on s_t^{R1} and u_t^{R1}, as is shown by the succession of grey dashed bold arcs[14] in Figure 6.13.

If we adopt a daily time step, the daytime hours and the nighttime hours are combined in a single step so that the inflow a_{t+1}^{R1} in the 24 hours depends on what was released in the first 12. A whiteness test on an historical series of a_{t+1}^{R1} would therefore show that the inflow

[14]Note that the arc between s_{t+1}^{R2} and s_t^{R2} is travelled in the opposite direction to the arrow: this denotes the fact that one must wait for a time step of 12 hours before s_t^{R1} and u_t^{R1} influence a_{t+1}.

process is not white,[15] but the network in Figure 6.12 shows us that we can describe it with a conditional probability distribution, which can easily be estimated from historical data, as follows

$$a_{t+1}^{R1} \sim \phi_t\big(\cdot|s_t^{R1}, u_t^{R1}\big)$$

Therefore, the disturbance takes the form of equation (6.25) and the policy can be designed with the following model, which has a daily step, and is obtained from equation (6.26)

$$\begin{array}{lll}
R1 & s_{t+1}^{R1} = s_t^{R1} + a_{t+1}^{R1} - R_t^{R1}\big(s_t^{R1}, u_t^{R1}, a_{t+1}^{R1}\big) & \\
 & u_t^{R1} \in U_t^{S1}\big(s_t^{S1}\big) & \\
 & r_{t+1}^{R1} = R_t^{R1}\big(s_t^{R1}, u_t^{R1}, a_{t+1}^{R1}\big) & \\
 & h_t^{R1} = h^{R1}\big(s_t^{R1}\big) & \\
H & q_{t+1}^{d} = \begin{cases} 0 & \text{if } (r_{t+1}^{R1} - q_t^{\text{MEF}}) < q^{\min} \\ \min\{(r_{t+1}^{R1} - q_t^{\text{MEF}})^+, q^{\max}\} & \text{otherwise} \end{cases} & \text{Generation} \\
 & G_{t+1}^{g} = \psi \eta^g g \gamma q_{t+1}^{d}\big[h_t^{R1}\big(s_t^{R1}\big) - h^{R2}\big] & \\
 & q_{t+1}^{r} = r_{t+1}^{R1} - q_{t+1}^{d} & \\
 & q_{t+1}^{o} = r_{t+1}^{R1} &
\end{array}$$

in which h^{R2} is considered to be a constant reference value.

6.4.3 Random disturbances per-period

In the description of the disturbance that we have adopted until this point the uncertain element is the value that the disturbance assumes at each time t and so it is called a *per-step description*. This is not, however, the only description possible, since sometimes the uncertain element might be the trajectory ω (we will call it a *scenario*) that the disturbance follows over an entire period

$$\omega = \begin{vmatrix} \boldsymbol{\varepsilon}_1 & \cdots & \boldsymbol{\varepsilon}_T \end{vmatrix}$$

The new description is called *per-period description* and is advantageous when the temporal correlation of the disturbances is very strong within a period, but weak between one period and the next. This happens for example with the flow of the Nile at Aswan (see the hydrograph in Figure 6.14): after the flood peak, which occurs every year in July, the hydrograph is essentially a succession of recession curves.

Per-period descriptions can be adopted both for stochastic and uncertain disturbances, but they are most commonly used for the second. This kind of description assumes that, at the beginning of each period and in a way that is completely unknown to us, nature chooses a scenario ω from the elements of a set Ξ (called the *reference set*) and successively, at every time t until the end of the period, applies a value $\boldsymbol{\varepsilon}_t$ defined by ω. At the end of the period it will choose the scenario to apply in the following period. If one knows the probability $\phi(\omega)$ with which each scenario is chosen, the description is a stochastic one.

[15] Given that s_t^{R1} is correlated to its values at the previous time steps.

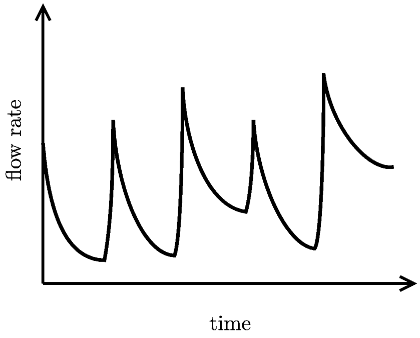

Figure 6.14: The qualitative behaviour of the flows in the Nile at Aswan.

Despite the apparently marked difference between the per-period and per-step description, it is possible to trace one back to the other, so that in the following we can simply consider the last without loss of generality.

Given a disturbance $\boldsymbol{\varepsilon}_{t+1}$, which has a per-period description, its transformation into an equivalent disturbance $\tilde{\boldsymbol{\varepsilon}}_{t+1}$ that has a per-step description is carried out by posing

$$\tilde{\boldsymbol{\varepsilon}}_{t+1} = |\boldsymbol{\varepsilon}_{t+1}, \eta_{t+1}|$$

where η_{t+1} is a new disturbance, and by introducing a new state z_t that represents the scenario index, i.e. a variable whose value identifies the current scenario. More precisely, having termed N the number of elements of Ξ and having established a one-to-one correspondence between the elements of Ξ and of the set $Z = \{1, \ldots, N\}$, the model of the new disturbance turns out to be the following

$$\tilde{\boldsymbol{\varepsilon}}_{t+1} \in \tilde{\Xi}_t(z_t) = \begin{cases} \{(\boldsymbol{\varepsilon}_1^1, 1), \ldots, (\boldsymbol{\varepsilon}_1^N, N)\} & \text{if } t = kT, \text{ with } k = 0, 1, \ldots \\ (\boldsymbol{\varepsilon}_{t+1}^{z_t}, z_t) & \text{otherwise} \end{cases} \tag{6.27a}$$

and the dynamics of the state z_t is described by the transition equation

$$z_{t+1} = \eta_{t+1} \tag{6.27b}$$

In this way the new disturbance $\tilde{\boldsymbol{\varepsilon}}_{t+1}$ formally has a per-step description, while it behaves like a disturbance per-period. If in fact at time kT the scenario index assumes the value $\bar{z}$, then for equation (6.27) z_t remains equal to $\bar{z}$ over the entire $(k+1)$th period and the disturbance $\boldsymbol{\varepsilon}_{t+1}$ follows the scenario

$$\omega^{\bar{z}} = \left| \boldsymbol{\varepsilon}_1^{\bar{z}} \quad \cdots \quad \boldsymbol{\varepsilon}_T^{\bar{z}} \right|$$

of the set Ξ.

We must conclude with an important observation. The model of the equivalent disturbance does have, however, a peculiar characteristic: *the state z_t is not measurable* and so the state of the global model will not be either. Since, as we explained in Section 10.1, the control policy requires that the state is measurable at every time instant, one can use the description of the disturbance per-period only if one can construct an estimator of z_t. This is not always possible and so, before adopting this description, it is advisable to verify that the estimator can actually be constructed.

Chapter 7

Identifying the optimal alternative

AC and RSS

In Part B we explained how to analyse the context of the Project in order to identify the aim, the Stakeholders that are involved, the actions that it is useful to consider, and that can be considered, the criteria, and the indicators to evaluate them; then, we comprehensively examined the process of identifying the model of the system. We have thus concluded the analysis of the preparatory *Reconnaissance* phase and the first three phases of the PIP procedure described in Section 1.3. Now we enter into the examination of the fourth, the phase of *Designing Alternatives*. We have already pointed out that, in planning practice, the alternatives considered are, quite often, only those that the Regulator's experience and the Stakeholders' proposals suggest. We have seen that on the one hand it is right to start with these alternatives, but on the other it is a mistake to limit the choice to this set alone. It is a better idea to consider all the alternatives obtained by combining the actions identified in Phase 1 in all the possible ways. The number of alternatives that are obtained in this way is, however, very high, and so it is necessary to choose the 'most interesting' ones. This choice should not be based on the Analyst's criteria, but made on the basis of the Stakeholders' criteria, which were identified in Phase 2. The phase of *Designing Alternatives* is therefore dedicated to the definition and solution of a *Design Problem* whose aim is to identify the *efficient alternatives* with respect to some appropriate *objectives*, which are defined on the basis of the *indicators* proposed by the Stakeholders.

From a mathematical point of view, defining and solving this problem is a complex task, which is further complicated by the plurality of objectives and the need to involve Stakeholders in its definition and solution. To simplify the presentation, we have decided to follow the teaching path adopted in Chapter 2. Firstly, we will examine the case in which the decision is made under the assumption of full rationality, when a single evaluation indicator is given[1]

$$i = i\big(\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_h; \mathbf{u}^p; \mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{h-1}; \mathbf{w}_0, \mathbf{w}_1, \ldots, \mathbf{w}_{h-1}; \boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \ldots, \boldsymbol{\varepsilon}_h\big) \quad (7.1)$$

[1] We will assume in the following that the indicator i is such that, given two alternatives $A1$ and $A2$, the inequality $i(A1) < i(A2)$ implies that alternative $A1$ is preferred to $A2$ (or vice versa), so that the optimal alternative is the one that minimizes (or maximizes) i. This condition does not always hold because it may be that the most satisfying value of the indicator is neither the maximum nor the minimum, for example when a wetland area is being considered (see Section 17.1). In that case, as we will show in Chapter 17, a value function must be applied to the indicator before it is used. However, in order to simplify the exposition, we do not consider this problem in Part C.

In practice, this case arises very rarely in a real problem, because its prerequisite is that there be just one Stakeholder in the system, certainly a very unusual situation. This case would only be relevant if the Decision Maker (DM) decided to exclude the Stakeholders from the decision, as would be the case, for example, if a Cost Benefit Analysis (CBA) were adopted. This last eventuality, however, does not concern us, given the central role of participation in the definition of our decision-making procedure. Nonetheless, by starting off with the analysis of this case it is easier to progressively explore the difficulties inherent in the Design Problem. Only in Part D will the hypothesis of dealing with a single indicator be removed, and a plurality of objectives be introduced, along with the consequent conflicts of interest.

7.1 Why the Problem is difficult

Once the presence of multiple Stakeholders, and thus multiple objectives, has been excluded, the decision-making problem is greatly simplified: the choice cannot but identify the *optimal alternative*, i.e. the alternative that provides the best value of the sole indicator considered. The reader might rightly wonder why the Design Problem is still so difficult. Actually, there are still three main difficulties: the presence of infinite alternatives; the uncertainty of the effects due to the randomness of the disturbances; and the presence of recursive decisions. Before looking in detail at the ways to overcome these difficulties in the following chapters, we would like to acquaint the reader with their nature.

7.1.1 Infinite alternatives

If the number of alternatives to examine is finite (and reasonably small), the optimal alternative can be identified with a simple exhaustive procedure: for each alternative the value of i is assessed and the optimal alternative is the one corresponding to the minimum value of i (assuming that the indicator represents a cost, as we always do).

If the number of alternatives is infinite (and in practice, even if it is finite but very high), the exhaustive procedure cannot be applied because it would never conclude or would require an unacceptably long time. One must therefore define a procedure that, by examining a finite number of alternatives, allows us to identify the optimal alternative, or at least an alternative that is reasonably close to it, within acceptable computation times.

7.1.2 Uncertainty of the effects

When there are random disturbances acting on the system, as equation (7.1) shows, the value of i is a random variable too. It follows that, except in very particular cases, it is no longer possible to rank the alternatives.

To fully appreciate this problem, consider a Project in which the decision concerns the height of a riverbank that should protect a city from flooding. Let the indicator i be the sum of future discounted damage plus the cost of constructing the river bank. If we knew that in the future there would not be any more floods, the optimal decision would clearly be not to construct the bank, since that would make the value of i zero. If, instead, we knew that the river would flood again, and we also knew the trajectory of future flows ahead of time, we could determine the optimal height. Its value would clearly not be zero, and would be different according to the trajectory of the flows, as one can easily understand by observing that for the same value of the bank's height, the value of i varies with the variations in the trajectory that occurs. Thus, when the river's flow is a random disturbance, and, as a consequence, the trajectory of the flows is not known in advance, one is not able to decide.

In this situation, to choose the optimal alternative we have to 'filter out the uncertainty', i.e. to associate a deterministic value with the cloud of values that i can assume in correspondence with each alternative. More precisely, we have to associate a value with the probability distribution of i, when disturbances are stochastic, or with the set of values assumed by i, when disturbances are uncertain. In the previous example one could choose the height of the bank that would minimize the expected value of i, but a more cautious person might suggest the height that minimizes the cost in the worst case. The choice depends then on the DM's *risk aversion* and this should be taken into account. There is a number of ways to do this, not only those mentioned above, and for each of them we should understand pros and cons.

7.1.3 Recursive decisions

In Section 2.1 we explained the difference between planning decisions and management decisions and how the second can be traced back to the first through a *regulation policy*. When regulation is included among the actions, in order to specify an alternative one must specify a regulation policy, which can be, in the simplest case, a periodic succession of *control laws* $m_t(\cdot)$

$$p = \left\{m_0(\cdot), \ldots, m_{T-1}(\cdot), m_0(\cdot), \ldots\right\}$$

each of which specifies the control $\mathbf{u}_t$

$$\mathbf{u}_t = m_t(\mathbf{x}_t) \tag{7.2}$$

that must be adopted at time t in correspondence with the state $\mathbf{x}_t$ of the system. To specify this policy it will therefore be necessary to design T functions and this is not a simple task. It is even more complicated when the policy is *set-valued* (see Section 2.1.1), or when it includes a compensation line as seen on page 42.

7.2 Organization of Part C

The presentation of the above topics follows the logical development that we have laid out in these pages.

We will begin by considering the Design Problem in Chapter 8 in a simplified case, in which only planning actions are considered and there are no stochastic disturbances that act on the system. In this way we will be able to concentrate on the difficulties posed by the presence of infinite alternatives. In Chapter 9 we will consider a case in which the disturbances are stochastic and study a way to take into account the risk aversion that, more or less manifestly, characterizes every DM's decisions. In this way we will conclude the analysis of the *Pure Planning Problem*, i.e. the problem in which there are no management decisions.

The latter are introduced in the following step, and, because of their complexity, are discussed in a number of chapters. Chapters 10 and 11 are dedicated to the formalization of the concept of policy, to the formalization of the *Pure Management Problem*, and, finally, to its integration in the Pure Planning Problem, to arrive at a more general formulation of the *Design Problem*. The solution of this Design Problem consists in nesting the algorithms that solve the Pure Management Problem in the algorithms that solve the Pure Planning Problem, which is studied in Chapter 8.

The following three chapters are dedicated to the search for these algorithms. In Chapters 12 and 13 algorithms for *off-line design* are studied, i.e. algorithms for designing a policy before it is used. Off-line design requires

(1) assuming the scenario (*design scenario*) in which the policy will operate; and

(2) defining the control laws (7.2) for all the states $\mathbf{x}_t$ that the system can assume.

Two disadvantages stem from this.

The first one is that the chosen policy depends on the design scenario and that, if that scenario does not actually occur, the effects that the policy induces will not be those expected and so the policy will not longer be optimal. To demonstrate this, think about a case in which one wants to design a regulation policy for a reservoir that feeds a hydropower plant. This would require establishing the power demand a priori, as well as hypothesizing that the plant is in service all the time. If at the moment of the decision the demand were less, or the plant were shut down for maintenance, the decision suggested by an off-line policy would not be optimal, because it would produce a supply that was either partially or totally useless.

The second disadvantage is that a considerable computation effort is made, even though it will be useless for the most part, since only a few states will actually be realized.

To overcome these two disadvantages, the policy can be re-designed every time that a decision is taken; i.e. the policy can be *designed on-line*, inheriting information from the off-line design about how to achieve the Project Goal in the long run. The on-line design algorithms will be described in Chapter 14.

At that point we will have considered the case in which the regulation policy operates on a system that contains reservoirs and, potentially, diversions. In Section 6.3, however, we explained that sometimes it is possible to identify a part of the system that contains diversions, and is without reservoirs: the so-called *distribution network*. When such a network exists, to reduce the computation time, it is advisable first to design its *distribution policy*, so that the distribution network can be embedded in the water system model as an *equivalent user*, and then to design the regulation policy for the whole system with the model so-obtained. In this way the computation times for the Design Problem are significantly reduced. The algorithms for the distribution policy design will be presented in Chapter 15 which closes this part.

After this section, we will finally be able to address the Design Problem in all its complexity, which arises when a plurality of Stakeholders is involved. The study of such a Design Problem is undertaken in Part D.

Once the policy has been designed, one is almost always interested in estimating the performances that it would provide given a particular scenario, and the corresponding probability distributions for some significant variables (releases, storages, ...). Such an estimate is obtained via simulation. We will deal with this topic in Chapter 19.

Chapter 8

Choosing among infinite alternatives

AC and RSS

In Chapter 7 we understood that, even when the Decision Maker (DM) is acting in conditions of full rationality, the design of the alternatives poses three difficulties: the presence of infinite alternatives; the presence of uncertainty produced by random disturbances; and the presence of recursive decisions. In this chapter, we begin to describe the tools to overcome these difficulties by analysing the Design Problem in the simplest of cases: the case in which the actions being considered are all planning actions (*Pure Planning Problem*), and the only disturbances acting on the system are deterministic. When this is the case, and the alternatives to examine are finite and few in number, we know that the procedure for solving the Problem is straightforward: the indicator (objective) is assessed in correspondence with each alternative, and the alternative that provides its best value is selected. When, instead, the number of alternatives is infinite, or finite but very large, this procedure is not practicable and we meet the first of the difficulties cited above. The aim of this chapter is to define a procedure to identify, with acceptable computation times, the optimal alternative, or at least an alternative very close to it.

8.1 The elements of the Planning Problem

Before we see how to formulate and solve a Pure Planning Problem, let us examine the elements that compose it.

8.1.1 The system model

A planning action $\mathbf{u}^p$ is an *una tantum* action, i.e. an action that is decided once and for all (see Section 1.1.4.2), without considering how it might influence an analogous decision made in the future. A typical example is the construction of a dam: in evaluating this action one certainly does not worry about how that dam might, in the future, influence the construction of another dam in that same place. The absence of dynamics in this type of decision does not imply, however, that the system for which it is made has to be a non-dynamical system. Thus it is described by the global model (6.21) that was introduced in Section 6.2, from which however, we will exclude the recursive decisions $\mathbf{u}_t$, which we have decided not to consider in this chapter. Therefore, the model has the following form

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{8.1a}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{8.1b}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{u}^p\right) \tag{8.1c}$$

$$g_{t+1} = g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \tag{8.1d}$$

$$\text{any other constraints} \tag{8.1e}$$

in which a step cost (g_t) appears as the only output. This is the result of two facts: first, we want to use the model for planning and thus we are interested only in the effects that are measured by the indicators; the other outputs do not interest us (if they did they would be indicators). Second, given that the problem has just a single objective, without loss of generality, we may assume that a single indicator is sufficient to express the DM's viewpoint. We will term this indicator *design indicator*.

Among the hypotheses posed at the beginning of the chapter, we included the absence of random disturbances $\boldsymbol{\varepsilon}_{t+1}$. This hypothesis can be fulfilled either because the system is actually not affected by random disturbances, or, more frequently, because we want to simplify the Problem by assuming that the disturbance's trajectory is known a priori. It is clearly unrealistic to think that we can know future rainfall patterns, for example, but just skim through the planning literature to see that, in order to make life easier, planners very often make this assumption. Moreover, for didactic reasons, it is useful to begin with an assumption which simplifies the Problem. Thus, let us suppose that we have a given trajectory of the disturbance, which we term *scenario*: it follows that the disturbance is deterministic and so we can include it in $\mathbf{w}_t$, thus increasing the number of components of this vector. As a consequence, the model (8.1) assumes the form

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \tag{8.2a}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{8.2b}$$

$$g_t = g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \tag{8.2c}$$

$$\text{any other constraints} \tag{8.2d}$$

As usual (see Section 4.11), *we assume that the system is periodic of period T*, i.e. we assume that the functions $f_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t)$ and $g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t)$ that appear in equation (8.2) are periodic. Exception is made for the trajectory of the disturbance that might be aperiodic.

8.1.2 The design indicator

The aim of the Problem is to find the optimal alternative from the DM's point of view, which is specified through a (design) *objective*. In Section 4.10.2 we defined an objective J as a *design indicator* i to which a *criterion* is applied to filter random disturbances. Since in this chapter we assume that there are not such disturbances, the criterion is useless: thus, design indicator and objective become synonymous and the symbols i and J are interchangeable.

In Section 4.10.1 we define the *indicator*

$$i = i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right) \tag{8.3}$$

as a functional of the trajectories[1] $\mathbf{x}_0^h$, $\mathbf{u}_0^{h-1}$, $\mathbf{w}_0^{h-1}$ and $\boldsymbol{\varepsilon}_1^h$ of the state, the control, and the disturbances that act on the system, over a time horizon $H = \{0, \dots, h\}$, which is termed

[1] The symbol $\mathbf{z}_0^h$ denotes the trajectory of the variable $\mathbf{z}_t$ in the horizon $\{0, \dots, h\}$, i.e. the succession $\{\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_h\}$.

design horizon. We have also seen that very often this functional is separable, in the sense that it can be expressed as a temporal aggregation[2] of h functions $g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$ called *step indicators* (or *step costs*)

$$i = \Psi\{g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}),\ t = 0, \dots, h-1;\ g_h(\mathbf{x}_h, \mathbf{u}^p)\} \tag{8.4}$$

where Ψ is a properly selected operator and the function $g_h(\cdot, \mathbf{u}^p)$ is called *penalty* and is not necessarily present.

Because of the hypotheses adopted in this chapter, the step indicator and the design indicator do not depend on the controls $\mathbf{u}_t$, or on the random disturbances $\boldsymbol{\varepsilon}_{t+1}$ either, so we can write

$$i = i(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{w}_0^{h-1}) \tag{8.5}$$

and

$$\Psi\{g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t),\ t = 0, \dots, h-1;\ g_h(\mathbf{x}_h, \mathbf{u}^p)\} \tag{8.6}$$

Let us consider equation (8.5) with more attention. The two trajectories $\mathbf{x}_0^h$ and $\mathbf{w}_0^{h-1}$ cannot be assumed arbitrarily since they are linked to the dynamics of the system: once the triple $(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1})$ is precisely given, the series $\mathbf{x}_1^h$ is univocally determined and can be calculated using equation (8.2a) recursively from 0 to $h-1$. Therefore, once the model is known, the value i of the indicator is expressed by the following expression

$$i = i(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}) \tag{8.7}$$

8.1.2.1 The design horizon

We must still clarify the role played by the horizon H. If we assume that the control $\mathbf{u}_t$ and the random disturbance $\boldsymbol{\varepsilon}_{t+1}$ appear among the arguments of the functions and their trajectories among the arguments of the functionals, what we will say is not only valid for Pure Planning Problems. It also holds for the other problems. For this reason, in the chapters that follow, we will often refer to the formulae described below.

The temporal aggregation operator Ψ can assume, in general, one or the other of two forms (see Section 4.10.1): the sum (the indicator is then termed *integral*) or the maximum (the indicator is termed *point wise* indicator). With the aim of making things clearer, in the text that follows we will use the operator Σ in the place of the inexpressive operator Ψ without loss of generality. Indeed, what we say may hold for the operator max, or for any other operator, by simply making the proper substitutions between the operators (for example by substituting Σ with max and '+' with max).

Two types of horizons can be considered in Pure Planning Problems: finite and infinite horizons. Clearly, the simpler of the two cases is the first: the performance of the system is considered along a *finite horizon* $t = 0, \dots, h$ and the indicator assumes the form

$$i(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}) = \sum_{t=0}^{h-1} g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t) + g_h(\mathbf{x}_h, \mathbf{u}^p) \tag{8.8}$$

[2] In Section 3.2.1.2 we mentioned that, in general, different levels of aggregation may exist and they can constitute a hierarchy, which is identified by the Stakeholders and/or the DM. The order of aggregation is usually fixed in such a way that the intermediate levels of aggregation are meaningful to them. When the Problem is a Pure Planning Problem, the order of aggregation is completely unessential, but, when the Problem is also concerned with the policy, the temporal aggregation must be at the highest level. For this reason this type of aggregation is the only one explicitly considered here and in the following.

The penalty $g_h(\mathbf{x}_h, \mathbf{u}^p)$ on the final state is somehow incoherent. If, in fact, there is no future after the final time h, why worry about which state the system will reach at the end of the horizon? When the system really ceases to exist after time h (an event which hardly never happens, but which is the only one that is coherent with the assumption that time finishes at h), the penalty $g_h(\mathbf{x}_h, \mathbf{u}^p)$ must be set equal to zero. In the opposite case, we must take into account the 'future costs' after h; these costs, by the very definition of the system, depend on the state $\mathbf{x}_h$ that is reached at the end of the horizon. However, it is often very difficult to define the penalty and so the temptation to assume it equal to zero, even when the life of the system is not really finite, can be strong. By doing so, one runs the risk that the unwanted system behaviours, which show up after time h, are not properly penalized and that the risk worsens increasingly with smaller values of h.

Logically, the simplest way to take into account the effects that an alternative produces in the long run is to consider an *infinite horizon*. It follows that the indicator is defined by a limit

$$i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^\infty\right) = \lim_{h\to\infty} \sum_{t=0}^{h} g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \tag{8.9}$$

which, to simplify the notation, we will denote with the following

$$i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^\infty\right) = \sum_{t=0}^{\infty} g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \tag{8.10}$$

However, this limit does not necessarily exist finite: when, for example, the minimum value g that the step cost can assume (see equation (4.29)) is positive, the value of i is sure to tend to infinity, i.e. it diverges, in the mathematical jargon. Therefore, equation (8.10) is not a good expression and it is advisable to modify it to be certain of never having to deal with infinite values, which not only can generate unpleasant and undesired paradoxes (as we know from the history of mathematics); but Decision Makers (DMs) are unlikely to accept them.

To guarantee that the indicator will converge we have two possible approaches. The first is to discount the future costs, and it is the approach almost always adopted when the indicator has an economic meaning. If the discount factor is indicated as γ, the indicator is defined in the following way

$$i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^\infty\right) = \sum_{t=0}^{\infty} \gamma^t g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \tag{8.11}$$

As is common practice, we will designate this form with the acronym TDC (*Total Discounted Cost*).

By discounting, the relevance of a given value of g_t diminishes with the distance in time at which it occurs. Nevertheless, this is not always acceptable. If, for example, the indicator represented the number of deaths caused by a given alternative (following a flood event, for example) we certainly could not discount it: the lives of those that live tomorrow do not have a lesser value than those that live today. We would therefore have to consider an *infinite horizon in correspondence with a zero discount rate*, i.e. for a discount factor[3] $\gamma = 1$. But we already know that this is not possible because equation (8.11) could diverge. In such

[3]The discount factor γ is related to the discount rate r by the relation $\gamma = 1/(1 + r)$.

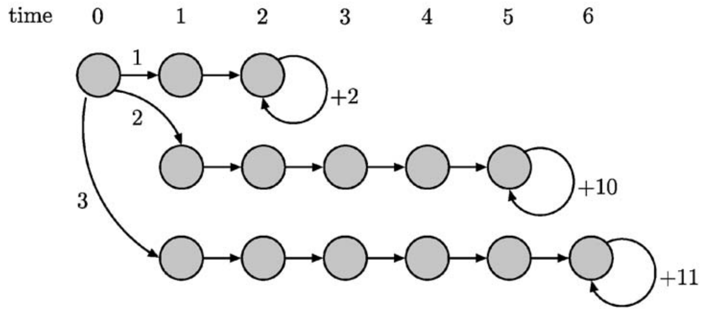

Figure 8.1: A deterministic automaton; all of the transitions have zero cost except for those represented by self-loops.

cases, we can revert to the second approach to guarantee convergence: define the indicator as the *average cost per step over an infinite horizon*

$$i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^\infty\right) = \lim_{h\to\infty} \frac{1}{h+1} \sum_{t=0}^{h} g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \tag{8.12}$$

Following consolidated praxis we will designate this form with the acronym AEV (*Average Expected*[4] *Value*). The AEV form presupposes that the behaviour of the system in the long-term is cyclostationary, i.e. that it tends to an equilibrium or a cyclostationary condition (cycle). Furthermore, although it is widely used, it has, nevertheless, the disadvantage that it does not distinguish between two alternatives that provide identical performances in the long-term (i.e. in steady-state conditions), but different in the short-term (i.e. in the transient). It is clear that the presence of the limit in equation (8.12) makes the costs in the transient insignificant with respect to the ones in steady-state conditions, while it is just as clear that between the two alternatives, in practice, we would lean towards the one that has lower costs in the transient.

To better understand the effects induced by the choice of time horizon, consider the example in Figure 8.1 from Kaelbling et al. (1996). The figure shows a deterministic automaton,[5] in which the three horizontal branches represent three planning alternatives, which are chosen at time 0 (the first node in the upper left) by assigning the values 1, 2 or 3 to the decision u^p. The transitions (arcs) from one state to another (the states are represented by circles) have an associated benefit, which is zero in the first n steps, where the value of n depends on u^p; then the benefit is positive and time-invariant, and the transitions are represented by a self-loop on the last state. Suppose that we adopt as indicator the total benefit over a finite time horizon with length h. When $h = 5$, the three alternatives produce the benefits 6.0, 0.0, 0.0, respectively, and therefore the optimal decision is $u^p = 1$. When h is greater than 5 and lower than 17, the optimal decision is $u^p = 2$. When the horizon is infinite and the discount factor γ is 0.9, the three alternatives provide the updated benefit

[4]Under the deterministic conditions assumed in this chapter, this attribute lacks of meaning. However, the acronym stems from the fact that the form (8.12) is generally used in the stochastic context and therefore, as we will see in Chapter 10, the operator $E[\cdot]$ has to be included (see equation (10.17)).

[5]A deterministic automaton is a system that has a finite number of states and whose transitions are not influenced by random disturbances.

16.2, 59.0 and 58.5, and therefore the optimal decision is still $u^p = 2$. Finally, when one considers the average benefit over the infinite horizon, the choice clearly goes to $u^p = 3$, which has an average performance of 11, as opposed to 2 and 10 of the first two. Therefore, it is evident that in reality the choice of the time horizon is critical, since it significantly influences the decision.

To conclude, it is important to underline that the indicator in the TDC form is more sensitive to what happens in the transient period than to what happens in steady-state conditions. In the AEV form it totally ignores what happens in a transient, however long it may be, and depends only on the steady-state performance. The example shows this very clearly. With the AEV form the best decision always turns out to be $u^p = 3$, no matter how long the transients with zero benefit are in the three alternatives. Instead, with the TDC form the decision $u^p = 2$ is the best only if the transients have the duration that is shown in the figure. The choice goes in fact to decision $u^p = 2$ because it begins to give a non-zero benefit at the fifth time instant, while with $u^p = 3$ this occurs at the sixth instant, but, if the transient period were shorter, $u^p = 3$ would be the best decision.

8.1.2.2 The step cost

The step indicator does not necessarily have to be the indicator associated to a single component; very often it is an aggregation (e.g. the sum or the maximum) of step indicators of one or more of the components. Resuming with the Piave example that we considered in Chapter 6, the indicator g_t could, for example, be defined as the sum of the energy G_{t+1}^{Hi} produced in each of the eight power plants

$$g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) = \sum_{i=1}^{8} G_{t+1}^{Hi}$$

or as the weighted sum of that value and the total water stress (clearly, with change of sign) of the six irrigation districts

$$g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) = \lambda \sum_{i=1}^{8} G_{t+1}^{Hi} - (1-\lambda) \sum_{i=1}^{6} g_t^{Ii} \tag{8.13}$$

with λ being a fixed *weight*. In Chapter 18 we will analyse how this idea can be fruitful.

8.1.3 The design scenario

In the previous section we saw that to compute the value of the objective it is necessary to have the trajectories $\mathbf{x}_0^h$ and $\mathbf{w}_0^{h-1}$ of the state and the deterministic disturbances over the entire design horizon H. We also said that the first is the trajectory that the state of the system follows in response to the triple $(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1})$ and that in order to determine it, it is sufficient to use equation (8.2a) recursively, once given that triple. We have not, however, explained the meaning of the second trajectory or clarified who provides it.

Together with the trajectories of all the parameters that define the model[6] and do not depend on the alternative being examined, this trajectory constitutes the so-called *design scenario*, which is the background that one assumes that the system will evolve in over the

[6] We do not mean just the trajectories of the time-varying parameters, but also the constant values of the time-invariant parameters, given that these can be seen as trajectories that are characterized by the same value at any time.

design horizon. For example, if the alternative concerns the construction of a reservoir, the scenario might include the trajectories of the water demand for irrigation and of the selling prices of the agricultural products, as well as the costs of the materials and the labour used to construct the dam, and lastly, the elevation at which flooding begins in the villages that are on the shores of the lake that will be created. Note, however, that, if the alternative includes the choice of crops in the irrigation district, the water demand trajectory for irrigation is not part of the scenario; just as if the decision to build a reservoir influences the markets in which agricultural products and building materials are sold and bought, the prices of agricultural products and the costs of materials are not part of the scenario. In these cases the model must provide these trajectories.

Thus the *design scenario is, by definition, constituted by the trajectory of all the variables that are not influenced, either directly or indirectly, by the alternatives being examined, and so do not depend upon the choice of the Decision Maker*. It follows that she can only speculate about which scenario would be best suited to evaluating the effects of the alternatives. As soon as possible, i.e. as soon as one gets sufficient information, these variables will be classified as random disturbances, rather than parameters or deterministic disturbances, and as such, they will not be described by trajectories, but through models, which are part of the global model. On closer consideration, therefore, also these models are part of the scenario, but it is common practice to use the term scenario to denote only the set of deterministic variables (parameters and disturbances) that are described by preassigned trajectories. Thus, it is correct to adopt a deterministic description for a variable only when the information available is so poor that one cannot derive a random description, or when one is sure of its value. Unfortunately, however, as we will see shortly, the scenario can be used incorrectly to simplify the mathematical formulation of the Problem, but this is to be avoided. In the case of extreme uncertainty[7] it is best to postulate several Design Scenarios and solve the Problem for each of them (see Section 1.3.3). By doing so, a sensitivity analysis of the optimal alternative with respect to the scenario is performed.

In the definition of the Design Scenario, particular attention must be paid to the definition of the trajectories of those variables, such as water demand, that seem to be periodic of period T, but which are not; we will call them *pseudo-periodic* variables. A variable of this type seems, at first glance, to be periodic, and suggests that in order to define its trajectory over the whole design horizon, let us say over a period of n years, it is sufficient to repeat the same series of T values n times (taking into account the 29th of February in leap years). Actually, this way of proceeding is not effective when the trajectory of the variable contains a secondary periodicity with a period Δ, which is shorter than the period T of the supposed primary periodicity, and when Δ is not a submultiple of T. To make this clearer, think about hydropower water demand: its value varies according to the day of the week; typically it is lower on Saturday and Sunday, when most industrial activities are suspended, and it reaches a peak on Wednesday. The values also change slowly, from one week to another, since the demand is higher in winter than in summer.[8] But one year is not a multiple of one week and so, even if a series of 365 demand values is aligned with the weekends of the first year of the design horizon, it will not be aligned with the weekends of the following years if it is repeated in direct succession to describe a longer time period. To preserve the correspondence between the original series and the days of the week in the following years, one has to shift it by a certain number of days before it is chained.

[7] Think for example of Mozambique which could, but not necessarily, adopt methanol as vehicle fuel, a possibility that cannot be ignored in the River Zambesi planning process, see page 51.

[8] When living standards do not include a heavy use of air conditioners.

The simplest way to fix a *chaining rule* is to define the 365 base values[9] with reference to what we will call a *standard year*: a non-leap-year, which begins on Monday. From this series, one can derive the trajectory of the design scenario by associating the current day with the demand value for the closest homonymous day of the standard year, i.e. the day of the standard year that has the same name (Monday, Tuesday, ...) and is the closest to the current day. In this way, two temporal indices are associated to each day: the first is the ordinal number (0–364/365) that labels it with respect to the first day of the year (day 0), which we call *natural date*; the second is the number that labels the closest homonymous week day in the standard year,[10] which we will call *anthropic date*. With this rule, it is easy to construct a trajectory, of any length, given the T values of the standard year and the period Δ of secondary periodicity (one week in the example).

A few words, finally, about the selection of the scenario values. Since we are planning for the future, the demand values and, more generally, the values of all the variables cannot be those that were recorded in the past, but those that are expected in the future, when the alternative will be implemented. To determine these values one makes use of suggestions provided by Experts, but also of information that can be provided by models, such as demographic and economic models. Hydrological models can also be used, catchment models for example, to generate a *synthetic series* (Section 6.4.1) of inflows. Remember the observations reported in Section 4.11 about the steady-state hypothesis that is often invoked in this context, especially with reference to inflows.

Sometimes, one encounters Projects for long-term planning (we will talk about these in the next section) in which the time horizon is one year and the deterministic disturbance scenario is the trajectory of the random disturbances (almost always the inflows) in the so-called *average year*[11] or in the *worst year*.[12] The aim of this *naïve* approach is to evaluate the average (or worst) performance of the system, but to do so, instead of computing the expected value (or worst value) of the indicator with respect to the trajectory of the disturbance, it evaluates the indicator in correspondence with the average (or worst) conditions that the system can encounter. However, the following identities

$$\mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,T}}\left[i\left(\mathbf{x}_0, \mathbf{u}^p, \boldsymbol{\varepsilon}_1^T\right)\right] = i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{T-1}\right)$$

$$\max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,T}}\left[i\left(\mathbf{x}_0, \mathbf{u}^p, \boldsymbol{\varepsilon}_1^T\right)\right] = i\left(\mathbf{x}_0, \mathbf{u}^p, \bar{\mathbf{w}}_0^{T-1}\right)$$

with $\mathbf{w}_t$ being the expected value and $\bar{\mathbf{w}}_t$ the worst value of the disturbance do not apply, except in the case where the indicator is linear. Since today we are able to formulate and solve the correct Problem, it is advisable to avoid this type of formulation, while in the past its use could be justified because of the modest computation potential available at that time. However, the concept itself can still be useful, and we will come across it again in Section 14.3.1 in another context.

[9] Here, to simplify the presentation, we hypothesize a daily time step; the general case is, however, easy to obtain.

[10] The standard year is labelled using the natural date system.

[11] Given the historical sequence $\boldsymbol{\varepsilon}_0^{(n+1)T-1}$ of the disturbance, the trajectory in the average year is defined by posing $\mathbf{w}_t = \sum_{k=0,\dots,n} \boldsymbol{\varepsilon}_{t+kT}/(n+1)$ for $t = 0, \dots, T-1$.

[12] The trajectory in the worst year is defined by posing $\bar{\mathbf{w}}_t = \min_{k=0,\dots,n} \boldsymbol{\varepsilon}_{t+kT}$ (or $\bar{\mathbf{w}}_t = \max_{k=0,\dots,n} \boldsymbol{\varepsilon}_{t+kT}$) for $t = 0, \dots, T-1$. The choice between min and max depends on the context of the Problem being considered.

8.2 Formulating the Problem

At this point we are able to formulate the Design Problem for the case of Pure Planning.

Let us consider the design objective (8.7) once again: given the initial state $\mathbf{x}_0$ and the trajectory of the deterministic disturbance $\mathbf{w}_0^{h-1}$, its value is a deterministic function of the planning decision $\mathbf{u}^p$ and, as such, it can be minimized[13] with respect to $\mathbf{u}^p$. Thus the most general form that the problem can assume is the following:

The Pure (Deterministic) Planning Problem: *To determine the* ***optimal alternative***[14] $\mathbf{u}^{p*}$, *i.e. the alternative that solves the following problem*

$$J^* = \min_{\mathbf{u}^p} i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}\right) \tag{8.14a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t\right) \quad t = 0, 1, \ldots, h-1 \tag{8.14b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{8.14c}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{8.14d}$$

$$\mathbf{x}_0 \textit{ given} \tag{8.14e}$$

$$\textit{any other constraints} \tag{8.14f}$$

The meaning and the implications of this formulation deserve a few remarks. In equation (8.14a) the objective is defined over a finite horizon, but nothing prevents it from being defined over an infinite horizon, in which case its definition would include a limit, as shown by equations (8.9) and (8.12). The constraints (8.14b)–(8.14f) specify that the objective must be evaluated over the trajectory of the state which, starting from the initial state $\mathbf{x}_0$, is produced by the trajectory $\mathbf{w}_0^{h-1}$ of the deterministic disturbance (two elements that contribute to define the Problem), and by the decision $\mathbf{u}^p$, which can be chosen only from among the feasible decisions, which are specified by the set $\mathcal{U}^p$. These constraints, which must always be included, are not the only ones possible; there can be others (for an example see Section 8.3).

The formulation of problem (8.14) is the most general one, because it considers all that happens within the system following the implementation of the alternative, from time 0 onwards; i.e. it considers the *transient* period, just as much as the *steady-state* conditions (see Section 3.2.2.1). However, the DM might not want to consider both of them: perhaps only the second is of interest for her.

To make this idea clearer, consider the Dead Sea Project,[15] whose purpose is to size (by determining the diameter u^p) a pipeline which transfers a flow from the Mediterranean

[13] Here, and in the following text, we maintain the convention that the design objective is to be minimized, coherently with the character of the 'cost' we attributed to it. The assumption does not carry any limitation, since each problem of maximum can be converted into a problem of minimum by simply changing the sign of the objective.

[14] If there is more than one optimal alternative, it might be interesting for the DM to determine all of them in order to choose the one that best satisfies any possible criteria that were not expressed in the formulation of the Problem.

[15] The Project is presented in Project P1 on the CD: here we will refer only to the information that is useful for the example.

Sea to the Dead Sea, to use it for hydropower purposes by exploiting the hydraulic head that exists between the two seas (about 390 metres). Just for the purposes of the example, assume that this flow is time-invariant.[16] The inflow w_t that feeds the Dead Sea from its tributaries is a stochastic cyclostationary process, but to conform to the deterministic context hypothesized for this chapter, we must describe it as a deterministic disturbance, specified by a periodic trajectory of period T, which is univocally defined by the knowledge of its behaviour w_0^{T-1} in the first period. Since the Dead Sea does not have effluents, the inflow is lost only through evaporation[17] and, given that evaporation does not necessarily follow the variations of the inflow, the level x_t of the sea is not constant, but fluctuates periodically, following a cycle of period T, which can be identified by solving the following system of algebraic equations[18]

$$\begin{aligned} \bar{x}_1 &= f_0(\bar{x}_0, 0, w_0) \\ &\vdots \\ \bar{x}_0 &= f_{T-1}(\bar{x}_{T-1}, 0, w_{T-1}) \end{aligned}$$

where $f(\cdot)$ is the function that describes the dynamics of the level of the sea and $u^p = 0$ denotes the absence of the pipeline (for more details see Project P1 on the CD). Once the pipeline were constructed and operating, the level of the sea would begin to rise and, as a consequence, its surface area would increase, given that the tectonic trench in which the sea lies does not have perfectly vertical walls. Because of this, the evaporated flow would also increase (see Section 5.1), until a new steady-state condition was established through which the inflowing water volume and the amount of evaporation would be equalized over the course of the year. In other words, from time 0, when the new pipeline begins operating, the system would enter a transient that would lead, after some time (probably a few years), to a new steady-state condition. If in the design of the pipeline[19] we were to adopt the form of problem (8.14), an infinite horizon and an indicator with the TDC form, we would give a lot of importance to the transient period with respect to the steady-state conditions, given that the first occurs before the second and so its costs would be discounted less than the ones of the other. It is probable that this emphasis on the short-term costs would not be appreciated by the DM, who might want to evaluate the alternatives only on the basis of their effects in the long-term (remember what was said in Section 3.2.2.1).

To evaluate the long-term (steady-state) effects alone, one must use the AEV form of the indicator, because, as we highlighted in Section 8.1.2.1, it totally ignores what happens in the transient, no matter how short it is, and considers only the long-term. Alternatively, a different formulation of the Problem can be adopted, which is completely equivalent to, i.e. defines the same optimal alternative of, problem (8.14) with the objective in the AEV

[16] By doing so the Problem becomes a Pure Planning Problem, otherwise it would be a mixed Problem given that the regulation policy for the pipeline would have to be defined.

[17] For this reason the waters of the Dead Sea are so salty that they do not support any form of life, from which its name.

[18] Note that, when $T = 1$, i.e. when the system is time-invariant, the algebraic system is reduced to the equation

$$\bar{x} = f(\bar{x}, 0, w)$$

that defines the state of equilibrium of the system.

[19] Note that $u^p = 0$ denotes the absence of the pipeline, so we are considering, among the various alternatives, also the Alternative Zero: not to construct anything.

form, but which, when the objective and the constraints have particular forms, allows us to determine the optimal alternative more quickly. This formulation considers the costs that are encountered in the long-term (steady-state conditions) and it is derived from problem (8.14) by defining the indicator over one cycle (the horizon $[0, \dots, T-1]$) and substituting the dynamical system (8.14b) with the algebraic system that defines the cyclic behaviour as a function of u^p. Then, the following is obtained:

The Long-term Pure (Deterministic) Planning Problem: *To determine the* ***optimal alternative*** $\mathbf{u}^{p*}$*, i.e. the alternative that solves the following problem*

$$J\big(\mathbf{u}^{p*}\big) = \min_{\mathbf{u}^p} i\big(\mathbf{u}^p, \mathbf{w}_0^{T-1}\big) \tag{8.15a}$$

subject to

$$\bar{\mathbf{x}}_1 = f_0\big(\bar{\mathbf{x}}_0, \mathbf{u}^p, \mathbf{w}_0\big), \ \dots, \ \bar{\mathbf{x}}_0 = f_{T-1}\big(\bar{\mathbf{x}}_{T-1}, \mathbf{u}^p, \mathbf{w}_{T-1}\big) \tag{8.15b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{8.15c}$$

$$\mathbf{w}_0^{T-1} \textit{ given scenario} \tag{8.15d}$$

$$\textit{any other constraints} \tag{8.15e}$$

Note that it is assumed that the value of the indicator i does not depend on the initial state, as is the case when the system is cycloergodic. As we will show in the next section, this new formulation is sometimes computationally useful, but it should be avoided by assuming that the scenario $\mathbf{w}_0^{T-1}$ is the trajectory of the random disturbances in the 'average year' or in the 'worst year', in order not to fall in the conceptual error that we pointed out at the end of the preceding section.

At the beginning of the chapter we observed that when the set $\mathcal{U}^p$ of alternatives to examine is finite and the number of its elements is small, the procedure for solving problem (8.14) is simple: the objective is evaluated for each alternative and the one that provides the best value for the objective is chosen. When the horizon is finite the meaning of the sentence is evident in practical terms: for every $\mathbf{u}^p \in \mathcal{U}^p$ the system is simulated, which means that problem (8.14b) is solved recursively, from 0 to h, to obtain the state trajectory $\mathbf{x}_0^h$. With this trajectory and with the triple $(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1})$ the value of the objective is then computed. Finally, the values obtained in this way are compared to find out the optimal alternative $\mathbf{u}^{p*}$. When the horizon is infinite, the same procedure would seem to be operationally impractical, since the simulation (and the evaluation of the objective) would require an infinite amount of time. We observe, however, that equation (8.11) discounts the costs increasingly, the further away they are in time, so that, if the computation of the summation that appears in it is done in parallel to the simulation, the variations of the value of the objective, sampled every T steps, become progressively smaller. The same thing happens with equation (8.12), if the system reaches a cyclostationary condition in the long-term, i.e. if it is cycloergodic. From an operational viewpoint it is now possible to stop the simulation when the difference between the value of the objective computed for $t = (k+1)T$ and that computed for $t = kT$ becomes less than a predefined threshold α. In this way, a *termination condition* is established for the simulation, which makes the procedure finite, even with the infinite horizon. Naturally, the bigger the value of α the shorter the simulation, but the more approximate the evaluation of the objective will be, and so also the evaluation of the optimal alternative.

The solution to problem (8.15), still in the case where the number of feasible decisions is finite, does not present these difficulties, but another, no less relevant, difficulty, which in the final analysis can be traced back to them: in correspondence with each value of $\mathbf{u}^p$ one must determine the cycle that the system reaches in the long-term by solving the algebraic system (8.15b). When $f(\cdot)$ is non-linear this task is not usually very easy and often the quickest way to carry it out is to simulate problem (8.14b) until the cycle has been determined (also in this case it is necessary to introduce a termination condition). In fact, using this method, the stability (more precisely the ergodicity) of the system is exploited, which naturally tends towards its equilibrium cycle. From this perspective problem (8.15) has no advantage over problem (8.14) with an AEV form objective.

When the number of feasible decisions is very high, or infinite, the preceding *exhaustive procedure* is no longer practicable, since it would require an infinite number of comparisons. So it is not possible, except in particular cases, to identify the optimal solution for problem (8.14) and we must be satisfied with a good suboptimal solution. In order to identify this solution, first of all note that problem (8.14) is a Mathematical Programming Problem,[20] since the variables for which we seek an optimal value are finite in number: they are in fact the components of the vector $\mathbf{u}^p$. The peculiarity of the Problem is that the evaluation of the objective requires a simulation and therefore it can be solved only with an algorithm from the class of *constrained evolutionary methods*. The basic idea of these methods is that the design objective $i(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1})$ is computed by simulating the system for a few[21] values of $\mathbf{u}^p$ in $\mathcal{U}^p$, after which the algorithm evolves in the set $\mathcal{U}^p$, until it finds an alternative that is sufficiently close to the sought optimal alternative. For instance, the methods that are based on the *steepest descent*, which can be adopted only when $\mathcal{U}^p$ is a continuous set, operate as follows: they determine the gradient of the objective, i.e. the direction in which it decreases most rapidly, for the current value of $\mathbf{u}^p$ (which is initially provided by the Analyst), and they look in that direction for a new solution, which would be better than the current one. This then becomes the current solution and the procedure is repeated. When it is no longer possible to proceed in this way, for example because the gradient is zero or because one would have to explore a direction that is orthogonal to a constraint, one has found an alternative which provides a minimum value of the objective. It is not possible, however, to know for certain if this alternative is an absolute minimum, and therefore it is the optimal alternative sought, or if it is a relative minimum. To ascertain this, it is advisable to repeat the procedure beginning from a different initial alternative, to see if a better alternative is identified. After a number of repetitions, one can consider that one has heuristically identified the optimal solution.

The computing time required by evolutionary methods to obtain a solution grows very rapidly with the number of elements in the vector $\mathbf{u}^p$, and if this number is high (in the order of hundred) the computing time becomes prohibitory. It is for this reason that problem (8.15) becomes interesting from a time-saving perspective. Equation (8.15b) is in fact equivalent to an evaluation over the infinite horizon, but it is constituted by only T relationships, each of which has the same form as one of the relationships that constitute equation (8.14b). When the objective and the constraints have a suitable form, it is possible to adopt very efficient Mathematical Programming algorithms, which exploit prior knowledge of those forms. For example, when the objective and the constraints are linear, problem (8.15) is a Linear Programming Problem, the set of the feasible decisions $\mathcal{U}^p$ is a convex polyhedron

[20] For the reader who has no notion of mathematical programming we have prepared Appendix A9.

[21] The number depends on the algorithm adopted.

with a finite number of vertices and the optimal solution can be proved to be one of them. The *Simplex Algorithm* (Dantzig, 1963) uses this last property and looks for the optimal alternative only in those vertices; it is so able to find it very quickly, even when the number of elements of the vector $\mathbf{u}^p$ is in the order of several thousands.

8.3 Example: the Sinai Plan[22]

In Chapters 1 and 3 we have already dealt with Egypt and the central role of water in its economy. Here we consider another example that has to do with this and dates back to twenty years before the Water Plan described in the chapters cited, but that nowadays, after the Water Plan, could be considered one of its detailed Plans: the one for the eastern Delta and the Sinai. The context in which the plan was conceived and the options for intervention that are proposed in it are described in the box on page 236.

8.3.1 Reconnaissance

After examining the specifics of the plan and interviewing people from the Ministry, one realizes that the Ministry poses the condition that the plan be elaborated in a non-participatory way, i.e. it requires that the Stakeholders are not to be involved in an active way, and the DM be the Ministry itself. The Ministry also decides that the evaluation method to be used must be the Cost Benefit Analysis (CBA) and expresses its interest in the long-term (i.e. steady-state) conditions. Furthermore, the initial examination of the system identifies farmers and other water users (mostly hydropower plants and industries) as the principal Stakeholders. The quality of the water is another attribute to be considered, since in some of the canals[23] the salinity Q_s [ppm] is such as to reduce the productivity of the crops. Thus, the possibility of mixing the water from the different canals to reduce the salinity has to be considered in the Plan.

8.3.2 Phase 1: Defining Actions

From the description in the box one deduces that the actions proposed by the Ministry regard the surface area to be reclaimed in each zone, the crop rotation and the irrigation techniques to use, the water volume to provide and its source. We can therefore quantify the possible actions with the following variables:

u^1_{ijh}: the area [feddan] to be dedicated to the rotation R_i in zone Z_j with the irrigation technique I_h;
u^2_{sj}: the annual water volume [m^3] to convey into the canal S_s in zone Z_j.

The indices can assume the values specified in Tables 8.1–8.4, but not all the combinations are possible: for example, flood irrigation ($h = 1$) is not allowed in sandy zones ($j = 1, 2, 4, 5, 6$), while drip irrigation ($h = 3$) is not at all practicable for rice crops ($i = 1$), which need flood irrigation. Each canal can serve only regions at a lower altitude, since

[22] Taken from Whittington and Guariso (1983); for didactic reasons, the features of the Plan have been simplified; however care has been taken to maintain and communicate the planning philosophy that was used.

[23] Particularly the canals Bahr Hadous and Bahr El Baqar (S_2 and S_3) which drain the wide irrigation areas that they cross.

Planning the Eastern Delta and the Sinai

The system Over the course of its history, Egypt has always been able to produce much more than was required by its population. This gave Egypt a central role in the economy of the African countries that look out over the Mediterranean and a higher standard of life than the others. Nevertheless, in the second half of the XX century, growing demographic pressure progressively reduced the water volumes devoted to irrigation and at the same time increased the demand for foodstuffs, so that in 1974, for the first time, Egypt had to import agricultural products. The desire to return to a condition of self-sufficiency pushed the Egyptian Government, in 1977, to elaborate a Reclamation Plan (Egypt's Ministry of Irrigation, 1977). For didactic purposes, we will consider only the part concerning the Eastern Delta and the Sinai peninsula.

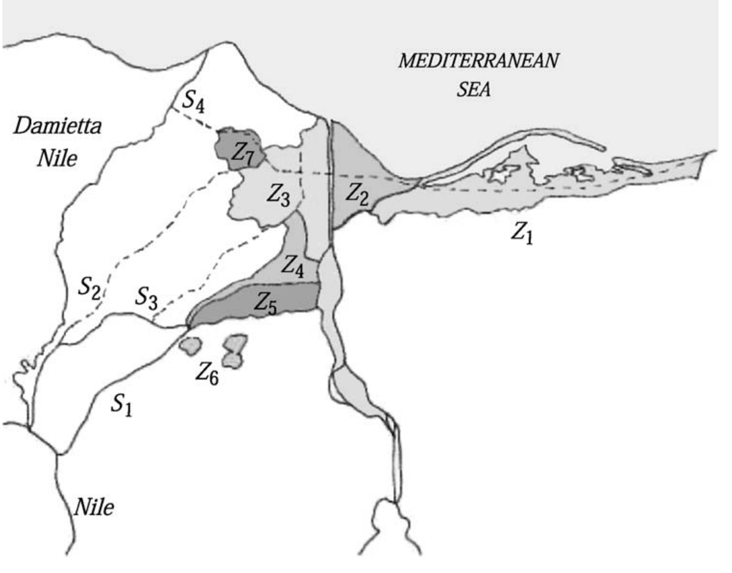

Figure 8.2: The regions of the Eastern Delta and the Sinai, with the reclaimable zones (Z_j) and the supply canals (S_s).

Project proposals The Reclamation Plan proposes to extend the cultivated surface by reclaiming areas of desert in the seven zones denoted with Z_j ($j = 1, \dots, 7$) in Figure 8.2; their names and the maximum reclaimable surface areas A_j are listed in Table 8.1. The Ministry of Irrigation estimated that they can be fed from the canals denoted with S_s ($s = 1, \dots, 4$) in Figure 8.2, the sth one of which can provide the average annual volume w_s (see Table 8.2).

Continued

The Ministry wants to evaluate a number of things: the area to reclaim in each region; the best canals to service these areas; the annual volumes to be delivered; the most efficient crop rotation[a] R_i among those listed in Table 8.3; and the irrigation technique I_h with which to irrigate them, from those in Table 8.4. Efficient crop rotation can permit a higher yield from the same water supply and a well-chosen irrigation technique can reduce water consumption while delivering the same amount to the plants.

Evaluation criteria The Ministry asks that the alternatives be evaluated on the basis of the net benefit that they can be expected to produce in the long-term.

[a] In the Egyptian climate crops grow quickly, so that it is possible to cultivate and harvest several crops in the same year from the same plot of land.

Table 8.1. The zones Z_j considered and the maximum reclaimable area A_j [feddan[24]] in each of them

j	Reclaimable zone Z_j	A_j
1	North-East coast of the Sinai	515 000
2	El Tina Plains	135 000
3	Territory West of the Suez Canal	235 000
4	Territory North of the Salhia Canal	102 000
5	Territory between the Ismailia and Salhia Canals	120 000
6	Territory South of the Ismailia Canal	45 000
7	Territory South of Lake Manzala	29 000

Table 8.2. The canals S_s that can serve the reclaimable areas and the corresponding average annual derivable volumes w_s [10^9 m^3]

s	Canal S_s	w_s
1	Ismailia	4.4
2	Bahr Hadous	2.4
3	Bahr El Baqar	1.4
4	Damietta–Salam	2.3

no water raising systems are considered. Therefore, we use F to indicate the set of all the feasible combinations of the indices i, s, j, h. To simplify the notation we will write $(i, j) \in F$ to denote the feasible pairs (i, j). We will use a similar notation for any other combination of indices. With this convention the vector $\mathbf{u}^p$ of the planning decisions is

[24] The feddan is the unit of measurement commonly adopted in Egypt for the area of agricultural land: a feddan is equal to 4212 m^2.

Table 8.3. The possible crop rotations R_i in the reclaimable areas and the corresponding unitary agricultural benefits b_i, expressed in Egyptian Pounds per feddan [EGP/feddan]

i	Crop rotation R_i	b_i
1	Clover–Rice	298
2	Clover–Cotton	374
3	Clover–Maize	152
4	Clover–Soya	158
5	Wheat–Maize	184

Table 8.4. The possible irrigation techniques I_h and the relative installation costs L_h [EGP/feddan]

h	Irrigation technique I_h	L_h
1	Flood	650
2	Sprinkler	1500
3	Drip	2500

expressed in the following way

$$\mathbf{u}^p = \left| u^1_{ijh} u^2_{sj} \right|_{(i,j,h)\in F;(s,j)\in F}$$

and has 55 components.

8.3.3 Phase 2: Defining Criteria and Indicators

The Project Goal, as requested by the Ministry, must be formulated as the CBA dictates: maximize the difference between the social benefits generated in the reclaimed areas and the social costs that the reclamation directly or indirectly imposes.

The social benefits B_{agr} are defined by the following expression

$$B_{\text{agr}} = \sum_{(i,j,h)\in F} b_i P_i(Q_j) u^1_{ijh} \tag{8.16}$$

where b_i represents the unitary social benefit [EGP/feddan], produced by the crop rotation R_i, whose values, estimated as the farmer's *willingness to pay*, are listed in Table 8.3. The benefit b_i is estimated in correspondence with a supply of fresh water that totally satisfies the water demand of the crops. However, productivity is reduced as the average salinity Q_j of the water supplied to the area A_j increases, as shown in Figure 8.3. The coefficient $P_i(Q_j)$ [%] that appears in equation (8.16) takes account of this effect.

The social cost is the sum of three addends:

- the installation cost C_{irr} for the irrigation systems

$$C_{\text{irr}} = \sum_{(i,j,h)\in F} L_h u^1_{ijh}$$

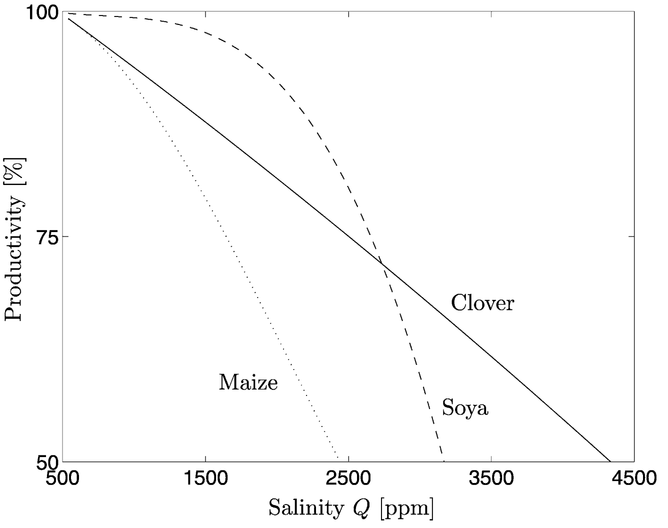

Figure 8.3: The productivity of three crops as a function of the salinity Q of the irrigation water.

where L_h is the installation cost [EGP/feddan] for the irrigation technique I_h (Table 8.4);

- the transport cost C_{tr} to bring the water from the canals to the reclaimed areas

$$C_{\text{tr}} = \sum_{(s,j)\in F} c_{\text{tr}} l_{sj} u_{sj}^2$$

where c_{tr} is the specific transport cost [EGP/m^3 km], which is assumed to be constant over the whole territory, and l_{sj} the distance [km] between the canal S_s and the zone Z_j;

- the opportunity cost C_{opp}

$$C_{\text{opp}} = \sum_{(s,j)\in F} o_s u_{sj}^2$$

where o_s represents the *opportunity cost*[25] [EGP/m^3] of the water from the canal S_s.

The design objective is thus defined as

$$i\left(\mathbf{u}^p, \mathbf{w}\right) = B_{\text{agr}} - C_{\text{irr}} - C_{\text{tr}} - C_{\text{opp}} \tag{8.17}$$

with $\mathbf{w}$ being the vector of the maximum water volumes w_s that can be withdrawn from the canal.

[25] The opportunity cost is the benefit that could be derived from the best alternative water use that is not agriculture. For example, the water taken from the Damietta–Salam canal (S_4) could be used to feed several hydropower plants and industries, but the water taken from the Bahr Hadous and Bahr El Baqar canals (S_2 and S_3), which is drainage water coming from irrigated lands upstream, has no alternative uses and so has a zero opportunity cost.

8.3.4 Phase 3: Identifying the Model

Provided that it is the long-term condition that must be evaluated, and that average annual withdrawable volumes w_s are given, the Design Problem is a Long-term Planning Problem and the time step adopted is one year. Clearly, with such a long time step, one obtains only a general evaluation of the benefits that the system might generate, and no indication about how it should be managed, but what is asked for is a Plan and not a Project. That said, it follows that the model has the form

$$\bar{\mathbf{x}} = f_0\big(\bar{\mathbf{x}}, \mathbf{u}^p, \mathbf{w}\big) \tag{8.18a}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{8.18b}$$

$$\mathbf{w} \text{ given scenario} \tag{8.18c}$$

$$\text{any other constraints} \tag{8.18d}$$

where equation (8.18a) defines the relationship between the value $\bar{\mathbf{x}}$ of state variables at the equilibrium and the pair $(\mathbf{u}^p, \mathbf{w})$. In our case the state is represented by the salinity of the water, and its value $\bar{Q}_j$ at the equilibrium is defined by the following expression

$$\bar{Q}_j = \frac{\sum_s [1 - \alpha l_{sj}] u^2_{sj} Q_s}{\sum_s [1 - \alpha l_{sj}] u^2_{sj}} \quad \forall Z_j \tag{8.19}$$

where α represents the unitary water loss[26] [km^{-1}] from the canals, which takes its own salt content with it.

Equation (8.18b) defines the feasible actions, through the set $\mathcal{U}^p$. In the case that we are considering here, as is almost always the case, that set is indirectly defined by a set of constraints that the components of the vector $\mathbf{u}^p$ must respect:

1. The overall volume derived from canal S_s cannot exceed the water availability w_s

$$\sum_{j \in \{j:\ \exists s:\ (s,j) \in F\}} u^2_{sj} \leqslant w_s \quad \forall s \tag{8.20a}$$

2. The overall reclaimed area in zone Z_j cannot exceed the maximum reclaimable area A_j

$$\sum_{(i,h) \in \{(i,h):\ \exists j:\ (i,j,h) \in F\}} u^1_{ijh} \leqslant A_j \quad \forall j \tag{8.20b}$$

3. The components of the vector $\mathbf{u}^p$ cannot assume negative values, since they would be meaningless

$$\big| u^1_{ijh} u^2_{sj} \big|_{(i,j,h) \in F; (s,j) \in F} \geqslant 0 \tag{8.20c}$$

The value zero, on the other hand, is allowed and the Alternative Zero corresponds to the identically zero vector.

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[26] Understood as loss from the bottom of the canal and not through evaporation from the water surface. The latter, despite the fact that we are in Egypt, constitutes a negligible portion compared to the first.

Equation (8.18c) establishes that the vector $\mathbf{w}$ is the vector of water volumes w_s that can be derived from the canals.

Equation (8.18d), finally, includes all the constraints that do not fall into the types that have already been considered. The reader will remember that the benefit b_i was estimated in correspondence with a water supply that totally satisfies the water demand of the crops. Therefore, in order that equation (8.16) provide a correct estimate, we must impose that the overall volume provided to a zone be greater than the demand of the crop rotations adopted in that zone

$$\sum_{s\in\{s:\ \exists j:\ (s,j)\in F\}} (1-\alpha l_{sj})u^2_{sj} - \sum_{(i,h)\in\{(i,h):\ \exists j:\ (i,j,h)\in F\}} W_{ih}u^1_{ijh} \geqslant 0 \quad \forall j \tag{8.21}$$

where W_{ih} is the specific demand[27] [m^3/feddan] of the rotation R_i, irrigated with the technique I_h.

8.3.5 Phase 4: Designing Alternatives

The Design Problem is therefore formulated with the form (8.15), and it is thus the following

$$J\left(\mathbf{u}^{p*}\right) = \max_{\mathbf{u}^p,\bar{Q}_1,\ldots,\bar{Q}_7} i\left(\mathbf{u}^p,\mathbf{w}\right) \tag{8.22}$$

where $i(\mathbf{u}^p,\mathbf{w})$ is defined by equation (8.17), subject to the constraints (8.19), (8.20) and (8.21).

Problem (8.22) defines a Non-linear Mathematical Programming Problem with 55 decision variables and 80 constraints. The non-linearity is found in function $P_i(\bar{Q}_j)$ (Figure 8.3) and in constraint (8.19); but if the values of $\bar{Q}_j$ $(j = 1,\ldots,7)$ were fixed it would be a Linear Programming Problem. This observation provides the key to its solution: adopt an evolutionary algorithm in the space $(\bar{Q}_1,\ldots,\bar{Q}_7)$ to maximize the following function

$$J(\bar{Q}_1,\ldots,\bar{Q}_7) = \max_{\mathbf{u}^p} i\left(\mathbf{u}^p,\mathbf{w}\right)$$
$$\text{subject to constraints (8.19), (8.20) and (8.21)} \tag{8.23}$$

whose value can be determined at each step of the evolutionary algorithm by solving problem (8.23) with the Simplex Algorithm.

The solution of the Problem provides the optimal alternative described in Table 8.5.

The solution was obtained by assuming an opportunity cost for the water of the canals equal to 10 EGP/1000 m^3 and establishes an overall reclamation of 444 000 feddan. The net annual benefit produced in the long-term is estimated at 80 million EGP. Note that, due to the transport cost and the opportunity cost, the solution does not foresee the reclamation of several zones, among which the North-East coast of the Sinai $(j = 1)$. The canals that are used the most are the drainage canals $(s = 2$ and $s = 3)$.

Once the optimal alternative has been determined, it is advisable to check its robustness with a sensitivity analysis (see Section A3.5 in Appendix A3). One can, for example, verify whether it is excessively sensitive to the estimates of the parameters in the model, such as the opportunity cost o_s of the water or the installation cost L_h of the irrigation systems. The analysis is carried out by simply checking the solution for different values of the parameters.

[27] Assumed to be identical in each zone, independently of the type of soil.

Table 8.5. The optimal alternative for the Sinai Plan

Zone	Reclaimed area [feddan]	Canals used (s)	Rotation (i)	Irrigation technique (h)
1	0	$*$	$*$	$*$
2	135 000	1, 3	5	2
3	235 000	2, 3	2	1
4	0	$*$	$*$	$*$
5	0	$*$	$*$	$*$
6	45 000	3	5	2
7	29 000	2, 4	2	1

Note lastly that the Plan is completely defined at the end of the fourth Phase of the PIP procedure, since we have only one DM who has only one objective.

Chapter 9

Dealing with risk and uncertainty

AC and RSS

In Chapter 8 we saw how a Pure Planning Problem can be formulated and solved when there are only deterministic disturbances that act upon the system. In this chapter we will remove that limitation, by considering the case in which there are also random disturbances.

9.1 Risk and uncertainty

Whenever the system is affected by a random disturbance, the value of the design indicator i is also random (see Section 4.10.1): it is a stochastic or uncertain variable according to whether the disturbance is of one type or the other. A ranking, however, can only be established among deterministic variables; therefore the indicator has to be necessarily converted to a deterministic quantity to allow the comparison among the alternatives. As we suggested in Section 4.10.1 this transformation can be done by filtering the randomness produced by the disturbance by means of a statistic applied to the random indicator i. This statistic is called a (*filtering*) *criterion* and the resulting deterministic indicator i is called *evaluation indicator* or *objective*. The most common criteria[1] are the *Laplace criterion*, which consists in applying the expected value operator ($E_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}[\cdot]$), and the *Wald criterion* (Wald, 1945, 1950), which considers the worst case and which is therefore expressed by the operator $\max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}[\cdot]$, when the indicator represents a 'cost', and by the operator $\min_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}[\cdot]$ in the opposite case. The application of a criterion, however, does not solve all of the problems and, sometimes, it creates new ones. The subject of this section is to explain and justify this last statement.

9.1.1 The Laplace criterion

The Laplace criterion is certainly the most known criterion and maybe the most intuitive: if the value of the indicator is stochastic, we consider its expected value. In many problems this criterion can effectively be a good choice, but are we sure that it always expresses the

[1] There are also other criteria that can be used to filter the disturbance, for example the so-called *min–min* (or *max–max*) criterion, which considers the best case and thus reflects the total optimism of the Decision Maker (DM); the Hurwicz criterion (Hurwicz, 1951), which considers a weighted combination (through a parameter that reflects the DM's risk aversion) of the worst case and the best case; the Savage criterion (Savage, 1951) which aims at minimizing the *regret* that the DM feels following a wrong decision. See French (1988).

attitude of the Decision Maker (DM) in a risky situation? To reply to this question, let us consider the following example, proposed in Keeney and Raiffa (1976): one must choose between four alternatives $A1$, $A2$, $A3$, and $A4$ which, given the same investment, produce the following results:

$A1$: €100 000 gain with absolute certainty;

$A2$: €200 000 or €0 gain, each with a probability of 0.5;

$A3$: €1 000 000 gain with a probability of 0.1 or €0 gain with a probability of 0.9;

$A4$: €200 000 gain with a probability of 0.9 or €800 000 loss with a probability of 0.1.

Note that the expected return of each of the four alternatives is €100 000: thus, they are equivalent for a DM who adopts the Laplace criterion. However, it is probable that the reader does not consider them as equivalent, and that alternative $A1$ appears to be the best choice, given that there is no risk in the outcome. If we were not using only the Laplace criterion as the basis for the decision, and from the alternatives that it suggests we were to choose the one for which the indicator has the minimum variance, $A1$ would emerge as the best alternative. Nevertheless, the choice of these cascade criteria is very arbitrary, and it does not completely solve the problem: $A3$ and $A4$ have the same variance, but presumably the reader would prefer $A3$ to $A4$, given that with the first (s)he would risk a much lower loss (i.e. only the investment).

The example clearly shows that the preference system that a DM adopts when there is risk involved cannot be described with statistics. Before we look at the means that were devised to overcome this difficulty, let us see if the Wald criterion is more robust than Laplace's.

9.1.2 The Wald criterion

In order to fully understand the way the Wald criterion operates, consider the following table (French, 1988), which shows the gain i produced by four different investment alternatives ($A1, \ldots, A4$), in correspondence with three different realizations of the disturbance (ε_1, ε_2 and ε_3).

	ε_1	ε_2	ε_3	min$[i]$	$E[i]$
$A1$	50	40	10	10	33.3
$A2$	35	45	15	15	31.7
$A3$	30	25	20	20	25.0
$A4$	40	35	5	5	26.7
$\phi(\varepsilon)$	1/3	1/3	1/3		

The Wald criterion associates the performance corresponding to the worst realization of the disturbance to each alternative and thus the values that it produces are those shown in the fourth column. With this criterion the best alternative is $A3$, which provides the highest gain in the worst case. This explains the name *min–max criterion* (or *max–min* in this case), often used to refer the Wald criterion. Note further that, once the alternative $A3$ is chosen, if the realization of the disturbance were not ε_3, but ε_1 or ε_2, the gain that would be obtained would be more than 20. Therefore 20 is the minimum performance that one can obtain by adopting the Wald criterion, and for this reason it is also called the *certain performance criterion*.

To better understand the properties of this criterion, let us examine the table more closely. If we knew that the disturbance would assume the value ε_1, the optimal alternative would be $A1$, because it is the one that, in such a case, would provide the highest gain. Similarly, $A2$ and $A3$ would be the optimal decisions if we knew that the disturbance would assume the value ε_2 or ε_3, respectively. If we do not know the value of the future disturbance, but only its probability distribution (for example if we knew that the three values were equally probable as indicated in the last line), the Laplace criterion suggests alternative $A1$, since its expected gain is 33.3, the highest among all the alternatives (see the last column of the table). Therefore, if we were sure that the values of disturbance were equally probable and we could repeat the investment many times, the alternative suggested by the Laplace criterion would surely be the best: in fact, statistics (the Law of Large Numbers) teaches us that as the number of repetitions increases, the relative frequency tends to the probability and thus in such a case the average gain from the investments would tend towards 33.3. Choosing, instead, the best alternative with the Wald criterion would bring about a reduction in the expected gain, from 33.3 to 25. However, if we were not certain of the disturbance probability distribution and/or we could make the investment only once, the Wald criterion would become interesting. It would, in fact, guarantee that in no case would we have a gain less than 20, while by adopting the Laplace criterion, and thus choosing $A1$, we would obtain only 10 if ε_3 actually occurred. The choice of certainty over maximization of the expected gain has a cost however: a lost gain of 20 $(50 - 30)$ if the disturbance ε_1 actually occurred and of 15 $(40 - 25)$ if ε_2 occurred. This should not be surprising: all of us pay our car insurance each year to protect ourselves from a big cash outlay in the regrettable case of an accident, and most of us have never had one.

9.1.3 Risk aversion and utility functions

The majority of individuals have a *risk aversion*, which is more or less concealed, and which makes them prefer certain returns to probable returns, even if the second are a little greater than the first; the value of the maximum difference by which one prefers the first over the second varies from individual to individual. Nevertheless, there are also individuals who, instead of being averse to risk, are *risk seeking*, in the sense that they obtain pleasure from choosing alternatives that offer the possibility of high gain, even if the probability is very small, and even if the expected value of the gain is negative. Those who love the thrill of roulette, for example, are risk seeking.

To remove any doubt about which criterion to adopt in the case of stochastic disturbances, Von Neumann and Morgenstern (1947), in their seminal work on Game Theory, were the first to propose a formalization of the DM's risk aversion. The approach is based on the identification (Keeney and Raiffa, 1976) of a function, called a *utility function*, which should be characteristic of each individual's preferences and completely define his/her risk aversion. We will discuss this assertion further on; for now we will concentrate on the definition of the function.

To introduce this function, let us suppose, as in the preceding examples, that the indicator i is a measure of the gain. Assume that i^0 is the minimum gain and i^* is the maximum gain that can be obtained with the set of alternatives available. Since the utility U is a relative measure and not an absolute one, we can arbitrarily set it at zero for i^0 and make it

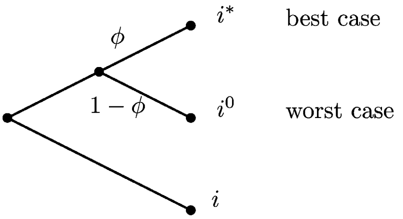


Figure 9.1: The decision tree shown to the DM to estimate her utility function. The value of ϕ is the subject of the question.

equal to 1 for i^*; i.e. we can assume

$$\mathrm{U}\big(i^0\big) = 0 \quad \text{and} \quad \mathrm{U}\big(i^*\big) = 1 \tag{9.1}$$

Then, for every other value of i between i^0 and i^*, we ask the DM to quantify the probability ϕ for which she feels indifferent between the following two alternatives: (1) participating in a lottery, whose result is i^* with a probability of ϕ and i^0 with a probability of $(1-\phi)$, and (2) gaining i with certainty (i.e. with zero risk). The decision tree that describes this question is shown in Figure 9.1. Because of the way that the question is posed, the utility of i is equal to the expected value of the lottery, i.e.

$$\mathrm{U}(i) = \phi \mathrm{U}\big(i^*\big) + (1-\phi)\mathrm{U}\big(i^0\big) = \phi$$

Therefore the utility of i is ϕ itself. Note that the definition is consistent with equation (9.1), since it implies $\mathrm{U}(i^0) = 0$ and $\mathrm{U}(i^*) = 1$.

If we define a DM who adopts the Laplace criterion as *risk neutral*, it is easy to show that her utility function is linear (Figure 9.2). In fact, such a DM is indifferent between the two alternatives when and only when the expected value of the lottery is equal to i; it follows that her utility function $\mathrm{U}^N(\cdot)$ is given for each i as

$$\mathrm{U}^N(i) = \phi^N(i) = \frac{i - i^0}{i^* - i^0}$$

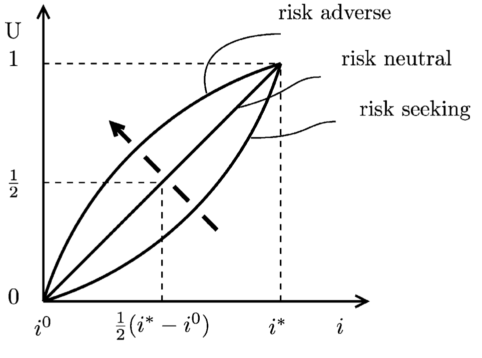


Figure 9.2: The utility function for different types of DMs. Risk aversion grows in the direction of the arrow.

For each value of i, the more risk averse the DM is, the higher the probability $\phi(i)$ must be with respect to $\phi^N(i)$, in order that she perceive the certain return and the lottery as equivalent. Therefore, her utility function is convex: the more averse she is to risk, the less linear it will be (Figure 9.2). When, instead, the DM is risk seeking her utility function $\mathrm{U}(\cdot)$ is concave.

Once the utility function has been correctly estimated, the best alternative (A^*), from the DM's perspective, is clearly the one corresponding to the maximum utility. More precisely: if the disturbance that influences the system is stochastic, it is the alternative corresponding to the maximum expected utility (Raiffa, 1968)

$$A^* = \arg\max_A \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} \left[\mathrm{U}(i)\right]$$

while if the disturbance is uncertain, it is the one corresponding to the maximum minimum utility

$$A^* = \arg\max_A \min_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\mathrm{U}(i)\right]$$

Therefore, the introduction of the utility function removes any doubt about which criterion to adopt: it is univocally defined by the type of disturbance. Consequently the objective assumes the following form

$$J\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}\right) = \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} \left[\mathrm{U}\left(i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right)\right]$$

when the disturbance is stochastic; and the form

$$J\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}\right) = \min_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\mathrm{U}\left(i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right)\right]$$

when it is uncertain.

Utility is thus an elegant and powerful mathematical tool. To apply it, it is necessary that:

(a) for a given individual there exists a utility function and that it is time-invariant;

(b) there exists at least one procedure to identify it.

Von Neumann and Morgenstern (1947) were the first to look for the behavioural axioms that imply the existence of a utility function, and their pioneer work was taken up and extended by many researchers; a good brief account of these developments can be found in French (1988). As for the identification procedures, they do exist, and they are based on the alternative posed to the DM in the utility definition; for their description see Farquhar (1984); in their application one must pay attention to avoiding 'behavioural bias' (Berkeley and Humphreys, 1982; Hershey et al., 1982).

Thus, we may conclude that an approach based on utility, even though it is fairly weak from the perspective of practical application (MacCrimmon and Larsson, 1979; Kahneman and Tversky, 1979; Scholz, 1983; Wright, 1984), is conceptually useful for expressing the DM's risk aversion. This approach suggests that we adopt the expected utility of the random indicator as indicator when the disturbances are stochastic. If the DM is risk neutral the approach coincides with the adoption of the Laplace criterion, because in that case the utility function is linear.

9.2 Chance constraints

In order to complete the description of the Design Problem we must define, if necessary, the 'any other constraints' which affect the variables in play. Clearly, they can have a great variety of forms and an exhaustive description is not possible; nevertheless, we would like to point out a very common error.

Except in the case where only deterministic disturbances act on the system, many, if not all, the system variables have a random nature: stochastic or uncertain. Therefore, it is necessary to avoid carefully the formulation of improper constraints that would make the Problem unsolvable.

Consider, for example, the Problem of designing a reservoir in which the inflow is treated as a disturbance and in which one wants to impose the constraint

$$s_t < \bar{s} \quad t = 0, \ldots, h-1 \tag{9.2}$$

in order to express the request to avoid exceeding the storage $\bar{s}$, above which undesirable effects are produced (e.g. flooding is caused). If the probability distribution of the inflow is Gaussian, constraint (9.2) cannot ever be satisfied: in fact, with a Gaussian distribution, the inflow has no upper bound, while the release r_{t+1} increases only if s_t increases (see equation (5.10)); therefore, there is a non-zero probability, even if it is small, that the storage of the reservoir could exceed $\bar{s}$ no matter how large that value might be. Imposing the constraint (9.2) thus renders the Problem unsolvable. The desire to avoid exceeding the storage $\bar{s}$ has to be expressed through a *chance constraint*, i.e. by imposing that constraint (9.2) be satisfied with a probability greater than a preassigned value α

$$\Pr(s_t < \bar{s}) > \alpha$$

Alternatively, one can transform the constraint (9.2) into an objective, by defining a suitable step indicator g_t and an associated functional J (see also Section 8.1.2). For example, one can set

$$g_t(s_t) = \begin{cases} 0 & \text{if } s_t < \bar{s} \\ 1 & \text{otherwise} \end{cases} \tag{9.3a}$$

and[2]

$$J = \mathop{E}_{\{\varepsilon_t\}_{t=1,\ldots,h}} \left[\sum_{t=0}^{h-1} g_t(s_t) \right] \tag{9.3b}$$

Notice that g_t is a Boolean variable, whose value denotes whether at time t the storage $\bar{s}$ has been exceeded or not; as a consequence, the value of J is the expected number of days of flooding over the horizon $[0, h-1]$. The minimization of J thus expresses just what we unconsciously wanted to ask through constraint (9.2), but in a way that was formally incorrect.

However, by adopting formulation (9.3) we leave full rationality behind: to solve the Problem, the DM is forced to express her opinion about the relative importance that she would assign to the new objective with respect to the original one. Thus, the solution to the problem depends on a subjective point of view, as inevitably happens whenever there is

[2] Clarifications about the ranges of variation of the index t in the following equation are contained in footnotes 4 on page 249 and 5 on page 251. The average value with respect to the sequence of disturbances is required because the storage s_t at time t is influenced by the sequence of disturbances $\{\varepsilon_\tau\}_{\tau=1,\ldots,t}$ which has occurred until that time.

more than one objective. To solve Problems in which the constraints are formulated through objectives, we will have to use the multi-objective methods that are presented in Chapter 18. In particular, it is interesting to note that one of these methods (the one presented in Section 18.3.4) requires a procedure that is just the reverse of the one described above: the transformation of some of the objectives into constraints. This is possible because constraints and objectives can always be freely exchanged.

When the disturbance is uncertain and the set Ξ_t has no upper bounds, the problem is similar to the preceding one and can be solved in a similar way. When, instead, the set Ξ_t has an upper bound it is not known a priori that the constraint (9.2) would result in the complete lack of solutions for the Design Problem and so the form (9.2) is correct.

One must not incorrectly conclude from the above that deterministic constraints imposed on stochastic variables always leave the Problem without solutions. Sometimes this does not occur, such as when the same random variables appear on both sides of the constraining inequality; the condition (5.6) is an example of this.

9.3 The Pure Planning Problem under risk or uncertainty

We now have all the tools we need to formulate the Planning Problem with random disturbances, or, more precisely, to formulate the Planning Problem *under risk* (when the disturbances are stochastic) or *under uncertainty* (when the disturbances are uncertain). Clearly, the random disturbances that we are talking about are 'per-step', because, as we proved in Section 6.4.3, the disturbances 'per-period' can be traced back to disturbances 'per-step'.

There are two possible formulations for the Planning Problem. The first uses a criterion to filter the randomness, as suggested in Section 9.1. When the Laplace criterion is adopted the Design Problem assumes the following form

The Pure Planning Problem with the Laplace criterion: *To determine the* ***optimal alternative***[3] $\mathbf{u}^{p*}$, *i.e. the alternative that solves the following problem*[4]

$$J^* = \min_{\mathbf{u}^p} \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E} \left[i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{9.4a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, \ldots, h-1 \tag{9.4b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{9.4c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{u}^p\right) \quad t = 0, \ldots, h-1, \tag{9.4d}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{9.4e}$$

$$\mathbf{x}_0 \textit{ given} \tag{9.4f}$$

$$\textit{any other constraints} \quad t = 0, \ldots, h-1 \tag{9.4g}$$

[3]If there is more than one optimal alternative, it might be interesting for the DM to determine all of them in order to choose the one that best satisfies any possible criteria that were not expressed in the formulation of the Problem.

[4]The notation $\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}$ denotes the sequence of disturbances $\{\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_h\}$, whose joint probability distribution is used to compute the expected value.

The length h of the horizon can be either finite or infinite; in the second case the definition of the indicator contains a limit, as in equations (8.11) and (8.12).

When the Wald criterion is adopted, the first formulation of the Planning Problem is, instead, the following:

The Pure Planning Problem with the Wald criterion: *To determine the* ***optimal alternative*** $\mathbf{u}^{p*}$*, i.e. the alternative that solves the following problem*

$$J^* = \min_{\mathbf{u}^p} \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{9.5a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, \dots, h-1 \tag{9.5b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{9.5c}$$

$$\boldsymbol{\varepsilon}_{t+1} \in \Xi_t\left(\mathbf{u}^p\right) \quad t = 0, \dots, h-1 \tag{9.5d}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{9.5e}$$

$$\mathbf{x}_0 \textit{ given} \tag{9.5f}$$

$$\textit{any other constraints} \quad t = 0, \dots, h-1 \tag{9.5g}$$

Note that Wald's form can be chosen even when the disturbance is stochastic, as long as the probability distribution $\phi_t(\cdot)$ is bounded at every instant. In such a case, in fact, the set Ξ_t of disturbances that have a non-zero probability of occurrence is bounded.

The second formulation of the Planning Problem uses the utility function, instead:

The Pure Planning Problem with utility functions: *To determine the* ***optimal alternative*** $\mathbf{u}^{p*}$*, i.e. the alternative that solves the following problem*

$$J^* = \max_{\mathbf{u}^p} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \mathrm{U}\left(i\left(\mathbf{x}_0, \mathbf{u}^p, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right) \tag{9.6a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, \dots, h-1 \tag{9.6b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{9.6c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{u}^p\right) \quad t = 0, \dots, h-1 \tag{9.6d}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{9.6e}$$

$$\mathbf{x}_0 \textit{ given} \tag{9.6f}$$

$$\textit{any other constraints} \quad t = 0, \dots, h-1 \tag{9.6g}$$

As in the deterministic case, also in presence of risk or uncertainty, one can formulate the Planning Problem in the long-term (steady-state conditions), but it is less interesting in this case, because it does not offer any computational advantage with respect to the formulations that we have just presented, except for one particular case: that in which the criterion is Laplace's, the indicator is separable and the step costs and the constraints are linear. In such conditions the following identities apply[5]

$$\underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E}\left[\sum_{t=0}^{h-1} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\right] = \sum_{t=0}^{h-1} g_t\Big(\underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t}}{E}[\mathbf{x}_t], \mathbf{u}^p, \mathbf{w}_t, E[\boldsymbol{\varepsilon}_{t+1}]\Big)$$

$$\underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E}[\mathbf{x}_{t+1}] = f_t\Big(\underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t}}{E}[\mathbf{x}_t], \mathbf{u}^p, \mathbf{w}_t, E[\boldsymbol{\varepsilon}_{t+1}]\Big)$$

and so the Problem can be formulated by substituting the variables $\mathbf{x}_t$ and $\boldsymbol{\varepsilon}_{t+1}$ with their expected values

$$\hat{\mathbf{x}}_t = \underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t}}{E}[\mathbf{x}_t] \quad \text{and} \quad \hat{\boldsymbol{\varepsilon}}_{t+1} = E[\boldsymbol{\varepsilon}_{t+1}]$$

The Planning Problem is thus led back from a stochastic environment to a deterministic one, which is much simpler to deal with. In real Problems, however, the indicator is very rarely linear and therefore in practice the long-term Planning Problem is useless. For this reason we will not consider it, nor will we consider its counterpart in the next chapter when we deal with the policy design. In so doing, we will not lose any generality, because the optimal alternative obtained with the long-term formulation is the same as the one we get from the above Problems when the indicator has the AEV form.

9.4 Solution

Since we have discarded the long-term formulation of the Problem, only *constrained evolutionary methods* can be adopted. Unlike what we saw in the preceding chapter, it is no longer possible to simulate the dynamics of the system for the alternative being evaluated in a deterministic way, for the simple reason that the trajectory that the disturbance will follow is no longer known. On the other hand, it would not be possible to consider a single trajectory either, since in order to evaluate the objective it is necessary to compute a statistic (expected or maximum value) of the indicator with respect to the trajectory of the disturbance and for this we must be able to determine the probability distribution or the set of values that the indicator can assume as the trajectory varies. Now we do not know what to do. Further on, in Sections 19.1 and 19.2, we will show how this difficulty can be overcome with two types of simulation, Markovian and Monte Carlo, which will be introduced in those sections. We will also show how to obtain the required statistic. Once we are able to compute it in correspondence with any value $\mathbf{u}^p$, the solution of the Planning Problem will be obtained by applying an evolutionary method, just as in the deterministic case (see Section 8.2).

[5] The reader should not be confused by the fact that in the following expressions the time intervals considered in the expected value and in the summation are not in phase with each other. This is a consequence of the fact that the disturbance $\boldsymbol{\varepsilon}$ has subscript $t+1$, but to compact the notation we preferred here and in the following to write $\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}$, instead of $\{\boldsymbol{\varepsilon}_{t+1}\}_{t=0,\dots,h-1}$.

Chapter 10

Planning the management

AC and RSS

In the two previous chapters we considered the Design Problem when the actions being considered are only planning actions, which means they are not recursive. We also settled on calling such a Problem a *Pure Planning Problem*. The dual case is when only management actions are considered, particularly regulation actions, so that the decisions to be made are exclusively recursive. This particular case of Design Problem will be termed *Pure Management Problem* or, more often, *Optimal Control Problem*, which we will shorten to *Control Problem*. In this chapter it would be logical to focus on this Problem, but since it is not difficult directly to introduce the Design Problem in its general form, i.e. the one in which both types of actions are considered, the Design Problem in its general form will be the object of our attention.

In Section 2.1 we saw what the essential difference is between planning decisions and management decisions and how the second can be traced back to the first by designing a *regulation policy*. We gave only an intuitive idea of the policy and therefore we need firstly to provide a more formal definition of this concept.

10.1 The policy

The most spontaneous way to choose the control $\mathbf{u}_t$ at every time t ($t > 0$) is to use all the information $\Im_t$ that is available at that time

$$\Im_t = [\mathbf{x}_0, \dots, \mathbf{x}_t, \mathbf{w}_0, \dots, \mathbf{w}_t, \mathbf{u}_0, \dots, \mathbf{u}_{t-1}]$$

i.e. to use the sequences of the states $(\mathbf{x}_0, \dots, \mathbf{x}_t)$ and the deterministic inputs $(\mathbf{w}_0, \dots, \mathbf{w}_t)$ measured from the initial time 0 to time t, and the sequence of controls $(\mathbf{u}_0, \dots, \mathbf{u}_{t-1})$ applied until that time. Clearly, at time 0 the available information is just the pair (x_0, w_0).

It follows that $\Im_t$ should be considered to be the argument of the control law a priori; however, it might be possible to obtain the same performance by considering less cumbersome information. This means that, when it is indicated, it is certainly advantageous to use it, since dealing with a smaller quantity of information means both saving computing time in the policy design and making it easier to use. For this reason, it is interesting to determine the minimum information required to produce the best possible performance: we will call that information the *sufficient statistics*. By definition, each piece of information that is not included in the sufficient statistics increases the costs of the design and use of the policy, without producing any benefit.

When the state of the system is measurable and no deterministic disturbances act upon it, the sufficient statistic is state $\mathbf{x}_t$, or it can be traced back to the state by redefining it, if necessary, in an appropriate way. When, on the other hand, the system is affected by deterministic disturbances, the sufficient statistic is (or can be traced back to) the pair $(\mathbf{x}_t, \mathbf{w}_t)$. Finally, when the state is not measurable, it is possible to obtain a system with a measurable state, through a suitable redefinition of the system. In this text, however, we will not show how this can be done: such an explanation would require advanced notions that would be impossible to provide within the limits of this book. The interested reader can find useful indications in Bertsekas (1976). We will simply note that the redefinition inevitably requires an enlargement of the state of the original system and therefore brings about, as we will see, an increase in the computing time required to solve the Design Problem. As a consequence, whenever possible, it is opportune to define the state of the system so that it is measurable. Therefore, in what follows we will assume that $\Im_t$ can be only the state $\mathbf{x}_t$, or the pair $(\mathbf{x}_t, \mathbf{w}_t)$.

Finally, we recall the terminology that we introduced in Chapter 2: we use the term *regulation policy* to indicate a policy for which we do not want to specify whether it is applied to a reservoir or a diversion while, more specifically, we use the term *release policy* in relation to reservoirs and the term *distribution policy* with reference to diversions.

10.1.1 Point-valued (PV) and set-valued (SV) policies

In Section 2.1.1.2 we saw that the most commonly used form for the *control law* is the *point-valued* (PV) form which, given the available information $\Im_t$ at time t, provides the release decision $\mathbf{u}_t$

$$\mathbf{u}_t = m_t(\Im_t) \tag{10.1}$$

where $m_t(\cdot) : \mathcal{S}_{\mathbf{x}_t} \to \mathcal{S}_{\mathbf{u}_t}$, $\mathcal{S}_{\mathbf{x}_t}$ and $\mathcal{S}_{\mathbf{u}_t}$ being the sets in which $\mathbf{x}_t$ and $\mathbf{u}_t$ assume their values (see Section 10.5 for their operational definition). Nevertheless, in this form, as experience shows (Section 2.1.1), the control law can be perceived as limiting by the Regulator, who would prefer a control law that provides, for every time t, a set $M_t(\Im_t)$ of controls, from which he can freely choose the control that he thinks is best at that moment. In other words, it is preferable to consider a *set-valued control law* (SV)

$$\mathbf{u}_t \in M_t(\Im_t) \tag{10.2}$$

where $M_t(\cdot) : \mathcal{S}_{\mathbf{x}_t} \to \{\text{subsets of } \mathcal{S}_{\mathbf{u}_t}\}$. The set $M_t(\Im_t)$ must be defined in such a way that all the controls $\mathbf{u}_t \in M_t(\Im_t)$ provide the same performances in the long-term. The definition of the SV control law is not classical and up to now it has been used very little in literature (the first known proposal is in Orlovski et al., 1983; see also Orlovski et al., 1984). Nonetheless, it is very interesting because it offers the Regulator all the freedom of choice permitted by the objective that he is pursuing.

Alternatively, a SV control law can be defined as a set of PV control laws

$$M_t(\cdot) \triangleq \{m_t(\cdot)\} \tag{10.3}$$

Definitions (10.2) and (10.3) are completely equivalent and interchangeable. In what follows, case by case, we will use the one that seems most appropriate in the particular context.

By doing so, there is no risk of ambiguity, because the relation $\mathbf{u}_t \in M_t(\Im_t)$ clearly defers to the SV control law, conceived as a set of controls, while the relation $m_t(\cdot) \in M_t(\cdot)$ reveals that it is conceived as a set of PV control laws. Ambiguity arises only in sentences such as "given a control law $M_t(\cdot)$", in which it is not clear if one is thinking of a set of controls or a set of PV control laws. Nevertheless, this should not create any difficulties, since the two meanings are completely equivalent.

A *point-valued policy* (PV)

$$p \triangleq \{m_t(\cdot);\ t = 0, 1, \ldots\} \tag{10.4}$$

is a finite or infinite sequence of PV control laws. The policy is said to be *periodic of period* T if $m_t(\cdot) = m_{t+kT}(\cdot)$, $t = 0, 1, \ldots, k = 1, 2, \ldots$. A PV policy is termed *feasible* if it satisfies the condition $m_t(\Im_t) \in \mathcal{U}_t(\mathbf{x}_t)\ \forall \Im_t$, which guarantees that the proposed control can be implemented physically at each time instant.

A *set-valued policy* (SV)

$$P \triangleq \{M_t(\cdot);\ t = 0, 1, \ldots\} \tag{10.5}$$

is a finite or infinite sequence of SV control laws. It is *periodic of period* T if $M_t(\cdot) = M_{t+kT}(\cdot)$, $t = 0, 1, \ldots$, and $k = 1, 2, \ldots$. A SV policy is said to be *feasible* if $M_t(\Im_t) \subseteq \mathcal{U}_t(\mathbf{x}_t)\ \forall \Im_t$.

Just like SV control laws, SV policies also have an alternative definition: rather than defining them as a sequence of SV control laws (10.5), they can be defined as a set of PV policies

$$P \triangleq \{p\} \tag{10.6}$$

It is easy to switch from one definition to the other: given a policy P defined by equation (10.6), the control laws $M_t(\cdot)$ that define it according to equation (10.5) are expressed by the following

$$M_t(\cdot) = \{\bar{m}_t(\cdot):\ \exists p \in P:\ p = [m_0(\cdot), m_1(\cdot), \ldots, \bar{m}_t(\cdot), m_{t+1}(\cdot), \ldots]\}$$

In an analogous way, if the policy P is defined by equation (10.5), the set $\{p\}$ that defines it according to equation (10.6) is the following

$$\{p = [\bar{m}_0(\cdot), \bar{m}_1(\cdot), \ldots, \bar{m}_t(\cdot), \bar{m}_{t+1}(\cdot), \ldots]:\ \forall (t, \mathbf{x}_t)\ \bar{m}_t(\mathbf{x}_t) \in M_t(\mathbf{x}_t)\}$$

In what follows, we will denote the subsequences of the PV or SV control laws, relative to the time interval $[t_1, t_2)$, with the symbols $p_{[t_1,t_2)}$ and $P_{[t_1,t_2)}$, i.e.:

$$p_{[t_1,t_2)} \triangleq \{m_t(\cdot);\ t = t_1, \ldots, t_2 - 1\}$$
$$P_{[t_1,t_2)} \triangleq \{M_t(\cdot);\ t = t_1, \ldots, t_2 - 1\}$$

10.1.1.1 Example: the Piave system policy

To clarify these ideas a little more, we will resume with the example of the Piave system, which was introduced in Chapter 6. The control vector of this system (Section 6.1.4) has eleven components

$$\mathbf{u}_t \triangleq \left| u_t^{R1} \quad u_t^{R2} \quad u_t^{R3} \quad u_t^{D1} \quad u_t^{D2} \quad \cdots \quad u_t^{D8} \right|^T \tag{10.7}$$

of which the first three are relative to reservoirs, and the remaining eight to diversions; while the state is the vector

$$\mathbf{x}_t \triangleq \left| \mathbf{c}_t^{C1} \quad s_t^{R1} \quad T_t^{C5} \quad \mathbf{c}_t^{C5} \quad s_t^{R2} \quad s_t^{R3} \right|^T \tag{10.8}$$

At every time t, its regulation policy is thus defined by a vector control law, with eleven components, each one of which specifies the control value (if the policy is PV) or the set of controls (if it is SV) for a reservoir or a diversion. For example, the first component has the form

$$u_t^{R1} = m_t^{R1}(\mathbf{x}_t)$$

and specifies the volume to release from the reservoir $R1$ as a function of the storage values of the three reservoirs (not just $R1$), of the states of the catchments $C1$ and $C5$, and of the air temperature in catchment $C5$. By reviewing Figure 6.2 one can understand why this is the most rational structure: the release from reservoir $R1$ feeds the power plants $H1$ and $H4$, the irrigation districts $I2$, $I4$ and $I6$, and the stretch of river $S8$; but all these components, with the exception of $H1$, can be fed also by $R2$, and $I6$ and $S8$ can be fed, in addition, by $R3$. If one wants to optimize the use of water, when making the release decision for $R1$ it is thus necessary to know the state of the entire system: for example, if $R2$ were spilling or if the high air temperature in $C5$ were causing snow to melt, with, as a consequence, a high inflow to $R2$, it might be sufficient, or appropriate, to satisfy the demands of $I2$, $I4$ and $I6$ by releasing more from $R2$ than from $R1$.

The fact that every control can depend a priori on each state component does not mean, however, that this dependence must necessarily exist: for example, if control u_t^{R3} depended only on the storage in $R3$, the control law $m_t^{R3}(\mathbf{x}_t)$ would give the same value for all the states $\mathbf{x}_t$ that are characterized by the same value s_t^{R3}. It is the task of the policy design to reveal which dependencies are useful and which are not.

10.1.2 On-line and off-line policies

It is useful to distinguish two types of policies: *on-line* and *off-line* policies, according to whether they are designed at time t (by defining and solving the Design Problem at that very moment) or a priori (by defining and solving the Design Problem before management begins).

Determining an off-line policy means determining a control law $m_t(\cdot)$ for each time t over the life of the system. Therefore, when a policy is designed in this way, the system model cannot contain deterministic inputs $\mathbf{w}_t$, since, by definition, their dynamics are not described. In other words, an off-line policy can be designed only when there are no deterministic disturbances that act on the system. In fact we know that, if the state is measurable, the sufficient statistic is $\mathbf{x}_t$, while if it is not, we can reformulate the system so that the new state is measurable. Thus, in all cases, the argument of the control laws for an off-line policy is always and only the state $\mathbf{x}_t$.

Between times t and $(t + 1)$ the dynamics of the system controlled with an off-line policy can be described in the following way:

- at time t the state $\bar{\mathbf{x}}_t$ is observed;
- if the policy is point-valued, the Regulator applies the control $m_t(\bar{\mathbf{x}}_t)$ (which we will indicate in the following with $\bar{\mathbf{u}}_t$ for short); while if the policy is set-valued the Regulator selects and applies a control $\bar{\mathbf{u}}_t$ from the set $M_t(\bar{\mathbf{x}}_t)$ of controls provided by

the SV policy; $m_t(\bar{\mathbf{x}}_t)$ and $M_t(\bar{\mathbf{x}}_t)$ can be determined by simply reading the element corresponding to $\bar{\mathbf{x}}_t$ in the table that defines the policy;

- nature generates the disturbance $\boldsymbol{\varepsilon}_{t+1}$, drawing it from the set Ξ_t, if required according to the probability distribution $\phi_t(\cdot)$;
- the system moves to the state $\mathbf{x}_{t+1} = f_t(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t, \boldsymbol{\varepsilon}_{t+1})$ and the cost $g_t(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t, \boldsymbol{\varepsilon}_{t+1})$ is incurred.

Determining an on-line policy requires, on the contrary, that a Design Problem be solved at every time t. More precisely:

- at time t the state $\bar{\mathbf{x}}_t$ is observed and, if required, the information $\mathbf{w}_t$ is acquired;
- once this information is known, the Regulator formulates and solves a Problem that provides the control $\bar{\mathbf{u}}_t$, or the set of controls $M_t(\bar{\mathbf{x}}_t)$, from which he chooses one; then the control is applied;
- nature generates the disturbance $\boldsymbol{\varepsilon}_{t+1}$, drawing it from the set Ξ_t, if required according to the probability distribution $\phi_t(\cdot)$;
- the system moves to the state $\mathbf{x}_{t+1} = f_t(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$ and the cost $g_t(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$ is incurred.

By proceeding in this way, as time passes, a *sequence of controls* $\{\bar{\mathbf{u}}_0, \bar{\mathbf{u}}_1, \bar{\mathbf{u}}_2, \ldots\}$ is determined, which is *not fixed a priori* but is established during the life of the system (*Adaptive Control*), i.e. it is an *on-line* policy. In this case, the policy is the very procedure that is used to solve the Design Problem at each time instant, and therefore its arguments are $\mathbf{x}_t$ and $\mathbf{w}_t$: a *closed-loop* policy *with compensation*.

The first difference between off-line and on-line policies is that the first cannot exploit *exogenous information* (i.e. the knowledge of deterministic disturbances), while the second can. The second difference is that in order to define an off-line policy, it is necessary to compute the control $\mathbf{u}_t$ a priori for all the possible occurrences of the state $\mathbf{x}_t$, while with an on-line policy, the control will be calculated only as a particular state occurs. The difference is not small: if, for example, at every time t the set $\mathcal{S}_{x_t}$, in which the state $\mathbf{x}_t$ assumes its value, contains 10 000 elements and the period T is 365 days, an off-line policy would require 3 650 000 control values to be calculated. Of these, assuming that the policy is used over the following 10 years, 3650 of them would be used at the most, i.e. one in a thousand. By adopting an on-line policy, instead, at the end of the decade 3650 values will have been calculated, i.e. only those that have been actually used. This kind of simplification has a cost: we must have enough computing power to solve the Design Problem at every time t; on the other hand, when the policy is off-line, at time t we only need to evaluate the function $m_t(\mathbf{x}_t)$, a task which, as we will see in the following (Section 10.5.6), can be reduced to simply reading a table.

10.1.3 When is a policy useful?

It is important to stress that a closed-loop policy is not always necessary. For example, if the inflows to the reservoir were deterministically known over the whole design horizon, it would be sufficient to design an open-loop control sequence (see Section 2.1.1.1): the trajectory of the state that this induces can be determined a priori with certainty. However, when the disturbance is random, this is not true and so we must decide what to do in every possible situation, i.e. for each of the states that could potentially occur: in this way a closed-loop policy is defined.

10.2 The elements of the Design Problem

To be able to formulate the Design Problem, we first must fix the time step and the form of the indicator.

10.2.1 The time step

The adoption of a policy requires the decision to be transformed from an irregular act, which is performed only when it becomes evident that the last decision that was made must be modified, into a periodic act. This in turn requires that the *decision step* Δ be defined. As we explained in Section 4.8, the step has to be selected by making a trade-off between two opposing needs: on the one hand it must be short enough to allow a timely adjustment of the decision as the state of the system varies; on the other hand, it should not be so short as to create social, economic or organizational difficulties for the Stakeholders. In fact, a policy that imposes a change in the decision too frequently would be perceived by the Stakeholders as unsuitable for their needs and thus abandoned. In Section 4.8 we provided some suggestions about how to find an acceptable trade-off between these two opposing requirements.

The decision step Δ must be time-varying and periodic whenever it is not a submultiple of the system's period T (see again Section 4.8, where this topic is dealt with in more detail).

10.2.2 The indicator

We also explained in Section 4.10.1 that the *indicator* i is a functional of the trajectories $\mathbf{x}_0^h$, $\mathbf{u}_0^{h-1}$, $\mathbf{w}_0^{h-1}$ and $\boldsymbol{\varepsilon}_1^h$ of the state, the control and the disturbances that act on the system over the *design horizon* $H = \{0, \ldots, h\}$

$$i = i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)$$

Very often, the functional $i(\cdot)$ is separable: which means that it can be expressed in the form

$$i = \Psi\left\{g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right),\ t = 0, \ldots, h-1;\ g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right\} \tag{10.9}$$

where Ψ is a suitable operator. Remember that the h functions $g_t(\cdot)$, are called *step indicators* or *step costs*, while the function $g_h(\cdot)$, which does not necessarily appear, is called a *penalty*.

When adopted in the *Designing Alternatives* phase an indicator of this type is termed *design indicator* and, as we saw in Chapter 9, it is transformed into an *objective* of the Design Problem by applying a criterion to it in order to filter the effect of the disturbance $\boldsymbol{\varepsilon}$. The forms that the indicator can assume were illustrated in Section 8.1.2 for the Pure Planning Problem. We should now extend those observations to the Pure Management Problem and, subsequently, to the general Design Problem. Given, however, that the difference between these last two Problems lies only in the presence of the planning decisions $\mathbf{u}^p$, to simplify the presentation and not to confuse the reader needlessly, we will introduce directly the general case.

What we explained in Section 8.1.2 for a Pure Planning Problem can be extended to the general Design Problem, by simply adding the control variables and the stochastic disturbance variables to all the formulae presented in that section. The modification is so simple that it is not worth rewriting all those formulae. Having said this, only two things remain to be clarified:

(1) the possible time horizons for the general case;

(2) the reason why the indicator cannot always be expressed by a separable function and what to do when this is the case.

10.2.2.1 The horizon

In Pure Planning Problems (Section 8.1.2) one may choose between two types of horizon, finite and infinite, and, in correspondence with the second, between two alternative indicator forms: TDC and AEV. This also holds for the general Design Problem. It is worth adding just one remark: when the indicator is defined on a finite horizon

$$i = \sum_{t=0}^{h-1} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) + g_h\big(\mathbf{x}_h, \mathbf{u}^p\big) \tag{10.10}$$

the resulting policy is often time-varying, even when the system and the step indicators are not. This is because, all conditions being equal, two time instants differ from one another by the distance in time that separate them from the end h of the horizon. More precisely, when the penalty $g_h(\mathbf{x}_h, \mathbf{u}^p)$ is zero, the policy is always time-varying; in the opposite case it might not be so: in Section 14.1 we will explain the conditions under which it is not.

When only the policy has to be designed, i.e. in Control Problems, there is also an interesting third type of time horizon: the *receding horizon*, which characterizes the on-line policy design. In a sense it is still a finite horizon, but it is defined with respect to time t at which the Problem is formulated

$$i = \sum_{\tau=t}^{t+h-1} g_\tau\big(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}\big) + g_{t+h}\big(\mathbf{x}_{t+h}\big) \tag{10.11}$$

The length h of the time horizon may vary with t. By adopting a receding horizon, when the system and the step indicators are time-invariant, the regulation policy will always be time-invariant.

10.2.2.2 Separability

An indicator is often written in a separable form (Section 4.10.1). When this occurs, the step indicator is the output transformation of one of the system components. We still must explain why, sometimes, the indicator is not separable, at least in the form in which it would be spontaneously defined.

Let us consider the case of a reservoir that feeds an irrigation district, where maize is grown. The evaluation criterion of the Farmers is the maximization of the production and the indicator that quantifies it is the biomass of the *harvest*. Let t_h denote the harvest time with respect to the first day of the year. It is fairly easy to understand that the harvest is a functional of the trajectory of the water supply to the district over the whole irrigation season. As such, it cannot be expressed as a combination of functions, each dependent only on the water supply at time t. As a more detailed analysis suggests, this is due to the fact that we are not considering the state of the maize (which, as a first approximation, we can assume to be its biomass m_t) as a component of the state of the system. If, on the other hand, the state of the maize is considered as a component of the system state, the harvest is naturally expressed as the sum of functions that are always zero, except for at harvest times (i.e. when $t_{\mathrm{mod}\, T} = t_h$), when they take on the value of the maize's biomass m_t.

How to obtain separability The above example can be generalized: it is possible to show that, by properly enlarging the state of the system, each indicator i can be traced back to a separable form. It is not always necessary, or convenient, to include states that have a precise physical meaning, as in the previous example; sometimes the state can just be used to 'remember' the value of some variables for a given number of steps.

The following example, taken from PRACTICE (Sections 4.3.4 and 6.9.2) shows this very well. As we explained on page 175 a proxy indicator for the loss of harvest in an irrigation district is the *average annual potential damage* (*stemming*) *from the stress*, defined by the following expression

$$i_{\mathrm{Irr}} = \frac{1}{N} \sum_{a=1}^{N} f(\digamma_a) \tag{10.12}$$

where $f(\cdot)$ is a function[1] expressing the potential damage from the stress and $\digamma_a$ denotes the maximum stress that occurred during the irrigation season of year[2] a. The latter, in turn, is given by

$$\digamma_a = \max_{t \in a} \frac{1}{\delta} \sum_{\tau = t-\delta+1}^{t} (w_\tau - q_{\tau+1})^+ \tag{10.13}$$

where the number of days δ considered in the sum depends on the field capacity (we will assume $\delta = 14$ in the following), and the deficit $(w_\tau - q_{\tau+1})^+$ is the difference between the water demand w_τ (which is clearly zero outside the irrigation season) and the supply $q_{\tau+1}$ provided in the interval $[\tau, \tau+1)$ to the district, when the first is greater than the second, while it is zero in the opposite case.[3] The indicator i_{Irr} is clearly non-separable in time. At first glance, the non-separability would seem to result from having considered the year a, and not the day t, as the index of the sum in equation (10.12). However, this is not so, since the indicator can be redefined with an expression in which the sum operates on the days

$$i_{\mathrm{Irr}} = \frac{1}{N} \sum_{t=0}^{h} g_t(\mathcal{L}_t) \tag{10.14}$$

provided that the step cost $g_t(\cdot)$ is defined as

$$g_t(\mathcal{L}_t) = \begin{cases} 0 & \text{if } t_{\mathrm{mod}\, T} \neq t_h \\ f(\mathcal{L}_t) & \text{otherwise} \end{cases} \tag{10.15}$$

[1]The reader who is interested in knowing the shape of this function will find it in Figure 4.22 of PRACTICE.

[2]In order to avoid making the notation dull reading, here and in what follows we will use the symbol a to denote both an individual year and the set of days that compose the irrigation season in that year.

[3]The operator $(\cdot)^+$ returns the value of its argument when it is positive, and returns zero in the opposite case.

where the definition of $\mathcal{L}_t$ is completely irrelevant for every time t, with the exception of $t_{\text{mod}\,T} = t_h$, at which time $\mathcal{L}_t$ must coincide with the value of F_a at the end of the irrigation season of the year to which t belongs. It is evident that also in this new definition the indicator is non-separable and that the very cause of the non-separability is the form of F_a. In fact, it is a function of all the deficits that occurred during the irrigation season and, therefore, it cannot be expressed as a function of variables relative to a single time t, as is required by equation (10.9). This is true when we assume the storage s_t as the state of the system, but it is no longer true if the state is defined by the following vector

$$x_t = \left|s_t, \mathcal{L}_t, z_t^0, \dots, z_t^{12}\right|^T$$

The dynamics of the new state components is, in fact, described by the following transition functions

$$\mathcal{L}_{t+1} = \begin{cases} 0 & \text{if } t_{\text{mod}\,T} = t_h \\ \max\left\{\left[\frac{1}{14}\left(\sum_{\tau=0}^{12} z_t^\tau + (w_t - q_{t+1})^+\right)\right], \mathcal{L}_t\right\} & \text{otherwise} \end{cases} \tag{10.16a}$$

$$z_{t+1}^0 = (w_t - q_{t+1})^+ \tag{10.16b}$$

$$z_{t+1}^\tau = z_t^{\tau-1} \quad \text{for } \tau = 1, \dots, 12 \tag{10.16c}$$

so that the value of $\mathcal{L}_{t_h}$ is equal to F_a, exactly as we wanted. It follows that the indicator defined by equations (10.14)–(10.16) is now separable, because the supply q_{t+1}, which appears in equation (10.16b), can be expressed as a function of the storage s_t and the inflow ε_{t+1} to the reservoir, as well as of the release decision $\mathbf{u}_t$, and these variables are all relative to the interval $[t, t+1)$.

The step indicator $g_t(\cdot)$, defined by equations (10.15)–(10.16), can be thought of as the output transformation[4] of a dynamical model that describes the irrigation district, whose state $|\mathcal{L}_t, z_t^0, \dots, z_t^{12}|$ is a surrogate of the state of the crops, and whose state transition function[5] is equation (10.16).

This conclusion can be generalized: by appropriately *enlarging* the state, it is always possible to express the indicator in a separable form and the step indicator that appears in it can be interpreted as the output transformation of a model that describes the component being considered. The state transition equation of such a model is the equation that describes the dynamics of the elements that were added to the 'old' state.

In Section 12.1.2 we will see that, when the indicator is separable, there is a powerful algorithm to solve the Design Problem, but the computing time it requires to identify the solution grows exponentially with the number of the state components. It follows that, even if it is always possible to define the Problem in such a way that there is an algorithm that solves it, the time required to achieve the solution can be so great as to render it, in practice, uncomputable. For example, the computing time needed to solve the Problem defined for the enlarged system in the previous example is 10^{14} times greater than the time needed to solve a Problem in which the state is only the storage s_t of the reservoir. To appreciate the dimension of this number, imagine that, if the solution of the second problem required just one second, the solution of the first would require almost 3.2 million years.

For this reason, very often the enlargement of the state is not feasible in practice, and we are left with only one possibility: substitute the non-separable indicator with an equivalent, separable, indicator. Note that, in order for the two indicators to be equivalent, it is sufficient

[4] See equation (4.5d).

[5] See equation (4.5a).

that they provide the same ranking of the alternatives, whichever set of alternatives is being considered, while it is not necessary that they provide the same value in correspondence with each alternative. There is no formal procedure that allows us to identify an indicator that is equivalent to another; its identification is entrusted to the perceptions of the Analyst, who is assisted by an analysis of the characteristics of each single case. An example of how this happens is described in Section 7.3.2 of PRACTICE.

Separability and filtering of the disturbances An *objective* is defined as a *design indicator* to which a criterion is applied for filtering the uncertainty produced by the random disturbance. The best way to carry out this operation is to define the objective J as the expected value of a utility function $\mathrm{U}(\cdot)$ applied to the indicator i

$$J = \underset{\{\boldsymbol{\varepsilon}\}_{t=1,\dots,h}}{E} \left[\mathrm{U}\left(i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h \right)\right)\right]$$

if the disturbance is stochastic, or as the minimum utility

$$J = \min_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\mathrm{U}\left(i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h \right)\right)\right]$$

when it is uncertain. The utility function is, however, non-linear, except for the case in which the Decision Maker (DM) is risk neutral, when it is completely superfluous. As a consequence, even when the indicator i is separable, the objective obtained in this way is non-separable.

When, instead, the uncertainty is filtered through a criterion, the objective is separable, as long as the Laplace criterion is applied when the time aggregation operator Ψ is the sum, and the Wald criterion is applied when the operator Ψ is the max. In fact, in the first case we get

$$\begin{aligned} J &= \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} \left[\sum_{t=0}^{h-1} g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) + g_h\left(\mathbf{x}_h, \mathbf{u}^p\right) \right] = \\ &= \sum_{t=0}^{h-1} \left[\underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t+1}}{E} \left[g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right]\right] + \underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,h}}{E} \left[g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right] \end{aligned}$$

and in the second

$$\begin{aligned} J &= \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \max\left\{ \max_{t=0,\dots,h-1} \left[g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right],\ g_h\left(\mathbf{x}_h, \mathbf{u}^p\right) \right\} = \\ &= \max\left\{ \max_{t=0,\dots,h-1} \left[\max_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t+1}} g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right], \max_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,h}} \left[g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right] \right\} \end{aligned}$$

Therefore, if one considers

$$\underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t+1}}{E} \left[g_t(\cdot)\right] \quad \text{or} \quad \max_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t+1}} g_t(\cdot)$$

as a step indicator and

$$\underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,h}}{E} \left[g_h(\cdot)\right] \quad \text{or} \quad \max_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,h}} \left[g_h(\cdot)\right]$$

as a penalty, the objective is separable. This proof is easily extendable to the case in which the design horizon is not finite.

For this reason, when one must design a policy, the objective is not defined through a utility function, but with a filtering criterion.

10.2.3 The objective

From the above the objective is, by construction, a deterministic function of the decision variables and can, therefore, be maximized or minimized with respect to them. In the following, we will assume that the aim is to minimize the objective, which is coherent with the nature of 'cost' that we have assumed for the indicator. Nevertheless, it must be clear that this last assumption is not limiting, since any problem of maximum can be traced back to a problem of minimum, by simply multiplying the indicator by -1.

As we explained in Section 4.10.1, the most frequent forms for the time aggregation operator are the sum and the maximum, while the most common criteria for filtering the disturbances are the expected value and the maximum (see Section 9.1).

From all this it follows that the objective can take four most common forms, but, as we will now show, they can be traced back to the following two, where the horizon h can be either finite or infinite:

- the Laplace criterion applied to an indicator with the sum as temporal operator

$$J = \frac{1}{h+1} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h+1}} \left[\sum_{t=0}^{h} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \right] \tag{10.17}$$

  or

$$J = \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h+1}} \left[\sum_{t=0}^{h} \gamma^t g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \right] \tag{10.18}$$

  according to whether one is considering the AEV or TDC form;

- the Wald criterion applied to an indicator with max as temporal operator

$$J = \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h+1}} \left[\max_{t=0}^{h} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \right] \tag{10.19}$$

The other two forms are the following:

$$J = \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h+1}} \left[\max_{t=0}^{h} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \right] \tag{10.20}$$

$$J = \frac{1}{h+1} \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h+1}} \left[\sum_{t=0}^{h} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \right] \tag{10.21}$$

They are of low interest in practical applications and can be traced back to the forms (10.17) or (10.19), as long as the state is properly enlarged. We will show how this is possible by tracing form (10.21) back to (10.19).

Let us define a new state variable G_t, whose dynamics is driven by the following transition equation

$$G_{t+1} = G_t + g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{10.22}$$

When $G_0 = 0$, G_t is the cumulative cost from the beginning of the horizon up to time t. Let us define then the following step cost

$$\tilde{g}_t\big(\mathbf{x}_t, G_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) = \begin{cases} G_h + g_h(\mathbf{x}_h, \mathbf{u}^p, \mathbf{u}_h, \mathbf{w}_h, \boldsymbol{\varepsilon}_{h+1}) & \text{if } t = h \\ 0 & \text{otherwise} \end{cases}$$

which is always zero, except at the last time step, when its value is the cumulative cost over the entire horizon. Finally, let us consider the following objective

$$\tilde{J} = \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h+1}} \Big[\max_{t=0}^{h} \tilde{g}_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\Big] \tag{10.23}$$

It is, by construction, equivalent to the objective (10.21) and has, as was desired, the form (10.19). Therefore, a Problem that is formulated with the objective (10.21) for a system having a state $\mathbf{x}_t$ can be reformulated as a Problem that has (10.23) as the objective and the pair $(\mathbf{x}_t, G_t)$ as the state. To intuitively understand why the state enlargement is necessary, let us consider that our objective is to minimize the expenses over the next month: we could attain this aim only by taking into account daily how much has been spent until this time.

Finally, it is useful to mention that, when the system is periodic of period T, the temporal operator sometimes assumes a mixed form: a first operator acts within the period, and a second between the periods, as in the following example

$$J = \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,kT}} \left[\max_{i=1}^{k} \sum_{t=(i-1)T}^{iT-1} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\right] \tag{10.24}$$

where J expresses the maximum cumulative cost in a period (maximum both with respect to the periods and the possible trajectories of the disturbances). This form appears spontaneously in many cases: for example, when g_t is the daily supply deficit and T is the year, equation (10.24) defines the maximum annual cumulative deficit, which is a common objective for many Regulators.

Depending on their form, the forms of this type can also be traced back to either form (10.17) or (10.19), by carefully choosing how to enlarge the state. For example, form (10.24) can be reduced to (10.19) by introducing a new state variable G_t, whose dynamics is driven by the following transition equation

$$G_{t+1} = \begin{cases} 0 & \text{if } t = iT - 1 \\ G_t + g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) & \text{otherwise} \end{cases} \tag{10.25}$$

G_t is thus the cumulative cost up until time t from the beginning of the period to which t belongs. Then, once

$$\tilde{g}_t\big(\mathbf{x}_t, G_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) = \begin{cases} G_t + g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) & \text{if } t = iT - 1 \\ 0 & \text{otherwise} \end{cases}$$

has been posed, objective (10.23), with $h = kT - 1$, is equivalent to objective (10.24).

10.2.4 The design scenario

As we explained in Section 8.1.3 the design scenario is the last element that characterizes the problem. All that was said in that section also holds good for the general Design Problem; we need only to add a remark about the pseudo-periodic variables.

As we will see in the following chapters, the algorithms for solving the Design Problem over an infinite horizon operate under the hypothesis that the system is periodic of period T. The pseudo-periodic variables, which are not periodic, may create a difficulty from this point of view. To understand how to overcome it, we can reconsider the example of the hydropower demand presented in Section 8.1.3. On the first of January the hydropower water demand does not have the same value every year, since if this year the first of January falls on a Sunday, the next year it will fall on a Monday, as long as the current year is not a leap year, in which case it will fall on a Tuesday, and we know that the hydropower demand varies during the week. To be able to assume that the system is periodic of one year we must be able to assume that all the years begin with the same day, e.g. Monday, i.e. we must assume that every year is a *standard year*, in the sense given to this term in Section 8.1.3 (to which we refer also for the definitions of all the other terms which appear in this section).

With this assumption the system turns out to be periodic and the algorithms can be used, but what happens to the solution obtained? The policy that is obtained will be synchronized with the sequence of Saturdays and Sundays of the standard year: this means, for example, that the control law that corresponds to day 7, from the beginning of the year (i.e. the day that has 7 as *natural date*), will be a law that takes account of the fact that that day is a Sunday. Clearly, it should not be used on the days that have natural date 7, but on those that have *anthropic date* 7 (Section 8.1.3), because, following the definition of anthropic date, these days are Sundays, and each of them, in its year, will be the Sunday that falls closest to the seventh day of the year. In this way, the policy will deal with the demand that it was designed for.[6]

10.3 The Design Problem with PV policies

Now we are able to formulate the Design Problem, but only when the policy is point-valued (PV). The definition of the Design Problem with a set-valued (SV) policy requires concepts that the reader does not yet have, and so it is postponed to the next chapter. As was anticipated in Section 6.2, we only consider the case in which the random disturbances are either all stochastic or all uncertain, since dealing with mixed cases goes beyond the limits of this book.

When one adopts the Laplace criterion, the Design Problem for a PV policy is the following:

The PV Design Problem with the Laplace criterion: *To determine*[7] *a policy*[8] p^* *and an optimal vector* $\mathbf{u}^{p*}$ *such that*

$$J^* = \min_{\mathbf{u}^p,\, p} \; \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} \left[i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{10.26a}$$

[6] Clearly, this is true under the hypothesis that the design scenario is effectively what happens in the future.

[7] If the optimal pair $(\mathbf{u}^{p^*}, p^*)$ is not unique, it might be interesting for the DM to consider all the pairs and choose the one that best satisfies any criteria that were not expressed in the formulation of the Problem. If the Design Problem is a Control Problem, the multiple solutions constitute the SV policy, which we introduced in Section 10.1.1 and whose design will be considered in the next chapter. We will not consider multiple solutions when the Problem is mixed, because this goes beyond the limits set to this work.

[8] When the initial state is given and the Design Problem is defined over a finite horizon, it is not necessary to determine the entire control law $m_0(\cdot)$, but only the control $m_0(\mathbf{x}_0)$. Nevertheless, in order not to weigh down the notation, we will not make this fact explicit in the formulation of the Problem, but we will exploit it when it is useful, for example in Algorithm 1 on page 299, and in the formulation of the on-line policy on page 343.

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \qquad t = 0, \ldots, h-1 \tag{10.26b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{10.26c}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \qquad t = 0, \ldots, h-1 \tag{10.26d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\big(\cdot|\mathbf{u}^p\big) \qquad t = 0, \ldots, h-1 \tag{10.26e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{10.26f}$$

$$\mathbf{x}_0 \textit{ given} \tag{10.26g}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \ldots, h-1\big\} \tag{10.26h}$$

$$\textit{any other constraints} \qquad t = 0, \ldots, h-1 \tag{10.26i}$$

The length h of the horizon can be either finite of infinite; in the second case the definition of the indicator embodies a limit, as equations (8.11) and (8.12) show.

When, instead, one adopts the Wald criterion, the Problem is the following:

The PV Design Problem with the Wald criterion: *To determine an optimal policy* p^* *and an optimal vector* $\mathbf{u}^{p*}$*, such that*

$$J^* = \min_{\mathbf{u}^p,\, p} \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \Big[i\big(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\big)\Big] \tag{10.27a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \qquad t = 0, \ldots, h-1 \tag{10.27b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{10.27c}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \qquad t = 0, \ldots, h-1 \tag{10.27d}$$

$$\boldsymbol{\varepsilon}_{t+1} \in \Xi_t\big(\mathbf{u}^p\big) \qquad t = 0, \ldots, h-1 \tag{10.27e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{10.27f}$$

$$\mathbf{x}_0 \textit{ given} \tag{10.27g}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \ldots, h-1\big\} \tag{10.27h}$$

$$\textit{any other constraints} \qquad t = 0, \ldots, h-1 \tag{10.27i}$$

Note that the Wald form can be chosen also when the disturbance is stochastic, provided that, at each time instant, the probability distribution $\phi_t(\cdot)$ is bounded. In this case it is, in fact, possible to derive the set Ξ_t of disturbances that have a non-zero probability of occurrence from $\phi_t(\cdot)$. Furthermore, since the random disturbances per-period can easily be transformed into random disturbances per-step, without losing generality, we can formulate the Design Problem only with the disturbances per-step.

The elements of the pair $(\mathbf{u}^{p*}, p^*)$ that solve problems (10.26) and (10.27) are called *optimal planning decision* and *optimal policy* respectively.

10.3.1 Characteristics of the solution

As the reader may easily understand, if the policy design is not required in the Project, problem (10.26) takes the form of a Pure Planning Problem, the solution of which we already dealt with in Chapters 8 and 9. In an analogous manner, when there are no planning decisions $\mathbf{u}^p$, the Design Problem is a Control Problem, the solution of which we will deal with in Chapters 12 and 13.

Problem (10.26) can be rewritten as a couple of cascade problems:

The Planning Problem:

$$J^* = \min_{\mathbf{u}^p} J^*\big(\mathbf{u}^p\big) \tag{10.28a}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{10.28b}$$

And

The Off-line Control Problem:

$$J^*\big(\mathbf{u}^p\big) = \min_{p} \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E} \Big[i\big(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\big)\Big] \tag{10.29a}$$

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \qquad t = 0, \dots, h-1 \tag{10.29b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \qquad t = 0, \dots, h-1 \tag{10.29c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\big(\cdot|\mathbf{u}^p\big) \qquad t = 0, \dots, h-1 \tag{10.29d}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{10.29e}$$

$$\mathbf{x}_0 \textit{ given} \tag{10.29f}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \dots, h-1\big\} \tag{10.29g}$$

$$\textit{any other constraints} \qquad t = 0, \dots, h-1 \tag{10.29h}$$

Note that the objective $J^*(\mathbf{u}^p)$ of the Planning Problem is the optimal value of the Off-line Control Problem as a function of the 'parameter' $\mathbf{u}^p$. Given a value for $\mathbf{u}^p$, problem (10.29) is a Control Problem defined with the Laplace criterion and, as such, it can be solved with one of the algorithms that we will describe in Chapters 12 and 13. The optimal value $J^*(\mathbf{u}^p)$ of the objective, which is obtained in this way, is therefore a function of $\mathbf{u}^p$ and that function is the objective of problem (10.28). The latter has the form of a Pure Planning Problem and can therefore be solved with the algorithms that we described in Chapter 8. Among these there are some evolutionary algorithms, which determine the solution by evolving in the space $\mathbf{u}^p$ and by evaluating the value of the objective $J^*(\cdot)$ at each step. The most efficient way to solve problem (10.28), and thus problem (10.26), to which it is completely equivalent, given (10.29), is to adopt one of those algorithms. By doing so, it is not necessary, in fact, to solve problem (10.29) in advance for all the values of $\mathbf{u}^p$, because it has to be solved only for those values that are, step by step, considered by the evolutionary algorithm. In other words, the algorithm that solves problem (10.26) is obtained by nesting an algorithm that solves the Control Problem (10.29) within an evolutionary algorithm, which solves the Pure Planning Problem (10.28). This is the reason why the two Problems can be studied separately.

What we have said until this point is clearly also valid for problem (10.27), as long as the proper algorithms are selected to solve the Control Problem when it is defined with the Wald criterion.

In both Problems, when the horizon is finite, the optimal policy is defined by a finite number of control laws (at the most h, but their number could be lower if some of the control laws are applied more than once). When, instead, h is infinite, the policy is defined by an infinite number of functions and so, in general, it cannot be determined. Nevertheless, it is possible to show[9] that when the system model, the disturbances that act upon it and the indicator are all periodic[10] of period T, the optimal policy will also be periodic of that same period. From this it follows that the optimal policy may be computable, since it is defined by a finite number (T) of functions. Lastly, since a periodic, time-invariant function can be considered to be a function that is periodic of period $T = 1$, from the above it follows that when the system, the disturbances, and the indicator are time-invariant, the optimal policy will also be time-invariant.

10.3.2 Approaches to the solution

10.3.2.1 Functional and Parametric Approach

Since the policy p is a succession of control laws and given that each of them is defined by a function, and so by infinite pairs $(\mathbf{x}_t, \mathbf{u}_t)$, determining the optimal policy p^* requires that an infinite number of values be determined, even when the number of control laws is finite, and so, with the exception of particular cases, requires an infinite computing effort. Therefore, determining exactly the optimal policy is generally impossible: we must settle for an approximation of it.

Nevertheless, one of the particular cases, to which we have just referred, occurs when the sets $\mathcal{S}_{\mathbf{x}_t}$, $\mathcal{S}_{\mathbf{u}_t}$, $\mathcal{S}_{\mathbf{w}_t}$ and $\mathcal{S}_{\boldsymbol{\varepsilon}_t}$, in which the state, control and disturbances assume their value (see the following Section 10.5), are finite for every t, i.e. they contain only a finite number of elements. In such a case, the model is said to be an *automaton* and each control law $m_t(\mathbf{x}_t)$ is defined by a look-up table (see Section 10.5.6), which can be specified by defining a finite number of elements. Therefore, when the Problem is defined over a finite time horizon or when, if the horizon is infinite, the model of the system, the disturbances and the indicator are periodic, as a consequence of what we saw at the end of the last section, the optimal policy p^* is defined by a finite number of elements. Hence, it cannot be ruled out a priori that the policy is determined in a finite time period: in Section 12.1 we will see that it can actually be determined when the model satisfies additional conditions, which we will specify in due course.

If the model is not an automaton, an approximation of it can be considered. This model is termed *discretized model*, or also *discretized system*, by interchanging the terms model

[9] We do not provide the proof because it requires higher mathematical knowledge.

[10] A system is periodic when the state transition functions, the output transformation functions, and all the functions that define its constraints, are periodic functions of time t. A disturbance is periodic when its description is periodic. In the case of a random disturbance this means that the probability distribution $\phi_t(\cdot)$ (or the set Ξ_t) repeats periodically with period T; in the case of a deterministic disturbance this means that the sequence $\{\mathbf{w}_0, \ldots, \mathbf{w}_{h-1}\}$ is a periodic sequence. N.B.: this is not equivalent to asking that the realization $\{\boldsymbol{\varepsilon}_1, \ldots, \boldsymbol{\varepsilon}_h\}$ of the random disturbance be periodic!

and system as is usual practice. The optimal policy that is obtained by using the discretized model will approximate the optimal policy that we are looking for with the same accuracy as the discretized model approximates the original model. In the last section of this chapter we will show one of the procedures through which the discretized model can be obtained, once the system model is known.

To this point, the approach that we have considered is a *Functional Approach*, since, conceptually, one is determining the optimal policy as a succession of control laws, upon which no conditions are imposed; in other words in the space of all the control laws, i.e. the space of all the functions, the best ones are looked for, those that minimize the objective. In practice, we have seen that this is not completely true, because the control laws are assumed to be discrete functions; nonetheless, the attribute ‘functional’ is retained, because by simply increasing sufficiently the number of elements of the sets $\mathcal{S}_{\mathbf{x}_t}$, $\mathcal{S}_{\mathbf{u}_t}$, $\mathcal{S}_{\mathbf{w}_t}$ and $\mathcal{S}_{\boldsymbol{\varepsilon}_t}$ over which the automaton is defined, the desired policy can be approximated at will. In Sections 12.1.4 and 13.4 we will study the algorithms based on the Functional Approach.

Alternatively, one can adopt a *Parametric Approach*: the class of functions to which the control law must belong is fixed a priori, so that a particular function of that class is defined by a finite number of parameters. It follows that also the policy is defined by a finite number of parameters. The policy design will therefore consist in identifying the values of the parameters which minimize the objective. We will consider algorithms of this type in Section 12.2.

10.3.2.2 Learning approach and model-free approach

Both of the above approaches to the solution first assume that the available information is collected and organized in a model of the system and the disturbances (the model of the disturbances defines the environment in which the system operates) and then look, with a suitable algorithm, for a policy that ‘satisfies’ the objective. However, this way of proceeding is time-consuming and costly, both economically and in terms of computation efforts. Furthermore, any error that is introduced in the description of the system and of the disturbances reflects on the quality of the policy that is identified. However, these two approaches are not the only ones possible: there are two other possibilities.

The first possibility is based on the observation that also a human being has ‘automatic’ (i.e. policy-driven) behaviours, such as grasping objects or walking. A child learning how to grasp objects or to walk is designing a ‘policy’, but the approach she adopts is quite different from the one considered so far. The ‘regulation policy’ for her hands or legs is learned through direct interaction with the environment: the child tries an action, evaluates the effect, and makes another attempt. The variety of situations that the child encounters, and the complexity of reactions that her interaction with the environment produces, generate information about the correspondence between action and effect, and this information, appropriately elaborated, allows her to individuate the ‘most suitable’ behaviours for the different situations in which she may find herself. This unconscious elaboration is what we call ‘learning’. The approaches to solve the Design Problem that have been considered until this point totally ignore this notion, but emulating the human learning process may be an appealing strategy to design a regulation policy. This strategy seems even more advantageous if one considers that the system and the environment in which it operates could slowly change over time (namely, they are neither stationary nor cyclostationary). Thus, a policy designed with the approaches that we have seen until now could become progressively obsolete, while if the training of the policy were to continue, even during its use, it would be modified as the system and its environment evolve.

Learning per se does not necessarily require direct interaction with the physical system, i.e. learning directly from the system, but can also come about through interacting with a model, namely with a simulator of the system and the environment. Whenever the wrong action could produce unsustainable consequences, one turns to this option: think for example about training pilots in a flight simulator or mountain climbers who learn to climb on an artificial wall. Both the simulator and the wall are models, the first mathematical and the second physical, of the environment in which the pilot or the mountain climber will actually be operating. In the field of environmental management it is unacceptable to perform direct experiments: for example, a reservoir regulation policy cannot be fine tuned by experimenting freely with actions that could produce floods or drought. For this reason, in a certain sense, we must train the policy in an artificial environment.

And here is the second possibility for an alternative approach: until now we have taken for granted that the artificial environment must be a model of the system, and in fact the model acts as a constraint in the Design Problem (see for example equations (10.26b), (10.26e)). However, in order for this model to be used, it must be identified through a previously collected time series of pairs of input–output measures. However, we can think of using this same series of measures directly in the definition of the Design Problem, thus obtaining the non-negligible advantage of saving not only the time and the expenses needed for calibrating the model, but also of retaining the entire richness of the information in the original series. This way of proceeding is termed *model-free* approach.

Finally, it is spontaneous, as has already emerged in the above phrases, to think about coupling the two approaches to carry out a *model-free learning* process. In particular, this is the only possible way when previously collected measurement series are not available. By coupling the two methods, the *automatic regulator*,[11] within which the learning algorithm is implemented, would begin to control the system (clearly taking into careful consideration the security problems that we noted earlier), even though it has not been specifically designed for that system, and would learn the most suitable policy through experience, and the passage of time. Clearly, at the beginning it would often propose mistaken controls, but the number of errors would diminish over time. One might thus consider that the approach is not profitable; however, if one reflects, one realizes that, in the absence of a priori information, it is not possible to do any better, unless one waits and collects a data series that is long enough to identify a model. However, given the nature of environmental systems, this would take several decades. On the other hand, what other alternative is available? The current solution is to entrust the decision to a human DM, who cautiously learns what to do. A learning algorithm can act in a similar way. Thus, to reduce the number of errors as much as possible, it is convenient to use all of the a priori information that is available: it is for this very reason that a new system is entrusted to an expert Regulator. A mixed approach is the best, i.e. an approach in which all that is known is described by a model, and a model-free description is used only for the part of the system for which no information is available. Since regulation facilities, such as reservoirs, are artificial, they are always well-known and can be modelled, so the parts that are not modelled are almost always the catchment and the meteorological system.

The class of algorithms that allows us to carry out what is described above will be illustrated in Section 13.4.

[11] By the term *automatic regulator* we mean an artificial system that implements a given policy, i.e. a system that, at every time instant, given the state $\mathbf{x}_t$ (or more generally the information $\Im_t$), provides the control value that the policy establishes. In the case we are examining here, the automatic regulator implements a learning algorithm which, once the learning has terminated, comes to be the policy itself.

10.4 A Law of Duality: from Laplace to Wald

By comparing problem (10.26) with problem (10.27) one can observe how the second can be derived from the first by substituting max in the place of E in equation (10.26a) and Ξ_t in the place of $\phi_t(\cdot)$ in equation (10.26e). This is a simple application of an empirical Law of Duality which can be formulated like this:

Law of Duality: *From a proposition, an algorithm or a formula that is valid for a Design Problem defined with the Laplace criterion, one can derive a proposition,*[12] *an algorithm or a formula that is valid for a Design Problem defined with the Wald criterion by making the following substitutions*:

$$\begin{aligned} \max_{\boldsymbol{\varepsilon}_{t+1}} &\longrightarrow \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \\ \max &\longrightarrow + \\ \Xi_t &\longrightarrow \phi_t(\cdot) \end{aligned}$$

Clearly, the inverse substitutions take us from a proposition, an algorithm or a formula that is valid for a Design Problem defined with the Wald criterion to a proposition, an algorithm or a formula that is valid for a Problem defined with the Laplace criterion. This rule allows us to deduce and enunciate the expression or the algorithms for only one of the two cases.

10.5 Discretization

As explained in Section 10.3.2 the functional Control Problem (10.29) can be solved when, at every moment in time, the sets $\mathcal{S}_{\mathbf{x}_t}$, $\mathcal{S}_{\mathbf{u}_t}$, $\mathcal{S}_{\mathbf{w}_t}$ and $\mathcal{S}_{\boldsymbol{\varepsilon}_t}$, in which state, control and disturbances take values (see Section 6.2), are finite sets.

If the original model does not comply with this condition, it is necessary to consider a model that is 'almost' equivalent, in which the sets of state, control and random disturbance are finite sets, i.e. a *discretized model*. This requires that a discretization procedure be adopted: the one described in this paragraph is just an example, but it is not the only one possible, nor is it always the best.

Since in Section 10.3.1 we saw that the optimal policy can be determined only when the model of the system, the disturbances that act upon it and the indicator are periodic of period T, *in the following we will assume that the sets* $\mathcal{S}_{\mathbf{x}_t}$, $\mathcal{S}_{\mathbf{u}_t}$, $\mathcal{S}_{\mathbf{w}_t}$ and $\mathcal{S}_{\boldsymbol{\varepsilon}_t}$ *are periodic*, i.e.

$$\begin{aligned} &\mathcal{S}_{\mathbf{x}_t} = \mathcal{S}_{\mathbf{x}_{t+kT}} \qquad \mathcal{S}_{\mathbf{u}_t} = \mathcal{S}_{\mathbf{u}_{t+kT}} \qquad \mathcal{S}_{\mathbf{w}_t} = \mathcal{S}_{\mathbf{w}_{t+kT}} \qquad \mathcal{S}_{\boldsymbol{\varepsilon}_t} = \mathcal{S}_{\boldsymbol{\varepsilon}_{t+kT}} \\ &\quad t = 0, 1, \ldots; \; k = 1, 2, \ldots \end{aligned}$$

10.5.1 Classes of state

For $t = 0, 1, \ldots, T-1$, given the state set $\mathcal{S}_{\mathbf{x}_t}$, define a rectangular set $\tilde{\mathcal{S}}_{\mathbf{x}_t} \subseteq \mathcal{S}_{\mathbf{x}_t}$

$$\tilde{\mathcal{S}}_{\mathbf{x}_t} = \left\{ \mathbf{x}_t \;\middle|\; \underline{x}_t^i \leqslant x_t^i \leqslant \bar{x}_t^i,\; i = 1, 2, \ldots, n_{\mathbf{x}} \right\} \quad t = 0, \ldots, T-1$$

[12] This always holds for propositions; however the conditions under which the convergence of the algorithms is ensured can be slightly different from the Laplace case.

where:

$n_{\mathbf{x}}$ is the dimension of the state;
x_t^i is the ith component of $\mathbf{x}_t$;
$\underline{x}_t^i, \bar{x}_t^i$ are respectively the lower and upper bounds of x_t^i, which must be selected in such a way that the probability of the event $\mathbf{x}_t \notin \tilde{\mathcal{S}}_{\mathbf{x}_t}$ is negligible.

Consider then a partition $(\tilde{\mathcal{S}}_{\mathbf{x}_t}^1, \tilde{\mathcal{S}}_{\mathbf{x}_t}^2, \dots, \tilde{\mathcal{S}}_{\mathbf{x}_t}^{N_{\mathbf{x}}})$ of $\tilde{\mathcal{S}}_{\mathbf{x}_t}$, i.e. consider $N_{\mathbf{x}}$ disjoint subsets $\tilde{\mathcal{S}}_{\mathbf{x}_t}^1, \tilde{\mathcal{S}}_{\mathbf{x}_t}^2, \dots, \tilde{\mathcal{S}}_{\mathbf{x}_t}^{N_{\mathbf{x}}}$ such that

$$\tilde{\mathcal{S}}_{\mathbf{x}_t} = \bigcup_{j=1}^{N_{\mathbf{x}}} \tilde{\mathcal{S}}_{\mathbf{x}_t}^j$$

For example, if for $i = 1, 2, \dots, n_{\mathbf{x}}$ each interval $[\underline{x}_t^i, \bar{x}_t^i]$ is divided into N_i adjacent intervals, the sets $\tilde{\mathcal{S}}_{\mathbf{x}_t}^1, \tilde{\mathcal{S}}_{\mathbf{x}_t}^2, \dots, \tilde{\mathcal{S}}_{\mathbf{x}_t}^{N_{\mathbf{x}}}$ can be defined as all the possible Cartesian products among the $n_{\mathbf{x}}$ subintervals, one for each state variable. Thus, $N_{\mathbf{x}} = N_1 \times N_2 \times \cdots \times N_{n_{\mathbf{x}}}$ sets are obtained.

Let $\hat{\boldsymbol{x}}^j \in \tilde{\mathcal{S}}_{\mathbf{x}_t}$ be a point that characterizes, in some suitable sense, the set $\tilde{\mathcal{S}}_{\mathbf{x}_t}^j$: for example, it could be the barycentre of the set, i.e. the point whose coordinates correspond to the middle points of the subintervals, the Cartesian product of which produces $\tilde{\mathcal{S}}_{\mathbf{x}_t}^j$. Then,

$$\hat{\mathcal{S}}_{\mathbf{x}_t} = \left\{\hat{\boldsymbol{x}}_t^1, \hat{\boldsymbol{x}}_t^2, \dots, \hat{\boldsymbol{x}}_t^{N_{\mathbf{x}}}\right\}$$

can be assumed as the state set of the discretized model, often called (*state*) *grid*.

10.5.2 Classes of control

In a completely analogous way, a rectangular set $\tilde{\mathcal{S}}_{\mathbf{u}_t} \subseteq \mathcal{S}_{\mathbf{u}_t}$ of controls can be defined

$$\tilde{\mathcal{S}}_{\mathbf{u}_t} = \left\{\mathbf{u}_t \mid \underline{u}_t^i \leqslant u_t^i \leqslant \bar{u}_t^i,\ i = 1, 2, \dots, n_{\mathbf{u}}\right\} \quad t = 0, \dots, T-1$$

from which, once a partition $(\tilde{\mathcal{S}}_{\mathbf{u}_t}^1, \tilde{\mathcal{S}}_{\mathbf{u}_t}^2, \dots, \tilde{\mathcal{S}}_{\mathbf{u}_t}^{N_{\mathbf{u}}})$ of it has been chosen, the set $\hat{\mathcal{S}}_{\mathbf{u}_t}$ of controls of the discretized model (often called (*control*) *grid*) is obtained

$$\hat{\mathcal{S}}_{\mathbf{u}_t} = \left\{\hat{\boldsymbol{u}}_t^1, \hat{\boldsymbol{u}}_t^2, \dots, \hat{\boldsymbol{u}}_t^{N_{\mathbf{u}}}\right\}$$

10.5.3 Classes of deterministic disturbance

In a completely analogous way, a rectangular set $\tilde{\mathcal{S}}_{\mathbf{w}_t} \subseteq \mathcal{S}_{\mathbf{w}_t}$ of deterministic disturbances can be defined

$$\tilde{\mathcal{S}}_{\mathbf{w}_t} = \left\{\mathbf{w}_t \mid \underline{w}_t^i \leqslant w_t^i \leqslant \bar{w}_t^i,\ i = 1, 2, \dots, n_{\mathbf{w}}\right\} \quad t = 0, \dots, T-1$$

from which, once a partition $(\tilde{\mathcal{S}}_{\mathbf{w}_t}^1, \tilde{\mathcal{S}}_{\mathbf{w}_t}^2, \dots, \tilde{\mathcal{S}}_{\mathbf{w}_t}^{N_{\mathbf{w}}})$ of it has been chosen, the set $\hat{\mathcal{S}}_{\mathbf{w}_t}$ of deterministic disturbances of the discretized model is obtained

$$\hat{\mathcal{S}}_{\mathbf{w}_t} = \left\{\hat{\mathbf{w}}_t^1, \hat{\boldsymbol{w}}_t^2, \dots, \hat{\boldsymbol{w}}_t^{N_{\mathbf{w}}}\right\}$$

10.5.4 Classes of random disturbance

Also for the random disturbance one proceeds in an analogous way: first a rectangular, bounded set

$$\tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t} = \left\{\boldsymbol{\varepsilon}_t \;\middle|\; \underline{\boldsymbol{\varepsilon}}_t^i \leqslant \boldsymbol{\varepsilon}_t^i \leqslant \bar{\boldsymbol{\varepsilon}}_t^i,\ i = 1, 2, \dots, n_{\boldsymbol{\varepsilon}}\right\} \quad t = 0, \dots, T-1$$

included in $\mathcal{S}_{\boldsymbol{\varepsilon}_t}$ is defined. Then, if the disturbance is uncertain and for every t the sets Ξ_t are bounded, the extremes $\{\underline{\boldsymbol{\varepsilon}}_t^i, \bar{\boldsymbol{\varepsilon}}_t^i\}_{i=1,\dots,n_{\boldsymbol{\varepsilon}}}$ can easily be determined. If, instead, the disturbance is stochastic, the extremes must be chosen in such a way that the probability of the event $\boldsymbol{\varepsilon}_t \notin \tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}$ is negligible. For example, if the components of the disturbance have a normal distribution and are statistically independent, it is sufficient to assume $\underline{\boldsymbol{\varepsilon}}_t^i = -\vartheta\sigma_{\text{Max}}$ and $\bar{\boldsymbol{\varepsilon}}_t^i = \vartheta\sigma_{\text{Max}}$, where ϑ is a sufficiently large positive integer (for example 3 or 4), and $\sigma_{\text{Max}} = \max\{\sigma_0, \sigma_1, \dots, \sigma_{T-1}\}$, i.e. it is the greatest of the mean square deviations of the disturbance.

Then a partition $(\tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}^1, \tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}^2, \dots, \tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}^{N_{\boldsymbol{\varepsilon}}})$ of $\tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}$ is defined, and thus the set $\hat{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}$ of the random disturbances of the discretized model is the following

$$\hat{\mathcal{S}}_{\boldsymbol{\varepsilon}_t} = \left\{\hat{\boldsymbol{\varepsilon}}_t^1, \hat{\boldsymbol{\varepsilon}}_t^2, \dots, \hat{\boldsymbol{\varepsilon}}_t^{N_{\boldsymbol{\varepsilon}}}\right\}$$

where $\hat{\boldsymbol{\varepsilon}}_t^i$ denotes a point that characterizes the set $\tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}^i$ (for example its barycentre). If the disturbance is stochastic, to every element of $\hat{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}$ a probability of occurrence

$$\hat{\phi}_t\left(\hat{\boldsymbol{\varepsilon}}_t^j\right) = \int_{\tilde{\mathcal{S}}_{\boldsymbol{\varepsilon}_t}^j} \phi_t(z)\,dz \quad j = 1, \dots, N_{\boldsymbol{\varepsilon}};\ t = 0, \dots, T-1$$

is associated, where $\phi_t(\cdot)$ is the probability distribution of $\boldsymbol{\varepsilon}_t$. Since the probability $\phi_t(\cdot)$ is periodic of period T, also the probability $\hat{\phi}_t(\cdot)$ is periodic, i.e.

$$\hat{\phi}_t(\cdot) = \hat{\phi}_{t+kT}(\cdot) \quad k = 1, 2, \dots;\ t = 0, \dots, T-1$$

10.5.5 The discretized model

Once the sets that define the state, control and disturbance variables have been defined, the discretized model is described by the following dynamical equation

$$\hat{\mathbf{x}}_{t+1} = \hat{f}_t\left(\hat{\mathbf{x}}_t, \mathbf{u}^p, \hat{\mathbf{u}}_t, \hat{\mathbf{w}}_t, \hat{\boldsymbol{\varepsilon}}_{t+1}\right) \tag{10.30}$$

where $\hat{\mathbf{x}}_t \in \hat{\mathcal{S}}_{\mathbf{x}_t}$, $\hat{\mathbf{u}}_t \in \hat{\mathcal{S}}_{\mathbf{u}_t}$, $\hat{\mathbf{w}}_t \in \hat{\mathcal{S}}_{\mathbf{w}_t}$, $\hat{\boldsymbol{\varepsilon}}_{t+1} \in \hat{\mathcal{S}}_{\boldsymbol{\varepsilon}_{t+1}}$, for $t = 0, \dots, T-1$ and $\mathbf{u}^p \in \mathcal{U}^p$. The function $\hat{f}_t(\cdot)$ is defined by posing

$$\hat{f}_t\left(\hat{\mathbf{x}}_t, \mathbf{u}^p, \hat{\mathbf{u}}_t, \hat{\mathbf{w}}_t, \hat{\boldsymbol{\varepsilon}}_{t+1}\right) = \hat{\mathbf{x}}^j \quad \text{when} \quad f_t\left(\hat{\mathbf{x}}_t, \mathbf{u}^p, \hat{\mathbf{u}}_t, \hat{\mathbf{w}}_t, \hat{\boldsymbol{\varepsilon}}_{t+1}\right) \in \tilde{\mathcal{S}}_{\mathbf{x}_{t+1}}^j$$
$$t = 0, \dots, T-1,\ j = 1, 2, \dots, N_{\mathbf{x}}$$

Similarly, the step cost function $\hat{g}_t(\cdot)$ is defined as

$$\hat{g}_t\left(\hat{\mathbf{x}}_t, \mathbf{u}^p, \hat{\mathbf{u}}_t, \hat{\mathbf{w}}_t, \hat{\boldsymbol{\varepsilon}}_{t+1}\right) = g_t\left(\hat{\mathbf{x}}_t, \mathbf{u}^p, \hat{\mathbf{u}}_t, \hat{\mathbf{w}}_t, \hat{\boldsymbol{\varepsilon}}_{t+1}\right) \quad t = 0, 1, \dots, T-1$$

Both the function $\hat{f}_t(\cdot)$ and the step cost $\hat{g}_t(\cdot)$ are periodic functions of period T, because $f_t(\cdot)$ and $g_t(\cdot)$ are such.

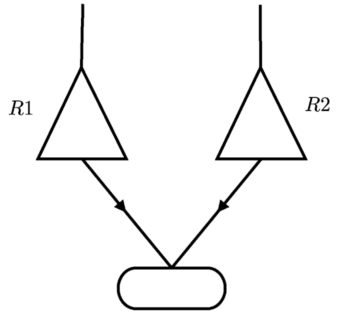


Figure 10.1: A system with two reservoirs.

A point-value policy

$$\hat{p} = \left\{\hat{m}_1(\cdot), \hat{m}_2(\cdot), \hat{m}_3(\cdot), \ldots\right\}$$

is a sequence of functions $\hat{m}_t(\cdot)$, each of which associates a control from the set $\hat{\mathcal{S}}_{\mathbf{u}_t}$ to each value of the new state $\hat{\mathbf{x}}_t$.

The so-defined system proves to be an automaton with an uncertain or stochastic input; in the second case it is a Markov chain (see Sections 4.7 and 11.1).

Once the discrete control laws $\hat{m}_t(\cdot)$, $t = 0, 1, \ldots, T-1$, have been determined for the discrete model (10.30), a control law for the original system can be deduced from each one of them by assuming that the control $\hat{m}_t(\hat{\mathbf{x}}_t^j)$ is applied to each $\mathbf{x}_t \in \tilde{\mathcal{S}}_{\mathbf{x}_t}^j$.

In the following chapters, to simplify the notation, we will omit the symbol ^ to denote the variables of the discretized model.

The discretization proposed is the most intuitive and produces completely uniform grids in the sets of variables. The *full uniformity* finds its *raison d' être* in the Functional Approach: since we cannot know a priori which form the functions that we are seeking have, i.e. the control laws that minimize the objective, see Section 10.3.2, it is useful for us to construct a uniform grid (which is sufficiently dense!) of the variables. This is not to run the risk of leaving unexplored significant areas of the range of these functions. However, the uniform discretization is not the only one possible, nor is it often the most appropriate. Others exist (Niederreiter, 1992), which can prove particularly useful from the computational point of view. We will return briefly to this subject in Section 12.2.3.4.

10.5.6 The control law as a matrix

When the system is an automaton, a control law $\hat{m}_t(\cdot)$ is spontaneously represented by a matrix m of $n_{\mathbf{x}}$ dimensions, whose generic element $m_{ijh\ldots}$ is the control vector $\hat{m}_t(\hat{\mathbf{x}}_t)$ that the law associates to the state $\hat{\mathbf{x}}_t = |x_t^1, x_t^2, \ldots|$ when its component x_t^1 assumes the ith of its possible values and the component x_t^2 assumes the jth, and so on. For example, if the storage values of the reservoirs R_1 and R_2 of the system in Figure 10.1 are discretized with three values $(1, 2, 3)$ and the releases are also discretized with three values $(0, 1, 2)$, we would have $\hat{\mathcal{S}}_{\mathbf{x}^1} = \hat{\mathcal{S}}_{\mathbf{x}^2} = \{1, 2, 3\}$ and $\hat{\mathcal{S}}_{\mathbf{u}} = \{0, 1, 2\}$. A time-varying PV control law is thus defined by the following matrix

		x_t^1: 1	2	3
x_t^2	1	(2, 0)	(2, 1)	(2, 2)
	2	(1, 0)	(1, 1)	(1, 2)
	3	(0, 0)	(0, 1)	(0, 2)

A time-invariant SV control law is also defined by a matrix, whose elements are sets, such as the following matrix

		x_t^1: 1	2	3
x_t^2	1	{(2, 0), (2, 1)}	{(2, 1)}	{(2, 2), (1, 2)}
	2	{(1, 0), (1, 1)}	{(1, 0), (1, 1), (1, 2)}	{(1, 1), (1, 2)}
	3	{(0, 0)}	{(0, 1)}	{(0, 2)}

Chapter 11

The Design Problem with SV policies

AC and RSS

In the previous chapter we formulated the Design Problem for the case in which the desired policy is point-valued (PV). Now we will explain the case in which the policy is set-valued (SV). When a PV policy is adopted, there is no uncertainty about the control that will be adopted: if the state is $\mathbf{x}_t$, the control is $m_t(\mathbf{x}_t)$. When the policy is set-valued, however, if the state is $\mathbf{x}_t$ the control can be any of those in the set $M_t(\mathbf{x}_t)$: it will be up to the Regulator to decide at time t which one to implement. We have to take account of this uncertainty when we define the Design Problem.

For this reason, and for other reasons that will become clear later, it is advisable not to describe the system with a mechanistic or empirical model, as we have done until now, but with a Markov chain instead. We anticipated the particular nature of this model in Section 4.7, but the notions there provided were only intuitive and not formalized. Therefore, in the first section of this chapter we will take up the subject again, to then use the notions that we have learned in the following parts of the chapter with the aim of formalizing the SV Design Problem. The algorithm that solves this Problem will be presented in Section 12.3, once the algorithms that solve the Design Problem for a PV policy have been explained.

11.1 Markov chains

The Markov chain was devised for the case of stochastic disturbances and so we will begin with it; then we will extend the idea to the case of uncertain disturbances.

11.1.1 Stochastic disturbances

When a system is stochastic, i.e. it only takes stochastic disturbances as inputs, its state is a stochastic variable and, as such, it is described by a probability distribution. If its model is an automaton with $N_{\mathbf{x}}$ states, the probability distribution of the state $\mathbf{x}_t$ at time t is described by a finite vector $\boldsymbol{\pi}_t$, whose components represent in order the probability that $\mathbf{x}_t$ assumes the first, the second, ..., the $N_{\mathbf{x}}$th of its possible values. Once the probability distribution $\boldsymbol{\pi}_t$ is known, it is easy to understand that the probability distribution at time $t+1$ is univocally defined by the following transition equation[1]

$$\boldsymbol{\pi}_{t+1}^T = \boldsymbol{\pi}_t^T B_t\big(m_t(\cdot)\big) \tag{11.1}$$

where $B_t(m_t(\cdot))$ is a matrix whose element $b_t^{ij}(m_t(\mathbf{x}_t^i))$ represents the probability that, having adopted the control $m_t(\mathbf{x}_t^i)$, the system will transit from state[2] $\mathbf{x}_t^i$, at time t, to state $\mathbf{x}_{t+1}^j$, at time $t+1$. Given that the elements of the ith row of B_t depend on the control $m_t(\mathbf{x}_t^i)$, and that the control changes as the row changes, the matrix B_t in its entirety is the function of the whole control law $m_t(\cdot)$. This dependence is highlighted in the form of equation (11.1). A system that is driven by such an equation is termed *Markov chain*. Matrix B_t can easily be computed, once the state transition equation (4.5a) of the system and the probability distribution $\phi_t(\cdot)$ of the disturbance $\boldsymbol{\varepsilon}_{t+1}$, which influences it, are known

$$b_t^{ij}\big(m_t\big(\mathbf{x}_t^i\big)\big) = \sum_{G_t^{ij}} \phi_t(\boldsymbol{\varepsilon}_{t+1}) \tag{11.2a}$$

where

$$G_t^{ij} = \big\{\boldsymbol{\varepsilon}_{t+1}\colon \mathbf{x}_{t+1}^j = f_t\big(\mathbf{x}_t^i, \mathbf{u}^p, m_t\big(\mathbf{x}_t^i\big), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\big\} \tag{11.2b}$$

is the set of disturbances that produce the transition from $\mathbf{x}_t^i$ to $\mathbf{x}_{t+1}^j$ when the control is $m_t(\mathbf{x}_t^i)$. Note that the properties of $\phi_t(\cdot)$ and the form of equation (11.2) guarantee that the elements of B_t are all non-negative and that their sum per line is equal to one, as must be, given that by beginning from any of the states one is sure to reach another. These two properties are enough to guarantee that, if the elements of the vector $\boldsymbol{\pi}_t$ are all non-negative and their sum is equal to one (as must be, given that $\boldsymbol{\pi}_t$ is a vector of probability), equation (11.1) provides a value of $\boldsymbol{\pi}_{t+1}$ which has the same properties.

Finally, it is interesting to observe that equation (11.1) defines a deterministic nonlinear system (with respect to the control), whose state is $\boldsymbol{\pi}_t$ and whose control is the vector $m_t(\cdot) = |m_t(\mathbf{x}_t^1), \dots, m_t(\mathbf{x}_t^{N_\mathbf{x}})|$. Since the system (11.1) is completely equivalent to the system (4.5a), it follows that a stochastic system with a state $\mathbf{x}_t$ can be described as a deterministic system with a state $\boldsymbol{\pi}_t$, i.e. as a system whose state is the probability distribution of the original state.

11.1.2 Uncertain disturbances

When a system is uncertain, i.e. when it takes uncertain disturbances as inputs, its state is an uncertain variable and, as such, it is described by the set of values that it can assume (this type of description is called *set-membership*). Such a set is termed *set of the reachable states*, or in short, *reachable set*. When the model of the system is an automaton with $N_\mathbf{x}$ states, the reachable set at time t can be described by a Boolean vector $\boldsymbol{\chi}_t$ with finite dimensions, whose ith component is 0, if the state value $\mathbf{x}_t^i$ cannot occur at time t, and 1 in the opposite case. Once the set $\boldsymbol{\chi}_t$ is known, the set of reachable states at time $t+1$ is univocally defined by the following transition equation

$$\boldsymbol{\chi}_{t+1}^T = \boldsymbol{\chi}_t^T \odot W_t\big(m_t(\cdot)\big) \tag{11.3}$$

[1] Remember that the superscript T denotes the transposition of the vector or the matrix to which it is applied.

[2] N.B.: $\mathbf{x}_t^i$ denotes the ith value that the vector $\mathbf{x}_t$ can assume, not its ith component.

where $W_t(m_t(\cdot))$ is a matrix whose element $w_t^{ij}(m_t(\mathbf{x}_t^i))$ is 1, if there is a disturbance that, in correspondence with the control $m_t(\mathbf{x}_t^i)$, transfers the system from state $\mathbf{x}_t^i$ to state $\mathbf{x}_{t+1}^j$; while it is 0 in the opposite case, i.e.

$$w_t^{ij}\big(m_t\big(\mathbf{x}_t^i\big)\big) = \begin{cases} 1 & \text{if } \exists \boldsymbol{\varepsilon}_{t+1} \in \Xi_t\text{: } \mathbf{x}_{t+1}^j = f_t(\mathbf{x}_t^i, \mathbf{u}^p, m_t(\mathbf{x}_t^i), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) \\ 0 & \text{otherwise} \end{cases} \tag{11.4}$$

The operator $\odot$ operates between the vector line $\boldsymbol{\chi}_t^T$ and the matrix W_t in such a way that the jth element of the resulting vector line $\boldsymbol{\chi}_{t+1}^T$ is given by

$$\chi_{t+1}^j = \max_i\big[\chi_t^i w_t^{ij}\big(m_t\big(\mathbf{x}_t^i\big)\big)\big]$$

The operation is thus formally similar to a row by column multiplication, except that, instead of summing the partial products, the maximum is considered.

Equations (11.3) and (11.4) express the obvious fact that the state $\mathbf{x}_{t+1}^j$ is reachable at time $t+1$ (and so the corresponding value of χ_{t+1}^j is 1) if and only if there is at least a state $\bar{\mathbf{x}}_t^i$ reachable at time t and at least a disturbance $\bar{\boldsymbol{\varepsilon}}_{t+1}$ such that, in correspondence with the control $m_t(\bar{\mathbf{x}}_t^i)$ established by the control law and the deterministic control $\mathbf{w}_t$, it follows that $\mathbf{x}_{t+1}^j = f_t(\bar{\mathbf{x}}_t, m_t(\bar{\mathbf{x}}_t^i), \mathbf{w}_t, \bar{\boldsymbol{\varepsilon}}_{t+1})$.

It is interesting to note the strict analogy between what we have just defined and what we saw in the preceding section: more precisely between the vector of probability $\boldsymbol{\pi}_t$ and the vector $\boldsymbol{\chi}_t$, between the matrices B_t and W_t, and between equations (11.1) and (11.3). For each state the vector $\boldsymbol{\pi}_t$ provides the probability that it occurs at time t; similarly, the ones and the zeros of the vector $\boldsymbol{\chi}_t$ tell us if it is or is not reachable. The same information can be deduced also from $\boldsymbol{\pi}_t$, by observing whether the probability associated to the state is different from zero or not. The information from $\boldsymbol{\pi}_t$ is greater, however, as it is a feature of stochastic description, since it reveals not only which states can occur, but also with what probability this happens.

In a similar way, the unitary elements of the matrix W_t show us which transitions are possible, while its analogue B_t provides not only this information (through its non-zero elements), but also the probability of every transition. Finally, equation (11.1) computes the jth element of the distribution as the sum of the products of the probabilities of the transitions $\mathbf{x}_t^i \to \mathbf{x}_{t+1}^j$ multiplied by the probability of the system being in state $\mathbf{x}_t^i$ at time t, and does so for all the states $\mathbf{x}_t^i$. Similarly, equation (11.3) defines the jth element of the vector $\boldsymbol{\chi}_{t+1}$, by observing whether there is a transition that terminates in that element, beginning from a state $\mathbf{x}_t^i$ that is reachable at time t.

The reader should observe how equation (11.3) can be formally obtained from equation (11.1) by applying the Law of Duality presented in Section 10.4.

As for stochastic systems, equation (11.3) defines a deterministic system, which is nonlinear (with respect to the control), whose state is $\boldsymbol{\chi}_t$ and whose control is the control law $m_t(\cdot)$. Once again uncertainty vanishes by considering, instead of the state $\mathbf{x}_t$, the vector $\boldsymbol{\chi}_t$ that describes the set of the reachable states.

11.2 The Design Problem with SV policies

11.2.1 From the PV to the SV Design Problem

11.2.1.1 An alternative representation of the objective

In order to introduce the Design Problem for an SV policy, it is above all necessary to rewrite the objective of the SV Design Problem in a form that will prove useful to our aim. For this reason note that in the previous chapters the system was always described by its 'natural' state variable $\mathbf{x}_t$, whose dynamics is controlled by the state transition function

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \tag{11.5}$$

When the disturbance is stochastic, we saw (Section 11.1.1) that it is spontaneous to associate a probability density $\boldsymbol{\pi}_{t+1}$, whose dynamics is described by equation (11.1), to the state $\mathbf{x}_{t+1}$.

Consider now the objective[3] of problem (10.26) for the case in which the indicator i is separable, i.e. it is expressed by equation (10.10)

$$J\big(\mathbf{x}_0, \mathbf{u}^p, p, \mathbf{w}_0^{h-1}\big) = \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E}\left[\sum_{t=0}^{h-1} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) + g_h\big(\mathbf{x}_h, \mathbf{u}^p\big)\right] \tag{11.6}$$

By using the probability $\boldsymbol{\pi}_t$ this objective can be expressed in a more attractive form. To obtain it observe, above all, that thanks to the operator E the objective can be rewritten as

$$\begin{aligned} J\big(\mathbf{x}_0, \mathbf{u}^p, p, \mathbf{w}_0^{h-1}\big) = \sum_{t=0}^{h-1} \underset{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=1,\dots,t+1}}{E}\Big[g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\Big] + \\ + \underset{\{\varepsilon_t\}_{t=1,\dots,h}}{E}\Big[g_h\big(\mathbf{x}_h, \mathbf{u}^p\big)\Big] \end{aligned} \tag{11.7}$$

The sequence of disturbances $[\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_{t+1}]$, which are arguments of the operator E, can be divided into two subsequences $[\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_t]$ and $[\boldsymbol{\varepsilon}_{t+1}]$. Given the initial state $\mathbf{x}_0$ and a feasible point-valued policy p, the first subsequence determines, at time t, a probability density $\boldsymbol{\pi}_t$. Therefore, in equation (11.7) the expected value with respect to $[\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_t]$ can be substituted by the expected value with respect to $\mathbf{x}_t$. By so doing, we obtain the following, new form of the objective

$$\begin{aligned} J\big(\mathbf{x}_0, \mathbf{u}^p, p, \mathbf{w}_0^{h-1}\big) = \sum_{t=0}^{h-1}\Big[\underset{\mathbf{x}_t \backsim \boldsymbol{\pi}_t}{E}\ \underset{\boldsymbol{\varepsilon}_{t+1} \backsim \phi_t}{E}\Big[g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\Big]\Big] + \\ + \underset{\mathbf{x}_h \backsim \boldsymbol{\pi}_h}{E}\Big[g_h\big(\mathbf{x}_h, \mathbf{u}^p\big)\Big] \end{aligned} \tag{11.8}$$

11.2.1.2 The system with an SV policy

In Section 10.1.1 an SV policy

$$P = \big\{M_t(\cdot);\ t = 0, 1, \dots\big\} \tag{11.9}$$

[3] The adoption of the Laplace criterion and of a finite time horizon is only illustrative: what will be shown in the following can easily be extended to the other forms of the objective, and also to the Wald criterion. This last extension is obtained by applying the Law of Duality presented in Section 10.4.

was defined as a finite or infinite sequence of control laws

$$M_t(\cdot) : \mathbf{x}_t \to \{\text{set of all the subsets of } \mathcal{S}_{\mathbf{u}_t}\}$$

each of which can be thought of not only as a set of controls that depend on $\mathbf{x}_t$, but also as a set of PV control laws, i.e.

$$M_t(\cdot) \triangleq \{m_t(\cdot)\} \tag{11.10}$$

Adopting an SV policy implies that at time t the Regulator chooses the control $\mathbf{u}_t$ from the set $M_t(\mathbf{x}_t)$ only on the basis of his own judgement. The system is thus affected by another source of uncertainty, since we do not know how the Regulator will make his choice. This uncertainty in the control $\mathbf{u}_t$ can be expressed by considering the control as an uncertain disturbance, namely by describing it with the *set-membership* relation $\mathbf{u}_t \in M_t(\mathbf{x}_t)$. It follows that, unlike what happens with a PV policy, given the probability $\boldsymbol{\pi}_t$, the probability $\boldsymbol{\pi}_{t+1}$ is no longer univocally determined. In other words, when an SV control law is adopted, in system (11.1) the control $\mathbf{u}_t$ plays a role analogous to the one that the disturbance $\boldsymbol{\varepsilon}_{t+1}$ plays in system (11.5). In Section 11.1.2 it was seen that when there is a set-membership disturbance, the state $\mathbf{x}_{t+1}$ belongs to a set $\boldsymbol{\chi}_{t+1}$ of reachable states. Similarly, when there is an SV policy, the probability that $\boldsymbol{\pi}_{t+1}$ will belong to a set $\varPi_{t+1}$ of probabilities (set of reachable probabilities) is given by

$$\varPi_{t+1} = \left\{\boldsymbol{\pi}_{t+1}\colon \exists \boldsymbol{\pi}_t \in \varPi_t,\ m_t(\cdot) \in M_t(\cdot)\colon \boldsymbol{\pi}_{t+1}^T = \boldsymbol{\pi}_t^T B_t\big(m_t(\cdot)\big)\right\} \tag{11.11}$$

where $\varPi_t$ is the set of the probability distributions of the state that can occur at time t.

Generally, at the initial time instant, the state $\mathbf{x}_0$ is known, so that the initial probability $\boldsymbol{\pi}_0$ is a vector with all of its elements equal to zero, except for the element that corresponds to the state $\mathbf{x}_0$ which is equal to one.

11.2.1.3 The objective of the SV Design Problem

When the design objective (11.8) depends on an SV policy, its structure must be suitably modified to take the nature of the policy. Two observations are relevant to this aim:

- Since the proposed control is not unique and the criterion adopted by the Regulator for choosing among the controls of $M_t(\mathbf{x}_t)$ is not known, it is advisable that the objective considers the worst case, i.e. the one in which the control $\mathbf{u}_t$ that produces the highest expected cost is chosen. As a consequence, *all the controls* $\mathbf{u}_t$ *that belong to an optimal policy provide the same value of the objective*[4]; in other words the optimal value of the objective does not depend on the choice of the Regulator.

- As we have already observed, when the policy is SV, the probability $\boldsymbol{\pi}_t$ of state $\mathbf{x}_t$ at time t is not univocally determined. We know only that it will belong to a set $\varPi_t$. As a consequence, in equation (11.8), in order to evaluate the worst case, the operator E with respect to $\mathbf{x}_t$ will have to be preceded by operator max with respect to $\boldsymbol{\pi}_t \in \varPi_t$. Clearly, a similar process of reasoning is possible also for the penalty.

[4] The formal proof of this statement requires a complex mathematical apparatus, but is easy to understand why this is true. Assume by absurd that in an optimal policy there exists a control law $M_t^*(\mathbf{x}_t)$, which contains a control $\bar{u}_t$ whose choice could produce, in particular circumstances, a cost $\bar{J}$ greater than the one that would result from adopting one of the other controls. Since the cost associated to the policy is the greatest expected cost with respect to the controls, $\bar{J}$ will also be the cost associated to the policy. Therefore, if $\bar{u}_t$ is excluded from $M_t^*(\mathbf{x}_t)$ the performance improves. But this contradicts the hypothesis that the policy is optimal.

On the basis of these observations, given an SV policy P, it is opportune to define the objective with which to evaluate the policy performance as follows

$$L\left(\mathbf{x}_0, \mathbf{u}^p, P, \mathbf{w}_0^{h-1}\right) = \sum_{t=0}^{h-1}\Big[\max_{\boldsymbol{\pi}_t \in \Pi_t} \mathop{E}_{\mathbf{x}_t \backsim \boldsymbol{\pi}_t} \max_{\mathbf{u}_t \in M_t(\mathbf{x}_t)} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1} \backsim \phi_t}\left[g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right]\Big] + \\ + \max_{\boldsymbol{\pi}_h \in \Pi_h} \mathop{E}_{\mathbf{x}_h \backsim \boldsymbol{\pi}_h}\left[g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right] \tag{11.12}$$

The extension of this definition to the other forms of objective presented in Section 10.2.3 is immediate.

Given an SV policy, it is possible to derive another SV policy from it by simply eliminating an element from one of the sets $M_t(\mathbf{x}_t)$ that define it. In a certain sense, we can say that the policy so obtained is smaller than the first. In our research we are clearly interested in finding 'the largest' optimal policy and so it is appropriate to define this notion precisely.

Let us say that the feasible policy

$$\bar{P} \triangleq \left\{\bar{M}_t(\cdot);\ t = 0, \dots, h-1\right\}$$

is *larger* than the feasible policy

$$\tilde{P} \triangleq \left\{\tilde{M}_t(\cdot);\ t = 0, \dots, h-1\right\}$$

if the following relation holds

$$\bar{M}_t(\mathbf{x}_t) \supseteq \tilde{M}_t(\mathbf{x}_t) \quad \forall t \text{ and } \forall \mathbf{x}_t \in \boldsymbol{\chi}_t(\tilde{P})$$

and there is at least one pair $(t, \mathbf{x}_t)$ for which the relation is a strict inequality. The script $\boldsymbol{\chi}_t(\tilde{P})$ denotes the fact that the comparison must be carried out only for those states that are reachable with policy $\tilde{P}$.

We will denote that $\bar{P}$ is larger than $\tilde{P}$ with the following notation

$$\bar{P} \supset \tilde{P}$$

11.2.2 Formulating the SV Design Problem

We are now finally able to formulate the Design Problem for an SV policy. We will do this only for the case in which the Laplace criterion is used, since the formulation with the Wald criterion can easily be obtained from the first, by applying the Law of Duality presented in Section 10.4.

The SV Design Problem: *To determine*[5] *a vector* $\mathbf{u}^{p*}$ *and the largest policy* $\bar{P}^*$ *that are optimal. That is*

$$L^* = \min_{\mathbf{u}^p, P} L\left(\mathbf{x}_0, \mathbf{u}^p, P, \mathbf{w}_0^{h-1}\right) \tag{11.13a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \qquad t = 0, \dots, h-1 \tag{11.13b}$$

[5] If the optimal pair $(\mathbf{u}^{p^*}, \bar{P}^*)$ is not unique, it might be interesting for the DM to consider all the pairs, so that she can choose the pair that best satisfies any criteria that were not expressed in the formulation of the Problem. We will not consider this Problem, however, because it goes beyond the limits posed to this work.

$$\mathbf{u}^p \in \mathcal{U}^p \tag{11.13c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq \mathcal{U}_t\left(\mathbf{x}_t, \mathbf{u}^p\right) \qquad t = 0, \dots, h-1 \tag{11.13d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{u}^p\right) \qquad t = 0, \dots, h-1 \tag{11.13e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{11.13f}$$

$$\mathbf{x}_0 \textit{ given} \tag{11.13g}$$

$$P \triangleq \left\{M_t(\cdot);\ t = 0, \dots, h-1\right\} \tag{11.13h}$$

$$\textit{any other constraints} \qquad t = 0, \dots, h-1 \tag{11.13i}$$

11.2.2.1 Some properties

In Section 10.1.1 we saw that an SV policy can be defined in two equivalent ways:

- as a succession of SV control laws: $P \triangleq \{M_t(\cdot);\ t = 0, 1, \dots\}$;
- as a set of SV policies: $P \triangleq \{p\}$.

So far we have used the first definition, as shown by the maximum with respect to $\mathbf{u}_t \in M_t(\mathbf{x}_t)$ which appears in the definition (11.12) of the Problem objective. One might ask what would happen if, instead, the objective were defined based on the second definition of the SV policy; more precisely, if, given an SV policy $P \triangleq \{p\}$, we were to evaluate its performance with the following objective

$$\tilde{L}\left(\mathbf{x}_0, \mathbf{u}^p, P, \mathbf{w}_0^{h-1}\right) = \max_{p \in P}\left[J\left(\mathbf{x}_0, \mathbf{u}^p, p, \mathbf{w}_0^{h-1}\right)\right] \tag{11.14}$$

where $J(\cdot)$ is defined by equation (11.6).

A question arises: if $L(\cdot)$ is substituted with $\tilde{L}(\cdot)$ in problem (11.13), has the new problem the same solution as the original one? If it were not so, we would be in difficulty, since the two alternative definitions of the SV policy seem to be absolutely equivalent. Fortunately for us, it is possible to prove (Aufiero et al., 2001) that the following relation

$$L\left(\mathbf{x}_0, \mathbf{u}^p, P, \mathbf{w}_0^{h-1}\right) = \tilde{L}\left(\mathbf{x}_0, \mathbf{u}^p, P, \mathbf{w}_0^{h-1}\right)$$

holds for every feasible policy P. From this relation, given equation (11.14), it follows that

$$L\left(\mathbf{x}_0, \mathbf{u}^p, P, \mathbf{w}_0^{h-1}\right) = \max_{p \in P}\left[J\left(\mathbf{x}_0, \mathbf{u}^p, p, \mathbf{w}_0^{h-1}\right)\right] \tag{11.15}$$

This last identity is particularly interesting, not only because it clears our doubts, but, above all, because it links the performance of an SV policy to the performances of the PV policies that constitute it: it states in fact, that the performance of the SV policy corresponds to the worst performance provided by the PV policies contained in the SV policy. This is coherent with the approach adopted in the definition of the objective: to consider the worst case with respect to the uncertainty introduced by the decisions of the Regulator (see equations (11.12) and (11.14)).

Note now that identity (11.15) clearly holds even when the policy P is optimal. In this case the PV policies contained in P in turn must be optimal policies for the PV Design Problem, so that in correspondence with them the functional $J(\mathbf{x}_0, \mathbf{u}^p, p, \mathbf{w}_0^{h-1})$ on the right-hand side of equation (11.15) takes the minimum possible value, i.e. the value J^* provided by the following problem associated to problem (11.13).

The PV Design Problem associated to the SV Problem:

$$J^* = \min_{\mathbf{u}^p, p} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\sum_{t=0}^{h-1} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) + g_h\big(\mathbf{x}_h, \mathbf{u}^p\big) \right] \tag{11.16a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \qquad t = 0, \dots, h-1 \tag{11.16b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{11.16c}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \qquad t = 0, \dots, h-1 \tag{11.16d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\big(\cdot|\mathbf{u}^p\big) \qquad t = 0, \dots, h-1 \tag{11.16e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{11.16f}$$

$$\mathbf{x}_0 \textit{ given} \tag{11.16g}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \dots, h-1\big\} \tag{11.16h}$$

$$\textit{any other constraints} \qquad t = 0, \dots, h-1 \tag{11.16i}$$

which is none other than the PV Design Problem defined by (10.26).

The following proposition (Aufiero et al., 2001) confirms that our guess is correct:

Proposition: *The optimal value L^* of the objective of the* SV *Design Problem* (11.13) *is equal to the optimal value J^* of the corresponding* PV *Design Problem* (11.16)*, i.e.*

$$L^* = J\big(p^*, \mathbf{u}^{p*}\big)$$

From this it follows that each optimal SV policy is constituted by a set of optimal PV policies and so *the largest optimal* SV *policy is constituted by the totality of the optimal* PV *policies*. In Section 12.3 we will begin right from this last proposition to introduce the algorithm that solves problem (11.13).

11.3 Cascade criteria

Until this point we have assumed that just one criterion is adopted, but in some cases this may be disadvantageous. To understand why, consider a case in which the Stakeholders could be seriously damaged even if only one negative event were to occur (e.g. a flood). In that case it makes sense to adopt the Wald criterion, but by doing so all but the worst events are considered as equivalent. For example, suppose we must design a regulation policy for a lake, using the Wald criterion, with the aim of mitigating the floods along its shores, and the indicator we adopt is the level of the lake. The optimal policy will be the one that minimizes the maximum level that the lake reaches over the design horizon, i.e. minimize the peak of the biggest flood. Let h^* be the optimal value of the objective, i.e. the value of the minimum maximum peak, and let $M_t^{w*}(\mathbf{x}_t)$ be the optimal policy. All the single-objective policies contained in $M_t^{w*}(\mathbf{x}_t)$ are considered to be equivalent, both those that keep the lake almost always at levels lower than the flood threshold and those that keep it nearly always at levels higher than the threshold: in fact, none of these policies produce flood peaks higher than h^*

(Orlovski et al., 1984). Clearly, the Stakeholders would judge the first to be better than the second, but the Wald criterion alone does not make it possible to distinguish them.

One possible solution is the approach proposed by Nardini et al. (1992), in which the Wald criterion and the Laplace criterion are used in sequence. More precisely, first of all an SVP Design Problem is formulated with the Wald criterion and is solved by obtaining the optimal policy $M_t^{w\hbar*}(\mathbf{x}_t)$. Then the problem is reformulated with the Laplace criterion (see (11.13)), by posing the following constraint in place of (11.13d)

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq M_t^{w*}(\mathbf{x}_t) \quad t = 0, \ldots, h-1$$

i.e. by imposing that the sought after policy must be completely contained in the optimal policy obtained with the Wald criterion. This approach expresses a marked risk aversion on the part of the DM, given that the criteria are arranged in sequence and the first is the Wald criterion. At the same time, however, it searches for the best policy, from the point of view of the Laplace criterion, among all the risk averse policies.

Obviously, what is said for the Wald criterion can be repeated for the Laplace criterion: of all the policies that minimize the expected value of the objective there are only a few that minimize its maximum value and these can be singled out by sequencing a Wald problem to a Laplace one.

Chapter 12

Off-line non-learning-based policies

AC and RSS

In the previous two chapters we defined the Design Problem, both for point-valued (PV) and for set-valued (SV) policies. In the present and the following two chapters we will deal with the solution of those problems and so we will illustrate the appropriate algorithms for this purpose. More precisely, the subject of this chapter is the off-line design of non-learning-based policies[1]; in the next one we will consider the design of learning-based policies, while the on-line design will be discussed in Chapter 14. We will first study the off-line design of PV policies, both with the Functional and the Parametric Approach; for the SV policies we will only examine the Functional Approach, as well-tested algorithms are available only for that. For each case we will consider whether or not it is possible to adopt a *model free* approach. In conclusion the materials are organized as follows

		APPROACH	LEARNING-BASED	
			NO	YES
Off-line design	PV policies	Functional	Section 12.1	Chapter 13
		Parametric	Section 12.2	
	SV policies	Functional	Section 12.3	
On-line design	PV policies	Functional	Chapter 14	

As one can see, we will not consider all the possible combinations: the reason is not only that the presentation of the omitted parts would require much more space than is possible, if we want to maintain a good balance among the different parts of the book, but also that the omitted subjects are distinctly less interesting, both from a theoretical point of view and for their practical utility.

In Section 10.4 we provided a Law of Duality by means of which it possible to transform the propositions, algorithms and formulae for Problems defined with the Laplace criterion into the corresponding propositions, algorithms and formulae for Problems defined with the

[1] The terms used in this introduction are defined in Sections 10.1 and 10.3.2.

Wald criterion. This is why, with no loss of generality, in this and in the following chapter we will mainly refer to Problems formulated with the first criterion. Then, remembering what was presented in Section 10.2.3, we will assume that the indicator to which the Laplace criterion is applied has a separable and integral form: the operator for temporal aggregation is thus the sum. Finally, we saw in Section 10.3.1 that the solution of the Design Problem in the general case can be traced back to the solution of a Control Problem (see problem (10.29)) encapsulated in a Pure Planning Problem (see problem (10.28)). Since the algorithms for the solution of the latter were the subject of Chapters 8 and 9, we must only deal with the algorithms that solve the first.

In conclusion: the off-line *Control Problem* that we consider in this chapter is an *Optimal Control Problem*, the objective of which is defined by applying the Laplace criterion to an indicator with a separable and integral form, and the initial state $\bar{\mathbf{x}}_0$ is given. Moreover, as already observed in Section 10.1.2, the system dynamics cannot depend on deterministic disturbances. If a finite time horizon is adopted, the Control Problem will thus have the following form

The Off-line Control Problem:

$$J^* = \min_{p} \mathop{E}_{\{\varepsilon_t\}_{t=1,\dots,h}} \left[\sum_{t=0}^{h-1} g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}\big) + g_h\big(\mathbf{x}_h, \mathbf{u}^p\big) \right] \tag{12.1a}$$

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}\big) \qquad t = 0, \dots, h-1 \tag{12.1b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \qquad t = 0, \dots, h-1 \tag{12.1c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, \dots, h-1 \tag{12.1d}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{12.1e}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \dots, h-1\big\} \tag{12.1f}$$

$$\textit{any other constraints} \qquad t = 0, \dots, h-1 \tag{12.1g}$$

Since, from our current viewpoint, $\mathbf{u}^p$ can be treated as an externally given parameter, it will be omitted in all the formulae in this and in the following chapter. In problem (12.1) the finite time horizon can be substituted by a receding or infinite horizon; in the second case two different formulations, TDC and AEV, are available (see Section 8.1.2.1).

As it has been decided to design the policy off-line, logic demands that the deterministic disturbances should not be considered, since they will only be known at instant t and a model which describes their dynamics is not available. Therefore, the formulation (12.1) is the only one which is rigorously coherent with the intention to design the policy off-line. This implies, however, as the (12.1c) has already stated, that the resulting control laws will have $\mathbf{x}_t$ as argument. However, we are not satisfied since we intuit that if at instant t the value of the deterministic disturbance $\mathbf{w}_t$ is known, ignoring it and deciding only on the basis of the state $\mathbf{x}_t$ is very likely to lead to suboptimal results. We would therefore like to have control laws which have as argument the pair $(\mathbf{x}_t, \mathbf{w}_t)$. To obtain them, the formally correct road is to give up the off-line design and postpone the definition of the policy to instant t, i.e. design the policy on line (as we will see in Chapter 14). As we do not wish to give up the off-line design the only possibility is to use a scenario $\mathbf{w}_0^{h-1}$ of the deterministic disturbance, almost always the historical series, to train a policy which permits us to decide also on the basis of $\mathbf{w}_t$. As we will see, the algorithms that permit this training are equivalent to assuming simple modelling hypotheses on the nature of the disturbance. In fact, therefore, not all the off-line policies are designed by solving problem (12.1), but from what has been said one understands why it is the formulation from which it is best to begin the presentation of the algorithms for the off-line design.

12.1 PV policies: Functional Design

The Functional Approach can be adopted only when the system (12.1b) is an automaton, i.e. a system in which, at each time instant, the state can only assume a finite number of values. When the system, as usually happens, does not have this property, one must consider an automaton that approximates it, as explained in Sections 10.3.2 and 10.5.

The following definition will be very useful later on: the constraint system of a Control Problem is called *separable*, if it can be partitioned into sets of constraints, each of which contains only variables relative to one single time interval $[t, t+1)$, with the exception, at most, of the constraints that express pure definitions, such as, for example, constraints (12.1e)–(12.1f). Given the structure of the model and the hypothesis that the random disturbance $\boldsymbol{\varepsilon}_{t+1}$ is generated by a white random process (as reflected by equation (12.1d)), the constraint system of problem (12.1) is separable if and only if the 'any other constraints' are separable.

The best algorithms nowadays available for the solution of this class of problem, and the only ones that we will consider, *can be applied only to Problems in which the water system is an automaton with a separable constraint system* and, clearly, without deterministic disturbances. From this last condition it follows that the state $\mathbf{x}_t$ is the only argument of the policy, i.e. the sufficient statistic.

12.1.1 The optimal cost-to-go

Depending on the time horizon adopted, different algorithms are available. Before introducing them formally, we will try to guess what might be their common characteristic. The standard Control Problem is a multi-stage (or sequential) decision-making process,[2] in which, at each stage, given the state $\mathbf{x}_t$, the control $\mathbf{u}_t$ must be chosen, as illustrated in Figure 12.1. The consequences of the choice are not deterministically known, but can be

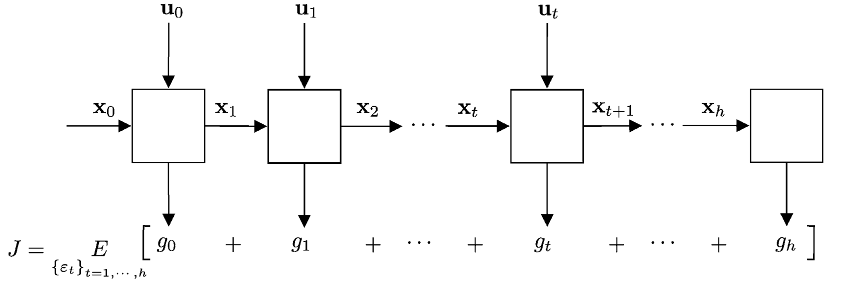


Figure 12.1: The multi-stage decision-making problem.

[2]The first to intuit the sequential nature of the decision-making process in the field of water resource management was Massé in 1946, in a study on the management of a hydropower reservoir (Massé, 1946).

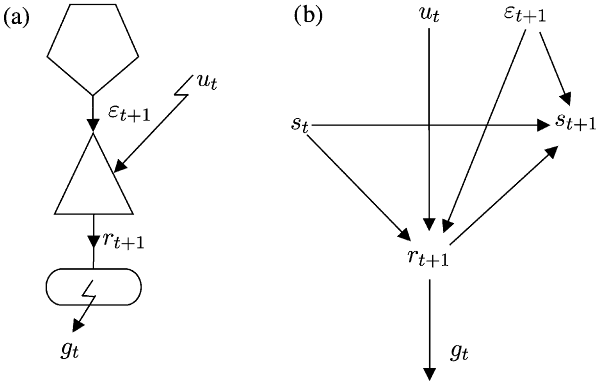


Figure 12.2: The block diagram (a) of the water system considered in the example and the corresponding causal network (b).

somehow anticipated before the next control must be chosen. Each control incurs an immediate cost $g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1})$, but also impacts, through the state $\mathbf{x}_{t+1}$ that it contributes to producing, the context in which the next control choice will be made and thus the effect that this latter will produce on all the future stages. Our aim is to identify a policy, i.e. a rule for choosing the control, that minimizes the total expected cost of all the stages, from the time the choice is made onwards: the key element is thus the interrelationship between immediate and future costs.

To better understand this point, consider again the simple example introduced in Section 2.1, in which we want to design a regulation policy for a reservoir that feeds an irrigation district. The system is described by the block diagram in Figure 12.2a, while its variables are interlinked as in the causal network of Figure 12.2b. At every time t we must explicitly establish the volume u_t to send to the irrigation district, but in doing so we implicitly establish also the volume s_{t+1} to 'send' to the next time instant. We can visualise this fact through the space–time plot in Figure 12.3. The volume u_t influences the crop stress in the interval $[t, t+1)$, which is measured by the cost[3] $g_t(s_t, u_t, \varepsilon_{t+1})$; while the storage s_{t+1} specifies the resource that will be available at time $t+1$ to satisfy the water demand in that step and in the following ones. This is the process described in general terms by the diagram in Figure 12.1, whose tth block contains the model of the system, as shown, for our example, in Figure 12.4.

When making the decision, i.e. when choosing the control, we must therefore weigh up the immediate cost $g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1})$ that will be incurred in the next transition, with the desirability of the state $\mathbf{x}_{t+1}$ that the system will reach: it is therefore necessary to find a way of measuring the latter. The most immediate one is to consider the total expected cost that one would incur in starting from $\mathbf{x}_{t+1}$ and adopting *optimal decisions in all the following stages*. We denote that cost with the symbol $H^*_{t+1}(\mathbf{x}_{t+1})$ and call it the *optimal cost-to-go*, omitting the attributes *total* and *expected* for sake of brevity.

If the optimal cost-to-go $H^*_{t+1}(\mathbf{x}_{t+1})$ were known for all the states $\mathbf{x}_{t+1}$, the optimal decision $m^*_t(\mathbf{x}_t)$ at time t could be identified by minimizing the expected value of the sum

[3] The cost g_t (crop stress) is an explicit function of r_{t+1}, which, in turn, is a function of s_t, u_t and ε_{t+1}. From here, the arguments of $g_t(\cdot)$.

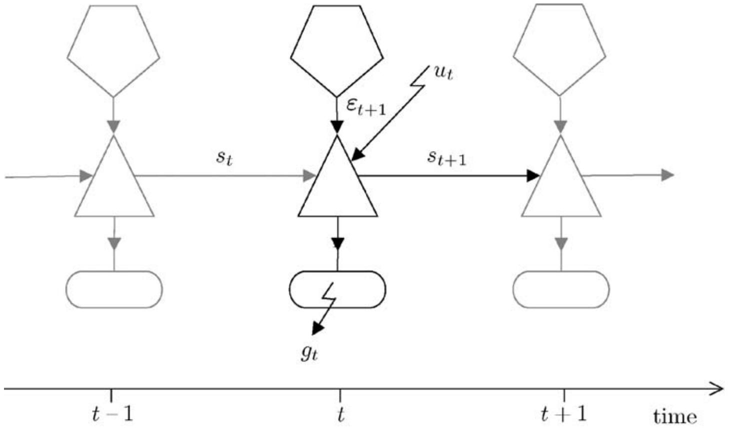


Figure 12.3: Space–time visualisation of the decision-making process in Figure 12.1 applied to the system in Figure 12.2.

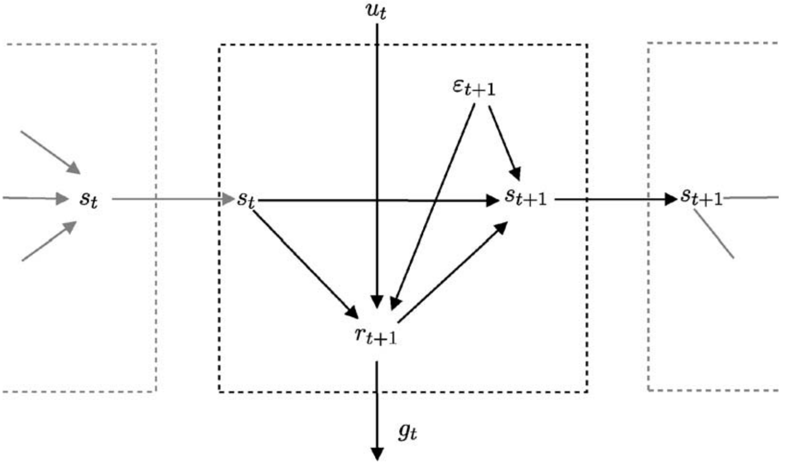


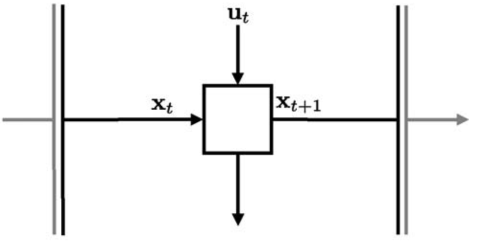
Figure 12.4: The causal network that describes the tth stage of the decision-making process in Figure 12.1 applied to the system in Figure 12.2.

of the immediate cost and the cost-to-go with respect to $\mathbf{u}_t$

$$\underset{\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot)}{E} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H^*_{t+1}(\mathbf{x}_{t+1}) \right]$$

We denote this fact with the following expression

$$m^*_t(\mathbf{x}_t) = \arg \min_{\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)} \underset{\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot)}{E} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H^*_{t+1}(\mathbf{x}_{t+1}) \right] \tag{12.2}$$



$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right]$$

Figure 12.5: The isolation of a stage in a multi-stage decision-making process through the optimal costs-to-go $H_t^*(\mathbf{x}_t)$ and $H_{t+1}^*(\mathbf{x}_{t+1})$.

where $\mathbf{x}_{t+1}$ is computed with equation (12.1b). As a consequence, the optimal cost-to-go $H_t^*(\mathbf{x}_t)$ associated to the current state will be given by the following equation

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot)} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right] \tag{12.3}$$

This equation, called *Bellman equation* (Bellman, 1957a, 1962), provides the optimal cost-to-go at time t, given the one at time $t+1$. It is, therefore, straightforward to think of using it in a recursive way, by proceeding backwards from the final stage (time instant) to the initial one. This is the core of Stochastic Dynamic Programming[4] (SDP): a family of algorithms that makes it possible to compute the optimal cost-to-go for every time t.

Equation (12.3) states that the multi-stage decision-making problem in Figure 12.1 can be solved by considering one stage at a time. Each stage can be isolated by cutting the two arcs that link it to the past and the future, as shown in Figure 12.5: cutting the first requires specifying the state $\mathbf{x}_t$, while cutting the second requires specifying the optimal cost-to-go $H_{t+1}^*(\mathbf{x}_{t+1})$. The decision u_t is then made with equation (12.2), because *the sequence of decisions over the entire horizon h can be optimal if and only if each decision is optimal for its own stage*. This statement is the so-called *Bellman's Principle of Optimality*.

The optimal policy is completely defined by equation (12.2) for every time t, when the function $H_\cdot^*(\cdot)$ is known. In turn, this function, the so-called *Bellman function*, is univocally defined by the optimal costs-to-go $H_t^*(\cdot)$ at each time t, which constitute its time slices. Therefore, *knowing the Bellman function is a sufficient condition for knowing the optimal policy*. Note also that the Bellman function makes it possible to avoid the anticipated, off-line computation of the policy for all the events $(t, \mathbf{x}_t)$, given that the optimal control $m_t^*(\mathbf{x}_t)$ can be computed *just in time* through equation (12.2). The Bellman function thus turns out

[4] In 1957 Bellman proposed the deterministic version of Dynamic Programming (DP) and only later, in 1962, did he devise the stochastic version (SDP). The approach immediately aroused interest in the field of the management of water resources, even if, for a long time, it was only experimented with in the deterministic version. We owe the first application of the DP to Hall and Buras (1961). Since then, the method has spread enormously and with success, mainly in the management of reservoirs for hydropower production (see, among others, Hall et al., 1968; Heidari et al., 1971; Fults and Hancock, 1972; Trott and Yeh, 1973; Turgeon, 1980). In 1989 DP was cited by Esogbue (Esogbue, 1989) as one of the techniques most used on the real management of reservoirs. Beginning in the early 1980s, interest also spread in the stochastic version (SDP) and its applications were extended to multi-purpose reservoirs and networks of reservoirs (see the reviews by Yakowitz (1982) and Yeh (1985), and the contributions by Gilbert and Shane (1982), Read (1989), Hooper et al. (1991), Vasiliadis and Karamouz (1994) and Tejada-Guibert et al. (1995)).

to be the parameterization that defines the optimal policy in the class of functions defined by equation (12.2).

Furthermore, note that equation (12.2) assumes that at time t the state $\mathbf{x}_t$ is known (*perfect state information*): if the state is not measurable, the policy can be designed, but it would be unusable.[5] The dependency of the policy on $\mathbf{x}_t$ is the reason why in Section 10.1 we stated that $\mathbf{x}_t$ is a sufficient statistic, in the sense that any additional information known at time t would not lead to an improvement of the objective.[6]

In a totally similar way, if the filtering criterion adopted in the Control Problem were the Wald criterion, the optimal control at time t would be provided by the following expression

$$m_t^*(\mathbf{x}_t) = \arg \min_{\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)} \max_{\boldsymbol{\varepsilon}_{t+1} \in \Xi_t} \Big[\max\big[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}),\ H_{t+1}^*(\mathbf{x}_{t+1})\big]\Big]$$

In fact, with the Wald criterion, the optimal decision must minimize the worst occurrence between the maximum immediate cost and the maximum cost-to-go with respect to the disturbance. As in the previous case, the optimal cost-to-go associated to the state $\mathbf{x}_t$ is provided by the Bellman equation, which now takes the form

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)} \max_{\boldsymbol{\varepsilon}_{t+1} \in \Xi_t} \Big[\max\big[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}), H_{t+1}^*(\mathbf{x}_{t+1})\big]\Big] \tag{12.4}$$

The reader should observe that, as we anticipated, the last two relationships can be derived from equations (12.2) and (12.3) by applying the Law of Duality introduced in Section 10.4.

12.1.2 Stochastic Dynamic Programming

In the previous section the Bellman equation was introduced intuitively. Now we would like to derive it in a formal way from the serial structure of problem (12.1).

On the basis of the definition of optimal cost-to-go, given the initial state $\bar{\mathbf{x}}_0$ and the planning decision $\mathbf{u}^p$, problem (12.1) can be rewritten as

$$H_0^*(\bar{\mathbf{x}}_0) = \min_p \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\sum_{t=0}^{h-1} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + g_h(\mathbf{x}_h)\right] \tag{12.5a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \qquad t = 0, \dots, h-1 \tag{12.5b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \qquad t = 0, \dots, h-1 \tag{12.5c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, \dots, h-1 \tag{12.5d}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \dots, h-1\big\} \tag{12.5e}$$

$$\text{any other constraints} \qquad t = 0, \dots, h-1 \tag{12.5f}$$

The objective (12.5a) can be rewritten by separating the cost relative to the time interval $[0, 1)$ from the total expected cost of all the subsequent intervals, which depend on the

[5] Heuristically, one could think of determining the state with a state estimator (see Appendix A4). However, only when the system is linear and the objective is quadratic (LQG framework, see 12.2.2), is this way of proceeding optimal. More precisely, in the LQG framework the optimal policy in the case of partial state information is obtained by coupling an optimal state estimator with the optimal policy designed by assuming perfect state information (*Separation Theorem for linear system and quadratic criteria*, see Bertsekas, 1976, page 133). Notwithstanding the optimality, the performance of such a controller will be inferior to the one that would be obtained in the case of perfect state information.

[6] If, instead, we knew the disturbance $\boldsymbol{\varepsilon}_{t+1}$, we could do better, but at time t the disturbance has not yet occurred and all that we know about it is the probability distribution $\phi(\cdot)$, which in fact is used in equation (12.2).

policy $p_{[1,h)}$ adopted

$$H_0^*(\bar{\mathbf{x}}_0) = \min_{\mathbf{u}_0} \Bigg\{ \mathop{E}_{\boldsymbol{\varepsilon}_1} \Bigg[g_0(\bar{\mathbf{x}}_0, \mathbf{u}_0, \boldsymbol{\varepsilon}_1) + \\ + \min_{p_{[1,h)}} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=2,\ldots,h}} \left(\sum_{t=1}^{h-1} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + g_h(\mathbf{x}_h) \right) \Bigg] \Bigg\} \tag{12.6}$$

Analogously, from the hypothesis that the constraint system (12.5b)–(12.5f) is separable, it follows that it can be partitioned in two sets: the set of constraints relative to the time interval $[0, 1)$ and the set of constraints relative to all the other intervals. As the probability distribution $\phi_t(\cdot)$ of the disturbance $\boldsymbol{\varepsilon}_{t+1}$ does not explicitly depend on the values $\boldsymbol{\varepsilon}_t, \boldsymbol{\varepsilon}_{t-1}, \ldots, \boldsymbol{\varepsilon}_0$ assumed in the previous time instants[7] (see page 208), this second set of constraints and the second term on the right-hand side of equation (12.6) define a Problem that is formally similar to problem (12.5), except for the fact that it concerns the horizon $[1, h]$, instead of $[0, h]$, and has $\mathbf{x}_1$ as initial state. Precisely

$$H_1^*(\mathbf{x}_1) = \min_{p_{[1,h)}} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=2,\ldots,h}} \left[\sum_{t=1}^{h-1} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + g_h(\mathbf{x}_h) \right] \tag{12.7a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \qquad t = 1, \ldots, h-1 \tag{12.7b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \qquad t = 1, \ldots, h-1 \tag{12.7c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 1, \ldots, h-1 \tag{12.7d}$$

$$p_{[1,h)} \triangleq \left\{ m_t(\cdot);\ t = 1, \ldots, h-1 \right\} \tag{12.7e}$$

$$\text{any other constraints} \qquad t = 1, \ldots, h-1 \tag{12.7f}$$

It follows that the optimal cost-to-go $H_0^*(\bar{\mathbf{x}}_0)$, defined by equation (12.5), is equivalently defined by the following expression

$$H_0^*(\bar{\mathbf{x}}_0) = \min_{\mathbf{u}_0} \mathop{E}_{\boldsymbol{\varepsilon}_1} \left[g_0(\bar{\mathbf{x}}_0, \mathbf{u}_0, \boldsymbol{\varepsilon}_1) + H_1^*(\mathbf{x}_1) \right]$$

$$\mathbf{x}_1 = f_0(\bar{\mathbf{x}}_0, \mathbf{u}_0, \boldsymbol{\varepsilon}_1)$$

$$\mathbf{u}_0 \in \mathcal{U}_0(\bar{\mathbf{x}}_0)$$

$$\boldsymbol{\varepsilon}_1 \sim \phi_0(\cdot)$$

$$\text{any other constraints relative to the interval } [0, 1)$$

The previous reasoning also applies to the computation of $H_1^*(\mathbf{x}_1)$, which is defined by equation (12.7), and subsequently for $H_2^*(\mathbf{x}_2)$, and so on. In general, the cost over the time horizon $[t, h]$ is therefore expressed by the following recursive equation

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right] \tag{12.8a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.8b}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \tag{12.8c}$$

[7] This condition is satisfied when the disturbance is a white process.

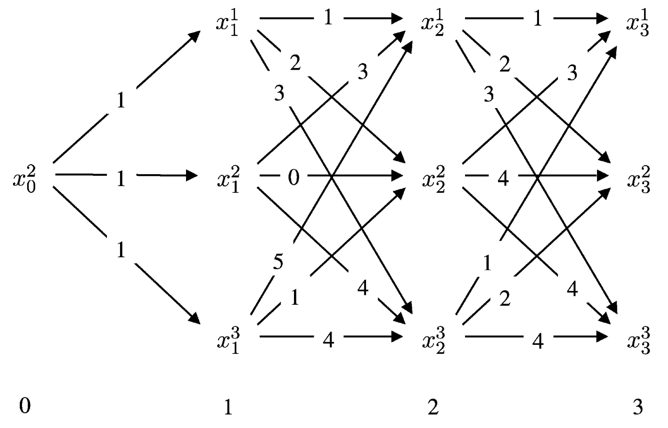


Figure 12.6: Three transitions of a system with three states starting from a given initial state (x_0^2).

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{12.8d}$$

$$\text{any other constraints relative to } [t, t+1) \tag{12.8e}$$

We note straightaway that equation (12.8) is the complete form of the *Bellman equation* introduced in the previous section in an intuitive way (see (12.3)). Therefore, we have not only tested it formally but also identified the constraints that it is subject to and the conditions for its validity: *the probability distribution $\phi_t(\cdot)$ of the disturbance $\boldsymbol{\varepsilon}_{t+1}$ must not depend explicitly on the values assumed by the disturbance in the previous time instants and the constraint system must be separable.*

From equation (12.8) it emerges that the optimal control $m_t^*(\mathbf{x}_t)$ is expressed, as we guessed it would be, by the following relation

$$m_t^*(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right] \tag{12.9}$$

subject to the constraints (12.8b)–(12.8e).

In conclusion, the necessary condition for problem (12.1)[8] to be solved with Stochastic Dynamic Programming is that the objective (12.1a) have a separable form, the constraint system (12.1b)–(12.1e) be separable and the probability distribution $\phi_t(\cdot)$ that describes the disturbance $\boldsymbol{\varepsilon}_{t+1}$ not depend explicitly on the values that the disturbance assumes in the previous time instants. When the disturbance is a white process this last condition is always satisfied.

12.1.3 Computational complexity

It is worthwhile to consider an example to demonstrate the potential and the strategy of SDP. Consider a deterministic system with three states, in each of which the control can assume three values; in each transition between states a given cost is incurred. We want to determine the sequence of controls that minimizes the total cost over three transitions, starting from a given initial state. Figure 12.6 describes the problem: node x_0^2 is the initial state, while nodes x_t^i, with $i = 1, \ldots, 3$ and $t = 1, \ldots, 3$, represent the three possible states in the three

[8]Where, remember, the system is assumed not to be affected by deterministic disturbance $\mathbf{w}_t$.

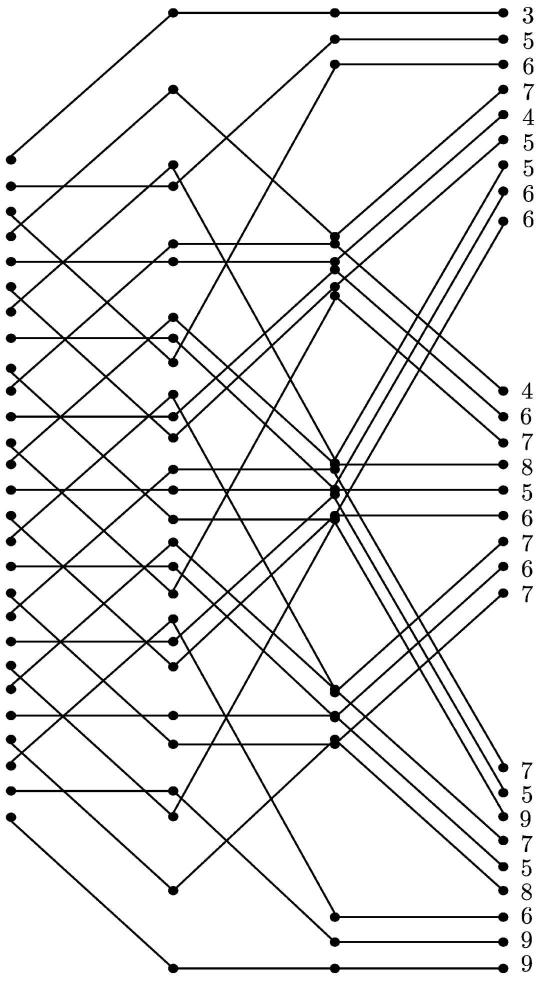


Figure 12.7: All the possible paths of the system in Figure 12.6.

following time instants. The arcs describe the possible transitions between nodes; note that there are three outgoing arcs from every node/state: as many as there are controls. The value on each arc represents the cost of the corresponding transition. The problem (we will denote it with P) can thus be visualised as the problem of identifying the minimum-cost path between x_0^2 and one of the three final states (x_3^1, x_3^2, x_3^3).

The most spontaneous way to solve it is to consider all the possible paths (*exhaustive procedure*) and to associate each one of them with its total cost (Figure 12.7). Comparing the 27 costs obtained in this way, the minimum-cost path is identified.

The strategy of SDP is different: it does not identify all the possible paths from the outset, to then choose among those on the basis of the total cost; instead, it progressively constructs the solution by solving a sequence of three problems, the same number of transitions considered in P. Each problem is defined on a single transition.

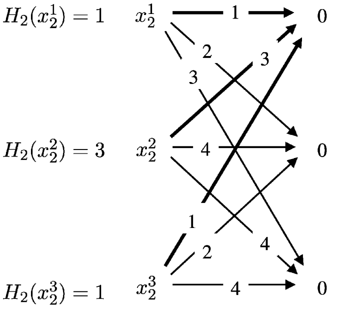


Figure 12.8: The graph of the problem P_2; in bold the minimum-cost paths.

The first problem (P_2) considers the transition from time 2 to time 3, and its aim is to determine the minimum-cost path from each of the three possible states at time 2 to the three possible final states, assuming, as implicitly stated by problem P, that there are no costs associated to reaching them. The subject of problem P_2 is visualised in Figure 12.8, where the last assumption is described by the three zeros that appear in the three nodes representing the final states. The solution of P_2 is obtained by comparing, for each of the three states at time 2, the costs of the three alternative paths that originate from it plus the cost-to-go (zero) of the final state. By doing so, the optimal cost-to-go associated to each of the three states (on the left of the state in the figure) and the best control (and therefore the transition, in bold in the figure) are determined. In total 9, comparisons are made.

The second problem (P_1) considers only the transition from time 1 to time 2, but, by associating the three previously determined costs-to-go (Figure 12.9a) to the three states at time 2, in practice it considers the graph in Figure 12.9b, which concerns the transitions from time 1 to time 3. Note that the number of paths in that graph is smaller than the one in Figure 12.6, because from time 2 onwards it considers only the optimal paths. Problem P_1 is solved with the same procedure as P_2, but here the optimal cost-to-go associated to the final states of the transition (the states at time 2) is no longer zero. The solution requires that another 9 comparisons be performed, with which the three optimal paths from the three states at time 1 to the states at time 3 are individuated. They are marked in bold in Figure 12.9b.

The third problem (P_0) considers only the transition from time 0 to time 1 (Figure 12.10a), but because of the way it is formulated it is equivalent to comparing the three paths in Figure 12.10b. The best path is the one that solves problem P, as it is easy to see by comparing Figure 12.10b with Figure 12.7. Note that problem P_0 concerns three transitions, just like P, but only 3 paths instead of 27: the previous problems in fact made it possible to exclude, thanks to the *Bellman's Principle of Optimality*, the paths that are certainly not optimal.

In total, 21 comparisons are made against the 27 required by the exhaustive procedure, without considering that each comparison is relative to a single transition instead of a series of three transitions.

In general, in a problem with a given initial state, if h denotes the transitions considered, $N_{\mathbf{u}}$ the number of controls and $N_{\mathbf{x}}$ the number of states, then the exhaustive procedure requires $(N_{\mathbf{u}})^h$ comparisons (in the example $3^3 = 27$), while the SDP requires $N_{\mathbf{u}} + (N_{\mathbf{x}} N_{\mathbf{u}})(h - 1)$ (in the example $3 + (3 \times 3) \times 2 = 21$). The number of comparisons

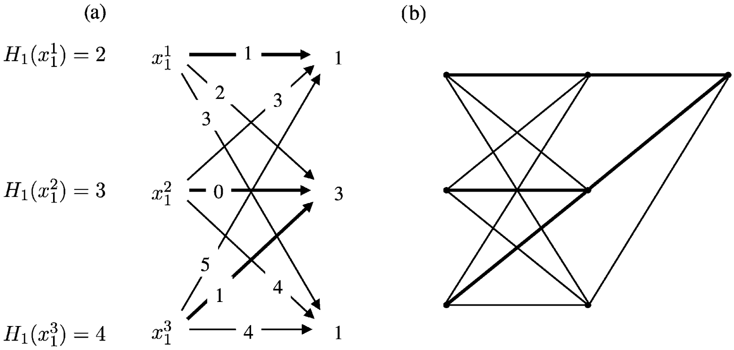


Figure 12.9: The graph of problem P_1 (a) and the paths that it examines (b); the minimum-cost paths are in bold.

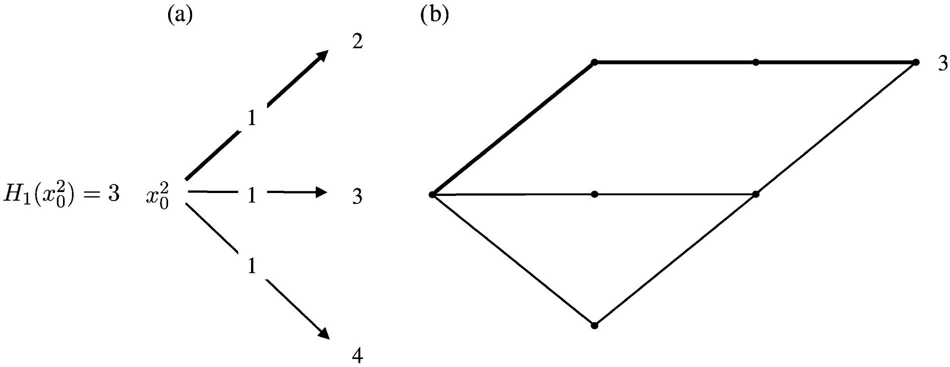


Figure 12.10: The graph of problem P_0 (a) and the paths that it examines (b); the minimum-cost paths are in bold.

thus increases exponentially with h in the exhaustive procedure and only linearly in SDP: the advantage of the second over the first rises very rapidly with the length of the time horizon.

Furthermore, it is important to note that, when the state and the control are vectors of dimensions $n_{\mathbf{x}}$ and $n_{\mathbf{u}}$ respectively and each of their components can assume N values, we obtain $N_{\mathbf{x}} = N^{n_{\mathbf{x}}}$ and $N_{\mathbf{u}} = N^{n_{\mathbf{u}}}$, from which it follows that *the computing time required for the solution of the Problem increases exponentially with the dimension of the state and the control*.

When the system is affected by a random disturbance, the previous considerations are still valid, except for the fact that $N_{\mathbf{u}}$ must be substituted with the product of the number of controls and the number of possible disturbances.

12.1.4 Algorithms

As we know, problem (12.1) can be defined over different time horizons. For each one we have now to identify the solution algorithm.

12.1.4.1 Finite horizon

When the horizon is finite, the indicator has the form (10.10), where h is the length of the time horizon. The solution of the Problem can be obtained with the following algorithm:

Algorithm 1 (*Finite horizon*):

Step 0 (*Initialisation*): *Let*

$$H_h^*(\mathbf{x}_h) = g_h(\mathbf{x}_h) \quad \forall \mathbf{x}_h \in \mathcal{S}_{\mathbf{x}_h}$$

Step 1: *For* $t = h-1, h-2, \ldots, 1$ *compute the costs-to-go* $H_t^*(\cdot)$ *with the following recursive equation*

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{12.10a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.10b}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \tag{12.10c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{12.10d}$$

$$\textit{any other constraints relative to } [t, t+1) \tag{12.10e}$$

Step 2 (*Termination test*): *Given* $t = 0$, *compute* $H_0^*(\mathbf{x}_0)$ *with equation* (12.10) *for* $\mathbf{x}_0 = \bar{\mathbf{x}}_0$. *The value obtained is the optimal value* J^* *of the objective of problem* (12.1), *and the h functions* $H_t^*(\cdot)$ *computed individuate the Bellman function* $H_\cdot^*(\cdot)$ *of the Problem.*

The optimal policy p^* is thus defined by equation (12.9), in which the values $H_{t+1}^*(\mathbf{x}_{t+1})$, for $t = 0, \ldots, h-1$, are provided by the h functions $H_{t+1}^*(\cdot)$ obtained in Steps 0 and 1, which are time slices of the Bellman function.

12.1.4.2 Receding horizon

This horizon is used only in the design of *on-line policies* and would be best considered in Chapter 14. However, since the algorithm that solves the Control Problem defined with it is strictly linked to the previous algorithm, we prefer to deal with it right away.

When the control is selected at every time t by solving a Problem defined over a receding horizon $[t, t+h]$ (see (10.11)), nothing changes with respect to the previous case, except for posing $t+h$ instead of h and t instead of 0. We denote with $H_{\cdot/t}^*(\cdot)$ the Bellman function that is obtained in this way. The optimal control for time t is thus obtained by equation (12.9), which takes the form

$$m_t^*(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1/t}^*(\mathbf{x}_{t+1}) \right] \tag{12.11}$$

The difference with respect to the previous case is that the control law $m_{t+1}^*(\mathbf{x}_{t+1})$, at time $t+1$, is not obtained by substituting in equation (12.11) $H_{t+2/t}^*(\mathbf{x}_{t+2})$ in place of $H_{t+1/t}^*(\mathbf{x}_{t+1})$, but by using in place of the last the function $H_{t+2/t+1}^*(\mathbf{x}_{t+2})$, which must be computed by solving the Problem for the horizon $[t+1, t+h+1]$, formulated at time $t+1$.

The solution is, therefore, much more laborious than in the previous case, since an entire Control Problem must be solved at each time instant (this is why the algorithm provides an on-line policy). Hence, it would only make sense to bear this greater computational burden if there were some advantage from the fact of re-formulating the Problem at each time instant. This might happen if the Problem were formulated at each time t on the basis of a system model updated with the new information obtained at time t and/or on the basis of the a posteriori probability distributions of the disturbance

$$\phi_{t/t}(\cdot), \ldots, \phi_{t+h/t}(\cdot)$$

instead of the a priori ones

$$\phi_t(\cdot), \ldots, \phi_{t+h}(\cdot)$$

and if the first were better than the second. Unfortunately, the second case can never occur because, by hypothesis, the disturbance is a white process, and, therefore, its a priori and a posteriori distributions coincide. Solving a new Problem at each time thus seems unnecessarily exhausting and should be done only to avoid, as explained in Section 10.2.2.1, the policy depending on the duration $(h - t)$ of the time interval that separates t from the final time instant h. Nevertheless, if the disturbance were not 'truly white ...', the receding horizon Problem might be interesting. We will meet it again in Chapter 14 when we discuss the suboptimal solutions. If the only reason for the adoption of a receding horizon is to avoid the dependency of the control law at each time t on $(h - t)$, it is a better idea to define the Problem over an infinite horizon with a TDC indicator (Section 8.1.2.1).

12.1.4.3 Infinite horizon: Total Discounted Cost

The *Total Discounted Cost* (TDC) indicator is defined as follows (see (8.11))

$$i\left(\mathbf{x}_0, \mathbf{u}_0^\infty, \boldsymbol{\varepsilon}_1^\infty\right) = \lim_{h\to\infty} \sum_{t=0}^{h} \gamma^t g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.12}$$

The solution of the Problem would be conceptually obtainable using Algorithm 1, if it were not for the fact that it becomes an infinite procedure. When the horizon is finite the algorithm starts, in fact, from the final time instant and proceeds backwards to the initial time instant. However, now the final time instant tends to infinity, and thus also the number of steps tends to infinity. Furthermore, what could 'starting from the infinite time instant' really mean in operative terms? When the Problem is defined for a periodic system[9] (i.e. it is a *periodic Problem*) this difficulty might be circumvented by starting from the zero time instant and proceeding backwards to infinity: the solution of a periodic problem, in fact, is not influenced by a shift in the origin of the time; however, also this procedure is infinite. One may only hope that, by proceeding towards $-\infty$, the cost-to-go tends towards a constant value (if the system is stationary[10]) or to a periodic value (if the system is periodic). In fact, the discount factor makes the costs that are very far away in time practically negligible and so, provided that one proceeds far enough into the past, the optimal cost-to-go could tend to a finite limit. If this actually happened, and we decided to stop the procedure when the differences in the estimate of the optimal cost-to-go for the different steps fell below a given level, the procedure would become finite.

[9] That is, the indicator and the constraint system are defined by periodic functions, all of the same period T, as assumed in Section 8.1.1.

[10] Remember that a stationary system is a periodic system of period $T = 1$.

Bertsekas (1976) proved the correctness of this guess: if the Problem is periodic of period T, the following algorithm converges under very broad hypotheses (which we will not describe here because they are too complex and, in practice, always satisfied for the water systems we consider).

Algorithm 2 (*Total Discounted Cost (TDC) over an infinite horizon*):

Step 0 (*Initialisation*): *Let* $\tau = 0$, *Term* $= 0$ *and*

$$H_t(\mathbf{x}_t) = \bar{H}(\mathbf{x}_t) \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t},\ t = 0, \dots, T-1$$

with $\bar{H}(\cdot)$ *being an arbitrary function, for example, identically zero.*

Step 1: *Set* $t = \tau - 1$ *and compute*

$$H_t(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma H_{t+1}(\mathbf{x}_{t+1}) \right] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{12.13a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.13b}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \tag{12.13c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{12.13d}$$

$$\textit{any other constraints relative to } [t, t+1) \tag{12.13e}$$

If

$$\left| H_{t+T}(\mathbf{x}_t) - H_t(\mathbf{x}_t) \right| < \alpha \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t}$$

with α *small and preassigned, increase Term by one*; *otherwise set it equal to zero.*

Step 2 (*Termination test*): *If Term* $= T$, *the algorithm terminates and the last* T *functions* $H_t(\cdot)$ *computed are a good approximation of the Bellman function* $H_{\cdot}^{*}(\cdot)$, *which proves to be periodic of period* T *and thus is univocally defined by* T *time slices* $H_{\varsigma}^{*}(\cdot)$, $\varsigma = 0, \dots, T-1$. *Therefore, one assumes*

$$H^{*}_{(t+\varsigma)_{\mathrm{mod}\,T}}(\mathbf{x}_{t+\varsigma}) = H_{t+\varsigma}(\mathbf{x}_{t+\varsigma}) \quad \forall \mathbf{x}_{t+\varsigma} \in \mathcal{S}_{\mathbf{x}_{t+\varsigma}},\ \varsigma = 0, \dots, T-1$$

Otherwise set $\tau = t$ *and return to Step* 1.

The parameter α controls the accuracy of the estimate, which is by nature asymptotic. The termination test verifies whether the Bellman function remained unchanged, up to α, in the last T steps, i.e. it checks whether convergence has been achieved. In general, the number of steps required before the termination test is passed depends on the value of the discount factor γ: the smaller it is, the faster the convergence.

The algorithm is based on three theorems (Bertsekas, 1976): the first shows that the Bellman function $H_{\cdot}^{*}(\cdot)$ of the Problem is a periodic function of period T; therefore, it is univocally defined once the T costs-to-go functions

$$H_t^{*}(\cdot) \quad t = 0, \dots, T-1$$

are known. The second theorem proves that, under very broad conditions, almost always satisfied in real cases, the Bellman function is the unique solution to the following system of Bellman equations (*Condition of Optimality*):

$$H_0(\mathbf{x}_0) = \min_{\mathbf{u}_0} \underset{\boldsymbol{\varepsilon}_1}{E}\big[g_0(\mathbf{x}_0, \mathbf{u}_0, \boldsymbol{\varepsilon}_1) + \gamma H_1(\mathbf{x}_1)\big] \quad \forall \mathbf{x}_0 \in \mathcal{S}_{\mathbf{x}_0} \tag{12.14a}$$

$$H_1(\mathbf{x}_1) = \min_{\mathbf{u}_1} \underset{\boldsymbol{\varepsilon}_2}{E}\big[g_1(\mathbf{x}_1, \mathbf{u}_1, \boldsymbol{\varepsilon}_2) + \gamma H_2(\mathbf{x}_2)\big] \quad \forall \mathbf{x}_1 \in \mathcal{S}_{\mathbf{x}_1} \tag{12.14b}$$

$$\dots$$

$$H_{T-1}(\mathbf{x}_{T-1}) = \min_{\mathbf{u}_{T-1}} \underset{\boldsymbol{\varepsilon}_0}{E}\big[g_{T-1}(\mathbf{x}_{T-1}, \mathbf{u}_{T-1}, \boldsymbol{\varepsilon}_0) + \gamma H_0(\mathbf{x}_0)\big] \quad \forall \mathbf{x}_{T-1} \in \mathcal{S}_{\mathbf{x}_{T-1}} \tag{12.14c}$$

subject to the constraints

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \quad t = 0, \dots, T-1 \tag{12.14d}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \quad t = 0, \dots, T-1 \tag{12.14e}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, \dots, T-1 \tag{12.14f}$$

$$\text{any other constraints} \quad t = 0, \dots, T-1 \tag{12.14g}$$

The third theorem proves that Algorithm 2 is a procedure for solving this system of equations; in other words, it proves that, when t tends to $-\infty$, the functions $H_t(\cdot)$ computed at Step 1 with equation (12.13) tend to the optimal cost-to-go $H_t^*(\cdot)$. This justifies the termination test in Step 2. For the same reason, once the convergence has been reached, the last T functions computed are an estimate of the Bellman function $H_{\cdot}^*(\cdot)$.

The optimal policy is defined, as usual, by equation (12.9), opportunely corrected with the discount factor γ, in which $H_{t+1}^*(\mathbf{x}_{t+1})$ is given by the Bellman function provided by Algorithm 2: precisely,

$$m_t^*(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} \underset{\boldsymbol{\varepsilon}_{t+1}}{E}\left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma H_{t+1_{\bmod T}}^*(\mathbf{x}_{t+1})\right] \tag{12.15}$$

It follows that also the control policy is periodic of period T.

12.1.4.4 Infinite horizon: Average Expected Value

As seen in Section 8.1.2.1 (see (8.12)), the *average cost per step* indicator (often called Average Expected Value, from which the acronym AEV that we will use in what follows) has the following form

$$i\left(\mathbf{x}_0, \mathbf{u}_0^\infty, \boldsymbol{\varepsilon}_1^\infty\right) = \lim_{h\to\infty} \frac{1}{h+1} \sum_{t=0}^{h} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.16}$$

The formula shows very well what we already explained on page 227: the average cost per step depends only on the costs incurred in the long-term, since the costs incurred in a transient of any finite length k cannot influence it, given that their contribution disappears when h tends to infinity

$$\lim_{h\to\infty} \frac{1}{h+1} \sum_{t=0}^{k} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) = 0$$

If Algorithm 2 were applied to the Problem defined with this indicator, the values of $H_t(\mathbf{x}_t)$ would diverge as t tends towards $-\infty$. This is due to the fact that the cost-to-go over an infinite horizon, in the absence of discount, is generally infinite. The cost-to-go thus loses its meaning and must be substituted by an indicator that does not diverge, but that still expresses the 'desirability' of the state. The simplest solution is to adopt as indicator, for every state $\mathbf{x}_t$, the difference $h_t(\mathbf{x}_t)$ between its cost-to-go $H_t(\mathbf{x}_t)$ and the cost-to-go $H_t(\bar{\mathbf{x}}_t)$ of a reference state $\bar{\mathbf{x}}_t$, arbitrarily chosen. This difference, in fact, is proved to remain finite as t tends to $-\infty$. The optimal difference

$$h^*_{t+1}(\mathbf{x}_{t+1}) = H^*_{t+1}(\mathbf{x}_{t+1}) - H^*_{t+1}(\bar{\mathbf{x}}_{t+1})$$

also has the property that, when used in equation (12.9) in place of the optimal cost-to-go $H^*_{t+1}(\mathbf{x}_{t+1})$, it provides the same decision that the latter would have provided: we immediately note that the optimal control $m^*_t(\mathbf{x}_t)$, defined by equation (12.9), is not influenced by the constant value $H^*_{t+1}(\bar{\mathbf{x}}_{t+1})$ subtracted from $H^*_{t+1}(\mathbf{x}_{t+1})$.

These simple considerations should allow us to understand the differences between Algorithm 2 and the following

Algorithm 3 (*Average Expected Value per step (AEV) over an infinite horizon*):

Step 0 (*Initialisation*): *Set*

$$h^0_t(\mathbf{x}_t) = \bar{H}(\mathbf{x}_t) \quad \forall \mathbf{x}_t \in \mathcal{S}_{x_t},\ t = 0, \ldots, T-1$$

with $\bar{H}(\cdot)$ being an arbitrary function, for example identically zero. Set $i = 1$ and choose an arbitrary state $\bar{x}_t \in \mathcal{S}_{x_t}$, for $t = 0, \ldots, T-1$.

Step 1: *For $t = T-1, \ldots, 0$ compute the T functions $H^i_t(\cdot)$ defined by the following recursive equation*

$$H^i_t(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H^i_{t+1}(\mathbf{x}_{t+1}) \right] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{12.17a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.17b}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \tag{12.17c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{12.17d}$$

$$\textit{any other constraints relative to } [t, t+1) \tag{12.17e}$$

with

$$H^i_T(\mathbf{x}) = h^{i-1}_0(\mathbf{x}) \quad \forall \mathbf{x} \in \mathcal{S}_{\mathbf{x}_0} \tag{12.17f}$$

Step 2 (*Termination test*): *Let*

$$h^i_t(\mathbf{x}_t) = H^i_t(\mathbf{x}_t) - H^i_t(\bar{\mathbf{x}}_t) \quad \forall \mathbf{x}_t \in \mathcal{S}_{x_t},\ t = 0, \ldots, T-1 \tag{12.18}$$

If

$$\left| h^i_t(\mathbf{x}_t) - h^{i-1}_t(\mathbf{x}_t) \right| < \alpha \quad \forall \mathbf{x}_t \in \mathcal{S}_{x_t},\ t = 0, \ldots, T-1$$

with α small and given, the algorithm terminates and the last T functions $H^i_t(\cdot)$ computed are a good approximation of the Bellman function $H^*_\cdot(\cdot)$, which proves to be periodic of period T and so is univocally defined by T time slices $H^*_t(\cdot)$, $t = 0, \ldots, T-1$. Thus, one *assumes*

$$H_t^*(\mathbf{x}_t) = H_t^i(\mathbf{x}_t) \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t},\ t = 0, \ldots, T-1$$

Otherwise increase i by 1 *and return to Step* 1.

The reader will certainly have understood that the differences $h_t^i(\mathbf{x}_t)$ defined by equation (12.18) are those very differences $h_t(x)$ that we introduced at the beginning of this section.

This algorithm is known as Successive Approximations Algorithm (SAA) and was first proposed by White (1963) for the stationary case and then generalized to the periodic one by Su and Deininger (1972). Like Algorithm 2, the SAA is founded on a series of theorems (Bertsekas, 1976) that we would like to summarize briefly. The first and the second theorems show that, under very broad conditions, almost always satisfied in real cases, the Bellman function $H_\cdot^*(\cdot)$ is periodic and that, together with the optimal value J^* of the Problem objective, it is the solution to the following system of Bellman equations (*Condition of Optimality*):

$$H_0(\mathbf{x}_0) = \min_{\mathbf{u}_0} \mathop{E}_{\boldsymbol{\varepsilon}_1}\big[g_0(\mathbf{x}_0, \mathbf{u}_0, \boldsymbol{\varepsilon}_1) + H_1(\mathbf{x}_1)\big] \quad \forall \mathbf{x}_0 \in \mathcal{S}_{\mathbf{x}_0} \tag{12.19a}$$

$$H_1(\mathbf{x}_1) = \min_{\mathbf{u}_1} \mathop{E}_{\boldsymbol{\varepsilon}_2}\big[g_1(\mathbf{x}_1, \mathbf{u}_1, \boldsymbol{\varepsilon}_2) + H_2(\mathbf{x}_2)\big] \quad \forall \mathbf{x}_1 \in \mathcal{S}_{\mathbf{x}_1} \tag{12.19b}$$

$$\cdots$$

$$JT + H_{T-1}(\mathbf{x}_{T-1}) = \min_{\mathbf{u}_{T-1}} \mathop{E}_{\boldsymbol{\varepsilon}_0}\big[g_{T-1}(\mathbf{x}_{T-1}, \mathbf{u}_{T-1}, \boldsymbol{\varepsilon}_0) + H_0(\mathbf{x}_0)\big] \quad \forall \mathbf{x}_{T-1} \in \mathcal{S}_{\mathbf{x}_{T-1}} \tag{12.19c}$$

subject to the constraints

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \quad t = 0, \ldots, T-1 \tag{12.19d}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \quad t = 0, \ldots, T-1 \tag{12.19e}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, \ldots, T-1 \tag{12.19f}$$

$$\text{any other constraints} \quad t = 0, \ldots, T-1 \tag{12.19g}$$

In other words, the value J^* and the T functions $H_t^*(\cdot)$ that solve this system of equations are the minimum average cost per step and the time slices that define the Bellman function, respectively. From the fact that J^* does not depend on $\mathbf{x}_0$ it follows that the minimum average cost per step is independent of the initial state, as the observation that opens this section suggested.

The third theorem proves that, as i increases, the functions $H_t^i(\cdot)$, $t = 0, \ldots, T-1$ provided by equation (12.17) tend towards a solution of system (12.19) and the difference $(H_t^i(\mathbf{x}) - h_t^{i-1}(\mathbf{x}))$ tends, for every t, to T times the minimum average cost per step J^*, i.e.

$$\lim_{i \to \infty} \big(H_t^i(\mathbf{x}_t) - h_t^{i-1}(\mathbf{x}_t)\big) = TJ^* \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \text{ and } \forall t \tag{12.20}$$

This result does not appears surprising when one observes that between $H_t^i(\cdot)$ and $h_t^{i-1}(\cdot)$ there are T steps, in each of which the average expected cost is J^*. A further observation: from equation (12.18) it follows that $h_t^{i-1}(\bar{\mathbf{x}}_t) = 0$ and it is, therefore, straightforward to deduce from equation (12.20) that

$$\lim_{i\to\infty} H_t^i(\bar{\mathbf{x}}_t) = TJ^* \quad \forall t \tag{12.21}$$

which is a very useful expression for determining J^*. Finally, from equations (12.18), (12.20) and (12.21) it follows that

$$\lim_{i\to\infty}\left(h_t^i(\mathbf{x}_t) - h_t^{i-1}(\mathbf{x}_t)\right) = 0 \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \text{ and } \forall t \tag{12.22}$$

from which the termination test in Step 2 is derived.

One last observation: the solution to system (12.19) is not unique, because, if $H_\cdot^*(\cdot)$ is a solution, so is $H_\cdot^*(\cdot) + \beta$, with any β. This justifies the arbitrariness in the choice of states $\bar{x}_t$ in Step 0.

The optimal policy is once again given by equation (12.9), in which $H_{t+1}^*(\mathbf{x}_{t+1})$ is the Bellman function provided by Algorithm 3. This function being periodic of period T, the control policy will also be periodic of the same period. Lastly, note that even if the Bellman function seems not to be completely defined, since it is defined up to an arbitrary constant, that arbitrariness does not reflect upon the policy, because, as already mentioned, it is invariant with respect to the addition of any given constant to $H_{t+1}^*(\mathbf{x}_{t+1})$.

12.1.4.5 Infinite horizon: maximum cost per step

When the Wald criterion is adopted, the indicators presented in the last two sections cannot be considered. The adoption of a discount factor openly contradicts risk aversion, which is the basis of the Wald criterion. Furthermore, it is no longer strictly necessary to discount the future, since, unlike the total expected cost, the maximum cost cannot diverge when the step costs have an upper limit, as hypothesized in equation (4.29). Lastly, since the time operator is the maximum (*max*), the average cost per step is of no interest. Therefore, with the Wald criterion, the only well defined indicator over an infinite horizon is the *maximum cost per step*

$$i\left(\mathbf{x}_0, \mathbf{u}_0^\infty, \boldsymbol{\varepsilon}_1^\infty\right) = \lim_{h\to\infty} \max_{t=0,\dots,h} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1})$$

It is easy to imagine that the Control Problem defined by applying the Wald criterion to this indicator can be solved with the algorithm that is obtainable by applying the Law of Duality (Section 10.4) to Algorithm 2, where $\gamma = 1$ is assumed (Piccardi, 1993a, 1993b). However, while the T functions $H_t^i(\cdot)$ obtained with Algorithm 2 are a good approximation of the Bellman function $H_\cdot^*(\cdot)$ for any choice of the function $\bar{H}(\cdot)$ in the Initialization Step (see page 299), in the Wald case the choice can affect the solution. For example, if all the values of $\bar{H}(\cdot)$ were set greater than the maximum step cost which the system may incur, the T functions $H_t^i(\cdot)$ provided by the algorithm would be the same T functions $\bar{H}(\cdot)$ adopted in the Initialization Step.

12.1.5 Policy and Bellman function

We have seen that the policy can be computed with equation (12.9), given the Bellman function $H_\cdot^*(\cdot)$. Since, by hypothesis, the system is an automaton, i.e. a system whose state can assume only a finite number of values at every time, the policy is described by a double entrance table, which associates the pair $(t, \mathbf{x}_t)$, called *event*, with the value $m_t^*(\mathbf{x}_t)$ of the optimal control. Likewise, the Bellman function is described by a table that associates each event with its optimal cost-to-go $H_t^*(\mathbf{x}_t)$.

The reader might rightly observe that to obtain the policy it is not strictly necessary to use equation (12.9): the control values might be stored in memory contextually to the optimal cost-to-go values, when the minimum in the recursive Bellman equation is determined. More precisely, when the Problem is formulated over a finite horizon, in Step 1 of Algorithm 1 the values of $\mathbf{u}_t$ that provide the minimum of equation (12.10a) for each pair $(t, \mathbf{x}_t)$ could be stored as they are computed. Note, in fact, that those values are none other than those defined by equation (12.9). With an infinite horizon, one may proceed in a similar way and, once the convergence has been reached, the last T tables computed would define the optimal policy.

This modification of the algorithms is completely licit, but is not used in practice for two reasons. The first, which will rapidly lose its force through technological progress, is that it increases the memory requirements: each of the matrices $m^*_\cdot(\cdot)$ and $H^*_\cdot(\cdot)$ has in general significant dimensions (in the order of Gbytes, as we will show in Section 12.2.3), while the number of elements of $m^*_\cdot(\cdot)$ that will actually be used is definitively small: at the most several thousand, i.e. the number of events that will occur in the life of the policy, which is unlikely to last ten years. Thus it is preferable to halve the memory requirements by using equation (12.9) to compute the control values as they are required, i.e. using equation (12.9) on-line.

The reader might object that, once the entire policy has been computed, it is no longer necessary to remember the Bellman function, which could thus be discarded. The second reason replies to this objection: while the policy can be used only to determine the control, the Bellman function can be used to design policies on-line, and so, once it is known, it is worth storing it.

In conclusion, the Bellman function provides the optimal cost-to-go and, along with the model, it defines univocally the optimal policy: it is thus richer in information than the latter and, therefore, one gives preference to it.

12.1.6 Examples

To understand the algorithms presented up to this point more thoroughly, it is advisable to examine their functioning in practice. For this reason, in Chapter E2 of the CD we present some simple numerical examples.

12.2 PV policies: Parametric Design

12.2.1 The curse of dimensionality

In the preceding paragraphs, we have learned how to organize and solve a Control Problem whose solution is the policy sought. The algorithms that we have proposed for its solution are based on Stochastic Dynamic Programming (SDP). These are very robust algorithms that are applicable under hypotheses so general as almost always to be satisfied: in fact, we required that the objective and the constraint system of the Problem be separable but we did not formulate any further hypothesis either on the form of the step costs which make up the first or on the characteristics of the state transition function (12.5b) and of the 'other possible constraints' (12.5f) which appear in the second; we required the causal disturbances to be a realization of a white process (a condition, note, which is only sufficient but not necessary), but we did not impose any condition on the class of probability distributions which describes this process. Such a wide theoretical applicability has in practice a high price: the algorithms based on the SDP in fact have significant computational and memory requirements. These are so significant that for many Problems a precise solution is inconceivable, at least with the computing power usually available today. The reason for this is what Bellman has picturesquely described as the curse of dimensionality: the computational and memory requirements grow exponentially with the system dimension. To appreciate this statement fully, consider the problem (12.5) in which, you will remember Section 10.5, the state has dimension $n_{\mathbf{x}}$, the control has dimension $n_{\mathbf{u}}$ and the disturbance has dimension $n_{\boldsymbol{\varepsilon}}$. As we know, in order to determine the function $H_{\cdot}(\cdot)$ with the SDP, the state set $\mathcal{S}_{\mathbf{x}_t}$ must be a discrete and finite set at every instant t. Suppose, for example, that every state component can assume 100 values at every instant: the set $\mathcal{S}_{\mathbf{x}_t}$ will contain $100^{n_{\mathbf{x}}}$ points. For each of these points minimization must be carried out numerically with respect to the control $\mathbf{u}_t$. This is performed with a minimum search algorithm in $\mathcal{U}_t(\mathbf{x}_t)$, which is a set in the space $\mathbb{R}^{n_{\mathbf{u}}}$. For every value of $\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)$ one must evaluate the corresponding expected cost-to-go; this leads to the computation of the expected value and thus requires a numerical integration in the space $\mathbb{R}^{n_{\boldsymbol{\varepsilon}}}$. All this proves very onerous, from the viewpoints both of computing time and of memory occupation. It follows that the algorithms of the SDP can, in fact, only be applied when the dimensions $n_{\mathbf{x}}$, $n_{\mathbf{u}}$ and $n_{\boldsymbol{\varepsilon}}$ are very small. To deal with Problems in which the dimensionality of the system is higher, it is necessary to have recourse to alternative strategies.

Of these letter in this section we illustrate the principal in detail. Before proceeding, however, we feel that it is useful to dwell a little on an observation that the reader will surely have made: if the curse of dimensionality is the price to pay for the generality of the hypotheses on which the SDP is based, the question naturally arises whether, by giving up this generality in part, it might not be possible to circumvent the problem of dimensionality. The answer that Control Theory gives is, at least in part, affirmative. When some particular hypotheses have been verified, the so-called LQG framework (see Section 12.2.2), the Control Problem admits a unique analytical solution and "dimensionality is not a curse any more". Nevertheless, however appealing this framework may be, it proves, unfortunately, to be of little practical utility, since the hypotheses which define it are decidedly restrictive and usually, in realistic operating conditions, the water systems that we deal with are far from satisfying them. For this reason in the following Section 12.2.2 we will confine ourselves to formulating the LQG framework and to analyzing its limits. We will pay greater attention, however, to much more interesting strategies, for our purposes, that approach the dimensionality problem from a different methodological perspective: if, in fact, the problem is a consequence of the large dimensions of the system, it is opportune to operate in such as way as to reduce the incidence of these on the computational and memory requirement of the algorithm. In this chapter we will describe the strategies conceived to combat the exponential growth of complexity with the state dimension, postponing what can be done with respect to the control until Chapter 15.

Among the first are found various techniques, which are valid above all in the deterministic case: for example, coarse grid approximation, the use of Lagrange's multipliers, approximation with Legendre's polynomials (see, for example, Bellman and Dreyfus, 1962; Larson, 1968; Kaufmann and Cruon, 1967), and special techniques (Wong and Luenberger, 1968; Luenberger, 1971a). These artifices, although they are useful in some Problems, are partial remedies and thus we will not examine them.

The same limits affect strategies based on the reduction of state dimensions through aggregations of the components of the system and decompositions of the Problem into sub-problems. For example, Turgeon (1981) proposes to decompose the Control Problem of a network of m reservoirs into m sub-problems with only two reservoirs: the first is a reservoir of the original Problem, the second an aggregated representation of reservoirs downstream from it. In this way, computing time increases only linearly with the number of reservoirs, i.e. with $n_{\mathbf{x}}$, and therefore very extensive networks can be considered. However, the sub-problems which describe the reservoirs further downstream correspond little to reality because the regulation of their inflows is not described. For this reason, Archibald et al., 1997, suggest a different decomposition: every sub-problem considers three reservoirs: one is a reservoir of the original Problem while the other two represent aggregates of reservoirs upstream of and downstream from the original reservoir considered. The computing time, however, increases as the square of the number of reservoirs, i.e. as a quadratic function of the state of the water system. However effective these approaches may be from the computational point of view, they still constitute partial remedies, because they are in fact applicable only to a restricted class of systems: networks of artificial reservoirs, fed by small catchments which serve non-dynamic users, as in the case of networks of artificial reservoirs exclusively for hydropower use. We will therefore devote no further space to these techniques.

The curse of dimensionality can be overcome on condition that one abandons the Functional Approach, which aims at determining the optimal policy in the space of all the feasible policies, and is content with a Parametric Approach, which limits itself to finding the best policy in a given class. There are two possible strategies. The first defines the class of policies indirectly, determining the class to which the functions expressing the cost-to-go must belong. In fact, we know that these, via (12.9), specify the policy. It is therefore still an SDP-based approach, but it makes it possible to reduce the discretization grid drastically. The second strategy determines a priori, up to a certain number of parameters, the form of the policy, to then identify the optimal values of the parameters by solving an opportune Mathematical Programming Problem. Abandoning the SDP framework, it obtains three significant advantages: it does not require the system to be an automaton; it allows the design of a policy for a system affected by deterministic disturbances (the reader will remember that this is impossible with the SDP); the policy designed is more intuitive and comprehensible for a DM than a policy defined by the (12.9).

We will examine separately these two strategies in Sections 12.2.3 and 12.2.4, referring constantly to the finite horizon Problem to simplify the explanation; the results are not, however, limited to this case, since in some cases it is possible to extend them to infinite horizon Problems, analogously to what was done in Section 12.1.4. First, however, as promised, let us briefly examine the LQG framework, referring, with no loss of generality, to the finite horizon Problem.

12.2.2 Linear Quadratic Gaussian[11]

As stated in the previous section, the LQG framework requires the system that one wishes to control to possess some particular properties. Precisely:

[11] This book does not aspire to supply an exhaustive and rigorous description of the LQG framework (Kalman, 1960a; Merrian, 1964) and of all its variants (e.g. Jacobson et al., 1980). The significant computational advantages that it offers, in fact, have made it one of the most commonly used and thus most documented Optimal Control frameworks in the literature of the sector. In this section we limit ourselves to supplying its standard formulation and corresponding solution, referring the interested reader to one of the numerous texts that deal with the subject more fully (for example, Bertsekas, 1976).

(1) that the state transition function be *linear*,

(2) that there not be other constraints either on the state or the control,

(3) that the objective of the Control Problem be *quadratic*, i.e. that it be expressed as the sum of quadratic step indicators in the state and the control,

(4) that stochastic disturbances be *Gaussian* white noise.

Let us formalize these four conditions and study their effects in the Control Problem (12.1).

The first condition is equivalent to assuming for equation (12.1b) the following expression[12] (see Appendix A3)

$$\mathbf{x}_{t+1} = \mathbf{F}_t \mathbf{x}_t + \mathbf{G}_t \mathbf{u}_t + \boldsymbol{\varepsilon}_{t+1} \quad t = 0, \ldots, h-1 \tag{12.23}$$

where $\mathbf{F}_t$ and $\mathbf{G}_t$ are time-variant matrices of dimensions respectively $n_{\mathbf{x}} \times n_{\mathbf{x}}$ and $n_{\mathbf{x}} \times n_{\mathbf{u}}$. The second condition corresponds to assuming that in (12.1c) the feasible control set $\mathcal{U}_t(\mathbf{x}_t)$ coincides with the entire space $\mathbb{R}^{n_{\mathbf{u}}}$ and that 'any other constraints' on the state and the control do not exist (equation 12.1g). The third condition requires that the form of the step indicator be the following

$$g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) = \mathbf{x}_t^T \mathbf{Q}_t \mathbf{x}_t + \mathbf{u}_t^T \mathbf{R}_t \mathbf{u}_t \quad t = 0, \ldots, h-1 \tag{12.24}$$

where $\mathbf{Q}_t$ and $\mathbf{R}_t$ are symmetrical matrices with $\mathbf{Q}_t$ positive semi-definite and $\mathbf{R}_t$ positive definite for every t. For the fourth condition the disturbance probability distribution is normal with zero mean and variance $(\sigma^{\boldsymbol{\varepsilon}_t})^2$, i.e.

$$\boldsymbol{\varepsilon}_{t+1} \sim N\big(0, \big(\sigma_t^{\boldsymbol{\varepsilon}}\big)^2\big) \tag{12.25}$$

The problem (12.1) thus assumes the form

The LQG Off-line Control Problem:

$$J^* = \min_p \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \left[\sum_{t=0}^{h-1} \big(\mathbf{x}_t^T \mathbf{Q}_t \mathbf{x}_t + \mathbf{u}_t^T \mathbf{R}_t \mathbf{u}_t\big) + \mathbf{x}_h^T \mathbf{Q}_h \mathbf{x}_h \right] \tag{12.26a}$$

$$\mathbf{x}_{t+1} = \mathbf{F}_t \mathbf{x}_t + \mathbf{G}_t \mathbf{u}_t + \boldsymbol{\varepsilon}_{t+1} \qquad t = 0, \ldots, h-1 \tag{12.26b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \qquad t = 0, \ldots, h-1 \tag{12.26c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim N\big(0, \big(\sigma_t^{\boldsymbol{\varepsilon}}\big)^2\big) \qquad t = 0, \ldots, h-1 \tag{12.26d}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{12.26e}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, \ldots, h-1\big\} \tag{12.26f}$$

in which, note, also the penalty is expressed in quadratic form, with $\mathbf{Q}_h$ a symmetrical and semi-definite positive matrix.

The problem (12.26) can be solved (Bertsekas, 1976) with Algorithm 1 on page 299. However, in computing the costs-to-go by equation (12.10a) it is not necessary to explore exhaustively the discrete state, control and disturbance spaces. One intuits why this happens by observing that from the fact that the step indicators are quadratic it follows that also the

[12] Consistently with what was stated on page 288 we omit u^p in the following formulae.

costs-to-go are so, and thus at every time t the minimum in the (12.10a) can be determined analytically, by solving a system of linear equations which expresses the optimality condition (i.e. first derivative with respect to u_t of the cost-to-go equal to zero). Thus one obtains h closed-loop control laws of the form

$$m_t^*(\mathbf{x}_t) = \mathbf{L}_t \mathbf{x}_t \quad t = 0, \ldots, h-1 \tag{12.27}$$

whose sequence $m_0^*(\cdot), \ldots, m_{h-1}^*(\cdot)$ constitutes the optimal policy p^* sought. The matrices $\mathbf{L}_t$ which define it are called *gain matrices* and are defined by the following expression

$$\mathbf{L}_t = -\big(\mathbf{G}_t^T \mathbf{K}_{t+1} \mathbf{G}_t + \mathbf{R}_t\big)^{-1} \mathbf{G}_t^T \mathbf{K}_{t+1} \mathbf{F}_t \quad t = 0, \ldots, h-1 \tag{12.28}$$

where $\mathbf{K}_t$ are symmetric positive semi-definite matrices recursively computable with the following equation

$$\mathbf{K}_h = \mathbf{Q}_h \tag{12.29a}$$

$$\mathbf{K}_t = \mathbf{F}_t^T \big[\mathbf{K}_{t+1} - \mathbf{K}_{t+1} \mathbf{G}_t \big(\mathbf{G}_t^T \mathbf{K}_{t+1} \mathbf{G}_t + \mathbf{R}_t\big)^{-1} \mathbf{G}_t^T \mathbf{K}_{t+1}\big] \mathbf{F}_t + \mathbf{Q}_t$$

$$t = 0, \ldots, h-1 \tag{12.29b}$$

known as *discrete matrix Riccati* equation. Finally, given the form of the control laws, the optimal value of the objective (12.26a) for $\mathbf{x}_0 = \bar{\mathbf{x}}_0$ is immediately computed

$$J^* = \bar{\mathbf{x}}_0^T \mathbf{Q}_t \bar{\mathbf{x}}_0 \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \left[\sum_{t=0}^{h-1} \boldsymbol{\varepsilon}_{t+1}^T \mathbf{K}_{t+1} \boldsymbol{\varepsilon}_{t+1} \right] \tag{12.30}$$

The computational advantages that the LQG framework can boast compared to SDP are therefore clear: the combination of a quadratic objective with a linear model allows the analytical solution of the Control Problem.[13] The dimensions of the state and control vectors thus do not affect the computational cost of the solution and the curse of dimensionality does not constitute a problem any more. Moreover, the control laws (12.27) are linear[14] and this makes the implementation of the controller particularly simple from an engineering point of view.

Unfortunately, however, the hypotheses of the LQG framework are unlikely to be satisfied by the characteristics of a water system. Let us again take as an example the simple

[13] There exists another more deep reason why the quadratic form of the objective is coupled with the linear model. Observe that it is very improbable that mechanical systems and industrial processes, for which the LQG framework was originally conceived, and still more the environmental systems that we discuss in this book, are really linear. Whichever linear model that we adopt for these therefore constitutes a simplified, or even rough, representation of a non-linear reality whose 'real model' (the reader will forgive the oxymoron) we do not know. We can, however, think that the linear model is the linearization (i.e. a Taylor expansion truncated at the first order, see Section A3.3.9) of the unknown 'real model'. The 'honesty' of the linear model thus depends on how much information is lost with the first order expansion. It is possible to demonstrate (see Athans, 1971) that the second order terms neglected really are proportional to the square of the state and of the control of the linear model: thus by minimizing a quadratic objective one minimizes the error implicitly committed in considering a linear model. Of course, this is true only if one supposes that the terms of an order superior to the second really are negligible: otherwise we will have to consider a cubic objective, or an objective of a superior order. In other words, if the 'real model' is only mildly non-linear, the quadratic objective guarantees that the linear model 'honestly' represents reality.

[14] Note also that while the optimum value depends on the characteristics of the disturbance $\boldsymbol{\varepsilon}_{t+1}$ (see (12.30)), the gain matrices L_t defined by (12.28) do not depend on them and are identical to those that would be obtained in the deterministic case (*Certainty Equivalence Principle*).

system of Figure 12.2, which is made up of a reservoir which feeds an irrigation district. As the reader will remember (see Section 5.1.2.3), the dynamic behaviour of a reservoir is never linear, owing to the non-linearity of the release function (equation (5.10)) due not only to the non-linearity of the storage–discharge function, but above all to the finiteness of the reservoir. When the storage–discharge function is approximately linear it could be thought that a linear reservoir model could be acceptable, when the storage is between the minimum and the maximum. This means, however, setting constraints on the state of the system and therefore the hypotheses of the LQG framework are no longer satisfied and it cannot be adopted. However, to be able to use it often the minimum and maximum storage constraints are ignored, thus assuming that the reservoir is infinite, a hypotheses which is almost always far from the truth.[15] The finiteness of the reservoir will be remembered, however, in the phase of *Estimating Effects* when it is impossible to ignore it, when the reservoir is filled beyond the maximum or drained beyond the minimum. For this reason, in these conditions, when simulating the behaviour of the regulated system, the decisions suggested by the policy are no longer applied. It follows that the policy which is in fact applied is not the optimum policy computed and from this it follows that this latter cannot be considered optimal a priori even between the maximum and the minimum.

There is little confirmation also for the hypothesis that step indicators are quadratic. Many indicators are, in fact, asymmetrical, for example not zero only when a certain variable is superior or inferior to a given threshold (possibly time-variant). Think of the water supply deficit (page 176 and Section 4.5 of PRACTICE) which we could use as indicator of the irrigation district in our example: it would make no sense to replace it with a quadratic function of the difference between supply and demand, since when supply is superior to demand crops are not damaged. The same thing happens for the flood indicator (4.1) introduced in Chapter 4 and in many other cases that the reader can find in Chapter 4 of PRACTICE. The only indicators for which a quadratic expression can prove significant are those of a technological nature, like the hydropower production. If, in fact, the network for which the energy is intended requires a certain power, we can consider that the cost incurred by not supplying exactly that value grows as the square of the difference between demand and supply. Observe, however, that this indicator is quadratic only if the hydraulic head is constant (as in the case of run-of-river power plants or when the variation in level produced by regulation can be considered negligible compared to the hydraulic head), but it is not so in the opposite case (Georgakakos, 1989a; McLaughlin and Velasco, 1990), which is much more interesting from the management point of view.

For all these reasons, the LQG framework has achieved little success[16] in the field of the management of water resources and there are really few applications documented in the literature (Wasimi and Kitanidis, 1983; Loaiciga and Mariño, 1985; McLaughlin and Velasco, 1990). We will speak no more of this. Let us move on to deal with a strategy of great effectiveness and more general validity to tackle the curse of dimensionality in the water field.

[15] It is significant that McLaughlin and Velasco (1990) observe that the LQG framework can be applied only in reservoirs of large dimensions, in which the volume of inflow is very small compared to the effective volume of the storage.

[16] Note that the well-known Extended Linear Quadratic Gaussian (ELQG) approach proposed by Georgakakos (1989a) should not be considered as an extension of the LQG framework, but rather as an on-line control scheme (see Section 14.3.2).

12.2.3 Fixed-class costs-to-go: Neural Stochastic Dynamic Programming

In Section 12.2.1 we saw that to identify the Bellman function $H^*_\cdot(\cdot)$ of a given Problem it is necessary to determine a very high number of control values: precisely, $TN_{\mathbf{x}}$ in the case of a periodic automaton of period T whose discretized state set $\mathcal{S}_{\mathbf{x}_t}$ contains $N_{\mathbf{x}}$ elements at every instant t ($N_{\mathbf{x}} = N^{n_{\mathbf{x}}}$ when each of the $n_{\mathbf{x}}$ state components is discretized in the same number N of values). The effort required by this determination is often so formidable as to exceed our computation capacity. For example the design of a daily control policy of a network of three reservoirs, the state of each of which can assume 100 values, requires the specification of 365 million values to define the Bellman function. Their determination requires a large computing time and to memorize them, even only with simple precision, it is necessary to have a RAM of about 1.5 Gbyte capacity. And if the possible state values were 1000, the values to specify would rise to 365 billion and the RAM capacity required to 1500 Gbyte. It is the high number of values required to specify the Bellman function that makes the problem beyond us. To avoid the difficulty, it can be assumed that the values of the Bellman function are not completely independent of each other, but that relations exist between them that are known, at least partially, a priori. This is equivalent to supposing that the function $H^*_\cdot(\cdot)$ can be approximated with a function $\tilde{H}(\cdot, \boldsymbol{\theta}_\cdot)$, where for every t the parameter (vector) $\boldsymbol{\theta}_t$ has dimension r. Once the class $\tilde{H}(\cdot, \boldsymbol{\theta}_\cdot)$ is known, the identification of a function contained in it requires the computation and storage of only rT values. This is a vastly smaller number compared to the previous one, even when the dimension r might be of the order of 100. Moreover, since it is no longer necessary to explore the entire space of the cost-to-go functions, but only the sub-space of those belonging to the class $\tilde{H}(\cdot, \boldsymbol{\theta}_\cdot)$, it is no longer necessary, as we will see, to determine the optimal control in correspondence with all the state values $N_{\mathbf{x}}$, as required by the Functional Approach, but it is sufficient to do it for a significantly lower number of these. Once the best approximation $\tilde{H}(\cdot, \boldsymbol{\theta}^*_\cdot)$ is known we will in fact have at our disposal the following suboptimal policy

$$\tilde{m}_t(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \tilde{H}(\mathbf{x}_{t+1}, \boldsymbol{\theta}^*_{t+1}) \right] \tag{12.31}$$

subject to the constraints (12.8b)–(12.8e), to compute the optimal control for any required value of $\mathbf{x}_t$. Compare it with (12.9). The function $\tilde{H}(\cdot, \boldsymbol{\theta}_t)$ is called *scoring function* or *approximate cost-to-go function* and, analogously, $\tilde{H}(\mathbf{x}_t, \boldsymbol{\theta}_t)$ is the *score* or *approximate cost-to-go* of the state $\mathbf{x}_t$. The form of $\tilde{H}(\cdot, \boldsymbol{\theta}_t)$ must be established a priori and generally the structure is assumed to be time-invariant, as in (12.31); the dependence on time is expressed only through the parameter $\boldsymbol{\theta}_t$. Once the value of the latter is established, it is essential that the evaluation of the function, in correspondence with a given value of the state $\mathbf{x}_t$, be very rapid, and thus one prefers scoring functions described by vectors $\boldsymbol{\theta}_t$ of low dimensionality, that is, in jargon, classes of functions described by *compact architecture*. To obtain good performance one determines the values of $\boldsymbol{\theta}_t$ so that, for every t, $\tilde{H}(\cdot, \boldsymbol{\theta}_t)$ is a good approximation of $H^*_t(\cdot)$. In conclusion, the identification of the scoring function takes place in two steps:

(1) choice of compact architecture;

(2) estimation of parameter $\boldsymbol{\theta}_t$ which minimizes, appropriately, the distance between $\tilde{H}(\cdot, \boldsymbol{\theta}_t)$ and $H^*_t(\cdot)$.

12.2.3.1 Architecture

Thus the first step is to identify a *good architecture* for the scoring function. If the form of the optimal cost-to-go were, for some reason, deducible a priori, i.e. without solving the Problem, a class that contains it would be the natural candidate, but this is very rarely possible. For this reason, one is forced to make the choice independently of the Problem under examination. Among the architectures which can be used, let us remember the linear polynomials (Bellman et al., 1963; Tsitsiklis and Van Roy, 1996), the cubic Hermite polynomials (Foufoula-Georgiou and Kitanidis, 1988) and the splines (Johnson et al., 1993). However, the most interesting architecture is that of the *Neural Networks* (NNs) (Castelletti et al., 2007), which we have already met in Section 4.4 and which is described in Appendix A8. One adopts the so-called multi-layer perceptron (MLP), which is a simple, compact and numerically stable architecture.

The validity of this architecture is further supported by the following result: via an MLP and a good choice of parameter $\boldsymbol{\theta}_t$ one can approximate arbitrarily well any function $H_t^*(\cdot) : \mathcal{S}_{\mathbf{x}_t} \to \mathbb{R}$ which is continuous on a closed and limited set of $\mathcal{S}_{\mathbf{x}_t}$, on condition that the number p of sigmoidal functions used in the hidden layers of the MLP is sufficiently large (see Cybenco, 1989; Funahashi, 1989; and Hornik et al., 1989).

12.2.3.2 Training

Having chosen the structure, the second step consists in identifying the weight of the network which best approximates the unknown Bellman function. The operation is called *network training*. If, but it is not our situation, m input/output pairs of the function to approximate were available, i.e. a set $\{(\mathbf{x}_t^i, H_t^*(\mathbf{x}_t^i)),\ i = 1, \ldots, m\}$ of observations, the training would be reduced to a simple least squares parametric estimation (see Section A4.2.2)

$$\boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \sum_{i=1}^{m} \left(H_t^*\left(\mathbf{x}_t^i\right) - \tilde{H}\left(\mathbf{x}_t^i, \boldsymbol{\theta}_t\right)\right)^2 \tag{12.32}$$

whose solution is particularly fast when $\tilde{H}(\cdot)$ is an MLP (see Appendix A8). Obviously the number m of observations must be much greater than the number r of the components of the vector $\boldsymbol{\theta}_t$.

In our situation, however, it is not possible to make use of measures $H_t^*(\mathbf{x}_t^i)$ and it is thus necessary to substitute them with estimates. Thus the training does not consist in the mere solution of problem (12.32), but is expressed by a more complex recursive procedure, which in the following we will indicate as *Neural Stochastic Dynamic Programming* (NSDP) (Bertsekas and Tsitsiklis, 1996). The NSDP is available for Problems defined both on a finite and an infinite horizon, but, in the second case, only with indicators of the TDC type, as with indicators of the AEV type the procedure could diverge. We will then illustrate this procedure in the case TDC, leaving it to the reader to derive the procedure to be used for Problems on the finite horizon.

The guiding principle is the following: if at a certain instant $t + 1$ we knew the optimal cost-to-go $H_{t+1}^*(\cdot)$, we could use the recursive equation

$$H_t^*\left(\mathbf{x}_t^i\right) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t\left(\mathbf{x}_t^i, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}\right) + \gamma H_{t+1}^*(\mathbf{x}_{t+1}) \right] \quad \forall \mathbf{x}_t^i \in \mathcal{S}_{\mathbf{x}_t}^m \tag{12.33}$$

to compute the optimal cost-to-go at time t in correspondence with a set $\mathcal{S}_{\mathbf{x}_t}^m = \{\mathbf{x}_t^i,\ i = 1, \ldots, m\}$ of values of the state $\mathbf{x}_t$ chosen a priori, for example distributed on a uniform grid in the state set $\mathcal{S}_{\mathbf{x}_t}$. We will thus have m values ($\mathbf{x}_t^i$, $H_t^*(\mathbf{x}_t^i)$) and we will thus be able to train a network for the time instant t by solving problem (12.32). We will thus obtain also the approximate cost-to-go $\tilde{H}(\cdot, \boldsymbol{\theta}_t^*)$.

If $\tilde{H}(\cdot, \boldsymbol{\theta}_t^*)$ were sufficiently similar to $H_t^*(\cdot)$, we could repeat the above for time $t-1$ as long as we substituted $\tilde{H}(\cdot, \boldsymbol{\theta}_t^*)$ for $H_t^*(\cdot)$ in the recursive equation (12.33); i.e.

$$\hat{H}_{t-1}\big(\mathbf{x}_{t-1}^i\big) = \min_{\mathbf{u}_{t-1}} \underset{\boldsymbol{\varepsilon}_t}{E}\big[g_{t-1}\big(\mathbf{x}_{t-1}^i, \mathbf{u}_{t-1}, \boldsymbol{\varepsilon}_t\big) + \gamma \tilde{H}\big(\mathbf{x}_t, \boldsymbol{\theta}_t^*\big)\big] \quad \forall \mathbf{x}_{t-1}^i \in \mathcal{S}_{\mathbf{x}_{t-1}}^m$$

If $\tilde{H}(x_t^i, \boldsymbol{\theta}_t^*)$ were sufficiently similar to $H_t^*(\cdot)$, then also the m estimates $\hat{H}_{t-1}(\mathbf{x}_{t-1}^i)$ should not be too different from the optimal costs-to-go $H_{t-1}^*(\mathbf{x}_{t-1}^i)$, which are unknown to us. We could thus assume the first in place of the second in problem (12.32), with which to train a network for the time $t-1$. We will thus obtain the approximation $\tilde{H}(\cdot, \boldsymbol{\theta}_{t-1})$. Proceeding iteratively, we could, therefore, determine an approximation $\tilde{H}(\cdot, \boldsymbol{\theta}_t^*)$ of $H_t^*(\cdot)$ for all the times t of the cycle.

Unfortunately, the procedure we have just set out cannot be applied, since no time instant exists for which the optimal cost-to-go is known. The only possibility that remains is that of initializing the procedure with an arbitrary cost and hoping that it converges. We thus obtain the following procedure

Algorithm 4 (*Total Discounted Cost (TDC) over an infinite horizon: NSDP*):

Step 0 (*Initialisation*): *Assuming a neural architecture $\tilde{H}(\cdot,\cdot)$, set $\tau = 0$, Term $= 0$ and*

$$\hat{\boldsymbol{\theta}}_t = \bar{\boldsymbol{\theta}}_t \quad t = 0, \ldots, T-1$$

with $\bar{\boldsymbol{\theta}}_t$ arbitrary, for example identically zero.

Step 1: *Set $t = \tau - 1$ and compute the parameter $\hat{\boldsymbol{\theta}}_t$ with the following procedure*:

1. *Determine m estimates $\hat{H}_t(\mathbf{x}_t^i)$ with the following recursive equation*

$$\hat{H}_t\big(\mathbf{x}_t^i\big) = \min_{\mathbf{u}_t} \underset{\boldsymbol{\varepsilon}_{t+1}}{E}\big[g_t\big(\mathbf{x}_t^i, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}\big) + \gamma \tilde{H}(\mathbf{x}_{t+1}, \hat{\boldsymbol{\theta}}_{t+1})\big] \quad \forall \mathbf{x}_t^i \in \mathcal{S}_{\mathbf{x}_t}^m \tag{12.34a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{12.34b}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \tag{12.34c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{12.34d}$$

$$\textit{any other constraints relative to } [t, t+1) \tag{12.34e}$$

where $\mathcal{S}_{\mathbf{x}_t}^m$ is a set of m preassigned states;

2. *Determine the value $\hat{\boldsymbol{\theta}}_t$ that solves the following optimization problem*

$$\min_{\boldsymbol{\theta}_t} \sum_{i=1}^{m} \big(\hat{H}_t\big(\mathbf{x}_t^i\big) - \tilde{H}\big(\mathbf{x}_t^i, \boldsymbol{\theta}_t\big)\big)^2$$

3. *If*

$$\left\| \tilde{H}(\cdot, \hat{\boldsymbol{\theta}}_t) - \tilde{H}(\cdot, \hat{\boldsymbol{\theta}}_{t+T}) \right\| < \alpha$$

with α small and preassigned, increase Term by one; otherwise set it equal to zero.

Step 2 (*Termination test*): *If Term $= T$ the algorithm terminates and the last T parameters $\hat{\boldsymbol{\theta}}_t$ computed are an estimate of the values of the parameters $\boldsymbol{\theta}_t^*$ which best approximate the Bellman function $H_\cdot^*(\cdot)$, which proves to be periodic of period T and is thus univocally defined by T time slices $H_\varsigma^*(\cdot)$, $\varsigma = 0, \ldots, T-1$. Therefore, one assumes*

$$H^*_{(t+\varsigma)_{\bmod T}}(\mathbf{x}_{t+\varsigma}) = \tilde{H}\left(\mathbf{x}_{t+\varsigma}, \boldsymbol{\theta}^*_{t+\varsigma}\right) \quad \forall \mathbf{x}_{t+\varsigma} \in \mathcal{S}_{\mathbf{x}_{t+\varsigma}}, \ \varsigma = 0, \ldots, T-1$$

Otherwise, set $\tau = t$ and return to Step 1.

Compare this algorithm with Algorithm 2. It was asked whether the second converged to the Bellman's function and Bertsekas (1976) proved that the answer is positive under very broad conditions. Now it is not possible to pose the same question for the present algorithm, for the simple reason that it is not certain that $H_\cdot^*(\cdot)$ belongs to the class of functions defined by the neural architecture which has been chosen. One must limit oneself to simpler questions, such as: does the algorithm converge? If the algorithm converges, does its limit possess some desirable properties? For example, is it close, in some sense, to $H_\cdot^*(\cdot)$? In the eventuality that it oscillates, do the oscillations stay in a bounded neighbourhood of $H_\cdot^*(\cdot)$? Bertsekas and Tsitsiklis (1996) showed that the answers to these questions are all affirmative under very broad hypotheses.

12.2.3.3 An example: the Piave policy

An example will permit us to appreciate better the advantages that the NSDP offers compared to the classical SDP. Consider again the Piave Project, described in the box on page 187, and let us suppose that we wish to evaluate the effects of the various combinations of MEF only on hydropower production.[17] This requires, as was explained in the box, the redesigning of the system regulation policy in correspondence with every combination of MEF. For this a Design Problem is formulated, assuming as objective the maximization of the economic value of the hydropower production. Since this is an economic variable, the form TDC imposed by the NSDP proves to be completely acceptable. The Problem must be solved in correspondence with every combination of the MEF values considered (three values for each of the three MEF = 27 combinations).

For a given combination of MEF, by adopting a grid $\mathcal{S}_{\mathbf{x}_t}$ of 420 points in the space state ($N_{\mathbf{x}} = 420$) at every time t, the SDP (Algorithm 2) has permitted the identification of the optimal policy in 41.37 hours of computation,[18] to which a value for the objective of €31.0 million corresponds (see the bars SDP$_{420}$ in the two charts of Figure 12.11). To evaluate the 27 combinations of MEF it is thus estimated that about 46 days of computation are necessary. To reduce this time, which is very high, we must consider a less dense grid, i.e. a set $\mathcal{S}_{\mathbf{x}_t}$ with a lesser number $N_{\mathbf{x}}$ of elements: assuming $N_{\mathbf{x}} = 54$, for the same MEF combination the computing time is reduced to 5.32 hours (6.0 days for all the work), but the

[17]The simplification is necessary, since we do not yet know how to formulate Problems with more than one objective. We will deal with this in Chapter 18.

[18]The computing time was measured on a Pentium II 350 Mhz processor.

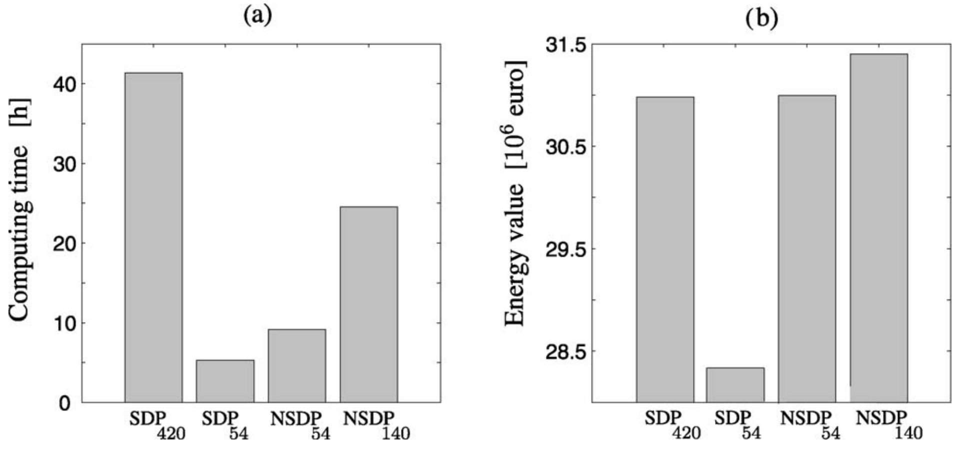


Figure 12.11: Design of the Piave regulation policy: comparison between classical SDP and NSDP for various discretizations of the space state and a give MEF combination: computing times (a) and performance obtained (b).

performance of the policy decreases to €28.3 million (bars SDP_{54} in the figure). This shows that the discretization is too sparse for the classical SDP. Adopting, however, the NSDP (Algorithm 4), still with $N_{\mathbf{x}} = 54$, the computing time rises to 9.13 hours (10.3 days for all the work), and the performance of the policy rises to €31.0 million (bars NSDP_{54}). The increase in time is due to the greater computational complexity of the NSDP compared to the SDP, while the better performance is explained by the intelligent interpolation performed by the neural network between the sparsely distributed points of the grid. To be sure that this discretization is enough to individuate the optimal policy one must try to solve the Problem with a denser discretization: letting $N_{\mathbf{x}} = 140$, the computing time rises to 24.53 hours (27.6 days) and the performance to €31.4 million (bars NSDP_{140}). It can thus be concluded that with $N_{\mathbf{x}} = 54$ the performance has not yet reached the maximum (the one which would be obtained with $N_{\mathbf{x}} = \infty$), but is quite close to it: in particular, it is significant that the same performance of €31.4 million would be obtained with the SDP by adopting $N_{\mathbf{x}} = 420$; but, with NSDP it is obtained with the 22% of the computing time required by the SDP.

12.2.3.4 Low-discrepancy sequences

The example of the Piave clearly shows how the NSDP allows the containment of the curse of dimensionality effect by acting on the number $N_{\mathbf{x}}$ of discretization points of the state: with a very sparse grid one can obtain results analogous to those that the SDP gives only with a much denser grid. Unfortunately, however, the NSDP does not circumvent the curse of dimensionality: the computational requirements of Algorithm 2 grow linearly with $N_{\mathbf{x}}$, but exponentially with the dimension $n_{\mathbf{x}}$ of the state ($N_{\mathbf{x}} = N^{n_{\mathbf{x}}}$). The NSDP is thus a partial remedy and we cannot use it when the state dimension is very high, indicatively superior to ten (Sharma et al., 2004). Fortunately, this is a decidedly large number, which does not preclude the adoption of the NSDP in most of the real applications and thus does not cancel its advantages. There remains, however, the problem of what to do when the state dimension exceeds this value. In the next paragraph we will show a strategy which enables us to avoid all the problem of dimensionality and that is, as a general rule, applicable to systems of any dimensions. In this paragraph we introduce a recently developed idea which allows the reduction of the growth of computing times of the NSDP from exponential to polynomial and thus extends its applicability to systems with dimensions greater than 10.

In Section 10.5, by introducing the discretization of the model, we mentioned the reasons which indicate the adoption of a *uniform discretization*. It assumes that all the $N^{n_\mathbf{x}}$ points on the grid[19] are equally important with the aim of obtaining a good approximation of the optimal costs-to-go: since we do not know their form, uniformity seems to be the most reasonable assumption. This intuition is confirmed by some Numerical Analysis results (Cervellera and Muselli, 2004) relating to the estimate of the value of a function $H_t(\mathbf{x}_t)$ defined in $\mathcal{S}_{\mathbf{x}_t}$ given the values that it assumes in P points $\mathbf{x}_t^i \in \mathcal{S}_{\mathbf{x}_t}$. In particular, the estimation error proves to be proportional to an index, called *discrepancy index*, which expresses the minimum density of the points $\mathbf{x}_t^i$ among all the subsets of $\mathcal{S}_{\mathbf{x}_t}$. Therefore, the smaller the undersampled regions the smaller is the error. A uniform discretization has a low discrepancy and thus is associated with a low estimation error. However, since the number P of points in a uniform grid is equal to $N^{n_\mathbf{x}}$, it is impossible to increase P continuously. For example, if $n_\mathbf{x} = 3$, P can assume the values $3^3, 4^3, 5^3, \dots$, but not intermediate values. The distance between the successive values of P increases exponentially with $n_\mathbf{x} - 1$: this is the origin of the curse of dimensionality. One therefore understands why it is important to individuate methods to be able to decide where to place a new point of discretization, given P points already placed. Clearly, it should be placed in such a way as to decrease as much as possible the discrepancy index, and various methods have been developed based on this idea (Fang and Wang, 1994; Niederreiter, 1992) which iteratively produce discretizations (called *low-discrepancy sequences*) such that their index decreases in an almost linear (polynomial) way with P. To understand the significance of this result, consider the case in which P points are already positioned and one wishes to add others so as to halve the discrepancy index and thereby halve the estimation error. By adopting low-discrepancy sequences, it is sufficient to add little more than P points, while with a uniform discretization it is necessary to add as many as $(2^{n_\mathbf{x}} - 1)P$. In other words, estimation errors being equal, the number of points of a low-discrepancy sequence no longer grows in an exponential way but in a polynomial way with the state dimension and this makes our Problem computationally possible. Some recent experiments (Cervellera et al., 2006; Baglietto et al., 2006) have demonstrated how by coupling NSDP and low-discrepancy sequences it is possible to solve Problems (on a finite or receding horizon) with a very high state dimension, even as high as 30 (in Cervellera et al., 2006 the system includes 10 reservoirs fed by as many catchment described by models AR(2)).

12.2.4 Fixed-class policies

In the preceding section we saw how the Parametric Approach offers significant computational advantages compared with the Functional one, since it limits the search for optimal cost-to-go functions within a class of functions whose form is defined up to a vector of parameters. Once the vector of parameters is individuated to which a good approximation of the optimal functions corresponds, the policy is defined by (12.9). However, the idea of performing the search in a parametric, rather than functional, space can be applied directly to the policy, as anticipated in Section 10.3.2. The idea is simple and promising: if, for every time instant t, we knew the class $\tilde{m}_t(\cdot, \boldsymbol{\theta}_t)$ of functions to which the control laws $m_t^*(\cdot)$ of

[19] To facilitate the explanation, we assume, as we have done until now, that all the state components are discretized in the same number N of values. However, what we are going to explain can easily be extended to the case in which every component is discretized in a different number of values.

the optimal policy $p^* \triangleq \{m_t^*(\cdot); t = 0, \dots, h-1\}$ belong, p^* would prove to be univocally defined by the sequence of vectors[20] $\{\boldsymbol{\theta}_0^*, \dots, \boldsymbol{\theta}_{h-1}^*\}$. In the following it will be useful to think of this sequence as a vector of vectors

$$\boldsymbol{\theta}^* = \left|\boldsymbol{\theta}_0^*, \dots, \boldsymbol{\theta}_{h-1}^*\right|$$

An estimate $\hat{\boldsymbol{\theta}}$ of the value of $\boldsymbol{\theta}^*$ could be obtained by solving the following Problem

The Off-line Control Problem (*Parametric Approach*):

$$J^* = \min_{\boldsymbol{\theta}} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\sum_{t=0}^{h-1} g_t\left(\mathbf{x}_t, \tilde{m}_t(\mathbf{x}_t, \boldsymbol{\theta}_t), \boldsymbol{\varepsilon}_{t+1}\right) + g_h(\mathbf{x}_h) \right] \tag{12.35a}$$

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \tilde{m}_t(\mathbf{x}_t, \boldsymbol{\theta}_t), \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, \dots, h-1 \tag{12.35b}$$

$$\tilde{m}_t(\mathbf{x}_t, \boldsymbol{\theta}_t) \in \mathcal{U}_t(\mathbf{x}_t) \quad t = 0, \dots, h-1 \tag{12.35c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, \dots, h-1 \tag{12.35d}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{12.35e}$$

$$\boldsymbol{\theta} = |\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_{h-1}| \tag{12.35f}$$

$$\textit{any other constraints} \quad t = 0, \dots, h-1 \tag{12.35g}$$

This Problem is derived from problem (12.1), but while the former is a functional problem, the latter is a Mathematical Programming Problem, which, as we know, is much simpler to solve. We also already know its solution algorithm, because problem (12.35) is completely analogous to problem (9.4), given that the vector of the parameters $\boldsymbol{\theta} = |\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_{h-1}|$ is completely equivalent to the vector $\mathbf{u}^p$, not only from the formal point of view, but also from the semantic point of view: the choice of $\boldsymbol{\theta}$ is, in fact, an act of planning. Finally, problem (12.35) can be solved without recourse to Bellman's recursive equation and thus it is no longer necessary for the system to be an automaton. The problem of dimensionality is thus completely overcome.

When the duration h of the horizon is greater than T it can be assumed that the control laws are periodic of period T. In this case it is possible to consider also the presence of a deterministic disturbance and describe its dynamics through a given scenario (almost always the historically measured sequence). The Problem thus becomes the following

The Off-line Periodic Control Problem (*Parametric Approach*):

$$J^* = \min_{\boldsymbol{\theta}} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \left[\sum_{t=0}^{h-1} g_{t \bmod T}\left(\mathbf{x}_t, \tilde{m}_{t \bmod T}(\mathbf{x}_t, \boldsymbol{\theta}_{t \bmod T}), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) + g_h(\mathbf{x}_h) \right] \tag{12.36a}$$

$$\mathbf{x}_{t+1} = f_{t \bmod T}\left(\mathbf{x}_t, \tilde{m}_{t \bmod T}(\mathbf{x}_t, \boldsymbol{\theta}_{t \bmod T}), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, \dots, h-1 \tag{12.36b}$$

$$\tilde{m}_{t \bmod T}(\mathbf{x}_t, \boldsymbol{\theta}_{t \bmod T}) \in \mathcal{U}_{t \bmod T}(\mathbf{x}_t) \quad t = 0, \dots, h-1 \tag{12.36c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_{t \bmod T}(\cdot) \quad t = 0, \dots, h-1 \tag{12.36d}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{12.36e}$$

[20] Naturally, $\boldsymbol{\theta}^*$ is the value of the parameter that specifies $m_t^*(\cdot)$ in the class $\tilde{m}_t(\cdot, \boldsymbol{\theta}_t)$.

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{12.36f}$$

$$\boldsymbol{\theta} = |\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_{T-1}| \tag{12.36g}$$

$$\textit{any other constraints} \qquad t = 0, \dots, h-1 \tag{12.36h}$$

In certain cases it is possible to reduce further the number of parameters by assuming that at every time instant the vector $\boldsymbol{\theta}_t$ is a known function $\boldsymbol{\theta}_t(\boldsymbol{\vartheta})$ of a vector $\boldsymbol{\vartheta}$ of non-time parameters, whose values can be estimated by solving problem (12.36) with respect to $\boldsymbol{\vartheta}$.

Note that problem (12.36) admits the existence of a deterministic disturbance and describes its dynamics through a given scenario. Nothing prevents the argument of the control law $\tilde{m}_t(\cdot, \boldsymbol{\theta}_t)$ being the pair $(\mathbf{x}_t, \mathbf{w}_t)$, instead of the single state $\mathbf{x}_t$.

A deterministic scenario can also be used to represent in a "model-free" way the components of the system whose outputs are not influenced by the control, for example the catchment. These components, in fact, can be represented by the recorded historical time series of their outputs, which assume the role of deterministic disturbance scenarios. Obviously, in this case, the control law must not depend on such disturbances, since they are not in fact known at time t. In this last case, it is no longer possible to compute in equation (12.36a) the expected value with respect to the disturbances. However, given that by hypothesis the system that generates the disturbances is cycloergodic, if the horizon h is sufficiently long, the expected value can be substituted by the mean value in this horizon.

To understand better the comment of the last paragraph, consider again the simple example introduced in Section 2.1. Knowing a time series $\mathbf{a}_1^{kT}$ of k years of inflows to the reservoir considered in the example, its policy is the solution of the following Problem

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \frac{1}{k} \sum_{i=0}^{k-1} \left[\sum_{t=iT}^{(i+1)T-1} g_{t \bmod T}\big(s_t, \tilde{m}_{t \bmod T}(s_t, \boldsymbol{\theta}_{t \bmod T}), a_{t+1}\big) \right] \tag{12.37a}$$

$$s_{t+1} = f_{t \bmod T}\big(s_t, \tilde{m}_{t \bmod T}(s_t, \boldsymbol{\theta}_{t \bmod T}), a_{t+1}\big) \qquad t = 0, \dots, kT-1 \tag{12.37b}$$

$$\tilde{m}_{t \bmod T}(s_t, \boldsymbol{\theta}_{t \bmod T}) \in \mathcal{U}_{t \bmod T}(s_t) \qquad t = 0, \dots, kT-1 \tag{12.37c}$$

$$a_1^{kT} \text{ given scenario} \tag{12.37d}$$

$$s_0 = \bar{s}_0 \tag{12.37e}$$

$$\boldsymbol{\theta} = |\boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_{T-1}| \tag{12.37f}$$

$$\text{other possible constraints} \qquad t = 0, \dots, kT-1 \tag{12.37g}$$

in which it is assumed that:

(1) $g_{t \bmod T}(\cdot)$ represents the stress of the maize on day t and is therefore a periodic function of period T, for example of the form[21]

$$g_{t \bmod T} = \left[\big(w_{t \bmod T} - R_{t \bmod T+1}\big(s_t, \tilde{m}_{t \bmod T}(s_t, \boldsymbol{\theta}_{t \bmod T}), a_{t+1}\big)\big)^+\right]^{12}$$

in which $w_{t \bmod T}$ is the water demand of the maize and $R_{t \bmod T+1}$ the release from the reservoir (see (5.10));

(2) the duration kT of the horizon is sufficiently long to allow us to assume the penalty on the final state to be zero;

[21] See Section 7.3.2 of PRACTICE.

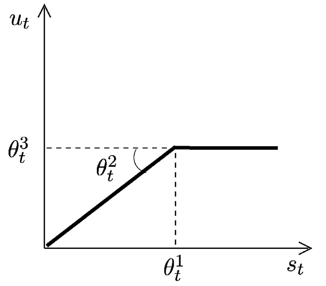


Figure 12.12: The family of control laws for the Problem in the example.

(3) s_t is the storage capacity of the reservoir, whose transition equation (12.37b) is expressed by (5.2).

Note how the control law $\tilde{m}_t(s_t, \boldsymbol{\theta}_t)$ cannot have as argument the inflow a_{t+1}, since this is not known at time t. Nothing prevents, however, one assuming $\tilde{m}_t(s_t, a_t, \boldsymbol{\theta}_t)$, thus treating a_t as the state of the catchment of which a *model free* representation is given with (12.37d).

To conclude, the approach seems to be brilliantly simple, but ... the difficulty lies completely in the premiss: "if one knew ..."; but how can one know the class $\tilde{m}_t(\cdot, \boldsymbol{\theta}_t)$, $t = 0, \dots, kT-1$, to which the control laws $m_t^*(\cdot)$ belong, if these are not known and we are seeking them? Clearly, if the class chosen were unsuitable, the policy would in any case be identified, but, in spite of being the best of the policies of that class, it could be very bad. It is advisable then to make use of a class of functions which is general enough to approximate the largest possible number of functions. In the preceding section we saw that of the classes that possess this property the class of Neural Networks (NNs), and in particular the MLP (see Appendix A8), undoubtedly constitutes the best choice. By utilizing the NNs, the sequence of control laws proves to be defined by a sequence of MLPs and the problems (12.35)–(12.37) become Non-linear Mathematical Programming Problems (due to the presence of the sigmoidal functions in the hidden layers of the perceptrons) which can be solved, for example, using some Direct Search Method (Zoppoli et al., in press) (see also Appendix A9). An example of the application of this approach is illustrated in Baglietto et al. (2006), where it is used to design the control policy of a network of ten reservoirs.

Not always is it necessary or opportune to have recourse to classes of functions which are as general as NNs: the choice of the number p of sigmoidal functions to use in the hidden layers is not always easy. When the state dimension of the system is small it can be easy for the Analyst to intuit the forms of the control laws. In the preceding example it is, for example, possible to intuit that, for every t, an appropriate class could be that of Figure 12.12. Given the form of the step cost it would be, in fact, short-sighted to release w_t every time that it is physically possible, since one could find oneself with the reservoir empty precisely when the demand is greatest.

Whether one uses the neural approach or one depends on the Analyst's intuition, there is no way to establish a priori if and to what extent the policy

$$\tilde{p} \triangleq \left\{ \tilde{m}_t(\cdot, \hat{\boldsymbol{\theta}}_t);\ t = 0, \dots, kT-1 \right\}$$

that has identified, which is the best in the fixed class, is a good policy. In both the cases it is possible to do this only a posteriori, comparing the results obtained with different classes on the basis of the value of the objective (12.35a) (or (12.36a) or (12.37a)) that they produce. This is, however, a completely subjective judgement, since we do not know the performance given by the optimal policy.

12.3 SV policies: Functional Design

Lastly, let us examine the case of SV policy design. We will only consider the Functional Approach, given that only for this are there well-tested algorithms.

The Problem to solve is now problem (11.13) and using the Proposition at the end of Chapter 11 we can identify a solution algorithm. The Proposition states: *the largest optimal* SV *policy is constituted by the totality of the optimal* PV *policies*. Therefore, to solve the Problem it is sufficient to determine the set of all the optimal policies that solve the corresponding PV Problem, defined by equation (11.16). At first glance, this might seem a very laborious task, but actually it is not so difficult: all the information relative to the optimal policies for the PV Problem is, in fact, embedded in its Bellman function. In Section 12.1.2 we saw that the controls provided by each optimal policy satisfy equation (12.9). It is thus logical to think that 'the set of all optimal policies' provides at time t all and only the controls provided by equation (12.9). This hypothesis is confirmed by the following theorem (Aufiero et al., 2001):

Theorem 1: *Given the Bellman function* $H_t^*(\cdot)$, $t = 0, \dots, h-1$, *associated to problem* (11.16), *for* $t = 0, \dots, h-1$ *consider the following* SV *control laws*

$$M_t^*(\mathbf{x}_t) = \Big\{\mathbf{u}_t\colon \mathbf{u}_t = \arg \min_{\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot)} \big[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*\big(f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1})\big)\big]\Big\} \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{12.38}$$

The policy

$$P^* \triangleq \big\{M_t^*(\cdot);\ t = 0, 1, \dots, h-1\big\}$$

is the largest SV *optimal policy, i.e. it is the solution to problem* (11.13).

The theorem is formulated for a Problem defined over a finite horizon, but can easily be extended to all the forms of objectives considered in Chapter 8. Clearly, the algorithm used to compute the Bellman function must be chosen from those presented in Section 12.1.4, taking into account the form of the indicator. Finally, by using the Law of Duality presented in Section 10.4, the theorem can be extended to the case in which the Wald criterion is adopted (Aufiero et al., 2002).

12.3.1 Examples

To thoroughly understand the algorithm it is advisable to examine how it works in practice. For this purpose, some simple numerical examples are presented in Chapter E2 of the CD. For an example of how SV policies can be useful in practice, see Weber et al. (2002) and Section 7.6.3 of PRACTICE.

Chapter 13

Off-line learning policies

AC and RSS

In the last chapter we described a class of algorithms, based on Stochastic Dynamic Programming (SDP), which solve the Control Problem with a Functional Approach, by posing very mild conditions to the characteristics of the problem. These algorithms, however, are affected by two serious limitations: the "curse of dimensionality" and their inability to deal with systems affected by deterministic disturbances. This second limitation is particularly problematic, since frequently the models do have a deterministic disturbance in their inputs: inflow models are fed by precipitation for instance. These models cannot be adopted in off-line designs, unless the disturbance is described either as a white process (thus, however, transforming the deterministic disturbance into a random disturbance), or with a dynamical model (but then the disturbance is no longer an exogenous variable for the global model, because the model turns out to be the union of the original model and the model of the disturbance). Furthermore, the first solution is rarely feasible without introducing significant distortions, since the process that generates a deterministic disturbance is hardly ever white; the second not only requires the construction of another model, but in addiction the resulting enlargement of the state makes the solution of the Problem very onerous in terms of computing time. In conclusion: to be able to apply the SDP-based algorithms to these systems, we must either assume that the disturbance is a white noise or drastically 'simplify' its model.

We are not willing to settle for either of these rough solutions. We would like to be able to design off-line policies whose arguments are not only the state variables of the system, but also the deterministic disturbances, such as the meteorological conditions or the inflow to a reservoir. From a theoretic viewpoint one could rightly argue that the requirement is absurd: by asking to include these variables among the arguments of the policy we implicitly affirm that they are state variables (in the two previous examples they describe the weather system or the catchment), but at the same time we do not intend to model them as such. In spite of this, we claim our right to be creative: we want this kind of policies, but we also want to compute them without having to model all the variables that appear among their arguments. This is our dream. The subject of this chapter is how we can make that dream reality.

The tool for realizing our dream is based on the ideas introduced in Section 10.3.2.2: it is an algorithm known in literature as *Q-learning* and was originally developed in the branch of the Artificial Intelligence (AI) which goes under the name of *Reinforcement Learning* (RL), but it is also a relative of SDP. For this reason, in the first of the next two sections we will summarize the main ideas of RL, and in the second we will show how to obtain Q-learning from SDP. Only in the third section will we return to the problem with which we started.

13.1 Reinforcement Learning

In the approach based on reinforcement learning, the system is left to evolve under the control of a suitable algorithm, which experiments with alternative controls until, by trial-and-error, it identifies the optimal policy. Each control that has been tested in a given situation is associated to a *factor*, i.e. a cost that quantifies the effects produced by that control. The factor is modified in time with learning. The algorithm usually chooses the control to apply from those whose factors have the lowest values, namely, those that produced the best effects; sometimes, nevertheless, it tries one of the others, since it must never stop exploring all the possibilities. At any time instant in its evolution it is possible to derive from the overall set of factors, the control policy that at that time appears to be the best one. This policy is modified as the tests proceed (i.e. as learning increases): sometimes for ever, sometimes until it becomes (cyclo)stationary. The first occurs when the system evolves slowly in time and so 'one never stops learning', the second when the system is (cyclo)stationary. When the latter is the case one says the policy "has been trained".

The Reinforcement Learning Algorithms[1] were born from the fusion of two research lines, which developed completely independently. The first concerns learning by trial-and-error and was initially the subject of psychological studies about animal learning, which then inspired many of the Artificial Intelligence ideas. The second is the search for the Optimal Control. Inevitably, the two lines proceeded totally independently for a long period: the first was linked to psychology, with strong philosophical connotations; the second to traditional engineering. It is worthwhile to describe briefly what happened, since this gives a historical perspective of what we have learned so far.

The term *Optimal Control* came into use in the late 1950s, more or less when Bellman (1957a), extending the theories proposed in the eighteen hundreds by Hamilton and Jacobi, devised a methodology (SDP) which hinged on the concepts of state and of optimal cost-to-go that we introduced in Section 12.1.1. Bellman (1957b) also extended SDP, which was initially defined only for the deterministic case, to the discrete and stochastic case,[2] which we presented in Section 12.1.2. In 1960 Howard proposed the SDP methods based on iteration in the policy space (*policy iteration*). Several studies (Bertsekas, 1982, 1983, 1996; White, 1969, 1985, 1993; and Lovejoy, 1991) developed and sharpened several aspects of SDP in an attempt to contain its computational burden (see Section 12.2.1).

The first to understand the essence of *trial-and-error learning* was probably Thorndike, who noted how actions which produce favorable (unfavorable) results tend to be reselected (substituted):

> *Of several responses made to the same situation, those which are accompanied or closely followed by satisfaction to the animal will, other things being equal, be more firmly connected with the situation, so that, when it recurs, they will be more likely to recur; those which are accompanied or closely followed by discomfort to the animal will, other things being equal, have their connections with that*

[1] To learn more about the history of this branch of AI and get an introductory review of the major algorithms that are part of it see Kaelbling et al. (1996) and Sutton and Barto (1998), from which the following is partly taken.

[2] This case is called *Markov decision process* (MDP) in the literature.

situation weakened, so that, when it recurs, they will be less likely to occur. The greater the satisfaction or discomfort, the greater the strengthening or weakening of the bond. (Thorndike, 1911)

Thorndike termed this process the *Law of Effect* because it describes the effect of the reinforcing events on the tendency to select actions. Although sometimes controversial, the Law of Effect is considered an evident principle underlying most behaviours. It contains the two most important aspects of learning by trial-and-error: first, it is selective, meaning that it chooses among the actions that have been already experienced, on the basis of their effects, and second, it is associative, since the actions that are selected are associated with particular situations (i.e. particular states). Natural selection in the evolution of species is an example of a selective process, but it is not associative; while supervised learning, in which the supervisor directs the pupil with a priori information about the outcomes of his/her choices, is associative, but not selective.

In other words, one can say that the Law of Effect is an elementary way to combine exploration and memory: exploration in the form of trying and selecting from many possible actions in each situation; memory in the form of recollection of those actions that had the best outcome in a given situation. This way of combining exploration and memory is the heart of the Reinforcement Learning Algorithms.

The idea of programming a computer, so that it will learn according to these principles, dates back to the first speculations about computers and intelligence (Turing, 1950). It was in that period that people guessed the importance that many psychological principles can have in the training of artificial systems.

The first to publish results of computer experiments with trial-and-error algorithms was Minsky (1954), who proposed a computer model for reinforcement learning and implemented it by designing an analog machine that he called SNARC (Stochastic Neural-Analog Reinforcement Calculator). SNARC was based on a model of the mental learning process: each agent, considered as a minimum constituent particle of the mind (Minsky, 1986), interacts with other agents; if agent A participated in stimulating agent B and this obtained a good result, it becomes more likely that A will stimulate B. SNARC was made up of 40 agents, each connected, more or less randomly, to many others through a reward system, which, activated after each positive outcome, increased the probability that each agent would re-stimulate the agents associated with the highest rewards.

The other big contribution was due to Samuel (1959), who, inspired by Shannon's argumentation about the possibility of programming a computer to play chess (Shannon, 1950), created a learning system that contained the outline idea for the Q-learning algorithm, which would take form thirty years later. In the following decade interest in the topic became weaker: the computers were too slow for the algorithms to provide meaningful results. In the seventies Klopf recovered the ideas developed by Samuel and Minsky in his works on 'generalized learning' (Klopf, 1972, 1975), but it was only at the beginning of the eighties, when the dramatic increase in computing capacity allowed the first significant experiments, that the real potential of artificial systems based on reinforcement learning was understood. In 1989 Watkins devised the Q-learning algorithm discussed in this chapter, an algorithm that applies the concepts of reinforcement learning to the resolution of Optimal Control Problems.

It is important to highlight how tight the relationship is between SDP and RL algorithms. The first requires a model of the system to be controlled and so linking it to Reinforcement Learning might appear unnatural; however, this would be a shortsighted conclusion. Not only because, as we have already observed in Section 10.3.2.2, Reinforcement Learning can be performed on a simulator, but also because RL algorithms, as well as SDP, operate in an incremental and iterative way, reaching the solution gradually, through successive approximations. On the other hand, also SDP 'performs' experiments on a simulator of the system and of the environment in which it operates, i.e. on the global model, proceeding backwards in time. We will explore this relationship more thoroughly in the next section.

13.2 From SDP to Q-learning

13.2.1 The Q-factor

In Section 12.1.1 the optimal cost-to-go $H_t^*(\mathbf{x}_t)$ was introduced as a measure of the desirability of the state $\mathbf{x}_t$, as it is the total cost that one expects to incur from t onwards when, starting from the state $\mathbf{x}_t$, an optimal policy is adopted. We also proved that when the optimal cost-to-go is known for all the values of $\mathbf{x}_{t+1}$, the optimal control $m_t^*(\mathbf{x}_t)$ at time t is the solution of the following problem

$$m_t^*(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \varepsilon_{t+1}) + \gamma H_{t+1}^*(\mathbf{x}_{t+1}) \right] \tag{13.1}$$

subject to the usual constraints, which, to simplify the notation, we will assume to be understood both here and in the following. Expression (13.1) constitutes a Control Problem with a TDC objective. Only this form will be considered because, as we will see, it is the only type of objective that Q-learning can deal with. The control law defined by equation (13.1) is referred to as *greedy*, because it pursues the optimal and nothing less than the optimal. A succession of greedy control laws constitutes a greedy policy. The greedy control law is completely coherent with the definition of $H_t^*(\mathbf{x}_t)$; we have in fact already seen that $H_t^*(\mathbf{x}_t)$ satisfies the following recursive equation (*Bellman* equation)

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma H_{t+1}^*(\mathbf{x}_{t+1}) \right] \tag{13.2}$$

which states that the optimal cost-to-go at time t is obtained by summing the cost produced in the interval $[t, t+1)$ (by adopting the control suggested by a greedy control law) to the optimal cost-to-go corresponding to the state which is produced as a consequence. Equation (13.2) is a recursive equation in $H_\cdot^*(\cdot)$, whose solution requires, for every time t, the evaluation of the costs produced by each feasible control $\mathbf{u}_t$ and the identification of their minimum value.

To this point there is nothing new with respect to what was presented in Section 12.1.2.

Note now that, instead of the optimal cost-to-go, which is associated to the state $\mathbf{x}_t$, we could consider a Q-factor associated to the state-control pair $(\mathbf{x}_t, \mathbf{u}_t)$, defined by the following expression

$$Q_t^*(\mathbf{x}_t, \mathbf{u}_t) = \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma H_{t+1}^*(\mathbf{x}_{t+1}) \right] \tag{13.3}$$

The function $Q_t^*(\mathbf{x}_t, \mathbf{u}_t)$ is known in the literature as a *Q-factor* and provides, given $\mathbf{x}_t$, the optimal cost-to-go at time t conditioned by the fact that the decision $\mathbf{u}_t$ is assumed at the first step and the greedy policy is adopted in the following steps.

The relationship between Bellman's optimal cost-to-go and the Q-factor can easily be inferred by comparing equation (13.2) to equation (13.3)

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t} Q_t^*(\mathbf{x}_t, \mathbf{u}_t) \tag{13.4}$$

From this relationship and equation (13.2) it follows that

$$Q_t^*(\mathbf{x}_t, \mathbf{u}_t) = \underset{\boldsymbol{\varepsilon}_{t+1}}{E}\left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{t+1}^*(\mathbf{x}_{t+1}, \mathbf{u}_{t+1})\right] \tag{13.5}$$

Like equation (13.2), equation (13.5) is also recursive, but in $Q_\cdot^*(\cdot)$, and its solution can be obtained with an algorithm that is analogous to the one used to solve equation (13.2). In fact, equation (13.4) allows the effects of the two theorems introduced in Section 12.1.4.3 (about the condition of optimality and the periodicity of the Bellman function) to be extended to the Q-factor. The first theorem, in particular, guarantees that $Q_\cdot^*(\cdot)$ is the sole solution to the following system of equations

$$Q_0(\mathbf{x}_0, \mathbf{u}_0) = \underset{\boldsymbol{\varepsilon}_1}{E}\left[g_0(\mathbf{x}_0, \mathbf{u}_0, \boldsymbol{\varepsilon}_1) + \gamma \min_{\mathbf{u}_1} Q_1(\mathbf{x}_1, \mathbf{u}_1)\right] \qquad \forall(\mathbf{x}_0, \mathbf{u}_0) \in \left(\mathcal{S}_{\mathbf{x}_0} \times \mathcal{U}_0(\mathbf{x}_0)\right) \tag{13.6a}$$

$$Q_1(\mathbf{x}_1, \mathbf{u}_1) = \underset{\boldsymbol{\varepsilon}_2}{E}\left[g_1(\mathbf{x}_1, \mathbf{u}_1, \boldsymbol{\varepsilon}_2) + \gamma \min_{\mathbf{u}_2} Q_2(\mathbf{x}_2, \mathbf{u}_2)\right] \qquad \forall(\mathbf{x}_1, \mathbf{u}_1) \in \left(\mathcal{S}_{\mathbf{x}_1} \times \mathcal{U}_1(\mathbf{x}_1)\right) \tag{13.6b}$$

$$\dots$$

$$Q_{T-1}(\mathbf{x}_{T-1}, \mathbf{u}_{T-1}) = \underset{\boldsymbol{\varepsilon}_0}{E}\left[g_{T-1}(\mathbf{x}_{T-1}, \mathbf{u}_{T-1}, \boldsymbol{\varepsilon}_0) + \gamma \min_{\mathbf{u}_0} Q_0(\mathbf{x}_0, \mathbf{u}_0)\right]$$
$$\forall(\mathbf{x}_{T-1}, \mathbf{u}_{T-1}) \in \left(\mathcal{S}_{\mathbf{x}_{T-1}} \times \mathcal{U}_{T-1}(\mathbf{x}_{T-1})\right) \tag{13.6c}$$

subject to

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \qquad t = 0, \dots, T-1 \tag{13.6d}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \qquad t = 0, \dots, T-1 \tag{13.6e}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, \dots, T-1 \tag{13.6f}$$

$$\text{any other constraints} \qquad t = 0, \dots, T-1 \tag{13.6g}$$

The solution to this system requires the identification of T functions $Q_t^*(\cdot)$, $t = 0, \dots, T-1$. From equation (13.1) we know that, once these functions are known, the optimal policy is given by

$$m_t^*(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} Q_t^*(\mathbf{x}_t, \mathbf{u}_t) \quad t = 0, \dots, T-1 \tag{13.7}$$

and so it is as periodic as $Q_\cdot^*(\cdot)$ is. The T functions $Q_t^*(\cdot)$ can be determined with the following algorithm, which is easily obtained by taking equation (13.4) into account, from Algorithm 2, which was proposed in Section 12.1.4.3.

Algorithm 1 (*Total Discounted Cost (TDC) over an infinite horizon: Q-learning*):

Step 0 (*Initialization*): *Set* $\tau = 0$, *Term* $= 0$ *and*

$$Q_t(\mathbf{x}_t, \mathbf{u}_t) = \bar{Q}_t(\mathbf{x}_t, \mathbf{u}_t) \quad \forall(\mathbf{x}_t, \mathbf{u}_t) \in \left(\mathcal{S}_{\mathbf{x}_t} \times \mathcal{U}_t(\mathbf{x}_t)\right),\ t = 0, \dots, T-1$$

with $\bar{Q}_t(\cdot)$ *being an arbitrary function, for example equal to zero.*

Step 1: *Given* $t = \tau - 1$ *compute*

$$Q_t(\mathbf{x}_t, \mathbf{u}_t) = \underset{\boldsymbol{\varepsilon}_{t+1}}{E}\left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{t+1}(\mathbf{x}_{t+1}, \mathbf{u}_{t+1})\right]$$

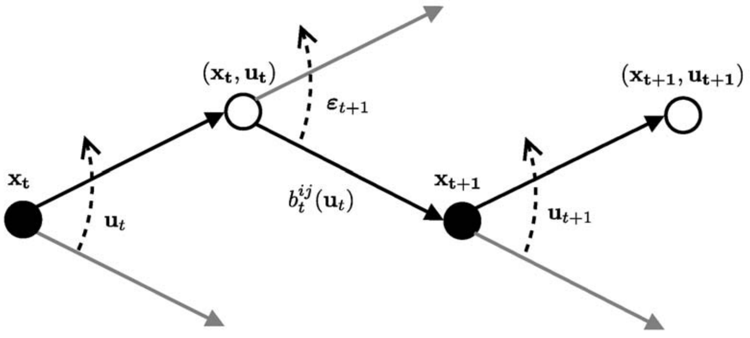


Figure 13.1: The graph of the transitions of the system.

$$\forall(\mathbf{x}_t, \mathbf{u}_t) \in \left(\mathcal{S}_{\mathbf{x}_t} \times \mathcal{U}_t(\mathbf{x}_t)\right) \tag{13.8a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{13.8b}$$

$$\mathbf{u}_{t+1} \in \mathcal{U}_{t+1}(\mathbf{x}_{t+1}) \tag{13.8c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{13.8d}$$

$$\textit{any other constraints relative to } [t, t+1) \tag{13.8e}$$

If

$$\left|Q_{t+T}(\mathbf{x}_t, \mathbf{u}_t) - Q_t(\mathbf{x}_t, \mathbf{u}_t)\right| < \alpha \quad \forall(\mathbf{x}_t, \mathbf{u}_t) \in \left(\mathcal{S}_{\mathbf{x}_t} \times \mathcal{U}_t(\mathbf{x}_t)\right)$$

with α being a given small number, increase Term by 1*, otherwise set it to zero.*

Step 2 (*Termination test*): *If Term $= T$ the algorithm stops and the last T computed functions $Q_t(\cdot)$ are a good approximation of the Q-factor $Q^*_\cdot(\cdot)$, which is periodic of period T and so it is univocally defined by T time slices $Q^*_\varsigma(\cdot)$, $\varsigma = 0, \ldots, T-1$. In other words one may assume*

$$Q^*_{(t+\varsigma)_{\mathrm{mod}\,T}}(\mathbf{x}_{t+\varsigma}, \mathbf{u}_{t+\varsigma}) = Q_{t+\varsigma}(\mathbf{x}_{t+\varsigma}, \mathbf{u}_{t+\varsigma})$$
$$\forall(\mathbf{x}_{t+\varsigma}, \mathbf{u}_{t+\varsigma}) \in \left(\mathcal{S}_{\mathbf{x}_{t+\varsigma}} \times \mathcal{U}_{t+\varsigma}(\mathbf{x}_{t+\varsigma})\right), \;\; \varsigma = 0, \ldots, T-1$$

Otherwise, set $\tau = t$ and go back to Step 1.

The difference between Algorithm 2 (page 301) and the one just defined is that in the latter the Q-factor $Q_t(\mathbf{x}_t, \mathbf{u}_t)$ is estimated at every step, but its minimum value with respect to $\mathbf{u}_t$ does not have to be determined: the operation is postponed to the next step, when the Q-factor $Q_{t-1}(\mathbf{x}_{t-1}, \mathbf{u}_{t-1})$ is computed and the values of $Q_t(\mathbf{x}_t, \mathbf{u}_t)$ have already been memorized for all the pairs $(\mathbf{x}_t, \mathbf{u}_t)$.

By observing system (13.6) it would seem that the control $\mathbf{u}_t$ was the second component of an enlarged state, defined by the pair $(\mathbf{x}_t, \mathbf{u}_t)$. This feeling is correct and makes it possible to deduce the condition of optimality (13.6) from equation (12.14) and to prove that a solution exists for such a condition. To understand this fact thoroughly, observe in detail how the system evolves (Figure 13.1). At time t, given state $\mathbf{x}_t$ we choose a control $\mathbf{u}_t$. By doing so we move to the pair $(\mathbf{x}_t, \mathbf{u}_t)$, from which we can reach a state $\mathbf{x}_{t+1}$ with a probability that depends on the pair $(\mathbf{x}_t, \mathbf{u}_t)$: this probability is, in fact, the element $b_t^{ij}(\mathbf{u}_t)$ of the Markov matrix, defined by equation (11.2), where i is the class of $\mathbf{x}_t$ and j the class of $\mathbf{x}_{t+1}$. In other words, first the system undergoes a deterministic transition from the state $\mathbf{x}_t$ to the pair $(\mathbf{x}_t, \mathbf{u}_t)$, then a purely stochastic (i.e. not controllable) transition from the pair $(\mathbf{x}_t, \mathbf{u}_t)$ to the state $\mathbf{x}_{t+1}$. The representation that was used in the previous chapters is based on the observation of the system whenever it is in a state such as $\mathbf{x}_t$ and it must therefore represent what occurs between $\mathbf{x}_t$ and $\mathbf{x}_{t+1}$. The Bellman function is associated to this representation. Nothing prevents us from observing the system when it is expressed by the pairs $(\mathbf{x}_t, \mathbf{u}_t)$, which means considering the evolution from $(\mathbf{x}_t, \mathbf{u}_t)$ to $(\mathbf{x}_{t+1}, \mathbf{u}_{t+1})$. The pair $(\mathbf{x}_t, \mathbf{u}_t)$ now appears as the new state, and by writing the Bellman equation for this system one obtains equation (13.5) itself. The Q-factor is none other than the optimal cost-to-go associated to the pair $(\mathbf{x}_t, \mathbf{u}_t)$. We may now intuitively understand why the optimality condition (13.6) might be completely equivalent to condition (12.14) and why the same theorems of uniqueness and existence of the solution that require the Problem to be periodic are also valid for it (see page 300).

Therefore, it would seem that the two representations are completely equivalent from a logical point of view, while from a practical point of view the classic representation (i.e. the one adopted by Bellman and used until this point) is certainly better, since it requires a smaller state dimension. This deduction is correct; however, we will see further on that the representation with the Q-factor is advantageous in few particular cases, but of great practical interest.

13.3 Model-free *Q*-learning

Algorithm 1 is only a reformulation of Algorithm 2 and as such they are completely equivalent. It does not contain any form of learning and, to be able to apply it, a system model (13.8b) and a model of the disturbance (13.8d) must be identified. We would now like to transform it into a learning algorithm that, as requested in Section 10.3.2.2, allows us to identify the best management policy, by learning it directly from experience, by trial-and-error, without a model of the system.

The idea is to assign a score, the Q-factor, to every possible event, which means to every triple $(t, \mathbf{x}_t, \mathbf{u}_t)$, so that we can discriminate between them. Let us formalize this idea to derive an algorithm. For that purpose suppose that at time t the system is in state $\mathbf{x}_t$ and that the control $\mathbf{u}_t$ is applied. In response to this control and to the disturbance $\boldsymbol{\varepsilon}_{t+1}$, the system evolves from $\mathbf{x}_t$ to $\mathbf{x}_{t+1}$. To evaluate the effect of the choice of $\mathbf{u}_t$ we can associate the sum of the immediate cost and the optimal cost-to-go of the state $\mathbf{x}_{t+1}$ to the pair $(\mathbf{x}_t, \mathbf{u}_t)$

$$g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{t+1}(\mathbf{x}_{t+1}, \mathbf{u}_{t+1}) \tag{13.9}$$

which constitutes the *reinforcement*[3] associated to the pair $(\mathbf{x}_t, \mathbf{u}_t)$. Nevertheless, this value is not very significant in itself, since it does not depend exclusively on the state-control pair $(\mathbf{x}_t, \mathbf{u}_t)$, but also on the disturbance $\boldsymbol{\varepsilon}_{t+1}$. In fact, in correspondence with the same pair

[3]More correctly, we should say "the penalty" and speak of "penalty learning", given that we are minimizing costs. Nevertheless, we prefer the term reinforcement, by now commonly used; after all, a penalty is nothing but a negative reinforcement.

$(\mathbf{x}_t, \mathbf{u}_t)$ the occurrence of a different disturbance could have produced a different reinforcement value. Thus, we must weigh up the experience just gained (i.e. the value (13.9) of the reinforcement just estimated) against the experience acquired in the past about this pair, or more precisely, about the event $(\bar{t}, \mathbf{x}_t, \mathbf{u}_t)$, with $\bar{t} = t_{\mathrm{mod}\, T}$, which is synthesized in the current estimate $Q_{\bar{t}}(\mathbf{x}_t, \mathbf{u}_t)$ of the Q-factor associated to it. In other words, we have to update the value $Q_{\bar{t}}(\mathbf{x}_t, \mathbf{u}_t)$ by taking into account the new value obtained for the reinforcement. The simplest way to do this is to assign[4] to it a new value that is the weighted sum of its current estimate and the reinforcement

$$Q_{\bar{t}}(\mathbf{x}_t, \mathbf{u}_t) \leftarrow (1-\lambda) Q_{\bar{t}}(\mathbf{x}_t, \mathbf{u}_t) + \\ + \lambda \Big[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{\overline{t+1}}(\mathbf{x}_{t+1}, \mathbf{u}_{t+1}) \Big] \tag{13.10}$$

where, to make the notation more compact, $t_{\mathrm{mod}\, T}$ is denoted with $\bar{t}$ and $(t+1)_{\mathrm{mod}\, T}$ with $(\overline{t+1})$. The coefficient λ is a *learning rate* that takes values between 0 and 1, and modulates the relative importance between the knowledge already acquired and the new experience. We will clarify its role better later on; for now just note that if λ were equal to one, the effects of past choices would not be taken into account, and so there would never be learning, just as there would be no learning if λ were zero, because it would not take the new experience into account and trust completely in what has already been learned. It is therefore clear that the value of λ must be equal to one only at the first attempt, i.e. when we have no experience. Then it must decrease progressively, tending asymptotically to zero, as experience grows. This is because, step by step, we are more certain and we are not impressed by unusual reinforcement values, which we are then able to recognize as such. Therefore, the value of λ must be a function of the number k of times in which the event $(\bar{t}, \mathbf{x}_t, \mathbf{u}_t)$ has been experienced, that is

$$\lambda = \lambda\big(k(\bar{t}, \mathbf{x}_t, \mathbf{u}_t)\big)$$

At this point, it is useful to introduce another aspect of human learning into the algorithm that we are conceiving: the possibility to learn 'by doing', i.e. without having prior knowledge about that particular reality, in other words, without a model: human learning is *model-free*. Similarly, we want the progressive estimate of the Q-factor to be performed through experiments on the physical system and not, as for Algorithm 1, by using a model of it (see equations (13.8b)–(13.8e)). It is surprisingly easy to obtain this result: if at time t we have applied the control $\mathbf{u}_t$ to the physical system, it is sufficient to wait for time $t+1$ to know the value assumed by $\mathbf{x}_{t+1}$ and $\boldsymbol{\varepsilon}_{t+1}$. It is therefore possible to update equation (13.10) without a model, provided that the training of the policy is carried out following the natural flow of time rather than working backwards from the future as Algorithm 1 does. On the other hand, this last condition is completely coherent: if we want to learn from the physical system, time becomes physical time, not simulated time, and so it can only flow from the past into the future.

At this point, we are able to formulate the Q-learning algorithm in its original *model-free* version, which was proposed by Watkins (1989).

[4] As is common practice in computer science, the symbol $\leftarrow$ denotes the operation of assignment.

Algorithm 2 (*Model-free Q-learning*):

Step 0 (*Initialization*): *Set* $\tau = 0$,

$$Q_t(\mathbf{x}_t, \mathbf{u}_t) = \bar{Q}(\mathbf{x}_t, \mathbf{u}_t) \tag{13.11a}$$

$$k(t, \mathbf{x}_t, \mathbf{u}_t) = 1 \tag{13.11b}$$

$$\forall(\mathbf{x}_t, \mathbf{u}_t) \in \big(\mathcal{S}_{\mathbf{x}_t} \times \mathcal{U}_t(\mathbf{x}_t)\big),\ t = 0, \dots, T-1$$

with $\bar{Q}(\cdot)$ *being an arbitrary function, for example, identically zero.*

Let $t = 0$ *be the initial learning time and* $\mathbf{x}_0^{\text{oss}}$ *the state of the system at that time.*

Step 1: *At time* t *arbitrarily choose and apply a control* $\bar{\mathbf{u}}_t \in \mathcal{U}_t(\mathbf{x}_t^{\text{oss}})$, *for example the greedy control*

$$\bar{\mathbf{u}}_t = \arg \min_{u_t \in \mathcal{U}_t(x_t^{\text{oss}})} Q_{\bar{t}}\big(\mathbf{x}_t^{\text{oss}}, \mathbf{u}_t\big)$$

but not always or only that one.[5]

Step 2: *Wait for the next time instant and observe the state* $\mathbf{x}_{t+1}^{\text{oss}}$ *that is reached and the disturbance* $\boldsymbol{\varepsilon}_{t+1}^{\text{oss}}$ *that occurred. On the basis of this new information update both the Q-factor and the value of* k

$$Q_{\bar{t}}\big(\mathbf{x}_t^{\text{oss}}, \bar{\mathbf{u}}_t\big) \leftarrow \big[1 - \lambda(\tilde{k})\big] Q_{\bar{t}}\big(\mathbf{x}_t^{\text{oss}}, \bar{\mathbf{u}}_t\big) + \\ + \lambda(\tilde{k}) \Big[g_t\big(\mathbf{x}_t^{\text{oss}}, \bar{\mathbf{u}}_t, \boldsymbol{\varepsilon}_{t+1}^{\text{oss}}\big) + \gamma \min_{\mathbf{u}_{t+1}} Q_{\overline{t+1}}\big(\mathbf{x}_{t+1}^{\text{oss}}, \mathbf{u}_{t+1}\big)\Big] \tag{13.12a}$$

$$\text{if} \qquad \tilde{k} = k\big(\bar{t}, \mathbf{x}_t^{\text{oss}}, \mathbf{u}_t\big) \tag{13.12b}$$

$$\mathbf{u}_{t+1} \in \mathcal{U}_{t+1}\big(\mathbf{x}_{t+1}^{\text{oss}}\big) \tag{13.12c}$$

and increase $k(\bar{t}, \mathbf{x}_t^{\text{oss}}, \mathbf{u}_t)$ *by* 1. *Remember that* $\bar{t} \triangleq t_{\text{mod}\,T}$ *and* $(\overline{t+1}) \triangleq (t+1)_{\text{mod}\,T}$.

Go to Step 1.

Note that the algorithm is without termination since, by developing itself on-line and so always acquiring new information, there is no reason to stop it. In other words 'one never stops learning'.

Watkins and Dayan (1992) proved that

(1) if for $t \to \infty$ every event $(\bar{t}, \mathbf{x}_t, \mathbf{u}_t)$ is visited an infinite number of times and

[5] There are several alternative ways of choosing the control. The simplest (Sutton and Barto, 1998) is to choose a control at random, once every so often, let us say with a small probability ϵ, assuming that the controls are distributed in a uniform way. In this case we are speaking of a ϵ-*greedy* control. Clearly, this way of proceeding implies that it is equiprobable to choose a worst or an optimal control. The most correct solution is to make the probability of the choice dependent on the value of the Q-factor (in this case $\bar{Q}(\cdot)$, in equation (13.11a), must be a constant). The greedy control is always the one that corresponds to the highest probability of the choice; nevertheless also the others can be chosen. This approach is known as the *softmax* choice of the control.

(2) if for $k \to \infty$, $\lambda(k)$ assumes progressively decreasing values, such that the following conditions are satisfied

$$\sum_{k=1}^{\infty} \lambda(k) = \infty \tag{13.13}$$

$$\sum_{k=1}^{\infty} \big[\lambda(k)\big]^2 < \infty \tag{13.14}$$

then $Q_{\bar{t}}(\mathbf{x}_t, \mathbf{u}_t)$, computed with Algorithm 2, tends, with probability 1, to the Q-factor $Q^*_{\cdot}(\cdot)$ to which Algorithm 1 converges when the Problem is periodic.

Observe that the first condition requires, in particular, that all the controls must be tried an infinite number of times, which means that not only must the controls suggested by the greedy policy be implemented, but it is necessary to test and re-test all the controls to infinity. Finally, note that Algorithm 2 also works if state $\mathbf{x}_t$ is substituted with pair $(\mathbf{x}_t, \mathbf{w}_t)$, when the system is affected by a deterministic disturbance,[6] or even with the information $\Im_t$ that we think is significant for controlling the system.

Before moving on, we will devote a little space to the choice of the learning rate.

13.3.1 The learning rate

The value of the learning rate λ is a function of the number $k(\bar{t}, \mathbf{x}_t, \mathbf{u}_t)$ of times that an event has been tested. It decreases as the number of tests grows, with the aim of progressively decreasing the weight given to recent experience and increasing the weight of past experience. Equation (13.13) guarantees that the values of $\lambda(k)$ are always large enough to overcome the effect of any unusual disturbance values or of a poor initialization of the algorithm (see equation (13.11a)), while condition (13.14) guarantees that sooner or later $\lambda(k)$ becomes small enough to assure the convergence of the Q-factor.

Therefore, the problem is to devise a function that meets these convergence conditions.

The simplest way[7] is to define $\lambda(k)$ with an expression in which the number k of visits appears in the denominator; the most common expression of this type is the following

$$\lambda(k) = \frac{a}{b+k} \quad \text{with } a, b \geqslant 1 \tag{13.15}$$

which is assumed to hold for $k > 1$, while for $k = 1$ λ is set to 1. The reason for the last setting is that the first time that an event is visited, there is no past experience (i.e. a current estimate of Q) that can be weighed against the results of that event. Note that when $a = b = 1$, equation (13.12a) provides the average value of the reinforcements obtained until that moment. In that case, the coefficients $(1 - \lambda)$ and λ, which weight past experience and the actual reinforcement, become $(k/(1 + k))$ and $(1/(1 + k))$, which are the same two coefficients that appear in the recursive estimator of the average.

Expression (13.15) satisfies equation (13.14) and so, as the number of times that an event is experienced increases, $\lambda(k)$ tends towards zero. However, since $\lambda(k)$ is the weight given to the recent experience (see equation (13.12a)), it follows that, as time goes on, the Q-factor becomes insensitive to new information. If, instead, $\lambda(k)$ were assumed to be constant, recent experience would always have the same weight and thus, if the system

[6] We know that in such a case the pair $(\mathbf{x}_t, \mathbf{w}_t)$ is the sufficient statistic: see Section 10.1.

[7] For other more complex expressions see Bertsekas and Tsitsiklis (1996) or Sutton and Barto (1998).

changed slowly through time, the policy would change to take this into account. The idea is correct, but unfortunately a constant λ does not satisfy condition (13.14) and thus does not guarantee the convergence of the algorithm. On the other hand, when the system changes in time such a convergence is not useful.

13.4 Partially model-free Q-learning

Algorithm 2 allows us to extend the principles of SDP to systems for which a model is not known and upon which a deterministic disturbance $\mathbf{w}_t$ acts. This extension is, however, feasible only by performing the learning process directly on the physical system and on-line, thus resulting in two disadvantages, which, in our case, are serious enough to render the algorithm useless:

1. In theory, in order to identify the optimal policy it is necessary to carry out, for each event, an infinite number of trials, i.e. evaluations of equation (13.12). Even settling for an approximation of the optimal policy requires carrying out at least several score. A water system is periodic with period one year, so any trial specific to time $t_{\text{mod}\,T}$ can be performed only once a year. Since, in a simple case, for every $\bar{t}$ the number of triples $(\mathbf{x}_{\bar{t}}, \mathbf{w}_{\bar{t}}, \mathbf{u}_{\bar{t}})$ is of the order of several thousand and, as we have seen, for each pair several score trials must be carried out, we can conclude that training a suboptimal policy for a water system would require many scores of thousands of years.[8]

2. During the training of a policy all of the controls have to be tested, so it is certain that controls that produce negative, if not disastrous, effects would also be tested. But since such controls must be actually applied to the physical system, adopting this algorithm would produce unsustainable costs.

Clearly, these two consequences were a problem in the context in which Q-learning was devised as well: Artificial Intelligence and in particular, the world of games, such as chess and back-gammon. Nevertheless, in that context these consequences are not so serious: if the algorithm plays against a human player it is able to make several hundred moves each day, but if the opponent is a copy of the same algorithm, as is almost always the case, the moves can be of the order of millions a day; in this case 'infinity' no longer seems so far away. As for mistaken decisions, the only consequence is losing the game. However, in the field of environmental decision-making the context is very different, and we cannot ignore this. It is thus necessary to modify the algorithm by retracing our steps: it would, in fact, be advisable to run trials on a simulator, rather than on the physical system. By doing so, both of the difficulties would be overcome in one hit: the damage would not be real and, being a simulated test, it would not be necessary to wait the actual time of the decision step to move from one decision to the next, and millions of trials could be performed each day.[9]

However, proceeding in this way seems like getting lost in a vicious circle: we want to avoid using a model and we return to requiring the use of one. This loop can be avoided, however. Note that in a water system $\mathbf{u}_t$ never influences the whole system, but only one of

[8] If the state can assume 100 values and the control 40, and one settles for 40 trials for each event, approximately $100 \times 40 \times 40$ trials must be carried out for each day t, but since the day t occurs just once a year, the policy training requires 160 thousand years.

[9] In the simple example mentioned in the last footnote the training would thus be completed in only 85 minutes.

its subparts, the network of reservoirs and distribution canals, while the catchment and the weather system are not influenced by it at all. Furthermore, these are the very two subsystems that are the most difficult to describe and about which we have the least information, while the dynamics of the reservoirs is described by very simple equations, whose parameters can be measured and about whose validity we have no doubts. In other words, in a water system there is a subsystem for which we have good a priori knowledge and, luckily for us, this is exactly the part of the system that is influenced by the control. It is from here that our idea (Castelletti et al., 2001) of an algorithm, which we called the *Q-learning planning* (QLP) algorithm, was born. This algorithm exploits the *model-free* approach only for the catchment and the weather system while using a model for the remaining part.

Let us formalize this idea. We assume that the system can be partitioned into two cascade subsystems: the first, which we will call the *upstream subsystem*, has an output vector $\mathbf{l}_{t+1}$ that influences the second, the *downstream subsystem*. Only this second subsystem is influenced by $\mathbf{u}_t$ and it is in it that the damage registered by the step cost $g_t(\cdot)$ is produced. At each time instant t one knows the deterministic disturbance $\mathbf{w}_t$, which always acts on the upstream subsystem and sometimes also on the downstream one. At the end of the interval $[t, t+1)$ the resulting output $\mathbf{l}_{t+1}$ from the upstream subsystem is also known. We have a model

$$\check{\mathbf{x}}_{t+1} = f_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{t+1}) \tag{13.16}$$

of the downstream subsystem, where the vector $\check{\mathbf{x}}_t$ describes the state. In other words, among the state variables of the global model (i.e. the model (13.8b) that in reality we do not know) only those variables whose dynamics we are able to describe, and that are influenced by the control $\mathbf{u}_t$, the deterministic disturbance $\mathbf{u}_t$ and the (random) disturbance $\mathbf{l}_{t+1}$, are elements of vector $\check{\mathbf{x}}_t$. About the latter, we know only that it also depends on the vector $\mathbf{w}_t$, even if we are not able to say how. The only thing we know about the upstream subsystem is a time series $\{(\mathbf{w}_t, \mathbf{l}_{t+1})\}_{t=0}^{h}$ of pairs $(\mathbf{w}_t, \mathbf{l}_{t+1})$. The traditional approach, the one followed until now, is based on the use of this series to identify a model of the upstream subsystem, which, together with model (13.16), constitutes the global model with which the policy is designed. Unlike in the traditional approach, we want to employ the series as it is and train the policy through it.

In order to clarify this idea, consider the following example: think of the usual system constituted by a catchment and a reservoir. The upstream subsystem is the catchment and w_t is the precipitation p_t measured in the interval $[t-1, t)$ (alternatively w_t might be the inflow a_t in the same period). The downstream subsystem is the reservoir and so the variable l_{t+1}, which links the upstream subsystem to the downstream one, is the inflow. Finally, model (13.16) is expressed by equation (5.2)

$$s_{t+1} = s_t + a_{t+1} - R_t(s_t, u_t, a_{t+1})$$

where $R_t(\cdot)$ is defined by equation (5.10). Note that this last model hardly ever needs to be calibrated, since it does not contain parameters and the functions of maximum and minimum release that appear in equation (5.10) are usually known by construction. We have a previously registered time series of pairs (p_t, a_{t+1}) that we want to use to train a policy. We must thus devise an algorithm to do this.

The algorithm that we seek should be, in a certain sense, a combination of Algorithms 1 and 2. From the first algorithm we should inherit the possibility to perform simulated trials, by using the model of the downstream subsystem, given that we cannot carry out physical experiments. Nevertheless, we cannot use this algorithm as it is, because we do not have a model of the upstream subsystem. We do have, however, a time series of input–output measures of that subsystem, which we should try to use by exploiting the ability of Algorithm 2 to operate with the time series gradually, as they come about.

To identify the new algorithm, we observe first of all that the global system (the upstream and downstream subsystems together) can be thought of as a system that has a state $\mathbf{x}_t$ (which we do not know) and the deterministic disturbance $\mathbf{w}_t$ as an input. We know that the argument of the regulation policy must be the pair $(\mathbf{x}_t, \mathbf{w}_t)$, from which it follows that the argument of the Q-factor should be the triple $(\mathbf{x}_t, \mathbf{w}_t, \mathbf{u}_t)$. Nevertheless, we do not know the vector $\mathbf{x}_t$, nor do we have its measures, we have only the measures of some of its components, namely, those that constitute $\check{\mathbf{x}}_t$. Therefore, all that we can do is regulate the system on the basis of pair $(\check{\mathbf{x}}_t, \mathbf{w}_t)$ and construct a Q-factor for this purpose based on the triple[10] $(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t)$. This is not the best way to proceed, because $\check{\mathbf{x}}_t$ is not the state of the global system, and so the policy obtained in this way will be suboptimal; however, it is the best that we can do.

The guideline is thus the following: since we must carry out simulated experiments, it is advisable to proceed backwards in time, as in Step 1 of Algorithm 1; however, the update of the Q-factor at time t cannot be done with equation (13.8a), which in our context would assume the form

$$Q_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) = \underset{\boldsymbol{\varepsilon}_{t+1}}{E}\left[g_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{t+1}(\check{\mathbf{x}}_{t+1}, \mathbf{w}_{t+1}, \mathbf{u}_{t+1})\right]$$
$$\forall(\check{\mathbf{x}}_{t+1}, \mathbf{w}_{t+1}, \mathbf{u}_t) \in \left(\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{S}_{\mathbf{w}_t} \times \mathcal{U}_t(\mathbf{x}_t)\right)$$

because we neither know anything about the random disturbance $\boldsymbol{\varepsilon}_{t+1}$ that influences the global system (note that $\mathbf{l}_{t+1}$ is not a disturbance, but only an internal variable of the global system), nor do we have a model to compute $(\check{\mathbf{x}}_{t+1}, \mathbf{w}_{t+1})$ from $(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t)$. We must therefore substitute this equation with (13.12a) which assumes the following form

$$Q_{\bar{t}}(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) \leftarrow (1-\lambda) Q_{\bar{t}}(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) + \\ + \lambda\left[g_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{\overline{t+1}}(\check{\mathbf{x}}_{t+1}, \mathbf{w}_{t+1}, \mathbf{u}_{t+1})\right]$$

because it allows us to estimate the expected value with respect to the disturbance $\boldsymbol{\varepsilon}_{t+1}$ through a time average and does not require that all the possible values of $\mathbf{w}_t$ be considered, but only those that occurred in the past. We could feed this equation with the time series $\{(\mathbf{w}_t, \mathbf{l}_{t+1})\}_{t=0}^{h}$ as if this were a sequence of experiments conducted on the physical system, because it allows us to know the input $\mathbf{w}_{t+1}$ that occurs in the upstream subsystem following event $(t, \mathbf{w}_t)$. Since, by hypothesis, the dynamics of this subsystem is not influenced by $\mathbf{u}_t$, nor by $\check{\mathbf{x}}_t$, at each time t we can explore all the pairs $(\check{\mathbf{x}}_t, \mathbf{u}_t) \in (\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{U}_t(\check{\mathbf{x}}_t))$, just as we did in equation (13.8a), given that we can evaluate the state $\check{\mathbf{x}}_{t+1}$ produced by the quartet $(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{t+1})$ using model (13.16).

However, in the algorithm it is not necessary, nor is it convenient for the convergence rate, to follow the chronological succession of the events $(t, \mathbf{w}_t)$. Instead, at each time t,

[10] In the following we will assume that $\mathbf{w}_t$ assumes its value in a finite discrete set $\mathcal{S}_{\mathbf{w}_t}$, which is periodic of the same period T as the system. This is required by the fact that the algorithm that we are formalizing derives from Algorithm 1, which in turn inherits from SDP, from which it descends, the condition that the system is an automaton (see Chapter 12). If, as is almost always the case in practice, $\mathbf{w}_t$ is not a discrete variable, it is necessary to discretize it with the procedure presented in Section 10.5.

it is better to consider all the events $(\bar{t}, \mathbf{w}_t)$ that we know could potentially occur at that time, because they have occurred in the past.[11] To identify all these events, T sets $\mathcal{S}_{\mathbf{q}_j}$, $j = 0, \ldots, T-1$ must be generated with the following procedure:

1. Set $j = 0$.
2. Go through the time series $\{(\mathbf{w}_t, \mathbf{l}_{t+1})\}_{t=0}^{h}$ from $t = 0$ to h. When $t_{\text{mod}\,T} = j$ add triple $\{\mathbf{w}_t, \mathbf{l}_{t+1}, \mathbf{w}_{t+1}\}$ to the set $\mathcal{S}_{\mathbf{q}_j}$.
3. Increase j by 1 and go to Step 2 if $j < T$.

Each of the T sets $\mathcal{S}_{\mathbf{q}_j}$ that are constructed thereby (e.g. the jth) has as many elements as there are times t, such that $t_{\text{mod}\,T} = j$ in the series $\{(\mathbf{w}_t, \mathbf{l}_{t+1})\}_{t=0}^{h}$. Moreover, each one of these elements is made up of the value that $\mathbf{w}_t$ assumes at that moment, of the output $\mathbf{l}_{t+1}$ which is obtained from it, and of the input $\mathbf{w}_{t+1}$ that occurs at the next time instant. Each set $\mathcal{S}_{\mathbf{q}_j}$ is thus a set of elements, ordered according to the order with which they were registered. All together, the sets constitute an embryonal form of a model, which, for each time j $(= 0, \ldots, T-1)$, provides the values of $\mathbf{w}_t$ that occurred and the consequences $(\mathbf{l}_{t+1}, \mathbf{w}_{t+1})$ that each of them produced. Since the elements of $\mathcal{S}_{\mathbf{q}_{\bar{t}}}$ are no longer associated to the times t at which they were registered, but only to the time $\bar{t}$, we will denote them in the following as $\{\mathbf{w}_{\bar{t}}, \mathbf{l}_{\overline{t+1}}, \mathbf{w}_{\overline{t+1}}\}$. With these sets and with the ideas that we presented earlier, we can formulate the following algorithm.

Algorithm 3 (*Q-learning-planning*):

Step 0 (*Initialization*): *Set* $\tau = 0$, *Term* $= 0$ *and*

$$\begin{aligned} Q_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) &= \bar{Q}_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) \\ k(t, \mathbf{w}_t) &= 1 \\ \forall(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) &\in \left(\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{S}_{\mathbf{w}_t} \times \mathcal{U}_t(\check{\mathbf{x}}_t)\right), \; t = 0, \ldots, T-1 \end{aligned}$$

with $\bar{Q}_t(\cdot)$ *being an arbitrary function, identically zero for example.*

Step 1: *Set* $t = \tau - 1$ *and for each pair* $(\check{\mathbf{x}}_t, \mathbf{u}_t)$ *that belongs to* $(\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{U}_t(\check{\mathbf{x}}_t))$ *execute the following procedure*:

(a) *Set* $Q^p(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) = Q_{\bar{t}}(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t)$ *and consider the first element* $\{\mathbf{w}_{\bar{t}}, \mathbf{l}_{\overline{t+1}}, \mathbf{w}_{\overline{t+1}}\}$ *of the set* $\mathcal{S}_{\mathbf{q}_{\bar{t}}}$.

(b) *Compute*

$$\begin{aligned} Q_{\bar{t}}(\check{\mathbf{x}}_t, \mathbf{w}_{\bar{t}}, \mathbf{u}_t) \leftarrow \left[1 - \lambda(\tilde{k})\right] Q_{\bar{t}}(\check{\mathbf{x}}_t, \mathbf{w}_{\bar{t}}, \mathbf{u}_t) + \\ + \lambda(\tilde{k})\left[g_t(\check{\mathbf{x}}_t, \mathbf{w}_{\bar{t}}, \mathbf{u}_t, \mathbf{l}_{\overline{t+1}}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{\overline{t+1}}(\check{\mathbf{x}}_{t+1}, \mathbf{w}_{\overline{t+1}}, \mathbf{u}_{t+1})\right] \end{aligned}$$

with

$$\tilde{k} = k(\bar{t}, \mathbf{w}_{\bar{t}})$$

[11] Clearly, by considering this set of events one implicitly assumes that the upstream subsystem is a cyclostationary system.

$$\check{\mathbf{x}}_{t+1} = f_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{\overline{t+1}})$$
$$\mathbf{u}_{t+1} \in \mathcal{U}_{t+1}(\check{\mathbf{x}}_{t+1})$$

Increase $k(\bar{t}, \mathbf{w}_{\bar{t}})$ *by* 1. *Remember that* $\bar{t} \triangleq t_{\bmod T}$ *and* $(\overline{t+1}) \triangleq (t+1)_{\bmod T}$.

(c) *If in* $\mathcal{S}_{\mathbf{q}_{\bar{t}}}$ *there is a subsequent element, go with it to point* (b)*, otherwise the procedure terminates.*

When all the pairs $(\check{\mathbf{x}}_t, \mathbf{u}_t) \in (\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{U}_t(\check{\mathbf{x}}_t))$ *have been considered, go to the next step.*

Step 2: *If*

$$\left| Q_{\bar{t}}(\check{\mathbf{x}}_t, \mathbf{w}_{\bar{t}}, \mathbf{u}_t) - Q^p(\check{\mathbf{x}}_t, \mathbf{w}_{\bar{t}}, \mathbf{u}_t) \right| < \alpha \quad \forall (\check{\mathbf{x}}_t, \mathbf{w}_{\bar{t}}, \mathbf{u}_t) \in \left(\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{S}_{\mathbf{w}_{\bar{t}}} \times \mathcal{U}_t(\check{\mathbf{x}}_t)\right)$$

with α *being a given small number, increase Term by* 1*, otherwise set it equal to zero.*

Step 3 (*Termination test*): *If Term* $= T$ *the algorithm terminates, and the* T *functions* $Q_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t)$, $t = 0, \ldots, T-1$, *are a good approximation of the Q-factor* $Q^*_\cdot(\cdot)$*, which is periodic with period* T*, and is thus univocally defined by* T *time slices* $Q^*_t(\cdot)$*. In other words one can assume*

$$Q^*_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) = Q_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t)$$
$$\forall (\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_{t+}) \in \left(\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{S}_{\mathbf{w}_t} \times \mathcal{U}_t(\check{\mathbf{x}}_t)\right), \; t = 0, \ldots, T-1$$

Otherwise, set $\tau = t$ *and go to Step* 1.

Note that k is not a function of $(\bar{t}, \check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t)$, as one might expect from equation (13.12), but only of $(\bar{t}, \mathbf{w}_t)$. The reason is that at every step of the algorithm the Q-factor is updated the same number of times for all the pairs $(\check{\mathbf{x}}_t, \mathbf{u}_t)$.

Observe also that the algorithm can be applied even when at each time instant only the output $\mathbf{l}_{t+1}$ of the upstream subsystem is registered and so no measurement for $\mathbf{w}_t$ is available: it is sufficient to note that, even if the symbols $\mathbf{w}_t$, $\mathbf{w}_{t+1}$ and $\mathcal{S}_{\mathbf{w}_t}$ are cancelled, the algorithm still works.

With the aim of better understanding the utility of this last observation, consider our simple example again and assume that the only information available about the catchment is the historical series $\{a_t\}_{t=0}^h$ of inflow values. We can treat this case in two different alternative ways:

1. Assume that no measure of w_t is available and set $l_{t+1} = a_{t+1}$. The Q-factor that is identified with Algorithm 3 has the pair (s_t, u_t) as its argument and so from it, through equation (13.7), a policy is derived, which has s_t as its only argument. Therefore, this policy has the same argument as the one that we would obtain if we were to use the historical series to identify an AR(0) model, defining a TDC problem with it, and solving this problem with Algorithm 2 (page 301). Such an approach would oblige us, however, to assume the probability distribution class that describes the inflows a priori, and the quality of the policy obtained would depend on the correctness of this assumption.[12] By adopting Algorithm 3, instead, such a hypothesis is not necessary. Pay close attention: this is an advantage as long as we are not sure about the

[12] We are not forgetting that the assumption is supported by a hypothesis test, but in many cases it is not univocally conclusive for a given distribution.

class of probability distribution; it becomes a disadvantage, however, once this class is known.[13] In this last case, in fact, we have a priori information that Algorithm 3 is not able to use while Algorithm 2 can (page 301). Therefore, the first algorithm might not always be better than the second.

2. Assume that at every time t the last registered inflow is known, i.e. set $w_t = a_t$ and $l_{t+1} = a_{t+1}$, and derive the series of pairs of inflows $\{(a_t, a_{t+1})\}_{t=0}^{h-1}$ from the historical series $\{a_t\}_{t=0}^{h}$. The Q-factor that is identified with Algorithm 3 has the triple (s_t, a_t, u_t) as an argument and from it, through equation (13.7), a policy whose argument is the pair (s_t, a_t) is derived. This policy has the same argument as the one that we would obtain by using the historical series to identify an AR(1) model, defining with it a TDC problem, and solving it with Algorithm 2 (page 301). The advantage of Algorithm 3 is that it does not require any hypothesis about the model that links a_{t+1} to a_t. The observation that we made in the closure of the last point is still valid, however.

Note that the choice between the two alternatives is not completely arbitrary: the second requires much more computation than the first, since the space of the arguments of the Q-factor is three-dimensional rather than two-dimensional and, as we know, the computing time increases exponentially with the number of arguments of Q. Furthermore, the second alternative can be used only when, during the system management, the measure a_t of the last inflow is available at every time t: without this information we would not know how to feed the policy that we obtained.

13.4.1 On-line learning

Observe that once the off-line policy has been identified with Algorithm 3 it is possible to continue to make it learn over the whole time it is used. Note that for this reason, at time $t+1$, once the output $\mathbf{l}_{t+1}$ and the input $\mathbf{w}_{t+1}$ are known, one gets a new triple of values $\{\mathbf{w}_t, \mathbf{l}_{t+1}, \mathbf{w}_{t+1}\}$. With this triple, one can thus update the Q-factor $Q_t^*(\cdot)$ for each pair $(\check{\mathbf{x}}_t, \mathbf{u}_t)$ belonging to $(\mathcal{S}_{\check{\mathbf{x}}_t} \times \mathcal{U}_t(\check{\mathbf{x}}_t))$ with the following expression

$$
\begin{aligned}
Q_{\tilde{t}}^*(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) \leftarrow & \left[1 - \lambda(\tilde{k})\right] Q_{\tilde{t}}^*(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t) + \\
& + \lambda(\tilde{k}) \left[g_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{t+1}) + \gamma \min_{\mathbf{u}_{t+1}} Q_{\overline{t+1}}^*(\check{\mathbf{x}}_{t+1}, \mathbf{w}_{t+1}, \mathbf{u}_{t+1}) \right]
\end{aligned} \tag{13.17}
$$

with

$$\tilde{k} = k(\bar{t}, \mathbf{w}_{\bar{t}}) \tag{13.18}$$

$$\check{\mathbf{x}}_{t+1} = f_t(\check{\mathbf{x}}_t, \mathbf{w}_t, \mathbf{u}_t, \mathbf{l}_{t+1}) \tag{13.19}$$

$$\mathbf{u}_{t+1} \in \mathcal{U}_{t+1}(\mathbf{x}_{t+1}) \tag{13.20}$$

Then increase $k(\bar{t}, \mathbf{w}_{\bar{t}})$ by 1.

13.5 SV policies

Lastly, we examine the case in which one wants to design an SV learning policy. The on-line design of an SV policy requires us to determine all the values of the control that minimizes

[13] This happens, for instance, when the hypothesis test univocally suggests a given distribution.

the optimal cost-to-go at every time instant. It comes naturally to extend this idea to the case in which learning is also required: from equations (12.38) and (13.7) it follows that at every time t the control law that defines the largest optimal SV policy is given by

$$M_t^*(\mathbf{x}_t) = \left\{\mathbf{u}_t\colon\ \mathbf{u}_t = \arg\min_{\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t)} Q_t^*(\mathbf{x}_t, \mathbf{u}_t)\right\} \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t}$$

Chapter 14

On-line policies

AC and RSS

When designing an off-line policy (Section 10.1.2) it is always necessary to assume a number of scenarios a priori, sometimes without even being aware of it. For example, the trajectory of irrigation demand and the obvious hypothesis, almost always implicitly assumed, that all the hydropower plants are continuously operational are both scenarios. We will call these scenarios the *nominal scenarios*.

During the application of the policy, however, it is possible that, due to an unusually cold spring, the irrigation season will be late and so the irrigation demand will be lower than the nominal one, or that a hydropower plant is under maintenance and therefore should not be fed. In these situations we do not really know what to do: in order not to waste the water resource, one would be tempted to reduce the release with respect to what the policy suggests; however, by doing so, the storage in the reservoir might become too high and, as a consequence, the risk of flooding would increase. Even the behaviour of random disturbances could deviate significantly from the outcome of the model used in the policy design: for example it is possible that in the month of February ten days of persistent Sirocco might cause a rapid snow melt despite the fact that this has never occurred before in the past. Again, the control suggested by the policy would be inadequate. But how can we choose a better one?

The answer to these questions is always the same: we must redesign the policy on-line (*On-line Adaptive Control*). As we will see, this does not mean that the old off-line design is useless, because its results will be used in the on-line design.

However, if we think for a moment we realize that an on-line policy is not necessary to solve these difficulties. The situations described above can easily be managed by an off-line policy, as long as the state of the model that describes the system is large enough to represent the events that we mentioned. For example, to take account of the possibility that a cold spring might modify the irrigation demand, it is sufficient to introduce a model of the crop. Thereby, the irrigation demand is no longer a scenario, but becomes an output of the global model; a particularly cold spring will produce an unusual state value for the crops, and the policy which has it as an argument will thus provide a release that takes account not only of the reduced irrigation demand, but also of an increased risk of flooding. Similarly, if the model of the catchment included the volume of accumulated snow among the components of its state and the model of the weather system described the evolution of the temperature according to the direction of the wind, a sudden snow melt produced by a persistent Sirocco would be managed by the off-line policy without any difficulty. Even the case of a hydropower plant under maintenance could be treated in a similar way: the state of the whole hydropower system is in fact described by a vector of Boolean variables that indicate the plants that are operational and those that are under maintenance.

A more careful analysis reveals that not only is the on-line policy useless, but it could even provide a worse performance. As we will see in a while, the On-line Design Problem considers a smaller state than the one we have just spoken of, and so the performance obtained cannot be better. Therefore the on-line policy is a *suboptimal* policy.

At this point the reader may be disoriented and wonder why we introduced the notion of on-line policy. The answer is simple and should now be clear: the optimal policy exists, we know how to formulate its Design Problem, we have the algorithms to solve it, but in practice they cannot be effectively applied because the computing time, which increases exponentially with the dimension of the state, can be prohibitive. Therefore, even if the off-line Design Problem is always theoretically solvable, it might not be in practice. It is in these cases that the on-line policy is useful: it is suboptimal with respect to the best policy that potentially exists, but it is better than the one that we can actually obtain off-line. Clearly, the borderline that determines in operational terms whether a Problem can be solved or not is a receding boundary, which shifts with technology advances and computer speed, but even though it moves, it will always continue to exist. It follows that policies which cannot be determined today may be determinable in the future; but, even then, there will be conditions in which it will be necessary to resort to an on-line policy, because it provides performances that are better than those of the best off-line policy computable at that time.

Therefore, an on-line design is undertaken with the sole aim of dealing with a *reduced state*, i.e. a state which is smaller than the one of the model that correctly describes the system and the problem. We will call this model (i.e. the one we want to consider, but cannot) the *Complete Model*. Correspondingly, we will call the Problem formulated with it the *Complete Problem*. In a dual manner, we will term *Reduced Model* the model based on the reduced state and *Reduced Problem* the Problem formulated on the basis of it.

We will begin by examining in greater detail the basic features of the on-line design and the criteria that are used for reducing the state. We will then show the condition that an on-line policy must satisfy to be interesting and illustrate some formulations of the Design Problem that satisfy this condition. We will terminate the chapter by considering firstly the SV policies, and then a new problem that can be traced back to the design of an on-line policy: how can we make the decision-making step more frequent, when necessary (e.g. for flood control), than the one adopted by the off-line policy which is being used?

14.1 On-line design and reduced state

As anticipated in Section 10.1.2, while an off-line policy is expressed by a table, an on-line policy is defined by formulating and solving a Control Problem at time t, i.e. at the same time that one has to apply the desired control. For this reason, given that it makes no sense that planning decisions would be implemented at that very moment, such a Problem is always, and only, a Control Problem, i.e. a Pure Management Problem, formulated over a h long finite horizon. Since the Problem is repeated at every instant, this latter turns out to be a receding horizon (see (10.11)).

In conclusion, the control $\bar{\mathbf{u}}_t$ defined by the on-line policy is the solution of the following On-line Control Problem, or, we could also say, *the policy is the following On-line Problem*

The On-line Control Problem:

$$\bar{\mathbf{u}}_t = \arg \min_{\mathbf{u}_t, p_{[t+1,t+h)}} \mathop{E}_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=t+1,\dots,t+h}} \left[\sum_{\tau=t}^{t+h-1} g_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}) + g_{t+h}(\mathbf{x}_{t+h}) \right] \quad (14.1a)$$

$$\mathbf{x}_{\tau+1} = f_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}) \qquad \tau = t, \dots, t+h-1 \quad (14.1b)$$

$$m_\tau(\mathbf{x}_\tau) \triangleq \mathbf{u}_\tau \in \mathcal{U}_\tau(\mathbf{x}_\tau) \qquad \tau = t, \dots, t+h-1 \quad (14.1c)$$

$$\boldsymbol{\varepsilon}_{\tau+1} \sim \phi_\tau(\cdot|\mathbf{I}_t) \qquad \tau = t, \dots, t+h-1 \quad (14.1d)$$

$$\mathbf{w}_t^{t+h-1} \textit{ given scenario} \quad (14.1e)$$

$$\mathbf{x}_t \textit{ given} \quad (14.1f)$$

$$p_{[t+1,t+h)} \triangleq \left\{ m_\tau(\cdot);\ \tau = t+1, \dots, t+h-1 \right\} \quad (14.1g)$$

$$\textit{any other constraints} \qquad \tau = t, \dots, t+h-1 \quad (14.1h)$$

The length h of the horizon can depend on the time t considered.

Equation (14.1d) shows that the description of the random disturbance can be conditioned to exogenous information $\mathbf{I}_t$, such as precipitation, temperature or barometric pressure. This information, being known at time t, is definitively a deterministic disturbance, but we prefer to denote it with a symbol that is different from the one ($\mathbf{w}_t$) that is usually adopted, not only because it explicitly influences the disturbance, something that up to now we have never admitted, but above all because $\mathbf{I}_t$ does not only influence $\boldsymbol{\varepsilon}_{t+1}$, but all the disturbances in the horizon $[t, t+h)$ and there is no dynamical description given for this information. Furthermore, in equation (14.1e) there appears also the scenario $\mathbf{w}_t^{t+h-1}$ of a deterministic disturbance, e.g. the trajectory of releases from a reservoir managed by a third party.

The proposed control $\bar{\mathbf{u}}_t$ thus proves to be a function not only of the initial state $\mathbf{x}_t$, but also of the information $\mathbf{I}_t$ and the scenario $\mathbf{w}_t^{t+h-1}$. The argument of the policy is thus the triple $(\mathbf{x}_t, \mathbf{I}_t, \mathbf{w}_t^{t+h-1})$ and, despite this, the state of the system is again only $\mathbf{x}_t$. This should provide the above mentioned computational advantages. We know very well that at time $t+1$ the value $\mathbf{I}_{t+1}$ of the exogenous information will generally be different from $\mathbf{I}_t$, as we know also that the scenario $\mathbf{w}_{t+1}^{t+h}$ that will be provided at that instant (and that, for this reason, more properly, we should denote with $\mathbf{w}_{t+1/t+1}^{t+h}$) could, in general, predict different values for $\mathbf{w}_{t+1}, \dots, \mathbf{w}_{t+h-1}$ than those predicted by the scenario $\mathbf{w}_{t/t}^{t+h-1}$. However, we will not deal with this, since at time $t+1$ we will not adopt, as the control, the value $m^*_{t+1/t}(\mathbf{x}_{t+1})$ provided by the control law $m^*_{t+1/t}(\cdot)$, which was obtained by solving the On-line Control Problem (14.1) at time t. This way of proceeding would actually be typical of an off-line policy. Instead, to obtain the control $\bar{\mathbf{u}}_{t+1}$, at time $t+1$ we will formulate and solve a new Problem with the new data. As the form of the On-line Control Problem (14.1) reveals, the result of an On-line Problem is not the policy $p_{[t+1,t+h)}$ that appears in it, but just the control $\bar{\mathbf{u}}_t$. Every other computed value is only instrumental, and, as such, is useless once the Problem has been solved.

The penalty $g_{t+h}(\cdot)$ plays a key role: it must assure that the choice of $\bar{\mathbf{u}}_t$ is not influenced by the finite, and often very short, time horizon. In order for this not to bias the choice of $\bar{\mathbf{u}}_t$, $g_{t+h}(\cdot)$ must express the costs that will be obtained from time $t+h$ onwards, as a function of the state $\mathbf{x}_{t+h}$ which will result, taking into account that after time $t+h$ the best possible policy will be still used. Therefore $g_{t+h}(\cdot)$ should be the cost-to-go function of the *Complete*

Problem, i.e. the Problem that has the complete state. Clearly, such a penalty is unknown, because, if it were known, we would already have the optimal policy and we would not have to worry about finding a way to approximate it. It thus follows that we can only look for an approximation of the penalty, which could be, for example, the slice $H^*_{t+h}(\cdot)$ at time $t+h$ of the Bellman function $H^*_\cdot(\cdot)$ corresponding to the following Problem (Nardini et al., 1994)

$$\min_p \lim_{h\to\infty} \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}} \frac{1}{h+1}\sum_{t=0}^{h} g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{14.2a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \qquad t = 0, 1, \dots \tag{14.2b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \qquad t = 0, 1, \dots \tag{14.2c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, 1, \dots \tag{14.2d}$$

$$\mathbf{x}_0 \text{ given} \tag{14.2e}$$

$$p \triangleq \left\{m_t(\cdot);\ t = 0, 1, \dots\right\} \tag{14.2f}$$

$$\text{any other constraints} \qquad t = 0, 1, \dots \tag{14.2g}$$

which is a Control Problem over an infinite horizon (which could also have the form TDC, rather than AEV), in which neither the deterministic scenario nor the exogenous information $\mathbf{I}_t$ are taken into account, because it is impossible, and in which the reduced state system is considered, i.e. the same system that appears in the On-line Control Problem (14.1).

14.1.1 State reduction

Finally, we can examine more closely what a *reduced* state is, how it can be obtained, and in which sense the form of the On-line Control Problem (14.1) follows from the state reduction.

To do this we will rely on the simple example introduced in Section 2.1, in which we consider a reservoir that is fed by a catchment. Suppose that the inflow from the latter has a log-normal distribution and it is an AR(1) process, i.e. it is perfectly described by an AR(1) model (see Appendix A6 on the CD). If the evaporation from the reservoir is negligible, the global model is the following

$$s_{t+1} = s_t + a_{t+1} - R_t(s_t, u_t, a_{t+1}) \tag{14.3a}$$

$$y_{t+1} = \alpha y_t + \varepsilon_{t+1} \tag{14.3b}$$

$$a_{t+1} = \exp\left(\sigma^a_{t+1} y_{t+1} + \mu^a_{t+1}\right) \tag{14.3c}$$

$$\varepsilon_{t+1} \sim N\left(0, \left(\sigma^\varepsilon\right)^2\right) \tag{14.3d}$$

where $R_t(\cdot)$ is defined by equation (5.10), which, both here and in the following, we take as being understood to simplify the expressions. Its state is thus the pair $[s_t, y_t]$. The design of the optimal policy for this system can be performed off-line, by solving a TDC (or AEV) Problem, with Algorithm 2 (or 3) presented in Chapter 12. According to the hypotheses made, this is the correct formulation of the Problem, which is thus the one we call the Complete Problem. It is clearly two-dimensional. Today, a two-dimensional Problem requires about a day of computation to be solved on a personal computer, but only eight years ago it was at the limits of solvability, as it required about a month of computation. To develop our example, let us assume that the computing time is prohibitive and so a state reduction is compulsory. Thus, we must eliminate one, or both, of the state transition equations. However, equation (14.3a) cannot be eliminated, otherwise we could not even express our aim, which is the reservoir regulation, and the Problem would be totally distorted. The only thing we can do is to remove the inflow dynamics, thus obtaining the following reduced model

$$s_{t+1} = s_t + a_{t+1} - R_t(s_t, u_t, a_{t+1}) \tag{14.4a}$$

$$a_{t+1} = \exp\left(\sigma^a_{t+1}\eta_{t+1} + \mu^a_{t+1}\right) \tag{14.4b}$$

$$\eta_{t+1} \sim N\left(0, \left(\sigma^\eta\right)^2\right) \tag{14.4c}$$

The reduced state, therefore, has a single component (s_t); equation (14.4a) becomes the only state transition equation; and the inflow a_{t+1}, which appears in it, plays the role of a disturbance, which we will call *reduced disturbance*.[1] This disturbance is described by its a priori distribution, which is expressed by equations (14.4b)–(14.4c). In these equations a disturbance η_{t+1} appears, whose standard deviation σ^ε is clearly greater than the deviation of the disturbance of the Complete Model (see equation (14.3d)). Note that the value of the reduced disturbance, i.e. the value of the inflow a_{t+1}, does not depend on its past values. The inflow is thus described as a white disturbance, even if we know that it is actually an autoregressive process of the first order: this loss of information is the price of reducing the state's dimension. By formulating and solving the AEV (or TDC) Problem with this model, a policy can be obtained; however, such a policy, even though it is optimal for model (14.4), cannot, in practice, provide very good performances, because the model does not correctly describe reality and so the policy 'ignores' that the inflow is a correlated process.

Observe, however, that, without increasing the state's dimension, it is possible to design a policy that is better than the last one, provided that this new policy is on-line. In fact, at time t the value I_t of the last measured inflow a_t is known, and thus, by letting

$$y_t = \frac{\ln I_t - \mu^a_t}{\sigma^a_t} \tag{14.5a}$$

we can use equations (14.3b)–(14.3d) to compute the a posteriori[2] probability distributions of $a_{t+1|t}, a_{t+2|t}, \ldots, a_{t+h-1|t}$

$$a_{t+1|t} = \exp\left(\sigma^a_{t+1}(\alpha y_t + \varepsilon_{t+1}) + \mu^a_{t+1}\right) \tag{14.5b}$$

$$\begin{aligned} a_{t+2|t} &= \exp\left(\sigma^a_{t+2}(\alpha y_{t+1} + \varepsilon_{t+2}) + \mu^a_{t+2}\right) = \\ &= \exp\left(\sigma^a_{t+2}\left(\alpha^2 y_t + \alpha\varepsilon_{t+1} + \varepsilon_{t+2}\right) + \mu^a_{t+2}\right) \end{aligned} \tag{14.5c}$$

$$\ldots$$

$$a_{t+h|t} = \exp\left(\sigma^a_{t+h}\left(\alpha^h y_t + \sum_{i=1}^{h} \alpha^{h-i}\varepsilon_{t+i}\right) + \mu^a_{t+h}\right) \tag{14.5d}$$

Let us assume that the last considered time instant ($t + h$) is sufficiently distant in time so that the a posteriori probability distribution[3] of a_{t+h} is practically indistinguishable from its a priori distribution (see Figure 14.1 in which $h = 4$). In other words, let us assume that h

[1] Note that this term does not mean that the new disturbance has a smaller dimension than the disturbance in the Complete Model.

[2] With respect to when I_t is known.

[3] Expressed by a Probability Density Function (PDF), when the inflow is described as a continuous variable, or a Discrete Density Function (DDF), in the opposite case (see Appendix A2).

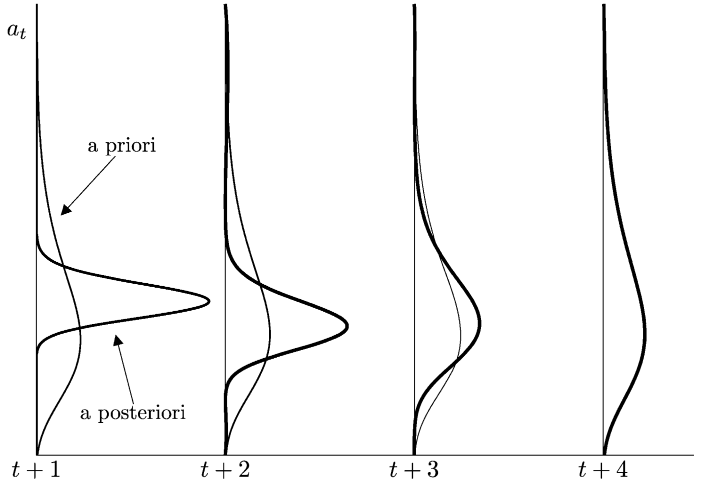


Figure 14.1: Moving forwards in time, the a posteriori probability distribution of a_t tends towards the a priori probability distribution.

is such that the knowledge of the last registered inflow value I_t is useless for forecasting the inflow value from $t+h$ onwards. Such a value for h exists for sure, since equations (14.5a)–(14.5c) show that the effect of I_t on $a_{t+h|t}$ decreases exponentially with h (remember that, due to stability, α is always less than 1). At time t we can thus formulate and solve the following On-line Control Problem, over the finite horizon $[t, t+h]$

$$\bar{u}_t^* = \arg \min_{u_t, p_{[t+1,t+h)}} \mathop{E}_{\{\varepsilon_\tau\}_{\tau=t+1,\dots,t+h}} \left[\sum_{\tau=t}^{t+h-1} g_\tau(s_\tau, u_\tau, \varepsilon_{\tau+1}) + H^*_{(t+h)_{\bmod T}}(s_{t+h}) \right] \quad (14.6a)$$

$$s_{\tau+1} = s_\tau + a_{\tau+1} - R_\tau(s_\tau, u_\tau, a_{\tau+1}) \qquad \tau = t, \dots, t+h-1 \quad (14.6b)$$

$$m_\tau(s_\tau) \triangleq u_\tau \in \mathcal{U}_\tau(s_\tau) \qquad \tau = t, \dots, t+h-1 \quad (14.6c)$$

$$a_\tau = \exp\left(\sigma_\tau^a \left(\alpha^{\tau-t} y_t + \sum_{i=1}^{\tau-t} \alpha^{\tau-t-i} \varepsilon_{t+i} \right) + \mu_\tau^a \right) \qquad \tau = t+1, \dots, t+h \quad (14.6d)$$

$$y_t = \frac{\ln I_t - \mu_t^a}{\sigma_t^a} \quad (14.6e)$$

$$\varepsilon_{\tau+1} \sim N\left(0, \left(\sigma^\varepsilon\right)^2\right) \qquad \tau = t, \dots, t+h-1 \quad (14.6f)$$

$$s_t \text{ given} \quad (14.6g)$$

$$p_{[t+1,t+h)} \triangleq \left\{ m_\tau(\cdot);\ \tau = t+1, \dots, t+h-1 \right\} \quad (14.6h)$$

$$\text{any other constraints} \qquad \tau = t, \dots, t+h-1 \quad (14.6i)$$

where $H^*_{(t+h)_{\bmod T}}(\cdot)$ is the slice at time $t+h$ of the Bellman function $H^*_\cdot(\cdot)$ associated to the following Problem

$$\min_p \lim_{h\to\infty} \mathop{E}_{\{\eta_t\}_{t=1,\dots,h}} \frac{1}{h+1} \sum_{t=0}^{h} g_t(s_t, u_t, \eta_{t+1}) \quad (14.7a)$$

$$s_{t+1} = s_t + a_{t+1} - R_t(s_t, u_t, a_{t+1}) \qquad t = 0, 1, \ldots \tag{14.7b}$$

$$m_t(s_t) \triangleq u_t \in \mathcal{U}_t(s_t) \qquad t = 0, 1, \ldots \tag{14.7c}$$

$$a_{t+1} = \exp\big(\sigma^a_{t+1}\eta_{t+1} + \mu^a_{t+1}\big) \tag{14.7d}$$

$$\eta_{t+1} \sim N\big(0, \big(\sigma^\eta\big)^2\big) \qquad t = 0, 1, \ldots \tag{14.7e}$$

$$s_0 \text{ given} \tag{14.7f}$$

$$p \triangleq \big\{m_t(\cdot);\ t = 0, 1, \ldots\big\} \tag{14.7g}$$

$$\text{any other constraints} \qquad t = 0, 1, \ldots \tag{14.7h}$$

The on-line policy defined by problem (14.6) will provide better performance than that produced by the policy designed by solving the AEV Problem formulated with the reduced model (14.4), and that (one hopes) is only a little inferior to those provided by the policy designed by solving the AEV Problem formulated with the Complete Model (14.3).

The pair of problems (14.6)–(14.7) is an example of the pair of problems (14.1)–(14.2). The information $\mathbf{I}_t$ that appears in the latter becomes the last measured value (a_t) of the inflow in this example; while the disturbance ε_{t+1} in equation (14.1b) becomes the inflow a_{t+1}, which acts as a disturbance in the state transition function equation (14.6b) and is defined by the triple (14.6d)–(4.6f) that depends on $\mathbf{I}_t$, just as ε_{t+1} does in equation (14.1d). The example shows that the reduction of the complete state $\mathbf{x}_t$ (the pair $[s_t, y_t]$) is performed by removing some of its components, chosen from among those that are not influenced by the control $\mathbf{u}_t$: e.g. we can eliminate either the components that represent the less significant parts of the system, or that have dynamics slower than the others, or that can be seen as disturbances in a simpler representation of the system (this last is the case in the example). The dynamics of the reduced state (s_t in the example) is described by a reduced model (14.6b), in which a reduced disturbance (a_{t+1}) appears. Generally, the reduced disturbance is not white, but in the formulation of the On-line Control Problem (14.1) we assume that it is and accept that its description can depend on the value assumed by some of the state variables that were eliminated, which then play the role of the exogenous information $\mathbf{I}_t$. The length h of the horizon of the On-line Control Problem can be fixed, or time-varying: e.g., it can be the duration of the time interval in which the a posteriori forecasts of the reduced disturbance, i.e. those conditioned to $\mathbf{I}_t$, are more significant than the a priori forecasts.

Whenever the dimension of the reduced state is such that it allows problem (14.2) to be solved, it is opportune to solve it in order to determine the Bellman function $H^*_\cdot(\cdot)$, from which the penalty $H^*_{(t+h)_{\bmod T}}(\cdot)$ of the On-line Control Problem is obtained.

14.2 Adaptive policies

Consider a Control Problem with a complete state, e.g. a problem defined over the finite horizon $[0, h]$

$$J^C = \min_p \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \left[\sum_{t=0}^{h-1} g_t(\mathbf{x}_t, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) + g_h(\mathbf{x}_h)\right] \tag{14.8a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) \qquad t = 0, \ldots, h-1 \tag{14.8b}$$

$$m_t(\mathbf{x}_t) \triangleq \mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \qquad t = 0, \ldots, h-1 \tag{14.8c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, \ldots, h-1 \tag{14.8d}$$

$$\mathbf{w}_t^{t+h-1} \text{ given scenario} \tag{14.8e}$$

$$\mathbf{x}_0 \text{ given} \tag{14.8f}$$

$$p \triangleq \left\{ m_t(\cdot);\ t = 0, \ldots, h-1 \right\} \tag{14.8g}$$

$$\text{any other constraints} \qquad t = 0, \ldots, h-1 \tag{14.8h}$$

The computing time required for its solution is considerable, because we want to determine the *closed-loop* optimal policy p^*. If the set of feasible controls $\mathcal{U}_t(\mathbf{x}_t)$ could be approximated with a set $\mathcal{U}_t$, which is independent of $\mathbf{x}_t$, and we settled for determining a sequence of optimal controls $\bar{\mathbf{u}}_0, \ldots, \bar{\mathbf{u}}_{h-1}$, rather than the optimal policy, the whole process would be simpler. We could, in fact, determine such a sequence by solving the following Problem

$$J^A = \min_{\mathbf{u}_0, \mathbf{u}_1, \ldots, \mathbf{u}_{h-1}} \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E} \left[\sum_{t=0}^{h-1} g_t(\mathbf{x}_t, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) + g_h(\mathbf{x}_h) \right] \tag{14.9a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) \quad t = 0, \ldots, h-1 \tag{14.9b}$$

$$\mathbf{u}_t \in \mathcal{U}_t \qquad t = 0, \ldots, h-1 \tag{14.9c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, \ldots, h-1 \tag{14.9d}$$

$$\mathbf{w}_t^{t+h-1} \text{ given scenario} \tag{14.9e}$$

$$\mathbf{x}_0 \text{ given} \tag{14.9f}$$

$$\text{any other constraints} \qquad t = 0, \ldots, h-1 \tag{14.9g}$$

which is easier to solve, because it is a parametric Problem and, as such, its solution can be obtained with Mathematical Programming techniques, e.g. with an evolutionary algorithm (see Appendix A9) that evolves in the space of the h controls $\mathbf{u}_0, \ldots, \mathbf{u}_{h-1}$. Note in fact that the Problem is of the same class as problem (9.4).

The sequence $\bar{\mathbf{u}}_0, \ldots, \bar{\mathbf{u}}_{h-1}$ that minimizes equation (14.9a) is called *open-loop optimal control sequence* and the value J^A that it produces is the *open-loop optimal cost*. Between the open-loop optimal cost J^A and the *closed-loop optimal cost* J^C there exists the following relationship

$$J^C \leqslant J^A$$

which is true for the simple reason that a sequence of open-loop controls is a particular case of policy: a policy in which the control laws provide the same control for every state. The difference $(J^A - J^C)$ quantifies the value of the information that is acquired during management and the aim of adopting a closed-loop policy is precisely to reduce the costs from J^A to J^C.

Thus, any suboptimal policy (e.g. an on-line policy) is acceptable, i.e. is of interest to us, only if the cost J^p that it produces satisfies the following inequality

$$J^C \leqslant J^p < J^A \tag{14.10}$$

If it does not, the information obtained by measuring the state at each time instant and by solving the problem on-line would be unfavourable, or useless, and it would be better to use an open-loop sequence of optimal controls. A control policy that satisfies the inequality

(14.10) is called *adaptive* policy, while it is called *quasi-adaptive* if the right side of the inequality is verified with the less-than-or-equal-to sign.

In conclusion, on-line policies are to be considered only if they are adaptive. Before adopting them, we must, therefore, ascertain whether they enjoy this property. In the following section we will see how.

14.3 Forms of On-line Problems

The form of problem (14.1) is the most complex of the forms that an On-line Control Problem can assume. Such forms were first studied by Bertsekas (1976) and we will now introduce them.

In the rest of the chapter, to simplify the description, we will omit the adjective 're-duced', taking it for granted that all the quantities and functions considered (state, disturbances, state transition functions, ...) refer to the reduced system, unless explicitly stated otherwise.

14.3.1 Naive Feedback Control

Naive Feedback Control (NFC) is the first and simplest of the forms that On-line Control Problems can assume. It is based on an idea that is often used in the field of automation: assume a given value $\bar{\boldsymbol{\varepsilon}}_{t+1}$ for the disturbances (e.g., the expected value), called *nominal value*, and then design the control policy for the deterministic system so obtained. Thereby, we are relying on the fact that, when the policy is applied, feedback can partly compensate for the effect of the disturbances, which are very unlikely to assume the nominal value at every time instant. Hence the NFC guideline: *at every time instant, apply the control that would be optimal if the disturbance assumed the nominal value at every future time*. The Problem considered in the NFC is therefore deterministic and, as such, an open-loop sequence of controls provides the same performances as a control policy (remember what we saw in Section 10.1.3). Since it is easier to determine a sequence of controls, we will pursue that option.

Let us denote the nominal values of the disturbance with $\bar{\boldsymbol{\varepsilon}}_{\tau+1}$, $\tau = t, \ldots, t+h-1$; the most common assumed value is $\bar{\boldsymbol{\varepsilon}}_{\tau+1} = E[\boldsymbol{\varepsilon}_{\tau+1}]_{|\mathbf{I}_t}$, i.e. the expected value of the disturbance conditioned to the exogenous information $\mathbf{I}_t$ available at time t. By adopting the NFC form the control $\bar{\mathbf{u}}_\mathbf{t}$ is determined at every time t by solving

The NFC On-line Control Problem:

$$\bar{\mathbf{u}}_t = \arg \min_{u_t, u_{t+1}, \ldots, u_{t+h-1}} \left[\sum_{\tau=t}^{t+h-1} g_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \bar{\boldsymbol{\varepsilon}}_{\tau+1}) + g_{t+h}(\mathbf{x}_{t+h}) \right] \tag{14.11a}$$

$$\mathbf{x}_{\tau+1} = f_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \bar{\boldsymbol{\varepsilon}}_{\tau+1}) \quad \tau = t, \ldots, t+h-1 \tag{14.11b}$$

$$\mathbf{u}_\tau \in \mathcal{U}_\tau \quad \tau = t, \ldots, t+h-1 \tag{14.11c}$$

$$\bar{\boldsymbol{\varepsilon}}_{t+1}^{t+h} \textit{ given scenario} \tag{14.11d}$$

$$\mathbf{w}_t^{t+h-1} \textit{ given scenario} \tag{14.11e}$$

$$\mathbf{x}_t \textit{ given} \tag{14.11f}$$

$$\textit{any other constraints} \quad \tau = t, \ldots, t+h-1 \tag{14.11g}$$

As we have already seen, whenever possible it is opportune, as penalty $g_{t+h}(\cdot)$, to assume the optimal cost-to-go $H^*_{(t+h)_{\mathrm{mod}\,T}}(\cdot)$ of the Problem defined over an infinite horizon. The optimal value of the objective that then appears on the right-hand side of equation (14.11a) will be denoted with J^{NFC}.

Problem (14.11) is a parametric and deterministic problem and, as such, it can easily be solved with Mathematical Programming techniques.

The NFC policy often provides good performance, and even the optimal performance when the objective is a quadratic function and the system is linear (this derives from the Certainty Equivalence Principle, see for example Bertsekas, 1976). Nevertheless, it can also be non-adaptive and so, before adopting it, it is always necessary to verify that it provides better performances than the open-loop sequence of optimal controls for the system and the problem that are being considered.

14.3.2 Open-Loop Feedback Control

The only difference between the *Open-Loop Feedback Control* (OLFC) Problem and the NFC one is due to the fact that in the former the uncertainty of the disturbances is taken into account explicitly in computing the control. The OLFC framework is thus more refined than the NFC's, and so the performances that it provides are presumably better.

By adopting the OLFC framework the control $\bar{\mathbf{u}}_t$ is determined, at every time t, by solving

The OLFC On-line Control Problem:

$$\bar{\mathbf{u}}_t = \arg \min_{u_t, u_{t+1}, \ldots, u_{t+h-1}} \mathop{E}_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=t+1,\ldots,t+h}} \left[\sum_{\tau=t}^{t+h-1} g_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}) + g_{t+h}(\mathbf{x}_{t+h}) \right] \tag{14.12a}$$

$$\mathbf{x}_{\tau+1} = f_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}) \quad \tau = t, \ldots, t+h-1 \tag{14.12b}$$

$$\mathbf{u}_\tau \in \mathcal{U}_\tau \quad \tau = t, \ldots, t+h-1 \tag{14.12c}$$

$$\boldsymbol{\varepsilon}_{\tau+1} \sim \phi_\tau(\cdot|\mathbf{I}_t) \quad \tau = t, \ldots, t+h-1 \tag{14.12d}$$

$$\mathbf{w}_t^{t+h-1} \textit{ given scenario} \tag{14.12e}$$

$$\mathbf{x}_t \textit{ given} \tag{14.12f}$$

$$\textit{any other constraints} \quad \tau = t, \ldots, t+h-1 \tag{14.12g}$$

The optimal value of the objective that appears on the right-hand side of equation (14.12a) will be denoted with J^{OLFC}.

Note that, for all the time instants between $(t+1)$ and $(t+h-1)$, the probability distributions $\phi_\tau(\cdot|\mathbf{I}_t)$ of the disturbances $\boldsymbol{\varepsilon}_{\tau+1}$ can be conditioned to the exogenous information $\mathbf{I}_t$ available at time t. As we have seen, this is particularly useful, because it allows us to generate such distributions through a forecaster. Although the OLFC Problem is more complex than the NFC one, it is still a parametric Problem, and, as such, it is solvable with Mathematical Programming techniques.[4] Bertsekas (1976) proved that OLFC policies are always

[4] It is important to note that, despite the name, the Extended Linear Quadratic Gaussian (ELQG) approach proposed by Georgakakos (1989a, 1989b), which encountered a wide diffusion and recognition in reservoir management practice, is precisely an example of OLFC. In fact, the LQG hypotheses on which it is based are not

quasi-adaptive. This means that the framework benefits from using the measures of state that become available in due course, even if, at each time instant, it 'assumes' that in the future no new measures of the state will be taken. It is not certain, however, that the OLFC policies always provide better performances than the NFC's: because of their suboptimal nature these two frameworks can produce counter-intuitive behaviours.

14.3.3 Partial Open-Loop Feedback Control

The *Partial Open-Loop Feedback Control* (POLFC) Problem is the most complex of on-line problems. It seeks a control policy at every time instant, because, unlike the OLFC problems, it 'assumes' that even in the future the state will be measured. Therefore, POLFC problems are functional problems and so they have a greater computational complexity than OLFC and NFC problems. However, one hopes that this is compensated by superior performances.

By adopting the POLFC framework the control $\bar{\mathbf{u}}_\mathbf{t}$ is determined at every time t by solving

The POLFC On-line Control Problem:

$$\bar{\mathbf{u}}_t = \arg \min_{\mathbf{u}_t, p_{[t+1,t+h)}} \mathop{E}_{\{\boldsymbol{\varepsilon}_\tau\}_{\tau=t+1,\dots,t+h}} \left[\sum_{\tau=t}^{t+h-1} g_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}) + g_{t+h}(\mathbf{x}_{t+h}) \right] \tag{14.13a}$$

$$\mathbf{x}_{\tau+1} = f_\tau(\mathbf{x}_\tau, \mathbf{u}_\tau, \mathbf{w}_\tau, \boldsymbol{\varepsilon}_{\tau+1}) \qquad \tau = t, \dots, t+h-1 \tag{14.13b}$$

$$m_\tau(\mathbf{x}_\tau) \triangleq \mathbf{u}_\tau \in \mathcal{U}_\tau(\mathbf{x}_\tau) \qquad \tau = t, \dots, t+h-1 \tag{14.13c}$$

$$\boldsymbol{\varepsilon}_{\tau+1} \sim \phi_\tau(\cdot|\mathbf{I}_t) \qquad \tau = t, \dots, t+h-1 \tag{14.13d}$$

$$\mathbf{w}_t^{t+h-1} \text{ given scenario} \tag{14.13e}$$

$$\mathbf{x}_t \text{ given} \tag{14.13f}$$

$$p_{[t+1,t+h)} \triangleq \left\{ m_\tau(\cdot);\ \tau = t+1, \dots, t+h-1 \right\} \tag{14.13g}$$

$$\text{any other constraints} \qquad \tau = t, \dots, t+h-1 \tag{14.13h}$$

which is none other than problem (14.1) that we introduced at the beginning of this chapter. The optimal value of the objective that appear on the right-hand side of equation (14.13a) will be denoted with J^{POLFC}.

Since the POLFC framework 'assumes' that it receives information in the future, it should provide performance that is not inferior to that provided by the OLFC framework, but it will probably be inferior to that provided by an off-line policy obtained by solving the Problem with the complete state. Clearly, POLFC and OLFC frameworks coincide when the horizon of the Problem is only one step ($h = 1$). The solution of problem (14.13) is carried out with Algorithm 1 (page 299), i.e. with SDP, with which it therefore shares the "curse of dimensionality".

Bertsekas (1976) proved that POLFC policies are quasi-adaptive and provide, generally, better performances than the OLFC ones, even if there are known examples in which this

introduced to compute an off-line feedback control law analytically, as in the traditional LQG approach (see Section 12.2.2); rather they are useful to solve an OLFC Problem with a very effective Mathematical Programming technique (based on the Newton method, see Section A9.3) to obtain an on-line open-loop policy.

does not occur. Nevertheless, because the three forms that we have examined are increasingly close to the form of the Closed-Loop Complete Problem, their performances are often ordered in the following way

$$J^C \leqslant J^{\text{POLFC}} \leqslant J^{\text{OLFC}} \leqslant J^{\text{NFC}} \leqslant J^A$$

where, as the reader will remember, J^C and J^A denote the performances obtained by solving the Closed- and Open-Loop Complete Optimal Control Problem respectively.

14.3.4 An example: application to the Verbano Project

An example will permit us to appreciate better the advantages that a POLFC scheme may offer in the daily management of a water system. Consider again the Verbano Project, described in the boxes on pages 74 and 78, and let us suppose that the planning process has been concluded and that the Italian and Swiss governments have agreed on the choice of the best compromise alternative. More precisely, we assume that among the reasonable alternatives they have chosen alternative $A34$ which foresees the excavation of the lake outlet (with an increase of 600 m^3/s of the outflow capacity), the modification of the regulation range (with the setting of its upper extreme to 1.5 m all through the year), and, finally, an optimal regulation policy p^{A34}, designed by taking into consideration, as objective of the Design Problem, a convex linear combination of flooding reduction around the lake and satisfaction of the downstream irrigation users[5] (for more details see Section 15.3.1 in PRACTICE).

We now hypothesize[6] that we are in the autumn of the year 2000 and that the Regulator is managing the system with the policy p^{A34}, i.e. that every day he decides to release a volume u_t^{A34} given by

$$u_t^{A34}(\mathbf{x}_t) = \arg\min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + \gamma H^{A34}_{t+1_{\text{mod}\,T}}(\mathbf{x}_{t+1}) \right] \tag{14.14}$$

where $H^{A34}_{t+1_{\text{mod}\,T}}(\cdot)$ is the optimal Bellman function corresponding to alternative $A34$, computed assuming the a priori probability distribution for the inflow $\boldsymbol{\varepsilon}_{t+1}$.

Starting from 19th September, three flood waves occur in succession (dot-dashed line in Figure 14.2a) that under the historic regulation produced two flooding events in correspondence with the second and third inflow peak (dotted line in Figure 14.2b). The level and release trajectories that would have been obtained with $A34$ are reported in Figure 14.2a and b (continuous line). Notice how the second flooding event (B) would have been completely avoided and the third (C) significantly reduced: this marked improvement is due to

[5] As we shall see in detail in Chapter 18, to solve a Multi-Objective Problem one first reduces it to a Single-Objective Problem (SOP) and then adopts one of the solution techniques that we have presented in this and in the preceding chapters. One of the most common methods for this reduction is the Weighting Method: the objective of the SOP is obtained as a weighted sum (with the sum of weights equal to one) of the individual objectives. Varying the weights, naturally one changes the regulation policy which solves the Design Problem. In the case of $A34$ the floods are weighted at 0.99 and the satisfaction of the irrigation users at 0.01. The aggregated step cost g_t is thus the following

$$g_t = 0.99 g_t^{\text{fl}} + 0.01 g_t^{\text{Irr}}$$

[6] Since none of the structural interventions prescribed by $A34$ has yet been realized, we can only study the behaviour of the system modified and controlled on the basis of this alternative by simulating its behaviour in the historical period. In this way we can use the performance of the historic regulation (*Alternative Zero*) as a standard of comparison.

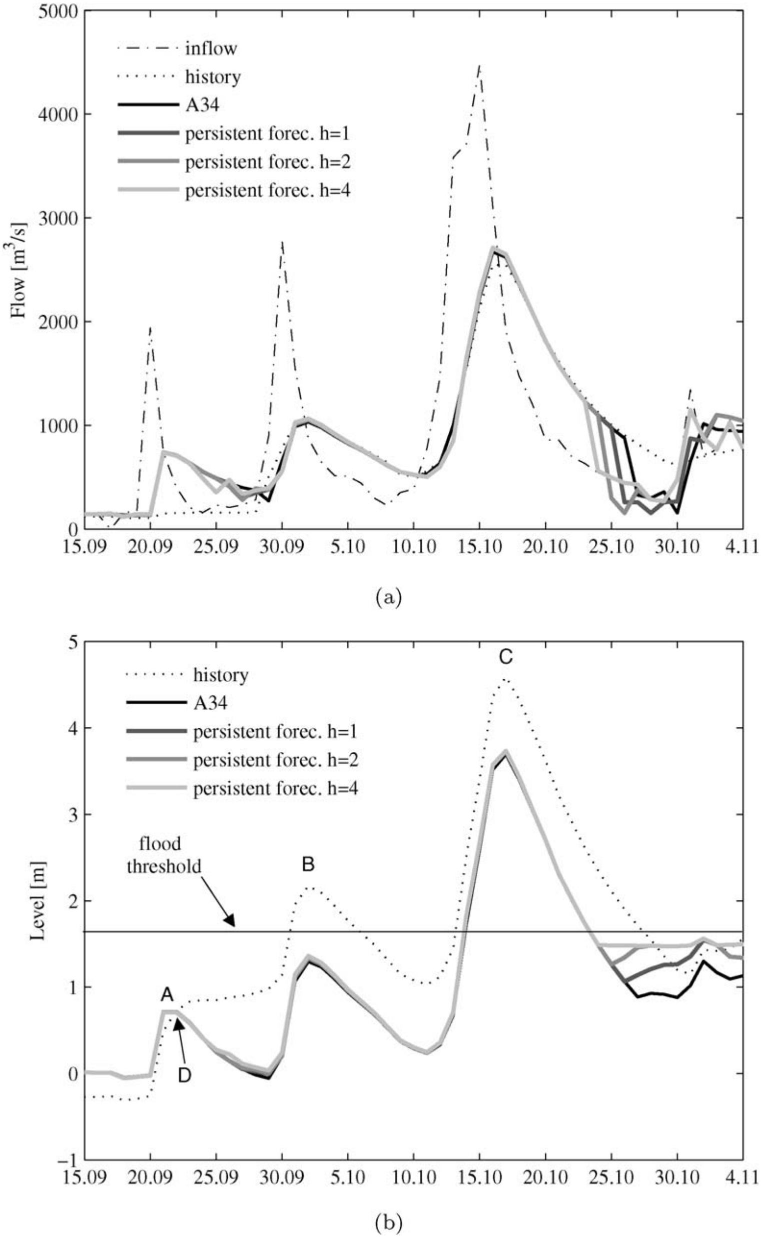


Figure 14.2: The flood event of the autumn of 2000. (a) Trajectories of the inflow, of the historical release and of the releases generated by $A34$ and by a POLFC fed by a persistent forecaster with $h = 1, 2, 4$. (b) Trajectories of the historical level and of the levels that would have been obtained with $A34$ and with a POLFC fed by a persistent forecaster with different forecasting horizon ($h = 1, 2, 4$).

the effect both of the excavation and of a more efficient regulation. The first allows, the levels being equal, the release of a larger flow and thereby a faster decrease of the level. To prove this to yourself, notice what occurs on 22nd September (point D in the figure), when the historic level and the one produced by $A34$ coincide: the historic release is 153 m^3/s (the maximum releasable given the lake level), while the release with the $A34$ is 709 m^3/s. It is this higher release that allows $A34$ to significantly reduce the lake level. As far as the second reason is concerned, the high expected costs associated with the high levels induce $A34$ to maintain, as far as possible, the reservoir level around zero (note that this is the value before the flood event). This level, in fact, is the one to which the minimum expected costs are associated in this season. That is the reason why, after 22nd September, $A34$ completely opens the dam gates, although the reservoir level is still within the regulation range. The historical regulation behaves differently: first it releases the maximum releasable volume, but later, in the following days, it maintains the release at the same value, in spite of the increase of the level. Finally, let us observe what happens at the end of the flood event. After 15th September, $A34$ encourages the Regulator to gradually increase the level, because in this period the thermal zero elevation generally rises, the flood probability lessens and as a consequence the interest in storing water for the spring irrigations prevails. That is the reason why p^{A34} tends to fill the lake in this period. Historical regulation has the same tendency: it reaches the full capacity on 4th November, while $A34$, which is more cautious, only reaches it on 9th November (both events are off the figure).

Since in the hydrometeorological sub-Alpine regime of the Lake Maggiore area the autumn is characterized by heavy rain and therefore by floods, it could be productive to use flow forecasters combined with a POLFC scheme to improve the performance of policy p^{A34}. Then, at every time step, a Problem of the form (14.13) has to be solved, on a horizon of h steps, assuming as penalty $g_{t+h}(\cdot)$ the Bellman function $H^{A34}_{(t+h)_{\mathrm{mod}\,T}}(\cdot)$ and using for the inflow the values $\hat{\varepsilon}_\tau$ supplied by a forecaster.

The simplest h-step-ahead forecaster that can be imagined is the *persistent forecaster*, which at every instant t forecasts a value $\hat{\varepsilon}_{t+\tau|t}$ for the inflow in the interval $[t+\tau-1, t+\tau)$ equal to the inflow ε_t which has occurred in the last time interval $[t-1, t)$, i.e.

$$\hat{\varepsilon}_{t+\tau|t} = \varepsilon_t \quad \tau = 1, \dots, h \tag{14.15}$$

In other words, the last inflow recorded at instant t is the only information which we exploit to predict the next h inflows; we limit ourselves to assuming that its value does not change through time. We shall also assume that the variance of $\hat{\varepsilon}_{t+\tau|t}$ is zero. The level trajectories obtained with the on-line policy (14.13), with the persistent forecaster and for different forecasting horizons ($h = 1, 2, 4$), are reported in Figure 14.2b (lines with different tones of grey). For $h = 1$, the trajectory practically overlaps that of the $A34$ while for higher values of h the peak levels (A) and (B) are higher than those produced by $A34$: the longer the forecasting horizon h, the higher the level. It is thus evident that it is dangerous to use the POLFC scheme when the forecaster is not a good one. It is therefore opportune to look for a better forecaster.

Intuitively, a good h-step-ahead inflow forecaster is the *perfect forecaster*: a forecaster able to predict reality exactly. For every instant τ between $t+1$ and $t+h$, it therefore supplies the value ε_τ, which effectively will occur, i.e.

$$\hat{\varepsilon}_{t+\tau|t} = \varepsilon_{t+\tau} \quad \tau = 1, \dots, h \tag{14.16}$$

Clearly, such a forecaster is not realizable in practice. However, the ∞-step-ahead forecaster is the best forecaster one can imagine adopting in a POLFC scheme, and therefore we may expect that with a h-step-ahead forecaster the performance of the on-line policy is not too far from the upper bound of the performance reachable with any other h-step-ahead forecaster[7]; thus, it allows one to estimate the maximum performance obtainable with an on-line policy.

The trajectories produced with the perfect forecaster on different forecasting horizons ($h = 1, 2, 4$) are reported in Figure 14.3. There is now an improvement compared with $A34$; the longer the horizon, the more marked it is. Observing in particular the first event (A) (see Figure 14.4), one notes that the bringing forward of the reservoir's spilling in order to buffer the first inflow peak corresponds precisely to the forecasting horizon used: for example, with the 4-step-ahead forecaster (lighter grey line) the policy begins to release more than the other policies on 15th September (4 days before the increase of the inflow) and this allows it to reduce the peak level on 21st by 0.40 m compared with $A34$. With all the forecasters, the improvement is marked on the first peak, more contained on the second (0.06 m with a 4-step-ahead forecaster and 0.03 m with a 1-step-ahead one) and zero on the third. This is easily explained by observing that from 22nd September the lake is in free regime with all the policies (also with the POLFC, which is driven by the penalty that it inherits from $A34$) and as a consequence of the asymptotic stability of the system all the trajectories tend to overlap as time goes on. Of course, they do not overlap the historic trajectory, because the historic system was not excavated and thus the behaviours of the two systems in free regime are different.[8]

Finally, let us observe what happens at the end of the flood event. Remember that both $A34$ and the historical regulation tend to fill the reservoir: the historical regulation reaches the full capacity on 4th November and $A34$ on 9th November. The on-line policies, which inherit the same tendency through the penalty, anticipate the filling because the more they see ahead the more they are able to establish that there is no more risk of floods.

As we have already emphasized a perfect forecaster actually is not realizable: the performance obtained with it are therefore only useful to have an idea of the upper bound of the performance that one expects from the use of an on-line policy. With a real forecaster, plausibly the improvement with respect to policy $A34$ will be less marked. And in fact with the best forecaster that we were able to create (1-step-ahead[9]) we earned only 0.02 m on peak B.

From this analysis we can conclude that the improvement of the off-line policy $A34$ by an on-line policy is, all in all, modest. The reason lies in the fact that the period considered (autumn) is a usual flood season and thus the off-line policy already takes due account of this. The advantage of the POLFC becomes more significant when an unexpected event occurs. In 1987, for example, surprisingly a flood took place in the month of July (a unique case in the hydrological series). Figure 14.5 shows that with a 1-step-ahead real forecaster the peak reduction compared with $A34$ is of 0.07 m with respect to 0.08 m of the 1-step-

[7]Notice that a POLFC with a h-step-ahead perfect forecaster may not always produce the best performance, because it assumes that the inflow probability distribution from time $t + h + 1$ onwards coincides with the a priori one (remember that it is driven by the penalty). As a consequence, when at time $t + h + 1$ the actual inflow is greater than the one expected a priori, a h-step-ahead forecaster that at time $t + h$ predicts an inflow greater than the one provided by the perfect forecaster can produce a better performance.

[8]Note that the difference between the levels of $A34$ and those produced by the POLFC policies is nearly constant and that the releases are all equal: a result foreseen by the theory, as explained in Section 3.1.3 of PRACTICE.

[9]Notice that the concentration time of the catchment is less than 24 hours.

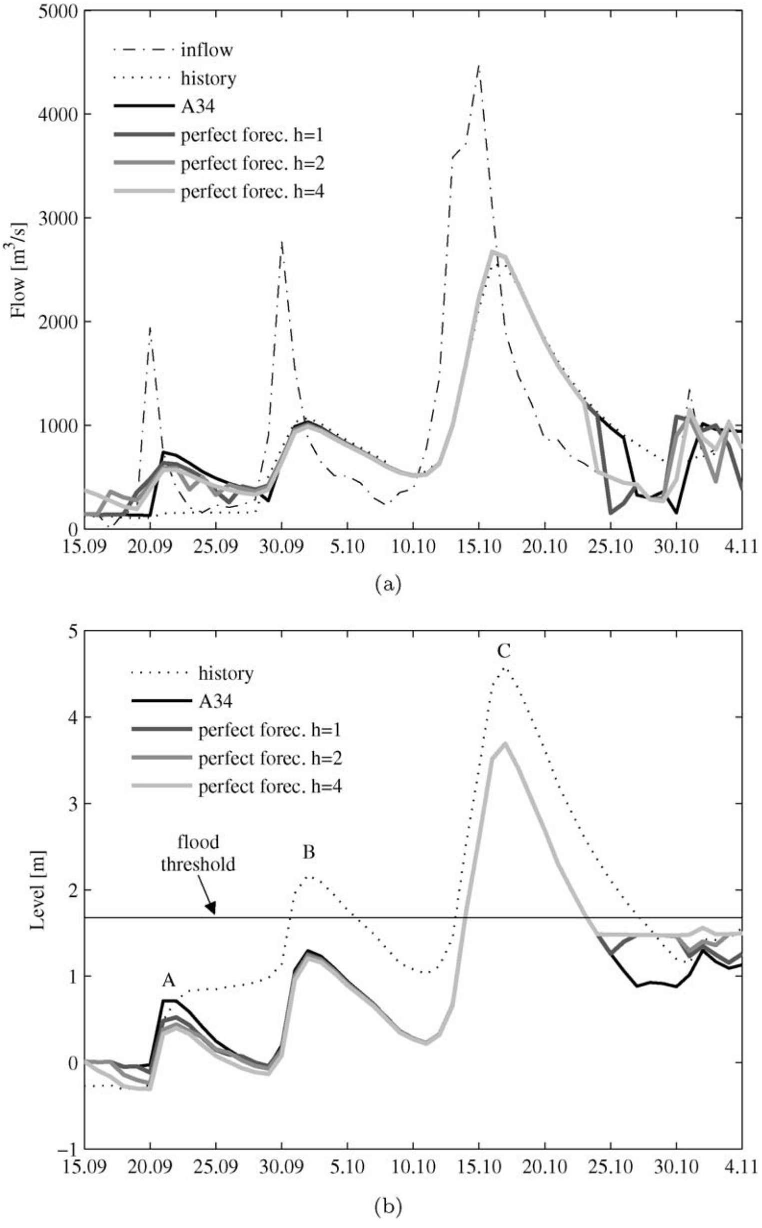


Figure 14.3: The flood event of the autumn of 2000. (a) Trajectories of the inflow, of the historical release and of the releases generated by $A34$ and by a POLFC fed by a perfect forecaster with $h = 1, 2, 4$. (b) Trajectories of the historical level and of the levels that there would have been with $A34$ and with a POLFC fed by a perfect forecaster with different forecasting horizon ($h = 1, 2, 4$).

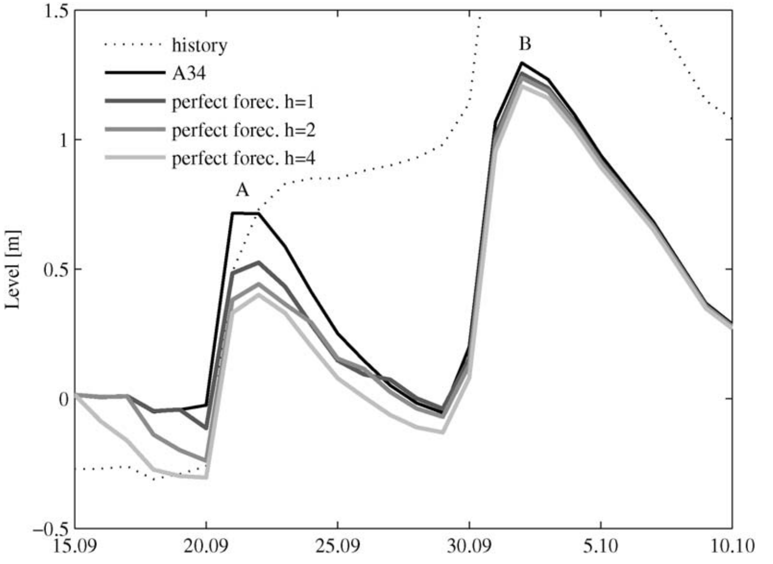


Figure 14.4: An enlargement of the first two flooding events reported in Figure 14.3b.

ahead perfect forecaster. The reduction may appear very modest, but notice that a reduction of 0.01 m corresponds to a reduction of 1 ha of the flooded area.

One last comment. From the figures it emerges that an increase in the forecasting horizon causes an increase in performance, but that increase is less than proportional to the increase of the forecasting horizon. On the other hand, the identification of a good inflow forecaster entails costs and these are the higher the longer the forecasting horizon. In the choice of forecaster it is thus necessary to find a trade-off between such costs and the resulting improvement in the performance. In this sense the information given by the perfect forecaster can constitute important information.

14.4 SV policies

As we explained in Section 11.2, the design of an on-line SV policy consists in finding not one, but all the control values that minimize the optimal cost-to-go at a given time. The way to find them for POLFC problems, whose solutions are determined with Algorithm 1, is evident. For NFC and OLFC problems it depends, instead, on the algorithm that is used, which, in turn, is chosen according to the form of the model's objective. When the algorithm is evolutionary, it is generally impossible to guarantee that all the controls that solve the problem have been found.

14.5 Variable-frequency regulation

As we explained in Section 4.8, the time step with which the Design Problem is defined must be sufficiently short to allow a timely adjustment of the control as the state of the system varies. In theory, this means that the time step must be shorter (by around one order of

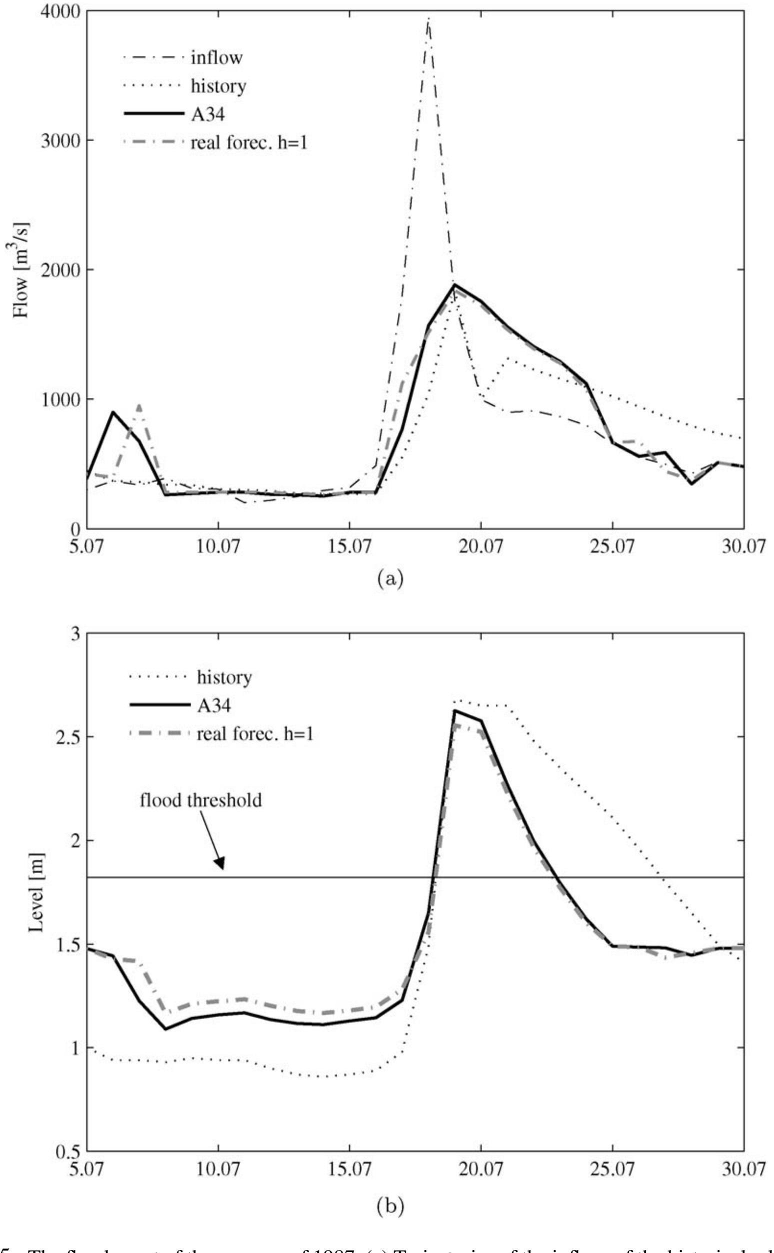


Figure 14.5: The flood event of the summer of 1987. (a) Trajectories of the inflow, of the historical release and of the releases generated by $A34$ and by a POLFC fed by a real forecaster with $h = 1$. (b) Trajectories of the historical level and of the levels that there would have been with $A34$ and with a POLFC fed by a real forecaster with $h = 1$.

magnitude) than the smallest time constant in the system (see Section 6.2.1), but since time constants are defined only for linear models, the definition of the time step generally refers to a 'normal' operational state of the system, around which the model is linearized to verify the satisfaction of the *Sampling Theory*. The most frequently adopted 'normal' condition is the one in which the storage values in the reservoirs are about half of the maximum storage. However, in abnormal conditions, e.g. in flood conditions, the time constant of the linearized system becomes shorter, since the slope of the maximum release curve increases when the spillway starts discharging (see Section 5.1.4). Therefore, it is necessary to control the system more frequently and, in fact, during a flood, the Regulator of a reservoir watches over the development of the event hour by hour.

On the other hand, it is not always possible to adopt a very short time step, because this would conflict with social needs (Section 4.8). The best solution would therefore be to adopt a step whose length varied according to the condition of the system. For the time being, however, there are no algorithms that are able to solve Problems defined with a step of this type.

The only possibility is to adopt a *heuristic solution*: a solution that cannot be proved to be optimal but that, in general, is satisfactory. The idea is, in normal conditions, to use a policy that was designed off-line (*normal policy*) with a time step suitable for such conditions, and move to an on-line policy in flood conditions. The latter is defined by an On-line Control Problem, in which:

- the decision-making step is the one used in flood conditions;
- the objective is defined taking current circumstances into account and so almost always is concerned only with flood damage;
- the final time instant $t+h$ is chosen so that it is a point on the time grid over which the normal policy is defined;
- the penalty $g_{t+h}(\cdot)$ is the optimal cost-to-go $H^*_{(t+h)_{\mathrm{mod}\,T}}(\cdot)$ obtained from the Bellman function $H^*_\cdot(\cdot)$ that solves the Design Problem and from which the normal policy was obtained.

Thereby decisions are guaranteed to be made in the short-term on the basis of the objectives that have been set and the information that is received, but the aim in the medium-term is to bring the system back to normal operational conditions.

This strategy can be adopted not only in flood conditions, but also in all the cases in which one might have particular information at particular times. Let us consider for instance the regulation of a lake used for irrigation purposes, which is fed by the inflow from a catchment, in which snow accumulates in winter. At the beginning of spring an estimate of the equivalent water volume accumulated in the snow pack is normally available. The best way to regulate this system would be to design an off-line policy, by formulating and solving a TDC (or AEV) Problem on the basis of a model of the lake and of the catchment, including the snow pack. However, the solution of such a problem could require a very long computing time. So one should proceed in a heuristic way as follows:

1. Formulate the Design Problem off-line, without taking the accumulation of snow into account and let p^{1*} and $H^{1*}_\cdot(\cdot)$ be the policy and the Bellman function that solve it.

2. At the beginning of spring (time 0), when the estimate of the equivalent water volume of the snow is obtained, compute a forecast of the inflow probability distribution trajectory for all the time steps in the subsequent year ($[0, T]$), using a suitable calibrated forecaster, in which the trajectory of the probability distribution of the air temperature is assumed a priori. Let $[0, \bar{h}_2]$ be the horizon over which this forecast is significantly different from the a priori forecast of the inflows and choose a time $h_2 > \bar{h}_2$ that coincides with one of the points of the time grid considered in the Problem at point 1.

3. With this forecast, define a POLFC Problem over the horizon $[0, h_2]$, by adopting the optimal cost-to-go $H^{1*}_{h_2}(\cdot)$ as the penalty. Determine the optimal policy p^{2*} and the Bellman functions $H^{2*}_{\cdot}(\cdot)$ that solve it.

4. Given policy p^{2*}, determine the sequence of the expected releases at each step by means of a Markovian simulation (see Section 19.1). This sequence can be very useful for the farmers in planning their activities.

5. Afterwards, at each time t, in the interval $[0, h_2]$, forecast the air temperature trajectory for the following days and determine the horizon $[0, \bar{h}_3]$ in which this forecast is significantly different from the a priori one. Then, forecast the inflow probability distribution trajectory, taking into account the new temperature forecast and any other information available at that time, e.g. the precipitation or the last recorded flow rates. On the basis of the computational power available on-line choose the most suitable form (i.e. NFC, OLFC or POLFC) for the on-line policy. Formulate and solve a new Control Problem, of that form, over the horizon $[0, h_3]$, with $h_3 > \bar{h}_3$ and corresponding to one of the points of the time grid used in the Problem formulated in point 3. Adopt $H^{2*}_{h_3}(\cdot)$ as the penalty of the new Problem. Let $\bar{u}_t$ and $H^{3*}_{\cdot}(\cdot)$ be the optimal control and Bellman function that solve it.

6. Apply the control $\bar{u}_t$ and go to step 5. Iterate until time h_2 is reached, after which the information obtained in springtime has no further effect. From that moment onwards the policy p^{1*} is adopted.

This procedure can be enhanced further in many ways: for example, by adopting more frequent decisions at step 6 whenever flood conditions occur; in this case the penalty of the Problem that defines this on-line policy must be derived from the Bellman function $H^{3*}_{\cdot}(\cdot)$.

The message to the reader should be clear by now: we can nest Control Problems into each other like *matryoshka*, and join them up through the corresponding Bellman functions. Each of these Problems can exploit particular information and be defined and solved only when such information is available. In the event that, at any time, the information necessary to formulate and solve a Problem at a given level is not available, note that it is always possible to adopt the policy obtained by the Problem at the preceding level. In this way the resulting control is robust with respect to the missing information.

Chapter 15

Distribution policies

AC, RSS and EW

In Chapter 12 we saw that the algorithms based on SDP suffer from the "curse of dimensionality" and illustrated how to mitigate its effects when they are caused by the state dimension. In this chapter we will present a solution that can be adopted when they are caused by the dimension of the control vector.

We introduce the idea upon which this solution is based with an example: consider the water system S in Figure 15.1a, in which a reservoir, affected by flooding on the shores, feeds a *distribution network*[1] D composed of two irrigation districts, two diversions and a stretch of river, whose environmental quality is threatened by the two diversions. In total, we must fix three controls at every time instant, as is shown by the causal network in Figure 15.1b.

Suppose that we must design the *regulation policy* p of the system with the aim of minimizing the overall damage produced by the water stress of the crops in the two irrigation districts, the water quality deterioration on the river stretch and the flooding on the reservoir shores. From what we have learned to this point, a regulation policy p for the system is defined at every time t by the following three control laws

$$u_t^R = m_t^R(s_t) \tag{15.1a}$$

$$u_t^{D1} = m_t^{D1}(s_t) \tag{15.1b}$$

$$u_t^{D2} = m_t^{D2}(s_t) \tag{15.1c}$$

where u_t^R is the *release decision*, u_t^{D1} and u_t^{D2} are *diversion decisions*, and the argument on the right-hand side is the state of the system, i.e. the storage s_t of the reservoir.

The last two laws do not have the form that one would expect, with the controls that are functions of the flow rate entering the diversions; in other words, one would expect that the control laws would have the following form

$$u_t^R = m_t^R(s_t) \tag{15.2a}$$

$$u_t^{D1} = m_t^{D1}(r_{t+1}) \tag{15.2b}$$

$$u_t^{D2} = m_t^{D2}\left(q_{t+1}^{v,D1}\right) \tag{15.2c}$$

[1] For the definition of this component see Section 6.3.

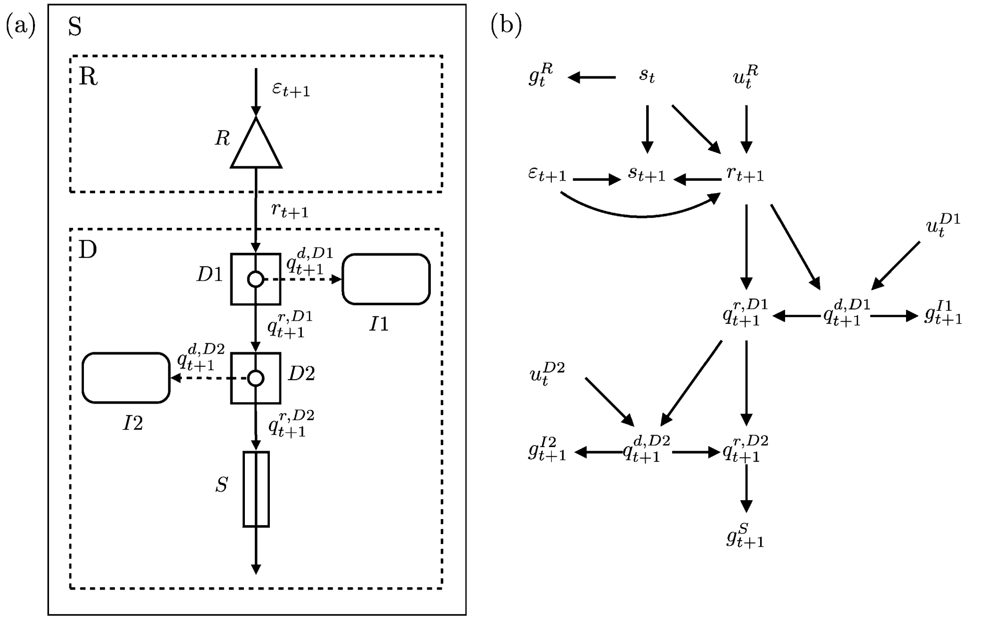


Figure 15.1: A water system composed of a reservoir and a distribution network D.

The succession of control laws of the form (15.2a) defines the *release policy* for the reservoir, while the succession of laws (15.2b)–(15.2c) specifies the *distribution policy* of the network D. Intuitively, we expect that the optimization of these two policies should be equivalent to the optimal policy p^* of the form (15.1), in the sense that it provides the same value of the objective.

The identification of p^* can be performed with one of the algorithms described in Section 12.1.4. These require that at every time t the optimal cost-to-go, which appears in the recursive Bellman equation (12.8), is minimized with respect to vector $\mathbf{u}_t = |u_t^R, u_t^{D1}, u_t^{D2}|$. We will denote this design procedure with the attribute *centralized*.

Alternatively, the policy design could be developed in three steps, with the following procedure, already outlined in Section 6.3, which we will call *decentralized* procedure:

Step 1: *Design a distribution policy of the form* (15.2b)–(15.2c) *for the network* D*, by exploring the space* $\mathbf{u}_t^{\mathrm{D}} = |u_t^{D1}, u_t^{D2}|$.

Step 2: *Derive the equivalent model of the network* D.

Step 3: *Use this model in the design of a regulation policy for the remaining system* R*, thus obtaining the release policy of the form* (15.2a).

If every control assumed 10 values and an exhaustive search were adopted to solve the Problem of Minimum within equation (12.8), the centralized procedure would require 10^3 evaluations of the optimal cost-to-go at every time t. Instead, the decentralized procedure would require only $10^2 + 10$: a number that is one order of magnitude smaller. The relationship between the two numbers increases exponentially with the difference between the dimensions of the control vectors of R and D.

The decentralized procedure therefore appears to be much more attractive, but unfortunately it is never (in theory) applicable. To understand the reason, observe, with greater attention, the control laws (15.2b)–(15.2c) that define the distribution policy: their arguments have the subscript $t + 1$. This means that the information required to make the diversion decision at time t is known only at time $t + 1$. Even if equation (15.2) proved to be equivalent to equation (15.1), the distribution policy would be unusable. The need to overcome this serious difficulty is the reason why the policies provided by the algorithms introduced in the previous chapters do not have the form (15.2), which we may find more natural, but the form (15.1): this form makes the diversion decision dependent on information (the state of the reservoir) that is known at time t.

Nevertheless, there is still one more possibility: if the system were deterministic, the arguments of equations (15.2b)–(15.2c) would be known at time t: then and only then, if the decentralized procedure allowed the identification of the optimal policy, it effectively would be much more useful than the centralized procedure.

In this chapter we will show that this does happen, but we cannot be too glad about it, because we know that natural deterministic systems do not exist. Therefore, even this most modest result gives the impression of being completely useless.

In Section 6.3 we noticed that it takes about 73 years of computation to determine the regulation policy for a system with the complexity of the Piave system. This means that, with today's machines, that policy cannot be designed. If, however, the system were deterministic, the disaggregated procedure would allow us to design the policy in just seven hours. Such a policy would certainly not be optimal, because the system is actually random, but it is certainly better than trusting the decision solely to the intuition of the Regulator.

We can do better than this, however: complete Steps 1 and 2 of the decentralized procedure assuming that the system is deterministic, but treat it as if it were random in Step 3, i.e. in the design of the regulation policy for system R. It can be proved that, by doing so, the regulation policy for R would be optimal, if equations (15.2b)–(15.2c) were usable. Unfortunately though, they are not, but we can avoid this difficulty in a heuristic way: by substituting r_{t+1} in equation (15.2b)–(15.2c) with its expected value at time t, for example; or with the value of the instantaneous flow rate measured at that time; or, when possible, by varying the diversion decision continuously over the course of the day, according to the total volume that has been released up to that moment.

We know that they are only heuristic solutions, but they are still better than leaving the Regulator to make a decision based on his intuition alone. It is therefore worth putting the idea that we outlined into practice. In the next section we will show how it is possible to formulate the design of the distribution policy so that such policy, together with the regulation policy of system R, are equivalent to the optimal policy p^* when the system is deterministic. In the following section we will develop the algorithm that solves such a problem.

15.1 Control Problem for distribution policies

The problem of designing the regulation policy for a water system over horizon $[0, h]$ is expressed by equation (10.26) and its solution is obtained with Algorithm 1 (page 299),

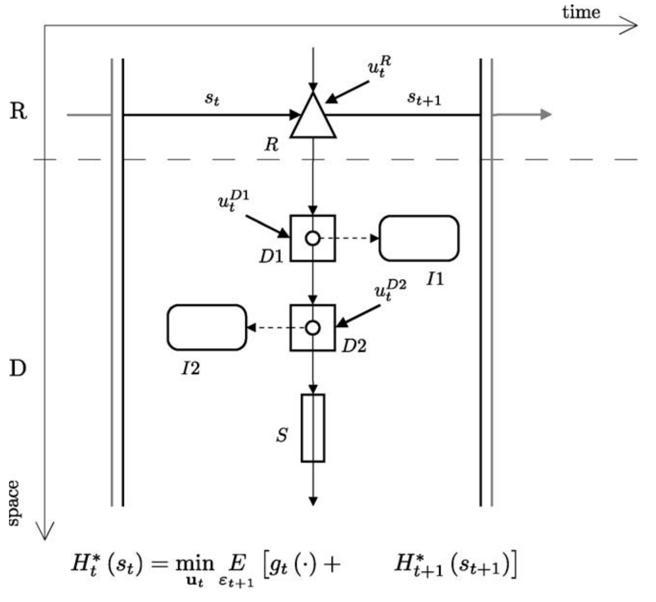


Figure 15.2: The Bellman Principle of Optimality applied to the system in Figure 15.1.

whose core is the solution of the following recursive equation

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}} \left[g_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{15.3a}$$

$$\mathbf{x}_{t+1} = f_t(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) \tag{15.3b}$$

$$\mathbf{u}_t \in \mathcal{U}_t(\mathbf{x}_t) \tag{15.3c}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \tag{15.3d}$$

$$\text{any other constrains relative to } [t, t+1) \tag{15.3e}$$

which is based on Bellman's Principle of Optimality (see page 292 and Figure 12.5). Figure 15.2 shows the graphic interpretation of this principle for the system in Figure 15.1.

When system S can be partitioned into a distribution network D and the residual system R, and the step cost $g_t(\cdot)$ is defined as the sum of the step costs $\breve{g}_t^i(\cdot)$ associated to the single components of the system (see Section 6.2), equation (15.3) takes the form

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t^{\mathrm{R}}} \min_{\mathbf{u}_t^{\mathrm{D}}} \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}} \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}} \left[\sum_{i \in \mathcal{N}^{\mathrm{R}} \cup \mathcal{N}^{\mathrm{D}}} \breve{g}_t^i(\mathbf{x}_t, \mathbf{z}_{t+1}, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1}) + H_{t+1}^*(\mathbf{x}_{t+1}) \right] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{15.4a}$$

$$\mathbf{x}_{t+1} = \breve{f}_t\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) \tag{15.4b}$$

$$\mathbf{z}_{t+1}^{\mathrm{R}} = \breve{z}_t^{\mathrm{R}}\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) \tag{15.4c}$$

$$\mathbf{u}_t^{\mathrm{R}} \in \mathcal{U}_t^{\mathrm{R}}(\mathbf{x}_t) = \prod_{i=1}^{\mathcal{N}^{\mathrm{R}}} \mathcal{U}_t^i(\mathbf{x}_t) \tag{15.4d}$$

$$\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}} \sim \phi_t^{\mathrm{R}}(\cdot) \tag{15.4e}$$

$$\text{any other constraints of R relative to } [t, t+1] \tag{15.4f}$$

$$\mathbf{z}_{t+1}^{\mathrm{D}} = \breve{z}_t^{\mathrm{D}}\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}, \mathbf{e}_{t+1}^{\mathrm{D}}\right) \tag{15.4g}$$

$$\mathbf{u}_t^{\mathrm{D}} \in \mathcal{U}_t^{\mathrm{D}} = \prod_{i=1}^{\mathcal{N}^{\mathrm{D}}} \mathcal{U}_t^i \tag{15.4h}$$

$$\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}} \sim \phi_t^{\mathrm{D}}(\cdot) \tag{15.4i}$$

$$\text{any other constraints of D relative to } [t, t+1] \tag{15.4j}$$

where $\mathcal{N}^{\mathrm{R}}$ and $\mathcal{N}^{\mathrm{D}}$ are the sets of the components of the systems R and D that we defined in Section 6.3, of which equations (15.4b)–(15.4f) and (15.4g)–(15.4j) constitute the relative models.[2] We remind the reader, because it is essential, that $\mathbf{e}_{t+1}^{\mathrm{D}}$ is the vector of the flows entering into D, i.e. the flow r_{t+1} in our example. Since $\mathbf{u}_t^{\mathrm{D}}$ and $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}$ influence only the step cost of D and the expected value is a linear operator, equation (15.4a) can be rewritten as

$$\begin{aligned} H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t^{\mathrm{R}}} \min_{\mathbf{u}_t^{\mathrm{D}}} \underset{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}}{E} \Bigg[\sum_{i \in \mathcal{N}^{\mathrm{R}}} \breve{g}_t^i\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) + \\ + \underset{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}}{E} \Bigg[\sum_{j \in \mathcal{N}^{\mathrm{D}}} \breve{g}_t^j\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}\right) \Bigg] + H_{t+1}^*(\mathbf{x}_{t+1}) \Bigg] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \end{aligned} \tag{15.5}$$

If (note that the phrase is hypothetical!) it were possible to change the position of the operators $\min_{\mathbf{u}_t^{\mathrm{D}}}$ and $E_{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}}$, equation (15.5) would become

$$\begin{aligned} H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t^{\mathrm{R}}} \underset{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}}{E} \Bigg[\sum_{i \in \mathcal{N}^{\mathrm{R}}} \breve{g}_t^i\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) + \\ + \min_{\mathbf{u}_t^{\mathrm{D}}} \underset{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}}{E} \Bigg[\sum_{j \in \mathcal{N}^{\mathrm{D}}} \breve{g}_t^j\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}\right) \Bigg] + H_{t+1}^*(\mathbf{x}_{t+1}) \Bigg] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \end{aligned} \tag{15.6}$$

which in turn we could rewrite as

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t^{\mathrm{R}}} \underset{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}}{E} \Bigg[\sum_{i \in \mathcal{N}^{\mathrm{R}}} \breve{g}_t^i\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) + \breve{g}_t^{\mathrm{D}}\left(\mathbf{e}_{t+1}^{\mathrm{D}}\right) + H_{t+1}^*(\mathbf{x}_{t+1}) \Bigg] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{15.7a}$$

$$\mathbf{x}_{t+1} = \breve{f}_t\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) \tag{15.7b}$$

$$\mathbf{z}_{t+1}^{\mathrm{R}} = \breve{z}_t^{\mathrm{R}}\left(\mathbf{x}_t, \mathbf{z}_{t+1}^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}\right) \tag{15.7c}$$

$$\mathbf{u}_t^{\mathrm{R}} \in \mathcal{U}_t^{\mathrm{R}}(\mathbf{x}_t) = \prod_{i=1}^{\mathcal{N}^{\mathrm{R}}} \mathcal{U}_t^i(\mathbf{x}_t) \tag{15.7d}$$

$$\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}} \sim \phi_t^{\mathrm{R}}(\cdot) \tag{15.7e}$$

$$\text{any other constraints of R relative to } [t, t+1] \tag{15.7f}$$

where

$$\breve{g}_t^{\mathrm{D}}\left(\mathbf{e}_{t+1}^{\mathrm{D}}\right) = \min_{\mathbf{u}_t^{\mathrm{D}}} \underset{\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}}{E} \Bigg[\sum_{j \in \mathcal{N}^{\mathrm{D}}} \breve{g}_t^j\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}\right) \Bigg] \tag{15.8a}$$

[2] For the definition of the variables that appear in equation (15.4) see Section 6.3.

$$\mathbf{z}_{t+1}^{\mathrm{D}} = \breve{z}_t^{\mathrm{D}}\left(\mathbf{z}_{t+1}^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}, \mathbf{e}_{t+1}^{\mathrm{D}}\right) \tag{15.8b}$$

$$\mathbf{u}_t^{\mathrm{D}} \in \mathcal{U}_t^{\mathrm{D}} = \prod_{j=1}^{\mathcal{N}^{\mathrm{D}}} \mathcal{U}_t^j \tag{15.8c}$$

$$\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}} \sim \phi_t^{\mathrm{D}}(\cdot) \tag{15.8d}$$

$$\text{any other constraints of D relative to } [t, t+1) \tag{15.8e}$$

Equation (15.8) would define (we remind the reader that we are in an hypothetical situation) the Control Problem for the distribution policy and its solution would provide a policy of the form

$$\mathbf{u}_t^{\mathrm{D}} = m_t^{\mathrm{D}*}\left(\mathbf{e}_{t+1}^{\mathrm{D}}\right)$$

which is the same form as equations (15.2b)–(15.2c) in our example. As we have already noted in the introduction, this policy would be unusable, because $\mathbf{e}_{t+1}^{\mathrm{D}}$ is not known at the decision-making time t. This surprising fact is the consequence of the inversion of the two operators that we assumed were feasible: by inverting the operators, it was implicitly assumed that the minimization of $\mathbf{u}_t^{\mathrm{D}}$ would come after the disturbance $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}$ had occurred, in which case $\mathbf{e}_{t+1}^{\mathrm{R}}$ would be known.

Inverting the two operators is not feasible and thus, when the system is affected by random disturbances, the distribution policy cannot be designed separately from the policy of system R. In other words, the decentralized procedure is not valid.

When the system is deterministic, however, the operator E disappears and so from equations (15.5) and (15.6) the following is obtained

$$H_t^*(\mathbf{x}_t) = \min_{\mathbf{u}_t^{\mathrm{R}}}\left[\sum_{i\in\mathcal{N}^{\mathrm{R}}} \breve{g}_t^i\left(\mathbf{x}_t, \mathbf{z}_t^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}\right) + \breve{g}_t^{\mathrm{D}*}\left(\mathbf{e}_t^{\mathrm{D}}\right) + H_{t+1}^*(\mathbf{x}_{t+1})\right] \quad \forall \mathbf{x}_t \in \mathcal{S}_{\mathbf{x}_t} \tag{15.9a}$$

$$\mathbf{x}_{t+1} = \breve{f}_t\left(\mathbf{x}_t, \mathbf{z}_t^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}\right) \tag{15.9b}$$

$$\mathbf{z}_t^{\mathrm{R}} = \breve{z}_t^{\mathrm{R}}\left(\mathbf{x}_t, \mathbf{z}_t^{\mathrm{R}}, \mathbf{u}_t^{\mathrm{R}}\right) \tag{15.9c}$$

$$\mathbf{u}_t^{\mathrm{R}} \in \mathcal{U}_t^{\mathrm{R}}(\mathbf{x}_t) = \prod_{i=1}^{\mathcal{N}^{\mathrm{R}}} \mathcal{U}_t^i(\mathbf{x}_t) \tag{15.9d}$$

$$\text{any other constraints of R relative to } [t, t+1) \tag{15.9e}$$

where

$$\breve{g}_t^{\mathrm{D}*}\left(\mathbf{e}_t^{\mathrm{D}}\right) = \min_{\mathbf{u}_t^{\mathrm{D}}}\left[\sum_{j\in\mathcal{N}^{\mathrm{D}}} \breve{g}_t^j\left(\mathbf{z}_t^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}\right)\right] \tag{15.10a}$$

$$\mathbf{z}_t^{\mathrm{D}} = \breve{z}_t^{\mathrm{D}}\left(\mathbf{z}_t^{\mathrm{D}}, \mathbf{u}_t^{\mathrm{D}}, \mathbf{e}_t^{\mathrm{D}}\right) \tag{15.10b}$$

$$\mathbf{u}_t^{\mathrm{D}} \in \mathcal{U}_t^{\mathrm{D}} = \prod_{j=1}^{\mathcal{N}^{\mathrm{D}}} \mathcal{U}_t^j \tag{15.10c}$$

$$\text{any other constraints of D relative to } [t, t+1) \tag{15.10d}$$

Note that now the internal variables $(\mathbf{z}_t^{\mathrm{R}}, \mathbf{z}_t^{\mathrm{D}}, \mathbf{e}_t^{\mathrm{D}})$ have the subscript t, which means they are known at the time in which the decision must be made. Equation (15.10) defines the *Control Problem for the distribution policy* that must be solved in Step 1 of the decentralized procedure. As we will see in the next section, its solution provides the control law $\mathbf{u}_t^{\mathrm{D}*} = m_t^{\mathrm{D}*}(\mathbf{e}_t^{\mathrm{D}})$ and, at the same time, the *equivalent model* $\breve{g}_t^{\mathrm{D}*}(\mathbf{e}_t^{\mathrm{D}})$ of the network (see Section 6.3). In Step 3 of the procedure, problem (15.9) is solved to obtain the regulation policy for the system R, which has the form

$$\mathbf{u}_t^{\mathrm{R}} = m_t^{\mathrm{R}*}(\mathbf{x}_t)$$

as does (15.2a) in our example. By solving the pair of problems (15.10)–(15.9) for $t = h-1, h-2, \dots, 0$, one obtains the succession $\{m_t^{\mathrm{D}*}(\mathbf{e}_t^{\mathrm{D}}), m_t^{\mathrm{R}*}(\mathbf{x}_t)\}_{t=0}^{h-1}$ of control laws that defines a regulation policy for system S, which, by construction, coincides with the optimal policy p^*. In this way we have proved the assumption formulated in the introduction, upon which the decentralized design procedure is based.

It is worth underlining that this procedure is valid only when the system is deterministic; if it is not, one could apply the procedure only by accepting a deterministic description of the disturbance, i.e. by using its expected value (if it is a stochastic disturbance), or by using one of its values (if it is uncertain). Clearly, the greater is the variance (or the variability of the disturbance), the greater will be the error made by doing so. As equation (15.8) shows, the disturbances that act on the distribution network, when the system is random, are $\mathbf{e}_{t+1}^{\mathrm{D}}$ and $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}$. The first is a function only of the decisions $\mathbf{u}_t^{\mathrm{R}}$, except when one of the reservoirs in system R starts overflowing or is completely emptied during the day. Since these events are extremely rare, the error incurred by assuming that the flow rate $\mathbf{e}_{t+1}^{\mathrm{D}}$ is deterministic is generally very small. The same cannot be said, however, for $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}$ and it is not possible to do anything to reduce the effects of its randomness. Instead, we can take into account the randomness of $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{R}}$ by designing the policy for R with problem (15.7), in place of (15.9). By doing so, in the example in Figure 15.1, in which $\boldsymbol{\varepsilon}_{t+1}^{\mathrm{D}}$ is not present, the system S would be incorrectly described only in the very rare cases in which reservoir R overflows or is completely emptied, given that there are no catchments in the distribution network. On the contrary, in the Piave water system (Figure 6.7) the error would be more relevant because the MP network contains several catchments.

15.2 Solution algorithms

To synthesize the algorithm that solves problem (15.8), re-examine the system in Figure 15.1. For this system, the Control Problem (15.10) for the distribution policy assumes the form (see Figure 15.1)

$$\breve{g}_t^{\mathrm{D}*}(r_{t+1}) = \min_{u_t^{D1}, u_t^{D2}} \left[\breve{g}_t^{I1}\left(q_{t+1}^{d,D1}\right) + \breve{g}_t^{I2}\left(q_{t+1}^{d,D2}\right) + \breve{g}_t^{S}\left(q_{t+1}^{r,D2}\right)\right] \tag{15.11a}$$

$$q_{t+1}^{d,D1} = \min\left\{u_t^{D1}, r_{t+1}, q^{\max,D1}\right\} \tag{15.11b}$$

$$q_{t+1}^{r,D1} = r_{t+1} - q_{t+1}^{d,D1} \tag{15.11c}$$

$$q_{t+1}^{d,D2} = \min\left\{u_t^{D2}, q_{t+1}^{r,D1}, q^{\max,D2}\right\} \tag{15.11d}$$

$$q_{t+1}^{r,D2} = q_{t+1}^{r,D1} - q_{t+1}^{d,D2} \tag{15.11e}$$

$$\mathbf{u}_t^{\mathrm{D}} \in \mathcal{U}_t^{D1} \times \mathcal{U}_t^{D2} \tag{15.11f}$$

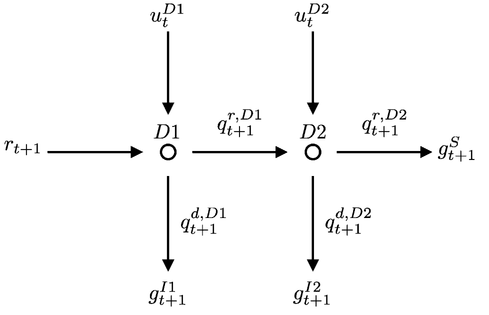


Figure 15.3: The distribution problem is a multi-stage decision-making problem, in which space behaves as time does in regulation problems.

while problem (15.7) becomes

$$H_t^*(s_t) = \min_{u_t^R} \mathop{E}_{\varepsilon_{t+1}} \left[\breve{g}_t^R(s_t) + \breve{g}_t^{\mathrm{D}*}(r_{t+1}) + H_{t+1}^*(s_{t+1})\right] \quad \forall s_t \in \mathcal{S}_{s_t} \tag{15.12a}$$

$$s_{t+1} = s_t + \varepsilon_{t+1} - R_t\big(s_t, u_t^R, \varepsilon_{t+1}\big) \tag{15.12b}$$

$$r_{t+1} = R_t\big(s_t, u_t^R, \varepsilon_{t+1}\big) \tag{15.12c}$$

$$u_t^S \in \mathcal{U}_t^S(s_t) \tag{15.12d}$$

Problem (15.11) provides the cost $\breve{g}_t^{\mathrm{D}*}(r_{t+1})$ of the distribution network as a function of the flow rate r_{t+1} that is conveyed to it, under the hypothesis of taking optimal diversion decisions: it is thus an optimal cost. From equation (15.9a) and Figure 15.2 we understand that $\breve{g}_t^{\mathrm{D}*}(r_{t+1})$ plays a role similar to that of $H_{t+1}^*(s_{t+1})$: the latter defines the optimal cost-to-go, i.e. the value associated to the volume s_{t+1} that is 'delivered' to the future, while the first represents the optimal spatial cost, i.e. the value associated to the release r_{t+1} that is 'delivered' to the network. Figure 15.2 allows us to fully appreciate the symmetry between space (network) and time, and shows how the Regulator's decision must take both into account. Equation (15.12) thus follows from the Bellman Principle of Optimality (Section 12.1.1), as does equation (12.8).

We can use this same principle to solve problem (15.11). Note the analogy between Figure 15.3, which represents network D in Figure 15.1 by highlighting the controls that act on the diversions, and Figure 12.1: the time that flows from left to right in the second corresponds to the water that flows downstream in the first, and the decision time instants correspond to the nodes of the network. Once again time and space appear to be symmetric. The solution to problem (15.11) can thus be determined with an algorithm that works in a way that is analogous to Algorithm 1 (page 299) moving upstream. More precisely, the following problem for the diversion $D2$ is solved at the outset

$$\breve{g}_t^{D2*}\big(q_{t+1}^{r,D1}\big) = \min_{u_t^{D2}} \big[\breve{g}_t^{I2}\big(q_{t+1}^{d,D2}\big) + \breve{g}_t^{S}\big(q_{t+1}^{r,D2}\big)\big] \tag{15.13a}$$

$$q_{t+1}^{d,D2} = \min\big\{u_t^{D2}, q_{t+1}^{r,D1}, q^{\max,D2}\big\} \tag{15.13b}$$

$$q_{t+1}^{r,D2} = q_{t+1}^{r,D1} - q_{t+1}^{d,D2} \tag{15.13c}$$

$$u_t^{D2} \in \mathcal{U}_t^{D2} \tag{15.13d}$$

thus obtaining the policy

$$u_t^{D2} = m_t^{D2*}\big(q_{t+1}^{r,D1}\big) \tag{15.14}$$

Then the following problem is considered

$$\breve{g}_t^{\mathrm{D}*}(r_{t+1}) = \min_{u_t^{D1}}\big[\breve{g}_t^{I1}\big(q_{t+1}^{d,D1}\big) + \breve{g}_t^{D2*}\big(q_{t+1}^{r,D1}\big)\big] \tag{15.15a}$$

$$q_{t+1}^{d,D1} = \min\big\{u_t^{D1}, r_{t+1}, q^{\max,D1}\big\} \tag{15.15b}$$

$$q_{t+1}^{r,D1} = r_{t+1} - q_{t+1}^{d,D1} \tag{15.15c}$$

$$u_t^{D1} \in \mathcal{U}_t^{D1} \tag{15.15d}$$

which provides the policy

$$u_t^{D1} = m_t^{D1*}(r_{t+1}) \tag{15.16}$$

It is easy to see that this last problem, with the definition of $\breve{g}_t^{D2*}(q_{t+1}^{r,D1})$ provided by problem (15.13), is equivalent to problem (15.11). Thus, we have obtained both the distribution policy, defined by the pair (15.16), (15.14), and the *equivalent model* $\breve{g}_t^{\mathrm{D}*}(r_{t+1})$ of the network, i.e. the function that describes it as an equivalent user.

At this stage the algorithm has only been guessed at: we should now formalize it and, above all, generalize it. Not always, in fact, do distribution networks have the sequential structure shown in Figure 15.1: more often moving from downstream upwards, one encounters branching, just as happens for example in the MP network in the Piave system (Figure 6.7) when, moving up from $D8$, one encounters the node $CP8$, in which the two branches on the right and the left join. We will nevertheless forego both generalizing and formalizing the algorithm because it would require a lot of space: it is sufficient that the reader understand how it is conceived.

Chapter 16

The decision-making process

AC, FC and RSS

In Section 2.3 we explained how the management alternatives that could be adopted by the Zambesi Water Authority (ZWA) could not be evaluated on the basis of a single criterion, and therefore a single indicator, because each of the countries involved would evaluate the alternatives with its own criterion, which would be different from the other countries' criteria. This situation is not exclusive to the Zambesi Project; it is common to most real world projects, and can be traced back to three main causes:

1. *Often it is impossible to formulate all the indicators using the same unit of measurement.*

   Think, for instance, of the design of a dam for hydropower production, in which the evaluation criteria that have been adopted concern the attainable profits, the health of the inhabitants on the shores of the lake that will be formed, who will be threatened by (malarial) anopheles mosquitoes, and the human lives put at risk by the possibility, even if remote, of the dam collapsing. A cost–benefit analysis (CBA), one of the most well-known decision-making methods in the field of economics, struggles to express each of these indicators in economic terms, monetizing not only profits, but also human health and human lives. In cases such as this the results can be largely disappointing, if not unacceptable. The reader can appreciate the reason for this by imagining that (s)he is one of the residents involved in the project and trying to define the economic value of his/her health or life.

2. *The different criteria, and therefore the different indicators, express the viewpoint of different Stakeholders.*

   In this case, even when the indicators can be all expressed with the same unit of measurement, aggregating them to obtain a single indicator makes no sense. The example of the Zambesi (Section 2.3) demonstrates this well: Zambia and Zimbabwe adopt the same indicator (the hydropower benefit) but we cannot deduce that the overall hydropower benefit in the Zambesi basin can be considered as a single indicator that expresses both viewpoints, because each of the two countries is only interested in its own benefit.

3. *Very often the indicators are in conflict with each other.*

   In other words, by improving the value of one indicator, the value of another is worsened: there is thus a conflict not only between the indicators, but also between the Stakeholders that express them. It follows that it is impossible to identify an 'optimal' solution for all of the Stakeholders, and that it is therefore necessary to look for a solution that will be a compromise between their opposing needs.

It is essential to identify possible conflicts as early as possible, and ideally before the decision-making process is concluded, because if they emerge while the decision is in the process of being implemented, it will be too late. The Stakeholders who feel they are penalized would then do everything possible to stop implementation. They would easily be able to thwart the process and, as a consequence, there would be a heavy increase both in financial and social costs due to the delay in the work, as well as the postponement of the expected benefits. Furthermore, the negotiations for finding a new solution would be launched in the worst way, among great tensions and reciprocal suspicions. An example of such a situation are the tensions that often emerge around the construction of power plants (both nuclear and other) between Power Companies and Environmentalists. The correct and timely identification of a conflict and its preventive management are thus the best ways to resolve it and to be able to make reliable decisions that last in time.

When one is faced with conflicting interests, sometimes the problem can be further complicated by the presence of more than one Decision Maker (DM). When this occurs, another conflict may arise among the DMs, because each of them has her own system of preferences. This is the case of the Zambesi. A more frequently occurring, analogous situation, even if formally it appears to be different, is when there is only one DM, but the Stakeholders, who are in conflict with each other, are able to mobilize pressure groups to influence the DM. This is what happened, for example, in the Vomano Project (see page 147). In cases such as these the Stakeholders should be considered as if they were *persons with a decision-making role* and, while keeping them distinct from the official DM(s), we can adopt the same procedure that is used to manage a plurality of DMs. For the sake of simplicity we will classify both of these cases with the label *Multiple Decision Makers*, even if, more correctly, we should distinguish the second from the first, by labelling it as *Multiple Persons with a Decision-Making Role*.

In both cases, the decision-making procedure must be able to manage the plurality of individuals involved in the decision and to mediate between conflicting interests, with the aim of reaching, when possible, consensus on one alternative. When this is impossible, the procedure must determine the alternatives on which the agreement is widest and, for each of them, identify the Stakeholders that support it.

It is in these conditions that the PIP procedure (see Section 1.3) demonstrates its reason for existence and all of its phases become significant: in conditions of full rationality, it is at the end of the phase *Designing Alternatives* that a decision is reached, while in cases of partial rationality, if there is only one DM and the Stakeholders are not involved, the decision is reached only at the end of the phase *Evaluation*. When Stakeholders are involved or there is more than one DM, the phases of *Comparison* and *Final Decision* must also be implemented.

Like the other chapters that introduce a new part of this book, this chapter, with which we open Part D, aims at providing an overview of the topics that are examined: the problems that arise when there is more than one evaluation criterion (Section 16.1) and/or there is more than one DM (Section 16.2). In jargon, the first is called *Multi-Objective* (MO) *Problem*,[1] and the second is a *Multi-Decision Maker* (MDM) *Problem*.

16.1 Multiple objectives: from Designing Alternatives to Evaluation

Here we consider a Design Problem in which there is more than one evaluation indicator, but there is only one DM. In this case, as we have already noted, the concept of *optimal alternative*, which we used to formulate the Design Problem under full rationality, loses its meaning. Indeed, the alternative that is optimal for one indicator is not necessarily optimal for the others.

As we anticipated in Section 2.3, the design of the alternatives can be traced back to the formulation of a Design Problem, with only one difference: it does not have just one objective, as does a problem formulated in conditions of full rationality, but one objective for each indicator being considered. However, the number of objectives is often so large that the problem cannot be solved, within acceptable computing times, with the computers available today. Therefore, one has to choose a subset of *design indicators* (also called *control indicators*) from the set of evaluation indicators, with which to formulate the objectives of the Problem (Section 18.6).

While under conditions of full rationality the solution of the problem is unique (the *optimal alternative*), under partial rationality the problem has many solutions (called *efficient* alternatives), which satisfy the objectives in the "best possible way" in the sense that will be specified in Section 18.2.

The lack of a single criterion with which to rank the alternatives identified in this way means that the phase *Designing Alternatives* must be followed by the phase *Evaluation*, the aim of which is to order the alternatives according to the 'value' that the DM gives to each one of them. In this way it is possible to determine the *best compromise alternative*, where the attribute 'best' is obviously understood to be 'from the DM's point of view'. This phase is not computationally demanding and so the number of indicators considered is not as critical as it is in the phase *Designing Alternatives*. For this reason the evaluation is always carried out taking into account all of the indicators and not only the design indicators.

As we go into more detail we note that the phase *Evaluation* must be preceded by the phase *Estimating Effects* that each efficient alternative produces on each of the indicators. This estimation is indispensable, since it is on the basis of the values assumed by the indicators that the system of the DM's preferences (i.e. the way in which she judges the alternatives) emerges. The first step of the *Evaluation* is to identify these preferences. To consider the structure of the DM's preferences means introducing elements of subjectivity into the decision-making process: these are the elements that justify the title given to the present part, *Decision-making under partial rationality conditions*. Since the decision is not free of subjective elements, the methodology must allow us to represent explicitly any possible conflict between the indicators and make the subjective elements that are introduced by the DM

[1] Besides Multi-Objective problems, the reader might have heard about Multi-Attribute and Multi-Criteria problems. The term 'Multi-Objective' is used to denote a problem in which the set of the alternatives is continuous (and thus their number infinite), while the term 'Multi-Attribute' refers to the case in which the set is discrete. Finally, the term 'Multi-Criteria' is used to denote both the cases. Since the problems considered in this book generally concern the design of a policy (and the space of the policies is continuous) we decided to adopt only the first term.

evident and transparent. There are three principle methodologies that respect these conditions: *Multi-Attribute Value Theory* (MAVT) (Keeney and Raiffa, 1976), *Analytic Hierarchy Process* (AHP) (Saaty, 1980) and the ELECTRE family methods (Roy and Bouyssou, 1993; Roy, 1996). We present all three of them in the next chapter to evaluate their pros and cons, with the aim of identifying the one that best responds to the needs of the IWRM paradigm. We will discover that it is the MAVT.

16.2 Multiple Decision Makers: Comparison and Final Decision

When there is only one DM, from a technical viewpoint the Design Problem could find its natural conclusion in the phase *Evaluation*. However, we have already pointed out a number of times that this procedure is inappropriate when there are multiple interests in play, because it excludes the Stakeholders from the decision. Therefore, whenever possible, these persons should be given an *active decision-making role*, by making the alternatives that obtain the widest agreement emerge from *negotiations* among them. For this reason, in the PIP procedure the *Evaluation* phase is followed by the *Comparison* phase and the DM's choice is postponed to the next phase of *Final Decision*. By doing so, when we get to this last phase, the DM knows which alternatives are interesting to the Stakeholders and who are the supporters and opposers of each of them.

Even when there is more than one DM, the phase *Final Decision* cannot be limited to adopting the results from *Evaluation*, because too often each DM would choose a 'best compromise alternative' that is different to the other DMs', given that each one has, generally, her own particular system of preferences, i.e. her own particular point of view. In this case, therefore, also the phase *Final Decision* is carried out through negotiations with the aim of reaching a *consensus* on one alternative, even if the individuals involved in the negotiations are the DMs. In Section 2.3 we provided an initial idea about how these negotiations might be conducted.

The phases *Comparison* and *Final Decision* are thus aimed at organizing the participation of the Stakeholders and supporting the DM(s)[2] in the choice of an alternative. In both, the pivotal point is the *negotiation* process, i.e. the discussion among the Stakeholders or the DMs, who must compare their positions and make compromises in order to reach an agreement on the adoption of an alternative that improves their situation.

16.3 Mitigation and Compensation

Often the alternatives that are being compared do not satisfy the DM (in a MO Problem) or consensus cannot be reached among the Stakeholders or among the DMs (in a MDM problem), so that the phase *Comparison* and the following *Final Decision* do not always conclude with the identification of the *best compromise alternative*. The negotiations, however, always highlight the reason for dissatisfaction or conflict, and so, before definitively concluding that there are no alternatives that achieve consensus of all the involved parties, one should:

[2] In the following, where it is not necessary to distinguish between DM and DMs we will use the generic term DM(s) with which we will refer to both meanings.

- look for *new alternatives* based on the actions suggested by the increased understanding of the conflict, then submit them to the phases *Estimating Effects*, and *Evaluation*, and finally compare them with the alternatives that are already on the table;

- look for *mitigation measures* (which are actions that partially modify the alternatives that obtained the widest agreement, with the aim of removing the negative effects reported by the dissatisfied Stakeholders) and/or *compensation measures* (actions which compensate, in an appropriate way, the disadvantages about which the Stakeholders complain).

This investigation is carried out in the phase *Mitigation and Compensation*, which is described in Chapter 22.

16.4 Organization of Part D

The general framework that we have just outlined is developed in the six chapters that make up Part D. In particular, under the assumption that there is only one DM, in Chapter 17 the main approaches for dealing with a plurality of criteria are presented and compared, with the aim of identifying the one that best meets our needs. We assume that both the set of alternatives to be evaluated and their effects are known. In practice, this can be true only when the number of alternatives considered is finite. If this is not the case, it is necessary to identify the 'most interesting' alternatives from the infinite set, by solving a Multi-Objective Design Problem that is formulated coherently with the chosen decision-making method. This poses a series of difficulties that will be analysed in Section 17.6.

In Chapter 18 we will explain how the Design Problem is formulated and solved when there are multiple objectives. We will focus our attention on the *Pareto Frontier*: the main tool for conflict analysis, an initial idea of which was provided in Section 2.3. Chapter 19 is dedicated to the techniques for *Estimating Effects* produced by an alternative. Chapter 20 describes the techniques for the *Evaluation* based on the MAVT method; Chapter 21 looks at techniques to adopt in the negotiations or when the DMs are more than one. Finally, Chapter 22 deals briefly with *Mitigation and Compensation*.

Chapter 17

Choosing the decision-making method

FC, FP and RSS

When there are m conflicting indicators, even in the case in which there is only one Decision Maker (DM), in order to determine the best compromise alternative it is first necessary to evaluate the alternatives. This evaluation can be executed with different methods, called *decision-making methods*, which provide a procedure for coming to a decision in a transparent and repeatable way. The choice of the decision-making method to be adopted is critical, because it can influence the final decision.

In this chapter we will compare several of the most commonly used decision-making methods: *Multi-Attribute Value Theory* (MAVT); *Analytic Hierarchy Process* (AHP); and the family of ELECTRE methods. In our presentation we will assume that the system is not affected by random disturbances, that the number of alternatives considered is finite and that the effects that each one of them produces are known. We will begin our analysis by considering what it means to 'choose' an alternative and which kind of characteristics the decision-making method and the DM must have. Then, we will describe the three methods one by one; we will give some general indications about how to choose among them, and we will look at the difficulties that arise when the number of alternatives to be compared is infinite and/or the system is affected by random disturbances. We will then identify the method most suited for our context and conclude that it is the MAVT.

17.1 Rankings and ordinal scales

Assume that the set $\mathcal{A} = |A0, \ldots, An_A|$ of the n_A alternatives from which the DM must choose and the effects that each one produces are known. These latter are expressed through the vector $\mathbf{i}$ of the indicators. Assume for the moment that the system is not affected by random disturbances and thus these vectors are deterministic; we will remove this hypothesis in the last section. On the basis of this information the DM must select the alternative that she prefers. To do so, she must rank the alternatives in set $\mathcal{A}$ according to decreasing values of her satisfaction: the first alternative in the order is the one chosen. Thus, choosing means ranking, and thus how to create the *ranking* is the object of our attention. A ranking can be expressed in various ways, according to various *scales* (or paradigms), and the most simple and direct one is the *ordinal scale*. According to this scale a ranking only expresses the

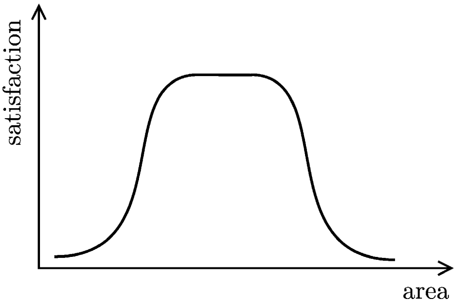


Figure 17.1: Satisfaction associated with the surface area of a wetland.

DM's[1] preferences about the alternatives. For example, given the alternatives $A1$, $A2$, $A3$ and $A4$, a (ordinal) ranking is expressed in the following form

$$\{A4,\ A1 \sim A3,\ A2\} \tag{17.1}$$

This states that $A4$ is preferred over $A1$ (in the following we denote this fact with $A4 \succ A1$), $A1$ is equivalent ($\sim$) to $A3$, which is preferred over $A2$. This is clearly sufficient to conclude that the chosen alternative is $A4$.

However, it is not easy for the DM to establish such an order, since the information that she has about each alternative is the corresponding vector of indicators. She must therefore compare these vectors, which is not a simple task for two reasons. The first reason is that the value of an indicator quantifies an effect in physical units and this is quite different from the level of *satisfaction* that the DM attributes to that effect: in fact, her satisfaction is not always proportional to the value that the indicator assumes. Consider for example a project in which the canalization of a river channel reduces the surface area of a wetland, which is potentially malarial, where migratory birds nest and mosquitoes reproduce. A good indicator of the contraction of the wetlands is its surface area. The environmentalists' satisfaction is not proportional to the area since, if its value is too high, there could be an annoying and dangerous swarm of mosquitoes; their satisfaction is not inversely proportional to the area either, because reducing it too much would pose a risk to the survival of the migratory birds. To conclude, their satisfaction is a bell-shaped function of the area (Figure 17.1), because the maximum satisfaction is obtained with intermediate values. It is thus necessary to find a way to associate a value of satisfaction to each value of the indicator. Once this has been achieved, it is possible to *rank* the alternatives with respect to the satisfaction that the indicator produces: in this way one obtains a *partial ranking* for each indicator.

Now the second difficulty arises: the partial rankings for the different indicators are generally different, in fact, the first alternative is rarely the same in all of them, because the evaluation criteria are usually in conflict. The task of the DM is thus to reduce this plurality of partial rankings, by defining, on the basis of these, a *global ranking*, in which the alternatives are listed from the best to the worst according to the 'global satisfaction' that each one produces. To define this satisfaction, value judgements come into play, which

[1] As already said, in this chapter we assume that there is just one Decision Maker and that the Stakeholders do not have a decision role. The person that conducts the evaluation is therefore the DM. The considerations that are developed, however, are also applicable to the case in which the evaluation is conducted by several people at the same time, for example a group of DMs or Stakeholders. In Chapter 21 we will explain how to recompose their independent evaluations in one ranking.

are, by nature, subjective. A transparent decision thus requires that these subjective aspects be well highlighted and kept separate from the objective aspects (the effects produced). So the *ordering rule*, i.e. the procedure with which the DM passes from the partial rankings to the global ranking, must be explicitly formulated.

17.1.1 Arrow's Theorem

An ordering rule does not only require an explicit formulation: it must also be *democratic*, because only in this case will it be accepted by the Stakeholders. But what does 'democratic' mean? An interesting definition of 'democratic' comes from Arrow,[2] according to whom an ordering rule is democratic if, in *non-trivial* cases, i.e. when there are at least three alternatives to be ordered, it satisfies the following conditions:

(1) *unanimity*: if in all the partial rankings $A1$ is preferred over $A2$ ($A1 \succ A2$), then in the global ranking there is $A1 \succ A2$;

(2) *non-imposition* (or *citizen sovereignty*): the global ranking depends exclusively on the partial rankings;

(3) *non-dictatorship*: the global ranking does not always coincide with one of the single partial rankings while ignoring all the others;

(4) *independence from irrelevant alternatives*: this condition is valid when the alternatives are designed, evaluated and compared in two or more subsequent phases. Suppose that in the first phase alternatives $A1$ and $A2$ are generated, and that the global ranking derived from the rule is the following: $\{A1, A2\}$. If in the second phase alternative $A3$ is generated, the global ranking of the three alternatives must be such that the preference between $A1$ and $A2$ remains unvaried: the global ranking could be $\{A1, A3, A2\}$, or $\{A1, A2, A3\}$, or $\{A3, A1, A2\}$, but it cannot be $\{A2, A3, A1\}$. This condition is important because it prevents alternatives from being inserted in the set $\mathcal{A}$ with the sole aim of manipulating the order of the others.

Different rules have been proposed for generating a global ranking from the partial rankings when they are expressed with ordinal scales. The most famous rules were elaborated in the second half of the 1700s by Borda (1781) and Condorcet (1785). The simplest is the rule of *simple majority*, which is the one upon which most electoral systems are based, and which we now illustrate to give an idea of how such rules can be formulated.

The *rule of simple majority* establishes that, given a set of alternatives, the relationship between a pair of alternatives in the global ranking is equivalent to the one they have in the simple majority of the m partial rankings. For example, consider a decision problem with three alternatives $A1$, $A2$ and $A3$ and three indicators relative to three evaluation criteria: flooding, irrigation and hydropower production. Suppose that, on the basis of values of the three corresponding indicators, the DM has expressed the following three partial rankings:

- flooding: $\{A1, A3, A2\}$;
- irrigation: $\{A2, A1, A3\}$;
- hydropower production: $\{A2, A3, A1\}$.

[2] Nobel Prize for Economics in 1971.

We note that

- $A1 \succ A3$ in the simple majority of the partial rankings (flooding and irrigation) and thus in a global ranking based on the rule of simple majority $A1 \succ A3$;
- $A2 \succ A1$ in the simple majority of the partial rankings (irrigation and hydropower production) and thus $A2 \succ A1$ also in the global ranking;
- $A2 \succ A3$ in the simple majority of the partial rankings (irrigation and hydropower production), and thus $A2 \succ A3$.

The global ranking will therefore be the following:

$$\{A2, A1, A3\}$$

In order to accept the rule of simple majority we must make sure that it is democratic, but this is useless, since in 1951 Arrow identified the following paradox, today known as *Arrow's Paradox* (Arrow, 1951), which shows that in some cases the rule of simple majority fails to produce a meaningful global ranking and therefore it is not an acceptable ordering rule. Let us suppose that in the previous example the DM expresses the following partial rankings:

- flooding: $\{A1, A2, A3\}$;
- irrigation: $\{A2, A3, A1\}$;
- hydropower production: $\{A3, A1, A2\}$.

By aggregating these orders with the simple-majority rule, the following relations are obtained:

- $A1 \succ A2$ in two of the three partial rankings (flooding and hydropower production): thus $A1 \succ A2$ in the global ranking;
- $A2 \succ A3$ in two of the three partial rankings (flooding and irrigation): thus $A2 \succ A3$;
- $A3 \succ A1$ in two of the three partial rankings (irrigation and hydropower production): thus $A3 \succ A1$

from which the following global ranking results

$$A1 \succ A2 \succ A3 \succ A1$$

Clearly this ranking is paradoxical, because it does not respect the property of transitivity.[3]

One might think that the problem is in the simple-majority rule and that a different ordering rule should be found; but this conjecture must also be abandoned, since in 1963 Arrow proved that when the ordinal scale is adopted, it is not possible to obtain a global ranking from the partial rankings by using democratic ordering rules, i.e., that an ordering rule that respects all four conditions on page 381 cannot be found when the ordinal scale is adopted (*Impossibility Theorem*) (Arrow, 1963).

[3] For a formal definition of this term see Section 17.2.

17.1.2 Absolute and interval scales

In the previous section we have seen that, when looking for a democratic (in the Arrow sense) ordering rule, ordinal scales are inadequate. Therefore, it is necessary to refer to scales that express not only qualitative preference judgements but also the intensity of the preference (*cardinal scales*), as, for example, the *absolute scale*.

These scales generate cardinal rankings that can be expressed with the following notation

$$\{18, 15, 18, 21\} \tag{17.2}$$

where the first number is the value attributed to alternative $A1$, the second to $A2$, and so on. Note that from the cardinal rankings an ordinal ranking can always be extracted; for example, from the cardinal ranking (17.2) it is possible to extract the ordinal ranking defined by (17.1). The adoption of an absolute scale requires us to fix the unit of measurement of the values and the zero, which represents total lack of value; this is not a simple task, since these elements are very subjective. Therefore, often an *interval scale* is preferred, i.e. a scale in which preference intensities are not related to the absolute values, but to the differences in value between the alternatives.

On an interval scale, the value can be defined in an infinite number of equivalent ways, each related to the other by a linear transformation with positive coefficients. For example, by adopting an interval scale, the ranking (17.2) is equivalent to the following

$$\{1810, 1510, 1810, 2110\} \tag{17.3}$$

which is obtained by applying the transformation

$$i' = 100i + 10$$

to (17.2); but it is also equivalent to the following

$$\{2.8, 2.5, 2.8, 3.1\} \tag{17.4}$$

which is obtained by applying the transformation

$$i' = 0.1i + 1$$

and so on.

For instance, temperature scales are interval scales: they associate different values to the two physical states of water freezing and water boiling, 0 and 100 in the Celsius scale (°C), and 32 and 212 in the Fahrenheit scale (°F). These values are linked by the following relation

$$F = 1.8C + 32$$

Zero degrees on the Celsius scale corresponds to 32 degrees on the Fahrenheit scale, but both measure the same effect: the freezing temperature of water. One could apply any other monotonic transformation with positive coefficients to the values of temperature indicators to produce an infinite number of other temperature scales, all of which would be equivalent.

In conclusion, by adopting an interval scale, one can accept that a value is a subjective quantity, without it influencing the course of the decision-making process. This is why interval scales are very suitable for expressing intensity of preference. The adoption of an interval scale also offers practical advantages, as we will see in Section 20.4 when describing more deeply the MAVT method, which adopts an interval scale.

17.2 Preference axioms

In the previous section we analysed *what* to make a choice means and we said that choosing means ranking; then, we introduced several ways in which ranking can be expressed. Now, we will consider *how* to make a choice. The aim of the evaluation, in fact, is to formalize the logical procedure through which the DM makes a choice. In this way the decision is made transparent, in the sense that the objective and subjective motivations at the basis of the decision are made explicit and repeatable. This is the equivalent of identifying the so called *preference structure* of the DM (we will return to this subject in Section 20.2). However, a DM does not necessarily have a preference structure: if, for example, she decides randomly, she would not have one. Therefore we must firstly establish under which conditions this structure exists. The majority of authors hold that the existence of preference structure requires the satisfaction of the following axioms:

Axiom 1 (*Completeness*): *Given two alternatives $A1$ and $A2$, the* DM *is always able to say which of the two she prefers, and by how much, or whether they are equivalent to each other.*

Axiom 2 (*Transitivity*): *Given three alternatives $A1$, $A2$ and $A3$, if the* DM *prefers $A1$ to $A2$ and $A2$ to $A3$, then she must necessarily prefer $A1$ to $A3$, i.e.*

$$A1 \succ A2 \quad and \quad A2 \succ A3 \quad \Rightarrow \quad A1 \succ A3 \tag{17.5}$$

Axiom 3 (*Independence from irrelevant alternatives*): *Given a set $\mathcal{A}$ of alternatives, the order of preference that the* DM *establishes between any two of them does not vary if $\mathcal{A}$ is enlarged with a set $\mathcal{A}'$ of new alternatives.*

We have already encountered this axiom among the conditions required for an ordering rule to be democratic (see page 381).

Axiom 2 adopts ordinal expressions and it is thus suitable when an ordinal scale is adopted. Therefore, when working with the interval scale, it must be substituted by the following axiom, which is its cardinal form:

Axiom 2b (*Consistency*): *Given three alternatives $A1$, $A2$ and $A3$, if the* DM *affirms that $A1$ is m times more preferable to $A2$ and $A2$ is n times more preferable to $A3$, then she must necessarily affirm that $A1$ is $m \cdot n$ times more preferable to $A3$, i.e.*

$$A1 \sim m\,A2 \quad and \quad A2 \sim n\,A3 \quad \Rightarrow \quad A1 \sim m \cdot n\,A3 \quad with\, m,n > 0 \tag{17.6}$$

Evaluation methods based on the hypothesis that the DM satisfies the *preference axioms* listed above are called *normative methods* (or prescriptive methods) (French, 1988). On the other hand, *descriptive methods* do not require a priori that the DM's preference structure satisfies any given axioms. Instead, they try to follow the DM's own way of reasoning. The price of the malleability of these methods is a loss of mathematical rigour; the advantage is a more realistic representation of the decision-making process. The choice between the two methods is left to the Analyst, who must assess whether a rigorous mathematical approach would be helpful to organize the information provided by the DM, or would be a strong constriction on her way of reasoning.

Table 17.1. The four forms of problem

Set of alternatives	Quantitative indicators	Quantitative and qualitative indicators
Discrete and finite	Form 1	Form 2
Continuous	Form 3	Form 4

The evaluation method should be chosen not only on the basis of the DM's needs: it should also take into account the characteristics of the problem that is being tackled. Certain methods can in fact be used only with problems of a given form and not with others. The characteristics that define the form of a problem are the nature of the set (either discrete-finite or continuous) of the alternatives to be considered and the nature of the indicators (either quantitative or quantitative–qualitative) used to estimate the effects. The four forms of problem that derive from these characteristics are classified in Table 17.1. Note that a real problem can be formalized in different forms, according to how one decides to formulate it (e.g. by adopting only quantitative indicators or not).

Now we will illustrate three decision-making methods: *Multi-Attribute Value Theory*, *Analytic Hierarchy Process* and ELECTRE methods. We chose these three both because they are the most widely employed, and because together they cover the first three forms of problem in Table 17.1, which are the only ones that can be solved with the knowledge we have today. Our description will not be exhaustive, but it will simply aim at providing the elements necessary for a comparison of the three methods. Since, as we will see in the following, Multi-Attribute Value Theory is the most suitable method for the problems we deal with, Chapter 20 will be dedicated to a more detailed description of it.

17.3 Multi-Attribute Value Theory

Multi-Attribute Value Theory (MAVT) (Keeney and Raiffa, 1976) is based on the identification of a *value function* $V(\cdot)$. This function, the argument of which is the vector of the indicators that quantify the effects of an alternative, returns a single value that expresses the DM's satisfaction for the alternative. Increasing values of V correspond to increasing levels of satisfaction. Once the value function has been identified, it is possible to order any set of alternatives. The value V is thus the Project Index that we introduced in Section 3.4. The MAVT can be applied to Form 1, 2 and 3 problems (see Table 17.2).

Since it is extremely difficult to identify the value function directly, the problem is led back to the definition of a *partial value function* for each one of the indicators. This is possible only if the DM's choices satisfy the axioms of Completeness, Consistency and Independence from irrelevant alternatives. The MAVT is thus a normative method. The (global)

Table 17.2. The forms of problem to which the MAVT can be applied

Set of alternatives	Quantitative indicators	Quantitative and qualitative indicators
Discrete and finite	♦	♦
Continuous	♦	

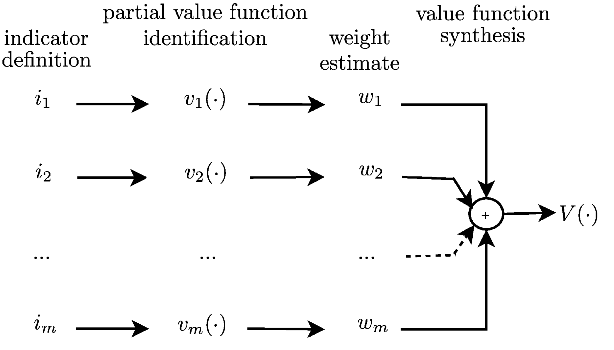


Figure 17.2: Steps in identifying the value function, given the values $i_1, \ldots, i_m$ of the indicators, according to the MAVT.

value function is then obtained through the composition of the partial value functions, generally through a weighted sum (see Figure 17.2). The values of the coefficient (*weights*) in the sum depend on the relative importance that the DM associates to each of the indicators. Notice that, by computing the value function as a weighted sum of the partial value functions, one implicitly assumes that a poor performance of one of the indicators can be balanced by good performances of others. In other words one implicitly assumes that *Compensation* is allowed among the criteria. The MAVT thus requires that compensation exists among all the indicators (with the exception of one at the most, Keeney and Raiffa, 1976).

In order to guarantee coherence among the values provided by the partial value functions and the weights, it is important that the latter are estimated, with the procedure that we will define in Section 20.6, on the basis of the values that each indicator can assume and not of abstract considerations, far from the specific context examined. Also the definition of the partial value functions requires a series of complex operations, in which the DM has to express her evaluation of the effects of an alternative in a formalized way. Any attempt at simplifying these operations by adopting quick hypotheses should be absolutely avoided: for example, by hypothesizing a priori, i.e. without due verification, that the partial value functions are linear. By doing thus in fact, the results are seriously invalidated. We will return to this discussion in Chapter 20.

17.4 Analytic Hierarchy Process

Just as in the MAVT, in the *Analytic Hierarchy Process* (AHP) (Saaty, 1980, 1992) the performance of an alternative is obtained through the weighted sum of the performances that it attains with respect to the single indicators. Therefore, compensation must be admitted among the criteria. However, the procedure used to define the weights and the performances, given the effects of each alternative, is very different from the one used in the MAVT: it follows that the types of problems to which it can be applied are not the same (see Table 17.3).

Once the hierarchy of criteria (Chapter 3) has been defined, the AHP requires that all the alternatives are appended to each one of the leaf criteria. For example, Figure 17.3 shows the hierarchy of the *Upstream Tourism* sector in the Verbano Project (see Chapter 4 of PRACTICE) in the form required by the AHP. The evaluation is then performed in two steps: *pairwise comparison* and *hierarchical recomposition*.

Table 17.3. The types of problem to which the AHP can be applied

Set of alternatives	Quantitative indicators	Quantitative and qualitative indicators
Discrete and finite	♦	♦
Continuous		

First, the DM is required to compare, for each criterion in the hierarchy, all the elements that depend on it at the immediately lower level. For example, when the leaf criterion *Reduced access to beaches* is considered in the hierarchy of Figure 17.3, the elements of the lower level are the alternatives; when the sector criterion *Loss of activity due to vacating tourists* is considered, the elements of the lower level are the three criteria *Reduced landscape aesthetics*, *Reduced access to beaches*, *Discomfort produced by mosquitoes*; etc. The comparison is aimed at identifying a *ranking vector* for each criterion, whose ith component quantifies how much the ith element of the lower level satisfies that criterion. The ranking vector can thus be used to rank the lower-level elements with respect to the criterion being considered.

The identification of the ranking vector for each criterion is performed by asking the DM to make pairwise comparisons of the elements at the lower level, filling in a square matrix (called *pairwise comparisons matrix*), whose elements a_{ij} express the intensity of preference for the element in the ith row compared to the element in the jth column, with respect to the criterion under examination. The ranking vector is then extracted from this matrix by means of a suitable algorithm. Note that, depending on the level in the hierarchy, the pairwise comparisons matrix expresses preferences either among alternatives or criteria. In the first case, the ith component of the resulting ranking vector measures how much the ith alternative satisfies a given criterion; in the latter, it measures the relative importance of the ith criterion with respect to the upper level criterion, i.e. the ranking vector is a vector of weights.

It is important to underline that, when filling in the pairwise comparison matrix, the DM might not satisfy the axiom of Consistency: the AHP, in fact, allows the ranking vectors to be derived even when there are slight[4] inconsistencies. Unlike the MAVT, the AHP is therefore

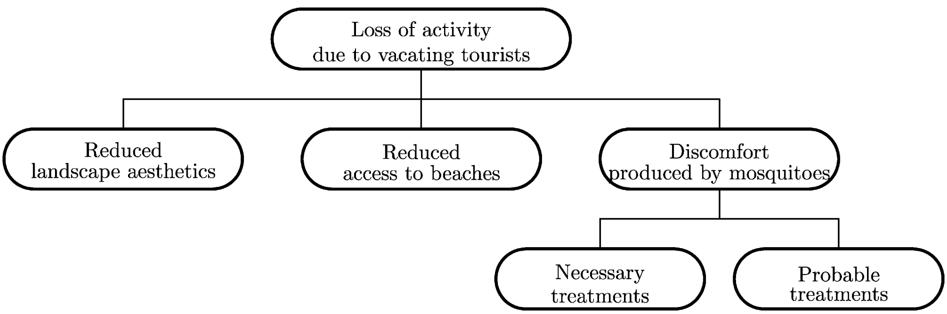


Figure 17.3: The hierarchy for the *Upstream Tourism* sector of the Verbano Project, in the case when three alternatives were to be compared.

[4] An inconsistency is slight when it does not cause a rank-reversal between two alternatives. For example, if the DM says that $A1 \sim 2 \cdot A2$ and that $A2 \sim 2 \cdot A3$, it is not necessary that she affirm that $A1 \sim 4 \cdot A3$ (as the axiom

a descriptive method. It requires only that the condition of *Reciprocity* be satisfied, i.e. that $a_{ji} = 1/a_{ij}$, a condition that is so obvious that it is almost always satisfied.

Now, once a pairwise comparison matrix has been filled in for every criterion in the hierarchy, it is possible to derive the ranking vectors of the alternatives with respect to each leaf criterion, and the ranking vectors of the criteria with respect to each upper-level criterion. Through a sequence of matrix products a final ranking vector of the alternatives with respect to the root criterion of the hierarchy is obtained; this operation is called *hierarchical recomposition*. Note that deriving the final ranking of the alternatives as a linear combination of the alternatives' ranking vectors with respect to the single leaf criteria, implies assuming compensation among the criteria.

With respect to the MAVT, the AHP makes the interaction with the DM simpler, because she is no longer asked to identify functions that express how her degree of satisfaction varies with the indicator values, but only to express her own preferences through a sequence of pairwise comparisons. The questions that are posed are always the same, regardless of the elements in the hierarchy that are taken into consideration: "Do you prefer criterion (alternative) X or criterion (alternative) Y, with respect to criterion Z? And what's the intensity of your preference?" Answers to such questions are usually expressed through Saaty's verbal scale, which allows nine possible judgments, ranging from "very weak" to "very strong" preferences, which are then transposed into numbers between 1 and 9. However, simplifying the interaction with the DM has three disadvantages:

(1) the number of questions to be posed increases: the DM must express her preference between all possible pairs of elements at every level of the hierarchy;

(2) the complexity of the method is increased from a mathematical point of view;

(3) there is the risk that the questions posed to the DM will be poorly formulated.

Let us analyse these issues one at a time.

1. The number of questions increases significantly as the number of alternatives and criteria increases. For example, consider a hierarchy with only two levels, n alternatives and m criteria: the number of questions that must be posed is given by the following relation[5]

$$\#\,\text{questions} = m \cdot \frac{n \cdot (n-1)}{2} + \frac{m \cdot (m-1)}{2} \tag{17.7}$$

   It is clear that, as n and m increase, it rapidly becomes impossible to pose all the questions.

2. The increase in mathematical complexity would not of itself be a problem but for the fact that it introduces degrees of freedom into the methodology which can be used by the Analyst but not by the DM (for example the choice of the algorithm used to obtain the scoring vector from the pairwise comparisons, in the event that the matrix provided by the DM is not consistent).

of Consistency would have done), but only that she judge $A1$ to be better than $A3$. This is the same as requiring that the axiom of Transitivity be satisfied.

[5] To derive it, remember that for the pairwise comparison matrices the condition of Reciprocity must be fulfilled and that the elements along the diagonal do not need to be provided by the DM, because they are equal to 1 by definition.

Table 17.4. The types of problem to which the ELECTRE methods can be applied

Set of alternatives	Quantitative indicators	Quantitative and qualitative indicators
Discrete and finite	♦	♦
Continuous		

3. Questions to the DM are poorly formulated because, when comparing the criteria, the DM does not refer to the range of values actually assumed by the indicators: she expresses her judgements in the abstract. To provide correct responses, instead, it would be necessary to consider the range of possible indicator values, since the importance of criterion X with respect to criterion Y might depend on the value of the indicator that measures the satisfaction of X. To clarify this point, consider again the hierarchy in Figure 17.3. If the indicator associated to the criterion *Reduced landscape aesthetics* takes on values between 'unacceptable' and 'sufficient', while the one associated to the criterion *Discomfort produced by mosquitoes* assumes values between 'good' and 'optimal', one naturally tends to give the first more importance, because it is the more worrying one. However, the judgement could be reversed if the alternatives would produce very positive effects for the first indicator and negative ones for the second.

Finally, note that unlike the MAVT, which foresees defining value functions that can be later on be used to order any set of alternatives, with the AHP only the ranking of the alternatives that were directly compared by the DM is obtained. This means that if a new alternative is introduced, in order to know its position in the ranking it would be necessary to ask the DM to compare it with all the alternatives that have already been considered. Furthermore, since the ranking provided by the AHP depends on the alternatives being considered, the method does not satisfy the axiom of *Independence from irrelevant alternatives*. Therefore, the introduction of a new alternative could change the order of the other ones. This effect could be exploited to manipulate the order itself: alternatives could be introduced with the specific aim of sabotaging a given alternative in the final ranking.

17.5 ELECTRE methods

The ELECTRE methods (*ELimination Et Choix Traduisant la REalité*) (Roy, 1978, 1996) are only applicable to problems with a discrete set of alternatives (see Table 17.4).

The literature speaks of ELECTRE *methods* because a number of methods have been developed from the same founding idea. They all intend to be as close as possible to the actual decision-making process, which means that they strive not to introduce any elements that oblige the DM to follow a line of reasoning that is not her own. ELECTRE methods are thus descriptive methods. As in the AHP, Consistency of preferences is not required; moreover, ELECTRE methods also admits *Incomparability* among the alternatives and *Lack of transitivity* in the rankings. As far as incomparability is concerned, note that both the MAVT and the AHP assume that an alternative's bad performance with respect to a specific criterion can be compensated for by good performances with respect to the other criteria, whatever the gravity of the bad performance. It follows that it is always possible to compare two alternatives and identify which is preferable: this is the very reason why these two methods satisfy the axiom of Completeness. On the contrary, in the ELECTRE methods it is possible for the DM to establish thresholds over which compensation among criteria is no longer possible: a circumstance that can often occur in reality. For example, think of a comparison between two cars, a utility passenger car and a limousine. The latter is for sure better than the former for many criteria, such as comfort, velocity, and perhaps also aesthetics. However, since it is much more expensive than the utility passenger car, it is not possible to express a clean preference for it. In other words, better performances on nearly all the criteria cannot compensate for the bad performance on the price criterion. In this situation, the DM is not able to express her preference between the two alternatives, which the ELECTRE methods would classify as incomparable among each other.

As far as transitivity is concerned with, the ELECTRE methods acknowledge that, being human, the DM has limited discrimination capacity. This means that, when the difference between two indicator values is lower than a certain *discrimination threshold*, the DM does not perceive those values as being different. The values of the discrimination thresholds, which differ from one indicator to another, must obviously be defined by the DM. For example, when the DM has to compare alternatives from the point of view of supplying water to an irrigation district, she might not perceive the difference between supplies that differ by less than 2 m^3/s, while the difference could become appreciable for values of 5 m^3/s and substantial over 10 m^3/s. However, allowing for limited discrimination capacity may lead to intransitivity, as in the case of the "coffee paradox". Think of two cups of coffee, of which one is bitter, the other is sweetened with one sugar grain only: since her discrimination capacity is limited, the DM is not able to express a preference among them and judges them as indistinguishable. Considering the sweetened cup and comparing it with a cup containing two grains of sugar, again the DM judges them undistinguishable. Thus, for transitivity, the bitter cup is judged undistinguishable from the sweetened with two grains cup. Keeping in comparing two cups which only differ for one sugar grain, the DM ends for comparing a completely saturated of sugar cup with a nearly saturated of sugar cup, and, again, judges them as undistinguishable. Applying transitivity to the whole chain of comparisons, the bitter cup results undistinguishable from the saturated of sugar cup. But this makes no sense, since they taste completely different. This paradox shows that accepting limited discrimination capacity implies the ranking might lack in transitivity.

From these considerations it follows that the ranking of the alternatives obtained with an ELECTRE method might be incomplete and/or non-transitive.[6] This means that, given an alternative, it might be impossible classifying all the other alternatives as better or worse than it, since some could be non-comparable and/or indistinguishable. In these conditions, it might be only possible understanding whether between two alternatives there is an *outranking relationship*, i.e. if one of them is clearly preferable to the other (Roy, 1991). If such a relationship exists, then the ranking reflects it by putting the outranking alternative in a better position than the one that is outranked (such as $A2$ with respect to $A4$ in Figure 17.4a). In order that an alternative outrank another, it is necessary that the reasons in its favour (or at least those that do not oppose it) be sufficiently strong with respect to the ones to the contrary. If an outranking relationship does not exist between two alternatives, they are either *indistinguishable* or *not-comparable* (for example $A2$ and $A1$ in Figure 17.4b).

The ELECTRE methods all share this framework. Some of them allow to identify only a core set of alternatives, which are judged incomparable or indistinguishable among each

[6] A ranking is said to be complete when all the alternatives being considered have a place in it and each position in the order is occupied by only one alternative (Figure 17.4); it is said to be transitive when it satisfies the axiom of Transitivity.

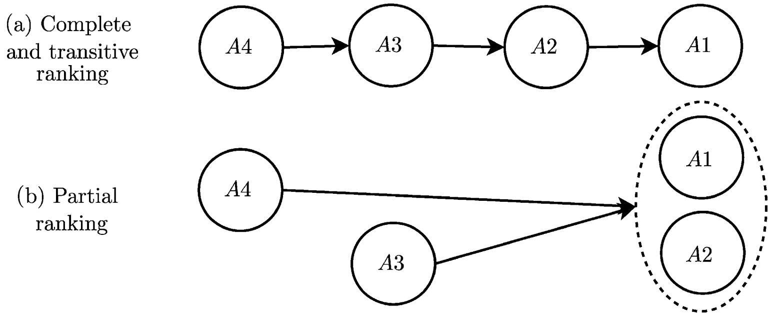


Figure 17.4: Comparison between a complete and transitive ranking (a) and a partial one (b). In the first, $A4$ is worse than $A3$, which in turn is worse than $A2$, and so on. In the second, $A3$ and $A4$ are worse than $A1$ and $A2$, and these latter are indistinguishable; nothing can be said about the relation between $A3$ and $A4$, which are not-comparable.

other. The so-called ELECTRE III method provides also a ranking of the alternatives, though partial. It requires the DM to provide a set of discrimination thresholds, one for each criterion considered, and a vector of weights for the different criteria. With this information, the ranking of the alternatives is produced through an automatic procedure, which we cannot describe here, but which the reader may find in Roy (1996).

We can thus conclude that:

1. Since the outranking relationships are derived for pairs of alternatives, the ranking obtained with the ELECTRE methods depends on the set of alternatives being considered. Just as in the AHP, Independence from irrelevant alternatives is not guaranteed.

2. The number of questions for the DM does not depend on the number of alternatives, but only on the number of criteria (both the thresholds and the weights are defined for the criteria). The introduction of a new alternative thus does not lead to new questions.

3. Even if the method was born with the aim of designing a procedure that was as close as possible to the real decision-making process, the algorithm for extracting the alternatives's ranking contains elements that are difficult for the DM to manage and understand.

17.6 Choice of the method

To facilitate the comparison of the three methods that have been illustrated, Table 17.5 lists, for each one, the preference axioms that it presupposes, while Table 17.6 lists the methods that are applicable for each type of problem. The choice of the method must take into account which axioms are satisfied by the preference structures of the actors involved (DM(s) and/or Stakeholders), and the characteristics of the problem.

Before adopting a method it is necessary to check if the axioms that it presupposes are verified in the context examined. ELECTRE methods do not pose any conditions for the DM's preference structure (see Table 17.5). The AHP requires that the axiom of *Completeness* be satisfied; the MAVT also calls for *Consistency* and *Independence from irrelevant*

Table 17.5. The axioms required by each method

	MAVT	AHP	ELECTRE
Completeness	♦	♦	
Transitivity	♦	♦	
Consistency	♦		
Independence	♦		

Table 17.6. Forms of problem and solution methods

Set of alternatives	Quantitative indicators	Quantitative and qualitative indicators
Discrete and finite	MAVT AHP ELECTRE	MAVT AHP ELECTRE
Continuous	MAVT	

alternatives. Both the MAVT and the AHP require that there can be compensation among the criteria and that the alternatives be comparable. The validity of these axioms can be checked by questioning the actors and analysing their responses.

If the Project involves multiple DMs, none of the methods listed above can be applied as such, because they were all derived in the context of a problem with only one DM. As we will explain in Chapter 21, however, some of the tools developed in the MAVT can be used as a starting point for negotiations.

When the Project foresees the identification of the alternatives in several steps, both the AHP and the ELECTRE methods are not very suitable, because the results that they provide depend on the set of alternatives considered. Often, in fact, to identify the whole set of alternatives, one goes through several steps, in each of which a subset of alternatives is designed, evaluated and compared (for an example see Chapters 11–14 of PRACTICE). The information that emerges during a comparison is used to identify the alternatives to be generated in the next step, for example, by identifying the mitigation actions for the most disfavoured sectors. In order that this procedure be concluded successfully, it is necessary that an alternative that is judged to be preferable to another in a given step continue to be so in the following steps; this is guaranteed only if the evaluation is performed with the MAVT method. If, instead, the AHP or the ELECTRE methods were used, at each new generation the evaluation would have to be repeated ex-novo with all the alternatives, not just those from the last generation; further, adding new alternatives could modify the order of those generated in the previous step, because the *Independence from irrelevant alternatives* is not guaranteed.

When there is a large number of alternatives to be evaluated, the AHP, although applicable in theory, cannot be applied in practice, because the number of questions that must be posed to the DM grows with the square of the number n_A of alternatives. A high number of alternatives does not constitute, however, a limit for the applicability of ELECTRE methods, as long as the number is finite. Finally, when using the MAVT, it is theoretically possible to deal even with an infinite number of alternatives.

Problems with an infinite number of alternatives are the ones most frequently encountered in practice: the number of feasible alternatives is, in fact, infinite when, for example, at least one component of the vector $\mathbf{u}^p$ of planning decisions assumes a value in an infinite set, or when one wants to design a management policy with a functional approach. In both the cases it is impossible to evaluate all the alternatives in an exhaustive way and so it is necessary to consider only the 'most interesting' ones. Then, the problem arises of how to identify these 'most interesting' alternatives. In Chapter 8 we explained that one can choose among infinite alternatives by formulating and solving an optimization problem (the Design Problem). However, in that chapter we dealt with full rationality conditions, i.e. with only one criterion, and therefore the concept of optimality was well defined. With more than one criterion that concept is, instead, no longer evident. We know, in fact, that, with conflicting criteria, the different viewpoints will consider different alternatives as optimal. It is thus clear that, in order to formulate the Design Problem, we must, first of all, define what we mean by 'optimal', i.e. we must define how to recompose the different viewpoints. For this it is, first of all, necessary to choose the evaluation method and then to formulate a Multi-Objective (MO) Design Problem according to the evaluation method adopted.

The presence of an infinite number of alternatives determines the choice of the evaluation method: Table 17.6 shows, in fact, that only the MAVT allows for the evaluation of infinite alternatives. This method also has an interesting extension, known by the acronym MAUT, i.e. *Multi-Attribute Utility Theory* (Keeney and Raiffa, 1976), by means of which it is possible to deal, in a theoretically rigorous way, also with random indicators. By adopting the MAUT, the value function $V(\cdot)$ is replaced by the *utility function* $U(\cdot)$, which was introduced in Section 9.1.3. Thanks to this function, both the DM's satisfaction and her risk aversion can be taken into account at the same time. The MAUT would therefore be the appropriate method to deal with the problems most frequently encountered, if it were not for two difficulties, which we have already met:

1. To identify the utility functions, the range of indicator values in correspondence with the alternatives to be compared must be known; but, to identify the alternatives to be compared, a Design Problem, which includes the utility functions, must be formulated. We talked about this vicious circle when dealing with full rationality conditions in Section 9.1.3, and there we proposed to break it through a recursive procedure. An analogous procedure can be applied in partial rationality conditions.

2. When there are management actions among the actions being considered and we want to solve the Design Problem with algorithms based on SDP, the objectives cannot be defined through the utility function. They must be defined with filtering criteria so that separability is guaranteed (see page 262). It follows that the Design Problem must be formulated and solved in the space of the objectives, i.e. the indicators that are obtained by applying a filtering criterion to the random indicators (see page 133). Thus the evaluation will be performed with the MAVT, however not at the same time as *Designing Alternatives*, but afterwards (and after *Estimating Effects*), when the values of the evaluation indicators are known.

In conclusion, for Pure Planning Problems the most appropriate method seems to be the MAVT, if no random disturbances act on the system, and the MAUT in the opposite case. For Pure Management (or Mixed) Problems it is always necessary to turn to the MAVT, and filter the randomness of the indicators beforehand, by applying filtering criteria. Since our interest is more oriented to this second class of Problems, in Chapter 20 we will go into greater detail about the application of the MAVT. The reader interested in the MAUT can usefully refer to Keeney and Raiffa (1976).

Chapter 18

Identifying efficient alternatives

AC, RSS and EW

We will begin this chapter by showing the form that the Design Problem assumes when there is more than one objective. Then we will develop an explicit visual representation of the conflict among the objectives, in order to make it evident to the Decision Maker (DM) that when she makes a given decision she is making a subjective choice; her decision is subjective in the sense that the choice involves value judgements that are, by nature, political and not objective. The representation that we will use is called the *Pareto Frontier*, and was introduced in Section 2.3.

Then we will see how the problem of individuating this Frontier can be traced back to the solution of a family of Single-Objective Design Problems, such as the ones we studied in Chapters 8, 10 and 11. Since often the number m of evaluation indicators is very high, we will conclude the chapter by illustrating some criteria for selecting a restricted number of evaluation indicators upon which to construct the q objectives of the Multi-Objective (MO) Design Problem.

18.1 Multi-Objective Design Problems

Given q objectives

$$J^i\left(\mathbf{u}^p, P\right) = \mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}\left[i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)\right] \tag{18.1}$$

with $i = 1, \dots, q$, Design Problem (10.26) generally assumes the following form:

The MO Design Problem:

$$\min_{\mathbf{u}^p, P}\left[J^1\left(\mathbf{u}^p, P\right), J^2\left(\mathbf{u}^p, P\right), \dots, J^q\left(\mathbf{u}^p, P\right)\right] \tag{18.2a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, 1, \dots \tag{18.2b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{18.2c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq \mathcal{U}_t\left(\mathbf{x}_t, \mathbf{u}^p\right) \qquad t = 0, 1, \dots \tag{18.2d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t\left(\cdot|\mathbf{u}^p\right) \qquad t = 0, 1, \ldots \tag{18.2e}$$
$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{18.2f}$$
$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{18.2g}$$
$$P \triangleq \left\{M_t(\cdot);\ t = 0, 1, \ldots\right\} \tag{18.2h}$$
$$\textit{any other constraints} \qquad t = 0, 1, \ldots \tag{18.2i}$$

We also assume that the objectives $J^1(\mathbf{u}^p, P), \ldots, J^q(\mathbf{u}^p, P)$ all have a separable form. As we have shown in Sections 18.3.2 and 18.3.3, this assumption is necessary when one wants to solve problem (18.2) with methods based on SDP which require that the objectives be separable. As we explained in Section 17.6, this forces us to define the objectives with filtering criteria, as in equation (18.1), and not with utility functions. Thus, for algorithmic reasons, except when problem (18.2) is a Pure Planning Problem, we work in the space of the control indicators instead of in the space of the utilities, as would be more correct.

Equation (18.2e) has the form shown above when the disturbance is stochastic, but it assumes the following one

$$\boldsymbol{\varepsilon}_{t+1} \in \Xi_t\left(\mathbf{u}^p\right) \quad t = 0, 1, \ldots \tag{18.3}$$

when the disturbance is uncertain. In that case, the criterion adopted for all the objectives can be only the Wald criterion. Lastly, the set Ξ_t or the probability distribution $\phi_t(\cdot)$ could also be conditioned to the exogenous information $\mathbf{I}_t$ which is available at the time t when the problem is formulated.

We should note that the problem does not necessarily relate to the design of a set-valued (SV) policy, as is assumed in problem (18.2); it could also be a point-valued (PV) policy. Similarly, one does not necessarily have to choose both the policy P, and the vector of planning decisions $\mathbf{u}^p$ (structural and/or normative), if only one of the two is really of interest.

Unlike the problems that we have considered up until now, problem (18.2) does not define an optimal alternative, because different objectives generally produce different rankings of the alternatives, and so the 'optimal alternative' will differ depending on which objective is considered. Therefore the concept of 'optimal' must be re-examined, and that is just what we will do in the next section.

18.2 Pareto Efficiency

To introduce this subject we ought to begin with a simple example. Consider two objectives $J^1(u^1, u^2)$ and $J^2(u^1, u^2)$, each to be minimized, whose values depend on a pair (u^1, u^2) of decisions,[1] for which the only admissible values are those that belong to a set $\mathcal{U}$, called *set of feasible decisions* (Figure 18.1). For convenience we will denote the pair of decisions with the vector $\mathbf{u}$. The problem is thus the following

$$\min_{\mathbf{u}}\left[J^1(\mathbf{u}), J^2(\mathbf{u})\right] \tag{18.4a}$$
$$\mathbf{u} \in \mathcal{U} \tag{18.4b}$$

[1] To simplify the example, think of planning decisions, such as the size of two non-regulated diversions; but they could also be two policies.

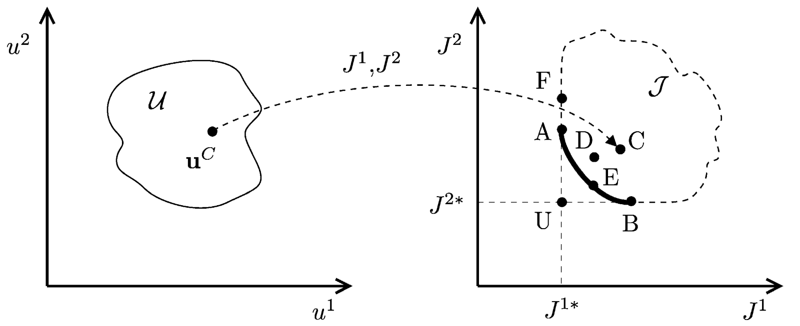


Figure 18.1: The decision space and objective space for problem (18.4).

For every decision $\mathbf{u}$, such as $\mathbf{u}^C$ in Figure 18.1, the two objectives $J^1(\mathbf{u})$ and $J^2(\mathbf{u})$ are the components of a vector $\mathbf{J} = |J^1(\mathbf{u}), J^2(\mathbf{u})|$ that individuates a point in the objective space, in the example, the point C. As $\mathbf{u}$ varies in $\mathcal{U}$ the vector $\mathbf{J}$ describes a set $\mathcal{J}$. Technically, this is expressed by saying that the objectives $J^1(\mathbf{u})$ and $J^2(\mathbf{u})$ 'map' the set $\mathcal{U}$ onto a set $\mathcal{J}$. The set $\mathcal{J}$ thus represents the set of the objectives' value pairs that can be obtained by considering all the feasible decisions, i.e. the decisions that belong to $\mathcal{U}$. The aim of the DM is to identify the decisions $\mathbf{u}$ within $\mathcal{U}$ that produce the 'best', in a sense that we must now define, value for the pair of objectives.

If one considers only the first objective, the best decision is undoubtedly a decision ($\mathbf{u}^A$) which minimizes the function $J^1(\mathbf{u})$, i.e.

$$\mathbf{u}^A = \arg\min_{\mathbf{u}\in\mathcal{U}} J^1(\mathbf{u})$$

for which $J^1(\mathbf{u})$ assumes the minimum value J^{1*} of all the values that J^1 assumes in the set $\mathcal{J}$ (point A in the objective space in Figure 18.1). Similarly, if one considers the second objective, the best decision is $\mathbf{u}^B$

$$\mathbf{u}^B = \arg\min_{\mathbf{u}\in\mathcal{U}} J^2(\mathbf{u})$$

which is represented in the objective space by a point (point B in Figure 18.1) for which $J^2(\mathbf{u})$ assumes the minimum value J^{2*}. Given that both the objectives must be considered, the ideal decision for the DM would be the one that produces a point in the objective space with the coordinates $|J^{1*}, J^{2*}|$, i.e. the point U: she could not, in fact, hope to obtain anything better. Often, however, as in the example that we are considering, in $\mathcal{U}$ there is no decision that produces point U: it is thus an unachievable dream[2] and for this reason it is called the *Utopia point*.

If the Utopia point cannot be reached, the DM is obliged to settle for something less. Consider a decision $\mathbf{u}^C$ which produces the point C with the coordinates $|J^1(\mathbf{u}^C), J^2(\mathbf{u}^C)|$ in the space (J^1, J^2). Let us suppose that another decision $\mathbf{u}^D$ exists, for which the values assumed by the objectives $J^1(\mathbf{u}^D)$ and $J^2(\mathbf{u}^D)$ (point D in Figure 18.1) are such that

$$J^1\big(\mathbf{u}^D\big) \leqslant J^1\big(\mathbf{u}^C\big) \qquad J^2\big(\mathbf{u}^D\big) \leqslant J^2\big(\mathbf{u}^C\big)$$

[2] The point U can be reached only when the two objectives are not conflicting, which means that the minimum of the first implies the minimum of the second. In that case it is often called *ideal point*.

where at most one of the two relationships is verified with the equal sign (if both relationships were verified with the equal sign the two decisions would produce the same performance). Clearly, the DM is not interested in $\mathbf{u}^C$ and for this reason $\mathbf{u}^C$ is called *dominated* decision. We can thus conclude that the only interesting decisions for the DM are those which produce points such as E in Figure 18.1, i.e. decisions which cannot be improved by another decision that provides better performances for both of the objectives. In other words, the interesting decisions are characterized by the fact that *to improve the value of one objective it is necessary to worsen another*. The decisions that satisfy this condition are called *Pareto-efficient*, or shortly *efficient*, from the name of Vilfredo Pareto, a 19th century Italian economist who was the first to define them. There can be more than one efficient decision: for example, in Figure 18.1 all the values of the objectives on the thick curvilinear segment between points A and B correspond to efficient decisions. Their set $\mathcal{P}$ is called *Pareto-Efficient Decision Set* and is formally defined as follows (see for instance Miettinen, 1999)

$$\mathcal{P} = \left\{\mathbf{u} \in \mathcal{U}\colon \nexists \tilde{\mathbf{u}} \in \mathcal{U}\colon \exists j \in \{1, \dots, q\}\colon\ J^j(\tilde{\mathbf{u}}) < J^j(\mathbf{u}) \text{ and } J^i(\tilde{\mathbf{u}}) \leqslant J^i(\mathbf{u})\ \forall i \neq j\right\}$$

The image of this set in the objective space is called *Pareto Frontier*, or, when the context allows, simply *Frontier*. Strictly, it is the set

$$\mathcal{F} = \left\{\left|J^1(\mathbf{u}), \dots, J^q(\mathbf{u})\right| \text{ with } \mathbf{u} \in \mathcal{P}\right\}$$

which, in the example in Figure 18.1, is the curvilinear segment between points A and B.

In summary, we can classify the points of the set $\mathcal{J}$ in three types:

- *dominated* points, for which there exist decisions that improve all the objectives, such as point C;
- *semi-dominated* points (also called *semi-efficient*), for which there exist decisions that improve some of the objectives without worsening the others, such as point F;
- (*Pareto-*)*efficient* points, for which there exist no decisions which improve an objective without worsening at least one of the others, such as points A, E and B.

From the above, it is reasonable to establish that the 'solution of the MO Design Problem', i.e. the solution of problem (18.2), is the Pareto-Efficient Decision Set $\mathcal{P}$. Therefore, the solution of the problem does not provide the decision to adopt, but a set of decisions, upon which the DM must focus her attention. The associated Pareto Frontier $\mathcal{F}$ shows that an increase in one objective's performance will inevitably result in decreasing the performance of another. In other words, the solution defines a trade-off curve among the objectives.

Note that the definition of the Pareto Frontier pivots on efficiency alone and makes no reference to equity. For example, if the problem is that two people A and B must share a roast chicken, efficiency requires the whole chicken to be eaten. However, it considers equivalent the following three alternatives: a decision is considered to be efficient if the whole chicken is eaten, and each of the following three decisions is equally efficient

(a) the whole chicken is eaten by A;

(b) the whole chicken is eaten by B;

(c) half of the chicken is eaten by A and the other half by B.

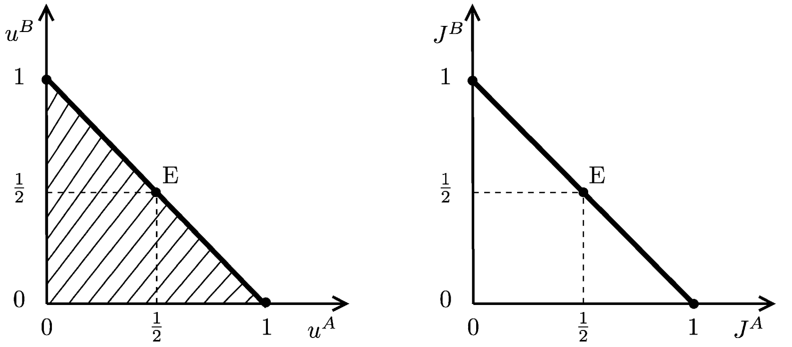


Figure 18.2: The decision and objective spaces in dividing a roast chicken.

More precisely, if we indicate the fraction of the chicken eaten by each person with u^A and u^B and we assume that each one's satisfaction J is measured by the fraction he ate (i.e. we assume $J^A = u^A$ and $J^B = u^B$), the set $\mathcal{U}$ of feasible decisions is the striped triangle in the left-hand diagram in Figure 18.2. The Pareto-Efficient Decision Set is the thick line segment in the left-hand diagram while the Pareto Frontier is the analogous segment in the right-hand one. If equity is used as a selection criterion, the only equitable decision that is also efficient is point E. Thus, Pareto Efficiency is a criterion that photographs the conflict, highlights it, but has nothing to do with equity. Note also that efficiency should not be idealized, because it is always relative to the objectives that are being considered.

By establishing that the solution to problem (18.2) is the Pareto-Efficient Decision Set, we have also established that the solution of the problem does not provide a particular decision. What we expect from this solution is that it excludes the decisions that can be ignored without expressing a value judgement, i.e. without bringing subjectivity into play. In fact, no DM (nor any Stakeholder) would ever be interested in the decisions that do not belong to the Pareto-Efficient Decision Set,[3] because they are dominated: in other words, there exist better decisions from all viewpoints, i.e. with respect to all the objectives. Making the decision is thus postponed to a later phase, when value judgements must necessarily come into play, i.e. when DM and/or Stakeholders can no longer avoid expressing their opinion about the relative importance of the objectives. This will happen in the phases *Evaluation* and *Comparison*, which we analyse in Chapters 20 and 21. However, before the alternatives can be evaluated and compared, we must address the problem of how to identify the Pareto Frontier[4] once problem (18.2) has been formulated.

18.3 Determining the Pareto-efficient decisions

As is often the case, the way to solve problem (18.2) is to trace its solution back to a procedure that is already known. After all, we have used this strategy since we were children, when we tried to persuade our friends to play the game that we were good at. Now, we are

[3] Clearly this is true only if the objectives being considered actually reflect the entire set of the DM's or Stakeholders' interests.

[4] More precisely, since we have to determine the Pareto-Efficient Decision Set and the Pareto Frontier, we should mention both; however, for the seek of brevity, we will always mention only the second.

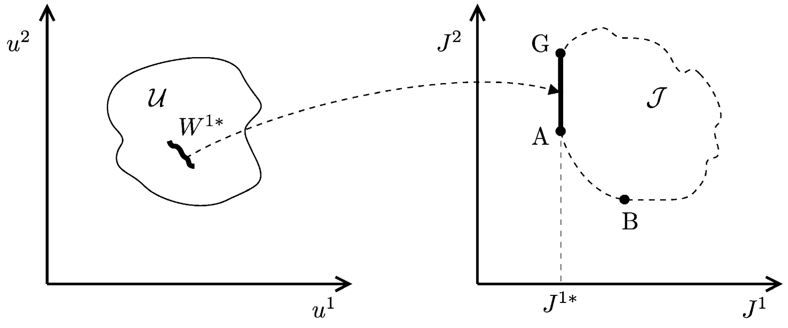


Figure 18.3: Determining a point on the Pareto Frontier for problem (18.4) with the Lexicographic Method.

good at solving Single-Objective Design Problems, such as problem (11.13). Therefore the common strategy is to reduce problem (18.2) to problem (11.13), and there are a number of ways this can be done, which are more or less appropriate depending on the situation; for alternative strategies see Lotov et al. (2004).

We will begin with the Lexicographic Method, which is actually a way of calculating a few very significant Pareto-efficient points, rather than the entire Frontier.

18.3.1 Lexicographic Method

The simplest strategy is to consider one objective at a time, but to do so we must sort the objectives according to decreasing importance, i.e. we have to establish a ranking rule, similar to the rule used to order the words in a lexicon, and it is just from the *lexicographic* order of the words that this method gets its name and inherits its evident arbitrariness.

With the aim of establishing a ranking of the objectives, it is plain that we are openly contradicting the assumption that we posed above, that in this phase we do not want to express value judgements; nevertheless, please allow us to be incoherent, just for the moment.

Before we formulate the method in general terms, we will illustrate it in the simple case of problem (18.4). Suppose we want to consider the ranking (J^1, J^2) of the objectives: let us consider the following problem (*Primary Problem* or *1st level Problem*)

$$\min_{\mathbf{u}} J^1(\mathbf{u}) \tag{18.5a}$$

$$\mathbf{u} \in \mathcal{U} \tag{18.5b}$$

We are looking for the set of all its optimal decisions: it will be given by the set $\mathcal{W}^{*1}$ of all the decisions $\mathbf{u} \in \mathcal{U}$ that map onto the minimum value (J^{1*}) that J^1 can assume. In our example $\mathcal{W}^{*1}$ is constituted by all the decisions that map onto the points of the segment G–A in Figure 18.3. If, as in the figure, the set $\mathcal{W}^{*1}$ contains more than one decision, we can look among them for those that minimize the second objective; strictly speaking, we can formulate and solve the following problem (*Secondary Problem* or *2nd level Problem*)

$$\min_{\mathbf{u}} J^2(\mathbf{u}) \tag{18.6a}$$

$$\mathbf{u} \in \mathcal{W}^{*1} \tag{18.6b}$$

whose solution[5] is the decision that corresponds to the lexicographic order (J^1, J^2) and is represented by point A in Figure 18.3. By construction, that point is an optimal decision with respect to J^1 and, conditioned to this, optimal for J^2. Therefore, it is an *extreme decision*: the best decision with respect to J^1, i.e. with respect to the first objective in the lexicographic order that we have assumed. Figure 18.3 shows in fact that A is actually one of the extreme points of the Pareto Frontier A–B.

Instead, if we assume the order (J^2, J^1), the same procedure takes us to point B, which is the best Pareto-efficient decision with respect to J^2.

So, each order corresponds to an extreme point on the Pareto Frontier and the Lexicographic Method allows us to identify all of them. Several of the points may coincide, as would happen if in our example the Utopia point were achievable. It should be evident that the extreme points are particularly interesting from the DM's perspective, and even more so from the Stakeholders' perspectives, when they are given an active role. Each of these points corresponds, in fact, to the best performance that the Stakeholders interested in the first objective of the ranking can hope to achieve.[6] In evaluating the alternatives, which, as negotiations proceed, will be presented to them, these Stakeholders will automatically refer to that performance. It is therefore essential to be able to determine all the extreme points of the Frontier before going into the negotiation process.[7] The Lexicographic Method allows us to obtain them.

We are now able to define this method in general terms, i.e. for problem (18.2). Assume that a ranking has been established among the objectives and denote it with $\{J^1, J^2, \dots, J^q\}$. This assumption does not imply any loss of generality, given that it is always possible to rename the objectives. Then, formulate and solve the following problem

The MO Design Problem (*Lexicographic Method* (*1st level*)):

$$\min_{\mathbf{u}^p, P} J^1\left(\mathbf{u}^p, P\right) \tag{18.7a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right) \quad t = 0, 1, \dots \tag{18.7b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{18.7c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq \mathcal{U}_t\left(\mathbf{x}_t, \mathbf{u}^p\right) \quad t = 0, 1, \dots \tag{18.7d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, 1, \dots \tag{18.7e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{18.7f}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{18.7g}$$

$$P \triangleq \left\{M_t(\cdot);\ t = 0, 1, \dots\right\} \tag{18.7h}$$

$$\textit{any other constraints} \quad t = 0, 1, \dots \tag{18.7i}$$

Hence, the minimum possible value (J^{1*}) for the first objective and the set $\mathcal{W}^{*1}$ of all the pairs ($\mathbf{u}^p$, P^1) that produce it are obtained. If $\mathcal{W}^{*1}$ contains only one pair, the desired lexicographic solution is obtained; if not, formulate and solve the following problem

[5] In general, the solution may not be a single decision, as in our example, but a set of decisions.

[6] Clearly, given the set of decisions $\mathcal{U}$ that is being considered. Things could change if a different set were considered.

[7] Only in this way can each Stakeholder know which is the best performance that (s)he can hope for.

The MO Design Problem (*Lexicographic Method* (*2nd level*)):

$$\min_{\mathbf{u}^p,P} J^2\big(\mathbf{u}^p, P\big) \tag{18.8a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \qquad t = 0, 1, \ldots \tag{18.8b}$$

$$\big(\mathbf{u}^p, P^1\big) \in \mathcal{W}^{*1} \tag{18.8c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq M_t^{P^1}(\mathbf{x}_t) \qquad t = 0, 1, \ldots \tag{18.8d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \qquad t = 0, 1, \ldots \tag{18.8e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{18.8f}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{18.8g}$$

$$P \triangleq \big\{M_t(\cdot);\ t = 0, 1, \ldots\big\} \tag{18.8h}$$

$$P^1 \triangleq \big\{M_t^{P^1}(\cdot);\ t = 0, 1, \ldots\big\} \tag{18.8i}$$

$$\textit{any other constraints} \qquad t = 0, 1, \ldots \tag{18.8j}$$

The objective considered in this problem is the second in the ranking, and the control $\mathbf{u}_t$ is constrained to belong to the control laws $M_t^{P^1}(\cdot)$. In turn, these laws belong to the policies P^1 which, for (18.8c), are solutions to problem (18.7), and therefore produce the minimum possible value (J^{1*}) for the first objective. The solution to the problem provides the set $\mathcal{W}^{*2}$ of all the pairs ($\mathbf{u}^p$, P) that minimize J^2. The procedure can be then iterated, considering the subsequent objectives according to the lexicographic order assumed, either until the last set of solutions found contains only one solution, or until all the q objectives have been considered. In both cases the desired lexicographic solution, and so an extreme point of the Pareto Frontier, is obtained.

By swapping the order in which the objectives are considered, the Lexicographic Method allows us to determine the q extreme points of the Pareto Frontier, some of which may coincide. However, it does not allow us to determine the other points of the Frontier. This is the aim of the following method.

18.3.2 Weighting Method

The central idea of the Weighting Method is to derive from problem (18.2) another problem with only one objective, which is parametric in one or more parameters, whose solution provides a Pareto-efficient point, and which, as the parameters vary, moves along the Pareto Frontier.

As in the previous section, we introduce the method by considering the example of problem (18.4).

If we ask the DM to express her opinion about the relative importance of the two objectives, she might say that she weights the first λ ($0 \leqslant \lambda \leqslant 1$) and, as a consequence, $(1 - \lambda)$ the second.[8] Once the weights are known, the two objectives can be combined in a single function that expresses the *score* with which the DM evaluates a decision

$$S(\mathbf{u}) = \lambda J^1(\mathbf{u}) + (1 - \lambda) J^2(\mathbf{u}) \tag{18.9}$$

[8] Commonly, even if it is not strictly necessary, it is assumed that the weights have a sum of one.

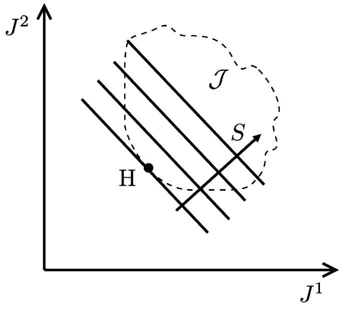


Figure 18.4: Determining a point on the Pareto Frontier for problem (18.4) with the Weighting Method.

Thanks to this function, it is possible to sort all the decisions with respect to the point of view expressed by the DM. The decision $\mathbf{u}^*$ that is preferred by her is thus the one which, among all the $\mathbf{u} \in \mathcal{U}$, minimizes S; formally, it is the solution to the following Single-Objective Problem

$$\min_{\mathbf{u}} \lambda J^1(\mathbf{u}) + (1-\lambda) J^2(\mathbf{u}) \tag{18.10a}$$

$$\mathbf{u} \in \mathcal{U} \tag{18.10b}$$

Consider the level curves of the scoring function $S(\cdot)$: they have the form

$$S = \lambda J^1(\mathbf{u}) + (1-\lambda) J^2(\mathbf{u}) \tag{18.11}$$

and correspond to the bundle of lines shown in Figure 18.4. Remember also that as $\mathbf{u}$ varies in $\mathcal{U}$ the vector $|J^1(\mathbf{u}), J^2(\mathbf{u})|$ describes the set $\mathcal{J}$ in the objective space. The solution to problem (18.10) will thus produce a value S^* of S such that the line (18.11) is the closest line to the origin (the objectives have to be minimized) among all the lines that have at least one point H in common with $\mathcal{J}$. If the Frontier is locally smooth, as in the figure, this line is tangent to it and H is the point of tangency. Decision $\mathbf{u}^*$, which maps onto H, i.e. the decision for which $|J^1(\mathbf{u}^*), J^2(\mathbf{u}^*)|$ coincides with H, is the solution we were looking for. It is a point of the Pareto-Efficient Decision Set, since by construction, there are no decisions $\mathbf{u} \in \mathcal{U}$ that improve both the objectives.

As the weight λ ($0 < \lambda < 1$) varies, the solution to the problem moves along the Frontier, even if all its points cannot always be generated. This last case happens when, as in Figure 18.5, the Frontier has a concave part: it is evident that all the points of the curvilinear segment B–C belong to the Pareto Frontier, but that none, with the exception of the two extreme points, can be determined by solving problem (18.10). In fact, as the value of λ varies from 1 to 0, first the points of the curvilinear segment A–B will be identified, then for a certain value of λ, the points B and C will be identified at the same time, and lastly, for smaller values, the points of the segment C–D will be identified.

We can now define the Weighting Method in general terms, i.e. for problem (18.2). Let us consider the following problem, parametric in the vector of weights $\boldsymbol{\lambda} = |\lambda_1, \dots, \lambda_q|$, such that $\sum_{i=1}^{q} \lambda_i = 1$ with $\lambda_i > 0 \; \forall i$

The MO Design Problem (*Weighting Method*):

$$\min_{\mathbf{u}^p, P} \lambda_1 J^1\big(\mathbf{u}^p, P\big) + \lambda_2 J^2\big(\mathbf{u}^p, P\big) + \cdots + \lambda_q J^q\big(\mathbf{u}^p, P\big) \tag{18.12a}$$

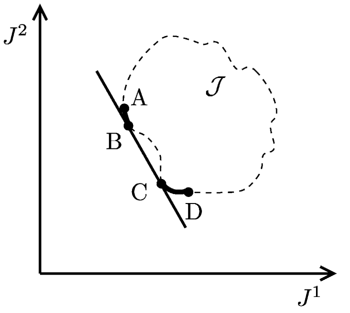


Figure 18.5: The solution with the Weighting Method when the Frontier has a concave part.

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \quad t = 0, 1, \ldots \tag{18.12b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{18.12c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \quad t = 0, 1, \ldots \tag{18.12d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, 1, \ldots \tag{18.12e}$$

$$\mathbf{w}_0^{h-1} \text{ given scenario} \tag{18.12f}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{18.12g}$$

$$P \triangleq \big\{M_t(\cdot);\ t = 0, 1, \ldots\big\} \tag{18.12h}$$

$$\text{any other constraints} \quad t = 0, 1, \ldots \tag{18.12i}$$

The set of its solutions is called *Set of Supported Points* (Lotov et al., 2004) and is a subset of the Pareto-Efficient Decision Set $\mathcal{P}$. More precisely it is that subset of $\mathcal{P}$ that maps into the union of the convex parts of the Frontier, each of which can be defined as a part of the Frontier such that each segment, whose extreme points belong to it, lies completely within $\mathcal{J}$. When the number of objectives is larger than three its inspection is not easy. It can be greatly facilitated by using the *Interactive Decision Maps* proposed by Lotov et al. (2004).

When the policy P is the object of the decision, we know that the necessary and sufficient condition for the problem to be solved with algorithms based on SDP is that the aggregated objective (18.12a) has a separable form and the constraints (18.12i) are separable. This latter condition means that their set can be partitioned in subsets, each one of which is constituted by constraints containing only variables relative to the same time interval $[t, t+1)$.

The first condition is satisfied if and only if all the objectives have a separable form, adopt the sum as a temporal operator and use the Laplace criterion. The last two characteristics are spontaneously coupled, as we explained in Section 10.2.3. The fact that these three characteristics are sufficient to guarantee the separability of the aggregated objective can easily be demonstrated with an example. Consider the case in which there are two objectives defined over a finite time horizon and with the three characteristics above

$$J^1(\mathbf{u}^p, P) = \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E}\left[\sum_{t=0}^{h-1} g_t^1(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})\right]$$

$$J^2(\mathbf{u}^p, P) = \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E}\left[\sum_{t=0}^{h-1} g_t^2(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})\right]$$

The objective of problem (18.12) thus assumes the form

$$\lambda \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E}\left[\sum_{t=0}^{h-1} g_t^1(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})\right] + \\ + (1-\lambda) \underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E}\left[\sum_{t=0}^{h-1} g_t^2(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})\right]$$

that can be rewritten as

$$\underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}}{E}\left[\sum_{t=0}^{h-1}\left[\lambda g_t^1(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) + (1-\lambda) g_t^2(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})\right]\right]$$

which is a separable objective defined by the following step indicator

$$g_t(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) = \lambda g_t^1(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) + (1-\lambda) g_t^2(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})$$

The fact that the three characteristics are necessary can easily be intuited by observing that extracting the operator that translates the criterion requires that it be a linear operator and that, among the operators that we consider, only the expected value has this last property. At the same time, the order of the temporal operator and the operator '+', which combine the objectives, can be inverted only when the temporal operator is the sum.

When the object of the decision is also the policy P, the Weighting Method is therefore not suitable for dealing with cases in which one or more of the objectives adopt the Wald criterion. A suitable method will be described in the next section.

The Weighting Method is often unconsciously adopted when, in the definition of a Design Problem, an objective is defined as the weighted sum of several indicators: two examples are the Piave Project, where the step indicator is defined by equation (8.13) as the sum of the hydropower and irrigation step costs, and the Sinai Plan, where the objective (8.16) is the sum of the benefits produced in each zone. In both cases a Design Problem, which is in fact multi-objective (hydropower vs irrigation, zone vs zone), was reduced, without explicitly declaring it, to a Single-Objective Problem, on the basis of weights that were assumed a priori by the Analyst. However, this must be declared and the Analyst must perform a sensitivity analysis with respect to the weights, which is equivalent to solving a MO Problem.

18.3.3 Reference Point Method

Like the Weighting Method, the Reference Point Method adopts the strategy of deriving a parametric problem from problem (18.2) which has only one objective, and whose solution is an efficient decision. Its image in the objective space moves along the Pareto Frontier as

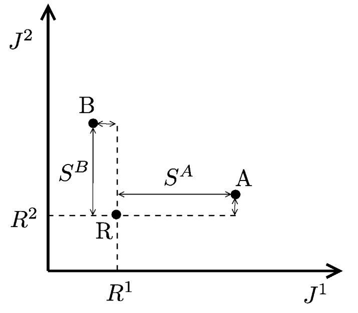


Figure 18.6: The scores S associated to different decisions.

the parameters vary. As for the previous methods, we will introduce this method by using problem (18.4) as an example.

When we ask the DM to illustrate her evaluation, she might respond by explaining that her system of preferences is not absolute – and as such it cannot be expressed by associating weights to the objectives – but relative, since it depends on the values that the objectives assume case by case. More precisely, she aims at finding a decision that produces a value R^i, $i = 1, \dots, q$, for each of the q objectives, since she judges such values as being very good. In other words, she would like to find a decision that maps onto the objective space at point R (Figure 18.6), which we will call the *reference point*, and whose coordinates are expressed by the values R^i. Correspondingly, she evaluates a decision $\mathbf{u}$ on the basis of the maximum deviation between the values that it produces for the objectives and the reference values R^i; which means that according to her judgement, the smaller the value that the decision produces for the following indicator, the better the decision

$$S(\mathbf{u}) = \max_i \left[J^i(\mathbf{u}) - R^i \right] \tag{18.13}$$

For example, for decision $\mathbf{u}^A$, which maps onto point A in the objective space in Figure 18.6, the value of S is equal to the deviation $S^A = J^1(\mathbf{u}^A) - R^1$, given that this is the larger of the two residuals, while for decision $\mathbf{u}^B$, which corresponds to point B in Figure 18.6, the value of S is the residual $S^B = J^2(\mathbf{u}^B) - R^2$. Thus the decision that the DM prefers is the decision $\mathbf{u}^*$ which minimizes S and which can be identified by solving the following Single-Objective Problem

$$\min_{\mathbf{u}} \max_i \left[J^i(\mathbf{u}) - R^i \right] \tag{18.14a}$$

$$\mathbf{u} \in \mathcal{U} \tag{18.14b}$$

Just as we did for the Weighting Method we can visualize the solution of this problem by considering the level curves of its objectives. Each curve consists in two half-lines, which angle off from a point on the line that passes through R at forty-five degrees (see Figure 18.7). The optimal value of S is the minimum value for which the corresponding level curve has at least one point in common with the set $\mathcal{J}$ (which, as the reader should remember, is generated by the vector $|J^1(\mathbf{u}), J^2(\mathbf{u})|$ as $\mathbf{u}$ varies in $\mathcal{U}$). Let H be such a point. The optimal decision $\mathbf{u}^*$ is the one which maps onto H. The point H is a point on the Pareto Frontier, because, by construction, there are no decisions $\mathbf{u} \in \mathcal{U}$ which improve both the objectives.

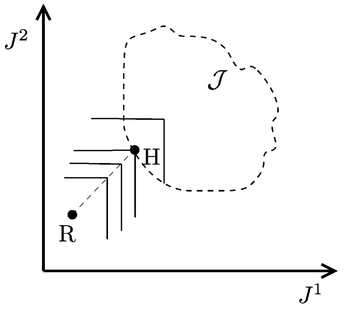


Figure 18.7: Determining a point on the Frontier for problem (18.4) with the Reference Point Method.

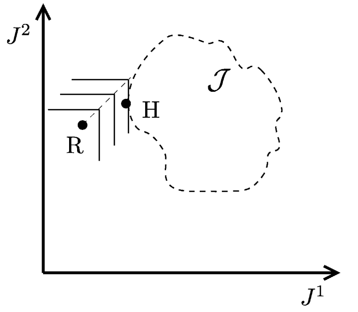


Figure 18.8: The solution to problem (18.14) is not always on the angular point of a level curve.

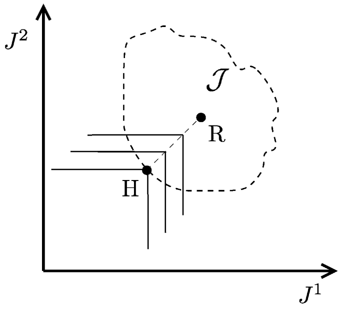


Figure 18.9: The solution to problem (18.14) when the point R belongs to $\mathcal{J}$.

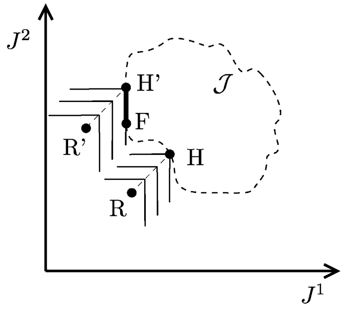


Figure 18.10: The solution to problem (18.14) when the Frontier is concave (point H) or when there are semi-dominated decisions (segment H'–F).

The point H is not always positioned on the angular point of a level curve: it could also be on one of its rectilinear segments as shown in Figure 18.8. The solution could also be associated to negative values of S (Figure 18.9): this occurs when the reference point R belongs to the set $\mathcal{J}$, i.e. when the DM's aspiration turns out to be more modest than what is actually possible to achieve.

As R varies, the solution to problem (18.14) describes the whole Pareto-Efficient Decision Set, including the points that map into any concave parts (e.g. point H in Figure 18.10), but when it provides multiple solutions, some might be semi-dominated decisions, as also shown in Figure 18.10 (segment H'–F).

Therefore, we can define the Reference Point Method in general terms, i.e. for problem (18.2). The points of the Pareto-Efficient Decision Set can be determined by solving the following problem, parametric in the coordinates $R^1, \ldots, R^q$ of the reference point R

The MO Design Problem (*Reference Point Method*):

$$\min_{\mathbf{u}^p, P} \max_i \left[J^i\left(\mathbf{u}^p, P\right) - R^i \right] \tag{18.15a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \quad t = 0, 1, \ldots \tag{18.15b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{18.15c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \quad t = 0, 1, \ldots \tag{18.15d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, 1, \ldots \tag{18.15e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{18.15f}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{18.15g}$$

$$P \triangleq \big\{M_t(\cdot);\ t = 0, 1, \ldots\big\} \tag{18.15h}$$

$$\textit{any other constraints} \quad t = 0, 1, \ldots \tag{18.15i}$$

When the policy P is the object of the decision, we know that the necessary and sufficient condition for the problem to be solved with algorithms based on SDP is that the aggregated objective (18.15a) has a separable form and that the constraints (18.15i) are separable. This latter condition means that their set can be partitioned in subsets, each one of which is constituted by constraints containing only variables relative to the same time interval $[t, t+1)$.

The first condition is satisfied only if all the objectives have a separable form, the operator max is used as a temporal operator and the Wald criterion is adopted. The last two characteristics are spontaneously coupled, as we explained in Section 10.2.3. The fact that the three characteristics are sufficient to guarantee the separability of the aggregated objective is easily demonstrated with an example. Consider the case in which there are two objectives defined over a finite time horizon that have the three characteristics above

$$J^1\big(\mathbf{u}^p, P\big) = \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \max_{t=0,\ldots,h-1} g_t^1\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)$$

$$J^2\big(\mathbf{u}^p, P\big) = \max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \max_{t=0,\ldots,h-1} g_t^2\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)$$

The objective of problem (18.15) takes the form

$$\max_{i=1,2}\Big[\max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \max_{t=0,\ldots,h-1} g_t^i\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) - R^i\Big]$$

which can be rewritten as

$$\max_{\{\boldsymbol{\varepsilon}_t\}_{t=1,\ldots,h}} \max_{t=0,\ldots,h-1}\Big[\max_{i=1,2}\big[g_t^i\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) - R^i\big]\Big]$$

which is a separable objective defined by the following step indicator

$$g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) = \max_{i=1,2}\big[g_t^i\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) - R^i\big]$$

The fact that the three properties are necessary can easily be perceived by observing that collecting the operator that translates the criterion is possible only when it is the maximum and so the criterion is the Wald criterion. At the same time, inverting the order of the temporal operator and the 'max' operator, which combine the objectives, is possible only when the temporal operator is the maximum.

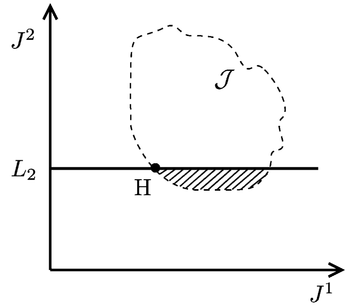


Figure 18.11: Determining a point on the Frontier of problem (18.4) with the Constraint Method.

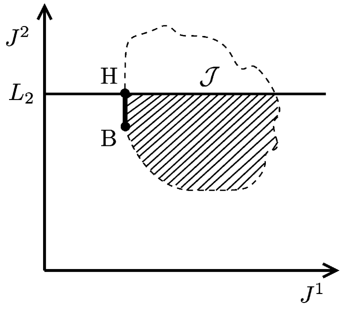


Figure 18.12: For certain values of L_2 the Constraint Method can provide semi-dominated solutions for problem (18.4).

When the policy P is the object of the decision, the Reference Point Method is therefore suitable to deal with the case in which the objectives are defined with the Wald criterion, but it is easy to see that it cannot deal with the case that uses the Laplace criterion.

18.3.4 Constraint Method

Just like the two above methods, the strategy of the Constraint Method is based on deriving, from problem (18.2), a parametric Single-Objective Problem, whose solution is an efficient decision. Its image in the objective space moves along the Pareto Frontier as the parameters vary. Once again we will introduce the method by considering problem (18.4) as an example.

When we ask the DM to illustrate her evaluation of the decisions, she might explain that she would like the smallest possible value for the first objective, with the condition that the value of the second does not exceed a threshold L_2 that she establishes. The best decision, from her point of view, is thus the solution of the following Single-Objective Problem

$$\min_{\mathbf{u}} J^1(\mathbf{u}) \tag{18.16a}$$

$$J^2(\mathbf{u}) \leqslant L_2 \tag{18.16b}$$

$$\mathbf{u} \in \mathcal{U} \tag{18.16c}$$

whose solution, as in the previous cases, can easily be visualized. The problem imposes that the best decision must map into the part of the set[9] $\mathcal{J}$ that satisfies the constraint (18.16b), i.e. in the striped set in Figure 18.11. Among all the points in this set, we will thus consider the ones that correspond, as required by the problem, to the minimum value of J^1. If such a point is unique (point H in Figure 18.11) the decision $\mathbf{u}^*$ that the DM prefers is the one which maps onto it and this decision is a point on the Pareto-Efficient Decision Set, because, by construction, there are no decisions $\mathbf{u} \in \mathcal{U}$ that improve both the objectives. Instead, if the points corresponding to the minimum value of J^1 are more than one (segment H–B in Figure 18.12), all but one of the corresponding decisions, are semi-dominated. As L_2 varies, the solution of problem (18.16) describes the whole Pareto-Efficient Decision Set.

[9] Remember that the set $\mathcal{J}$ is obtained by the vector $|J^1(\mathbf{u}), J^2(\mathbf{u})|$ as $\mathbf{u} \in \mathcal{U}$ varies.

Thus, in general, the Constraint Method requires that one of the q objectives be kept as the objective of the new problem (without losing generality, we will assume that it is the first) and that the thresholds $L_2, \dots, L_q$ – which the remaining objectives must respect – be fixed. The points of the Pareto-Efficient Decision Set can now be determined by solving the following problem, parametric in $L_2, \dots, L_q$

The MO Design Problem (*Constraint Method*):

$$\min_{\mathbf{u}^p, P} J^1\big(\mathbf{u}^p, P\big) \tag{18.17a}$$

subject to

$$\mathbf{x}_{t+1} = f_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big) \quad t = 0, 1, \dots \tag{18.17b}$$

$$\mathbf{u}^p \in \mathcal{U}^p \tag{18.17c}$$

$$\mathbf{u}_t \in M_t(\mathbf{x}_t) \subseteq \mathcal{U}_t\big(\mathbf{x}_t, \mathbf{u}^p\big) \quad t = 0, 1, \dots \tag{18.17d}$$

$$\boldsymbol{\varepsilon}_{t+1} \sim \phi_t(\cdot) \quad t = 0, 1, \dots \tag{18.17e}$$

$$\mathbf{w}_0^{h-1} \textit{ given scenario} \tag{18.17f}$$

$$\mathbf{x}_0 = \bar{\mathbf{x}}_0 \tag{18.17g}$$

$$P \triangleq \big\{M_t(\cdot);\ t = 0, 1, \dots\big\} \tag{18.17h}$$

$$J^j\big(\mathbf{u}^p, P\big) \leqslant L_j \quad j = 2, \dots, q \tag{18.17i}$$

$$\textit{any other constraints} \quad t = 0, 1, \dots \tag{18.17j}$$

A critical point in the application of the Constraint Method is the choice of the thresholds $L_2, \dots, L_q$ as it is apparent from Figure 18.11: a value of L_i (L_2 in the figure) too close to the minimal feasible value of J^i might result in high sensitivity of the solution to the constraint and even in the absence of solution.

When the policy P is the object of the decision, we know that the necessary and sufficient condition for the problem to be solved with algorithms based on SDP is that the objective (18.17a) have a separable form and the constraints (18.17i)–(18.17j) be separable, i.e. that their set can be partitioned in subsets, each one of which is constituted by constraints containing only variables relative to the same time interval $[t, t+1)$. This last condition cannot be satisfied, however, because the constraint (18.17i) is not separable. Therefore, the Constraint Method can never be used in designing policies; it is only useful for Pure Planning Problems.

18.3.5 Choosing the method

We have seen that the extreme points of the Pareto Frontier can be obtained with the Lexicographic Method, while the search for the other points on the Frontier can be carried out with the other three methods for Pure Planning Problems. In Pure and Mixed Management Problems one must use the Weighting Method, when all the objectives are defined with the Laplace criterion, and the Reference Point Method, when they are defined with the Wald criterion. Both the methods provide a means of determining efficient decisions, and thus the Pareto-Efficient Decision Set, but not the corresponding values of the objectives, and thus the Pareto Frontier cannot be determined. Since knowing these values is essential in the phases of *Evaluation* and *Comparison*, we must find a way to determine them. This will be the subject of the next chapter.

Finally, when dealing with Pure and Mixed Management Problems, if the objectives of the problem are not separable and/or they are not all defined with the same criterion, we do not have an algorithm to solve the Single-Objective Problem to which the Multi-Objective Problem is reduced. We will dedicate Section 18.6 to looking for a way to overcome this difficulty, but first we examine an example of an MO Problem.

18.4 Preferences among the objectives

There are cases in which the law or the norms establish that some objectives have priority over others: for example, in some countries satisfying domestic water supply is given priority over supplying agriculture, which in turn is given priority over supplying hydropower production. In these cases the set of objectives

$$\mathcal{J} = \left\{ J^1\left(\mathbf{u}^p, P\right), J^2\left(\mathbf{u}^p, P\right), \dots, J^q\left(\mathbf{u}^p, P\right) \right\}$$

is partitioned into subsets $\mathcal{J}'$, $\mathcal{J}''$, ... which are ordered by decreasing priority. The Design Problem can then be solved spontaneously with the Lexicographical Method, but at each level the Problem is still a multi-objective problem, which must therefore be solved with one of the methods described above. For an example see Chapter 7 of PRACTICE or Weber et al. (2002).

18.5 An example: the Sinai Plan

Let us consider the example of the Sinai Plan that we introduced in Section 8.3. In that section we assumed that the Ministry of Agriculture had only one evaluation criterion, in order to get a single objective, as required by the assumption that underlies Part C of this book. Actually, there were two criteria: an economic criterion, the one already introduced, and a socio-political criterion. The latter comes from the fact that at the time the Plan was conceived the Egyptian Government expected that in the near future the Sinai peninsula would return under its sovereignty (this was in 1977, just before the Camp David agreements, on the basis of which the Sinai was returned to Egypt from Israel, which had occupied it in the 'six day war'). The Ministry wanted the peninsula to be as populated as possible, with the condition that development be balanced among the various areas.

The phases of the Project that are described in Section 8.3 are still valid with two objectives, except for some differences that emerge in Phase 2 (*Defining Criteria and Indicators*) and Phase 4 (*Designing Alternatives*). These phases are therefore re-formulated below, and the reader can refer to Section 8.3 for the meaning of the variables.

18.5.1 Phase 2: Defining Criteria and Indicators

The new criterion and its corresponding indicator have to be added to the economic criterion and indicator described in Section 8.3. We can translate the new criterion into an objective by saying that the smallest fraction of reclaimed area among all the zones should be maximized, or more precisely

$$\max \min_{j \in \{j:\ \exists (i,h) \in F:\ (i,j,h) \in F\}} \sum_{(i,h) \in \{(i,h):\ \exists j:\ (i,j,h) \in F\}} \frac{u^1_{ijh}}{A_j}$$

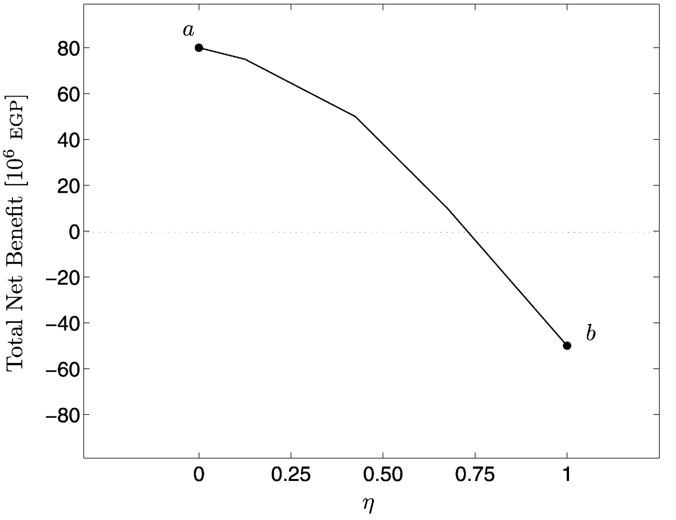


Figure 18.13: The Pareto Frontier of the Sinai Plan (elaborated by Whittington and Guariso, 1983).

Operationally, we ought to reformulate this objective by first saying that the reclaimed area in each zone j should not be inferior to a fraction η of the maximum reclaimable surface area A_j in that zone, and then by maximizing η. In mathematical terms

$$\max \eta \tag{18.18}$$

subject to

$$\sum_{(i,h)\in\{(i,h):\ \exists j:\ (i,j,h)\in F\}} \frac{u^1_{ijh}}{A_j} \geqslant \eta \quad \forall j \tag{18.19}$$

18.5.2 Phase 4: Designing Alternatives

The new Design Problem is obtained by adding equations (18.18) and (18.19) to problem (8.22), which thus becomes the following two-objective problem

$$\max_{\mathbf{u}^p, Q_1,\dots,Q_7} \left[i\left(\mathbf{u}^p, \mathbf{w}\right),\ \eta \right] \tag{18.20}$$

where $i(\mathbf{u}^p, \mathbf{w})$ is defined by (8.17), subject to constraints (8.19), (8.20), (8.21) and (18.19).

Its solution is easily obtained with the Constraint Method, which in this context is the natural choice: a value for η is fixed and problem (18.20) is solved taking into account only the first objective. The problem is thus completely analogous to problem (8.22), except for the fact that it has seven more constraints, which are defined by equation (18.19). It can therefore be solved with the algorithm defined in Section 8.3.

As η varies, the Pareto Frontier shown in Figure 18.13 is obtained. Note that the alternative found in Section 8.3 corresponds, clearly, to an extreme point of the Frontier, namely the point a, which corresponds to the maximum net benefit. For the reader's interest, following the Camp David agreements, and thanks to a considerable grant from the US

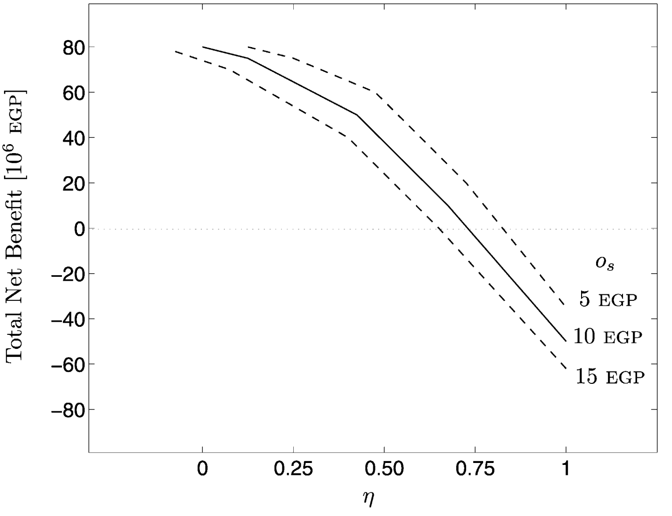


Figure 18.14: The sensitivity of the Frontier to the opportunity cost o_s of the water in the Damietta–Salam canal (elaborated by Whittington and Guariso, 1983).

Government, the Ministry chose point b. This means that the Ministry decided to consider only the socio-political criterion, with the aim of maximizing the population of the Sinai and the jobs created, even though these jobs had to be subsidized (the net benefit of point b is negative).

Figures 18.14 and 18.15 show the sensitivity of the Frontier to variations in two parameters: the opportunity cost o_s of the water in the Damietta–Salam canal and the unitary loss α from the canals. The sensitivity curves were obtained by solving problem (18.20) for different values of such parameters. In particular, from Figure 18.15 we can deduce that, if it is decided to seal the canals, the cost of the operation should not exceed the difference between the values of the two curves (the segment Δ in the case of the point b chosen by the Government), if the intervention is to be efficient.

18.6 Choosing the objectives

In real problems the number of evaluation indicators[10] is often very high. For example, in the Verbano Project 51 indicators are considered (see Chapter 4 PRACTICE). When there are so many indicators it is not possible to consider them all when designing the alternatives, because the computation time required to individuate the Pareto-Efficient Decision Set would be prohibitive. On the other hand, it is easy to guess that it is not always necessary to consider all of them: if, in fact, two objectives are linearly and positively correlated, it is evident that the alternative which produces an optimal value for one will also produce an optimal value for the other. Therefore, it is completely useless to consider both of them, and the choice between one or the other is indifferent.

[10] Remember that when there are random disturbances, by the term 'evaluation indicator' we do not denote the random indicator, but rather the statistic which is obtained from it by filtering the disturbances with a suitable criterion (see page 133).

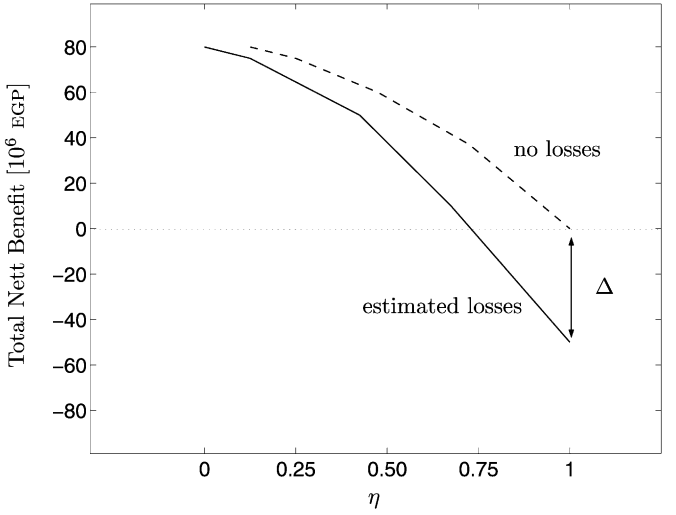


Figure 18.15: The sensitivity of the Frontier to the losses α from the canals (elaborated by Whittington and Guariso, 1983).

This is a prolific idea, since it is the starting point for a procedure that allows us to identify a reduced number of indicators with which to formulate the Design Problem. We will call these *design indicators*.

1. Construct a partition[11] $\{\mathcal{Z}^1, \ldots, \mathcal{Z}^q\}$ of the set $\mathcal{Z}$ of evaluation indicators $\mathfrak{i}_1, \ldots, \mathfrak{i}_m$ so that each subsystem $\mathcal{Z}^j$ is constituted by indicators that are positively and significantly correlated between each other. Then consider q indicators $\mathfrak{i}_1, \ldots, \mathfrak{i}_q$ of $\mathcal{Z}$, so that each one of them belongs to a different set $\mathcal{Z}^j$, i.e. so that $\mathfrak{i}_j \in \mathcal{Z}^j$, $j = 1, \ldots, q$. They are the indicators to be used in formulating the objectives of the Design Problem.

In fact, by considering only objectives based on the indicators $\mathfrak{i}_1, \ldots, \mathfrak{i}_q$, the efficient alternatives that are obtained are the same ones that would be obtained by considering the entire set $\mathcal{Z}$ of indicators if the correlation between all the pairs of indicators of each set $\mathcal{Z}^j$ were sufficiently high.

When designing a policy the following conditions must also be considered when choosing the q objectives:

2. All the objectives must have a separable form.

3. All the objectives must be defined with the same criterion, i.e. all of them must adopt either the Laplace criterion or the Wald criterion. Further to what we explained in Section 10.2.3, they can always be formulated so that, in the first case, the temporal operator will be the sum for them all, while, in the second case, it will be the maximum.

Points 2 and 3 must be kept in mind only when one must design a policy. As we saw in Sections 18.3.2 and 18.3.3, it is only then that the separability of the indicators and the

[11] That is, decompose $\mathcal{Z}$ into q subsystems $\mathcal{Z}^j$ such that $\bigcup_{j=1}^{q} \mathcal{Z}^j = \mathcal{Z}$ and $\mathcal{Z}^j \cap \mathcal{Z}^i = \emptyset \; \forall i \neq j$.

uniformity of the criterion are necessary conditions for solving the problem with SDP-based algorithms.

A doubt may arise however: to construct the partition $\{\mathcal{Z}^1, \ldots, \mathcal{Z}^q\}$ we must have a certain number of n-tuples of values of the indicators of $\mathcal{Z}$, but how it is possible to obtain them, given that the alternatives have not yet been defined? Certainly the existence of a positive correlation between the two indicators cannot, except in particular cases, be ascertained only by inspecting their definitions. However, the doubt is unfounded, because it is not difficult to obtain alternatives for which the values of the indicators of $\mathcal{Z}$ can be calculated via simulation, with techniques that will be illustrated in the next chapter. In fact, the alternatives do not have to be efficient; it is sufficient that they cover the action space in a fairly uniform way.

When in a subset $\mathcal{Z}^j$ there are several indicators with a separable form, theoretically, any one of them can be chosen as an objective. However, when it is possible, it is convenient to adopt an indicator that is defined by continuous functions in order to avoid numerical problems in the solution of problem (18.2): an example of this kind of problem is given in Section 7.3.1 of PRACTICE.

The procedure that we sketched out is not complete however, because

(a) it is not always possible to individuate the objectives in such a way that they are all separable;

(b) it is not always possible to individuate the objectives in such a way that they all adopt the same criterion;

(c) the number q of objectives that are obtained might still be too high, making it necessary to reduce it further.[12]

We will examine these difficulties in order.

(a) In the case in which one must design a policy and there is a subset $\mathcal{Z}^j$ characterized by indicators that are all non-separable, it is necessary to find an equivalent objective that has a separable form and produces the same ranking of the alternatives as the objective that it is to substitute. Identifying this equivalent objective must be based on guesswork and the validity of the choice can be verified with the procedure illustrated in Section 7.2.2 of PRACTICE. If one cannot find an equivalent objective, the indicators in $\mathcal{Z}^j$ cannot be represented by an objective in the Design Problem, unless the latter is solved with algorithms that are not based on SDP. The pros and cons of these two possibilities have to be examined case by case.

(b) When one must design a policy and the objectives are not all defined with the same criterion, it is necessary to choose one criterion and substitute all the objectives that do not use it with equivalent objectives. An 'equivalent objective' is an objective that is defined with the chosen criterion and that produces the same ranking among the alternatives as the objective that it is to substitute. The simplest solution is almost always to choose an objective defined with the Laplace criterion and substitute the objectives that are defined with the Wald criterion. In fact, the ranking produced

[12]The same number of objectives might or might not be too high depending on the time required to solve a Single-Objective Design Problem, like for example problems (18.12) and (18.15). What is of interest is, in fact, the total time required to explore the Pareto Frontier. This is the reason why visualization methods are very useful.

by the objective

$$J\big(\mathbf{u}^p, P\big) = \max_{\{\varepsilon_t\}_{t=1,\dots,h}} \max_{t=0,\dots,h-1} \big[g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\big]$$

is identical to the one produced by the following

$$J\big(\mathbf{u}^p, P\big) = \lim_{\alpha\to\infty} \mathop{E}_{\{\varepsilon_t\}_{t=1,\dots,h}} \sum_{t=0}^{h-1} \big[g_t\big(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\big)\big]^\alpha$$

In practical applications, the limit with respect to α in the above equation can be substituted with a very big value of α. One can easily perceive why the two objectives are equivalent: the first considers the worst case among all the step costs and all the sequences of disturbances that can occur. The second considers the expected value of the cumulated cost, but, since the costs are raised to a very high power, in practice, the value of J is influenced only by the biggest cost, given that in all the other time instants, the costs are of many orders of magnitude smaller. From the last sentence we also understand that the swapping is feasible only when the worst value that the step cost can produce is definitively bigger than one.

(c) If the number q of the objectives is too high, it is absolutely necessary to discard some of them. The discarding criterion is clearly subjective; thus it should not be established by the Analyst alone, but should be shared with the Stakeholders. However, no generally applicable procedure exists to select the objectives to discard, so the choice must be made on the basis of case by case examination.

In practice, the sectors are often identified right from Phase 2 (see page 93). When this is the case, it is opportune to try to identify an objective for each sector, instead of identifying an objective for each of the sets $\mathcal{Z}^j$ defined at point 1 on page 414. The procedure explained in Section 7.2 of PRACTICE is useful for this. If, notwithstanding this, the number of objectives is still too high, one must turn to ad hoc considerations, as shown in Section 5.2.3 of PRACTICE.

18.7 Discretizing the alternatives

In theory, problem (18.12) (or problem (18.15)) should be solved for all the possible values of the vector of weights $\boldsymbol{\lambda}$ (or of the reference point $\mathbf{R}$), because only in this way is the entire Pareto-Efficient Decision Set determined. It is evident, however, that this is impossible in practice and one must settle for solving the problem only for a few values of the parameter, thus identifying only a finite number of points on the Set, and therefore a finite number of alternatives. To conduct an accurate analysis we ought to make sure that the grid of points that is generated on the Pareto Frontier is sufficiently dense and regular. Determining such a grid is not a simple operation, since we cannot choose the alternatives to be generated directly onto the Pareto Frontier, but only in the space of the parameter $\boldsymbol{\lambda}$ (or of $\mathbf{R}$), and often an even spread of weights will not result in an even spread of points on the Pareto Frontier (Das and Dennis, 1997). We must therefore adopt an adaptive selection of weights. For example, we can begin by fixing a grid with a constant mesh in the space of $\boldsymbol{\lambda}$ (or of $\mathbf{R}$), as shown in Figure 18.16 (where, to simplify the representation, we suppose that the objectives were only two). Once the corresponding problem (problem (18.12) or (18.15))

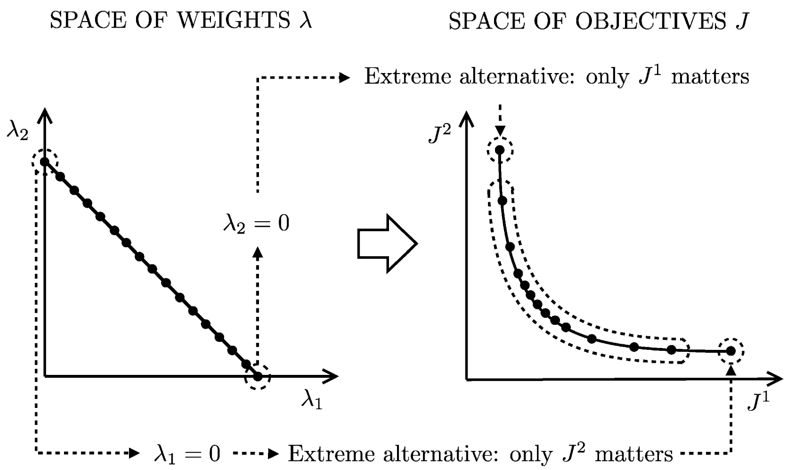


Figure 18.16: A uniform grid in the space of the weights and the images of corresponding alternatives in the objective space.

has been solved, the alternatives that have been found could be concentrated more in certain zones of the objective space and less in others (as occurs in Figure 18.16). This is due to the fact that the curvature of the Pareto Frontier is not generally constant. We should then discard some values of $\boldsymbol{\lambda}$ (or of $\mathbf{R}$) and/or add others, so that the alternatives cover the Frontier in a relatively uniform way. More sophisticate techniques can be found in Lotov et al. (2004).

Chapter 19

Estimating Effects

AC, RSS and EW

In the previous chapter we saw that a Multi-Objective (MO) Design Problem is solved by tracing it back to a Single-Objective Parametric Problem, which, under suitable conditions, can be solved with the SDP based algorithms presented in Chapters 12–14. The transformation of the MO Problem into a Single-Objective Problem is done with one of the methods presented in the previous chapter, by which it is then possible to identify the efficient alternatives. Pay close attention: these methods allow the efficient alternatives to be determined, in the sense that they provide, for every value of the parameter that they contain (weight $\boldsymbol{\lambda}$, reference point $\mathbf{R}$ and threshold $\mathbf{L}$), the pair $(\mathbf{u}^{p*}, P^*)$ that solves the Single-Objective problem.[1] However, they do not provide the values of the individual objectives that correspond to this pair. In fact, knowing the optimal value of the single (aggregated) objective does not allow the determining of the values of the individual objectives that compose it: for example, if the method adopted is the Weighting Method it is evident that knowing the value S^* of the score and the relation

$$S^* = \lambda_1 J^1\big(\mathbf{u}^{p*}, P^*\big) + \lambda_2 J^2\big(\mathbf{u}^{p*}, P^*\big) + \cdots + \lambda_q J^q\big(\mathbf{u}^{p*}, P^*\big)$$

it is not possible to obtain the values $J^1(\mathbf{u}^{p*}, P^*), \ldots, J^q(\mathbf{u}^{p*}, P^*)$ for each single objective. In other words, those methods provide the Pareto-Efficient Decision Set, but not the Pareto Frontier.

This is not always the only reason why, once the Single-Objective Problem has been solved, we must proceed with estimating the effects: as we explained in Section 18.6, it may be that the evaluation indicators are not all translated into objectives of the Design Problem or that some of the indicators are substituted by an equivalent indicator. Knowing the effects produced by the alternatives on the whole set of evaluation indicators, and not only on the design indicators, is of paramount importance in the subsequent phases of the PIP procedure. Think for example of the negotiation process among the Stakeholders: it can be carried out only if every Stakeholder is able to evaluate the effects that each of the efficient alternatives produces on his/her interests, and (s)he can evaluate the effects only by knowing the values of the indicators that (s)he him/herself suggested for this very purpose (Chapter 3). Therefore, one must identify not only the values of the objectives, but also the

[1] In this chapter we assume that an SV policy has been designed, since a PV policy can be thought of as a particular case of an SV policy.

values of all the evaluation indicators.[2] The procedure to be adopted for this purpose is the subject of this chapter.

It might seem that identifying the indicator values would be an easy task. When dealing with a Pure Planning Problem, to obtain these values for a given efficient alternative we only have to compute the functions that define each indicator in correspondence with the vector $\mathbf{u}^{p*}$: that's it. This might suggest that the same procedure can be successfully applied to the Pure Management Problem (or to the Mixed Problem): given that we know the control policy P^*, it should be sufficient to simulate the system regulated by P^* over the horizon (*evaluation horizon*) on which the indicators are defined (remember that they all share the same horizon). Once the state and control trajectories have been obtained, it should be easy to compute the values assumed by the different indicators. The idea is sound and this is exactly the approach that we will use, but three difficulties arise:

1. According to what we have seen so far, the system can only be simulated if a realization of all the disturbances, both deterministic and random, has been specified. Furthermore, since the definition of an evaluation indicator includes a statistic with respect to the random disturbances (Section 4.10.2), all their possible realizations must be considered. Given that the number of such realizations can be huge, if not infinite, the simulation for all of them is impossible.

2. If the time horizon is infinite, the simulation is conceptually well defined, but actually impracticable.

3. At every time instant, SV policies propose more than one control value, among which the Regulator must choose the one to implement and, thus, it seems impossible to simulate the effects of this type of policy.

These are the difficulties to overcome.

The first difficulty comes from the unconscious idea that the deterministic simulation is the only one possible, while, actually, there are two other possibilities. Let us explore them.

19.1 Markov simulation

For each value of the parameter (weight $\boldsymbol{\lambda}$ and reference point $\mathbf{R}$)[3] the solution to the Single-Objective Design Problem provides an alternative A, which is defined by the pair $(\mathbf{u}^{p*}, P^*)$. At every time instant t, given the state $\mathbf{x}_t$ (or more in general the information $\Im_t$), the policy P^* provides the Regulator with the set $M_t^*(\mathbf{x}_t)$ (or $M_t^*(\Im_t)$) of all the controls $\mathbf{u}_t$ that produce the 'best' system performance, according to the viewpoint adopted in designing the policy. This viewpoint is expressed by the value adopted for the parameter $\boldsymbol{\lambda}$ (or $\mathbf{R}$) in the Single-Objective Design Problem. The effects that alternative A produces are quantified by indicators, whose values can be computed, given all the possible trajectories of the state and the system inputs. To determine these trajectories, the evolution of the system must be *simulated*, by assuming an *evaluation scenario* and by assuming that alternative A has been

[2] Remember that when there are random disturbances, by the term (evaluation) indicator we denote not the random indicator i, but the statistic i which is obtained from it by filtering the disturbances with a suitable criterion (see page 133). Obviously, when the system is not affected by random disturbances, the indicator i is not a random variable but a deterministic one, and thus it is the evaluation indicator i.

[3] Given what was said on page 410, the Constraint Method cannot ever be used in the policy design.

implemented. The evaluation scenario (see Section 19.3) is the analogue of the design scenario and, like it, denotes the set of deterministic variables (parameters and disturbances) that are described by preassigned trajectories. As we explained in Section 8.1.3, the scenario does not include random disturbances, because these are described through models, which are part of the global model. The simulation begins from a time instant, conventionally called time 0, and from a state $\mathbf{x}_0$. It is performed over a horizon (evaluation horizon) that, almost always, has the same length as the design horizon. Therefore, when the design horizon is infinite, the evaluation horizon should also be infinite, but this unlimited length obviously creates some practical difficulties.

The system upon which we operate is characterized by the presence of stochastic and/or uncertain variables, produced by two different causes: the first is the disturbances, which can assume either form (never at the same time because of the hypothesis that we introduced in Section 6.2); the second is the behaviour of the Regulator, who chooses the control that he considers to be most suited to the current situation from $M_t^*(\mathbf{x}_t)$, and thus in a way that is not known a priori. Therefore, the simulation with which the indicators are computed must not ignore the presence of uncertainty and stochasticity and so it should not be a deterministic simulation.

Let us then examine the causes that generate uncertainty and stochasticity, in order to see what can be done.

19.1.1 The Regulator's behaviour

To better understand the effect of the Regulator's choice, let us suppose for a moment that the system is deterministic, i.e. that it is not affected by any random disturbance.

Given the state $\mathbf{x}_0$, the state $\mathbf{x}_1$ is determined by the state transition function of the model

$$\mathbf{x}_1 = f_0\big(\mathbf{x}_0, \mathbf{u}^{p*}, \mathbf{u}_0, \mathbf{w}_0\big) \tag{19.1}$$

The equation shows that, in the absence of random disturbances, $\mathbf{x}_1$ is not univocally determined, because the control $\mathbf{u}_0$ chosen by the Regulator from $M_0^*(\mathbf{x}_0)$ is not known. We can, however, determine the set χ_1 of all the states $\mathbf{x}_1$ that can be produced as the Regulator's choice varies. We call this set *the set of the reachable states*, or in short, *the reachable set*. The same observation can be repeated at time $t = 1$, because nothing changes if we substitute $\mathbf{x}_0$, which is deterministically known, with $\mathbf{x}_1 \in \chi_1$. The same applies in the subsequent time instants. Thus, the simulation of a deterministic system that is managed by an SV policy does not provide the trajectory of the system state, which is uncertain, but it can provide the trajectory $\chi_0, \chi_1, \ldots$ of the set of the reachable states. This may remind the reader of the uncertain version of Markov chain that we dealt with in Section 11.1. The uncertainty about the choice that the Regulator will make is none other than a particular form of uncertainty, and so it seems plausible that the mathematical framework, introduced in that section for treating uncertain disturbances, would also be suitable for this form of uncertainty.

As we explained in that section, given that the model of the system is an automaton with $N_\mathbf{x}$ states, the reachable set at time t can be described by a Boolean vector[4] $\boldsymbol{\chi}_t$ with a finite dimension, whose ith component equals 0 if the value $\mathbf{x}_t^i$ of the state cannot occur at time t, and equals 1 if it can. Once the vector $\boldsymbol{\chi}_t$ is known, the set of reachable states at time $t+1$

[4] In the following, we will often use the vector $\boldsymbol{\chi}_t$ also to denote the set χ_t. It is an acceptable simplification, since both the set and the vector describe the same reality, the reachable states, in different ways.

is univocally defined by equation (11.3), which we rewrite below

$$\boldsymbol{\chi}_{t+1}^T = \boldsymbol{\chi}_t^T \odot W_t\big(M_t^*(\cdot)\big) \tag{19.2}$$

where $W_t(M_t^*(\cdot))$ is a matrix whose element $w_t^{ij}(M_t^*(\mathbf{x}_t^i))$ equals 1, if in $M_t^*(\mathbf{x}_t^i)$ there exists a control that transfers the system from the state $\mathbf{x}_t^i$ to the state $\mathbf{x}_{t+1}^j$, and equals 0, if such a control does not exist, i.e.

$$w_t^{ij}\big(M_t^*(\mathbf{x}_t^i)\big) = \begin{cases} 1 & \text{if } \exists \mathbf{u}_t \in M_t^*(\mathbf{x}_t^i)\text{: } \mathbf{x}_{t+1}^j = f_t(\mathbf{x}_t^i, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t) \\ 0 & \text{otherwise} \end{cases}$$

and the operator $\odot$ is defined in Section 11.1.

Given that the initial state $\mathbf{x}_0$ is known, we know the initial set χ_0, which is described by the vector $\boldsymbol{\chi}_0$, whose elements are all zero, with the exception of the element that corresponds to the state $\mathbf{x}_0$, which is equal to 1. Given $\boldsymbol{\chi}_0$ we can use equation (19.2) to compute $\boldsymbol{\chi}_1$, and so on recursively. This procedure is called *Markov simulation*.

Now it is clear why the trajectory of the system is uncertain, but we have also understood that, once the trajectory $\boldsymbol{\chi}_0, \boldsymbol{\chi}_1, \ldots$ of the reachable sets, which contains all the trajectories that the system can follow, is known, the uncertainty is completely characterized and it becomes possible to determine the worst case with respect to the Regulator's behaviour, as required by the objective of the SV Design Problem (see equation (11.12)).

When the indicator i_k is a separable function and represents a cost,[5] it is easy to compute the maximum step cost of the kth indicator:

1. When the evaluation horizon is finite with a length h, once the trajectory $\boldsymbol{\chi}_0, \ldots, \boldsymbol{\chi}_h$ is known, the maximum step cost (with respect to the Regulator's behaviour) is given by the following formula

$$\max_{\{\mathbf{u}_t \in M_t^*(\mathbf{x}_t)\}_{t=0,\ldots,h}} \left[\mathrm{i}_k\big(\mathbf{u}^{p*}, P^*\big)\right] \max\left\{ \max_{t=0,\ldots,h-1}\left[\boldsymbol{\chi}_t^T \odot \Big(\max_{\mathbf{u}_t \in M_t^*(\mathbf{x}_t)} \mathbf{g}_t^k\Big)\right], \left[\boldsymbol{\chi}_h^T \odot \mathbf{G}_h^k\right]\right\} \tag{19.3a}$$

where $\mathbf{g}_t^k$ and $\mathbf{G}_h^k$ are vectors constituted by the $N_\mathbf{x}$ values that the step cost $g_t^k(\cdot)$ and the penalty $g_h^k(\cdot)$ assume in correspondence with each of the $N_\mathbf{x}$ values of the system state

$$\mathbf{g}_t^k = \left|g_t^k\big(\mathbf{x}_t^1, \mathbf{u}^{p*}, \mathbf{u}_t, \mathbf{w}_t\big), \ldots, g_t^k\big(\mathbf{x}_t^{N_\mathbf{x}}, \mathbf{u}^{p*}, \mathbf{u}_t, \mathbf{w}_t\big)\right|^T \tag{19.3b}$$

$$\mathbf{G}_h^k = \left|g_h^k\big(\mathbf{x}_h^1, \mathbf{u}^{p*}\big), \ldots, g_h^k\big(\mathbf{x}_h^{N_\mathbf{x}}, \mathbf{u}^{p*}\big)\right|^T \tag{19.3c}$$

The operator $\odot$, defined in Section 11.1, performs a sort of 'multiplication' between the two factors to which it is applied. The result b_t of the multiplication is defined as

$$b_t = \max_{j=1,\ldots,N_\mathbf{x}}\left[\chi_t^j \max_{\mathbf{u}_t \in M_t^*(\mathbf{x}_t^j)} g_t^k\big(\mathbf{x}_t^j, \mathbf{u}^{p*}, \mathbf{u}_t, \mathbf{w}_t\big)\right] \tag{19.4}$$

In other words, b_t is the worst cost that can occur at time t, taking into account all the states that are reachable at that moment and all the controls that the Regulator could choose in correspondence with each one of the reachable states. The

[5] When the indicator i_k represents a benefit, the worst value is the minimum, which can be determined with a formula analogous to equation (19.3), in which all the operators max are substituted by min.

values of the pairs $(\mathbf{x}_t, \mathbf{u}_t)$ that appear in equation (19.3b) are recursively determined through the simulation model, which is defined by equations (11.13b)–(11.13d), (11.13f)–(11.13i).[6] Note how equation (19.3) expresses the indisputable fact that the maximum step cost is the maximum among the transition costs (considering only feasible transitions) and the costs of the reachable final states. The non-trivial aspect of equation (19.3) is the way of computing the possible transitions and the reachable final states, both of which are univocally individuated by the trajectory $\boldsymbol{\chi}_0, \ldots, \boldsymbol{\chi}_h$.

2. When the horizon is infinite, since the system is by hypothesis a periodic automaton with period T, it is possible to prove, under very broad hypotheses,[7] that the trajectory of the reachable states set converges towards a cycle[8] $\bar{\boldsymbol{\chi}}_0, \ldots, \bar{\boldsymbol{\chi}}_{T-1}$ in a finite number of steps. This is equivalent to saying that the deterministic system (19.2) has a unique stable cycle. It follows that it is not necessary to simulate over a infinite horizon, but that the simulation can be stopped when the cycle has been reached. Note that ergodicity guarantees that the choice of the initial state of the simulation is not critical, since it is forgotten as time goes on. Once the cycle has been determined, it is easy to compute the value of the indicators. In fact, when the initial state belongs to the set $\bar{\boldsymbol{\chi}}_0$, the maximum step cost of the kth indicator is given by the following formula

$$\max_{\{\mathbf{u}_t \in M_t^*(\mathbf{x}_t)\}_{t=0,1,\ldots}} \left[i_k\left(\mathbf{u}^{p*}, P^*\right)\right] = \max_{t=0,\ldots,T-1} \left[\bar{\boldsymbol{\chi}}_t^T \odot \left(\max_{\mathbf{u}_t \in M_t^*(\mathbf{x}_t)} \mathbf{g}_t^k \right)\right] \quad (19.5)$$

where $\mathbf{g}_t^k$ is given by equation (19.3b). When, instead, the initial state does not belong to the set $\bar{\boldsymbol{\chi}}_0$, it might happen that the maximum cost occurs during the transient period that takes the state into cycle: this must be verified by applying equation (19.5) to the transient period.

When the indicator is not a separable function, there is no algebraic formula to compute its worst case value quickly: the only way to determine it is to simulate all the possible choices of the Regulator, namely to explore the entire decision tree. In real cases, however, the number of possible combinations is almost always huge: for example, if at every time instant the set $M_t^*(\mathbf{x}_t)$ even contained only 10 elements, i.e. if the Regulators' choice were to be made among just 10 controls, and the evaluation horizon were just one year, the possible combinations would be 10^{365}. To appreciate the size of this number, consider that the number of seconds from the Big Bang to today is in the order of 10^{21}. The computing time required is therefore huge (millions of billions of times the age of the universe). As a consequence, when the indicator is not separable, the worst case can almost never be determined. We must rely on an intuitive estimation, which we will explain how to obtain in Section 19.2.2. First we should examine how to deal with disturbances.

19.1.2 Uncertain disturbances

When the system is affected by uncertain disturbances, the Design Problem is always formulated with the Wald criterion and thus, as in the previous case, the worst system performances must be determined for each alternative whose effects we want to estimate. The

[6] Equation (11.13e) is not part of the model, given that we are under the hypothesis of a deterministic system.

[7] They are the same hypotheses under which Algorithms 2 and 3 (see pages 301 and 303) converge.

[8] Note that when the system is time invariant, i.e. when T is equal to 1, the cycle is none other than an equilibrium set.

procedure proposed in the previous section can be applied in this case as well, with a difference however:

1. When the policy is point-valued (PV), the only sources of uncertainty are the disturbances; the matrix $W_t(M_t^*(\cdot))$ that appears in equation (19.2) must thus be substituted by the matrix $W_t(m_t^*(\cdot))$, whose elements are calculated with the following expression

$$w_t^{ij}\big(m_t^*(\mathbf{x}_t^i)\big) = \begin{cases} 1 & \text{if } \exists \boldsymbol{\varepsilon}_{t+1} \in \Xi_t\colon\ \mathbf{x}_{t+1}^j = f_t(\mathbf{x}_t^i, \mathbf{u}^p, m_t^*(\mathbf{x}_t^i), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) \\ 0 & \text{otherwise} \end{cases} \tag{19.6}$$

2. When, instead, the policy is set-valued (SV), uncertainty is produced also by the Regulator's behaviour and so this must be taken into account in determining the elements of matrix $W_t(M_t^*(\cdot))$. For this reason it is assumed that $w_t^{ij}(M_t^*(\mathbf{x}_t^i))$ equals 1 if for the state $\mathbf{x}_t^i$ there is at least one disturbance-control pair that takes the system from state $\mathbf{x}_t^i$ to $\mathbf{x}_{t+1}^j$; otherwise it equals 0. Precisely

$$w_t^{ij}\big(M_t^*(\mathbf{x}_t^i)\big) = \begin{cases} 1 & \text{if } \exists (\boldsymbol{\varepsilon}_{t+1}, \mathbf{u}_t) \in (\Xi_t \times M_t^*(\mathbf{x}_t^i))\colon \\ & \qquad :\mathbf{x}_{t+1}^j = f_t(\mathbf{x}_t^i, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}) \\ 0 & \text{otherwise} \end{cases} \tag{19.7}$$

19.1.3 Stochastic disturbances

We have seen how to deal with uncertain disturbances; we will now consider stochastic ones. To simplify the explanation, we will only consider the case in which there is no uncertainty, which is equivalent to assuming that the policy is point-valued. For the general case the procedure could be obtained by following the line of reasoning presented in Section 11.2.1.2, but the resulting formulae are very complex. For this reason, and for the reasons that will be presented at the beginning of the next chapter, they are very rarely used and so we do not present them here.

By analogy from what is presented in the previous section, it is easy to understand that the presence of stochastic disturbances make the state $\mathbf{x}_t$ stochastic. It is thus advisable to describe it with a vector $\boldsymbol{\pi}_t$, whose components represent the probability that it assumes the first, the second, ..., the $N_{\mathbf{x}}$th of its possible values. Once the probability distribution $\boldsymbol{\pi}_t$ is known, we know that the probability distribution at time $t+1$ is univocally defined by equation (11.1), which we rewrite

$$\boldsymbol{\pi}_{t+1}^T = \boldsymbol{\pi}_t^T B_t\big(m_t^*(\cdot)\big) \tag{19.8}$$

where the matrix $B_t(m_t^*(\cdot))$ is defined by equation (11.2).

Since the initial state $\mathbf{x}_0$ is known, also the initial distribution $\boldsymbol{\pi}_0$ is known and has the form

$$|0 \quad 0 \quad \cdots \quad 0 \quad 1 \quad 0 \quad \cdots \quad 0 \quad 0|^T$$

where the value 1 is assigned to the component that corresponds to the value of $\mathbf{x}_0$ that occurs. Through equation (19.8) it is possible to compute first $\boldsymbol{\pi}_1$, then, from that, $\boldsymbol{\pi}_2, \ldots$ and so on, thus obtaining the trajectory of $\boldsymbol{\pi}_t$ over the whole evaluation horizon.

Once this trajectory is known, when the indicator i_k is a separable function, it is easy to compute the desired statistic of the kth indicator, with the procedure which we will now present:

1. When the time horizon is finite with a length h:
   (a) When the statistic considered is the worst case (Wald criterion) and i_k represents a cost, equation (19.3) is used, in which $m_t^*(\mathbf{x}_t)$ substitutes $M_t^*(\mathbf{x}_t)$ and the vector $\boldsymbol{\chi}_t$, which describes the set of states that have a non-zero probability of occurring, is obtained from the vector $\boldsymbol{\pi}_t$, by setting all its non-zero components to 1.

   (b) When the statistic considered is the expected value (Laplace criterion), it can be computed with the following expression

$$\underset{\{\boldsymbol{\varepsilon}_t\}_{t=1,\dots,h}}{E}\left[i_k\left(\mathbf{u}^{p*}, p^*\right)\right] = \sum_{t=0}^{h-1} \boldsymbol{\pi}_t^T \underset{\boldsymbol{\varepsilon}_{t+1}}{E}\left[\mathbf{g}_{t+1}^k\right] + \boldsymbol{\pi}_h^T \mathbf{G}_h^k \tag{19.9a}$$

   where

$$\mathbf{g}_{t+1}^k = \left|g_t^k\left(\mathbf{x}_t^1, \mathbf{u}^{p*}, m_t^*\left(\mathbf{x}_t^1\right), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right), \dots, g_t^k\left(\mathbf{x}_t^{N_\mathbf{x}}, \mathbf{u}^{p*}, m_t^*\left(\mathbf{x}_t^{N_\mathbf{x}}\right), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right|^T \tag{19.9b}$$

$$\mathbf{G}_h^k = \left|g_h^k\left(\mathbf{x}_h^1, \mathbf{u}^{p*}\right), \dots, g_h^k\left(\mathbf{x}_t^{N_\mathbf{x}}, \mathbf{u}^{p*}\right)\right|^T \tag{19.9c}$$

   In fact, the expected value of the indicator with respect to the disturbances is equal to its expected value with respect to the state, whose distribution is induced by the stochasticity of the disturbances. This second expected value is given by the sum of the expected costs at each step, each one of which, in turn, is the sum, extended to all the states $\mathbf{x}_t^i$, of the values of the expected costs $E_{\boldsymbol{\varepsilon}_{t+1}}[g_t^k(\mathbf{x}_t^i, \mathbf{u}^{p*}, m_t^*(\mathbf{x}_t^i), \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1})]$ of the transition that originates from $\mathbf{x}_t^i$, weighted with the probability $\boldsymbol{\pi}_t^i$ of being at time t in the state $\mathbf{x}_t^i$.

2. When the evaluation horizon is infinite, given that the system is by hypothesis a periodic automaton of period T, it is possible to prove that, under very broad hypotheses, the trajectory $\boldsymbol{\pi}_0, \boldsymbol{\pi}_1, \dots,$ converges asymptotically towards a cycle[9] $\bar{\boldsymbol{\pi}}_0, \dots, \bar{\boldsymbol{\pi}}_{T-1}$. This is equivalent to saying that the deterministic system (19.8) has only one stable cycle, a property that statisticians express by saying that the process is *cycloergodic*. From a theoretic viewpoint this does not help us in any way, since, unlike the case in which the disturbances are uncertain, the convergence is asymptotic and does not occur in finite time. In practical terms, however, we can assume that the cycle is reached when a properly defined stopping condition has been satisfied. Thanks to ergodicity, the choice of the initial probability $\boldsymbol{\pi}_0$ is not critical, because it is forgotten as time goes by. Once the cycle has been determined, it is easy to compute the value of the indicators:
   (a) When the statistic considered is the worst case (Wald criterion) and i_k is a cost, equation (19.5) is used, in which $M_t^*(\mathbf{x}_t)$ is substituted with $m_t^*(\mathbf{x}_t)$ and the vector $\bar{\boldsymbol{\chi}}_t$, which describes the set of states that have a probability of occurring greater than zero, is obtained from the vector $\bar{\boldsymbol{\pi}}_t$, by setting all the non-zero components of that vector to 1. When the initial state does not belong to the set $\bar{\boldsymbol{\chi}}_0$, it is necessary to ascertain that the maximum cost does not occur during the transient.

[9] Note that when the system is time invariant, i.e. T is equal to 1, the cycle is an equilibrium distribution.

(b) When the statistic considered is the expected value (Laplace criterion), it can be computed as follows:
   (i) when the Design Problem has the TDC form, the value of the kth indicator is given by

$$\mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,2,\dots}}\left[i_k\left(\mathbf{u}^{p*}, p^*\right)\right] = \sum_{t=0}^{\infty} \gamma^t \bar{\boldsymbol{\pi}}_t^T \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}}\left[\mathbf{g}_{t+1}^k\right] \tag{19.10}$$

   where the probability distributions $\bar{\pi}_t$ are those of the cycle. In practice, the sum is stopped when the variation of its current value produced by the addition of a new term is less than a given threshold;

   (ii) when the Design Problem has the AEV form, the value of the kth indicator is given by

$$\mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,2,\dots}}\left[i_k\left(\mathbf{u}^{p*}, p^*\right)\right] = \frac{1}{T} \sum_{t=0}^{T-1} \bar{\boldsymbol{\pi}}_t^T \mathop{E}_{\boldsymbol{\varepsilon}_{t+1}}\left[\mathbf{g}_{t+1}^k\right] \tag{19.11}$$

   In both of these formulae $\mathbf{g}_{t+1}^k$ is given by equation (19.9b).

This method, however, does not allow us to compute a statistic of the indicators when they are non-separable. To understand the reason for this, suppose, for example, that we have to calculate the statistic of an indicator whose value depends on the state in two consecutive time instants $t-1$ and t. The two marginal probabilities $\bar{\boldsymbol{\pi}}_{t-1}$ and $\bar{\boldsymbol{\pi}}_t$ would not be enough: we also need the joint probability density $\bar{\boldsymbol{\pi}}_{t-1,t}$ of the states $\mathbf{x}_{t-1}$ and $\mathbf{x}_t$. Therefore, it is not always possible to compute the evaluation indicators using analytical formulae when the disturbances are stochastic variables. We must look for another way.

19.2 Deterministic and Monte Carlo simulations

Conceptually, Markov simulation is the most correct way of simulating a system that is affected by randomness, since it provides a complete description of what might happen. Nevertheless, even if the simulation is always possible, we are not able to use its results to compute indicators with a non-separable form. So how can their value be determined? There is only one practicable way: assume a realization of the disturbances and use it to simulate the trajectory followed by the state and input variables of the system over the evaluation horizon; then, in correspondence with this trajectory, compute the values of the indicators. This means that a *deterministic simulation* must be carried out, or at the most we can run many deterministic simulations, a method that is called *Monte Carlo simulation*.

To run a deterministic simulation it is necessary to fix a trajectory of the random disturbances and 'simulate' how the Regulator chooses the control. We will examine these two steps in order, but only after having highlighted the advantages that the deterministic and Monte Carlo simulations offer with respect to the Markov one:

1. We have seen that, to limit the computing times in the Policy Design, the model with the smallest possible dimension of state must be used. This issue concerns the Markov simulation as well, because, as in SDP, computing times increase exponentially with the dimension of the state, given that the number of the elements in the

vectors $\boldsymbol{\chi}_t$ and $\boldsymbol{\pi}_t$ grows with the same law. Deterministic simulation, however, does not suffer from the "curse of dimensionality" and thus it can be performed with complex, even distributed-parameter, models. It is for this reason that, often, in the literature one reads about models for the initial 'screening' of the alternatives (an expression often used to indicate the phase *Designing Alternatives*) and evaluation models for *Estimating Effects*. Deterministic simulation is intrinsically useful, because it makes it possible to ascertain whether the simplifications introduced in the *screening model*, to reduce the state dimension, eliminated phenomena which are important from the point of view of the effects that we want to evaluate.

2. Just like SDP, Markov simulation must necessarily assume that the disturbances that act on the system are white disturbances. It is, in fact, upon this assumption that equations (19.2) and (19.8) are based. We know, nevertheless, that this hypothesis is often strained, in the sense that the whiteness test rejects the hypothesis that disturbances are white, but nonetheless we accept it as such to avoid enlarging the state. In this case Markov simulation inherits the approximation and can thus easily underestimate the frequency of extreme events. The same may happen with the Monte Carlo simulation, but not with a simple deterministic simulation if it is run on the historical time series. We will return to this point in the next section.

19.2.1 Choosing the simulation series

The disturbance time series adopted in a deterministic simulation must necessarily have a finite length. While in the phase *Designing Alternatives* this may constitute a serious problem, it does not in the phase *Estimating Effects*: in fact, in the first of these phases, when the indicators are defined over a finite horizon, no value is attributed to the resource after the end of the *design horizon* (see Section 8.1.2.1) and, as a consequence, close to the end of the horizon the controls proposed may over-exploit the resource. In estimating the effects, however, there is no risk of this because the policy is given.

To select the series of disturbances, the most immediate solution is to adopt a sample from the *historical series*, over N years, during which the process of the disturbances can be reasonably regarded as *cyclostationary*. In this way, the N years may be treated as N equiprobable one-year-long realizations and the statistic of the random indicators, with respect to the disturbances, can be approximated by using the corresponding sampling statistic computed on the time series. For example, for an AEV indicator defined with the Laplace criterion, one may set

$$\mathop{E}_{\{\boldsymbol{\varepsilon}_t\}_{t=1,2,\ldots}}\left[\lim_{h\to\infty}\frac{1}{h+1}\sum_{\tau=0}^{h} g_\tau\big(\mathbf{x}_\tau,\mathbf{u}^p,\boldsymbol{\varepsilon}_{\tau+1}\big)\right]=\frac{1}{NT+1}\sum_{\tau=0}^{NT} g_\tau\big(\mathbf{x}_\tau,\mathbf{u}^p,\boldsymbol{\varepsilon}_{\tau+1}\big)$$

This way of doing things is based on the hypothesis that the process of the disturbance is ergodic, since ergodicity means that the temporal mean of a realization of the process is equal to the expected value of the process.

The estimate obtained in this way can be regarded as an estimate of the *effects that would be produced by the alternative considered, if in the future the same series of disturbances should occur*; while, in general, it cannot be considered as an estimate of the effects that would have been obtained in the past if the alternative had been implemented before the series occurred. As we will explain in Section 19.3, this last interpretation is precluded by the fact that in the *evaluation scenario*, just as in the design scenario, we do not consider the 'historical characteristics' of the users (e.g. their water demand), but rather their expected future characteristics. We will call this first solution *historical simulation*.

The second possibility requires that we identify a (stochastic) model of the process that generates the disturbances. This is often more complicated than the white process description which is used in designing the alternative, given that now the complexity of the model no longer causes the computing time to increase prohibitively. With this model, first L *synthetic series of disturbances* are generated; then, the alternative is simulated for each of them and, finally, the values of the indicators are computed, by evaluating the statistic (mean or maximum) on the sample of the L simulations. This is the *Monte Carlo simulation*.

Let us examine advantages and disadvantages of the two possibilities:

- First of all, we note that among the alternatives to be simulated, the *Alternative Zero* ($A0$) is always included, i.e. the alternative that assumes that nothing is done and everything remains as it is (Section 1.3). However, a formal expression of the regulation policy that was adopted in the past is rarely available and it is not always possible to identify it from interviews with the Regulator or the Stakeholders. Without this policy it is impossible to perform a Monte Carlo simulation, or even a Markov simulation for $A0$, but it is possible to run a simulation over the historical series. If, in fact, the policy is not changed, and no planning actions are implemented, as required by $A0$, the historical trajectory of the reservoir storage inevitably follows from the historical trajectory of the disturbances and then, because the policy has not changed, from this trajectory the historical release trajectory will follow. The historical simulation of $A0$ is, therefore, useless, because it would provide trajectories that are already known. Generally, only the distribution at the diversions must be recomputed, since it might be different from the historical one, given that the users' demand values are those expected in the future and not the historical ones. Nevertheless, given that the sequence of past releases is available and the historical distribution rule can be estimated through interviews,[10] it is easy to simulate the distribution and compute the indicators.

- By using historical simulation, it is easy for Experts and Stakeholders to compare the effects that the alternatives would produce in the future with what has been observed in the past. This gives significance to the indicator values and simplifies their evaluation.

- It is possible that even the best available model of the process that generates the disturbances insufficiently reproduces the temporal correlations that characterize the real process, because the input of this model must be a white process. As a consequence, with the Monte Carlo simulation, as with Markov one, the probability of extreme phenomena, such as prolonged draughts or floods, might be underestimated and, therefore, their effects might be too. This risk is excluded with the historical simulation.

- When the number N of years in the historical series is small, the indicator values provided by a historical simulation might not be representative of what could happen in the future. This problem does not arise, however, with the Monte Carlo simulation,

[10] For an example see Section 6.7.3 of PRACTICE.

because it is possible to generate an arbitrarily large number L of synthetic series, so that the estimated indicator values effectively reflect the characteristics of the disturbance process. The problem does not arise also when the historical series has a good record of the variety of events that can occur.

- Finally, a strong drawback in adopting the historical simulation is that the historical series has already been used to identify the water system model. The significance of the indicator estimates is thus greatly reduced, since the alternative 'is on home ground', in the sense that it implicitly knows what will happen. One could object that, in a certain sense, we hope that this is true whichever series is considered, because we use models for the very reason that they learn from the past and tell us what we can expect from the future; this observation is completely correct but, if the 'future' is exactly the same as the past, it is easy to obtain good performances. Obviously, this observation is not valid when the historical series is long enough to be split into two parts: one for the identification of the model and one for the estimation of the effects.

From the previous points it emerges that there are many arguments in favour of using the historical simulation, and thus often it is the preferred solution. Before definitively adopting it, however, we must ascertain that the series contains a succession of N years, where N is sufficiently large, during which the process that generates the disturbance can be reasonably considered cyclostationary. In fact, only in this case:

(1) can each year be considered as an experiment that is independent from the previous years and that can thus be used to obtain a sample estimate of the statistic (expected or maximum value) which defines the indicators;

(2) does it makes sense to assume that the series being considered could represent one of the possible future realizations of the inflow process.

The reader can find suggestions about how to ascertain these points in the second part of Section 8.2 of PRACTICE.

19.2.2 The Regulator's Model

A model that describes how, at time t, the Regulator chooses a control $\mathbf{u}_t$ from the set $M_t^*(\mathbf{x}_t)$ of controls that an SV policy proposes to him is called *extractor*. We have already stated (see page 281) that all the controls of this set provide the same value for the objective of the Control Problem. Therefore, when there is only one objective, the choice of the control, and thus of the extractor, is irrelevant from the point of view of computing the value of the objective. On the contrary, when there is more than one objective, the above statement should be interpreted in the sense that all the controls of the set $M_t^*(\mathbf{x}_t)$ provide the same value for the scoring function, i.e. for the single objective of the Problem into which the multiple objectives were transformed; however, this does not imply that each individual objective must have the same value. Furthermore, the statement is true as long as the disturbances affecting the system are generated by the model used in the policy design (i.e. by equation (11.13e)). When the disturbances are generated by another model, or belong to the historical series, the statement may not be true, even if, heuristically, we expect that it is not seriously contradicted when equation (11.13b) describes reality reasonably well. It is therefore necessary to establish how much the estimates obtained with the historical or

Monte Carlo simulation are influenced by the choice of a particular extractor. This requires a sensitivity analysis of the estimate of the indicators with respect to the extractor.

Such an analysis is conducted by repeating the estimation with different extractors, which reflect some of the Regulator's possible selection criteria. These criteria vary from case to case and so an extractor of general validity cannot be defined. We can only give a couple of examples of the most frequently used forms:

1. *Reference extractors*: they operate with respect to a given reference $\bar{\mathbf{u}}_t$. First of all, the subset $\tilde{M}_t^*$ of controls that 'satisfy', in a sense that depends on the context being considered, the reference is extracted from the set $M_t^*(\mathbf{x}_t)$; for instance, when the aim of the regulation is supplying water to the downstream users of a reservoir, the set $\tilde{M}_t^*$ can be defined as

$$\tilde{M}_t^* = \left\{\mathbf{u}_t\colon\ \mathbf{u}_t \in M_t^*(\mathbf{x}_t),\ \mathbf{u}_t \geqslant \bar{\mathbf{u}}_t\right\} \tag{19.12}$$

   Finally, the control $\mathbf{u}_t$ that is 'closest' to the reference is chosen from $\tilde{M}_t^*$; in the previous example we could assume

$$\mathbf{u}_t = \arg \min_{\mathbf{u}_t \in \tilde{M}_t^*} (\mathbf{u}_t - \bar{\mathbf{u}}_t)^2 \tag{19.13}$$

   If the set $\tilde{M}_t^*$ is empty, the control that is closest to the reference is identified from the set $M_t^*(\mathbf{x}_t)$. When there is more than one downstream user we may have different reference values. For example, in the Verbano Project, three references are considered: one for irrigation, one for hydropower, and the last for the environment (see Section 8.4.1 in PRACTICE).

2. *Superior extractors*: they choose the 'maximum' control $\mathbf{u}_t$ from $M_t^*(\mathbf{x}_t)$. Usually, the meaning of 'maximum' depends on the context: for example, if are concerned with flooding, the 'maximum' could be the control that has the maximum absolute value.

The extractor must always provide one and only one control and so, when the selection criterion that has been specified does not guarantee this, as in the above, it is necessary to follow it up with a sequence of other criteria, the last of which is the random choice of a control from the ones selected until that point.

The sensitivity analysis is carried out first by simulating different alternatives with each of the chosen extractors, then by computing the values assumed by the indicators for each simulation, and, finally, by comparing the results.

To conclude we must return to a point that was left unresolved in Section 19.1.1, where we explained that when the indicator is not separable the worst case cannot be determined in an exact way and one has to rely on an educated guess. This is left to the Analyst, who must estimate which scenario/extractor pair would produce it and compute the corresponding indicator values using a deterministic simulation.

19.3 The evaluation scenario

The series of deterministic disturbances is not the only element that defines the scenario used to estimate the effects, called *evaluation scenario*; generally, many other elements (e.g. the users' demand values and the flood threshold) contribute to its definition. Some of them must coincide with the elements that define the *design scenario* (Section 8.1.3), because it is part of the evaluation scenario: more precisely, it is the part that is required to estimate the *design indicators*, which are in fact a subset of the *evaluation indicators*. In particular, the scenarios of the pseudo-periodic variables must be constructed with the concatenation rule presented in Section 8.1.3 and the policies must be implemented correspondingly by following the anthropic date rather than the natural one, as explained in Section 10.2.4.

The reader will find an example of the definition of an evaluation scenario in Section 8.3 of PRACTICE.

19.4 Validating the indicators

Once the indicator values have been estimated, we must verify that they are adequate for evaluating and comparing the alternatives. For this reason, two aspects must be examined for each indicator: the significance of the values obtained and its representativeness.

The necessary condition for the values obtained by an indicator to be significant, and therefore to be adequate for comparing the alternatives, is that the difference between the best and worst values of the indicator among all the alternatives be more than double the uncertainty with which the values of the indicator have been estimated. Such uncertainty can be derived from the results obtained by validating the models used to compute the indicators.

The representativeness of an indicator is verified by ascertaining, in collaboration with the Expert who defined it, if it achieves the aim for which it was created: to quantify the leaf criterion that it is associated to and allow the ranking of the alternatives with respect to that criterion. The verification is carried out by choosing a set of alternatives and presenting the Expert with the trajectories of the hydrological variables produced by each one of them. On the basis of these trajectories the Expert must rank the alternatives by decreasing value of satisfaction. The technique used to obtain the ranking is based on the pairwise comparison of the alternatives. We have already encountered this technique in Section 3.2.4. Clearly, the Expert can only rank the alternatives if there are not too many; on the other hand, they should be numerous enough to represent most of the situations that could come about. Once the Expert's ranking has been obtained, the Analyst ranks the same alternatives by decreasing value[11] of the indicator that quantifies the leaf criterion. The indicator is considered to be representative when the two rankings correspond sufficiently well.

Whenever the validation of an indicator fails, the indicator should not be used to evaluate the alternatives and should be substituted by another indicator, but not always may the latter be identified. Thus, at the end of phase *Estimating Effects* the set of criteria can be divided into two subsets: a set of criteria that are quantified by indicators and a set of criteria that have not been or cannot be thus quantified. With respect to the first the evaluation of the alternatives is carried out by analysing the estimated indicator values, so that the complexity of the evaluation procedure depends on the number of the criteria being considered. With respect to the second set, the evaluation requires the direct analysis of the alternatives, through the comparison of the trajectories of the hydrological variables that are produced by

[11] We are assuming that the indicator represents a cost; it would be an increasing order if the indicator expressed a benefit. Nevertheless, there are more complicated cases, in which the function that expresses the satisfaction is not monotonic in the value of the indicator. See Chapter 20.

them; in this case the complexity of the evaluation procedure depends on the number of alternatives to evaluate. In the following we will assume that all the indicators are significant, and refer the reader to PRACTICE to see how to operate if they are not.

19.5 Matrix of the Effects

Once the values $\mathrm{i}_j(A)$ of the m evaluation indicators ($j = 1, \ldots, m$) have been estimated for a given alternative A, they are organized in a vector $\mathbf{i}(A)$. The set of the vectors $\mathbf{i}(A)$, which were calculated for each of the n_A alternatives Ak that one wants to examine ($k = 0, \ldots, n_A$), forms a matrix, called *Matrix of the Effects* (or Evaluation Matrix)

$$\left|\mathbf{i}(A0) \ \cdots \ \mathbf{i}(An_A)\right| = \begin{vmatrix} \mathrm{i}_1(A0) & \mathrm{i}_1(A1) & \cdots & \mathrm{i}_1(An_A) \\ \mathrm{i}_2(A0) & \mathrm{i}_2(A1) & \cdots & \mathrm{i}_2(An_A) \\ \cdots & \cdots & & \cdots \\ \mathrm{i}_m(A0) & \mathrm{i}_m(A1) & \cdots & \mathrm{i}_m(An_A) \end{vmatrix}$$

in which the rows correspond to the indicators and the columns to the alternatives. The element i_{hk} is thus the value $\mathrm{i}_h(Ak)$ that the statistic of the random indicator i_h assumes, in correspondence with the alternative Ak. This value is almost always estimated with the procedures that are described in this chapter, but it could also be provided by an Expert on the basis of his experience, as we mentioned in Section 3.2.

The Matrix of the Effects is the starting point for the subsequent phase *Evaluating Alternatives*, which we deal with in the next chapter.

Chapter 20

Evaluation

FC, EL, FP and RSS

In Section 16.1 we described the purpose of the *Evaluation* phase and in Chapter 17 we presented some of the methods most commonly used to conduct this evaluation. At the end of that chapter we explained that the Multi-Attribute Value Theory (MAVT, Keeney and Raiffa, 1976) is the most suitable method for dealing with the problems considered in this text. We are now prepared to describe it in greater detail. Here we consider the simplest case: there is just one Decision Maker (DM), who is interested in a plurality of criteria, and the Stakeholders do not have an active decision-making role. In the next chapter we explain how the instruments developed for this case can be usefully employed also in the case the Stakeholders have an active role and/or there is more than one DM.

20.1 MAVT: basic assumptions

First of all, let us reiterate the founding assumptions of the MAVT:

1. The MAVT generates *cardinal rankings* (see Section 17.1.2), which means that it allows us not only to determine if one alternative is preferred over another, but also to give the preference a numeric value that expresses its intensity. In particular, an *interval scale* is adopted and therefore, for a given alternative, it is not the absolute value of its preference that counts, but the *difference* between that value and the ones associated to the other alternatives. As we will explain in the following sections, by using this scale it is possible to overcome the limit posed by Arrow's Theorem (Section 17.1.1), i.e. to derive a *democratic* ordering rule (for the definition of this attribute see page 381).

2. The MAVT is a *normative* method, which means it requires that the DM's preference structure satisfies the axioms of *Completeness*, *Transitivity*, *Consistency* and *Independence from Irrelevant Alternatives*.

3. Lastly, the MAVT requires *compensation* among the criteria, i.e. that the worsening of an alternative's performance with respect to one criterion can be compensated with the improvement of its performance with respect to another.

20.2 MAVT: utility functions and value functions

The aim of the MAVT is to formalize the DM's *preference structure*, i.e. the information that is sufficient for ranking the set $\mathcal{A} = |A0, \ldots, An_A|$ of the alternatives, with respect to the satisfaction the DM gets from them; the first alternative in the ranking is the *best compromise alternative*, having taken into account the m evaluation criteria. In Section 17.3 we anticipated that the DM's satisfaction with the effects of an alternative A is expressed through a *value function*, whose argument is the vector $\mathbf{i}(A)$ of the evaluation indicators. The components of this vector are the m values $\mathrm{i}_j(A)$ of the statistics of the m random indicators $i_j(A)$, which were filtered with a suitable criterion. In the phase of *Estimating Effects* the vectors $\mathbf{i}(A)$ are obtained for each of the n_A alternatives and are organized in the Matrix of the Effects (Section 19.5). On the basis of this information the DM must choose the best compromise alternative.

20.2.1 The global value function

One of the ways to define a preference structure is to define a function $V(\cdot)$, called *global value function*, which associates to each alternative A a *value* that expresses the DM's satisfaction with it. Since satisfaction depends on the effects that the alternative produces, this function is actually a function of the value that the vector $\mathbf{i}(A)$ of the indicators assumes in correspondence with alternative A

$$V\big(\mathbf{i}(A)\big) = V\big(\mathrm{i}_1(A), \ldots, \mathrm{i}_m(A)\big) \tag{20.1}$$

The value $V(\mathbf{i}(A))$ is also called the *Project Index* (see page 90) and enjoys the property that

$$\begin{aligned} V\big(\mathbf{i}(A1)\big) > V\big(\mathbf{i}(A2)\big) \quad &\Longleftrightarrow \quad \mathbf{i}(A1) \succ \mathbf{i}(A2) \\ V\big(\mathbf{i}(A1)\big) \geqslant V\big(\mathbf{i}(A2)\big) \quad &\Longleftrightarrow \quad \mathbf{i}(A1) \succcurlyeq \mathbf{i}(A2) \end{aligned}$$

where $\mathbf{i}(A1) \succ \mathbf{i}(A2)$ denotes the fact that the DM prefers the effects of alternative $A1$ to those of alternative $A2$ and so the first alternative to the second; while $\mathbf{i}(A1) \succcurlyeq \mathbf{i}(A2)$ denotes that she weakly prefers the first to the second or considers them to be equivalent. It follows that the best compromise alternative A_c is the alternative that maximizes the value of $V(\mathbf{i}(A))$

$$A_c = \arg \max_{A \in \mathcal{A}} V\big(\mathbf{i}(A)\big)$$

20.2.2 Indifference curves

The global value function is not the only way to represent a preference structure. It can be univocally defined also by partitioning the m-dimensional space of the indicators space into disjoint sets, called *indifference curves*, whose points correspond to vectors of indicators that the DM judges to be equivalent, i.e. among which she has no preference. Consider, for example, a case where there are only two indicators ($m = 2$), the indifference curves have the form shown in Figure 20.1, and the alternatives to be analysed are three ($A1$, $A2$ and $A3$). From Figure 20.1 it is easy to infer that the order of the alternatives is $\{A3 \succ A1 \approx A2\}$. Often indifference curves are decreasing and concave, as in Figure 20.1, because, if the value of one of the indicators increases (e.g. i_1), the satisfaction can remain constant only

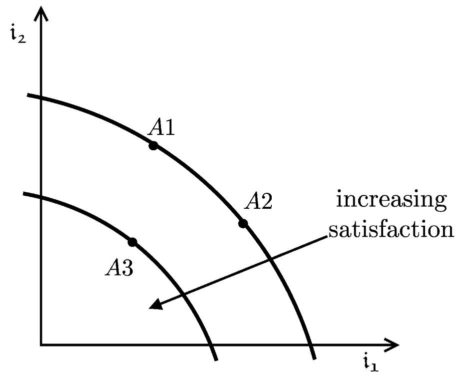


Figure 20.1: Indifference curves in a two-dimensional space.

if the value of the other ($\mathfrak{i}_2$) decreases sufficiently (assuming that the indicators are to be minimized), and, the increase of $\mathfrak{i}_1$ being equal, the reduction of $\mathfrak{i}_2$ will be greater the higher is the value of $\mathfrak{i}_1$.

By definition, the global value function $V(\cdot)$ must be constant along an indifference curve, i.e. it must be

$$\mathbf{i}(A1) \approx \mathbf{i}(A2) \quad \Longleftrightarrow \quad V\big(\mathbf{i}(A1)\big) = V\big(\mathbf{i}(A2)\big)$$

Therefore, the indifference curves can be interpreted as the projection of the level sets of the global value function onto the indicator plane. However, a moment's reflection is enough to realize that the relationship between the preference structure, and thus the indifference curves, and global value function is not one-to-one: given a value function, there is only one corresponding preference structure, but the reverse is not true. A preference structure can in fact be represented by a multiplicity of value functions, because each linear transformation (with positive coefficients) of a value function represents the same preference structure as the original function, since all the value functions so obtained produce the same ranking, no matter which set $\mathcal{A}$ of alternatives is examined. We say that two value functions are *strategically equivalent* if there exists a linear transformation with positive coefficients that allows us to transform one into the other. The existence of strategically equivalent functions permits us to arbitrarily choose the zero and the scale of the values of the function, and this makes the identification of the value function easier.

20.2.3 Identifying the preference structure

At this point it is easy to perceive that to identify the DM's preference structure it is necessary to interview her, asking questions designed to elicit her indifference curves, given which it is possible to identify her global value function. It is easy to guess, however, that the operation is not simple, given that one must work in an m-dimensional space. It is now advisable to see if there is a particular representation of $V(\cdot)$ that allows us to simplify the task. The most interesting form of $V(\cdot)$ is the *separable* one, which means that $V(\cdot)$ is defined by an appropriate composition $f[\cdot]$ of *m partial value functions* $v_1(\cdot), \dots, v_m(\cdot)$

$$V(\mathbf{i}) = V(\mathfrak{i}_1, \dots, \mathfrak{i}_m) = f\big[v_1(\mathfrak{i}_1), \dots, v_m(\mathfrak{i}_m)\big] \tag{20.2}$$

each of which depends on one indicator only, i.e. on a single component of the vector $\mathfrak{i}$ of indicators.[1] The advantage of this form is that, when interviewing the DM, it is possible to concentrate on the single partial value functions, that is, to reason on one indicator at a time.

To specify the global value function $V(\mathfrak{i})$ it is thus necessary to specify the m partial value functions and the function $f[\cdot]$; the simplest case is the one where this last function is *additive*, and so equation (20.2) assumes the following form

$$V(\mathfrak{i}) = V(\mathfrak{i}_1, \ldots, \mathfrak{i}_m) = v_1(\mathfrak{i}_1) + \cdots + v_m(\mathfrak{i}_m) \tag{20.3}$$

Equation (20.3) implicitly defines an ordering rule for combining the partial rankings, which are defined by the functions $v_i(\cdot)$, to form a global ranking. It is easy to see that this rule is democratic. In fact *unanimity*, the first condition that defines a democratic ordering rule (see page 381), is satisfied, because if the value of alternative $A1$ is greater than the value of alternative $A2$ in each one of the m partial value functions $v_i(\cdot)$, i.e. if $A1$ is preferred to $A2$ in all the partial rankings, from equation (20.3) it follows that $V(A1) > V(A2)$, which means that $A1$ is preferred to $A2$ also in the global ranking. By reading their definition it is easy to see that the second condition (*non-imposition*) and the third condition (*non-dictatorship*) are satisfied, while the fourth (*independence from irrelevant alternatives*) is satisfied because the position of any new alternative in the global ranking would be obtained by applying the global value function, whose definition does not depend on a particular set of alternatives (in fact, only the range of variation of the indicator values was considered for its identification).

Unfortunately, however, as we will now explain, only a few preference structures allow a representation of the form of equation (20.3), but, luckily for us, they are those which are most frequently encountered in real cases.

It is not difficult to intuit, in fact, that equation (20.3) implicitly assumes that the DM's preference structure satisfies the following two conditions:

1. The satisfaction (i.e. the value) that she associates to a given indicator value (e.g. $\mathfrak{i}_1$) must be independent of the values $\mathfrak{i}_2, \ldots, \mathfrak{i}_m$ assumed by the other indicators. This characteristic is called *mutual preferential independence*. We will formally define it in the next section, but for now we will just get an idea of its meaning with a couple of examples and note the limit that it poses to the validity of equation (20.3). Imagine that we want to buy a digital camera from an on-line retailer. The WEB site's search engine shows all of the models available, ordering them on the basis of two criteria: the *cost* and the *resolution*. By comparing the cameras when they are ranked by price, for each given price, we prefer the camera with the highest resolution; and this criterion is valid whatever the price. Vice versa, if we rank them according to resolution, for each given resolution, we prefer the camera with the lowest price, whatever the resolution. This approach respects preferential independence: the rule with which the alternatives are ranked according to an indicator does not change with variations in the value assumed by the other indicator. However, sometimes the indicators are not independent: suppose, for example, that we are choosing the menu for a dinner by adopting the criteria *wine* and *main course*, which can assume the values: 'white' or 'red' and 'meat' or 'fish' respectively. In this case, the ranking according to the criterion *wine* depends on the value of *main course*: if its value is 'meat', then the ranking for wine is {red, white}, while if its value is 'fish' the order

[1] The partial value function constitutes the *index* introduced in Sections 3.2 and 3.2.2.

is inverted. In the same way the ranking of the two values of *main course* depend on the value assumed by *wine*.

2. The second condition is that the DM believes that the bad performances that an alternative provides with respect to some indicators can be compensated by good performances with respect to others, a characteristic called *complete compensation* (we introduced this concept in Section 17.3). However, this characteristic cannot be taken for granted: it might be that very poor performances with respect to some criteria cannot be made acceptable by good performances with respect to other criteria, no matter how good they are.[2]

20.3 Mutual preferential independence

Now we would like to explain how to verify whether the DM's preference structure can be expressed with an additive global value function, i.e. a function of the form (20.3). First we consider the case where there are only two indicators, then the case where there are more than two.

20.3.1 Two indicators

In a problem with two indicators i_1 and i_2, the first indicator (i_1) is said to be *preferentially independent* from the second (i_2) if and only if, for any pair i_1' and i_1'' of values of i_1 such that $(\mathrm{i}_1', \mathrm{i}_2') \succsim (\mathrm{i}_1'', \mathrm{i}_2')$ for some values i_2', the condition $(\mathrm{i}_1', \mathrm{i}_2) \succsim (\mathrm{i}_1'', \mathrm{i}_2)$ holds for every i_2.

Indicators i_1 and i_2 are said to be *mutually preferentially independent* when i_1 is preferentially independent of i_2 and i_2 is preferentially independent of i_1.

The last definition establishes how to verify whether the property subsists. As we intuited above, it can be proved that mutual preferential independence is a necessary condition for the global value function to be additive, but unfortunately it is not sufficient. However, the following theorem holds (Keeney and Raiffa, 1976)

Theorem 1: *The preference structure between two indicators can be represented by an additive global value function*

$$V(\mathrm{i}_1, \mathrm{i}_2) = v_1(\mathrm{i}_1) + v_2(\mathrm{i}_2)$$

if and only if the corresponding trade-off condition holds between indicators i_1 *and* i_2.

The *corresponding trade-off condition* is defined by the following procedure (see Figure 20.2), through which it can be also verified:

1. Consider a generic alternative $A1$ and ask the DM to specify the increase[3] b of i_2 that she would be willing to accept in order to obtain a decrease a of i_1, i.e.

$$V\big(\mathrm{i}_1(A1), \mathrm{i}_2(A1)\big) = V\big(\mathrm{i}_1(A1) - a, \mathrm{i}_2(A1) + b\big)$$

[2] For an example see Section 9.5 of PRACTICE.

[3] Remember that in this chapter the DM's satisfaction is assumed to decrease as the value of the indicators increases.

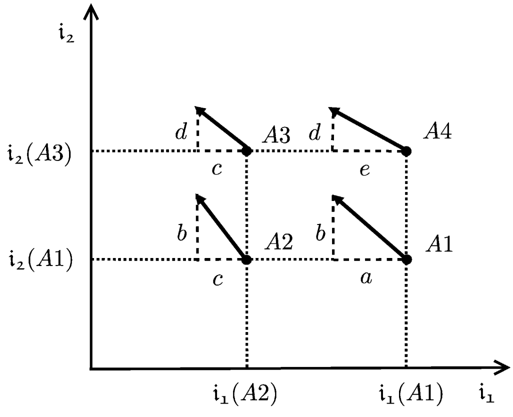


Figure 20.2: The corresponding trade-off condition holds if e is equal to a.

2. Choose an alternative $A2$, such that $\mathrm{i}_1(A_2) < \mathrm{i}_1(A_1)$ and $\mathrm{i}_2(A_2) = \mathrm{i}_2(A_1)$. Ask the DM to specify the decrease c of i_1 that, in her opinion, would compensate an increase b of i_2, i.e.

$$V\big(\mathrm{i}_1(A2), \mathrm{i}_2(A1)\big) = V\big(\mathrm{i}_1(A2) - c, \mathrm{i}_2(A1) + b\big)$$

3. Then choose an alternative $A3$, such that $\mathrm{i}_1(A3) = \mathrm{i}_1(A2)$ and $\mathrm{i}_2(A3) > \mathrm{i}_2(A2)$. Ask the DM to specify the increase d of i_2 that she would be willing to accept to obtain a decrease c of i_1, i.e.

$$V\big(\mathrm{i}_1(A2), \mathrm{i}_2(A3)\big) = V\big(\mathrm{i}_1(A2) - c, \mathrm{i}_2(A3) + d\big)$$

4. Finally, choose an alternative $A4$, such that $\mathrm{i}_1(A4) = \mathrm{i}_1(A1)$ and $\mathrm{i}_2(A4) = \mathrm{i}_2(A3)$. Ask the DM to specify the decrease e of i_1 that in her opinion would compensate an increase d of i_2, i.e.

$$V\big(\mathrm{i}_1(A1), \mathrm{i}_2(A3)\big) = V\big(\mathrm{i}_1(A1) - e, \mathrm{i}_2(A3) + d\big).$$

5. The corresponding trade-off condition holds if and only if $e = a$.

20.3.2 Many indicators

Let us now consider the case in which there are m indicators, with $m > 2$. First of all it is necessary to define what is meant by mutual preferential independence in this case.

Let $\mathbf{i}_1$ and $\mathbf{i}_2$ be two sub-vectors of $\mathbf{i}$ such that their union is $\mathbf{i}$, i.e. they are a partition of $\mathbf{i}$.

The vector $\mathbf{i}_1$ is said to be *preferentially independent* of the vector $\mathbf{i}_2$ if, given the value of $\mathbf{i}_2$, the DM's preference structure in the space of $\mathbf{i}_1$ does not depend on the value assumed by $\mathbf{i}_2$.

Formally, this means that for any pair $\mathbf{i}'_1$ and $\mathbf{i}''_1$ of values of $\mathbf{i}_1$ such that $(\mathbf{i}'_1, \mathbf{i}'_2) \succcurlyeq (\mathbf{i}''_1, \mathbf{i}'_2)$ for some values $\mathbf{i}'_2$,the condition $(\mathbf{i}'_1, \mathbf{i}''_2) \succcurlyeq (\mathbf{i}''_1, \mathbf{i}''_2)$ holds for every value $\mathbf{i}''_2$.

Then, the indicators $\mathrm{i}_1, \ldots, \mathrm{i}_m$ are said to be *mutually preferentially independent* if every subset $\mathbf{i}_1$ of $\{\mathrm{i}_1, \ldots, \mathrm{i}_m\}$ is preferentially independent from its complementary set $\mathbf{i}_2$.

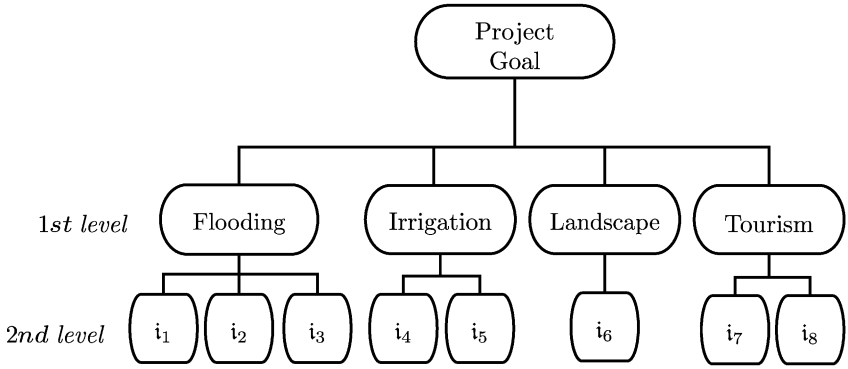


Figure 20.3: A simple example of a Project Hierarchy.

The following theorem (Keeney and Raiffa, 1976) holds:

Theorem 2: *The preference structure among m indicators* $\mathrm{i}_1, \ldots, \mathrm{i}_m$*, with* $m > 2$*, can be represented by an additive global value function of the form*

$$V(\mathrm{i}_1, \ldots, \mathrm{i}_m) = \sum_{i=1}^{m} v_i(\mathrm{i}_i)$$

if and only if mutual preferential independence holds among the indicators $\mathrm{i}_1, \ldots, \mathrm{i}_m$.

The theorem asserts that, in a multidimensional case, mutual preferential independence is a necessary and sufficient condition for the global value function to be additive. This would seem to simplify things, with respect to the previous two-dimensional case, but in fact it does not: verifying whether or not the mutual preferential independence condition holds, by applying the definition, is much more onerous and, in practice, it becomes impossible if m is higher than just a few units.[4]

Fortunately, some theorems (Keeney and Raiffa, 1976) prove that the existence of mutual preferential independence among the components of the vector $\mathbf{i}$ can be confirmed by ascertaining that this condition holds only among appropriate subsets. These theorems use the *Project Hierarchy*, which we introduced in Section 3.2.1.2. A very simple example of a hierarchy of this type is shown in Figure 20.3, where the Project Goal is resolved into four first-level criteria (*Flooding*, *Irrigation*, *Landscape*, *Tourism*), each of which is in turn decomposed into a number of second-level criteria. Thanks to these theorems, the existence of the mutual preferential independence condition can be tested through the following multi-step procedure: in the first step, one verifies whether the condition holds among the first-level criteria; if the test is positive, for each first-level criterion one ascertains whether

[4]The punctual verification of the condition for every possible partition of the set $\{\mathrm{i}_1, \ldots, \mathrm{i}_m\}$ is impossible in practice even for very small values of m: $m(m-1)/2$ checks are required just to verify the independence of sub-vectors constituted by pairs of indicators from their respective complements. Once this verification has been completed, one must verify the independence of the sub-vectors constituted by triples of indicators from their respective complements; then of sub-vectors constituted by quadruples from their respective complements; and so on.

the condition holds among the second-level criteria that derive from it; the procedure continues in the same way until either the condition is not satisfied or the leaf criteria are reached. Referring back to the example in Figure 20.3, if for example, it was determined that mutual preferential independence does not subsist among the triple (*Flooding*, *Irrigation*, *Tourism*) and the (complementary) criterion *Landscape*, then the cited theorems allow us to conclude right away that it does not subsist among the second-level criteria either. If, instead, the condition were verified for all of the possible partitions of the first-level criteria, one would move on to check it in each of the groups of second-level criteria. By operating in this way, the checks are much easier, not only because they are fewer in number, but also because the criteria considered are more easily compared.

20.4 Identifying partial value functions

Once it has been ascertained that the global value function can be expressed in an additive form, each one of the partial value functions must be identified. Many alternative procedures are available for this purpose. We will illustrate two of the most common ones: the *midvalue splitting method* and the *significant points method*.

20.4.1 The midvalue splitting method

Given an indicator i_k the procedure is the following:

1. Define the domain $[a, b]$ of the function $v_k(\cdot)$, on the basis of the values in the Matrix of the Effects and the Decision Maker's (or the Expert's) experience. The domain must be sufficiently wide to contain all the cases that can occur and not just those that appear in the Matrix of the Effects, since in the future it might be necessary to design and evaluate other alternatives; at the same time it should not be so wide as to confuse the DM (we will return to this point in the following).

2. Ask the DM if she wants to maximize the indicator. There are three possible responses:
   (a) the DM wants to maximize the indicator: the partial value function is monotonically increasing (see Figure 20.4a) and thus the values of $v_k(\cdot)$ at the two extremes of the domain $[a, b]$ are set as follows
   $$v_k(a) = 0 \qquad v_k(b) = 1 \tag{20.4}$$
   (b) the DM wants to minimize the indicator: the partial value function is monotonically decreasing (see Figure 20.4b) and thus the values of $v_k(\cdot)$ at the two extremes of the domain $[a, b]$ are set as follows
   $$v_k(a) = 1 \qquad v_k(b) = 0$$
   (c) the DM neither wants to maximize nor to minimize the indicator: the function is not monotonic. In this case it is not possible to associate a value to the points a and b, and one passes directly to step 3.

3. Let us suppose that we have ascertained that the function is increasing (a) or nonmonotonic (c); the extension to case (b) is left to the reader. Ask the DM to indicate

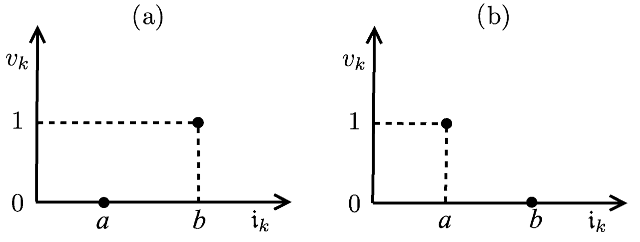


Figure 20.4: The initial operations for identifying the partial value function with the midvalue splitting method for a monotonically increasing function (a) and a monotonically decreasing function (b).

the greatest value $\mathrm{i}_k^0 \in [a, b]$ for which the value of the function is zero and the smallest value $\mathrm{i}_k^1 \in [a, b]$ for which its value is equal to one (see Figure 20.5 for case (a), Figure 20.6 for case (c)).

4. Then consider the interval $[\mathrm{i}_k^0, \mathrm{i}_k^1]$, and within this interval ask the DM to specify the value i_k^2 such that

$$v_k\left(\mathrm{i}_k^2\right) = 0.50$$

5. In the interval $[\mathrm{i}_k^0, \mathrm{i}_k^2]$ ask the DM to specify the value i_k^3 such that

$$v_k\left(\mathrm{i}_k^3\right) = 0.25$$

and in the interval $[\mathrm{i}_k^2, \mathrm{i}_k^1]$ to specify the value i_k^4 such that

$$v_k\left(\mathrm{i}_k^4\right) = 0.75$$

6. ...and so on, continuing to halve the intervals of the value function.

7. When a sufficient number of points have been determined, the function $v_k(\cdot)$ can be derived by interpolation.

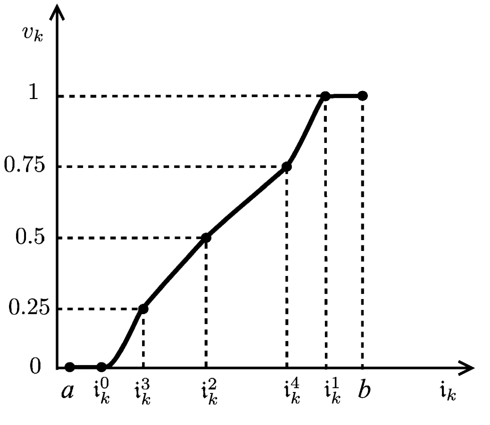


Figure 20.5: Identifying the partial value function with the midvalue splitting method for a monotonically increasing function.

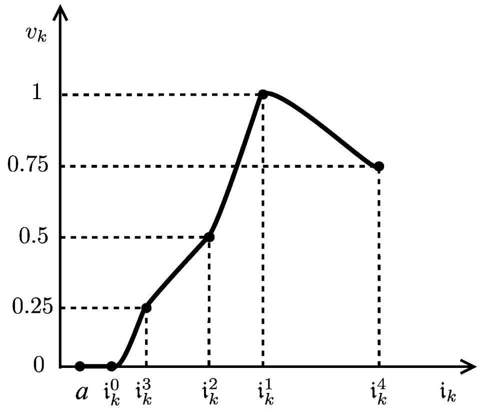


Figure 20.6: Identifying the partial value function with the midvalue splitting method for a non-monotonic function whose first part is increasing.

8. To do this, the DM is asked to associate values of the indicator to a number of values of the function that were not used in its identification; her responses are then compared with the values provided by the estimated function. If necessary, the function should be refined, until the DM is satisfied and believes that it mirrors her preferences.

20.4.2 The significant points method

Sometimes, mostly when working with an Expert, it can be more productive to ask for the significant values of the indicator and the associated values of the function; in other words, instead of starting with the values of $v_k(\cdot)$ and working up to the values of i_k, one starts from i_k and associates $v_k(\mathrm{i}_k)$. Having termed $\mathcal{I}$ the domain of the function and set $\mathcal{I} = [a, b]$, the procedure is as follows:

1. Ask the DM, or the Expert, to identify in $\mathcal{I}$ the subset $\mathcal{I}_{\min}$ of values of i_k to which the minimum value (0) is associated and the subset $\mathcal{I}_{\max}$ to which the maximum value (1) is associated. Clearly, the first, or the second, or both of the subsets $\mathcal{I}_{\min}$ and $\mathcal{I}_{\max}$ could contain just a single element.

2. Ask the DM/Expert to choose a significant point $\bar{\mathrm{i}}_k \in \mathcal{I} - (\mathcal{I}_{\min} \cup \mathcal{I}_{\max})$ and to associate it with a value, taking into account the values associated to the points that have already been defined. Then ask her to choose a new point and proceed in this way until a set $\mathcal{V}$ of sufficiently numerous pairs (i_k, v_k) has been defined. Clearly, there are many criteria with which the DM/Expert could choose the new point to propose; for this reason the Analyst should guide her by asking that she first identify the points for which she believes that the function $v_k(\cdot)$, or its first derivative, would present discontinuity.

3. Interpolate the points of $\mathcal{V}$, to obtain the function $v_k(\cdot)$.

4. Finally, run validation tests as for the previous method and, if necessary, refine the estimate until the DM/Expert is satisfied, i.e. she believes that the function identified mirrors her preferences.

This method is fruitful when the DM/Expert has an excellent knowledge of the sector and is able to reason in quantitative terms without difficulty.

20.4.3 Remarks

So far, we have assumed that all the partial value functions $v_k(\cdot)$ assume values in the interval [0, 1]: this choice may appear fairly arbitrary. However, it is justified by the presence of strategically equivalent value functions (see page 435), i.e. by the fact that there exist an infinite number of value functions that can equivalently represent the DM's preference structure. It is thus legitimate to fix the extreme values at 0 and 1, provided that all the other values are determined in a way that is coherent with this choice. This is the advantage offered by the interval scale that we have adopted: the value function does not represent the intensity of the DM's preferences through its absolute value, but through the differences in value between different points (Section 17.1.2). Two functions $v_k(\cdot)$ and $v'_k(\cdot)$ generate the same ranking of the alternatives when they are related to each other by a linear transformation

$$v'_k = \alpha_k + \lambda_k v_k$$

in which the coefficients α_k and λ_k are positive. In fact, for each pair i_k^1 and i_k^2, the two functions express value differences that are proportional, i.e.

$$v'_k(\mathrm{i}_k^1) - v'_k(\mathrm{i}_k^2) = \alpha_k + \lambda_k v_k(\mathrm{i}_k^1) - \alpha_k - \lambda_k v_k(\mathrm{i}_k^2) = \lambda_k\big[v_k(\mathrm{i}_k^1) - v_k(\mathrm{i}_k^2)\big]$$

It follows that the values of any two points can be fixed arbitrarily. This freedom can be exploited to fix the values of the minimum and maximum so that the interval of values of v_k is equal to one.

The definition of the domain $[a, b]$ of the partial value function $v_k(\cdot)$ also deserves comment. The set of values that the indicator i_k can assume may or may not be known a priori, i.e. independently of the alternatives that are designed. The first case occurs, for example, when i_k is a percentage: then the domain is clearly [0, 100]. The second case is not so simple. The values of i_k presented in the Matrix of the Effects individuate an interval $\mathcal{I}_1$ of values that is a function of the type and number of alternatives designed. If from the *Comparison* phase the need emerges to design new alternatives, these could produce values of i_k external to $\mathcal{I}_1$; if the value function has only been assessed in $\mathcal{I}_1$, one would then be forced to assess it again. To avoid this risk, it is advisable to take this eventuality into account when defining the domain $\mathcal{I}$. For example, in Figure 20.7 the domain $\mathcal{I}$ adopted for the partial value function associated to the indicator *Average annual number of necessary treatments* from the Verbano Project (see Section 9.2 of PRACTICE) is compared with the indicator values produced by the alternatives. As the Figure clearly shows, the interval $\mathcal{I}_1$ served only as a support for the definition of $\mathcal{I}$.

One last remark: it is important to ascertain whether the DM's preference structure is time-invariant. For this reason, the partial value function should be validated again in one or more sessions subsequent to the one in which it was identified. If the validation fails, it could be that the indicator i_k does not provide sufficient information to evaluate the effects of the alternatives with respect to the leaf criteria for which it was defined. In this case, one would have to return to Phase 2 of the PIP procedure to define a new indicator. The validation could fail, however, also when the DM does not take the completion of her task seriously, because she does not believe in it or is poorly motivated. If, instead, the validation proves positive, one can conclude that the function $v_k(\cdot)$ satisfactorily reflects the DM's preferences.

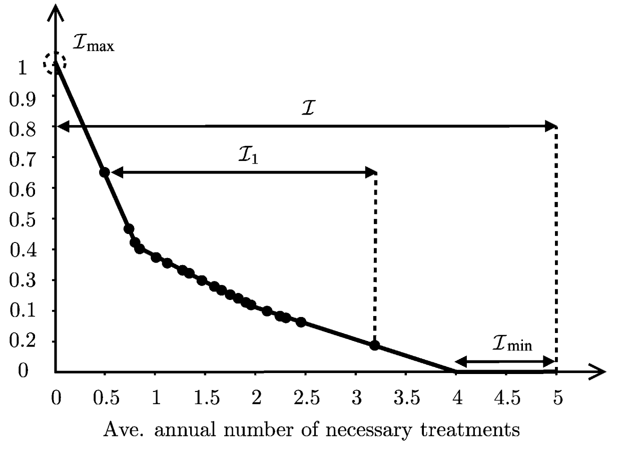


Figure 20.7: The partial value function associated to the indicator *Average annual number of necessary treatments* from the Verbano Project.

20.5 Excluding dominated alternatives

Before proceeding any further, any dominated alternatives must be identified and excluded.

This affirmation might surprise the reader: the alternatives considered in the *Evaluation* phase are, in fact, efficient for the Design Problem, i.e. not dominated. However, the concept of dominance is a relative concept and much attention should be paid to the context in which it is used. In solving the Design Problem, the efficiency of the alternatives is evaluated with respect to the *design indicators*, which do not necessarily coincide with the *evaluation indicators* for two reasons:

(1) in order to satisfy the condition for separability required by the algorithm adopted for the solution of the Design Problem (Section 18.6), the mathematical formulation of a design indicator could differ from the formulation of the corresponding evaluation indicator;

(2) a design indicator can be used in place of several evaluation indicators (see for example, the case presented in Section 7.3.2 of PRACTICE).

However, the main reason why dominance must be ascertained just at this point is that dominance has now to be defined with respect to the values of the partial value functions, and not with respect to the values of the evaluation indicators. The latter, in fact, do not represent the DM's preference structure; they are only a means through which it can be identified. Now that this structure is known, the alternatives to be considered are only those that are non-dominated according to this structure, i.e. according to the partial value functions.

20.6 Identifying the global value function

Once all the partial value functions $v_k(\cdot)$ have been identified, the global value function $V(\mathrm{i}_1, \ldots, \mathrm{i}_m)$ is defined by their sum (remember equation (20.3)). However, the partial value functions are not actually univocally defined, since any other definition obtained by applying a linear transformation with constant coefficients would be equivalent. As a consequence, the most general form of the global value function, whose value, we remind the reader, is the *Project Index V*, is the following

$$V(\mathrm{i}_1, \ldots, \mathrm{i}_m) = \alpha_1 + \lambda_1 v_1(\mathrm{i}_1) + \cdots + \alpha_m + \lambda_m v_m(\mathrm{i}_m) \tag{20.5}$$

Thus, to assess the global value function it is not sufficient to know the m partial value functions; it is also necessary to determine the values of the $2m$ coefficients (*weights*) $\alpha_1, \ldots, \alpha_m$ and $\lambda_1, \ldots, \lambda_m$.

Just as in the case of partial value functions, the adoption of an interval scale offers us two degrees of freedom that we can use to simplify the identification as much as possible. As usual, we first consider a simple two-dimensional case and then extend the results to a more general case.

20.6.1 Two-dimensional value functions

The global value function is two-dimensional when it is the composition of *two* partial value functions $v_1(\cdot)$ and $v_2(\cdot)$, whose arguments are the indicators i_1 and i_2 respectively. As Figures 20.5 and 20.6 show, these two functions are, by construction, such that

$$v_k\left(\mathrm{i}_k^0\right) = 0 \quad v_k\left(\mathrm{i}_k^1\right) = 1 \quad k = 1, 2$$

It follows that

$$V\left(\mathrm{i}_1^0, \mathrm{i}_2^0\right) = \alpha_1 + \alpha_2$$
$$V\left(\mathrm{i}_1^1, \mathrm{i}_2^1\right) = \alpha_1 + \lambda_1 + \alpha_2 + \lambda_2$$

The interval scale allows us to fix the extreme values of the function $V(\mathrm{i}_1, \mathrm{i}_2)$ as we like; for consistency, we can fix them at 0 and 1. So we have

$$V\left(\mathrm{i}_1^0, \mathrm{i}_2^0\right) = 0 \qquad V\left(\mathrm{i}_1^1, \mathrm{i}_2^1\right) = 1$$

from which it follows that

$$\alpha_1 + \alpha_2 = 0 \qquad \alpha_1 + \lambda_1 + \alpha_2 + \lambda_2 = 1$$

and, since the coefficients α_i and λ_i cannot be negative,

$$\alpha_1 = \alpha_2 = 0 \qquad \lambda_1 + \lambda_2 = 1$$

Therefore, it must be

$$V(\mathrm{i}_1, \mathrm{i}_2) = \lambda_1 v_1(\mathrm{i}_1) + \lambda_2 v_2(\mathrm{i}_2)$$
$$\lambda_1 + \lambda_2 = 1$$

In conclusion, in the global value function there appear only the parameters λ_i, called *scale coefficients* or *weights*, the values of which must be fixed so that the function reflects the DM's preference structure. In the two-dimensional case, for estimating these values it is sufficient to identify two pairs $(\mathrm{i}_1', \mathrm{i}_2')$ and $(\mathrm{i}_1'', \mathrm{i}_2'')$ of indicator values that the DM considers to be equivalent, i.e. that satisfy the following condition

$$V\left(\mathrm{i}_1', \mathrm{i}_2'\right) = V\left(\mathrm{i}_1'', \mathrm{i}_2''\right)$$

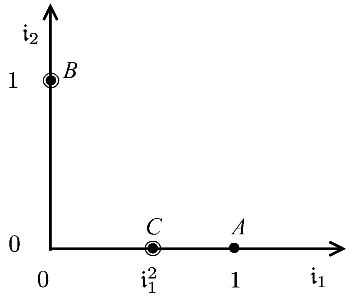


Figure 20.8: Searching for a pair of equivalent points for estimating the weights λ_i.

It follows that

$$\lambda_1 v_1\big(\mathrm{i}_1'\big) + \lambda_2 v_2\big(\mathrm{i}_2'\big) = \lambda_1 v_1\big(\mathrm{i}_1''\big) + \lambda_2 v_2\big(\mathrm{i}_2''\big) \tag{20.6}$$

which, when associated to the constraint

$$\lambda_1 + \lambda_2 = 1 \tag{20.7}$$

defines a system of two equations in the two unknowns λ_1 and λ_2, whose solution provides the values we are looking for.

To identify the two pairs of values of i_1 and i_2 that the DM considers to be equivalent, the following procedure can be adopted:

1. Consider two pairs $A = (\mathrm{i}_1^1, \mathrm{i}_2^0)$ and $B = (\mathrm{i}_1^0, \mathrm{i}_2^1)$ (Figure 20.8), for which it is known that

$$\begin{aligned} v_1(A) &= v_1\big(\mathrm{i}_1^1\big) = 1 \qquad & v_2(A) &= v_2\big(\mathrm{i}_2^0\big) = 0 \\ v_1(B) &= v_1\big(\mathrm{i}_1^0\big) = 0 \qquad & v_2(B) &= v_2\big(\mathrm{i}_2^1\big) = 1 \end{aligned} \tag{20.8}$$

   i.e. it is known that the DM is very dissatisfied with the performances of A from the point of view of indicator i_2 and is very dissatisfied with B from the point of view of indicator i_1.

2. Ask the DM if she prefers A or B. If the response is A, it means that the weight of i_1 in defining the global value function is greater than the weight of i_2, and so we deduce that it must be

$$\lambda_1 > \lambda_2$$

   If the response is B, the conclusion is the opposite. This information suggests the indicator about which the next question will be posed.

3. If the response was A, ask the DM to indicate the point C on the axis i_1 (Figure 20.8) that she considers equivalent to B, i.e. such that $V(C) = V(B)$. This point certainly exists since, by moving from A to the left, the DM's satisfaction decreases. The pair $(\mathrm{i}_1^2, \mathrm{i}_2^0)$ so determined is by definition equivalent to the pair $(\mathrm{i}_1^0, \mathrm{i}_2^1)$ that specifies B. Thus the two pairs of values $(\mathrm{i}_1, \mathrm{i}_2)$ that are equivalent for the DM have been found

and therefore equation (20.6) becomes

$$\lambda_1 v_1\big(\mathrm{i}_1^2\big) + \lambda_2 v_2\big(\mathrm{i}_2^0\big) = \lambda_1 v_1\big(\mathrm{i}_1^0\big) + \lambda_2 v_2\big(\mathrm{i}_2^1\big)$$

from which, due to equation (20.8) and remembering the constraint (20.7), the following system is obtained

$$\begin{aligned} \lambda_1 v_1\big(\mathrm{i}_1^2\big) &= \lambda_2 \\ \lambda_1 + \lambda_2 &= 1 \end{aligned}$$

Clearly, when the response to the question posed in point 2 is B, point C has to be identified on the axis i_2.

In order that the global value function effectively mirror the DM's preference structure, the partial value functions must be identified and the weights λ_1 and λ_2 must be estimated very carefully. The partial value functions are identified on the basis of several points and then validated, while the weights are estimated on the basis of just one pair of points. To increase the reliability of this estimate, it is advisable to analyse the sensitivity of the weights with respect to the position of point C. To do this, one or more points around C are chosen and the value of weights is estimated for each one of them. If the estimates do not differ too much, they can be considered reasonably reliable; if they do, it is advisable to repeat the estimation of the weights starting with a pair of points different from (A, B).

In conclusion, it is important to notice that the above procedure provides estimates of the weights that are essentially based on the actual values that the indicators assume in the Project being considered and not on preconceived abstract ideas about their relative importance.

20.6.2 Multidimensional value functions

Now let us consider the identification of the global value function $V(\cdot)$ when there are more than two indicators. By repeating the reasoning used in the two-dimensional case, it is easy to show that, when the conditions for additivity are satisfied, the function $V(\cdot)$ has the form

$$V(\mathrm{i}_1, \ldots, \mathrm{i}_m) = \sum_{i=1}^{m} \lambda_i v_i(\mathrm{i}_i) \tag{20.9a}$$

with

$$\sum_{i=1}^{m} \lambda_i = 1 \tag{20.9b}$$

and λ_i not negative.

The estimation of the weights λ_i is not as easy as in the previous case: the DM must identify $m-1$ pairs of equivalent points in the space of the values. With these pairs, $m-1$ equations can be defined from which, given equation (20.9b), the estimate is obtained. However, as the reader might easily imagine, the identification of these pairs becomes more and more difficult as the number m of indicators increases.

The estimate can be validated by identifying new pairs of equivalent points and verifying that the corresponding values of function (20.9a) are equal, when λ_i have the estimated values. If this does not occur, the DM must re-examine the equivalent pairs that she proposed. This confirms beyond any doubt, if it is still necessary, that a correct identification of the function $V(\cdot)$ essentially depends on the willingness of the DM to collaborate with the Analyst.

The methods described are not the only ones possible, but they are the most frequently used in practice. The reader should bear in mind that the procedures presented are only indicative. In a real problem, in fact, it is often possible, and advisable, to adapt them to the specific requirements of the case at hand. Thereby the Analyst's capacity to ask the DM the 'right' questions is fundamental. This is why we can say that the identification of the value functions is *more of an art than a science*.

20.6.3 Pairwise comparison and hierarchies

The identification of $m-1$ pairs of equivalent points in the space of the values is conceptually well defined; but, in practice, the greater the number m of indicators, the harder it is. To simplify the identification procedure, we can adopt an empirical method for estimating the vector $\boldsymbol{\lambda}$ of the weights: the *pairwise comparison method*. This method requires the DM to specify the relative importance between pairs of leaf criteria. More precisely, it requires the DM to fill in a square matrix, the *pairwise comparison matrix*, whose rows and columns are constituted by the m leaf criteria.[5] Each element of the matrix represents the result of the comparison of the criterion associated to that row with the one associated to that column, and it expresses how many times the first is preferred over the second. For example, given two criteria $crit_1$ and $crit_2$, the DM's response might be the following

$$crit_1 \sim 3\,crit_2$$

i.e. $crit_1$ is preferred 3 times over $crit_2$ (the notation was introduced in Section 17.2).

In theory, it is not necessary that the DM specify the relative importance of all the pairs of criteria: it is sufficient that she does so for $m-1$ pairs, provided that each criterion is considered at least once. In practice, however, to validate the result by ascertaining the coherence of the estimates provided by the DM, it is advisable that she makes more than $m-1$ comparisons.

A number of algorithms have been proposed to derive an estimate $\boldsymbol{\lambda^c}$ of the vector $\boldsymbol{\lambda}$ of the weights from the pairwise comparison matrix. In our opinion, the most interesting is the one proposed by Saaty (1980, 1986) in the context of the Analytic Hierarchy Process (see Section 17.4). By adopting this algorithm, $\boldsymbol{\lambda^c}$ is the dominant eigenvector of the pairwise comparison matrix. Once $\boldsymbol{\lambda^c}$ has been obtained, it is used in equation (20.9a): each λ_i^c multiplies the partial value function that corresponds to the indicator associated to the ith leaf criterion.

The risk with this method is that the DM might make the comparison in an abstract way, i.e. without considering the actual values assumed by the indicators, while, as we have already mentioned above, the relative importance among the criteria depends on these values (see the example on page 389).

A further difficulty arises when the project is characterized by a high number of indicators. In this case, pairwise comparison can be difficult because the DM cannot easily perceive the relation between a given pair of criteria. For example, in a project like Verbano, how could the DM reply when she is asked the relative importance of the *Distribution costs for the irrigation supply* with respect to the *Erosion in the lake's reed beds*?

[5] The leaf criteria are, by construction, as many as the indicators.

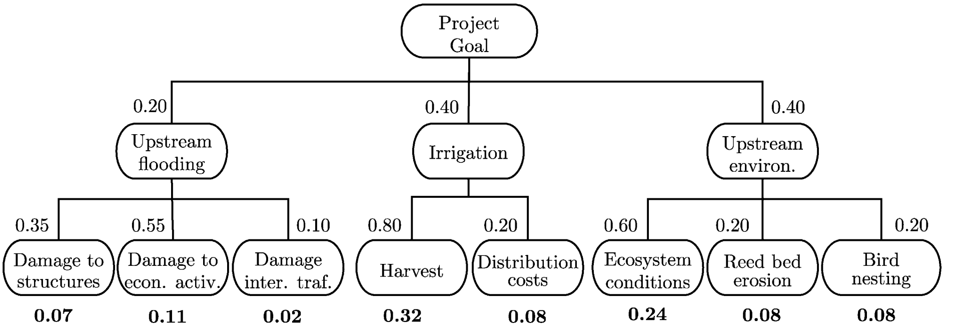


Figure 20.9: An example of a Project Hierarchy. The weight (bold) assigned to a leaf criterion is obtained by multiplying the weights placed along the branch to which it belongs.

This last difficulty is due to the fact that we are trying to determine the weights associated to the leaf criteria at one fell swoop. However, it can be overcome by adopting a two-step procedure that takes advantage of the Project Hierarchy introduced in Section 3.2.2. At the first step, for each criterion of the Hierarchy that has at least one child criterion, the pairwise comparison method is applied to the set of criteria that have it as the parent criterion. Once all the criteria at all the levels have been considered, the weight of a given leaf criterion is computed by multiplying the weights associated to all the criteria of the branch to which it belongs. An example may clarify this: in Figure 20.9, the weight (0.07) associated to the indicator *Damage to structures* is obtained by multiplying the weight of the criterion *Upstream flooding* (0.20) by the relative weight (0.35) of *Damage to structures* with respect to the other criteria which descend from *Upstream flooding*. This last weight (0.35) is obtained by applying the pairwise comparison method to the three children of *Upstream flooding*; the first weight (0.20) by applying the method to the three criteria at the first level, which descend from the Project Goal.

In this way, the DM must only compare criteria at the same level, which are, by nature, more easily comparable. Furthermore, the estimate of the weights can be delegated to different figures according to the level being considered. For example, the weights at the leaf level can be estimated by Experts; moving up the hierarchy, the comparison becomes less technical and more political, and the comparison is reserved for the DM. In the example in Figure 20.9, the comparison between *Irrigation* and *Upstream Environment* has a political character, while the comparison between *Damage to economic activities* and *Damage caused by interruption to traffic* is technical.

The defect of this procedure is that a clear way of comparing criteria other than leaf criteria does not exist, since indicators with specific units of measurement are associated only to the latter. Therefore, the comparison at the higher levels of the hierarchy tends to become aprioristic, which is precisely what we would like to avoid. Despite this defect, in real problems with many indicators, pairwise comparison is the only method that can deal with their complexity.

In conclusion, the estimation of the weights through the identification of pairs of equivalent points in the space of the values, although in theory more correct than pairwise comparison, it is rarely done in practice. In general, the definition of the Project Index $V(\cdot)$ is subdivided into several steps and is based on the concept of *sector index* introduced at page 93. For example, consider the case of a hierarchy with two levels like the one in Figure 20.9 and assume that the criteria at the first level are sector criteria. The sector indices I_h are defined first as

$$I_h(\mathrm{i}_1, \ldots, \mathrm{i}_{k_h}) = \sum_{j=1}^{k_h} \tilde{\lambda}_j^h v_j^h(\mathrm{i}_j) \quad h = 1, \ldots, k_h \tag{20.10a}$$

where $\tilde{\lambda}_j^h$ are the weights of k_h leaf criteria into which the hth sector criterion is resolved. Subsequently, the Project Index $V(\cdot)$ is defined as a linear combination of the sector indices

$$V(I_1, \ldots, I_s) = \sum_{h=1}^{s} \gamma_h I_h \tag{20.10b}$$

and the weights γ_h are fixed. The definition (20.10) of the sector index is completely equivalent to equation (20.9), provided that the weights λ_i are set to be equal to the products $\tilde{\lambda}_j^h \gamma_h$ with appropriate i, j and h.

20.7 Uncertainty in the Evaluation

The identification of the global value function (*Project Index*) permits us to reduce a Multi-Objective decision-making problem to a Single-Objective one: the maximization of the DM's satisfaction. The *ranking* of the alternatives is thus obtained by ordering the alternatives by decreasing values of this index. Simply the construction of this ranking and the consequent choice of the first alternative as *best compromise alternative* is not, however, completely satisfying for the DM. The procedure by which the ranking was obtained is, in fact, characterized by more or less significant *arbitrariness* and *uncertainty*, which are seen in the estimate of the effects, in the identification of the partial value functions and in the attribution of the weights. Regarding the estimate of the effects, in Section 19.5 we saw that this is made through models or the judgement of Experts. In the two cases the estimate is subject both to intrinsic uncertainty, since one estimates the effects of something that does not yet exist, and to a partial discretion both in the choice of models and of the evaluation scheme. Also the identification of the partial value functions is more or less arbitrary, according to the indicators considered: for some there exist forms which are by now well defined in the literature, while for others the choice is more subjective. Lastly, the attribution of weights is one of the most delicate points of the entire evaluation. The weights, in fact, must reflect the DM's value judgment on the relative importance of the indicators; however, while it is quite easy to express qualitative comparisons between them, their conversion into quantitative values involves wide margins of discretion and uncertainty.

These uncertainties and this discretion ensure that the DM perceives the ranking obtained as very fragile: i.e. she has the feeling that if only she had replied slightly differently to some of the questions, the ranking would have been changed. It is therefore necessary to ascertain whether this feeling of hers is well founded, i.e. to ascertain whether small changes in answers and assumptions made can really modify the ranking. For this purpose a sensitivity analysis is carried out on the estimate of the effects, on the partial value functions and on the weights.

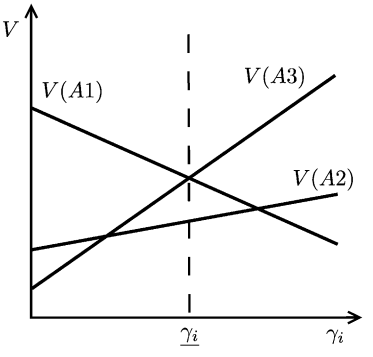


Figure 20.10: Sensitivity analysis with respect to the weight γ_i: for values of γ_i inferior to $\underline{\gamma_i}$, the alternative $A1$ obtains the best value V and thus it gets the first position in the ranking; for values of γ_i superior to $\underline{\gamma_i}$ a *rank reversal* takes place and the alternative $A3$ becomes the first.

20.7.1 Sensitivity analysis

The sensitivity analysis (see Section A3.5) aims to individuate the variations in the values of the indicators, of the parameters defining the partial value functions and of the weights that produce a modification in the ranking of the alternatives (*rank reversal*), concentrating mostly on the alternatives in the first positions in the ranking. One thus obtains an assessment of the robustness of the ranking: if the region is large, the DM knows that the answers given and the assumptions made, by her and by the Experts, should be modified significantly before the best compromise alternative changes and she thus becomes sure of the result; while if the region is small she is certain that the ranking is critical and she will thus re-examine carefully all the answers and assumptions.

The sensitivity analysis can be carried out with substantially different approaches. The simplest way is to re-calculate the ranking to verify the effects

- of the uncertainty between the two estimates of the value of an indicator;
- of the assumption of a partial value function between two or more alternative forms for the same indicator;
- of the uncertainty between two values of a weight.

It is better to adopt this approach when the number n_A of the alternatives to evaluate is not too great. Otherwise one can have recourse to analytical methods, developed ad hoc, easily derivable when the uncertainty can be formalized in terms of variation compared to the value of a single parameter. The simplest case is when the doubt arises whether one has overestimated or underestimated the value of a weight, or of an indicator: in this case the Project Indices of the alternatives can be expressed analytically as a function of the undefined parameter, and then, also analytically, it is possible to find how wide the differences from the initial value of the parameter must be to obtain a *rank reversal*. Figure 20.10 shows the value $\underline{\lambda_i}$ of the uncertain weight λ_i for which a rank reversal between the alternatives $A1$, $A2$ and $A3$ is determined. To obtain this result, equation (20.9a) has to be inverted in order to express λ_i as a function of V and, by considering all the values of λ_i, both superior and inferior to the one considered, the values $\underline{\lambda_i}$ which provoke a rank reversal are sought.

20.8 Beyond MAVT

So far the MAVT has been presented in the context in which it was originally proposed by its authors, i.e. when there is only one DM. However, the projects that we are interested in are characterized both by the possibility that there is more than one DM, and by the desire to give Stakeholders an active decision-making role. In this context, the Project Index defined by the MAVT is not appropriate, because it does not allow us to capture how the effects of the alternatives are distributed among the different Stakeholders, and thus it does not signal the emergence of potential conflicts between them. For this reason, the MAVT must be modified or substituted.

To guess how this can be done, firstly note that the operations described in this chapter formalize the way in which a single DM expresses her preferences and makes a decision. When there is more than one DM and/or the Stakeholders play an active decision-making role, each of them would like to act as if (s)he were the only DM, and therefore (s)he carries out the aforesaid operations in order to identify the best compromise alternative with respect to his/her own interests. Remember that each DM and each Stakeholder has his/her own preference structure, which in the MAVT approach is formalized through the partial value functions and the vector of weights (s)he expresses. As a consequence, the partial value functions of a given indicator may sometimes be differently defined by each of them, while the weights λ_i almost always differ, because each subject tends to assign a low weight to the criteria that are of not interest for him/her, and a high weight to the others. As a consequence, each person defines his/her own personal Project Index, which is different from the others', and therefore identifies different best compromise alternatives. It is thus necessary to find a way to mediate between the different viewpoints: this is the very aim of negotiations, which we deal with in the next section.

Chapter 21

Comparison, negotiations and the Final Decision

FC, EL, FP and RSS

When there is only one Decision Maker (DM), the best compromise alternative is identified in the *Evaluation* phase and therefore with this phase the decision-making process ends. In Section 16.2, however, we have explained that usually this is not the best approach, and that the decisions are made after the *Comparison* phase in which also the Stakeholders are involved. It is, in fact, not advisable that the DM take the responsibility for deciding alone, given that many interests are at stake: excluding the Stakeholders from the choice can generate tensions and opposition to the implementation of the chosen alternative. It is for this very reason that the IWRM paradigm requires that the decision emerge from *negotiations* that involve all the interested parties. Therefore, whenever possible, it is opportune to give the Stakeholders an active decision-making role (*co-deciding*) (Mostert, 2003; Hare et al., 2003), and encourage them to negotiate with each other to identify the alternatives that get the widest agreement. Only after acquiring a deep understanding of all the aspects of the problem that emerge in the negotiation process can the DM begin the *Final Decision* phase with full awareness.

When there is more than one DM, the application of the evaluation techniques almost always leads to identifying different best compromise alternatives for each DM, given that each of them generally has her own preference structure, i.e. her own particular point of view. In this case, even the *Final Decision* phase is carried out through negotiations among the DMs, the aim of which is to reach *consensus* on one alternative. In Section 2.3 we have already given a first idea of how such negotiations can be conducted.

This chapter is dedicated to the study of negotiation methods and procedures. Following the usage that is widespread among negotiation theorists, we will use the term *Parties* to refer to those who negotiate (whether they are Stakeholders or Decision Makers, according to the phase that is being considered) and *Group* to refer to them collectively. We can thus say that the subject of the chapter is *group decision*, which some authors call *joint* or *cooperative decision*. In what follows, just as we did in Section 16.2, wherever it is not explicitly necessary to distinguish between a single DM and several DMs, we will write 'DM(s)'.

21.1 How to negotiate

In this section we will introduce some general aspects of negotiations. These are illustrated in more detail in Appendix A10, where a presentation of the psychological and cultural aspects of negotiations can also be found. In order to get a deeper understanding of negotiation process dynamics, we suggest reading it before proceeding any further with this chapter.

The negotiation process is assisted by a *Facilitator*, a neutral third party (often the Analyst if he has the capacities), who defines the phases of the process and manages it in such a way that the Stakeholders proceed constructively towards the building of a consensus around one (or a few) alternatives. Sometimes the Facilitator acts as a *Mediator*, which means that he assumes a more active role: he does not only facilitate, but also structures the process, governing the agenda with care to identify the critical points, and using tools such as reformulating, active listening and open questions, as well as his analytical abilities. He can meet with some of the Stakeholders separately, if he wants or if they request, but he must keep what he learns from these private discussions absolutely confidential, unless they explicitly authorize him to disclose what was said. He can suggest solutions, but by doing so he risks losing the neutrality that is essential to his role because some of the Stakeholders might believe, either rightly or wrongly, that his proposal is partisan.

Rarely negotiations are necessarily competitive: almost always the Stakeholders, by coming to an agreement, can 'enlarge the pie' significantly before competing to establish which slice belongs to each of them. For example, in the case of the Verbano Project (see PRACTICE), by being willing to negotiate about raising the regulation range, the Swiss stakeholders opened the door to alternatives that were advantageous to everyone (*win–win* alternatives) and that otherwise would never have been put on the table. Nonetheless, very often the Stakeholders behave as if the negotiation process were purely competitive: they consider each other opponents, instead of feeling like cooperative problem solvers of a problem which involves everyone, even if for different reasons. We do not live in a 'zero-sum' society (i.e. it is not true that one's gain necessarily means another's loss); the problem is that often we behave as if this were true (Raiffa et al., 2002). The duty of a good Facilitator is to help the Parties to identify actions that enlarge the pie, i.e. enlarge the *zone of possible agreement* (ZOPA, see Lewicki et al., 1999) and encourage them to think creatively, keeping inventing alternatives well separate from decision-making.

Nevertheless, even when a cooperative attitude is adopted, not all the problems are solved: it is obvious that one wants to obtain a big slice of the pie that s(he) contributes to creating with the others, but the tactics for creating a larger pie (for example the open exchange of information) might be in conflict with the tactics aimed at claiming the biggest slice. In every negotiation process 'creating value' is inextricably related to 'claiming value': this is the negotiator's dilemma (Raiffa et al., 2002).

Since negotiating requires that a Party give something up (e.g. a very high value for a criterion that is close to his/her heart) in exchange for something else (usually the possibility to implement an alternative that improves other criteria that are of interest for him/her), it is essential that the viewpoint and motivations of each Party be known to the entire Group. This is why exchange of information, discussion and reflection in the Group must be suitably organized (Section 21.8). When the Parties share and exchange information about the problem, the system and their own interests, it is said that a paradigm of *Full*, *Open*, *Truthful Exchange* (FOTE) (Raiffa et al., 2002) is adopted. The PIP procedure is inspired by this paradigm: in fact, before the evaluation and comparison of the alternatives, it foresees the exchange of information (Phase 0), the cooperative definition of the actions (Phase 1), the enunciation of the interests (Phase 2) and the identification of a shared model (Phase 3). The DSS that we propose (see Chapter 24) makes all of this information available to the interested Parties on the WEB. Sometimes, however, the Parties do not want to communicate their interests at the very beginning of the process but reserve the right to reveal them only if and when they believe it most advantageous. When this occurs, the negotiation process is much more complicated, because many of the proposals put forward by the Parties will have the sole aim of forcing the other Parties to reveal their information and true interests. We will not deal with this kind of problem, referring the interested reader to specialist books (for an exhaustive bibliography see Raiffa et al., 2002).

In the negotiation process each Party has his/her own interests in mind and looks for an agreement that satisfies him/her as much as possible. Almost always, at the beginning of the process (s)he will hope that the alternative that (s)he considers to be the best will be implemented. This alternative is called his/her *vision*. One cannot take for granted that the vision can be implemented in practice, given the interests of the other Parties. For this reason, each Party considers at least another alternative, which may be called BATNA (*Best Alternative To a Negotiation Agreement*, see Fisher et al., 1991) that (s)he could implement by him/her self, in the event an agreement with the other Parties cannot be reached (see Appendix A10). To identify a compromise, each Party puts forward *options*, or proposals, to the other Parties and evaluates the ones that they offer, by comparing them with his/her own *vision* and with the BATNA. (S)he thus accepts an option only if it offers better performances that his/her BATNA. Therefore the first strategy of each Party is usually to try to strengthen the BATNA as much as possible.

The negotiation process is thus one in which the Parties put forward proposals, evaluate them and elaborate counterproposals. When the negotiation process is focussed on *visions*, i.e. positions (*Position-based negotiations*), it is carried out mostly through expressions of power, or tends to stall. To avoid this, it is advisable to shift the focus of negotiations to the interests (*Interest-based negotiation*), by elaborating options that try to satisfy the interests of several Parties at the same time (Fisher et al., 1991).

In the PIP procedure, negotiations usually occur during the *Comparison* and *Final Decision* phases. However, many of the decisions that are made in the preceding phases, such as the choice of the indicators and of the model that describes the system, determine the alternatives about which the Group will negotiate and the way in which that will be done. Thus the Parties should be invited to participate to all these phases and possibly to negotiate joint decisions, as illustrated in the next section.

21.2 What to negotiate

When we speak of negotiations, i.e. of a joint decision, our thoughts run immediately to the final decision: will an agreement be reached? The final choice is of crucial importance, but a joint decision does not come out of nothing: it is the result of a long negotiation process of which it is only the final step. The whole decision-making process is filled with joint decisions. This can be realized by examining the phases that constitute the process according to the PIP procedure, as we will do in the following for a typical situation: when there is one Decision Maker (DM) and several Stakeholders.

Phase 0 – Reconnaissance

The Parties must establish:

- the *Project Goal*;
- the *definition*, in time and space, of the boundaries of the system being considered;
- the decision-making *procedure* to be adopted: the PIP procedure, for example.

These points are often taken for granted, but they are actually the subject of a joint decision: sometimes it is an easy, implicit and non-formalized decision, but on other occasions it is not.

It is always possible to obtain the consensus of the Group about the Project Goal, by simply expressing it in sufficiently generic terms, like for example “to manage the regulated lake with the aim of satisfying the irrigation demand in the downstream districts, while guaranteeing the safeguarding of the fish species in the lake’s emissary”. Obtaining agreement on such a generic Goal, however, is not useful and can even be damaging, because it postpones the moment when the conflict will emerge, and thus postpones the search for an agreement. Divergences may arise when the objectives are specified in space and time: for example, what do we mean by “safeguarding the fish species in the lake’s emissary”? How far away from the barrier can we assume that the fish populations are affected by the regulation of the lake? Which species should be safeguarded? Are they only the species that would be present naturally, or also those which were artificially introduced?

While agreeing on the space–time definition of the system, it is of paramount importance to obtain an agreement about the elements that define the scenario, i.e. the variables that depend upon factors which the Group cannot, or believes it cannot, influence: for example, the amount and the space–time distribution of the inflows to the lake and the water demand from the irrigation districts.

Finally, the very procedure with which the decision-making process will be conducted must be chosen: it is in fact essential that the procedure be shared by the whole Group. Some of the Parties might not accept the PIP procedure, because they think it is too rigid and that it might affect the expression of their preferences in a negative way. The Facilitator must explain to them why he believes that this is not true and, if he is not able to convince them, negotiate another procedure.

Phase 1 – Defining Actions

The joint decision concerns the identification of the *interventions* aimed at reaching the Goal and the *actions* with which to execute them. Any disagreement can almost always be settled if the Facilitator proposes to accept every intervention and every action that are suggested by the Parties.

Phase 2 – Defining Criteria and Indicators

The critical moment is the choice of the *evaluation criteria* and of the *indicators*. Regarding the *criteria*, any conflicts can easily be resolved, just as in the previous Phase, by suggesting that all the evaluation criteria that are proposed should be considered. There is no real conflict, since there is no reason why one Party should reject the criterion proposed by another if his/hers is accepted. Accepting an evaluation criterion does not imply that it must be attributed with a non-zero weight in the *Evaluation* phase; it just means recognizing that the alternatives can be judged from that point of view as well.

As for the indicators, each Party usually concentrates only on the identification of the indicators associated to the criteria that (s)he is interested in. For those criteria which affect only one Party, conflicts are not likely to arise; for those criteria which affect more than one Party, conflicts can arise about the choice of the most suitable indicators for quantifying them. It is opportune to try to resolve these conflicts and arrive at definitions that are accepted by everyone. Nevertheless, if this is not possible, the conflicts can be overcome by agreeing to consider all the indicators that are proposed. Just as before, the Party who believes that an indicator is not significant would simply attribute a zero weight to it when the partial value functions are aggregated.

Phase 3 – Identifying the Model

It is not likely that the conflicts that arise in this phase cannot be resolved, because they involve purely technical questions. The choice of Experts that are trusted by the Stakeholders is generally sufficient to guarantee that all the Parties accept the modelling choices that are adopted. In the rare case in which this does not happen, one can let each Party adopt the model (s)he mostly believes in, and use it for estimating the effects of the alternatives.

This is why it is essential for the MODSS to allow the models to be substituted quickly and easily (see Section 24.3).

Phase 4 – Designing Alternatives

Here the joint decision concerns the choice of the *objectives* of the MO Design Problem and possibly the choice of the 'screening' model (Section 19.2). An agreement on the screening model can always be reached with the same method used in the previous phase, i.e. by considering a variety of models and designing the alternatives with each of them, though this should be avoided if possible. As for the objectives, in theory an agreement could be found by accepting all the objectives proposed; in practice, however, this could result in an unmanageable increase in the computing time required to determine the Pareto Frontier.[1] When this happens, the difficulty can be overcome by proceeding as follows.

Consider only a few objectives, selected by the Analyst; design the alternatives on the basis of these objectives; estimate the effects of the alternatives on all the proposed objectives; finally, check whether the objectives that were discarded are strongly correlated with those that were considered (see Section 18.6). If they are, the difficulty has been overcome. In the opposite case there is no way to reduce the number of objectives and therefore we suggest conducting the negotiation process with the Elementary Negotiation Procedure described in Section 21.5, with the further expedient of designing the alternatives during the negotiations themselves, in response to requests. Thereby, the computing time is significantly reduced because only the alternatives that are actually necessary are designed. However, the negotiation process takes more time because at each step it is necessary to wait while an alternative with the required characteristics is designed. To overcome this difficulty, either the negotiations are performed with the help of an MODSS that, operating on the WEB, does not require that the Parties meet physically in the same location (see Section 24.4.1) or, alternatively, the negotiations are split into steps and the alternatives designed between the steps, as in the Verbano Project (see Chapters 10–14 of PRACTICE).

[1] More precisely the Pareto-Efficient Decision Set.

Phase 6 – Evaluation

In this phase, each Party contributes to the construction of the sector indices that (s)he is interested in, by defining the partial value functions and the weights with which to aggregate them. Often a sector is of interest to only one of the Parties, in which case there is no conflict. Otherwise, the Parties interested in the same sector must look for an agreement about the definition of the index, possibly through negotiations: remember, in fact, that in Chapter 3 we have defined a sector as a sub-tree of the hierarchy of criteria, for which one can hypothesize that in the *Evaluation* phase a shared definition of the index can be reached. If on the contrary the Parties are not able to come to an agreement, the sector must be broken down into more sectors.

The choice of weights to attribute to the sector indices is on the other hand almost always a cause of conflict, because it is very subjective. This is why assigning weights to sectors is not included in the (technical) phase of *Evaluation* but, instead, in the (political) phase of *Comparison*.

Phase 7 – Comparison

This is the core of the participatory decision-making process. In this phase the Parties (Stakeholders) must attribute weights to the sector indices, but this operation can easily generate disagreements, and the only way to overcome them is to negotiate. In this chapter we propose some techniques and procedures which in our experience proved very useful for this purpose.

Phase 9 – Final Decision

Just as in the previous phase, disagreements among the Parties (Decision makers) arise when weights have to be attributed to the sector indices. To search for an agreement, the same techniques and procedures introduced for the *Comparison* phase can be used.

21.3 Step-by-step negotiations vs negotiations on rankings

From the examination above, one can appreciate that for group decision the negotiation process among the Parties is not necessarily limited to the *Comparison* and *Final Decision* phases. The process can be approached in two different ways:

1. *Step-by-step negotiations*: the presence of more than one Party is taken into account in all the phases of the decision-making process. The setting up and conclusion of each phase is negotiated and the following phase is not begun until a consensus has been reached about the current one. Sometimes ‘consensus’ means only that everyone recognizes the existence of differences of opinion; think of a case in which no agreement is found about the model to use and it is decided that several will be used. From the point of view of an external observer who looks at the results obtained in each phase, it seems as if the decision-making process were carried out by one DM, but from an internal point of view … what a negotiation effort to reach consensus on each phase!

2. *Negotiations on rankings*: by consulting with the Parties, but without being limited by their views, the Analyst develops the decision-making process to design a set of alternatives for evaluation; then, he collaborates with each Party, so that (s)he can evaluate the alternatives autonomously and rank them from his/her point of view. Only at this point does the negotiation begin, with the aim of combining the plurality

of rankings obtained by the different Parties into a single ranking. Note that it is not necessary that all the Parties adopt the decision-making procedure that we have described so far: they are free to use the methods that they prefer in order to construct their ranking and even create it on the basis of their personal convictions, without any support from the Analyst.

The two approaches are not equivalent. The first brings out the conflicts about the values and the interests in play, which are systematically made explicit: in this way the conflict is analysed in every conceivable aspect and the decision-making process is managed effectively, so that solidity and consensus are guaranteed for the final decision. On the contrary, the conflicts that emerge with the second approach only concern the results: therefore the debate can be very limited and the final decision might not be shared and consensual. Thus, since step-by-step negotiations produce the most solid and lasting results, generally one begins with that approach and turns to negotiations on rankings only if the first fails (see Figure 21.1).

When the *Comparison* phase is reached through step-by-step negotiations, the comparison of the alternatives can be carried out with *negotiations on weights* (Section 21.4). The Group, in fact, shares the definition of the sector indices and therefore, according to the MAVT paradigm (Chapter 20), to complete the decision-making process, it only remains to choose the vector of weights. If the Stakeholders agree on the value of the vector of weights, a ranking is obtained whose first alternative is the one that they judge to be the *best compromise alternative*. It will be proposed as such to the DM(s) in the subsequent *Final Decision* phase, along with the first few alternatives that follow it in the ranking, so that the DM(s) can understand why it was preferred. If, instead, the Stakeholders do not agree upon a vector of weights, the comparison proceeds with *negotiations on thresholds* (Section 21.5).

It is not essential that negotiations conclude with the identification of the *best compromise alternative*, which would require the complete agreement of all the Stakeholders. Indeed, negotiations on thresholds usually lead to the identification of a *set of reasonable alternatives*, i.e. of Pareto-efficient alternatives, each of which is supported by at least one Party. Even if the best compromise alternative is not identified, the negotiation process should not be regarded as useless, because it helps to screen the alternatives that are of no interest to anyone and to define the Stakeholder's positions about the reasonable alternatives: information that is precious to the DM(s). More in general, it is not necessarily true that the negotiation process is successful only if it concludes with a consensus: often one is satisfied when the Parties' positions have moved closer together. If it seems that negotiations have failed, because the Parties' preferences are spread out over alternatives that are very different, the Facilitator must try to enlarge the field of choices. To do so he can ask the Analyst to design new alternatives that include the new actions suggested by the process of social learning activated by negotiations, as possible ways to enlarge the agreement or mitigation actions (Chapter 22). The phases *Designing Alternatives*, *Evaluation* and *Comparison* thus occur recursively; an example is provided by the Verbano Project (Chapters 12, 13 and 14 of PRACTICE).

In the *Final Decision* phase it is the DM(s)' duty to choose an alternative; clearly she (they) must be allowed to choose not only among the reasonable alternatives, provided that she (they) explains the choice with the same instruments (indicators and criteria) that were defined by the Stakeholders in the course of the process. If the process were conducted well, any choice of the DM(s) different from the Stakeholders' should be explained only by the different relative importance that the DM(s) give to the sector.

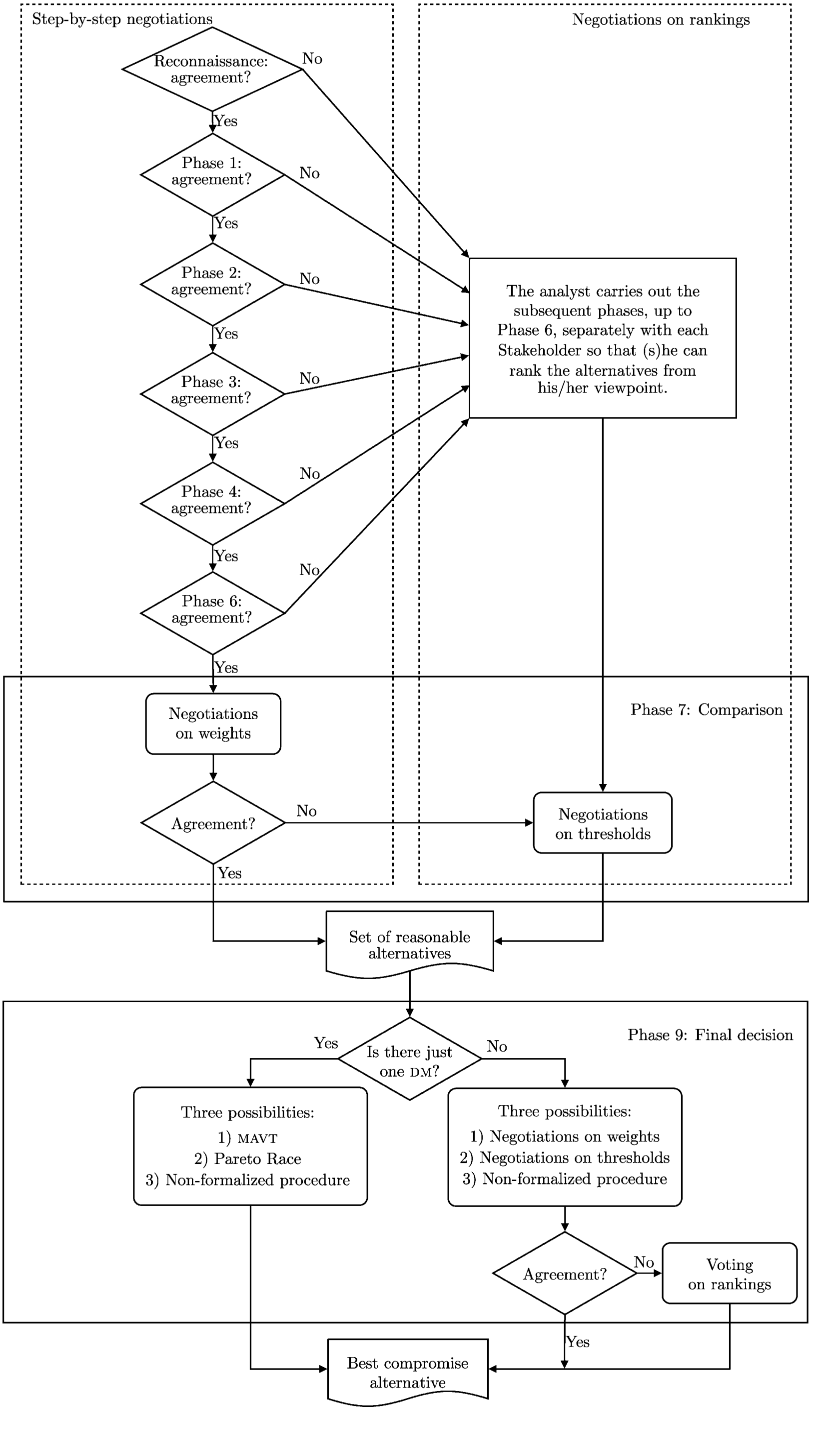


Figure 21.1: The negotiation process.

When there is only one DM, the *Final Decision* phase can be carried out with the MAVT techniques, by inviting the DM to specify her own vector of weights for the sector indices, and then, by means of this, evaluating the reasonable alternatives chosen by the Stakeholders. Otherwise the *Final Decision* can be carried out with the support of a procedure similar to the ENP (see Section 21.5.1). Sometimes, the DM prefers to choose with less formal methods, but in that case it is advisable, for reasons of transparency, that she makes the reasons for her choice explicit. When there are two or more DMs, the *Final Decision* is technically a negotiation process, and as such it can be conducted as *negotiations on weights* or *negotiations on thresholds*. In this situation, nevertheless, it is very likely that the decision is taken with less formalized methods, which depend upon the local customs and culture. It is rare that the *Final Decision* phase concludes without a decision: generally an alternative must be chosen. For this reason, if all the negotiation approaches fail one must resort to *voting on rankings* (Section 21.6).

21.4 Negotiations on weights

Negotiations on weights are adopted at the end of step-by-step negotiations, after the Stakeholders have defined the sector indices $(I_1, \ldots, I_s)$ for each of the s sectors. The *Project Index*, see equation (20.10b), is a linear combination of these indices

$$V(I_1, \ldots, I_s) = \sum_{h=1}^{s} \gamma_h I_h \tag{21.1}$$

and the purpose of the negotiation process is to establish the vector $\boldsymbol{\gamma} = |\gamma_1, \ldots, \gamma_s|$ of the weights that define it, given the vectors $\boldsymbol{\gamma}^k$ $(k = 1, \ldots, p)$ proposed by the p Parties. Of the various procedures that have been conceived to negotiate the value of $\boldsymbol{\gamma}$ we will illustrate two which are very similar, and which in our experience have proved to be useful. We term them the *asynchronous process* and the *synchronous process*.

The *asynchronous process* begins by elaborating the set of vectors $\boldsymbol{\gamma}^k$ with the aim of identifying first the barycentric vector $\boldsymbol{\gamma}^b$ of the set, and then the Party whose vector is the furthest from it. The barycentric vector $\boldsymbol{\gamma}^b$ is defined as the vector for which the sum of the distances between it and the vectors $\boldsymbol{\gamma}^k$ is minimal, and can thus be determined by solving an optimization problem. Once it has been found, it is easy to find the vector $\boldsymbol{\gamma}^f$ that is the furthest from it, and also the Party who expressed it. This Party is asked to illustrate his/her reasons for proposing the vector of weights $\boldsymbol{\gamma}^f$, and then if (s)he is willing to partially modify $\boldsymbol{\gamma}^f$ in order to make it closer to the barycentric vector and reduce the dispersion of the Group. With the same aim, the other Parties are also asked if they are prepared to modify their own vectors on the basis of the reasons provided by the first Party. If at least one vector is modified, the above procedure can be iterated with the new set of vectors.

The procedure is therefore iterative and at each step attention is concentrated on just one Party (from which the name asynchronous). It is stopped either when a consensus is reached about a vector of weights or when no Party is willing to modify his/her vector, or finally, when the thresholds of consensus, which we will define later, have been satisfied.

The *synchronous process* is different from the previous one only because, once the barycentric vector has been identified, each of the Parties is invited to modify his/her own vector of weights, declaring the reasons why (s)he is willing or not to do so.

Adopting one or the other procedure depends on the reactivity of the decision-making Group: it is advisable to begin with the synchronous process and, if there is little reactivity, to move to the asynchronous process.

The progress of negotiations can be monitored using a *conflict matrix*, whose generic element d_{ij} is the distance between the vectors of the Parties i and j. From this matrix many indicators of conflict can be derived: the number of Parties that have vectors $\boldsymbol{\gamma}^k$ that are different from the others' (it is equal to half of the number of elements $d_{ij} \neq 0$), an *average conflict index* (the average of d_{ij}) and a *maximum conflict index* (the maximum value of the d_{ij}). With these elements it is also possible to fix a *consensus threshold* for the average conflict index, or for the maximum conflict index, below which one considers that a consensus about the vector $\boldsymbol{\gamma}^b$ has been reached.

21.4.1 Definition of distance

In the description of these procedures we used the concept of distance $d(\boldsymbol{\gamma}', \boldsymbol{\gamma}'')$ between two vectors $\boldsymbol{\gamma}'$ and $\boldsymbol{\gamma}''$, without explaining how it is defined. In the literature various definitions can be found (Bogart, 1973, 1975), but all of them, except for one, are unsuitable for our aim: to measure the distance between vectors of weights.

To understand why, observe that a vector is generally defined by assigning a direction (ϑ) and a length (ρ). However, let us think for a moment about how a vector of weights is defined: it is the expression of a preference among sector indices. A preference such as "index I_1 is 3 times better than index I_2" can be expressed with an infinite number of vectors, for example {3, 1}, or {36, 12}, or {1, 1/3}, or {3/2, 1/2} and so on. The information is the same in all cases; only the representation rule changes. Note that a preference specifies a direction in the index space and all the vectors that lie on the half-line that characterizes that direction are equivalent expressions of it. It is because of this very fact that not all the definitions of distance between vectors are acceptable.

An example may clarify this: consider a simple case in which there are two sectors, and therefore two indices (I_1 and I_2), and three Parties. The Parties express the following vectors $\bar{\boldsymbol{\gamma}}^k$ ($k = 1, 2, 3$) of weights, by adopting the rule that the sum of their components must always be equal to one

$$\bar{\boldsymbol{\gamma}}^1 = \left\{\frac{1}{2}, \frac{1}{2}\right\} \qquad \bar{\boldsymbol{\gamma}}^2 = \left\{\frac{4}{11}, \frac{7}{11}\right\} \qquad \bar{\boldsymbol{\gamma}}^3 = \left\{\frac{2}{3}, \frac{1}{3}\right\}$$

Alternatively, the Parties could adopt the rule that the first component must always be equal to one. In this case they would express the following three vectors

$$\bar{\bar{\boldsymbol{\gamma}}}^1 = \{1, 1\} \qquad \bar{\bar{\boldsymbol{\gamma}}}^2 = \left\{1, \frac{7}{4}\right\} \qquad \bar{\bar{\boldsymbol{\gamma}}}^3 = \left\{1, \frac{1}{2}\right\}$$

Each one of them expresses the same preference of the homonymous vector of the first group: for example the vectors $\bar{\boldsymbol{\gamma}}^1$ and $\bar{\bar{\boldsymbol{\gamma}}}^1$ express a judgement of equivalence between the two indices, while the vectors $\bar{\boldsymbol{\gamma}}^3$ and $\bar{\bar{\boldsymbol{\gamma}}}^3$ say that the first has two times the importance of the second. Let us now assume the most common definition of distance: the *Euclidean* distance

$$d_e(\boldsymbol{\gamma}', \boldsymbol{\gamma}'') = \sqrt{\sum_{h=1}^{s} \left(\gamma_h' - \gamma_h''\right)^2}$$

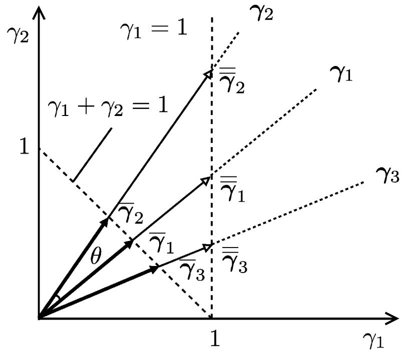


Figure 21.2: The graphic representation of three vectors, when their components are determined with the rules 'unitary sum' and 'first element equal to 1'.

where s is the number of the components of the vectors, which is equal to the number of sectors. The values of the distance between the vector $\overline{\boldsymbol{\gamma}}^1$ and the vectors $\overline{\boldsymbol{\gamma}}^2$ and $\overline{\boldsymbol{\gamma}}^3$ are thus the following

$$d_e(\overline{\boldsymbol{\gamma}}^1, \overline{\boldsymbol{\gamma}}^2) = 0.19$$
$$d_e(\overline{\boldsymbol{\gamma}}^1, \overline{\boldsymbol{\gamma}}^3) = 0.24$$

while the distances between $\overline{\overline{\boldsymbol{\gamma}}}^1$ and the vectors $\overline{\overline{\boldsymbol{\gamma}}}^2$ and $\overline{\overline{\boldsymbol{\gamma}}}^3$ are

$$d_e(\overline{\overline{\boldsymbol{\gamma}}}^1, \overline{\overline{\boldsymbol{\gamma}}}^2) = 0.75$$
$$d_e(\overline{\overline{\boldsymbol{\gamma}}}^1, \overline{\overline{\boldsymbol{\gamma}}}^3) = 0.50$$

Note that in the first triple $\overline{\boldsymbol{\gamma}}^1$ and $\overline{\boldsymbol{\gamma}}^2$ are the closest vectors, while in the second their equivalents $\overline{\overline{\boldsymbol{\gamma}}}^1$ and $\overline{\overline{\boldsymbol{\gamma}}}^2$ are the furthest away! We must conclude that the distance between the preferences proves to be dependent on the representation that is adopted, but this is clearly unacceptable.

To understand how to overcome this difficulty, we can examine the two groups of vectors with the help of Figure 21.2. The vectors $\overline{\boldsymbol{\gamma}}^1$, $\overline{\boldsymbol{\gamma}}^2$ and $\overline{\boldsymbol{\gamma}}^3$ characterize three directions ($\boldsymbol{\gamma}^1$, $\boldsymbol{\gamma}^2$ and $\boldsymbol{\gamma}^3$), along which also the vectors $\overline{\overline{\boldsymbol{\gamma}}}^1$, $\overline{\overline{\boldsymbol{\gamma}}}^2$ and $\overline{\overline{\boldsymbol{\gamma}}}^3$ lie. This confirms our previous observation that the preference characterizes a direction. Given the three preferences ($\boldsymbol{\gamma}^1$, $\boldsymbol{\gamma}^2$ and $\boldsymbol{\gamma}^3$), defining the vectors of weights with a unitary sum means assuming the vector individuated by the intersection of the half-lines that express the preferences and the line $\gamma_1 + \gamma_2 = 1$. Defining the vectors by imposing the first element as always equal to 1 means, instead, assuming the vectors individuated by the intersection of the half-lines and the line of equation $\gamma_1 = 1$. The figure clearly shows why the distance between the vectors depends on the representation that is adopted.

This is why Colorni et al. (2001a) proposed defining the distance between two vectors as the angle θ between the vectors $\boldsymbol{\gamma}'$ and $\boldsymbol{\gamma}''$: with this definition the distance clearly does not depend on the representation adopted. The definition of distance to adopt in negotiations on weights is therefore the following

$$d_\theta = \theta = \arccos \frac{\sum_{h=1}^{s} \gamma_h' \cdot \gamma_h''}{\sqrt{\sum_{h=1}^{s}(\gamma_h')^2}\sqrt{\sum_{h=1}^{s}(\gamma_h'')^2}}$$

21.5 Negotiations on thresholds

Unlike negotiations on weights, negotiations on thresholds are not aimed at defining a Project Index for the ranking of the alternatives; instead, they aim at directly identifying the alternatives that obtain a wide agreement. Also they are carried out with an iterative process, which, at each step, establishes *acceptability thresholds*[2] for the sector indices and examines only the alternatives that satisfy them.

The process is inspired by the Constraint Method (page 409) and by the *Pareto Race* method, proposed in the eighties (Korhonen and Laakso, 1986) to solve Multi-Objective Problems with only one DM. The latter is a method for generating points along the Pareto Frontier in real time, by solving a sequence of Design Problems formulated on the basis of the preferences that the DM expresses as she gradually gets the results of the previous problems. It is thus an interactive and iterative method defined by the following procedure:

1. As for the Reference Point Method (page 405), the DM expresses her ‘dream’ by specifying a *reference point* R in the index space.[3]

2. Given R, the Analyst defines the Design Problem (18.14) and determines its solution. As we explained, the solution is an efficient alternative and its image, in the space of the objectives, is the point of the Pareto Frontier which is ‘closest’ to R in the sense specified by equation (18.13).

3. With a bar chart the values of the indices that correspond to that alternative are shown to the DM.

4. She is asked whether she is ‘satisfied’ with those values. If the answer is positive, the procedure terminates. If it is negative, she is asked which indices she would like to improve; in this way a new reference point R is obtained and one goes back to Step 2.

Unlike other methods, for which the whole Pareto Frontier must be identified in advance, thus leaving the choice of the best compromise alternative to the DM, the Pareto Race has the advantage of identifying only the alternatives that the DM thinks are interesting. As the procedure develops, the DM has the feeling of moving along the Pareto Frontier (from which the name *Pareto Race*). She thus acquires knowledge about the conflicts between the different sectors, because she sees that some of their indices increase while others decrease, and can thus appreciate the possibilities for compromise. In this way, the best compromise slowly takes form in her mind. This is why the procedure ends when the DM is ‘satisfied’.

In water management problems it is not always possible to adopt the Pareto Race method, especially when the design of a regulation policy is required, because the great amount of computing time required to solve the Control Problem does not allow the design

[2] Negotiations’ theorists call them *reservation values*.

[3] The Reference Point Method considers the objectives space, but for the DM indices play the role of objectives.

to be completed in real time,[4] following the preferences that are expressed by the DM step by step. If the generation of the responses is too slow, it prevents the DM from 'perceiving' the compromise. In fact, it is only when the questions and answers follow each other in quick succession that the DM can 'acquire knowledge' about the compromise, since this is an acquisition process that is based on short-term human memory. There is a second difficulty: the Pareto Race requires there to be only one DM. The method, in fact, encompasses the three phases *Designing Alternatives*, *Estimating Effects* and *Evaluation*.

The way to overcome the first difficulty is simple: it is just because of the considerable computing time required for the Design Problem that in the PIP procedure the phases *Designing Alternatives*, *Estimating Effects* and *Evaluation* are conducted off-line (i.e. before the *Comparison*). Therefore the difficulty is overcome if in Step 2 the 'closest' alternative to the DM's 'dream' is not determined by solving a Design Problem (in the whole alternatives space), but by exploring the set $\mathcal{A}^c$ of the alternatives previously designed.

The second difficulty remains. However, we can solve it by adopting the procedure described in the following paragraph.

21.5.1 The Elementary Negotiation Procedure

The Elementary Negotiation Procedure (ENP) presumes that each of the p Parties that make up the Group has completed the *Evaluation* phase (possibly autonomously). Therefore, each Party, say the kth, has defined his/her own Project Index[5] V^k, through the definition of the vector of weights $\boldsymbol{\gamma}^k$ which expresses his/her preferences among the sector indices that appear in equation (21.1).

The ENP operates on the project indices V^k (which in the following we will call simply 'indices' for brevity's sake). By establishing acceptability thresholds for them, the procedure identifies the subset of reasonable alternatives within the set $\mathcal{A}^c$ of efficient alternatives to be compared. It consists in the following steps:

1. Given an initial alternative $A_{\text{chosen}} \in \mathcal{A}^c$, the Facilitator shows the values of the indices $V^1, V^2, \ldots, V^p$ that this alternative produces by means of a bar chart. Then he asks each Party to declare how they position themselves in relation to it: i.e. if they *support* it, *accept* it, or *oppose* it.

2. The Facilitator asks the Parties to identify the one they believe is the most disfavoured; this Party is denoted with P^m and, accordingly, his/her Project Index with V^m. Note that P^m is not necessarily the Party whose index has the lowest value, because it makes no sense to compare the values of the indices of different Parties. Let $\mathcal{A}^m$ be the subset in $\mathcal{A}^c$ of the alternatives that increase the value of V^m with respect to A_{chosen}, i.e.
$$\mathcal{A}^m = \left\{A \in \mathcal{A}^c \colon V^m(A) > V^m(A_{\text{chosen}})\right\}$$

3. The Facilitator asks each Party, except for P^m, if (s)he is willing to lower the value of his/her index with respect to the value obtained in correspondence with A_{chosen}. If the answer is positive, the Facilitator asks the Party to indicate the lowest value (*acceptability threshold*) that (s)he can accept. Otherwise, a stalemate has been reached,

[4] The computing time required for solving a single Optimal Control problem is usually greater than several minutes. For example, in the Verbano Project it varied, according to the alternative, from a minimum of 35 minutes to about 20 hours, on a Pentium III processor, 600 Mhz, with SCSI architecture.

[5] Remember that the Project Index is a value function and so we denote it with the symbol V.

because, since all the alternatives in $\mathcal{A}^c$ are efficient, it is not possible to improve the value of V^m without worsening the value of at least one other index. In the case of stalemate, A_{chosen} is a *reasonable alternative*, in the sense that it is supported by at least one Party and it is efficient. If, instead, at least one of the Parties is willing to accept a lower value of his/her index, identify the subset $\mathcal{A}^{\text{acc}}$ of the alternatives that satisfy the acceptability thresholds. The intersection of the sets $\mathcal{A}^m$ and $\mathcal{A}^{\text{acc}}$ provides the set $\mathcal{A}^{\text{exp}}$ of alternatives to be explored in the search for an alternative that gets a wider agreement than A_{chosen}.

4. If $\mathcal{A}^{\text{exp}}$ contains at least one alternative, proceed with the next step. If, instead, it is empty, the Facilitator proposes that the Parties review the acceptability thresholds: if at least one of them accepts, go back to Step 3; if all refuse, A_{chosen} could be a reasonable alternative. It actually will be one only if, by following all the other pathways from the branching point, no other alternative emerges that gets a wider agreement. In that case the procedure terminates.

5. To search for a reasonable alternative within $\mathcal{A}^{\text{exp}}$, the Facilitator asks the Parties to define a suitable criterion to rank the alternatives of this set. For example, the criterion could be one of the following:
   (a) *maximum value of the index*: the alternatives $A_i \in \mathcal{A}^{\text{exp}}$ are ranked by decreasing values of the index $V^m(A_i)$, i.e. the index of the most disfavoured Party;

   (b) *minimum overall dissatisfaction*: the alternatives $A_i \in \mathcal{A}^{\text{exp}}$ are ranked by decreasing values of the following function

   $$\sum_{k=1}^{p} \alpha_k \big(V^k(A_i) - V^k(A_{\text{chosen}})\big)^-$$

   where $(\cdot)^-$ is an operator that returns the value of the argument when this is negative and zero in the opposite case. The coefficients α_k, $k = 1, \ldots, p$ define the relative weights of the Parties; in practice, they are almost always used as Boolean variables to define the Parties to be considered;

   (c) *compensation among the differences*: the alternatives are ordered by decreasing values of the following function

   $$\sum_{k=1}^{p} \alpha_k \big(V^k(A_i) - V^k(A_{\text{chosen}})\big) \tag{21.2}$$

   where the weights α_k have the same meaning as in criterion (b). Note that, unlike criterion (b), with this criterion the decrease of an index can be compensated by improvement in others.

   Let A_{current} be the first alternative of the ranking obtained.

6. Using a bar chart like the one shown in Figure 21.3, the Facilitator shows the values of the indices of A_{current} compared to those of A_{chosen} and asks each of the Parties to declare whether (s)he *supports*, *accepts* or *opposes* A_{current}. A_{current} is said to get a *wider agreement* than A_{chosen} if all the Parties who support A_{chosen} support A_{current} and at least one of the following case occurs:

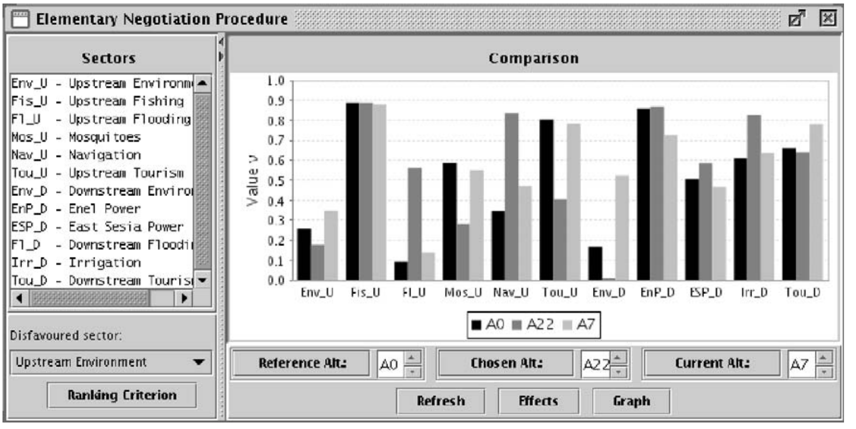


Figure 21.3: An example of bar chart of the sector indices from the Verbano Project.

- one Party who accepts A_{chosen} supports A_{current};
- one Party who opposes A_{chosen} accepts or supports A_{current}.

If A_{current} gets a *wider agreement* than A_{chosen}, replace A_{chosen} with A_{current} and go back to Step 2.

7. Otherwise, if there exists an alternative that follows A_{current} in the ranking, assume it as A_{current} and go back to Step 6. If such an alternative does not exist because the whole ranking has already been examined, A_{chosen} is a *reasonable alternative*, since within $\mathcal{A}^{\text{exp}}$ an alternative that gets a wider agreement has not been found. Then the procedure terminates.

When the negotiation process involves Stakeholders, for each reasonable alternative it is important to remember which Stakeholders *support* it, which of them *accept* it and which *oppose* it, because this information is very useful to the DM(s) in the phase of *Final Decision*.

Often in Step 2 it is not possible to get an agreement among the Parties to designate the most disfavoured Party P^m: if a conflict exists it emerges right now. When this occurs, the procedure must be split into two or more branches (we term this moment *branching* point). Denote the alternative A_{chosen}, i.e. the alternative in correspondence to which the branching occurs, with A_{bra}. Among the Parties proposed as P^m the Facilitator chooses, without appeal, the one whose index has to be improved, and proceeds until a reasonable alternative is found. Then, he goes back to the branching point and takes up the procedure again by considering another of the Parties proposed as P^m and setting $A_{\text{chosen}} = A_{\text{bra}}$. The procedure is repeated again and again until all the Parties proposed as disfavoured at the branching point have been considered.

Note that, by following the different pathways that come off the branching point, different reasonable alternatives can be found. An interesting case is when, by following a branch, the alternative A_{bra} itself emerges as a reasonable alternative. Bear in mind that A_{bra} can be considered reasonable only if it proves to be a reasonable alternative also by following all the other branches. In fact, if from at least one of them a different alternative emerges, by construction, this alternative obtains a wider agreement than A_{bra} and, as a consequence,

A_{bra} cannot be reasonable. Therefore, in correspondence with a branching point, the above Step 4 must be substituted by the following:

4. If $\mathcal{A}^{\text{exp}}$ contains at least one alternative, proceed with the next step. If, instead, it is empty, the Facilitator proposes that the Parties review the acceptability thresholds: if at least one of them accepts, go back to Step 3; if all refuse, A_{chosen} could be a reasonable alternative. It actually will be a reasonable alternative only if, by following all the other pathways from the branching point, no other alternative emerges that gets a wider agreement.

One after the other, all the branching points that are encountered are examined in this way.

To complete the description of the ENP, we must define a criterion for the choice of alternative A_{chosen} in Step 1. Remember that no discrimination among the Parties is allowed in the *Comparison* phase and that it is essential that each Party support at least one of the reasonable alternatives and that each reasonable alternative be supported by at least one Party. To guarantee that this occurs, it is sufficient that the ENP is repeated p times (i.e. as many times as there are Parties) and that, each time, a different Party selects the alternative A_{chosen} to start the procedure. The condition of *wider agreement* (Step 6) ensures that the Party that chooses A_{chosen} at Step 1 will support all the reasonable alternatives that are obtained from it.

21.5.1.1 The least-bad alternatives

When the Group consists of Stakeholders the final choice is not theirs, but of the DM(s), who can choose among the reasonable alternatives that have emerged from negotiations, taking account of the Stakeholders' positions. However, the set of reasonable alternatives is often composed of alternatives that have the support of some Stakeholders and are opposed by others, while alternatives that are accepted by many are rare. As a consequence, it is difficult for the DM(s) to choose among these alternatives, because her (their) choices will be very satisfying to some while very dissatisfying to others. The DM(s) would thus appreciate it if, besides the reasonable alternatives, other alternatives were submitted to her (them) which the dissatisfied Stakeholders are willing to abide by, even if reluctantly. This is why it is useful, at the end of the ENP, to use the following procedure to identify a set of alternatives that we will call the *least-bad alternatives*:

(a) Divide the set of reasonable alternatives into groups of alternatives characterized by the same types of actions. Then take the following steps for each of the groups for which there is at least one Stakeholder that opposes all the alternatives in the group(om).

(b) Ask each of the Stakeholders that support or accept at least one of the alternatives in the group to what extent (s)he is willing to diminish the value of his/her index, with respect to the worst alternative (for him/her), before passing to the opposition. In this way a set of acceptability thresholds is obtained.

(c) Determine the set $\mathcal{A}^s$ of the alternatives in $\mathcal{A}^c$ (the set of alternatives to be compared) which satisfy the acceptability thresholds and which are characterized by the same type of actions as the group of alternatives being examined. If the set $\mathcal{A}^s$ is empty, it must be enlarged by neglecting in turn one or more of the acceptability thresholds.

(d) Each of the Stakeholders who oppose all the alternatives in the group is asked to indicate the alternatives that (s)he prefers in $\mathcal{A}^s$. These will be catalogued as *least-bad* for him/her. If the Stakeholder refuses all the alternatives in $\mathcal{A}^s$, this set must be enlarged by neglecting in turn one or more of the acceptability thresholds.

(e) For each least-bad alternative obtained in this way, each of the other Stakeholders is asked to state how (s)he positions him/herself with respect to it, namely whether (s)he supports, accepts, suffers or opposes it.

By construction, the least-bad alternatives are characterized by less opposition than the reasonable alternatives identified with the ENP, since they were identified taking into consideration the preferences of the opposers as much as possible. Thus, it is advisable that they be included in the set of alternatives submitted to the *Final Decision* phase.

21.6 Voting on rankings

Finally, let us look at the procedure used when, for the reasons explained in Section 21.3, the DMs have to vote for the ranking of the alternatives. Note that, by voting, an alternative is concretely chosen but it is unlikely that the conflict is resolved: the Parties remain in their positions and the opposition can emerge later, in an active way. Voting is not negotiating and it does not stimulate a social learning process. For this reason it must be adopted only when every other possibility has already been precluded (see Figure 21.1).

If DMs turn to voting on rankings, it is because every attempt to negotiate has failed: this means that also the negotiations on weights have already been tried and so each DM has expressed a vector of weights $\boldsymbol{\gamma}^k$ that defines her preference structure among the sector indices. Just as for negotiations on weights, the problem is to derive from these vectors a single vector $\boldsymbol{\gamma}$ so that the Project Index can be defined, but this result must be achieved without negotiating. The simplest way is to attribute a weight δ_k to each DM and define $\boldsymbol{\gamma}$ as the weighted sum[6] of the vectors $\boldsymbol{\gamma}^k$, i.e.

$$\boldsymbol{\gamma} = \sum_{k=1}^{p} \delta_k \boldsymbol{\gamma}^k$$

The problem is how to attribute the weights δ_k to each DM: who can express them?

If there were a higher authority, the attribution of these weights would be her job, but if the Project involves many DMs, according to our definition, this authority does not exist. Thus one can turn to a *voting system*: each DM (j) attributes a weight β_{kj} to each one (k) of the others and, in some methods, even to herself. These weights are then combined with those expressed by the other DMs, to obtain the desired weights δ_k (with $k = 1, \ldots, p$). The weights β_{kj} can be combined with two different methods: the *average method* and the *eigenvector method*.

With the *average method*, the weight δ_k is obtained as the average of the values β_{kj} expressed by the DMs (including any weight β_{kk} that the kth DM attributed to herself), i.e.

$$\delta_k = \sum_{j=1}^{p} \frac{\beta_{kj}}{p}$$

[6] This implies that the weights are chosen so that their sum is equal to one.

From a logical standpoint, however, the method is incoherent because the operation of averaging values implies the assumption that all DMs have equal relevance, while the different weights are being estimated for the very reason that this is not believed to be true.

With the *eigenvector method* the weights δ_k are assumed to be equal to the components of the dominant eigenvector[7] of the square matrix of elements β_{kj}. The method is not affected by the logical contradiction that affects the previous one because, in the extreme case where all the DMs attribute a zero weight to one of them, the vector of weights expressed by this latter would not influence the results. Nonetheless, the method seems to generate another paradox, but fortunately it is only apparent and can easily be eliminated. Consider the following example, taken from Colorni et al. (2001b): 4 DMs P^1, P^2, P^3 and P^4 are asked to attribute a weight to each of the others, but not to herself. The weights β_{kj} the DMs expressed are shown in the following matrix

Voted DM k \ Voting DM j	P^1	P^2	P^3	P^4
P^1	0	0	0	0
P^2	0	0	0	1
P^3	0	0	0	0
P^4	1	1	1	0

in which the elements along the diagonal are zero because the DMs were asked not to attribute a weight to themselves. By examining the columns of the matrix, we see that DMs P^1, P^2 and P^3 believe that only P^4 is relevant, while P^4 believes that only P^2 is relevant. It follows that P^1 and P^3 are believed to be insignificant and, in fact, the rows that correspond to them contain only zeros. One would therefore expect that the weights δ_1 and δ_3 would be zero and that δ_4 would be higher than δ_2, given that P^4 is preferred by three of the four DMs. On the contrary, the dominant eigenvector is the following: $\boldsymbol{\delta} = |0.0, 0.5, 0.0, 0.5|$. As expected, P^1 and P^3 are assigned a zero weight; but P^2 and P^4 have equal importance. This seems paradoxical at first, but a moment's reflection shows it is not: given that P^1 and P^3 are insignificant, the relevance that they attributed to P^4 does not count and so P^1 and P^4 received an equal number of preferences: this is why δ_4 and δ_2 are equal.

Nevertheless, even if the result is logically understandable, the apparent paradox is disturbing and, to avoid it, the weights β_{kj} are usually required to be strictly positive.

In conclusion, one can demonstrate that this method of deriving the Project Index (which is in practice an ordering rule) is democratic and does not contradict Arrow's Theorem, since it adopts a cardinal scale.

21.7 Mediation suggestions

Often the Facilitator is asked to suggest a solution, i.e. propose a compromise. Accepting could be risky for him, not just because it would transform the negotiations into a mediation (see Appendix A10), but above all because his response might compromise his credibility as a neutral party, if any of the Parties believed, either rightly or wrongly, that his proposal was partisan. However, often the Facilitator cannot refuse and thus the problem of identifying an

[7] The eigenvector associated to the maximum eigenvalue of the matrix.

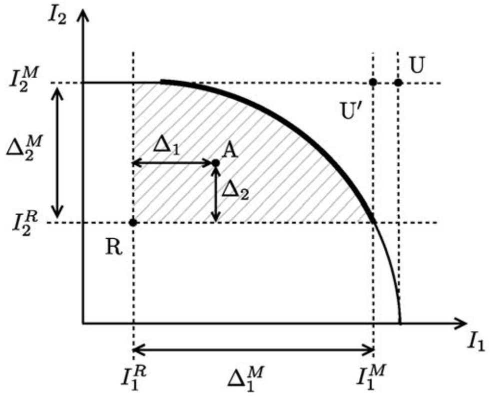


Figure 21.4: The negotiation space between two Parties P^1 and P^2 and the elements that characterize it: the Feasible Region (hatched area) and the Pareto Frontier $\mathcal{F}$ (bold line).

impartial compromise, acceptable to all the Parties, arises. But what does 'impartial' mean? Is there a way to identify a 'fair' proposal? As we will see in this section, there are not straightforward answers to these questions, but different criteria can be used to find a point of compromise.

Before we present these criteria, let us briefly examine the elements that make up the working context. We will refer to Figure 21.4, which represents the negotiation space between two Parties P^1 and P^2 in the joint choice of an alternative. The axes represent the two Parties' indices I_1 and I_2. In the figure the following elements are also shown:

- the *reservation* point (R) defined by the two Parties' *reservation values*(I_1^R, I_2^R), which are the minimum index values that each of them is willing to accept. The reservation value of a Party is at least equal to the value of his/her index for alternative $A0$, i.e. the satisfaction (s)he gets from the status quo; it is higher when the Party has a BATNA that is better than $A0$;

- the *Feasible Region*, which is the image of the set $\mathcal{R}$ of alternatives that provide performances better than reservation values;

- the *Pareto Frontier* $\mathcal{F}$ which is the image of the set $\mathcal{P}$ of the efficient alternatives within $\mathcal{R}$, i.e. the image of the Pareto-Efficient Decision Set (see page 398);

- the *maximum feasible values*, I_1^M and I_2^M, that each Party can aspire to, given the alternatives in $\mathcal{R}$;

- the *potentials* Δ_1^M and Δ_2^M, defined as the difference between the maximum feasible values and the reservation values;

- the point U′ with coordinates (I_1^M, I_2^M), called the *Utopia* point[8] or *Bliss point* of $\mathcal{R}$;

[8] Note that this definition of the *Utopia* point does not coincide with the one given in Section 18.2 (which individuates the U point in Figure 21.4), because different feasible regions are considered in the two cases: in that section, the region was defined only by technical limits, here it is limited further by the introduction of reservation values.

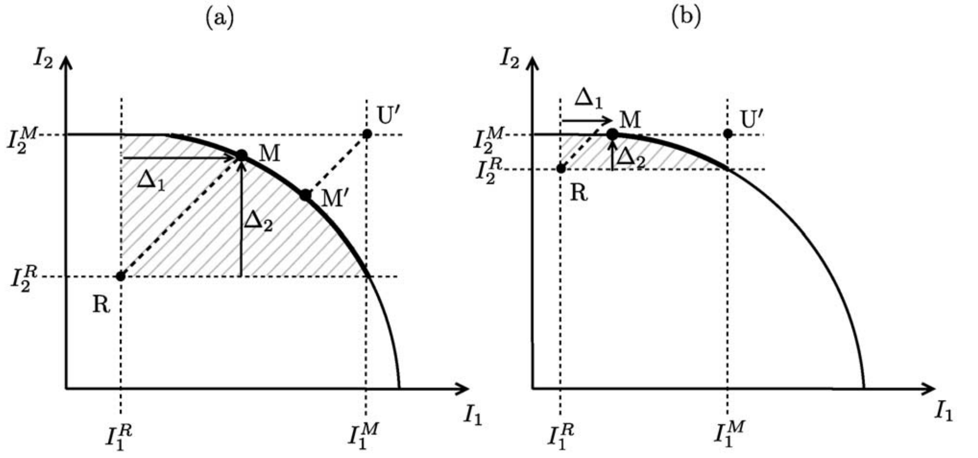


Figure 21.5: The *Maimonides* point, indicated with M, in the two cases described in the text.

- a generic alternative A in the Feasible Region, which produces the performance $(I_1(A), I_2(A))$;
- the *excesses* $\Delta_1(A)$ and $\Delta_2(A)$ of the Parties' indices, equal to $(I_1(A) - I_1^R)$ and $(I_2(A) - I_2^R)$ respectively, which they would obtain if they agreed upon the alternative A.

The ratio $\Delta_i(A)/\Delta_i^M$ between excess and potential is termed the *Proportion of Potential* (POP).

With these elements we can introduce the alternatives that can be proposed as 'fair' compromises. Obviously, they must be efficient, that is they must belong to the Pareto Frontier. For simplicity's sake, we will refer to a case in which there are only two Parties; the generalization to the case with p Parties is left to the reader except when it is not straightforward.

21.7.1 Maimonides point

The alternative that corresponds to the concept of impartiality in the simplest and most intuitive way is the one that attributes, *in principle*, the same excess to all Parties. The point that corresponds to it in the negotiation plane is called the *Maimonides* point by Raiffa et al. (2002), from the name of the Israelite philosopher Maimonides (1135–1204), who proposed the homonymous rule for solving disputes between creditors in cases of bankruptcy. An excellent discussion of this rule can be found in Aumann[9] and Maschler (1985). The Maimonides point can be determined with the following procedure: identify the point at which the bisector of the first quadrant centred on R intercepts the boundary of the Feasible Region. If that point is efficient, it is the Maimonides point (see Figure 21.5a); if it is not (see Figure 21.5b), the point so determined is semi-dominated and the index of one of the two Parties (I_2 in the figure) cannot be increased any further. Then, move along the

[9]Nobel prizewinner for Economics in 2005.

boundary in the direction that increases the other Party's index,[10] until an efficient point is identified: this is the Maimonides point. In this case, the excesses of the two Parties are not equal (this is why in the initial definition the expression 'in principle' appears), but even so, P^2 has nothing to object about if P^1 obtains more, given that his/her index has reached its maximum possible value.[11]

The Maimonides point is also the point on the Pareto Frontier $\mathcal{F}$ which maximizes the minimum excess, i.e.

$$\max_{A \in \mathcal{P}} \left\{ \min_{k=1}^{p} \left\{ \Delta_k(A) \right\} \right\}.$$

In some problems the Parties look at the result of the negotiation in terms of gain (excess) with respect to the reservation value, in others, in terms of loss with respect to the Utopia. In the second case the Maimonides rule requires that all the Parties, in principle, are attributed with the same loss. The point identified in this way (point M′ in Figure 21.5a) generally does not coincide with the one obtained by imposing that the excess be equal. It is interesting to note that when the excesses are considered, the Maimonides point is such that the POP obtained is greater for the Party that has the least potential. The opposite occurs when losses are considered.

21.7.2 Equipop point

The Maimonides rule assumes 'equality of the excesses' as the criterion of impartiality, but this assumption may not be acceptable to everyone. Some may think that the attribution of the same POP to all Parties would be fairer. The point in the negotiation space that is identified by this new rule is called the *Equipop* point. It can be obtained graphically by intersecting the Pareto Frontier with the segment that joins the point R with the *Utopia* point U′ (see Figure 21.6a).

There is a special case in which this graphic procedure does not seem to provide an *Equipop* point. This case is exemplified in Figure 21.6b, in which point R is a semi-dominated point and so the Utopia point U′ belongs to the Feasible Region (in particular, to the Pareto Frontier). In this case, according to the procedure, U′ is the compromise point, which is reasonable. Is this an *Equipop* point? It may not seem so, given that P^1 gets a positive excess while P^2 gets no excess. Actually, once the characteristics of the Feasible Region are known, P^1 certainly requests that point R be moved into U′ and P^2 must accept since in both cases his/her index has the same value.[12] With these new reservation values, the point U′ is *Equipop*, because it attributes a zero POP to both the Parties.

Unlike the Maimonides point, the *Equipop* point stays the same regardless of whether (proportional) gains or (proportional) losses are considered. This can be understood by observing Figure 21.6a.

21.7.3 Balanced-Increment point

The search for a compromise can also be developed by steps. The Mediator agrees with the Parties upon the direction in which to move in the negotiation space, starting from the R

[10] If there are more than two Parties, one must move in a direction that increases all of the other indices equally.

[11] This is based on the hypothesis that Party P^2 is not more satisfied if Party P^1 obtains less, i.e. that the satisfaction of one Party does not decrease as the satisfaction of the other increases. In other words it is based on the hypothesis that envy is not present. This hypothesis implicitly underlies the very concept of Pareto dominance (see page 398).

[12] As we already noted, it is assumed that envy does not influence the Parties' satisfaction.

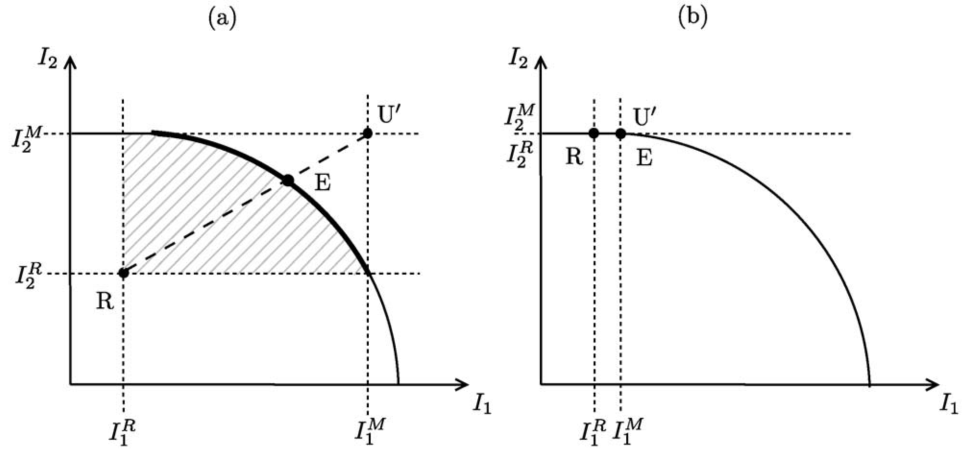


Figure 21.6: The *Equipop* point E in the two cases described in the text.

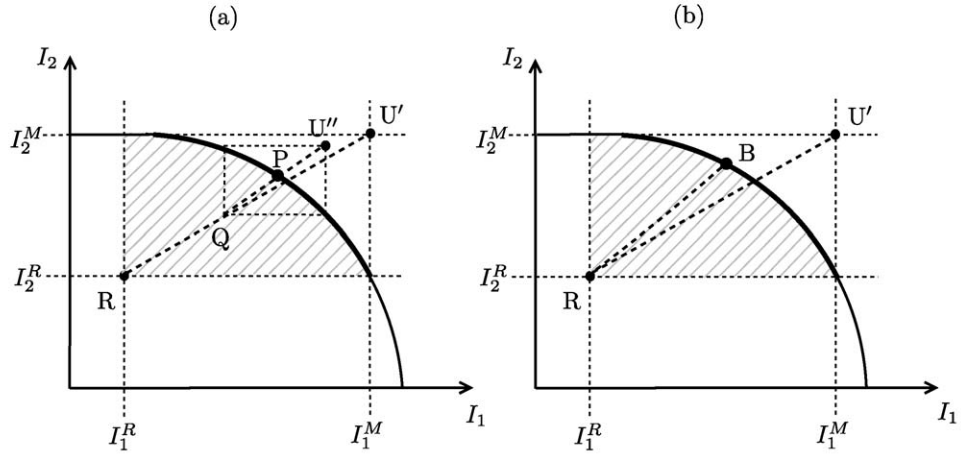


Figure 21.7: A compromise point (P) reached with two finite steps (a); the *Balanced-Increment* point (B) reached with infinitesimal steps (b).

point. One moves in that direction until both Parties are happy with their respective excesses. If one of the two Parties comes to a point where (s)he no longer accepts the direction that was taken, the direction is discussed again. For instance, in Figure 21.7a the result of the following procedure is shown: at the outset the Parties agreed to move in the direction that kept their POPs equal. Once they had reached a provisional compromise at point Q, they redefined the direction of the search: again with the criterion of equality of the POPs, but by assuming the point Q as the new reservation point and determining a new Utopia point U″ with respect to it. With these assumptions, the direction that kept their POPs equal changed; this new direction was followed until the intersection with the Pareto Frontier and the final compromise was found in point P. In general, the choice of the first direction is not critical: when one is far from the Frontier, all the Parties have a wide margin for improving their indices and they easily come to an agreement about which way to move; when they are close to the Frontier, the conflict emerges and the space for manoeuvring is reduced.

The procedure that was just illustrated can be automated in the following way. Take the direction in which the POP is equal for both Parties and move along it by one infinitesimal step. Adopt the point that is reached as the new reservation point R, determine the corresponding Utopia point U′, and thus the new *equipop* direction, and move by an infinitesimal step along this direction. Iterate until the Pareto Frontier is intersected. The point so identified (see Figure 21.7b) is called the *Balanced-Increment* point (Raiffa, 1953).

21.7.4 Nash point

Probably the most famous criterion for determining a compromise point is the one proposed by Nash[13] (1950). We will show its philosophy in a simple case, with only two Parties, before giving its general definition.

Consider a generic alternative A, to which the point $(I_1(A), I_2(A))$ and the excesses $\Delta_1(A)$ and $\Delta_2(A)$ correspond on the negotiation plane (see Figure 21.8a). In the following we will omit the argument (A) for brevity's sake. Let dI_i be an infinitesimal variation of the index I_i and dI_i/Δ_i the corresponding proportional variation of Δ_i (with respect to its value). This latter is a gain or a loss according to the sign of dI_i.

According to the Nash criterion, the move from point A, with coordinates (Δ_1, Δ_2), to point B, with coordinates $(\Delta_1 + dI_1, \Delta_2 + dI_2)$, is acceptable if one Party's proportional gain is greater than the other's proportional loss (as an absolute value). For example, if P^1 were the Party that is advantaged by the move, the move from A to B is acceptable if $dI_1/\Delta_1 > -dI_2/\Delta_2$. Therefore, according to this criterion, the point of equilibrium at which negotiations end, the *Nash* point, is the point in the Feasible Region that maximizes the product of the excesses $[\Delta_1 \cdot \Delta_2]$.[14]

The Nash point is thus the point at which one of the equilateral hyperboles inscribed in the first quadrant centred on R is tangential to the Pareto Frontier (see Figure 21.8a).

When there are p Parties the Nash criterion is defined as follows: a move is acceptable if the sum of the proportional gains is greater than the sum of the losses (as absolute values). The Nash point still corresponds to the point that maximizes the product of the excesses.

Nash demonstrated that this point satisfies the Axiom of *independence from irrelevant alternatives* (see Section 20.1) and on this property he founded its legitimacy. Here, we will

[13] Nobel prizewinner for Economics in 1994.

[14] The proof is easy. At the Nash point the following condition must be satisfied

$$dI_1/\Delta_1 = -dI_2/\Delta_2$$

which is equivalent to requiring that

$$\Delta_2 \cdot dI_1 + \Delta_2 \cdot dI_2 = 0$$

or

$$d(\Delta_1 \cdot \Delta_2) = 0$$

The latter condition states that the product $\Delta_1 \cdot \Delta_2$ must be at its maximum at the Nash point.

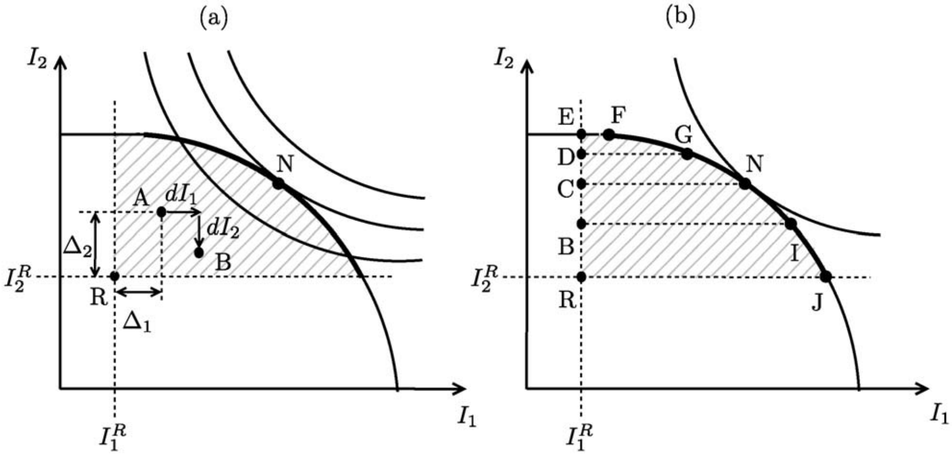


Figure 21.8: The *Nash* point, indicated with N (a); the Axiom of *independence of irrelevant alternatives* (b).

not give the proof but will be satisfied with an intuitive idea of it. Observe Figure 21.8b. If, instead of the set REFJ, the Feasible Region were the set of vertices RDGJ, the Nash point (N) would remain the same. This would continue to be true even if we reduced the region to RCNJ. By reducing it even further, to RBIJ for instance, the Nash point would move to the upper right vertex of the new region (point I). This behaviour respects the Axiom of *independence from irrelevant alternatives* which can be stated as: if two different regions $\mathcal{R}$ and $\mathcal{S}$ are considered, with $\mathcal{S} \subset \mathcal{R}$, and the optimal solution in $\mathcal{R}$ falls in $\mathcal{S}$, then it is also the optimal solution in $\mathcal{S}$; in other words, if the optimal solution in $\mathcal{R}$ is not the optimal solution in $\mathcal{S}$, then it must belong to the set $\mathcal{R} \setminus \mathcal{S}$ (for a more intuitive definition see page 381).

21.7.5 Solutions from Game Theory

Thus far, we have considered only intuitive rules for selecting a compromise point. It is possible, however, to reformulate the problem into a coalition problem,[15] which can be tackled with Game Theory. This makes it possible to use solutions proposed by Game Theory, such as the Kernel, the Nucleolus, and Shapley's value (Osborne and Rubinstein, 1994). However, the mathematics required to present these solutions exceeds the limits of this work and so the interested reader should refer to the literature.

All the compromise points that we have defined are sensitive to variations in the position of the point R. In particular, as one Party's reservation value increases, the index value that (s)he can obtain also increases (e.g. observe in Figure 21.9 how the *Maimonides* and *Equipop* points move due to a variation in I_2^R). If, for example, point R represents the present situation, the Party which is most advantaged by the status quo will be the most advantaged in the negotiation process. If the reservation values correspond to the BATNA of each Party, the one that has the best BATNA will be favoured. In general, the 'negotiating power' of the Parties is not equal: it depends on their reservation values, or in other words

[15] This requires formalizing the dynamics with which negotiations are carried out and/or the rule with which it can be established whether a compromise has been reached.

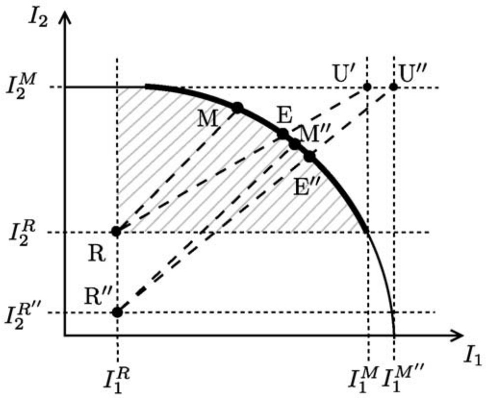


Figure 21.9: The movement of the *Equipop* point, from E to E″, and the *Maimonides* point, from M to M″, as the Reservation Value point varies from R to R″.

on 'the way they present themselves at the negotiating table'. This is why the compromise that is found might be considered not at all fair. To overcome this problem, the procedures suggested in this section can still be used, provided that point R is set at the origin of the axes, in order to start from scratch. In this way one avoids the possibility that the Parties declare reservation values higher than the ones they actually have, in order to obtain a more advantageous solution (*strategic behaviour*). However, the risk of making the original reservation values equal to zero is that, if the compromise that is found attributes a Party with an index value inferior to his/her reservation value and if that value corresponds to a BATNA that can really be implemented, it is likely that that Party will refuse the agreement and abandon the negotiations.

The position of the *Maimonides*, *Equipop* and *Balanced-Increment* points depends not only on the position of point R, but also on the position of the Utopia point and on the overall form of the Feasible Region. In particular, the *Balanced-Increment* point is very sensitive to any variation of that form. On the other hand, the position of the *Nash* point is only sensitive to local properties of the Frontier, so that if a part of the Feasible Region which does not contain it is eliminated, it remains unchanged (see Figure 21.8b). This fact can be difficult for the Parties to understand and may lead to mistrust of a compromise point obtained with the *Nash* criterion. Nonetheless, from a theoretical point of view, this criterion is probably the most rigorous. The *Equipop* point has the simplest definition and thus it is the most understandable to the Parties; this is why often, once the Parties have identified it, they are not likely to exchange it for another.

Lastly, it is important to note that, when the Pareto Frontier is symmetrical to the bisector of the first quadrant centred on R, all the criteria identify the same point: the intersection between the bisector and the Frontier. On the other hand, the more asymmetrical the Frontier, the greater the difference between the different points and so the more critical the choice of the criterion. Unfortunately, theory does not help us to come to a conclusion about which point would represent the best compromise and so the choice is referred to the Mediator, who must decide by considering the specific context in which he is working.

21.8 Organizing the meetings

We have already mentioned that the negotiation process is not the only activity that characterizes the *Comparison* and *Final Decision* phases, even though it is central to them. Given that it aims at consensus building and social learning, it is essential that, before beginning negotiations, each Party knows the reasons behind the viewpoints and the preference structures of the others. This is why the Facilitator must organize activities for enhancing the creation of a knowledge base that is shared by the Parties. These activities are carried out at the beginning of the process and must be organized in such a way as to leave a lot of space for telling their story, so that the Parties can freely express their point of view, make their positions explicit, verbalize their fears, and, feeling that they are being listened to, release the emotions and tensions accumulated, giving the other Parties the elements to understand their points of view. These activities thus play both an informative role and a psychological role, the one played by 'storytelling' in analytical group therapy. A practical example of how one may proceed is presented in Sections 12.1–12.3 of PRACTICE.

The way in which the Facilitator presents himself to the Group is another determining factor for the success of the negotiation process and this is why it is the object of study for a corpus of methodologies called ADR (*Alternative Dispute Resolution* or *Appropriate Dispute Resolution*), which makes use of contributions from Decision Making Theory, Psychology and Sociology; some mention of all this is made in Appendix A10.

Lastly, it is important to underline that negotiations, just like the activities of sharing and disseminating information, are greatly facilitated by having an information tool, an MODSS, which, besides providing the *support* for these activities, also guarantees the *transparency* of the results and the *repeatability* of the process with which they were obtained (Thiessen et al., 1992; Kronaveter and Shamir, 2006). We will discuss these and other features that are essential to a good MODSS in Chapter 24.

Chapter 22

Mitigation and Compensation

FC, EL and RSS

The search for mitigation and compensation measures (i.e. actions) is the last possibility to be explored in the attempt to identify a *best compromise alternative* that obtains the agreement of all the Parties involved. One explores this possibility after having identified the *reasonable alternatives*, i.e. the alternatives that are supported by at least one Party and, generally, obtain agreement from many.

Given the highly practical nature of this phase, we will simply clarify the difference between mitigation and compensation and illustrate this difference with simple examples, which will sometimes be taken from contexts other than the management of water resources. A detailed and well developed example is presented in Chapter 14 of PRACTICE.

22.1 Mitigation measures

Every mitigation measure is specifically linked to a particular reasonable alternative, and it is an action, or a mix of actions, specifically aimed at modifying those effects of the alternative that make it unsatisfactory for the Parties that oppose it.[1] Mitigating the effects of an alternative means, in fact, modifying some of its 'minor' characteristics, with the aim of increasing the satisfaction of the Parties that are unsatisfied.

To make the definition clearer, consider a Project aimed at satisfying the demand for transport between two cities. Suppose that at the end of the *Comparison* phase, two compromise alternatives have emerged: a high-speed railway line ($A1$) and a motorway ($A2$). The two alternatives were evaluated with respect to the following criteria:

(1) effectiveness in satisfying the transport demand,

(2) cost of implementation,

(3) atmospheric pollution,

(4) noise pollution,

(5) safety.

[1] When the reasonable alternative has been identified through the ENP (page 465), the reasons why one or more Parties are opposed to it are known.

They proved to be equivalent from the viewpoint of the first two criteria, while $A1$ (railway) is preferred over $A2$ (motorway) with respect to the third and the fifth criteria, but is opposed by those who live close to the proposed railway because of the high levels of noise pollution that it would produce. The preference of these people thus goes to alternative $A2$. Therefore, in order to get agreement from everyone about the railway line, its acoustic impact must be reduced and thus the introduction of anti-noise barriers is the mitigation measure to be considered. The new action 'anti-noise barriers' must be included in alternative $A1$, it must be suitably sized and the effects of the alternative so-obtained must be estimated; then the alternative must be evaluated and compared with the current reasonable alternatives to verify if it does effectively obtain a broader agreement.

22.2 Compensation measures

Just like mitigation measures, compensation measures aim at increasing the satisfaction of unsatisfied Parties but, instead of reducing the effects that they find disagreeable, these measures try to raise satisfaction on aspects that do not belong to the specific problem at hand. Compensation is, in fact, based on the idea that a Party may be motivated to accept an alternative that thwarts him/her as long as (s)he is compensated on another level, i.e. by conceding something to him/her or reducing other negative effects that afflict him/her.

Think, for example, of the decision-making process for localizing a waste dump: having a dump in one's own territory certainly is not pleasant, and so, whichever place is chosen for it, the local government will be opposed. This is a typical manifestation of the NIMBY syndrome (*Not In My BackYard*), a term that is used to describe the opposition of a community to the placing, in their local area, of facilities considered as unpleasant (such as dumps, incinerators and power plants). To overcome the NIMBY syndrome, compensation measures can be used: without modifying the negative effects caused by the dump in any way, one can propose to the local government that a park be created for children, or a swimming pool be built, or that degraded buildings be restored, in exchange for the approval for creating the dump. It is a *do ut des* exchange: after an agreement has been reached, both the unpleasant intervention and the agreed compensating interventions will be carried out.

Compensation is not as efficient as mitigation, especially from environmental viewpoint: calling the Parties' attention to the compensation, instead of the proposed alternative, reduces the awareness of the negative effects that this alternative generates. Thereby, instead of pushing for an efficient waste dump, well designed and as respectful as possible of the environment and the landscape in which it is inserted (a social objective which a mayor should pursue), the mayor involved might concentrate only on negotiating the compensation, neglecting the potential future negative effects produced by the dump.

In some cases relying on compensation measures can even increase the negative effects: this is the case of *subsidies*. To understand the reason why, consider an example in the field of water pollution. Let us suppose that the introduction of a law increasing the severity of quality standards for discharges, is being evaluated. Since such a law causes widespread discontent among the companies, it is coupled with a compensation measure: an annual subsidy granted to the companies. Since this is intended as a *partial reimbursement* of the expenses of purification, it will be given only to the companies that install a water treatment system or modify the one they already have to observe the new standard. In this way, because of the subsidy, companies are not motivated to respect the standard by acting on the production cycle. The compensation measure, therefore, first favours the creation of negative effects (production of pollution) and then reduces them at a future time, rather than eradicating them.

A similar effect can be observed in the context of water resource management: think, for example, of a regulated lake in which flood events seem to be getting more frequent as time passes, perhaps because of climate changes that are underway. Since any modification of the regulation policy aimed at reducing floods would produce negative effects in the irrigation districts fed by the lake, the Government decides to resolve the conflict between the shoreline communities and the farmers by reducing the taxation on the first group. In the long-term, the effect of the measure is perverse because, instead of allowing a spontaneous reduction of the pressure for building on the lake shores (as a consequence of the increased risk of flooding) the subsidy works to maintain or even increase this pressure. This is due, in fact, to the perception that a potential buyer of a building area on the shores has of the relationship between advantages and disadvantages: if the buyer underestimates the risk of flooding, for example because for several years there have been no floods, and (s)he is attracted by the tax reduction, the pressure increases. The damage that will be produced in the next flooding will be greater than the damage that would have occurred in the absence of the subsidy; in addition, the public takings will have been reduced. From a global perspective, however, these effects are partly compensated by the fact that the agricultural supply, and as a consequence the harvest, will not have diminished since the regulation was not modified. Nevertheless, as the reader can intuit, by reasoning only from the viewpoint of reducing flood damage, the best choice is the opposite one: to give tax concessions to those who decide to move home or activities from the lake shores.

22.3 Mitigation and Compensation in the decision-making process

From the first chapter of this book, we have highlighted that a good, participatory decision-making process cannot be carried out in a sequential way but is characterized by recursions that bring the participants back to the same phases several times. This is not because mistakes have been made, but because the understanding of the problem grows as the process goes on, and, with the increased understanding, new ways of approaching and solving the problem are discovered.

To understand this point well, consider an example: a Project in which several Stakeholders are involved. Based on the considerations that they express through interviews, the criteria and the indicators (Phase 2) and the value functions (Phase 6) are identified. Let us suppose that in Phase 7 negotiations on weights method are adopted and that they result in a wide, but not unanimous, agreement on a value $\bar{\boldsymbol{\gamma}}$ of the vector of weights. This value, through equation (21.1), defines a ranking O of the alternatives. The agreement is not unanimous because the alternatives in the first positions of the ranking O produce very negative effects for the kth sector, and thus the Stakeholders interested in it propose a vector of weights different from $\bar{\boldsymbol{\gamma}}$. This may happen because the weight $\bar{\gamma}_k$ is low and so alternatives with poor performances on the kth sector can reach the first positions in the ranking O. To prevent these alternatives from being selected, the Stakeholders of the kth sector, who have not the power to increase the weight $\bar{\gamma}_k$, propose the introduction of an *acceptability threshold* on the value of their index. Posing an acceptability threshold means posing a veto on the choice of the alternatives that do not exceed it. This implies that the alternatives that have poor performances for the kth sector cannot be chosen. However, in such conditions, the choice would fall on the first alternative in O that gets over the acceptability threshold, but this alternative might provide worse performances than those that are obtainable with the alternatives discarded for the very sectors to which $\bar{\boldsymbol{\gamma}}$ assigns the higher weight. Therefore, the Stakeholders of these sectors reject the proposal of an acceptability threshold and the negotiation process is stalled.

The solution lies not in excluding the first alternatives by introducing thresholds, but in trying to *modify the alternatives* in such a way that their performances are bearable, or still better acceptable, even for the Stakeholders of the kth sector. This means that mitigation measures must be designed for the kth sector, in a way not to reduce the performances of the other sectors.

To identify these measures, it is necessary to go back to re-examine the possible actions in light of the information that the negotiations produced. Unlike Phase 1 (*Defining Actions*), in which the role of the Analyst has to be limited to the coordination of the discussion, in the phase of *Mitigation and Compensation* her role must be as active as possible: she should not limit herself to taking note of the Stakeholders' ideas but, with the strong multi-facetted understanding of the problem that she has gained over the course of the decision-making process, she should propose ad hoc measures.

Once the mitigation actions have been identified, they must be sized in the context of each alternative to which they will be associated (Phase 4), the effects of the new alternatives obtained in this way must be estimated (Phase 5), the alternatives evaluated (Phase 6) and brought to the negotiation table (Phase 7). In Phase 6 the design and evaluation indices, as well as the value functions, both partial and global, do not have to be redefined. It is not necessarily sufficient to run through this cycle of phases only once: sometimes it might be necessary to go through these phases more than once before the Analyst and the Stakeholders can conclude that it is not possible to enlarge the agreement any further.

Only when this conviction is shared and the agreement is not unanimous is it opportune to examine possible compensation measures.

Chapter 23

How to cope with uncertainty: recapitulation

FP

In the previous chapters we have described in detail a decision-making procedure, the PIP procedure, for the planning and management of water resources. In this chapter we will go through that procedure once again, in order to verify if and where uncertainty can arise and affect the decision-making process. The reasoning will be conducted in general terms, with regards to either Single or Multi-Objective Problems and Single or Multi-Decision Makers. This is why the first section of this chapter is devoted to a general classification of decision-making problems; then, in Section 23.2, we will see how potential sources of uncertainty can be classified and how different type of uncertainty be described; finally, in the last section, we will consider how uncertainty can be handled in decision-making problems.

Given the breadth and the complexity of the subject, our survey will be far from being exhaustive. The aim is just to give the reader a global view of the main research directions in the field and, more importantly, awareness of the problem. The role and impact of uncertainty in environmental assessment and decision-making is well recognized in the literature. However, in real applications the most commonly used approach is that of trying to 'squeeze out' uncertainty as much as possible from the decision-making process, and carry it out as if the assumption of deterministic environment were satisfied.

23.1 Decision-making problems: a classification

In very broad terms, a decision-making problem can be formulated as the problem of choosing, among a number of options, the 'most preferable' one. The elements that characterize a decision-making problem are:

- the number p of subjects that make up the decision, accordingly called *Decision Makers* (DM);
- the number q of *criteria* upon which the decision is made (also referred to as *objectives*);

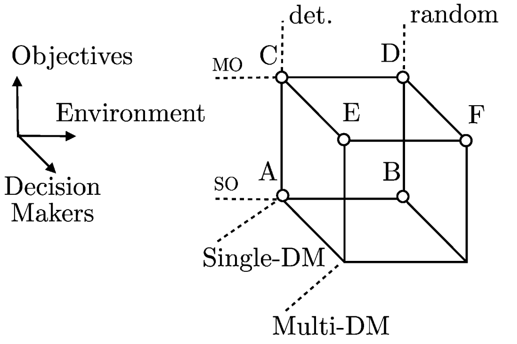


Figure 23.1: The space of the decision-making problems. Marked vertices are commented in the text.

- the type of *environment* in which the decision is made, i.e. *deterministic* if all information about the consequences of each possible alternative is deterministically known, *random* otherwise.

Bear in mind that the above definition is very general and a variety of problems can be reduced to it. In the context of water resources planning and management, we have seen that the options under study are suitable combinations of actions (interventions), designed to meet the goals of sustainable development and environmental protection. We called *alternative* such a combination of interventions and thus the decision-making problem consists of choosing the most preferable alternative.[1]

However, other problems can be formulated in the same way. For example, identifying a model can be viewed as a decision-making problem, provided that the structure of the model (the *meta-model*, see Section 4.9) has been chosen. In fact, once the meta-model has been chosen, the identification problem boils down to a parameter estimation problem, which can be seen as a decision-making problem where the options under study are the values to be attributed to the parameters of the model, and the criterion according to which the decision is made is the minimization of a given performance measure (e.g., the sum of the squared errors).

Decision-making problems can be classified as follows. If finding a solution is the task of one DM ($p = 1$), the decision-making problem is called a *Single*-DM *Problem*; otherwise ($p > 1$) it is called a *Multi*-DM *Problem*. Similarly, if the alternatives are judged by considering only one criterion ($q = 1$) the problem is called a *Single-Objective* (SO) *Problem*; otherwise ($q > 1$) it is known as a *Multi-Objective* (MO) *Problem*.

A Single-Objective Single-DM Problem ($p = q = 1$, see points A and B in Figure 23.1) can be formulated in mathematical terms as the problem of finding the alternative that minimizes a given objective,[2] which expresses the criterion used to judge the alternatives. Depending on the type of environment, this Problem can be solved by means of *Deterministic* or *Stochastic Optimization*.

As for Multi-Objective Single-DM Problems ($p = 1$, $q > 1$, see points C and D), because of the presence of many conflicting[3] criteria, the very concept of optimality loses its

[1]This is why in the following we will often use the term *alternative* instead of *option*.

[2]Here and in the following objectives are considered to express costs; however, the case in which they express benefits can be handled in the same way, provided that minimization is replaced with maximization.

[3]The criteria must be conflicting: otherwise the MO Problem boils down to a SO Problem.

meaning and must be replaced by the concept of *Pareto efficiency* (see Section 18.2). This means that an *optimal alternative*, i.e. an alternative that minimizes all objectives at the same time, cannot be found but the set of 'candidate' alternatives can be reduced to the set of *efficient alternatives*, i.e. alternatives that can not be improved with respect to a given criterion, without worsening the performances with respect to at least one of the others. The final choice of the *best compromise alternative* from the efficient alternatives is subjective, since it requires specifying the relative importance of the criteria. However, it is important that the procedure through which the decision is reached be formalized, with the aim of ensuring the awareness of the DM and the transparency and repeatability of the decision. For this purpose, many different methods have been proposed; in Chapter 17 we described three of the most commonly used: *Multi-Attribute Theory* (Keeney and Raiffa, 1976), *Analytic Hierarchy Process* (Saaty, 1980), ELECTRE *methods* (Roy, 1991). Analytic Hierarchy Process (AHP) and ELECTRE methods have been conceived for the deterministic environment, while Multi-Attribute Theory has been developed both for the deterministic and for the stochastic environment (points C and D respectively); in the former case, the method is called *Multi-Attribute Value Theory* (MAVT), in the latter it is called *Multi-Attribute Utility Theory* (MAUT).

Multi-DM Problems (points E and F) are studied in *Group Decision Theory* and *Negotiation Theory*. The first studies rules that can be used to aggregate the preferences expressed by the single DMs (e.g., simple majority rule or any other voting system). Negotiation Theory studies techniques to support the DMs in exploring the set of alternatives and finding an agreement on the best compromise alternative; if negotiations fail, it is possible to turn to *Mediation*, i.e. a neutral third party assists the disputants in reaching an agreement by submitting 'fair' proposals to the parties. All these disciplines have been developed mainly with regard to the deterministic case (some of these concepts have already been introduced in Chapter 21). Increasing attention is being paid to the management of Multi-DMs decision-making problems in a random environment.

23.1.1 Sources of uncertainty in decision-making problems for water resources planning and management

We have already said that problems of planning and management of water resources can be viewed as decision-making problems where the options under study are different alternatives, i.e. suitable combinations of actions. The Project Goal can be translated into a criterion or, more often, into a number of criteria, which are the basis for comparing the alternatives and finding the 'most preferable' one.

23.1.1.1 Single-DM Single-Objective Problem

A decision-making procedure for the Single-DM Single-Objective Problem is sketched out in Figure 23.2. A model of the environmental system is used to simulate the system behaviour subject to each of the possible alternatives. Simulation requires specifying the trajectories of the system inputs over the evaluation horizon $[0, h]$. In Section 4.1.2 we classified such inputs as deterministic disturbances ($\mathbf{w}_t$), random disturbances ($\boldsymbol{\varepsilon}_{t+1}$) and controls ($\mathbf{u}_t$). Depending on the alternative under examination, the trajectory $\mathbf{u}_0^{h-1}$ of the controls can be different; also the planning decision $\mathbf{u}^p$, which appears among the parameters of the model, varies with the alternative under study. Conversely, each alternative A is

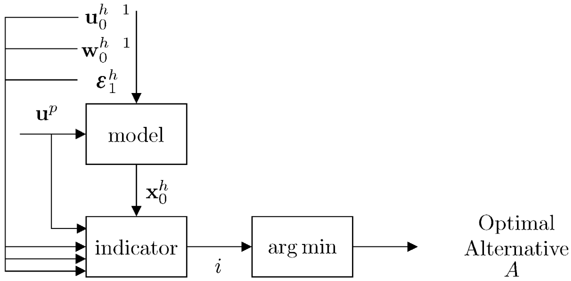


Figure 23.2: The procedure for comparing the alternatives in the Single-DM Single-Objective Problem.

univocally defined by the pair

$$\{\mathbf{u}^p, p\}$$

where p is the management policy

$$p := \{m_0(\cdot), m_1(\cdot), \ldots, m_{h-1}(\cdot)\}$$

i.e. the sequence of control laws $m_t(\cdot)$ from which the control $\mathbf{u}_t$ is derived at each time instant

$$\mathbf{u}_t = m_t(\mathbf{x}_t)$$

By simulating the system for a given pair $\{\mathbf{u}^p, p\}$, it is possible to estimate the trajectory $\mathbf{x}_0^h$ of its state, which provides a full description of the behaviour of the system over the horizon $[0, h]$ subject to alternative $A := \{\mathbf{u}^p, p\}$. Now, assume that the set $\mathcal{A}$ of the alternatives to be examined is known. Simulation can be repeated for each of the N alternatives in $\mathcal{A}$ and the N state trajectories obtained. However, comparing the alternatives on the basis of the state trajectories is almost impossible; the problem is simplified if each state trajectory is associated with a scalar measure that synthesizes the effects associated to that alternative. For this, we define

- a step-cost function $g_t(\cdot)$, whose value $g_t(\mathbf{x}_t, \mathbf{w}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1})$ expresses the performances of the system in the transition from time t to time $t+1$;
- a penalty function $g_h(\mathbf{x}_h, \mathbf{u}_t)$, which expresses the cost associated to the final state of the evaluation horizon;
- and a functional $i(\cdot) : \mathbb{R}^{h+1} \mapsto \mathbb{R}$, whose value

  $$i = \Psi\left\{g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right),\ t \in [0, h-1];\ g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right\}$$

  provides a synthetic measure of the 'global' cost over the horizon. We called such a global cost function *indicator* (see Section 4.10.1).

The most frequently used forms for the operator Ψ are the sum and the maximum, which correspond to defining the indicator respectively as

$$i = \sum_{t=1}^{h-1}\left(g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right) + g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)$$

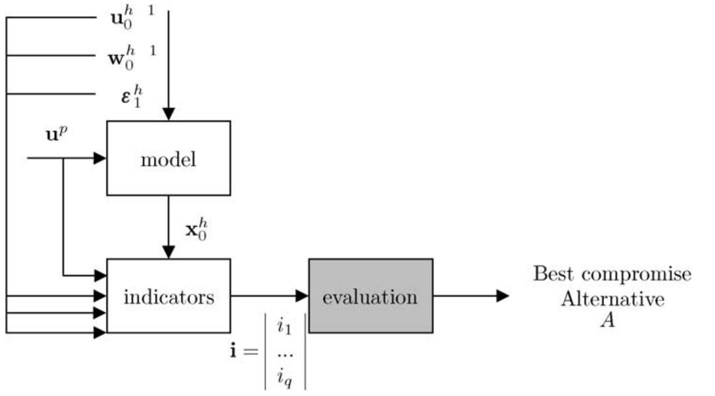


Figure 23.3: The procedure for comparing the alternatives in the Single-DM Multi-Objective Problem.

$$i = \max\left\{\max_{t=1}^{h-1}\left\{g_t\left(\mathbf{x}_t, \mathbf{u}^p, \mathbf{u}_t, \mathbf{w}_t, \boldsymbol{\varepsilon}_{t+1}\right)\right\}, g_h\left(\mathbf{x}_h, \mathbf{u}^p\right)\right\}$$

The time horizon can be either finite or infinite (i.e., $h \to \infty$). In the latter case, it is no longer necessary to define the penalty function $g_h(\mathbf{x}_h, \mathbf{u}_t)$, while the definition of the indicator as a sum of step-costs needs to be slightly changed in order to guarantee that it will converge: this can be done for example by introducing a discount factor or by turning to the mean instead of the sum (see Section 8.1.2.1).

Note that the indicator value, through its dependency on the trajectory of the step costs $g_t(\cdot)$, is a function of the trajectory of state, disturbances and control, and of the planning decision, i.e.

$$i = i\left(\mathbf{x}_0^h, \mathbf{u}^p, \mathbf{u}_0^{h-1}, \mathbf{w}_0^{h-1}, \boldsymbol{\varepsilon}_1^h\right)$$

Therefore, the indicator value depends on the alternative $A = \{\mathbf{u}^p, p\}$ under examination: $i = i(A)$. Then the 'most preferable' alternative can be found by comparing the values that the indicator assumes for the different alternatives. If i expresses a cost, the optimal alternative is the one that minimizes it, i.e.

$$A^* = \arg\min_{A}\left\{i(A)\right\} \tag{23.1}$$

23.1.1.2 Single-DM Multi-Objective Problem

In the environmental context we more frequently meet the case that many criteria must be considered, due to the coexistence and interconnections between environmental, social, economic and political interests. In this case (Figure 23.3) the following steps must be followed:

- The physical (or economic) effects produced by the alternative are synthetically measured by means of a number of indicators $i_j(A)$ (collected in the vector $\mathbf{i}$).

- The alternatives are compared based on the vector of indicator values that each of them produces. If the MAVT/MAUT method is used, each alternative is associated with an index I that expresses the DM's satisfaction with the effects produced by that

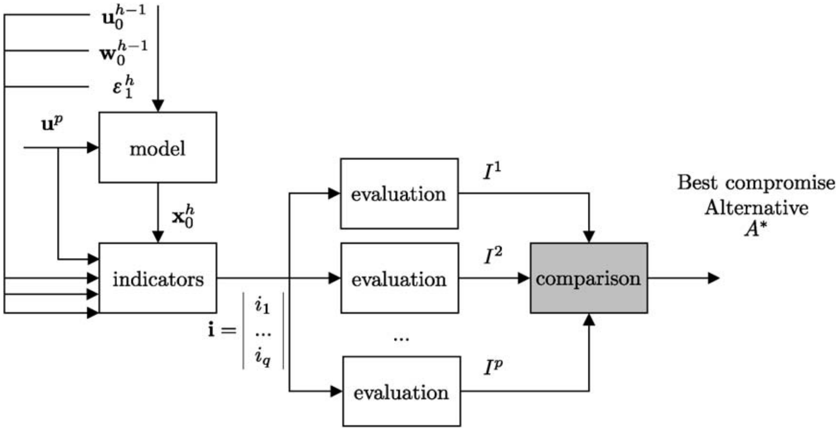


Figure 23.4: The procedure for comparing the alternatives in the Multi-DM Multi-Objective Problem.

alternative: the index is computed as a function of the indicator values

$$I(A) = f\big(i_1(A), \dots, i_q(A)\big) \tag{23.2}$$

- Once all the alternatives have been associated with an index I, the best compromise alternative is the one that maximizes this index

$$A^* = \arg\max_A \big\{I(A)\big\} \tag{23.3}$$

In some cases, it is possible to compare the alternatives by considering one criterion at a time (this happens when the property of *Mutual Preferential Independence* holds among the indicators $i_1, \dots, i_q$, see Keeney and Raiffa (1976) and Section 20.3). This means that the function (23.2) assumes the following additive form

$$I(A) = \lambda_1 I_1\big(i_1(A)\big) + \cdots + \lambda_q I_q\big(i_q(A)\big) \tag{23.4}$$

where $I_j(\cdot)$ is a function that provides the satisfaction associated to alternative A with respect to the jth criterion, and λ_j are positive coefficients (called *weights*) that express the relative importance of the criteria.

Note that also in this case it is not advisable to compare the alternatives by directly considering the indicator values $i_j(A)$; it is preferable to define and use the indices $I_j(A)$. The indicators, in fact, can be very different in their meaning, range of values and units of measurements, making it difficult (if not impossible) to compare their values and find trade-offs among the criteria. Moreover, the advantage of this approach is that it helps to maintain a clear distinction between objective elements (the effects produced by the alternatives) and subjective elements (the satisfaction associated to those effects).

23.1.1.3 Multi-DM Multi-Objective Problem

If there is more than one DM (Figure 23.4), the best compromise alternative is different for each of the DMs, depending on how each one has carried out the evaluation of the alternatives. In fact, even if all DMs used the same methodology, results could be different

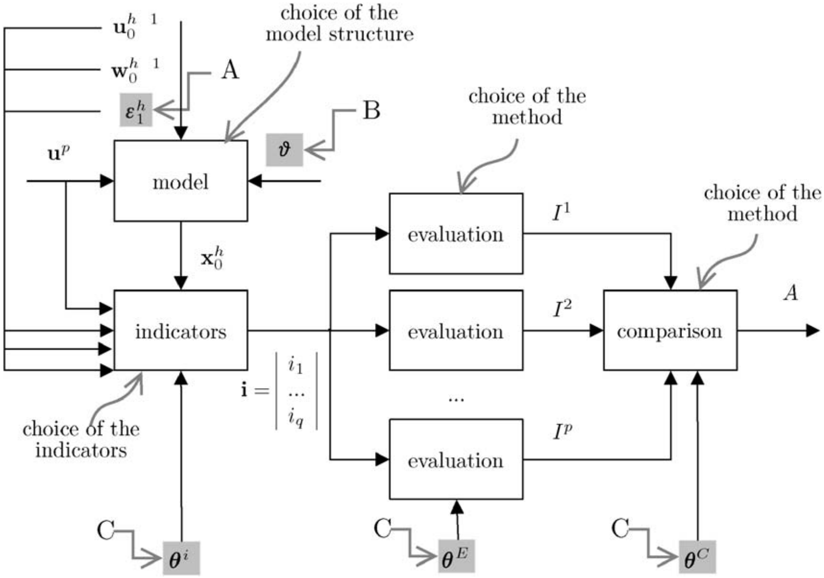


Figure 23.5: Sources of uncertainty in the decision-making procedure.

because evaluation is a way of formalizing the DM's preference structure and thus subjective judgments are allowed to come into play. For example, weighting the criteria is very subjective.

The evaluation phase must therefore be followed by a phase of *Comparison*, aimed at finding a best compromise alternative that is shared by all. This can be done in different ways. One approach is to define a rule that, given the p partial rankings expressed by the different DMs, provides one global ranking of the alternatives, or at least the first alternative in it. Otherwise, the best compromise alternative can be found through negotiations among the DMs.

23.1.2 Sources of uncertainty

Decisions regarding planning and management of water resources are typically made in a random environment. The principal sources of uncertainty in the decision-making procedure (see Figure 23.5) are associated to:

1. The system inputs: many of the variables that influence the system cannot be deterministically known. We have collected these variables in the vector $\boldsymbol{\varepsilon}_{t+1}$ of random disturbances.

2. The model of the system: the description of the processes acting in the system is not exact. Approximations in the description of the system's dynamics are frequent and often significant in environmental modelling, because of the complexity of the processes under study and, often, of the poverty of data that can be used for identifying the model. As a consequence, both the choice of the model structure and the estimation of the parameters appearing in the model are affected by strong uncertainty.

3. The indicators: selecting which are the indicators most suited for measuring the performances of the alternatives can be a very subjective operation. Moreover, once the indicators have been chosen, the mathematical definition of the functionals used to compute their values involves the same difficulties as those associated with model identification (choice of the functional and parameter estimation).

4. The evaluation method: evaluation can be carried out with different methods ranging from simple normalization of the indicators, aimed at reducing different measures to a common dimensionless scale, to more complex evaluation methods (see Chapter 17). The choice of the evaluation method obviously influences the results. Moreover, once the methodology has been chosen, further uncertainty is introduced in the way it is applied. For example, if the MAVT method is chosen, uncertainty is associated to the definition of the value functions.

5. The comparison method: just as for evaluation, comparison too can be carried out with different methods, and both the choice of the method and its application are sources of uncertainty.

As a result of the propagation of the input uncertainty and of the new uncertainties introduced in the course of the decision-making procedure, the system outputs, the values of the indicators and those of the indices are not deterministic values but are affected by uncertainty.

23.2 Classifying and modelling uncertainty

Classifying uncertainty is not an easy task (a scheme is proposed for example by Walker et al., 2003). Following Norton et al. (2005) a basic distinction can be made between *quantifiable* and *non quantifiable* uncertainty.

23.2.1 Quantifiable uncertainty

Quantifiable uncertainties are those associated to the values

- of the system variables (see marker 'A' in Figure 23.5);
- of the parameters appearing in the model of the system (marker 'B' in Figure 23.5) and of the parameters appearing in the functions used to compute indicators and indices (marker 'C' in Figure 23.5).

Note that a difference exists between the uncertainty associated to the variables and parameters appearing in the model of the physical system (A and B) and the uncertainty associated to the parameters appearing in the definition of indicators and indices (C). The quantification of the first type of uncertainty is somehow 'objective': it is based on the comparison between the values of the system variables computed with the model of the physical system and those measured in reality. Here, uncertainty is due to a lack of capability in modelling the system, i.e. it expresses the error that is made in assuming an approximate description of reality. Instead, the quantification of the second type of uncertainty, associated to the definition of indicators and indices, is 'subjective': we know that our synthesis of the physical effects on the system ($\mathbf{i}$) and of the satisfaction of each DM (I) is not perfect, but we do not know what the 'exact' values of $\mathbf{i}$ or I should be. This influences the choice of the model to be adopted for describing uncertainty and the possibility of validating such a model, as illustrated in the following.

23.2.1.1 Random input

'Objective' quantifiable uncertainties are usually modelled by describing the variable affected by uncertainty as a random variable. In Section 5.7 we classified random disturbances as *stochastic* or *uncertain*, and we said that they can be described by associating them respectively with a probability distribution and with a membership set.

Note that if a variable is function of random variables, it becomes random too. This means that, due to the presence of the random disturbance $\boldsymbol{\varepsilon}_{t+1}$ among the system's inputs, the step-costs $g_t(\mathbf{x}_t, \mathbf{w}_t, \mathbf{u}_t, \boldsymbol{\varepsilon}_{t+1})$ are also random, as are the indicators $\mathbf{i}$ and the indices I. More precisely, $\boldsymbol{\varepsilon}$ and $g_t(\cdot)$ can be described as random processes and $\mathbf{i}$ and I as random variables. The probability distribution (or set-membership) of $g_t(\cdot)$, $\mathbf{i}$ and I can be derived starting from the probability distribution (or set-membership) of $\boldsymbol{\varepsilon}$ and propagating uncertainty through analytical formulae or, more often, via numerical computation (*Monte Carlo simulation*).

23.2.1.2 Random parameters

As anticipated above, randomness is also introduced by the presence of unknown parameters into the model of the system. Following the Bayesian approach to the parameter estimation problem (see Appendix A4), parameters too can be viewed as stochastic variables, whose probability distributions are identified during the calibration process. Therefore, once the probability distributions of the parameters have been defined, the propagation of the uncertainty associated to the parameter values can be analysed with the same tools as for random inputs.

23.2.1.3 Example

Consider a water system composed of a hydrological basin, a water reservoir and several water users (Figure 23.6). Randomness in the system is associated to the description of the inflow to the reservoir. The inflow process can be described by means of an empirical model like, for example, an autoregressive model (see Appendix A6). Then the model of the basin takes the following form

$$a_{t+1} = \alpha^1 a_t + \cdots + \alpha^p a_{t-(p-1)} + \varepsilon_{t+1} \tag{23.5}$$

while the model of the reservoir is as described in Section 5.1. The parameters appearing in model (23.5) can be identified from data, i.e. from a sample $\{\bar{a}_{t+1}\}_{t=1}^{N}$ of inflow measures $\bar{a}_{t+1}$, by means of a suitable parameter estimation method (for example, the least squares method, see Appendix A6). Once the model has been calibrated, it is possible to simulate the system over the estimation horizon $[1, N]$ and, for each time instant t, compute the model error defined as

$$e_{t+1} = \bar{a}_{t+1} - \hat{a}_{t+1}$$

where $\bar{a}_{t+1}$ is the system output (measured inflow) and $\hat{a}_{t+1}$ is the model output (estimated inflow). In this way a sample $\{e_{t+1}\}_{t=1}^{N}$ of the model error is obtained. With this sample, the autocorrelation of the error process e can be computed; if e proves to be white, it coincides with the random disturbance ε that acts on the system. Therefore the sample of the

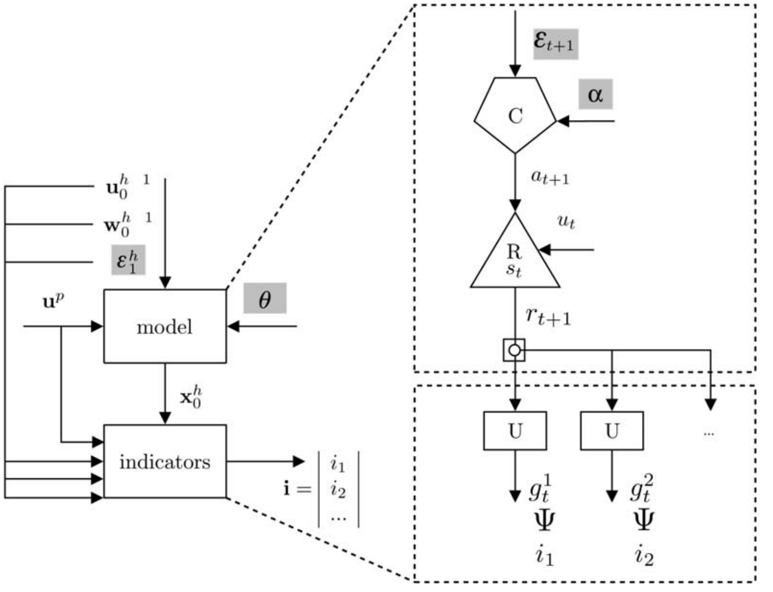


Figure 23.6: Example of a water system.

model error can be used to identify[4] the probability distribution of ε_{t+1}, which constitutes a mathematical description of the uncertainty associated with the modelling of a.

Note that if model (23.5) is calibrated by using a parameter estimation method based on the Bayesian approach (Appendix A4), it is possible to extend the above approach so as to include also the description of the uncertainty in the model parameters. In fact, let $\boldsymbol{\alpha}$ be the vector of the model parameters

$$\boldsymbol{\alpha} = \left[\alpha^1, \dots, \alpha^p\right]^T$$

According to the Bayesian approach, parameters are assumed to be stochastic variables with a given probability distribution $p_\alpha(\cdot)$, and the value they are attributed in the model is the expected value conditional upon the available data (the sample $\{\bar{a}_{t+1}\}_{t=1}^N$). Then a procedure for the quantification of all the uncertainty associated to model (23.5) is as follows:

1. Define the probability distribution $p_\alpha(\cdot)$ of the model parameters: for example, assume that they are Gaussian stochastic variables and use least squares to estimate their expected value $\hat{\boldsymbol{\alpha}}$ and covariance matrix.

2. Extract K realizations from $p_\alpha(\cdot)$ and obtain K parameter vectors $\boldsymbol{\alpha}_k$.

3. For $k = 1, 2, \dots, K$, simulate the system with model (23.5) and $\boldsymbol{\alpha} = \boldsymbol{\alpha}_k$, and obtain a trajectory of the model output. By comparing this trajectory with the sample $\{\bar{a}_{t+1}\}_{t=1}^N$, a sample of the model error can be derived.

[4]Methods for identifying the probability distribution of a stochastic variable from a sample are described in Appendix A2.

4. The K samples of model error obtained at the previous step are used to identify the probability distribution of the model error e. Such a disturbance encompasses both randomness in the model input and the uncertainty associated to the parameter values.

23.2.1.4 Sensitivity analysis

The procedure outlined in previous paragraph for the quantification of uncertainty associated to the model parameters is based on the key idea of *sensitivity analysis*: studying the variation in the model outputs depending on the variation of one of its inputs (in the previous example, the parameter vector).

Sensitivity analysis is probably the most suitable tool also for assessing the impact of the uncertainty associated to the values of indicators and indices. We have said that the quantification of this type of uncertainty is 'subjective', in the sense that it is not possible to define the error committed when a value of $\mathbf{i}$ (or I) is assumed instead of another, because no 'reference value' (e.g. a measure) is available. But the values of $\mathbf{i}$ (or I) influence the decision since they are used to obtain the ranking of the alternatives and thus the optimal (or best compromise) alternative (see equations (23.1) and (23.3)). Therefore, in this context sensitivity analysis can be used for identifying the variations in the parameters appearing in the definition of the step cost function (or of the index) that produce a rank-reversal, i.e. a modification in the ranking of the alternatives (and especially in the first position of the ranking). Note that this approach is not aimed at providing a general description of the uncertainty that affects the decision-making procedure, which in fact is impossible, but is aimed at assessing how uncertainty influences the decision, which is what we are actually interested in.

We have already spoken about sensitivity analysis in Section 20.7.1, for the case of a satisfaction index computed by means of an additive value function. A wide literature exists about sensitivity analysis for indices of additive form (equation (23.4)) and much attention is paid to the assessment of uncertainty in the values of the weights, i.e. the parameters λ_j (see for example Hyde et al., 2005). However, the same approach can be extended, at least in principle, to the analysis of any set of parameters appearing in the definition of either $\mathbf{i}$ and I.

23.2.2 Non-quantifiable uncertainty

Not quantifiable uncertainties are those associated to the choice of the model, of the functionals used to compute the indicators, of the evaluation and comparison method. Describing this type of uncertainty is very difficult and constitutes a challenge for future research (Maier and Ascough II, 2006). One possible way to assess the impact of non quantifiable uncertainty on the decision is that of repeating the decision-making procedure while changing some assumptions (e.g. using a different definition of the indicators, a different model or evaluation method) and check whether results change and how. Note that this can be seen as a sort of non-automatized ('trial-and-error') version of sensitivity analysis.

23.3 Handling uncertainty in the decision-making process

In the previous section we have seen how uncertainty can be described. In this section we will see how the presence of uncertainty can be taken into account when facing a decision-making process, i.e. which procedures can be used in order to make decisions with explicit awareness of the presence of uncertainty.

23.3.1 Single-DM Single-Objective Problem

When there is only one objective and one DM, the decision-making process stops with the solution of the optimization problem

$$\min_{A}\left\{ i(A) \right\} \tag{23.6}$$

In the presence of uncertainty, we have seen that the indicator i turns into a random variable, because it is a function of the trajectory $\boldsymbol{\varepsilon}_{t=1}^{h}$ of the random process $\boldsymbol{\varepsilon}$. Different approaches can be used to take this uncertainty into account; however, they are all based on the idea of 'pre-filtering' uncertainty and reduce the decision-making problem under uncertainty to a problem in a deterministic environment. To do this, a deterministic variable is derived from the random variable i and the decision is taken according to equation (23.6), where i is replaced by the deterministic variable.

For example, uncertainty can be filtered by replacing the random variable i with a suitable statistic i of it (e.g. its expected value or its maximum) or by defining a utility function $U(i)$ that expresses the DM's attitude towards risk and computing its expected (or maximum) value. The interested reader can find more details about these approaches and their theoretical foundations in Chapter 9.

23.3.2 Single-DM Multi-Objective Problem

When there is more than one objective, the alternatives under examination must be subjected to a process of evaluation, aimed at supporting the DM in identifying the alternative that realizes the best compromise between the different criteria. This can be done according to different methods, some of which are described in Chapter 17.

When choosing the evaluation method to be adopted, it must be considered, among other elements, whether uncertainty in the indicator values is going to be explicitly considered or not. For example, the evaluation method of *Multi-Attribute Value Theory* (MAVT), which has been described in Section 17.3 and in Chapter 20, has been proposed by Keeney and Raiffa (1976) for the case when the indicators are deterministic variables. This means that when the DM defines her partial value functions and weights, she does this based on a matrix[5] of deterministic indicator values. If, instead, the indicators are random variables, *Multi-Attribute Utility Theory* (MAUT) should be used (Keeney and Raiffa, 1976). The difference lies in the fact that value functions are replaced by utility functions: the former express the DM's evaluation of a given set of (deterministically known) indicator values, while the latter express the DM's evaluation of a possible realization of random indicators. Utility functions, in fact, express both the DM's satisfaction and her attitude towards risk. Assessing the utility functions is not easy: the method proposed by Keeney and Raiffa is based on the use of lottery questions, but this methodology is rather difficult to apply in the environmental context, where the uncertain effects of the alternatives that the DM has to judge can strongly impact the environment, social groups, etc. Moreover, in Section 17.6 we described some technical difficulties that arise in the application of the MAUT method to decision-making

[5] The indicator values can be displayed in a $(q \times N)$ matrix, where q is the number of criteria and N the number of alternatives (*Matrix of the Effects*).

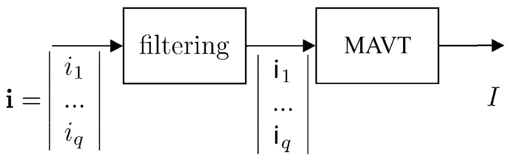


Figure 23.7: Handling random indicators by using the MAVT and pre-filtering uncertainty.

problems where a management policy p must be identified. Therefore, the approach that we propose, and that we followed throughout Chapters 17–20, is to pre-filter uncertainty by applying a filtering criterion to the random indicators i_j, and then to use the MAVT method over the indicator statistics i_j (Figure 23.7).

23.3.2.1 Evaluation as a source of uncertainty

So far, we have seen whether and how uncertainty deriving from the phase of *Estimating Effects*, i.e. uncertainty deriving from an approximate description of the physical system, can be handled in the evaluation phase. However, as shown in Figure 23.5, the evaluation process itself introduces new uncertainty. In fact, the results of this process may vary depending on the choice of the method to be followed (e.g. MAVT, AHP or ELECTRE, see Chapter 17) and on the way each method is applied. For example, if the MAVT is chosen, results depend on the values of the satisfaction indices associated to each alternative and thus on the definition of the value functions that were used to compute them. The uncertainty due to the choice of the method is difficult to quantify, while the impact of the uncertainty introduced when implementing the method can be assessed a posteriori, for example by means of sensitivity analysis (with reference to MAVT, see Section 20.7.1).

However, sometimes it might be opportune that the presence of this type of uncertainty be considered also during the evaluation process itself: in other words, we would like the DM to express the uncertainty, if any exists, in her judgments and be aware of such uncertainty when she takes her decision. Descriptive methods (see Section 17.2) like the AHP and the ELECTRE methods do allow for this, in the sense that, by relaxing some of the constraints required by normative methods, they do not force the DM to express her preferences in exact terms, if she is not able to do so. For example, we have said that AHP allows for slight inconsistencies in the pairwise comparison of the alternatives (see Section 17.4): this is a way of taking into consideration the uncertainty in the formalization of the DM's preference structure. Deriving the ranking vectors from a non-consistent pairwise comparisons matrix by means of a suitable algorithm is a way to come to a decision notwithstanding (and without removing) uncertainty in the judgments.

23.3.3 Multi-DM Multi-Objective Problem

We have seen that when there are many objectives, the alternatives under examination must be subjected to a process of evaluation. Such a process requires that the DM express judgments on the effects of the alternatives with respect to each criterion, and on the relative importance of different criteria. Since these judgments are inherently subjective, in the presence of many DMs the results of the evaluation, i.e. the choice of the best compromise alternative, can be different for each. Therefore, in the Multi-DM case, evaluation does not conclude the decision-making process, but must be followed by a phase of *Comparison* aimed at finding a compromise between the DMs' preferences.

Traditional approaches to the Multi-DM decision-making problems can be divided into two groups:

- Methods that focus on the preference structures and/or rankings expressed by the DMs, seeking to extract one 'group' (or 'social') ranking from the 'individual' preferences. These methodologies are often grouped under the name of *Group Decision Theory*.

- Methods that focus on the alternatives proposed by the DMs, suggesting interactive procedures aimed at finding the alternative that obtains the widest agreement. These methodologies are often grouped under the name of *Negotiation Theory*.

Group decision-making procedures can be applied if the DMs are willing to share all information and reveal their own interests and preference structures. Such conditions may occur when the multiple DMs are different authorities within the same country or groups of Stakeholders involved in a participatory decision-making procedure. However, when the DMs involved in the decision are the authorities of different countries (e.g. their governments), as happens for example in the management of trans-boundary catchments, the assumption of complete information exchange is often not verified. In this case, the solution to the Multi-DM Problem can be found by focusing on alternatives and having the DMs negotiate, i.e. submit proposals and counter-proposals until they come to an agreement. Alternatively, mediation can be used, i.e. a neutral third party, called a *Mediator*, can be asked to propose a 'fair' solution to the parties.

Some of these concepts, as well as a description of several approaches to the *Comparison* phase, have already been introduced in Chapter 21. In that chapter two methods were proposed for deriving one 'group' index (called a *Project Index*) from p indices expressed by the p DMs. We used two assumptions: (1) that all the individual indices had additive form, i.e. were a linear combination (with coefficients γ_h) of partial satisfaction indices,[6] and (2) that all the partial satisfaction indices were equal for all DMs. With these assumptions, the problem of aggregating the individual preferences boils down to the problem of finding a 'group' vector $\boldsymbol{\gamma}$ of weights, starting from the p individual vectors $\boldsymbol{\gamma}^k$. In Sections 21.4 and 21.6 two methods were presented for deriving $\boldsymbol{\gamma}$: the first is based on the idea of minimizing the average distance among the 'individual' vectors $\boldsymbol{\gamma}^k$ of weights, while in the second the 'group' vector $\boldsymbol{\gamma}$ is derived by linearly combining vectors $\boldsymbol{\gamma}^k$ with suitable coefficients δ_k.

As for negotiations and mediation, in Sections 21.5 and 21.7 some methods were proposed.

23.3.3.1 Uncertainty in Multi-DM Multi-Objective Problems

All the above-cited methods, as well as most of the methods proposed in the literature, have been conceived with reference to the deterministic context, i.e. assuming that all the DMs are able to express their preferences in exact terms. This does not mean that no uncertainty affects the decision-making process, but that any uncertainty has been filtered during the phase of *Evaluation*. Uncertainty introduced in the evaluation process is neglected during

[6] More precisely, we assumed that the linear combination of partial indices (equation (23.4)) could be broken down into two steps. In the first, the indices relative to the q leaf-criteria are partially aggregated to define s sector indices (with $s \ll q$); in the second, the sector indices are aggregated in a single index (see Section 20.6.3). We used the symbol γ_h to denote the coefficients appearing in the second linear combination, which is the subject of the present discussion.

the comparison, as well as any new uncertainty introduced during the comparison itself. As usual, the impact of these uncertainties can be assessed a posteriori by means of sensitivity analysis. For example, it is possible to compute the minimum variation in the value of $\boldsymbol{\gamma}^k$ or in the value of δ_k that produce a rank reversal (the former accounts for uncertainty in the evaluation, the second for uncertainty in the comparison).

A few techniques have been proposed to handle comparison by explicitly accounting for the uncertainty introduced during evaluation. Prodanovic and Simonovic (2003) propose a methodology based on fuzzy theory for explicitly considering uncertainty in group decision-making. The method allows for both expressing fuzzy individual preference relations during evaluation, and aggregating the individual preferences with fuzzy rules during comparison. The limit to this approach is that the fuzzy rules used to aggregate individual preferences require introducing a number of parameters, whose value is arbitrary. Again, sensitivity analysis could be used a posteriori to assess the impact of the choice of such values on the final decision.

Kronaveter and Shamir (2006) present a Negotiation Support System for a Two-DM Multi-Objective Problem. It provides a solution to the problem of the optimal allocation of water, supports the DMs in the evaluation of different allocation alternatives and implements a negotiation procedure based on the Nash bargaining solution (see Section 21.7). The negotiation procedure allows for taking into consideration the uncertainty introduced in the evaluation phase, when the DMs are asked to express the weights attributed to their different objectives. Let $I^k(A)$ be the satisfaction of the kth DM for alternative A (with $k = 1, 2$), obtained by linearly combining q partial satisfaction indices $I^k_j(i_j(A))$ through the weights λ^k_j, as in equation (23.4) (in Kronaveter and Shamir, 2006, the indices $I^k(A)$ are obtained by means of the AHP, but the method they propose can be applied also in the case when the indices were obtained by using Multi-Attribute Value Theory). According to Nash, the equitable solution of the decision problem is the efficient alternative that maximizes the product $I^1(A) \cdot I^2(A)$, i.e.

$$A^* = \arg\max_A \left\{ I^1(A) \cdot I^2(A) \right\}$$

This solution is computed by the Negotiation Support System and proposed to the DMs as a compromise alternative. Kronaveter and Shamir note that if the values of the weights λ^k_j are affected by uncertainty, they can be described by a range of values $[\underline{\lambda}^k_j, \bar{\lambda}^k_j]$ instead of a single value, and the Nash solution can be refined by solving the minimization problem

$$\max_{\boldsymbol{\lambda}^1, \boldsymbol{\lambda}^2} \left(\sum_{j=1}^{q} \lambda^1_j \cdot I^1_j\big(i_j(A^*)\big) \right) \cdot \left(\sum_{j=1}^{q} \lambda^2_j \cdot I^2_j\big(i_j(A^*)\big) \right)$$

subject to

$$\lambda^k_j \in \left[\underline{\lambda}^k_j, \bar{\lambda}^k_j\right] \quad j = 1, \ldots, q;\ k = 1, 2$$

Note that the presence of uncertainty, by adding degrees of freedom to the problem, expands the domain of admissible solutions in the space of satisfaction indices. In other words, uncertainty provides a chance to enlarge the zone of possible agreement and increase the DMs' satisfaction with the compromise alternatives, thus playing a positive role in Multi-DM decision-making.

Chapter 24

Software architecture of MODSSs

AR, RSS and EW

We have now finished reading the four preceding sections, and we have a map (the PIP procedure) for the management of a Participatory and Integrated Project; we also know the methods and techniques to adopt in each phase of the PIP procedure. All of this constitutes a useful body of knowledge, which is nevertheless quite hard to manage and govern unless, at least in some rudimentary form, a computer system is available: an MODSS that allows us to translate what we have learned into practice. This book could not be considered complete without some mention of the software architecture of an MODSS.

Therefore it is necessary to leave the world we have been explored until now, the world of hydrology, economics, the management of water resources, and their formal representations (the models), and to move into the world of software engineering, which the reader at first may find very abstract and dry, because all the entities and concepts we have discussed so far (Stakeholders, reservoirs, models, etc.) are mapped into their software counterparts as data structures and numerical procedures. Nevertheless, it is just this abstraction, such a vision of the problem from a higher standpoint, that allows us to create an MODSS that is not a simple translation of the PIP procedure and its phases into an electronic implementation, but something more vital and flexible, able to evolve and adapt itself to the ever-evolving reality (laws change, along with points of view, necessities and the physical system ...) in which it must operate. As Brooks (1995) rightly observes: only unused software ceases to change. For this reason, in this chapter we focus on the software development process of an MODSS, with particular regard to the requirement analysis and the overall software architecture. How the technical implementation is performed may change, according to the currently available technologies, but the general concepts will still hold.

The software development of an MODSS, as that of any software project, requires six phases:

(1) the feasibility study,

(2) the analysis and the specification of requirements,

(3) the design of a software architecture that satisfies them,

(4) the programming and testing of the individual modules,

(5) the integration and testing of the complete system,

(6) maintenance.

The phases do not necessarily have to be carried out in a *cascade life cycle*, in the sense that each must be terminated before the next one can begin, even if this is the most spontaneous and simple approach. For complex systems, as an MODSS, it is more appropriate to adopt an *evolutionary life cycle* (Royce, 1998), according to which the development occurs in an incremental way, with successive prototypes. In effect, the architecture that we will describe and our TWOLE system that implements it were born in that way. What we will present is the third version of TWOLE (for the earlier versions see Soncini-Sessa et al., 1990, 1999). We will not talk about all of the six phases, however, given that several of them concern the effective production of the MODSS and lie outside the area of interest for the majority of the readers of this book. We will examine two of them only: the phases of Requirement Analysis and Architectural Design.

The Requirement Analysis begins with a study of the typology of Projects that the application must help to deal with. We have carried out this study in Parts A–D of this book. From it the *functional requirements* emerge (more briefly the *functionality*): *what* the software must allow the Analyst to do; we will considered them in the next section together with some non-functional requirements concerning the services and functions of a MODSSs. We will consider only requirements that pertain to the MODSS as a whole, not to its individual modules.

Given the set of requirements, we must decide *how* to satisfy them. This is the phase of Architectural Design. It is not, however, possible to speak of architecture if one has no idea of the materials and structures with which to realize it (an analogous interpretation of the above terms connected to the construction will render their meaning). Therefore we will preface Section 24.3, which is dedicated to the architecture, with a section that delivers a concise excursus on data structures and software components.

The chapter is closed by a brief overview of what the future might hold for MODSSs in the light of the very rapid changes in software and communications technologies.

24.1 Requirements

The question is: what can we expect from a MODSS? The best way to identify the answer is to go back to what we have seen so far.

In Section 2.7.1 we noted that the decision-making structure of a water resources planning and management Agency suggests that a MODSS should have *levels* (Figure 2.14), which correspond to the first two of the three decision-making levels of the Agency: the planning level, the management level and the operative control level (Figure 2.13) (Anthony, 1965). We also noted that the operative control level can be delegated to an automatic regulator, as at this level human intervention is almost superfluous, while at the management level the intervention of the Regulator is very often required and at the planning level the one of the Decision Maker is essential. These two levels must therefore be supported by a MODSS.

24.1.1 Planning level requirements

At the planning level the MODSS, or more precisely the MODSS/P of Figure 2.14, is required to provide support to the actors (Stakeholders, Analyst, Decision Maker(s)) in the course of the decision-making process. To understand precisely what this means, let us examine one by one, how the MODSS can support the phases of the PIP procedure.

24.1.1.1 Reconnaissance, Defining Actions, Defining Criteria and Indicators

In these three phases *Project Goal*, the (spatial and temporal) boundaries of the system being considered and the possible *actions* are identified (see Chapter 3). The Stakeholders then specify their own *evaluation criteria* and translate them into *indicators*. The definition of the water system requires cartographic and graphic data processing, while the other tasks require the creation and manipulation of a number of *hierarchies of criteria*. In general, it is neither a simple nor an immediate task to define these hierarchies, and recursions often occur (like, for example, when the translation of criteria into indicators fails and so it is necessary to go back to the definition of criteria and identify a different hierarchy). Therefore the MODSS must provide software tools that allow the Analyst to manage the creation of the hierarchies in a simple and interactive way.

The indicators are defined on the basis of the leaf criteria and the MODSS must therefore provide tools that allow the definition of indicators, not only quantitative ones, but also qualitative ones. These latter must be translated into numeric values through conversion scales (*dictionaries*) which are defined by the user. This task can be supported by predefined *semantic networks*[1] to reduce the risk of ambiguity in the use of terms. Such semantic networks can be provided by means of *ontologies* (see Section 2.2).

Finally, it must be possible to estimate the value of the indicators for given scenarios, the historical one for example, to allow for their *validation*. Simulation will provide a tool to perform such estimations.

Remark: We will see that it is useful to think of a hierarchy of criteria as a *descriptive model* that evaluates the performance of an alternative from the standpoint of the root criterion of the hierarchy. These hierarchies are the branches of the overall (*Project Hierarchy*) (Section 3.2.1.2) that structures the whole set of the Project's evaluation criteria and whose root is the Project Goal. The operation of aggregating the single hierarchies into the Project Hierarchy is logically analogous to the aggregation of the component models into the aggregated model.

24.1.1.2 Identifying the Model

In this phase (Chapters 4–6) the MODSS/P must help the Analyst in the identification of the model for each one of the components of the water system (*component models*), the most common of which were presented in Chapter 5. For this reason, first of all it is necessary to associate each component with input and output data series which describe it and/or with the parameters that characterize it, to specify the actions which can be effected on it, and the norms to which it has to adhere.

Identification encompasses both model structure identification and parameter estimation (calibration). For identification MODSS/P must provide a library of model structures[2] and specifications on the conditions for their applicability. The identification can be facilitated by adopting *Integrated Modelling Environments* (Rizzoli and Argent, 2006) and *Model Management Systems* (Potter et al., 1992). For calibration appropriate algorithms must be offered that use input and output data series and model validation tools. Finally, MODSS/P

[1] A semantic network describes knowledge through a direct graph in which nodes represent concepts, and the arcs represent the relationships between them (e.g. degrees Celsius ↔ unit of measure ↔ temperature).

[2] Model equations (state transition and output transformation), model interface (inputs (disturbances and controls), outputs and parameters).

must support the aggregation of the component models to create the *aggregated model* of the water system (Chapter 6). All these operations (identification, calibration and validation experiments) must be carried out in such a way as to guarantee ease of execution, transparency and retraceability.

Remark: We may think of these models as *descriptive models* and the algorithms that operate upon them as *tools*.

24.1.1.3 Designing Alternatives

The MODSS/P must allow the Analyst to formulate the *Design Problem*, defining the objectives, the considered actions and the constraints (Chapters 8–11, 18), and to solve it with the appropriate algorithms (Chapters 12–15), identifying the optimal policy or the efficient ones and/or planning decisions. The Design Problem is formulated as a Mathematical Programming Problem or an Optimal Control Problem, according to the specific case. Therefore tools must be offered to support the definition of the Problem and algorithms for its solution.

Remark: It is useful to think of the Design Problem as a *decision model*, whose solution is the optimal alternative(s) or the efficient one(s). Thereby it is possible to interpret the solution algorithms as *tools* that, applied to a decision model, provide the solution to it. This can be, for example, a Bellman function that, together with the decision model for which it is the solution, constitutes a *prescriptive model*, from which one obtains a policy by applying a tool that solves an equation of the form of equation (12.9). Similar tools can be considered in the eventuality that the Project adopts policies of other types.

24.1.1.4 Estimating Effects

The aim of this phase is the estimation of the values that the indicators assume in correspondence with a given alternative, which all together constitute the *Matrix of the Effects* (Chapter 19). The tool to estimate the elements of that matrix is the *simulator*. When the water system is described by a non-dynamical model, the latter computes the indicator values by simply evaluating the output transformation equation of the aggregated model. In the opposite case, i.e. when the model is dynamical, it first computes the trajectories of the state variables and then, on their basis, the output transformation equation.

24.1.1.5 Evaluation

The aim of the phase is to transform the values of the indicators into *sector indices* and/or a *Project Index* that expresses the level of the Stakeholders' and/or Decision Maker(s)' satisfaction with each alternative. This requires the identification of the partial and global value functions (Chapter 20). The MODSS must provide a guided procedure for the estimation of the former, while the identification of the latter must be supported by graphic tools that allow each Stakeholders and/or each Decision Maker first either to specify the n-tuple of equivalent indicators or to pair-compare couples of alternatives; then to generate the consequent ranking of the alternatives from his/her point of view.

Remark: Once again, it is helpful to think of these functions as *descriptive models* of the structure of the Stakeholders' and Decision Maker(s)' preferences and to think of the procedures for their identification as *tools* that operate upon them.

In this, as well as in the following phase of Comparison, it is quite useful to have the possibility of recalling the steps of the decision-making process that have already been taken, in order to evaluate the effect of a modification made to some information or decision that was used or taken in it ("What if?" or *sensitivity analysis*). Consider, for example, the case where we are negotiating the regulation policy of a reservoir, which feeds, among other users, a hydroelectric user. The Facilitator has just shown the effects of a given efficient policy, which represents a compromise between the various sectors, when the hydroelectric user ask to know how his/her index can be influenced by a change in the price of energy from 0.05 to 0.06 €/kW · h. To answer, the only way is to modify the price as requested and to repeat all the computations effected on the basis of this until the required index is re-computed. It is therefore opportune for the MODSS to do all this computations automatically, while maintaining the current situation unchanged, i.e. on a copy of the Project. The automatic creation of a copy could be useful, not only in cases like the previous example, but also in many other cases: e.g. to explore the effects produced on a decision by the use of different models for a given component when the Stakeholders do not agree on the choice of model.

24.1.1.6 Comparison

The result of this phase is the set of *reasonable alternatives*. According to the method adopted, this set can be derived from the *global ranking* of the alternatives, or identified through negotiations among the Stakeholders. The tools that the MODSS must offer vary according to the method used. The methods for *voting on the order* and *negotiating the weights* require the same tools as the *Evaluation* phase, while the method for *negotiating the thresholds* requires *ad hoc* tools (Chapter 21).

24.1.1.7 Mitigation and Compensation

This phase does not strongly require particular computer support because it is a *brainstorming* phase, at the end of which the Analyst proceeds either to the *Definition of the actions* or to the *Project of alternatives* (Chapter 22).

24.1.1.8 Final Decision

This phase uses the same tools as the *Comparison* phase (Chapter 21).

24.1.1.9 Transfer

Once the *best compromise alternative* has been chosen, the user must transfer the set of models with which it was designed from the MODSS/P to the MODSS/M, along with the regulation policy that characterizes it.

24.1.2 Management level requirements

At the management level the MODSS, MODSS/M in Figure 2.14, is required only when either the *regulation policy* is set-valued (SV) or an on-line policy is adopted. In the first case, by simulating the water system model, the trajectories of its output variables can be observed for some control values $\mathbf{u}_t$, chosen by the Regulator in the set $M_t(\mathbf{x}_t)$ suggested by the policy (Section 10.1.1), and for different scenarios of deterministic and random disturbances, generated by forecast models. In the second case, the same tools that MODSS/P offers for dealing with the *Designing Alternatives* phase allow the Analyst to define and solve the Design Problem that specifies the on-line policy (Chapter 14).

Note that MODSS/M does not replace the Regulator, but only assists her in the evaluation of the consequences of her choices. The policy embeds the knowledge about the water system and the decisions taken at the planning level, taking account of which the Regulator must act. Her task is to take account of those elements or events that could not be considered in the policy design, because there was too much detail or they were unforeseeable. For this reason the model of the water system (*evaluation model*) often has greater detail than the model (*screening model* used in the MODSS/P for the policy design, Section 19.2). The MODSS/M enables one to make the most of the Regulator's experience in the management of the water system.

24.1.3 Review of the above requirements

Two levels are required in a MODSS: the MODSS/P, for planning with the PIP procedure, and the MODSS/M, for helping the Regulator to take account of the decisions taken at the planning level.

The main operations carried out by a MODSS can be interpreted as creation or manipulation of models, which can be classified as: descriptive, decision and prescriptive models.

A *descriptive model* describes a sector (for example through the corresponding hierarchy), a component or the preference structure of a Stakeholder or a Decision Maker (through the partial and global value functions). They can be component or aggregated models.

A *decision model* defines a Design Problem, as, for example, the design of a regulation policy. There are four types: *planning models* (that implement the Pure Planning Problems, see Chapters 8 and 9), *control models*, *distribution models* and *mixed models* (which are composed of a *planning model* in which a *control model* is nested, see Section 10.3). The last three types[3] can be in turn subdivided into *off-line* models (for off-line policy design, Chapters 12 and 13) and *on-line* models (for on-line policy design, Chapter 14).

A *prescriptive model* implements a policy and it is always associated with the descriptive model of the water system or of the distribution network for which the policy that it implements was designed.

The manipulations of models are carried out through tools and we will use the term *experiment* to denote each elementary manipulation of this type. Therefore a decision-making process can be thought of as an organized sequence of experiments.

The user of a MODSS is, according to the phase, any one of the actors (Analyst, Stakeholder, Decision Maker, Regulator) or even more than one actor at a time. In the following, for simplicity, we will use the generic term *user* to denote any of them.

24.1.4 Content management requirements

A MODSS must also have some of the functions of a *Content Management System* (CMS) in order to guarantee two properties that are fundamental for the support of a decision-making process: *transparency* and *traceability*, and *repeatability*.

24.1.4.1 Transparency and traceability

The *transparency* and *traceability* of the decision-making process requires that the actors, at any moment, be able to examine the decisions taken until that moment and the reasons that

[3] Yes: even the third type! Why and how? The reply is left to the reader.

supported them. This means that for each phase of the procedure there must be associated documentation, whose format can be quite varied, but that must allow anyone to know by whom and when it was produced or modified.

24.1.4.2 Repeatability

At any time, any actor must have the possibility to repeat all the computations and verify all the decisions that have been assumed up to that moment (*Repeatability*).

24.1.5 Distributed access requirements

Technological progress and the evolution of Internet have changed the way in which we can think about the requirements of a MODSS. The advent of personal computers has shifted the modality of computation from central data processing to local data processing, but the entrance of Internet on the scene allows one to pass from localized data processing to distributed data processing. A MODSS can exploit these new possibilities to great advantage:

- it can utilize remote data and models, for example data available in the data bases of the Irrigation Consortia of the water system or the National Hydrographic Service, or the models available in a major research institute;

- it can carry out the time-consuming computation required for the solution of Design Problems by distributing the work among computers that have computer time available at that moment;

- it allows the actors a *distributed participation*. For example, the Stakeholders could easily examine the alternatives without moving from their offices before negotiations; and even negotiations could be conducted without the physical presence of the Stakeholders in a single location. This last possibility is for example particularly interesting when negotiations are carried out as an *asynchronous process* (Section 21.4) or when *negotiations on thresholds* are adopted (Section 21.5), and the alternatives are designed during the negotiations.

24.1.6 Architectural requirements

The requirements considered until now specify the services to be provided by the MODSS from the user's point of view (*functional requirements*). From the MODSS designer's point of view there are other important requirements (*non-functional requirements*). In order that the software development and maintenance be economic, it is necessary for the software to be reusable, where possible, and easily maintainable. Two requisites take account of these constraints: separability and orthogonality.

By *separability* we mean the organization of the software architecture in *modules*[4] that communicate through well-defined interfaces. The modules must be strongly cohesive internally and loosely connected between each other (Parnas, 1972). Separability permits the MODSS designer to circumscribe the zones of intervention for software maintenance and update, or to assign the development of different modules to different developers in large projects. For example, a clear separation between Graphic User Interface (GUI) and algorithms allows the software designer to ensure that there will be no interference with the GUI if in the future the algorithms need to be modified.

[4]With the term *module* we mean a general software entity, that can be an application, a software component, an object, or a package, according to the context.

Orthogonality offers the possibility to develop modules that are able to interoperate to reach a certain result. Think of the solution of a Design Problem: it would be very useful to have alternative solution algorithms (*solvers*) for the same decision model. Being able to create, for the decision model and for each solver, separate software entities that can interoperate permits great elasticity of use, easy maintenance, and reusability of the software.

24.2 Design

In the previous section we examined the requirements of a MODSS, both from the user's point of view and from the designer's point of view. Given these requirements and considering the significant effort that is needed to implement them, one must exclude the possibility that a MODSS could be produced by the extemporary work of a single person. This statement may seem trivial, but very often the software implementations of mathematical models are the result of the work of a single researcher, who is expert in modelling but often with little knowledge of software engineering. As long as the software complexity can be managed by a one-person team, we are faced with products that do work well, even if they may have followed a winding development path. However, when the complexity increases, the quality of an un-engineered product diminishes in a more than proportional way, until it becomes unacceptable.

The object of *Software Engineering* is to increase the productivity of the development process and to check the quality of the software product, through the study and implementation of appropriate methodologies. It was born in response to the 'software crisis' in the 70s, when it became evident that the majority of software projects exceeded their cost estimates. The 80s and 90s demonstrated that all of the solutions that were proposed as 'definitive' actually were not universally applicable. Only recently did it clearly emerge that the methodology must be chosen according to the problem characteristics. For this reason, in the following, we present several software engineering concepts that target the specific software architecture of a MODSS.

24.2.1 Objects and classes

The most traditional approach to software design is the *procedural approach*, which concentrates on the logical steps of the procedure followed by the user when (s)he carries out his tasks. If we were to adopt this approach, the software implementation of a MODSS/P would tend to reproduce the diagram in Figure 1.4. This approach is fine from a functional point of view, but it does not take into account important architectural considerations such as the need for reusability and separability of the modules, and it does not allow an easy check of the quality of the product.

The *Object-Oriented Programming* (OOP) approach switches the focus from the process to the entities that play a role in the procedure, as for example descriptive and decision models, and to the manipulations they are affected by in the decision-making process.

According to the definition by Wegner (1987), a language is object-oriented when it makes *objects*, *classes* and *inheritance* available to the programmer. According to the definition by Booch et al. (1999), an *object* has its own *identity* (which distinguishes it from other objects), has a *state* specified by values (that can be either objects or links to other objects) of its *attributes*, and exhibits a *behaviour* that is described by the operations (*methods*) that can be performed upon it.

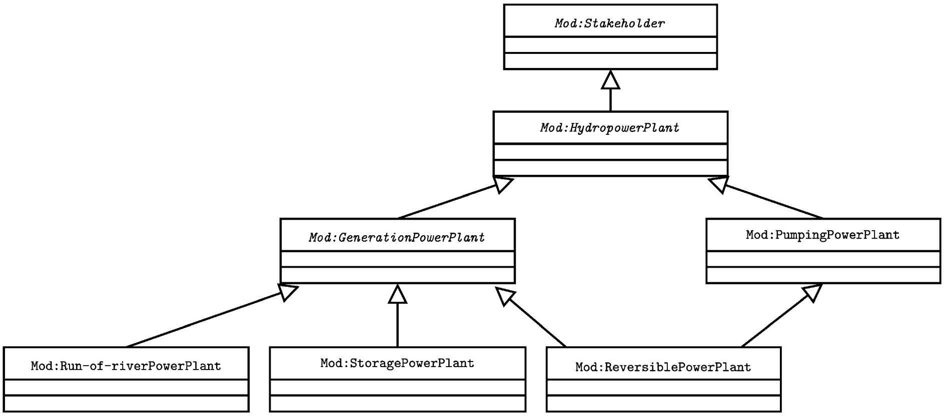


Figure 24.1: Inheritance hierarchy of *Mod:HydropowerPlant*.

An object is created according to the specifications provided by a *class*. A class can therefore be seen as a data type, and an object is an instance of that data type. The methods are defined in the class and therefore are the same for all the objects that belong to that class. Objects of the same class are distinguishable only by the values assumed by their attributes.

For example, in the *Identifying the Model* phase of the Piave Project (Chapter 6), when adopting the OOP approach, the object that describes the hydroelectric plant at Mis will be created (*instantiated* in technical jargon) according to the class *model of storage-power-plant*, whose characteristic is to have as attributes the parameters that appear in equation (5.32) (whose values are yet to be specified) and as methods those defined by the functions of equation (5.32). The Mis plant model is instantiated by assigning to its attributes the values of the parameters that characterize it. In the phase of *Estimating Effects* a simulation tool could ask the Mis plant model to carry out the methods that implement equations (5.32a) and (5.32c) in order to compute the energy produced by the plant.

The declaration of a class, apart from defining the nature of the attributes and the methods that its objects possess, requires the specification of the position that it occupies inside the *lattice* that defines the *inheritance hierarchy*. A class may not have parents (it is then called *root class*), or it can descend from another class. If the class B descends from the class A, one says that B *specializes* A, because B adds new attributes and methods to those that it inherits from A and so becomes more specific. For example (see Figure 24.1), the class `Mod:StoragePowerPlant` describes the model of a storage power plant (see page 172) and inherits the characteristics (the attributes and methods) of the progenitor class *Mod:GenerationPowerPlant*. This latter defines the common characteristics of the three classes that describe the models of run-of-river power plants (see equation (5.31)), of storage power plants (see equation (5.32)) and of reversible power plants (see equation (6.26)), as, for example, the fact that all three classes implement the method corresponding to equation (5.32b). In its turn, this class descends from the class *Mod:HydropowerPlant*, which describes the generic model of a hydropower plant, of both the generation and the pumping type, that it is a particular case of the class *Mod:Stakeholder* that describes the model of a generic Stakeholder. The class `Mod:StoragePowerPlant` differs from its progenitor *Mod:GenerationPowerPlant* both in the attributes it possesses (for example, it has a link to the reservoir to which the plant is hydraulically connected and has a control variable), and in the methods it implements (for example the method corresponding to equation (5.32a)), which differentiate it from its two sister classes and that therefore its ancestors[5] cannot possess.

The classes are not necessarily organised in a tree, since they can be *polymorphic*; that is, they can derive their characteristics from more than one class. This property is very useful for describing objects that can be seen from different points of view: e.g. a reversible power plant (`Mod:ReversiblePowerPlant`) is a pumping plant (`Mod:PumpingPowerPlant`), but it is also a generation plant (*`Mod:GenerationPowerPlant`*), see Figure 24.1.

From the point of view of a MODSS designer, the OOP Paradigm is very interesting because it allows for the efficient implementation of the properties of separability and orthogonality of the code. Just one example: the transfer of models and of policies between the two levels of the MODSS becomes immediate if the models and the policies are defined as objects: their *classes* just have to be declared at both levels and then they will know how to manipulate them.

24.2.1.1 Modelling software with UML

In the preceding parts of this book we have used the mathematical formalism of differential equations and the Systems Theory approach to write the models that describe the behaviour of the various components of the water system. Even the design of a software system is a modelling activity, since it requires the description of the software's behaviour during its execution, through appropriate models. Therefore a formalization is needed to describe these models and UML (*Unified Modelling Language*) (Booch et al., 1999) is universally adopted to design object-oriented systems. The UML formalism allows us to represent a software system from different points of view. It is useful to describe both the dynamical interaction of the objects that exchange messages (the calls of methods) during the execution of the program, and their static relationships: the classes, their structure (methods and attributes) and their hierarchical relationships. We will discuss only this second aspect in more detail.

An UML model can be represented both with a text document, and with diagrams. For example, the *class diagram* (Figure 24.2) allows us to associate graphic elements of immediate interpretation with the different entities. A class is represented with a rectangle partitioned in three zones (look at the block `Dom:Reservoir`). In the first zone the name of the class is reported (`Dom:Reservoir`), in the middle there is the list of its attributes (e.g. `Capacity`, `ReleaseFunction`, ...), in the last there is the list of its methods (`Compute_NextStorage()`, `Compute_Release()`,[6] ...).

It is not always necessary to express the full list of attributes and methods. The modeler may report only those that are of interest.

A triangular-pointed arrow between the classes represents an *inheritance relationship* between the descendent class (from which the arrow leads) and the progenitor class (to which the arrow points), e.g. the `Dom:WaterSystem` is a descendent of *`Dom:Aggregated`*. An arrow with a rhomboidal point represents an *aggregation relationship*: an object of the class indicated by the arrow is constituted by the composition of as many objects of the class from which the arrow leads as the number (or interval) that appears next to the arrow's tail. In the same way, an object of the class from which the arrow leads can belong to as many objects of the class to which the arrow points as the number (or interval)

[5] I.e. its progenitor and the progenitors of its progenitor.

[6] The parenthesis after the name indicates that it is the name of a method rather than of an attribute. It reminds us that a method, just as every function, may require arguments.

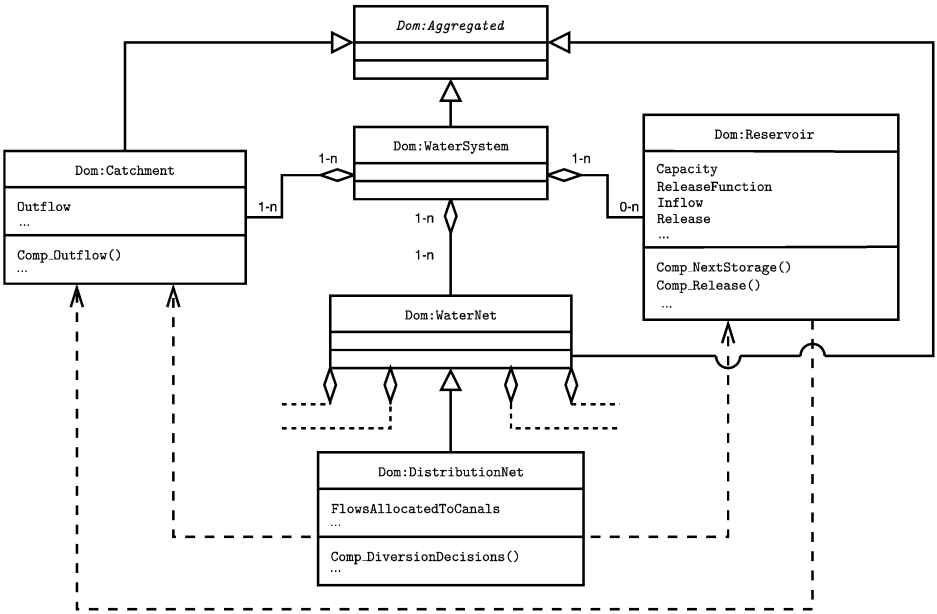


Figure 24.2: A part of the class diagram that defines the domain of a water system `WaterSystemDomain`.

that appears next to it. For example `Dom:WaterSystem` is composed of 0 to n objects of the class `Dom:Reservoir`, while an object `Dom:Reservoir` can belong to one or more $(1-n)$ objects of `Dom:WaterSystem`. A dashed arrow describes an *association relationship*: the class from which the arrow leads makes use of information provided by the class to which the arrow points. For example `Dom:Reservoir` must know the Outflow from `Dom:Catchment` to compute the `Release`, while `Dom:DistributionNet` needs the `Release` from a `Dom:Reservoir` and/or the `Outflow` from a `Dom:Catchment` to compute the `FlowsAllocatedToCanals` through the `Compute_DiversionDecisions()` method that implements a distribution policy.

To define, only once, attributes and methods that are common to several classes, particular classes, called *abstract*, are used from which objects cannot be instantiated. An example is the class *`Dom:Aggregated`* in Figure 24.2, which allows us to define the attribute 'map of the links' that is used by all its descendants to interconnect some of the attributes of the basic and/or aggregated domains of which its descendants are constituted (e.g. the link that specifies the identity between the 'Outflow' of `Dom:Catchment`) and the 'Inflow' of `Dom:Reservoir` in a `Dom:WaterSystem` that describes a water system composed of a reservoir and the catchment that feeds it). It is standard practice to write the names of abstract classes in italics, as we have done through this paragraph.

24.2.2 Software components

The search for higher productivity in the process of software development has identified the necessity to produce reusable software modules, in the same way as integrated electronic components are. This is the need that led to the creation of the term *software components*.

The need was highlighted by the advent of Internet, which requires that software be independent from the operation platform. Szyperski et al. (2002) defines the properties of a *component* as follows:

- it can be distributed as a stand-alone module;
- it can be combined with other components;
- its state should not be observable from the outside.

The first property requires the component to be clearly separated from the software environment in which it is used and for it not to be decomposable into constituents. The second requires that it communicate with the program in which it is utilized and with other software components through a clear, well-defined interface. The third distinguishes it from an object, which is characterized, instead, by the accessibility of its attributes. A component can act upon an object by modifying its state through the methods peculiar to the class to which the object belongs.

Object-Oriented Programming and Component-Oriented Programming go hand-in-hand, since a software component is basically a software object which can be accessed only through its interface. The interface of a software component is the set of methods that allow the component to interact with other software entities.

For the MODSS designer the components represent a valid solution for the implementation of its tools, because they allow him/her to separate the models from the algorithms that operate upon them. The possibility of this independence was theorized for the first time by Dolk and Kottemann (1993) who, influenced by the research of Geoffrion (1987) on structured modelling, introduced the concept of *model integration*, and were the first to identify the necessity of separating models, data, and solution algorithms.

24.2.3 Ontologies

In the previous paragraphs we have seen how Object-Oriented Programming and Component-Oriented Programming enhance software reusability and modularity. This can be very advantageous when complex integrated models need to be organised in smaller and simpler modelling units, as is commonplace in software frameworks for integrated modelling. However, the effective integration of the MODSS software modules, across different spatial and temporal scales, still remains a challenging task, which requires a great deal of pre-existing knowledge. The reason is easy to grasp, the interface of a software component can provide a detailed specification of the data type used to represent a model variable, but it does not tell us anything about the *meaning* of it. The name of the variable can be self-explicative, such as `Outflow`, but we do not know where the outflow has been measured, which dimensional unit is used to express it, which measurement unit, and which sampling rate, has been adopted to measure it. This list can be even longer. Therefore, unambiguous documentation becomes essential when we really want to integrate the MODSS software modules.

Software engineering techniques need then to be supported by knowledge representation ones, in order to associate semantics with model management procedures, to support all the phases of modelling, from the construction to linkages of model chains and the execution of experiments such as simulations, calibration and validation, sensitivity analyses and optimisation. All these activities are knowledge intensive and they must be adequately supported by the software infrastructure. The *Semantic Web* (Antoniou and van Harmelen,

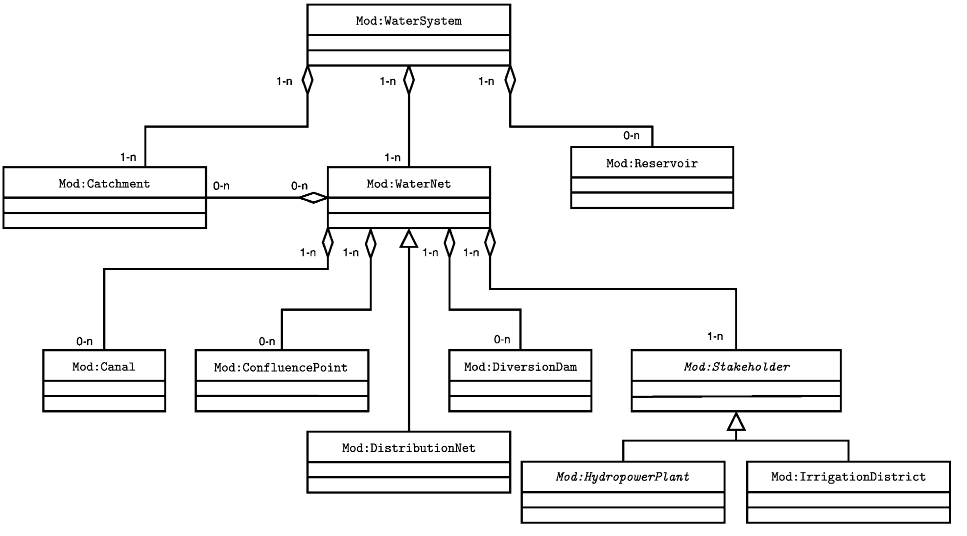


Figure 24.3: The class diagram that expresses the inheritance relationships present in the ontology of `Mod:WaterSystem`; the association relationships are not shown to make the figure simpler.

2004) initiative has spawned recent developments in knowledge representation and its use in semantically enriching the content of WEB sites. Ontologies have thus become a popular way to represent knowledge and a number of software tools for the creation and management of ontologies are being developed and gaining an increasingly wide diffusion.

An *ontology* is a model of a knowledge domain. The OpenCyC project[7] defines an ontology as "that part of the system which specifies what things exists and what is true about them". An ontology is expressed through an exhaustive and rigorous formalism that specifies *concepts*, *individuals*, *relationships*, *functions* and *axioms*. The concepts are the set of abstractions used to describe the objects of the knowledge domain to be described (in some sense they are analogous to classes of OOP). The individuals are instances of concepts (and are analogous to objects). The relationships are defined on the concepts, and the functions are applied to concepts and must return concepts. Finally, the axioms define a set of first-order, logic predicates, which constrain concepts, relationships and functions (Gruber, 1993).

Several attempts have been tried to use UML as a graphical model to define an ontology. We can draw a parallelism between ontologies and UML, since a class diagram can be used to represent the concepts and their relationships of inheritance, aggregation and association. For example, the diagram in Figure 24.3 explains the composition of the `Mod:WaterSystem` and so constitutes part of its definition, just as the diagram in Figure 24.1 can be seen as a definition of *`Mod:HydropowerPlant`*. Nevertheless, a class diagram is not sufficient to provide a complete semantical description of a knowledge domain, i.e. to express an ontology, since it cannot represent the rules that govern the interactions among the individuals and their attributes. Baclawski et al. (2002) list the compatibilities and incompatibilities between ontologies and UML.

[7] http://www.opencyc.org, visited September 2006.

Ontologies can play a fundamental role in the specification and formalization of the knowledge required to successfully use and operate a MODSS.

- Ontologies can be used to *declaratively* represent a mathematical model, as opposed to the imperative representation we are used to when coding the model equations into a programming language. The advantage of a declarative representation is that the fundamental concepts characterizing a model (equations, variables and parameters) are expressed independently from the algorithm used to produce results from them (such as a numerical integration routine). For example, an ontology may provide the definition of the storage and inflow concepts, and a model is then declared by defining specific storage and inflow variables with their names, initial values and the equations expressing their relationship. Such a declaration contains enough information to enable a software execution environment to simulate the behaviour of the model over a temporal and spatial horizon. Moreover, thanks to the rich meaning made possible by current ontology formalism, the execution environment can be programmed to connect properly models to data, and feed quantities computed by simulation to other models in the same environment, without risking the error of matching variables to data that specify different natural-world entities.

- Ontologies have also proven a valuable formalism to represent the structure of a database, thus exposing the metadata in a semantically rich way (Berners-Lee, 1998). Tools, such as D2R (Bizer, 2003) have been developed to automatically map database schemes into ontologies. The semantic annotation of databases becomes an essential tool to facilitate the preparation of data for models.

- Ontologies can be used to define the workflow of a participated and integrated decision-making process. All the steps of the PIP procedure can be formalized in an ontology, in order to deliver an unambiguous formalization of the workflow to be followed in the process development (Bowers and Ludascher, 2004). A valuable example of such an application is provided by MOST (Kassahun et al., 2004), the Modelling Support Tool, developed in the context of the HarmoniQUA EU project, which provides a guided workflow to the different stages of model development (data collection, model identification, etc.), where the workflow has been formalized and structured thanks to a dedicated ontology.

In conclusion, ontologies are powerful knowledge representation formalisms which are gaining momentum thanks to the fast development of new tools for their management and their use within the context of the Semantic Web initiative. They provide an open standard for knowledge representation, and they promise to deliver an essential infrastructure for the development of integrated modelling applications, which are in the heart of MODSSs.

24.3 Architecture

Until now we have presented the functionality that a MODSS must provide to the user and the software structures that are appropriate for its design. We can now describe a software architecture that achieves all this. It has been inspired by the pioneering contribution of Guariso and Werthner (1989), who were the first to structure a specialization of DSSs dedicated to environmental problems: the EDSSs (*Environmental Decision Support Systems*).

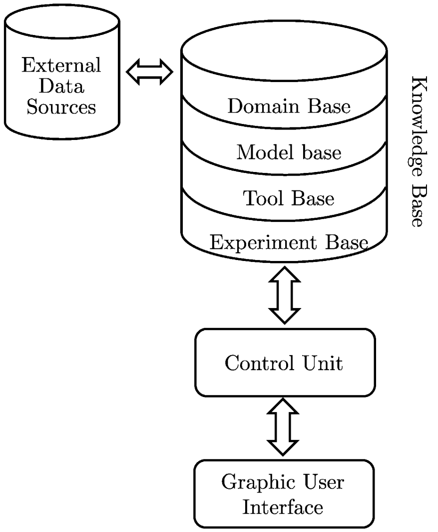


Figure 24.4: The architecture of a MODSS.

The MODSS we will present is a specialization of an EDSS to water resources and thus their architectures are partly similar.

Figure 24.4 shows how the user interacts with a MODSS through the *Graphic User Interface* (GUI), which in turn controls and is controlled by the *Control Unit* (CU). This latter has access to the *Knowledge Base* (KB), which can possibly access *External Data Sources* (EXS). In its turn the *Knowledge Base* is composed of the *Domain Base*, the *Model Base*, the *Tool Base* and the *Experiment Base*. Let us look at these units one at a time.

24.3.1 External Data Sources

The MODSS must access heterogeneous data sets, kept in databases that are external to its architecture and that contain raw data, such as measurements, news, time series, geographic maps and characteristic parameters. The MODSS does not provide the instruments to manage these data directly, but instead it interfaces with the external sources through software components such as ODBC and JDBC (*Open DataBase Connectivity* and *Java DataBase Connectivity*) that allow the management of data inside the MODSS independent of the platforms that host the External Data Sources.

24.3.2 The Knowledge Base

In the light of what we have learned in the first four Parts of this book, we can understand how prior knowledge, and knowledge that is progressively produced by the decision-making process, can be classified into four main categories: domains, models, tools and experiments.

24.3.2.1 Domains

A *meta-domain* is a structure that organizes the data describing the characteristics of an entity (e.g. a component of the water system), the potential time series related to it, recorded by someone or by a measuring station. Creating an instance of a meta-domain (in short *domain*) is the first level in modelling reality, which does not yet require the expression of any assumption on the forms of the mathematical relationships that exist between the variables that describe the entity being considered, but only the statement of which data are available and the specification of how they are represented.

Meta-domains should be organised according to a classification hierarchy, following the OOP Paradigm. This means, for example, that a particular historical time series is always associated with a measurement station, whose attributes are not only the time series itself but also geo-referenced information, which can be visualized on a GIS (*Geographical Information System*), and the name and telephone number of the person responsible for its management. The measurement stations belong to the domain of the catchment basin in which they are located, which in turn is part of the water system. Each of these domains (objects) is defined by a meta-domain (class), which specifies its type (for example water system, catchment basin, canal, reservoir, Stakeholder, measurement station) and by the operations which can be carried out upon its attributes.

The domains are subdivided into basic and aggregated. The *basic domains* are utilized to memorize the data that the user of the MODSS considers to be 'basic' (e.g. the series measured in a station, a Stakeholder's data, etc.) and they constitute the building blocks with which the entire water system and the Project itself can be represented. The first step towards the formalization of the system is to create instances of the basic domains. An *aggregated domain* is made by the aggregation of basic domains. Examples of composite domains are a catchment (with its measurement stations), a distribution network (with its sub-catchments, canals and diversion dams) and a water system.

24.3.2.2 Models

The domains, representing an embryo of a model (as they structure the attributes in a way that prefigures future modelling choices), perform the task of logically separating the physical world from the mathematical one, in order that different models can be associated with the same domain.

As with the domains, it is useful to define the classes of models (*meta-models*) and the *model instances* (in short *models*). An instance of a model is created when a meta-model is associated with the instance of a domain, by identifying its structure and calibrating its parameters with the data contained in the domain. A model is an object that encapsulates in its attributes the parameters and in its methods the functions of the mathematical model that it implements. The usefulness of the logical division between Domains and Models is evident when we observe that it allows us to represent the same domain with different models, so that it is easy to experiment with alternative modelling solutions, minimizing at the same time the risk of making mistakes when connecting the models to the data. For example (Figure 24.5), to model the domain of the Toce (`Dom:ToceCatchment`) one could adopt two alternative meta-models: an AR(1) or an ARX$(1,1)$, which are instantiated by estimating the values of their parameters.

24.3.2.3 Tools

We have seen how all the operations on domains and models (both meta and instantiated) are performed by *tools*. They are software components that implement procedures or algorithms and operate on meta-domains (meta-models), or on instantiated domains (models),

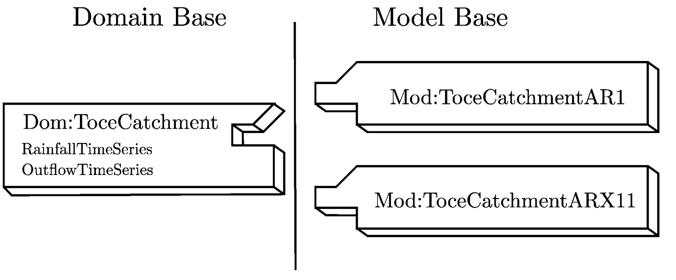


Figure 24.5: Associating different models to the same domain.

in order to create new domains (models) or to modify existing domains (models). For example, Algorithm 1 (page 299) can be implemented as a tool that acts on an instantiated decision model and produces a prescriptive model, from which, at each time instant t, one can obtain the control u_t^* (by applying it a tool that implements equation (12.9)) or the set of controls $M_t^*(x_t)$ (by applying it a tool that implements equation (12.38)).

A similar approach is adopted by commercial software for the solution of mathematical problems (linear and non-linear programming, etc.) such as GAMS[8] and AMPL.[9] These programs use software components, provided by third parties, that implement the instruments, or the algorithms, for the solution of mathematical problems.

24.3.2.4 Experiments

The *experiment* metaphor allows us to describe and prescribe the interaction between the user and the MODSS. The user, to carry out an experiment, first chooses the necessary ingredients (domains and/or models), then, from those available, (s)he selects the tool that is appropriate for the activity that (s)he wants to conduct. For example, to design the regulation policy for a water system, the user must first of all define a decision model that expresses the Policy Design Problem. For this, through the instrument `DecisionModelDefinition`, (s)he establishes which model of that water system to use, defines the objectives and the scenarios (which means to choose data series or models for the disturbances). The decision model that is so obtained is then stored in the Model Base for documentation or future use. Thereby the decision model definition experiment is concluded. Then the user chooses the tool (algorithm) to apply to the model that has just been defined. For example, if (s)he wanted to obtain a set-valued, off-line and adverse to risk policy, (s)he would have to choose the solver of an off-line, set-valued Wald Problem. In this way (s)he would obtain a prescriptive model (i.e. a pair decision model, Bellman function), that would be stored in the Model Base so that it could be used in the future. Another experiment is thus concluded.

A *meta-experiment* contains:

- the information about how an experiment should be conducted (for example the conditions to be satisfied for calibrating an ARX model of a catchment), and which results must be stored in the Knowledge Base at its completion for documentation or successive uses;

[8] GAMS Home Page, http://www.gams.com, last visited September 2006.

[9] AMPL Modelling Language for Mathematical Programming, http://www.ampl.com, last visited September 2006.

- the information about how the sequence of experiments will take place.

A meta-experiment therefore contains the procedural information that the Control Unit (see below) uses to guide the user in the execution of his/her tasks, so that it acts as a *Workflow Management System*, easily modifiable with the evolution of knowledge.

An *instance of an experiment* (in short *experiment*) instead contains information about how it was conducted (for example, in the case of a model calibration experiment, the catchment concerned, the data series used, the order of the model, who carried out the calibration and when, and which model was produced).

Thank to experiments it is easy to satisfy the requirements of transparency and retraceability.

24.3.2.5 A brief review

The *Knowledge Base* (KB) contains:

- The domains and the meta-domains, organized in the *Domain Base*. They provide the basic infrastructure for structuring data and knowledge that will be used by the models.

- The models and meta-models that describe the functionality of the components and aggregates of the water system. They are organized in the *Model Base*.

- The tools that, operating on the models, allow us to produce new domains (models) or to modify the existing ones. They are organized in the *Tool Base*.

- The procedures that define the sequence of the operations that tools carry out on domains and models (meta-experiments) and the documentation on the experiments that have been carried out. This knowledge is organized in the *Experiment Base*.

24.3.3 The Control Unit

The Control Unit (CU) can operate in two different ways:

- in the *execution environment* it
  - supervises the execution of the experiments following the instructions contained in the meta-experiments of the Experiment Base;
  - suggests to the user the correct algorithm or the alternative ones for completing a given task;
  - checks the integrity and the consistency of the data being used;
  - memorizes the choices that have been made during an experiment in the Experiment Base,

  in order to support both the planning level (in which case the CU instantiates what the user perceives as MODSS/P) and the management level (in which case the CU instantiates what the user perceives as MODSS/M).

- In the *development environment* an expert Analyst can define new classes, either domains (meta-domains), or models (meta-models), or experiments (meta-experiments) or new tools. It is thanks to the presence of this environment that the MODSS is flexible and can be easily extended to consider components that have not yet been defined, new tools or even modifications to the decision-making procedure.

The CU is therefore the heart of the MODSS. It manages the exchange of information with the user through the User Interface, and guarantees transparency, repeatability of the decision-making process, a high level of flexibility and a simple maintenance of the MODSS.

24.3.4 The Graphic User Interface

The Graphic User Interface (GUI) is the window through which the user sees the MODSS. It should help the user to apply the system without error, giving advice when (s)he is in doubt, and explanations of the logic underlying certain structural and implementation choices. It must also permit the user to perceive that (s)he is in control of the MODSS and all of its operations.

In order to reduce development times, the GUI should be realized in *development environments* that utilize beans (for example, Eclipse SDK[10] and Microsoft's Visual Studio .NET[11] provide such an environment).

We believe that it would be useless and too vague to speak about GUI just in abstract terms, and it is didactically more efficient to describe one. So we will present TWOLE's GUI, i.e. the GUI of the MODSS used in the Verbano Project (see PRACTICE), even if, since it was developed in 1998–1999, it does not yet allow several of the possibilities that have been presented in this chapter.

24.3.4.1 TWOLE's GUI

To start a work session, the user must identify him/herself (so that all of the experiments that will be conducted are attributed to him/her) and identify the environment and level at which (s)he wishes to operate. Let us suppose that (s)he chooses the execution environment and the planning level. (S)he will be asked to select a Project from those available or to define a new one, because the MODSS/P is conceived for working on several Projects, since its typical use is by Water Agencies.

In response to the user's choice, TWOLE opens its *desktop*, which is inspired by the *toolbox and folder* metaphor. All the operations (experiments) are carried out by specifying an element (a class or an object) in a (*folder*) and in response TWOLE presents the 'tool box' that can be applied to it. A navigation-tree is visualized on the left of the desktop (see Figure 24.6), as in a classical *file-system* manager, with the difference that, in general, each node does not represent only a folder, but a class or an object too. The user may act on an object by double-clicking on the corresponding node, or open the associated folder by clicking on it. For example, in Figure 24.6 there appear the nodes of the first level of the navigation-tree, which represent meta-domains. The folder of the catchment meta-domain (`Catchments`) is open and shows its contents: the domains of the catchments that have already been instantiated. The folder associated to the domain of the River Toce's catchment `Dom:ToceCatchment` is open and shows its contents: the two models that have already been instantiated for that catchment. If the user chooses the model `Mod:ToceCatchmentARX11` by double-clicking on it, in response, TWOLE presents him/her with a form that shows the attributes of the model and offers him/her the possibility of validating it or using it for forecasting. If, instead, (s)he wants to identify a new model, (s)he has to click on `Dom:ToceCatchment` and a form appears to him/her to select the model type. If (s)he selects a model of the ARMAX class, TWOLE provides him/her with the form in Figure 24.7, where (s)he may specify the attributes of the model (e.g. the order and the periodicity) and start its calibration.

[10] http://www.eclipse.org/platform, last visited September 2006.

[11] http://msdn.microsoft.com/vstudio/, last visited September 2006.

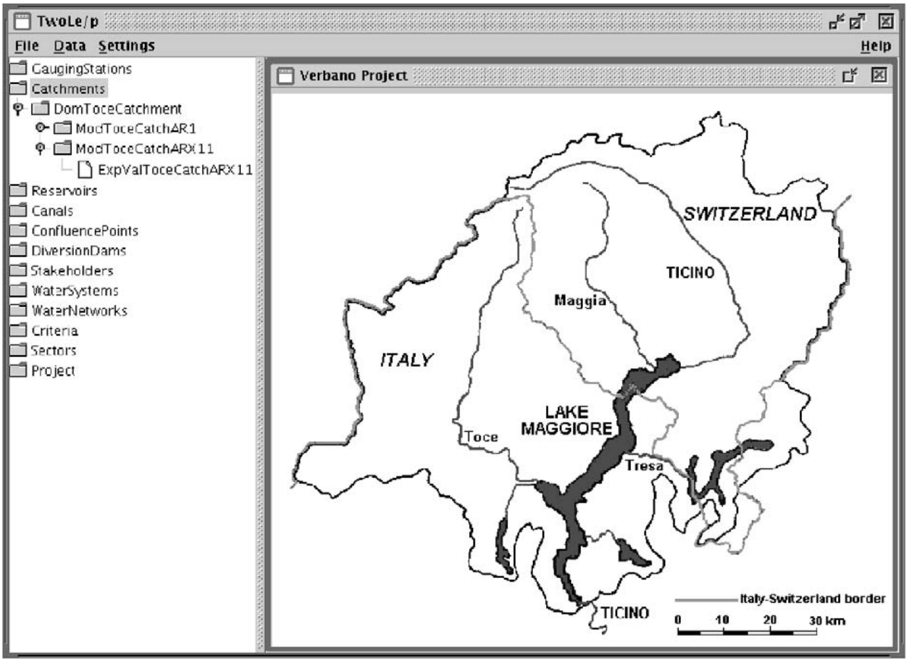


Figure 24.6: The folder navigation-tree and the menu for choosing an instrument.

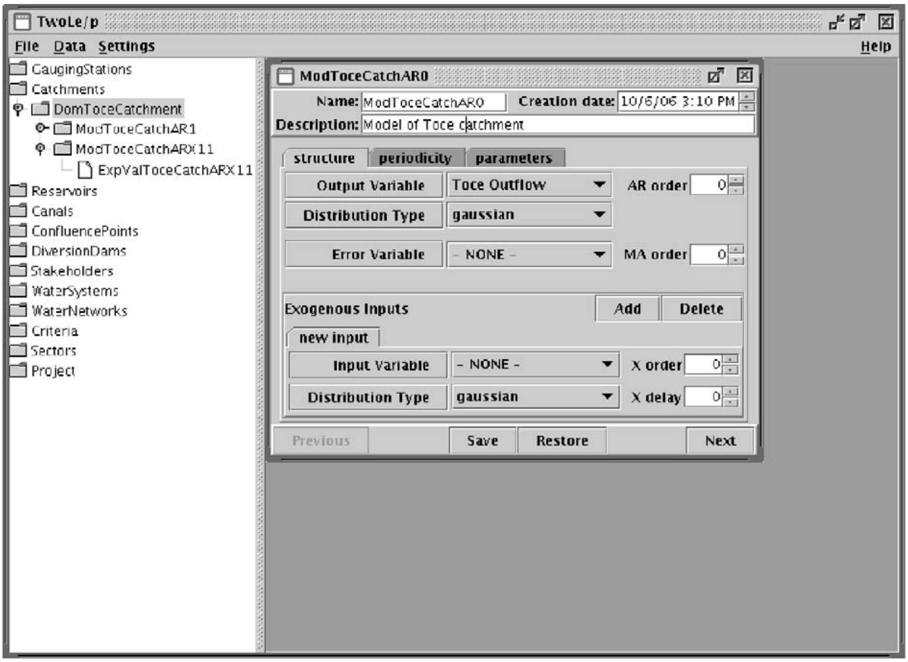


Figure 24.7: The tool for the identification of an ARMAX model of a catchment.

Table 24.1. The levels in the TWOLE's navigation-tree

Levels				
1 Meta-domain	2 Instances of domain	3 Instances of model	4	5
Basic	Basic	Basic	Validation exper. Prediction exper.	
Aggregated	Aggregated	Aggregated	Simulation exper. Decision model	...
Sector	Sector	Eval. Hierachy		
Project	Project	Project Hierachy		

The order in which the levels of the folder navigation-tree follow one another is described in Table 24.1. The nodes at the first level are folders representing meta-domains, each of which envelopes the domains already instantiated of its class (e.g. the folder of the class `Reservoirs` contains the domains of the reservoirs already instanced in the Project considered). The node of the second level represents domains and, if models have been instantiated for one of them, the associated folders contain them. In turn (third level) the folders associated one of these latter contain, if they have already produced, other entities that, as shown in the table, differ according to the type of model considered. For example, the folder associated to an instance of `Mod:Aggregated` (e.g. a model of a water system or a distribution network) contains simulation experiments or decision models (`Mod:Decision`). The folder associated to these latter, in turn, contain policies (`Mod:Prescriptive`), and the folders of one of these, in turn, simulation experiments conducted with it or .... We stop here because the reader should have got the idea by now.

The experiments in which a new object is instantiated (e.g. creating a new domain, calibrating a model, solving a Design Problem, ...) are one-to-one related with the object itself and thereby they are stored in the node that represents the object. On the contrary, the experiments in which an object is used to produce new information (e.g. validating and simulating) are stored in the folder associated with that object (e.g. the validation of a model appears in the folder that represents that model). Their list can be seen by clicking on the node that represents that object and the results produced by one of them can be inspected by double-clicking on the node that represents it. In reply, TWOLE presents the tools that are suited for the inspection. For example, by double-clicking on a node that represents the solution of an off-line, Optimal-Control Problem (a decision model) a visualizer for *Bellman functions* is presented (Figure 24.8), by means of which the user can explore the solution of the Problem (which, remember, is a Bellman function).

The navigation tree permits us to satisfy many of the requisites detailed in the preceding paragraphs:

- it is the user's point of view on the KB;
- it makes the system transparent, traceable and repeatable because it makes clear the links between the different objects, which data (or objects) each of them uses and who created it, when and how;

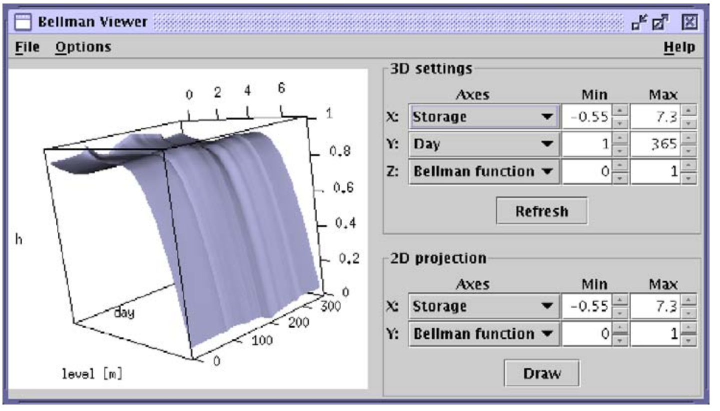


Figure 24.8: The visualizer of the Bellman function.

- it allows one easily to effect Sensitivity Analysis, in order to reply to questions such as "What if?", as in the example proposed on page 505. In that example, to respond to the hydroelectric user's request, the Analyst must simply double right click first on the node which represents the result (experiment) of the simulation with which the effects of the policy under examination have been estimated and then on the node which represents the domain of the user's hydropower plant. In reply TWOLE shows the path that links the two nodes, which, starting from the domain of the power plant, reaches the experiment: this path will move via the model of the power plant, the model of the water system, the decision model with which the policy has been designed, and the prescriptive model based on that policy, to finally reach the experiment. At the same time TWOLE creates a working copy of all these objects. The Analyst then open the power plant domain and changes the price per kilowatt-hour in it. With a simple command (s)he finally ask TWOLE to repeat all the experiments along the path. The new value of the user's index is thus computed, together with estimates of all the other effects that the price change induces.[12]

24.4 Some prospects on distributed architectures

The fundamental characteristic of a MODSS is its *multidisciplinarity*, which is imposed by the multiplicity of the objectives to be considered, each of which represents sectors which are often far apart. Think about the conservation of wildlife areas and the production of hydroelectric energy. The MODSS represents a valid support tool when its Knowledge Base contains appropriate information, such as specialized models, but this requires that many Experts participate in its development. As a consequence, the cost of its realization turns out to be high and, often, exceeds the available budget. Thus one must be satisfied with 'home-made' models, but this inevitably induces a loss of quality. The solution would be to access external Model Banks from which to draw the desired model. The architecture that we have presented makes this a concrete possibility if the specifications of the interface

[12] This service is not yet implemented in the present version of **twole** (v.2.beta, September 2006), but it will be implemented in the one that is currently being developed.

were to be defined in a standard way at an international level. In 2003 the European Union launched a programme (Harmoni-CA[13]) with the aim of exploring this possibility.

The difficulties do not finish here, however. Even if the Design Problem had been formalized by assembling models provided by third parties, its solution could be too onerous for the machine that the user is equipped with, but the possibility to distribute the computations on a set of computers through the WEB opens new prospects. Lastly, the negotiations, which would require the contemporary presence of all the actors, could greatly benefit from the use of ICT.

We will look at these three points in greater detail: model integration on the WEB, distributed computing, and distributed participation.

24.4.1 Model integration on the WEB

Thanks to software architecture based on WEB services, it is possible to access models or tools (we will indicate them in general as applications) that are installed on remote sites. The first software architecture that was explicitly conceived to make applications on the net available was CORBA (*Common Object Request Broker Architecture*[14]), from which development environments J2EE[15] from Sun Microsystems and .NET[16] from Microsoft took inspiration. They permit one to create applications that are installed on an *application server*: a machine connected on-line that makes them available to remote clients.

In a context of this type, the MODSS becomes a remote client of the *application server*. This latter makes public the interfaces of the application that it offers by using the XML format[17] which allows the MODSS to interpret the required input and output formats and dialogue automatically with the *application server*, without having a programmer that has to write complicated *drivers* for converting data. This is possible by defining non-ambiguous communication interfaces through ontologies. An example will help clarify the idea. Let us suppose that, on the WEB, we have bought a module for the solution of Planning Problems. To integrate it in a MODSS we need a programmer to examine the nature of the data that the model uses and map them into a class of the MODSS. This operation is complex, boring and repetitive for a human being, but it can be made automatic if the designer of the module defines the semantics of each one of the data used through an ontology. Automatic conversion software could then establish the necessary connections without requiring human intervention. Research into these solutions today is very active and recently the *Semantic Web*[18] was defined, with the aim of establishing shared ontologies in many scientific fields.

24.4.2 Distributed computing

In the preceding section we saw how it would be possible to 'delegate' the execution of specialist computation to external services, in a completely transparent way. Notwithstanding the fee to pay for the service required, this would cost less than acquiring the specialist software that would be otherwise necessary. The advantages that the evolution of Internet would bring do not finish here, however. It will be possible to parcel out onerous computation to

[13] http://www.harmoni-ca.info, visited September 2006.
[14] OMG's CORBA WEB site, http://www.corba.org/, visited September 2006.
[15] Java 2 Platform, Enterprise Edition, http://java.sun.com/j2ee/, visited September 2006.
[16] Microsoft .NET, http://www.microsoft.com/net, http://java.sun.com/j2ee/, visited September 2006.
[17] Extensible Markup Language, http://www.w3.org/XML/, visited September 2006.
[18] SemanticWeb.org, http://www.semanticweb.org, visited September 2006.

external machines to multiply the computational power and, as a consequence, reduce the waiting time to obtain the solution. This computation method is called *grid computing* and its diffusion is advancing rapidly, even if today it is limited to a few specialized applications (search for prime numbers,[19] folding proteins,[20] etc.). Actually, this technology is coming out of research laboratories to reach new user categories, thanks to software architectures[21] that facilitate the development and the implementation of *clusters* of computers.

For the application of a MODSS for the management of water resources, the possibility to carry out the Design Project in a distributed way is particularly attractive when the water system model has an elevated dimensionality, e.g. when the management of a network of reservoirs is considered. Today the solution of this latter problem is almost impracticable on the machines that the users normally have, when the networks contain more than three reservoirs.

24.4.3 Distributed participation

Distributed participation is interesting above all in the *Evaluation* and *Comparison* phases. To allow communication among several people, the architecture of the MODSS must be of a *client–server* type. Such a MODSS will allow the Decision Makers and the Stakeholders to connect to the *server*, which houses the MODSS_SERVER through the MODSS_CLIENT, through which they can carry out local operations on the objects downloaded from the *server* to which they will then upload the updated objects. The MODSS_SERVER will take care of the coordination of the decision-making process. For example, TWOLE/M implements this architecture for managing the daily decision-making process and the Stockholders can participate remotely in decision-making by means of ‘services’ offered by the TWOLE/M_SERVER through different media: desktop computers or cell-phones.

Negotiation Once the Analyst has designed the alternatives and made her effects public on the *server*, negotiations can begin and all of the Stakeholders will have access to the same information. The negotiation process is developed as a virtual meeting in which the Decision Makers and Stakeholders participate in two alternative ways: synchronous or asynchronous (Section 21.4). In the first case they are connected at the same time in video conference, or simply in a virtual conference room (*chat room*); in the second, each one can connect at the time (s)he prefers.

Managing access The management of the decision-making process that involves several actors requires the definition of the ways with which they can access the objects and components of the MODSS. The system must therefore include an infrastructure that assigns a precise role to each actor (Stakeholders, the Analyst, and the Decision Makers), to which there correspond operations permitted and objects to which (s)he has access.

24.5 Conclusion

In this chapter we have examined how we can, and should, design software systems that support participatory and integrated projects. A software system becomes an essential tool

[19] Mersenne Prime Search, http://www.mersenne.org/prime.htm, visited September 2006.

[20] Folding @ Home, http://folding.stanford.edu, visited September 2006.

[21] The Beowulf Cluster Site, http://www.beowulf.org; Apple ACG Xgrid, http://www.apple.com/acg/xgrid/; DataGrid Project, http://web.datagrid.cnr.it, visited September 2006.

to support the decision-making process in all its phases, and it is therefore important to clearly specify its requirements, its design and software architecture. Modern software engineering techniques greatly facilitate the development of such software systems, but the current practice is to develop a new system for each new project class. For instance, while the TWOLE can be easily customized to solve new water management projects, it would require considerable changes to be able to solve agricultural management problems. Yet, water management and agricultural management are close relatives, and a great part of the knowledge contained in TWOLE could be very useful for agricultural systems managers. This is the challenge for the future, to make available knowledge (models, data, processes) which are really re-usable across domains. This will pave the way to a new breed of software systems for developing Integrated Assessment Projects.

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